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Permutations and Combinations

Lecture no. 01

Counting principle $\Rightarrow$

1) Fundamental Principle of Addition:

If we have to perform only one of the jobs at a time then we use this principle & its keywords are: "Either or / only one / Any one"

2) Fundamental Principle of Multiplication:

If we have to perform all the jobs at a time then we use this principle & its keywords are: "AND / Both / All"

eg) if there are 10 Boys and 8 Girls in a class then in how many ways we can select:

1) Either a boy or a girl

$\Rightarrow B_1, B_2, B_3, \dots, B_{10}, G_1, G_2, G_3, \dots, G_8$

$\Rightarrow 10 \text{ ways} + 8 \text{ ways} = 18 \text{ ways}$

2) A boy & a girl =

$B_1, G_1, B_2, G_1, B_3, G_1, \dots, B_8, G_1$

$B_1, G_2, B_2, G_2, B_3, G_2, \dots, B_8, G_2$

$B_1, G_3, B_2, G_3, B_3, G_3, \dots, B_8, G_3$

$\vdots$

$\vdots$

$B_{10}, G_8, \dots, B_1, G_8$

80 ways are possible

$\Rightarrow 10 \text{ ways} \times 8 \text{ ways} = 80 \text{ ways}$

Q.) There are 3 student appearing for math scholarship test 4 student " Physics and 5 student appearing for Chemistry then in how many ways we can award these scholarships?

Sol.) 3 ways, 4 ways, 5 ways = 60 ways

2.) one of these scholarship can be awarded!

Sol: 3 ways + 4 ways + 5 ways = 12 ways

Q.) There are two friends living in a hostel and sharing the almirah there are 5 shirt & 4 skirt in there wardrobe if they both want to dress up for a party then in how many ways this can be done?

Sol.) Total ways to dress up = $\left(\begin{matrix} 5 \\ 1 \end{matrix} \right) \times \left(\begin{matrix} 4 \\ 1 \end{matrix} \right)$
 $= (5 \times 4 \text{ ways}) \times (4 \times 3 \text{ ways})$
 $= 20 \times 12$
 $= \underline{240 \text{ ways}}$

Q.) There are 5 letter & 4 letter boxes then in how many ways these letter can be posted?

Sol.) Total ways of posting letter = $\frac{4!}{1!} \times \frac{4!}{1!} \times \frac{4!}{1!} \times \frac{4!}{1!} \times \frac{4!}{1!}$
 (R.A.)
 $= 4^5$ Ans. =

study time

Chapter 11

Q) There are 3 rings and 4 fingers then in how many ways we can wear these rings if

(i) There is no restriction $\Rightarrow$ (RA)

$$\frac{4}{P_1} \times \frac{4}{P_2} \times \frac{4}{P_3} = 4^3 \underline{64}$$

(ii) Atmost one ring in a single finger = (RNA)

$$\frac{4}{P_1} \times \frac{3}{P_2} \times \frac{2}{P_3} = 24 \text{ ways}$$

Q) How many 3 digit number can be formed using odd digit if

(i) Repetition is allowed: $\frac{5}{P_1} \times \frac{5}{P_2} \times \frac{5}{P_3} = 125$

(ii) RNA = $\frac{5}{P_1} \times \frac{4}{P_2} \times \frac{3}{P_3} = 60$

Q) How many 3 letter words can be formed using vowels? with or without meaning if

(i) RA = $\frac{5}{P_1} \times \frac{5}{P_2} \times \frac{5}{P_3} = 5^3 = 125 \text{ ways}$

(ii) RNA = $\frac{5}{P_1} \times \frac{4}{P_2} \times \frac{3}{P_3} = 60$

Q) In how many ways 5 person can be seated on 8 chair? (RNA)

sol.) Total seating arrangement = $\frac{8}{P_1} \times \frac{7}{P_2} \times \frac{6}{P_3} \times \frac{5}{P_4} \times \frac{4}{P_5} = 6720$
 $\therefore 8P_5 =$

Q.) In how many ways 8 person can be seated on 8 chairs?

Sol.) Total seating arrangement = $\frac{8!}{8!} = 1$

Q.) In how many ways 8 person can be seated on 5 chairs?

Sol.) Wrong Question Ans = 0

⇒ Combination (only selection)

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

⇒ Permutation = (Selection & Arrangement both)

$${}^nP_r = \frac{n!}{(n-r)!} = \frac{n!}{(n-r)! \times r!} \times r!$$

$${}^nP_r = {}^nC_r \times r!$$

selection

Arrangement

Strong Points

- if $n > r$ & RNA the M. Rule = Perm. Rule
- if $n = r$ & RNA the M. Rule = Perm. Rule - factorial
- if RA then we have only use M. Rule.
- i.e. the concept of nC_r , nP_r , $r!$ is applicable only when RNA.
- Number of different arrangement of n different things at n places = $n! = n!$

only one

permutation

Lecture no. 02

⇒ Combination - if question is based on subsection only, then to count total no. of ways we use this principle as nC_r where

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

e.g. ${}^{11}C_3 = \frac{11 \times 10 \times 9}{3 \times 2 \times 1} = 165$

e.g. ${}^{12}C_4 = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} = 495$

e.g. ${}^{22}C_{15} = {}^{22}C_7 = \frac{22 \times 21 \times 20}{3 \times 2 \times 1} = 1540$

Note) ${}^nC_r = {}^nC_{n-r}$, ${}^nC_0 = {}^nC_n = 1$

e. ${}^nC_1 = {}^nC_{n-1} = n$

e.g. ${}^6C_1 = {}^6C_5 = 6$

⇒ Ex of combination.

Q:) if in a party there are 5 person then find possible no. of handshake in that party

(n-1)

= 10

ii) various handshake

(P_1, P_2)	(P_1, P_3)	(P_1, P_4)	(P_1, P_5)	(P_2, P_3)	(P_2, P_4)	(P_2, P_5)	(P_3, P_4)
(P_2, P_3)	(P_2, P_4)	(P_2, P_5)	(P_3, P_4)	(P_3, P_5)	(P_4, P_5)		
(P_3, P_4)	(P_3, P_5)	(P_4, P_5)					

(M-II) $\therefore$ counting is based on selection of two person at a time so total combinations

$$= {}^5C_2 = \frac{5 \times 4}{2 \times 1} = \frac{20}{2} = 10$$

Q:1) How many 2 letter words can be formed using without repetition using vowels

M-I

2nd.) = vowels: a, e, i, o, u

various 2 letter words

(ae) (ai) (ao) (au) = 20

(ea) (ei) (eo) (eu)

(ia) (ie) (io) (iu)

(oa) (oe) (oi) (ou)

(ua) (ue) (ui) (uo)

M-II Total 2 letter words (PNA)

$$= \frac{5}{P_1} \times \frac{4}{P_2} = 20 \text{ ways/words}$$

$$\Rightarrow = {}^5P_2 = 5 \times 4 = 20 \text{ words}$$

$$\Rightarrow = {}^5C_2 \times 2! = 20 \text{ words}$$

$\Rightarrow$ Concept of factorial is defined only for natural number.

#, Permutation: When counting is based on selection as well as arrangement also then to count total ways use nP_r where

$$n P_r = \frac{n!}{(n-r)!} = {}^n C_r \times r!$$

e.g) ${}^{11}P_3 = 11 \times 10 \times 9 = 990$

e.g) ${}^{21}P_4 = 21 \times 20 \times 19 \times 18 = 175,560$

i.e Perm = selection & Arrangement.

$\Rightarrow {}^nP_0 = 1, {}^nP_n = n!, {}^nP_1 = n$

$\Rightarrow$ Ex of Combination:

1) Question based on Handshake

2) " " " number of straight line

3) " " " number of Δ 's

4) " " " formation of committee

5) " " " of team, football team, cricket

$\Rightarrow$ Ex of permutation.

Whenever in a Question there is a feeling of interchanging things then use permutation.

$\Rightarrow$ For e.g.

formation of words.

" " number.

" " signals.

Counting of photographs.

Counting of seating arrangements.

Q:) 5 persons entered in a lift at ground floor in an 8 floor house then in how many ways they can leave the lift at

(i) any floor: $\frac{7}{1}, \frac{7}{2}, \frac{7}{3}, \frac{7}{4}, \frac{7}{5} = 7^5$ ways

(ii) at different floor:

$${}^7P_5 = {}^7C_5 \times 5! = 2520 \text{ ways}$$

Q:) How many straight lines can be formed using 8 non-collinear points and Δ

Ans: $\left\{ \begin{array}{l} \text{no. of straight line} = {}^8C_2 = 28 \\ \text{no. of } \Delta's = {}^8C_3 = 56 \end{array} \right.$

Q:) How many straight line & Δ 's can be formed using 8 points in which 5 are collinear.

Ans: $({}^5C_1 \times {}^3C_1) + ({}^3C_2) + 1$

$$= 15 + 3 + 1 = 19$$

M-2 No. of straight line

$${}^8C_2 - {}^5C_2 + 1 = 19 \Delta's$$

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(ii) No. of D's = (Two points from set 1 & one point from set 2.)
 or
 (1 point from set 1 & two points from set 2)
 or
 (3 points from set 2)

$$= \left({}^5C_2 \times {}^3C_1 \right) + \left({}^5C_1 \times {}^3C_2 \right) + \left({}^3C_3 \right) \quad \text{Set 2}$$

$$30 + 15 + 1 = 46 \quad \dots \dots \dots \text{Set 1}$$

M-2 Req. D's = ${}^8C_3 - {}^5C_3 = 46$

Q.) Out of 6 men & 4 ladies a committee of 5 person is to be formed then find the total no. of committees that can be formed if

(i) There is no restriction = ${}^{10}C_5 = 252$ ways.
 (RNA)

(ii) ~~At least~~ Exactly 2L are there = ${}^4C_2 \times {}^6C_3 = 120$

(iii) At least 2L are there = (2L & 3M) or (3L & 2M) or (4L & 1M) or (5L & 0M)
 $\left({}^4C_2 \times {}^6C_3 \right) + \left({}^4C_3 \times {}^6C_2 \right) + \left({}^4C_4 \times {}^6C_1 \right) = 186$

(iv) At most 2L are there =
 $\left({}^4C_2 \times {}^6C_3 \right) + \left({}^4C_1 \times {}^6C_4 \right) + \left({}^4C_0 \times {}^6C_5 \right)$
 = 186

Atleast one 'L' = Total - No lady

$$= {}^{10}C_5 - ({}^{02}C_5)$$

$$= 252 - ({}^4C_0 \times {}^6C_5) = 246$$

(v) Gents and majority :-

$({}^6C_5 \times {}^4C_0)$ same as part 4 = 156

(vi) Ladies are in majority :-

$({}^4C_4 \times {}^6C_1) + ({}^4C_3 \times {}^6C_2) + {}^4C_2$

= 66 committee

(vii) Atleast one ladies will be there.
 (Do your self.) = 246

= Lecture no. 03

Q:) In a party there are 66 handshakes then find the no of person is that party?

Sol:) $nC_2 = 66 = \frac{n(n-1)}{2 \times 1} = 66$

$= n(n-1) = 132 = 12 \times 11$

$= n(n-1) = 12 \times 11$

$n(n-1) = 12(12-1)$

$n = 12$

Note:) Atleast one = Total - None

Q:) Three dice are thrown simultaneously then find number of outcome in which at least one dice show digit '4'!
 (RA)

study time

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$$3.8.) \frac{6}{O_1} \times \frac{6}{O_2} \times \frac{6}{O_3} = 6^3 = 216$$

Total possible outcome

3.9) No. of outcome in which no die show digit

$$4.) \frac{5}{O_1} \times \frac{5}{O_2} \times \frac{5}{O_3} = 125$$

Number of outcome in which atleast one die show

Total - no die show digit 4

$$216 - 125 = 91 \text{ Ans.}$$

Q: How many 3 digit no. are there b/w 100 & 1000 s.t atleast one of the digit is '7'.

$$3.10.) \text{Total 3 digit no. b/w 100 \& 1000} = 900 \text{ Nos.}$$

No. in which none of the digit is '7' 8, 8, 8

$$(0, 1, 2, 3, 4, 5, 6, 8, 9) = 8 \times 8 \times 8$$

$$(0 \& 7 \text{ are not possible}) = 648$$

So, Req. 3 digit no.s in which atleast one digit

is '7' = Total - none

$$= 900 - 648$$

$$= 252 \text{ Ans.}$$

Summation of Numbers =

$$\text{Sum of } n \text{ digit nos} = (\text{Sum of digit}) \times \underbrace{(1111 \dots 1)}_{n \text{ times}} \times (n-1)$$

(formed by w/o repetition)

Note) This result also applicable when one of the digit is zero (but with proper logic).

Ex.) Find the sum of all 3 digit no that can be formed using 1, 2, 3 w/o Repetition!

Sol:) Total 3 digit no (P.N.A) = ${}^3P_3 = 3! = 6$ numbers

Various nos: 123
132
213
231
312
+ 321

M-2

$$\text{Req. sum} = (1+2+3) \times (111) \times (3-1)!$$

$$= 1332$$

Req. sum = 1332 ✓

Q:) Find the sum of all 5 digit no that can be formed using 1, 3, 5, 7, 9 w/o repetition!

Sol:) Total 5 digit no (P.N.A) ${}^5P_5 = 5! = 120$

13579
13597
13957
15793
97531

M-2

$$\text{Req. sum} = (1+3+5+7+9) \times (11111) \times (5-1)!$$

$$= 25 \times 11111 \times 24$$

$$= 6666600$$

Ans = 11

Q:) Find the sum of all 5 digit no that can be formed using the digit 0, 1, 2, 3, 4 w/o Repetition.

Sol:) Total no of 5 digit = 4 4 3 2 1
= 96 nos.

$$\text{Req. sum} = 2666640 - 66660$$

$$= 2599980$$

Derangement:

When no one goes at right place assigned for him then such types of arrangement are called Derangement.

⇒ If there are n person & n distinct places then

(i) Total no. of arrangements = $n!$ ways
(whether wrong or right) PMA

(ii) All correct arrangements = 1 ways

(iii) No. of Derangements = $n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + (-1)^n \frac{1}{n!} \right]$

⇒ Take care: These are applicable when PMA.

(eg) if there are 3 letter & 3 address envelopes then ~~total~~ arrangements will be = $3! = 6$
 $n=3$ so Derangement = $3! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right] = 2$ of.

Q. 6 student are appearing in an examination in which there are 6 directed chairs then find the no. of arrangement if

(i) There is no restriction.

$$= 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 \quad (\text{PMA})$$

$$= 720$$

(ii) Atleast two students are seated in a wrong chair.

sol.) Total arrangement - No. of arrangement in which ^{none} of the student is seated on a wrong chair.

$$= \text{Total} - \text{all correct}$$

$$720 - 1 = 719.$$

(iii) All the student are seated on their own
Chain = 1 way.

(iv) All the student are seated on a wrong
Chain = 1

No of De-arrangement.

Sol: 1.
$$6! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} \right]$$

$$= 360 - 120 + 30 - 6 + 1$$

$$= 265 \text{ Ans.}$$

Chart of wrong arrangement:

1) for $n=2$, No of De-arrangement = $2! \left[1 - \frac{1}{1!} + \frac{1}{2!} \right] = 1$

2) for $n=3$, " " " = $3! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right] = 2$

3) for $n=4$, " " " = $4! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right] = 9$

4) for $n=5$, " " " = " " " = 44

5) for $n=6$, " " " " " " = 265

(v) Find the no. of amagement in which exactly 4 person are to be seated on wrong chain

Sol: $({}^6C_4 \times 9) \times 1 \times 1 = 135$

(vi) Find the no. of amagement in which at least 4 person are to be seated on wrong chain.

Ans. $({}^6C_4 \times 9) \times 1 \times 1 + ({}^6C_5 \times 44) \times 1 + ({}^6C_6 \times 265) = 664$

Q.) There are 5 balls & 5 distinct cells then find the no. of arrangement in which Ball B_i is not placed in cell i & No cell remains empty?

Sol.) For $n=5$, No. of Derangement = 44

Note $\Rightarrow$ Above 44 Derangement do not include following types of derangement.

$\Rightarrow$ Repetition seems to be allowed so we can not count it using formula.

Lecture no. 04

$\Rightarrow$ Some imp points $\rightarrow$ (Based on Experience):

1.) Always together (Not separated) $\rightarrow$
Assume them as one unit with in bracket.

2.) All never together / All do not come together $\rightarrow$
Total - Always together.

3.) No two girl seated together $\rightarrow$
first arrange Boys

4.) Alternately (linear case) $\rightarrow$ two cases have ans.

Alternately (circular case) $\rightarrow$ only one case will give

Particular / fixed $\rightarrow$ No need to select & No need to arrange.

6) At least = Total - None.

7) At most $\rightarrow$ Include none also.

$\Rightarrow$ Fundamental Question 1:

Q:) 4 B & 4 G are to be seated in row then find the possible no. of arrangement if

(i) there is no restriction = $8!$ $\approx$ Max Ans

(ii) Boys are together & girls are together:

$$\frac{[4444]}{2} \frac{[GGGG]}{2} = 2! \times 4! \times 4! = 1152$$

(iii) All girls are not separated: -
girls want to be seated always together

$$\begin{matrix} B_1 & [4444] & B_2 & B_3 & B_4 \\ P_1 & P_2 & P_3 & P_4 & P_5 \end{matrix} = 5! \times 4! = 2880$$

(iv) All girls ^{are} never together:

$$\begin{aligned} & \text{Total - Always together} \\ & = 8! - (5! \times 4!) = 37440 \end{aligned}$$

(v) No two girls are seated together -
first arrange boys

$$\begin{matrix} B & B & B & B \\ P_1 & P_2 & P_3 & P_4 \end{matrix} = 4! \times {}^5P_4 = 2880$$

(Boys) (Girls)

i.e. favourable places for girls are 5
but girls are only 4 so they can arrange
by 5P_4 ways

(vi) Find the no. of arrangement in which boys & girls are seated Alternately.

Alternately $\begin{cases} \text{Case 1: } B G B G B G B G \rightarrow 4! \times 4! \\ \text{Case 2: } G B G B G B G B \rightarrow 4! \times 4! \end{cases}$

$$\text{Req. Ans} = 2 \times 4! \times 4! = 1152$$

Note: Part 5 is a particular case of part 4
 $\therefore$ Part 4 restriction are applicable only for 4 girls not for 3G & 2G

But in Part 5 restriction restriction are applicable not only for 4G but also of 3G & 2G.

$\Rightarrow$ Part 6 is a particular case of part 5 (I)

vii) if in 4G there are 2 Sisters and 1 Brother then find two sisters wants to be seated together =

$$\begin{array}{ccccccc} G & O & (S_1, S_2) & G & O & O & O \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} = \frac{7!}{0!} \times \frac{2!}{0!} = 10080$$

(G) (S)

viii) There is exactly one ^{Person} boy b/w two sisters =

$$\begin{array}{ccccccc} (S_1, P, S_2) & P & P & P & P & P & P \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} = \frac{6!}{0!} \times \frac{2!}{0!} \times \binom{6}{1} \times 1 = 8640$$

(ix) There is exactly one boy b/w two sisters =

$$\begin{array}{ccccccc} O & G & (S_1, O, S_2) & G & O & O & \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} = \frac{6!}{0!} \times \frac{2!}{0!} \times \binom{4}{1} \times 1 = 1760$$

(x) Two sisters are separated by one particular girl:-

$$\begin{matrix} 0 & G & (S_1 & G & S_2) & 0 & 0 & 0 \\ 1 & 2 & & 3 & & 4 & 5 & 6 \end{matrix}$$

$$= \frac{6!}{2! \times 2! \times 2!} = 6! \times 2! = 1440$$

(xi) Brother always wants to be seated in b/w sister:-

$$\begin{matrix} 0 & G & 0 & G & (S_1 & 0 & S_2) & 0 \\ 1 & 2 & 3 & 4 & 5 & & 6 \end{matrix}$$

$$= 6! \times 2! = 1440$$

" no need to select & arrange brother because brother is particular person & he knows it

(xii) Elder & younger sister wants to be a seated first and last position respectively:-

$$S_1 \begin{matrix} G & G & 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} S_2 = 6! \text{ ways}$$

(xiii) Two sisters are to be seated at extreme pos:-

$$\begin{matrix} S_1 & (G & G & 0 & 0 & 0 & 0) & S_2 \\ 1 & 2 & 3 & 4 & 5 & 6 & & \end{matrix} = \frac{6!}{2!} \times 2! = 1440$$

(xiv) Two sister have to be seated at extreme position but together:-

$$\begin{aligned} \text{case 1: } & (S_1, S_2) \begin{matrix} (G & G & 0 & 0 & 0 & 0) \\ 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} = 6! \times 2! \\ \text{case 2: } & \begin{matrix} G & G & 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} (S_1, S_2) = 6! \times 2! \end{aligned} \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{Req. Ans} = \\ 2 \times 6! \times 2! \\ = 2880 \end{array}$$

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Q. 16 Person are to be arranged along a long table with 8 chair on each side. Four particular person want to be seated on one particular side & 2 on the other side then find the no of possible no of arrangement.

$$\text{Sol.} \quad \left({}^8P_4 \times 4! \right) + \left({}^8P_2 \times 2! \right) \times 10! =$$

$${}^8P_4 \times 4! + {}^8P_2 \times 2! \times 10! =$$

Note: if Total arrangements for no restriction is
 $= 16!$

$$\text{Req prob} = \frac{\text{favorable}}{\text{Total}} = \frac{{}^8P_4 \times 4! \times 10!}{16!}$$

(ii) if side is not particular then answer will be -

$$\left({}^8P_4 \times 4! \right) + \left({}^8P_2 \times 2! \right) \times 10! =$$

(iii) if 6 particular person want to be seated on one particular side

$$= \left({}^8P_6 \times 6! \right) \times 10! =$$

Lecture no. 05

Circular Permutation:

if there are n person and they want to be seated at n places around a circular table

then no of circular arrangement = $(n-1)!$

eg let $n=3$ then

linear arrange = $3! = 6$

circular " = $(3-1)! = 2!$

⇒ Fundamental Question 2 →

Q: 4 B & 4 G are to be seated around a circular table for tea party then find the possible no of arrangement if

i) There is no restriction = $(8-1)! = 7! = 5040$
(ANA)

ii) All boys are not separated:


$$= (5-1)! \times 4! = 4! \times 4! = 576$$

iii) All boys do not come together
 Total - Always together
 $7! - 4! \times 4! = 960$

iv) No two boys are together:

$$(4-1)! \times 4! = 6 \times 24 = 144$$



v) Boys & girls are to be seated alternately

$$3! \times 4! = 6 \times 24 \quad \{\text{same as above}\}$$

$$(4-1)! \times 4! = 144$$

vi) Two sister are always separated by exactly one girl



$$= (6-1)! \times 2! \times 2! \times 1 = 120 \times 2 = 240$$

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(vii) There is a particular boy b/w two sister.

(5, 0, 5, 0)



$$= (6-1)! \times 2!$$

$$= 240$$

(viii) Both the sister wants to be seated at the adjacent sides of host?

(5, 1, 5, 1)



$$= (1, 2, 1) \times (6-1)! \times 2! = 240$$

No need to select host because host is a particular person he or she knows.

Permutation of alike item:

(eg) How many 4 digit nos can be formed using the digit 5, 5, 6, 6?

M-1 Manual counting: various nos are (RNA)

(5566), (5656), (5665), (6655), (6565), (6556)

i.e. Ans = 6

M-2 Req four digit nos = $\frac{4!}{2! \cdot 2!} = 6$

Q) How many 4 letter words can be formed using the letter a a a b? (RNA)

M-2 Req words = $\frac{4!}{3!} = 4$

- Number of linear arrangements of n things in which p are alike of 1st kind, q are alike of 2nd kind, r are alike of 3rd kind & rest are different

$$= \frac{n!}{p! q! r!} \quad \text{where } p + q + r + \text{Rest} = \text{Total } (n)$$

Note:- Number of linear arrangement of n different things = $n!$
 Number of circular " " " " = $(n-1)!$

- Q:- In how many ways 2 Hindi Books, 3 English books, 4 maths books, 1 history book, 1 geography book & 1 physics book can be arranged in a shelf?

Sol:- HH, EEE, MMMM, Hist, G, P = Total = 12

So req. Arrangement = $\frac{12!}{2! 3! 4!} = 168000$

- Q:- How many 9 letter words can be formed using the letter of the word 'ALLAHABAD'?

Sol:- So req. 9 letter words = $\frac{9!}{4! 2!} = 7560$

- Q:- How many 11 letter words can be formed using the letters of the word MATHEMATICS

(i) if there is no restriction = $\frac{11!}{2! 1! 1!} = \text{Max}$

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Permutation

(ii) Vowels are not separated -

$$= \frac{8!}{2! 2!} \times \frac{4!}{2!} = 12960 \text{ of}$$

(iii) Vowels are never together -

$$\frac{11!}{2! 2! 2!} - \left(\frac{8!}{2! 2!} \times \frac{4!}{2!} \right) =$$

(iv) No two vowels are together -

~ First arrange consonant

$$M T H M C T S = \frac{7!}{2! 2!} \times \frac{5P_4}{2!} = 2520 \text{ of}$$

(v) Vowels and consonants are alternately?

Senseless Question.

(vi) Vowels occupied even places -

$$\frac{5P_4}{2!} \times \frac{7P_3}{2! 2!}$$

of	"									"
1	2	3	4	5	6	7	8	9	10	11

Vowels = A, E, I & Consonant = M, T, H, C, S
4 7

(vii) Vowels are together consonant are together -

$$(A E I) (M T H C S) = \frac{2!}{2!} \times \frac{4!}{2!} \times \frac{7!}{2! 2!}$$

Q. In how many ways 7 + sign & 5 - sign are to be arranged in a row so that no two minus sign are together?

Sol: No two (-) sign are together.
first arrange (+) sign

$$+ + + + + - - - - - = \frac{7P_7}{7!} \times \frac{5P_5}{5!}$$

$$= \dots$$

Lecture no. 06

Distribution of different items →

• No formula exist, only use Multi Rule

e.g.) In how many ways 10 different pencils can be distributed among 4 children if

(i) There is no restriction:

$$\frac{10}{P_1} \times \frac{9}{P_2} \times \frac{8}{P_3} \times \frac{7}{P_4} \times \frac{6}{P_5} \times \frac{5}{P_6} \times \frac{4}{P_7} \times \frac{3}{P_8} \times \frac{2}{P_9} \times \frac{1}{P_{10}} = 10$$

✓

(ii) 1st child gets exactly 3 pencil ÷

$$\left({}^1C_1 \times {}^{10}C_3 \times 1 \right) \times \text{Remaining pencil} = 7$$

Remaining child = 3

$$= \left({}^1C_1 \times {}^{10}C_3 \times 1 \right) \times \frac{7!}{1! \cdot 1! \cdot 1! \cdot 1! \cdot 1! \cdot 1! \cdot 1! \cdot 1!} = {}^{10}C_3 \times 7! \text{ ways}$$

(iii) 1st child gets 2 particular pencils
 $({}^1C_1 \times {}^3C_2 \times 1) \times 3^7 = 3^7 \text{ ways}$

(iv) one child gets 3 particular pencils:
 $({}^4C_1 \times {}^3C_3 \times 1) \times 3^7 = 4 \times 3^7 \text{ ways}$

(v) Exactly one child gets exactly 3 pencils:
 ~~$({}^4C_1 \times {}^{10}C_3 \times 1) \times 3^7 = \text{similar question}$~~

Q. In how many ways 3 prizes can be awarded among 4 children if

(i) A child may get any no. of prizes?

(ii) No child gets more than one prize?

(iii) No child gets all the prizes =

(i) $\frac{4}{C_1} \times \frac{4}{C_1} \times \frac{4}{C_1} = 64 \text{ ways}$

(ii) $\frac{4}{P_1} \times \frac{3}{P_2} \times \frac{2}{P_3} = 24 \text{ ways}$

(iii) There are 4 ways of distributing all prizes to child

Total case - unfavorable case = $64 - 4 = 60$

Q:) In how many ways birthdays of 6 different person will fall in calendar months if

(i). There is no restriction = (6!)

$$\frac{12}{1} \times \frac{11}{2} \times \frac{10}{3} \times \frac{9}{4} \times \frac{8}{5} \times \frac{7}{6} = 6! \text{ ways}$$

Q:) (ii) fall in exactly 2 calendar months!

Sol:) Total ways of falling 6 day in 12 months = ${}^{12}C_2 \times 2^6$
 exactly 2 months = ${}^{12}C_2 \times (2^6 - 2) = 4002$

Distribution of Identical items :-

e.g:) In how many ways 4 identical can be distributed among 2 children. if

(i). Blank groups are allowed.

(ii). Blank groups are not allowed.

(i) $m-1$
 Various distribution are:
 (3,1), (1,3), (2,2), (4,0), (0,4)
 i.e. Ans = 5 ways.

$$\frac{m-2}{4+2-1} = {}^4C_2 = 5$$

(ii) $m-1$
 Various distribution are:
 (3,1), (2,2), (1,3)
 So Ans = 3 ways

$$\frac{m-2}{4-1} = {}^3C_2 = 3$$

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$$\frac{n+r-1}{r-1}C_{r-1}$$

Q.8) In how many ways 4 identical pencils can be distributed among 3 kids? if

(i) Blank groups are not allowed.

$$\frac{n-1}{r-1}C_{r-1} = \frac{4-1}{3-1}C_{3-1} = 3C_2 = \frac{3 \times 2}{1 \times 1} = 3 \text{ ways}$$

(ii) Blank groups are allowed.

$$\frac{n+r-1}{r-1}C_{r-1} = \frac{4+3-1}{3-1}C_{3-1} = 6C_2 = \frac{6 \times 5}{2 \times 1} = 15 \text{ ways}$$

Q.9) In how many ways 10 identical pencil can be distributed among 4 kids of

(DNT)

$$\frac{n-1}{r-1}C_{r-1} = \frac{10-1}{4-1}C_{4-1} = 9C_3 = 84 \text{ } \checkmark$$

P.T.O

⇒ if n identical items are to be distributed among r person then possible number of Distribution are

(i) if Blank groups are allowed =

$$\boxed{\frac{n+r-1}{r-1}C_{r-1}} \quad \left\{ \begin{array}{l} \text{(DNT)} \\ \text{formula starts} \\ \text{from 0} \end{array} \right.$$

(ii) if Blank groups are not allowed =

$$\boxed{\frac{n-1}{r-1}C_{r-1}} \quad \left\{ \begin{array}{l} \text{(BNT)} \\ \text{formula starts} \\ \text{from 1} \end{array} \right.$$

- (i) A child may get any number of pencils = ?
(BA)
- (ii) A child may get nothing = ?
(NA)
- (iii) Each child gets one or more pencil = ?
(BA)
- (iv) " " " atleast one pencil = ?
(NA)
- (v) Every child get something =
(BA)

$$(i) \quad n+r-1 \underset{r-1}{C} = 10+4-1 \underset{4-1}{C} = 13 \underset{3}{C} = 286$$

$$(ii) \quad 13 \underset{3}{C} = 286$$

$$(iii) \quad n-1 \underset{r-1}{C} = 10-1 \underset{4-1}{C} = 9 \underset{3}{C} = 84$$

$$(iv) \quad 9 \underset{3}{C} = 84$$

$$(v) \quad 9 \underset{2}{C} = 36$$

Q: In how many ways 16 mangoes can be distributed among 3 begger if each begger gets atleast 3 mangoes?

$$\text{Sol: } n=16, r=3 \quad n=2, r=2 \quad \text{Tab 1 (BA)}$$

first we will distribute 3-3 mango to each this can be done by one way
Now remaining mangoes are = 7 begger are still
we can distribute those 7 mangoes without any restriction. so $\text{Tab 1} \times \text{Tab 2} = 1 \times {}^9C_2 = 36 \text{ ways}$

Note: Set of non-negative integers $\{0, 1, 2, 3, 4, \dots\} = \mathbb{N}$ (natural no.)
 Set of non-negative integers $\{0, 1, 2, 3, 4, \dots\} = \mathbb{N}$ (natural no.)
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Q2: $x \geq 0, x+y+z=16; x \geq 3, y \geq 3, z \geq 2$

$(x-1) + (y-1) + (z-1) = 16-3, (x-1) \geq 0$

$u+v+w=7, u \geq 0, v \geq 0, w \geq 0$

$n=7, z=3$ (S.S.)

$n+2-1 C_{z-1} = 7+3-1 C_{3-1} = 9 C_2 = 36$ ways

Q3: How many non-negative integral solution are there of $x+y+z+w=100$

Sol: $x+y+z+w=100, x \geq 0, y \geq 0, z \geq 0, w \geq 0$

$n=100, z=4$ ways

$n+2-1 C_{z-1} = 100+4-1 C_{4-1} = 103 C_3$

Q4: Find the no. of ways of 0's & 1's of straight lines that can be formed using 8 points in which 5 are collinear?

Sol:  No. of lines = 2 points from set 1 & 1 point from set 2

 3 points from set 1, 1 point from set 2

$= (3 C_2 \times 5 C_1) + (3 C_1 \times 5 C_2) \times (3 C_1)$
 (reverse)
 $= 46$

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Chapter - 02Probability# Lecture, no, 01

$\Rightarrow$ Random Exp: When we are not sure about the outcome of an experiment then such type of experiment are known as R. Exp.

(eg) Tossing a coin, Throwing a Die.

$\Rightarrow$ Sample space: Set formed by possible outcome of random exp is called S. space.

(eg) for coin $S_c = \{H, T\}$, for Die $S_d = \{1, 2, 3, 4, 5, 6\}$
for family, $S_f = \{B, G\}$.

$\Rightarrow$ Event: Any subset of S. space is called event.

(eg) for Die $S_d = \{1, 2, 3, 4, 5, 6\}$
let $E_1 = \{1, 3, 5\}$, $E_2 = \{2, 4, 6\}$, $E_3 = \{1, 2, 3, 4\}$..
= odd num. = even num. = No. 5 occurs

Note: If no. of elements in any set $A = n$ then
total no. of subset $= 2^n$
so, total no. of event associated with random
S. space having n element $= 2^n$
 $=$ Total subset of S. space $= 2^n$

(eg) let $S = \{1, 2, 3\}$. Then total subset $= 2^3 = 8$
There are given that $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}$
 $\phi, \{1, 2, 3\}$.

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⇒ Impossible event: $\{\emptyset\}$: $\emptyset \subseteq S \Rightarrow \emptyset$ is also an event and it is called impossible event &
 $P(\emptyset) = 0$

⇒ Sure event / Certain event: $\because S \subseteq S \Rightarrow S$ is also an event & it is called sure event
 $P(S) = 1$

Note: 1) $0 \leq P(E) \leq 1$ 2) $P(\text{nothing occurs}) = 0$

3) $P(\text{something occurs}) = 1$ 4) $P(\text{given statement}) = 1$

5) $P(\text{Death}) = 1$ 6) $P(\text{god}) = 1$

⇒ Some concept of set theory:

Let A & B are two event associated with sample space S.

- only A = $A \cap \bar{B} = A - (A \cap B)$
- only B = $\bar{A} \cap B = B - (A \cap B)$
- Neither A nor B = $\bar{A} \cap \bar{B}$ = None of A & B
- Either A or B or Both = at least one of A or B
 $= A \cup B = (A \cap \bar{B}) \cup (\bar{A} \cap B) \cup (A \cap B)$
- Simultaneous occurrence A & B = Both A & B
 $= A \cap B$



⇒ Addition theorem:

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

⇒ Multiplication theorem:

$$P(A \cap B) = P(A|B) \cdot P(B).$$

$$* P(\text{neither A nor B}) = 1 - P(\text{either A or B})$$

$$P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$$

$$* P(\text{either A or B}) = 1 - P(\text{neither A nor B})$$

$$P(\text{at least one of A or B}) = 1 - P(\text{none of A or B})$$

⇒ Mutually Exclusive Event ⇒

Events are called ME they can't occur simultaneously.

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

$$P(E_1 \cup E_2) = 0.4 + 0.3 = 0.7$$

(If E_1 & E_2 are two ME events then $E_1 \cap E_2 = \emptyset$
 $P(E_1 \cap E_2) = 0$)

eg. $S = \{1, 2, 3, 4, 5, 6\}$ $E_1 = \{1, 3, 5\}$, $E_2 = \{2, 4, 6\}$
 $E_3 = \{1, 2, 3, 4\}$, $E_4 = \{5, 6\}$

$$\because E_1 \cap E_2 = \emptyset \Rightarrow E_1 \text{ \& E_2 are ME } E_1 \cup E_2 = S$$

$$\because E_1 \cap E_3 \neq \emptyset \Rightarrow E_1 \text{ \& E_3 are not ME}$$

$$\because E_2 \cap E_3 \neq \emptyset \Rightarrow E_2 \text{ \& E_3 are not ME}$$

$$\because E_1 \cap E_4 = \emptyset \Rightarrow E_1 \text{ \& E_4 are ME } E_1 \cup E_4 \neq S$$

* It is not necessary that union of ME event is sample space.

⇒ Independent Event:

if occurrence ^{or} non occurrence of one event does not affect the occurrence or non-occurrence of other event then events are called Ind. ev.

In case of ind event we can multiply the respective prob in order to find their simultaneous prob i.e

if A & B are ind event $P(A \cap B) = P(A) \cdot P(B)$

e.g. $S_0 = \{H, T\}$ $S_0 = \{1, 2, 3, 4, 5, 6\}$
 $A = \{\text{Head occur}\}$ $B = \{\text{No.} \leq 4 \text{ occur}\}$
 $P(A) = \frac{1}{2}$ $P(B) = \frac{4}{6} = \frac{2}{3}$

By common sense A & B are ind events.

$$\therefore P(A \cap B) = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3} \quad \Delta$$

v.v. note: i) ME event are associated with Same S space while ind event associated with Different S space.

$\therefore$ The concept of intersection will be considered only when events are associated with same S space.

$\Rightarrow$ Exhaustive Event $\Rightarrow$ Events are called exhaustive if their union is equal to S space.
 i.e.

Let E_1, E_2, E_3 are three events s.t. $E_1 \cup E_2 \cup E_3 = S$
 $\Rightarrow E_1, E_2, E_3$ are exhaustive events.

$$P(E_1 \cup E_2 \cup E_3) = 1$$

e.g.) $S_0 = \{1, 2, 3, 4, 5, 6\}$, $E_1 = \{1, 2, 3\}$, $E_2 = \{4\}$, $E_3 = \{5, 6\}$
 $\therefore E_i \cap E_j = \emptyset \quad \forall i \neq j$ so E_1, E_2, E_3 are ME
 Now $E_1 \cup E_2 \cup E_3 = \{1, 2, 3, 4, 5, 6\} = S \Rightarrow E_1, E_2, E_3$ are exhaustive events.

only one

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⇒ Nature of element in a S space ⇒
 If our Exp. is repeated n times then elements of S space are in the form of ordered n -tuples.

eg) A Die is thrown once:

$$S_0 = \{1, 2, 3, 4, 5, 6\} \Rightarrow n(S) = 6$$

eg) A Die is thrown twice:

$$S_0 = \left\{ \begin{array}{cccccc} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ 21 & 21 & 23 & 24 & 25 & 26 \\ 31 & 32 & - & - & - & 66 \end{array} \right\}$$

$$= n(S) = \frac{6}{D_1} \times \frac{6}{D_2} = 6 \times 6 = 36 \text{ pairs}$$

eg) A coin is tossed 5 times

$$S = \{HHHHH, HHHHT, \dots, (TTTTT)\}$$

$$n(S) = \frac{2}{C_1} \times \frac{2}{C_2} \times \frac{2}{C_3} \times \frac{2}{C_4} \times \frac{2}{C_5} = 2^5 = 32 \text{ results}$$

eg) A couple has 3 children then $S = 2 \times 2 \times 2 = 8$

Note) A coin is tossed thrice =
 or
 three coins are tossed simultaneously = } Both have same sample space

⇒ favourable case = {which is required should be assumed as fav. case}

⇒ Method of solving question.

Approach 1: By listing all the elements of -
favourable event (E) & S. space (S)

$$\text{i.e. } P(E) = \frac{n(E)}{n(S)}$$

Approach 2: if it is not easy to write favourable event and S. space in terms of set -
then directly calculate favorable no. of cases and total no. of cases by using the concept of permutation & combination and then
= $\text{req. prob} = \frac{\text{fav. case}}{\text{total case}}$

Approach 3: By using some standard result & standard Definition.

= Whenever in a question given info is in the form of prob then use approach 3.

⇒ operation	P & C	Prob		M.C	Ind.
= Either OR	Add	union	$P(A \cup B) = \text{Add}$	$P(A \cup B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
AND	Multiply	Intersection	$P(A \cap B) = \text{Multiply}$	$P(A \cap B) = 0$	$P(A \cap B) = P(A) \cdot P(B)$

Note: Individual elements of S. space are m.e

Lecture no. 02

Q.) A Die is rolled twice then write its S space.

$$S = \left. \begin{array}{cccc} (1,1) & (1,2) & (1,3) & \dots & (1,6) \\ (2,1) & (2,2) & (2,3) & \dots & (2,6) \\ (3,1) & (3,2) & (3,3) & \dots & (3,6) \\ (4,1) & (4,2) & (4,3) & \dots & (4,6) \\ (5,1) & (5,2) & (5,3) & \dots & (5,6) \\ (6,1) & (6,2) & (6,3) & \dots & (6,6) \end{array} \right\} \frac{6}{n_1} \times \frac{6}{n_2} = 36 \text{ pairs}$$

(i) Find the prob that sum of the outcome will be 5

Appl

$$\text{Soln)} A = \{(1,4), (2,3), (3,2), (4,1)\} = 4$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

(ii) Find the prob of that sum of the outcome will be 4

$$\text{Soln)} B = \{(1,3), (2,2), (3,1)\} = 3$$

Appl

$$P(B) = \frac{n(B)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

Appl

(iii) Find the prob that sum is both 8 & 9

$$\text{Soln)} P(A \cap B) = 0 \Rightarrow A \& B \text{ are ME} \& P(A \cap B) = 0$$

Appl

(iv) Find the prob that sum is either 8 or 9

$$\begin{aligned} \text{Soln)} P(A \cup B) &= P(A) + P(B) - P(A \cap B) & A \& B \text{ are ME} \\ &= \frac{4}{36} + \frac{3}{36} - 0 = \frac{7}{36} = \frac{1}{5.14} \end{aligned}$$

(v) Find the prob that sum is neither 8 nor 9.

sol: $= 1 - P(\text{sum is either 8 or 9})$

i.e. $P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$
 $= 1 - \frac{1}{4} = \frac{3}{4}$ Ans.

(vi) Find the prob that add the both the outcome are identical.

sol: $C = \{\text{identical outcome}\} = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$
 $= \frac{6}{36} = \frac{1}{6}$

(vii) Find the prob that Product of the 2 outcome will be perfect square

sol: Let $D = \{\text{Product is a perfect square}\}$
 $\{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (1,4), (4,1)\} = 8$
 $= \frac{n(D)}{n(S)} = \frac{8}{36} = \frac{2}{9}$ Ans.

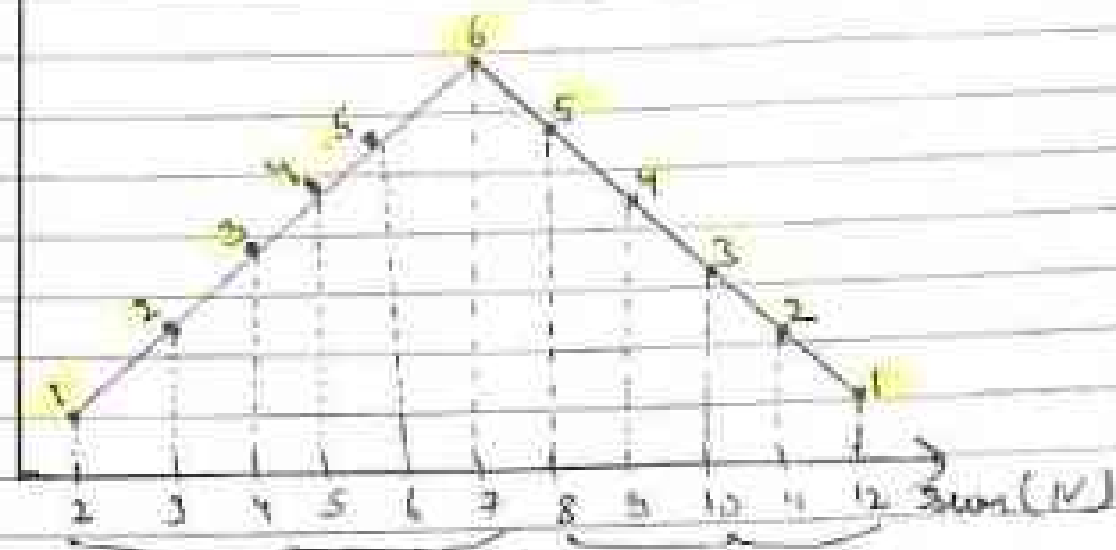
(viii) Find the prob that sum of the outcome will be a prime number. ($\frac{15}{36}$)

sol: $\{(1,1), (2,1), (3,2), (2,3), (1,2), (4,1), (1,4), (5,2), (2,5), (6,5), (5,6), (1,6), (6,1), (4,3), (3,4)\}$
 for condition = 15
 so, req prob = $\frac{15}{36}$ Ans.

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Show that:

No. of
0 pairNo. of 0 pair = $(N-1)$ No. of 0 pair = $(13-N)$

$$1) P(\text{sum is 8}) = \frac{13-8}{76} = \frac{5}{76}$$

$$2) P(\text{sum is 9}) = \frac{13-9}{76} = \frac{4}{76}$$

$$3) P(\text{sum is prime no.}) = 2, 3, 5, 7, 11 \\ = \frac{1+2+4+6+(13-1)}{76} = \frac{15}{76}$$

$$4) P(\text{sum is divisible by 4}) = 4, 8, 12 = 3+5+1 \\ = \frac{9}{76}$$

$$5) P(\text{sum is exceeds 9}) = P(N=10, 11, 12) \\ = 3+2+1 = \frac{6}{76}$$

Prob 4 Dice are throw simultaneously then find the prob that there sum is 22.

$$S = \left\{ \begin{matrix} (1,1,1) & (1,1,2) & (1,1,3) & \dots & (1,1,6) \\ (2,1,1) & (2,1,2) & (2,1,3) & \dots & (2,1,6) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ (6,6,6) & (6,6,6) & (6,6,6) & \dots & (6,6,6) \end{matrix} \right\}$$

$$= \frac{6}{6} \times \frac{6}{6} \times \frac{6}{6} = 6^3 = 1296$$

fav event $(E) = \{\text{sum is 21}\} =$

$\{ (664) (646) (466) (665) (656) (566) (556) (565) (655) (566) (556) (565) (566) (556) (565) (566) \}$

$\therefore n(E) = 10$ Outcomes

$$P(E) = \frac{10}{1296}$$

App-2: Total Outcomes = $6 \times 6 \times 6 \times 6 = 1296 = 6^4$

fav Outcomes = $\left\{ \begin{array}{l} (666) \dots \\ (665) \dots \end{array} \right\} \frac{4!}{1!1!1!} = 4$
 $\frac{4!}{1!1!1!} = 6$
 10 case

So req prob = $\frac{\text{fav Outcomes}}{\text{Total Outc}} = \frac{10}{1296}$ Q.

Ex-4:

(i) Four Dice are thrown simultaneously & P (sum is 21) = $\frac{x}{1296}$ then $x = ?$

(ii) Six dice are thrown simultaneously then find the prob of that:

(i) All of them will show same phase
 Total outcome = $6 \times 6 \times 6 \times 6 \times 6 \times 6 = 6^6$

sol: $= {}^6C_1 = 6 = \text{Req prob} = \frac{\text{fav}}{\text{Total}} = \frac{6}{6^6} = \left(\frac{1}{6^5} \right)$

(ii) All of them will show different face

$$= {}^6P_6 \times 6! = 1 \times 6! = 6!$$

$$= \frac{720}{6^6} = 0.0154320988$$

prob prob

PROBABILITY

(iii) Find the prob that exactly 3 will show the same phase & 4 remaining will show different phase.

sol) fav outcome = ${}^6C_3 \times {}^5C_4 \times \frac{6!}{1!5!} = 0.15 + 32.056$

Req prob = $\frac{{}^6C_3 \times {}^5C_4 \times \frac{6!}{1!5!}}{6^6} = \frac{6 \times 10 \times 120}{6^6} = \frac{7200}{46656}$

(iv) Find the prob that atleast 9 will show the same phase & remaining will show different show phase?

sol) $\left({}^6C_1 \times {}^5C_1 \times \frac{6!}{4!1!1!} \right) + \left({}^6C_2 \times {}^4C_1 \times \frac{6!}{3!1!1!1!} \right) + \left({}^6C_3 \times \frac{6!}{2!1!1!1!1!} \right)$

so Req prob = $\frac{\text{fav}}{\text{total}} = 6^6$

Q) 7 surgical strikes occurred in a week from India on Pakistan then find the prob that all will occur on a same day?

sol) App - 6 Total ways occurring = $\frac{7 \times 7 \times 7 \times 7 \times 7 \times 7 \times 7}{1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1}$

= 7^7 ways.

fav ways of occurring = $7C_1 = 7$ ways

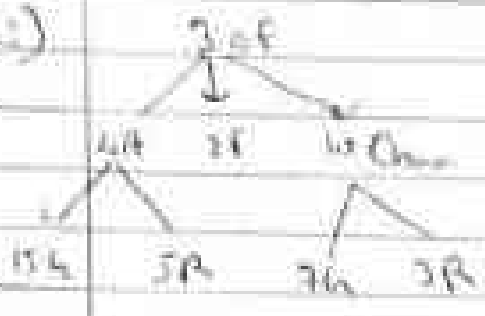
Req prob = $\frac{7}{7^7} = \frac{1}{7^6}$ Ans.

Q) A basket contain 20 apple & 10 oranges in which 5 apple and 3 oranges are rotten. if 2 fruits are chosen at random then find the prob that either both are apple or Both are good?

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sol:)



Total ways of selecting 2fr ${}^{30}C_2 = 435$

$A = \{ \text{One fruit is apple} \}$

$$P(A) = \frac{n(A)}{n(S)} = \frac{21C_1}{435} =$$

$B = \{ \text{Both fruits are orange} \}$

$$P(B) = \frac{n(B)}{n(S)} = \frac{9C_2}{435} =$$

$A \cap B = \{ \text{Both fruits are orange apple} \}$

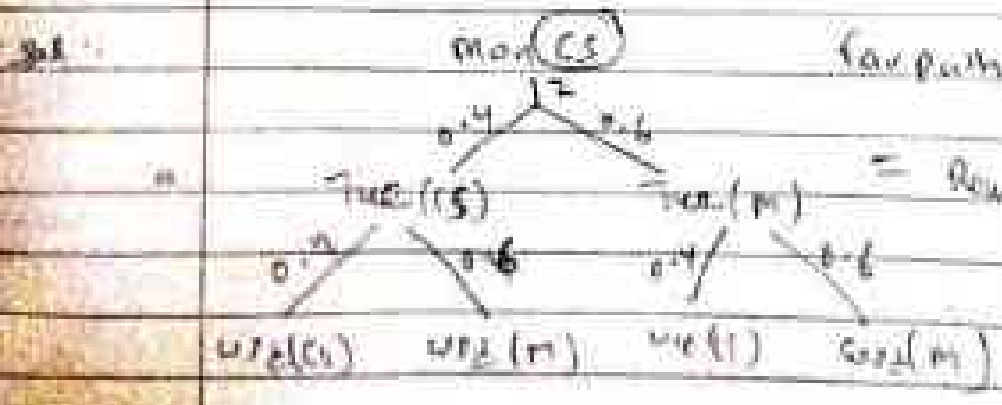
$$P(A \cap B) = \frac{{}^{15}C_2}{435}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{{}^{21}C_1}{435} + \frac{{}^9C_2}{435} - \frac{{}^{15}C_2}{435} = \frac{21 + 36 - 105}{435} = \frac{52}{145}$$

i.e. this question is mixed question of App II & App III

Q.) Rishwanya studies either CS or maths on everyday if she studies CS on a day then the prob of studying maths on next day is 0.6 & if she study CS is 0.4 given that Rish studies CS on Monday then find the prob of studying CS on Wednesday



fav path = (CS, CS, CS) or (CS, M, CS)

$$= \text{fav prob} = 1 \times 0.4 \times 0.4 + 1 \times 0.6 \times 0.4 = 0.4$$

Lecture no. 03

Fundamental Question 2:

Q:) A coin is tossed 6 times then write it's S space

$$S = \{ \overset{c_0}{(HHHHHH)}, \overset{c_1}{(HHHHHT)}, \overset{c_2}{(HHHHTT)}, \overset{c_3}{(HHHTTT)}, \overset{c_4}{(HHTTTT)}, \overset{c_5}{(HTTTT)}, \overset{c_6}{(TTTTTT)} \}$$

$$n(s) = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6 = 64 \text{ outcomes}$$

• $P(H) = P(T) = \frac{1}{2}$, All tosses are independent.

∴ no. of elements in S space = 64.
i.e. there are 64 outcomes and ME.

(i) Find the prob that both H and T appear at least once?

∴ unfavourable outcome = $\{(HHHHHH), (TTTTTT)\} = 2$ outcomes

∴ favourable outcome = $64 - 2 = 62$ outcomes

$$\text{so req prob} = \frac{62}{64} = 0.96 \quad \underline{\underline{2}}$$

(ii) Find the prob that head appears at least once?

∴ unfavourable outcome = $\{(TTTTTT)\} = 1$ outcome

favourable outcome = $64 - 1 = 63$ favourable

$$\text{so req prob} = \frac{63}{64} = 0.98 \quad \underline{\underline{2}}$$

Ans II

$$\begin{aligned} P(\text{at least one H}) &= 1 - P(\text{no head}) \\ &= 1 - P(\text{all T}) \\ &= 1 - P(\text{TTTTTT}) \\ &= 1 - \left(\frac{1}{2}\right)^6 \end{aligned}$$

$$= \frac{63}{64}$$

∴

All tosses are ind.

(iii)

Find the prob that all the outcomes are identical?

Sol.)

fav case = {HHHHHH, TTTTTT} = 2 cases

$$\therefore \text{req prob} = \frac{f}{T} = \frac{2}{64} = 0.03125$$

Ans 3

Req prob = $P(\text{HHHHHH}) + P(\text{TTTTTT})$

$$= \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6 = \left(\frac{1}{2}\right)^5$$

(iv)

Find the prob that head & tail appear after

Sol.)

fav = {HTHTHTHTHT, THTHTHTHTH}

$$\text{Req prob} = \frac{f}{T} = \frac{2}{64} = \frac{1}{32} = 0.03125$$

Ans 2

Req prob = $P(\text{HTHTHTHTHT}) + P(\text{THTHTHTHTH})$

$$= \left(\frac{1}{2}\right)^8 + \left(\frac{1}{2}\right)^8 = \left(\frac{1}{2}\right)^7$$

+

Individually these tosses are independent & in joint form these outcome are fig.

study time

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1) Find the prob that H & T appears equal no of times?

sol) App 2:) fav case = {exactly 3 H & exactly 3 T}
= {HHHTTT, TTTTHH, ...}

= Not easy to count.

App 2:) fav cases = {exactly 3 H occurs} = {HHTHHH, ...}
= $\frac{6!}{3!3!} = {}^6C_3 = 20$

So, req prob = $\frac{{}^6C_3}{64} = \frac{20}{64} = 0.3125$ Ans

(vi) Find the prob that exactly 3 H appears?

sol) ${}^6C_3 = \frac{{}^6C_3}{64} = \text{same as part (v)}$

(vii) Find the prob of getting exactly 2 H?

sol) App 2:) Not easy to count.

App 2:) fav case = {HTHTHT, ...} = $\frac{6!}{4!2!} = {}^6C_2 = 15$

Req prob = $\frac{15}{64} = \frac{{}^6C_2}{64} = \frac{15}{64} = 0.234375$ Ans

App 3:) Using Binomial Distribution

(viii) Find the prob that **only** first two tosses produce Head?

Sol: 1) App 1: fav case = {only first two tosses produce Head}
 $= \{HH TTTT\} = 1$

So, required Prob = $\frac{\text{fav}}{\text{Total}} = \frac{1}{64} = 0.015625$ or

(ix) Find the prob that 1st two produce Head

Sol: 1) App 1: fav case = $\{(HHHHHH), (HHHHHT), \dots\}$
 $=$ Not easy to count.

App 2: fav case = $\{(\underbrace{H \ H}_{1 \times 1} \underbrace{T \ T \ T \ T}_{2 \times 2 \times 2 \times 2}) - \dots\}$
 $= 16$ tuples

So, req prob = $\frac{\text{fav}}{\text{Total}} = \frac{16}{64} = \frac{1}{4}$ or.

(x) if 1st two tosses are H, H then find the Prob of getting T when coin is tossed again?

Sol: 1) App II: n=5 Req prob = $P(\text{getting H in 3rd toss}) = \frac{1}{2}$

n=2 Req prob = $P[H \ H \ T \ \text{something occurs}]$

$= 1 \times 1 \times \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$ or

X = $\frac{32}{64}$ or

⇒ we can not generalize find the Ans. if Question was based on conditional prob.

(2i) if 1st two tosses are both heads then find the prob of occurring tail in remaining 3 tosses

Sol: Req. Prob = $P\left[\begin{matrix} H & H & T & T & T & T \\ 1 \times 1 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \end{matrix}\right]$

$$= 1 \times 1 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16} \neq \frac{4}{24}$$

Q:) A coin is tossed 7 times then find the Prob that H & T appears equal no. of times.
Senseless Question so Ans = 0

(i)
Q:) A coin is tossed 10 times then find the Prob of getting exactly 3 Heads.

Sol: $S = \{(HHHHHHHHHT), (HHHHHHHHT), \dots, (HTHTHTHT)\}$

$$n(S) = 2^{10} = 1024 \text{ tuples.}$$

Ans: few case = $\{(HTTHTTHTT), \dots\} = \frac{10!}{7!3!} = 10 \times 9 \times 8$

$$\text{So Req. prob} = \frac{{}^{10}C_3}{1024} = \frac{120}{1024} \approx 0.116545117$$

Using Binomial Distribution:

(ii) Find the Prob that 4th H appears in 8th toss

Sol: App 3

$$\text{fav case} = \{4^{\text{th}}\}$$

$$= \text{Req prob} = P[4^{\text{th}} \text{ H in } 8^{\text{th}} \text{ toss}]$$

$$= P[\text{getting exactly 3 H in } 1^{\text{st}} 7 \text{ tosses}] \times P[4^{\text{th}} \text{ H in } 8^{\text{th}} \text{ toss}] + P[\text{getting more}]$$

$$= \left(\frac{{}^8C_3}{2^8} \right) \times \frac{1}{2} \times 1 = \frac{56}{512} \times \frac{1}{2}$$

App 2

$$\text{fav cases} = \left\{ \text{eg } \frac{\text{exactly 3H}}{2} - \frac{4}{8} \frac{1}{10} \right\}$$

$${}^8C_3 \times 1 \times 2$$

$$\therefore \text{So Req prob} = \frac{\text{fav}}{\text{Total}} = \frac{{}^8C_3 \times 1 \times 2}{2^{10}} = \frac{56}{512}$$

Concept of ME & Independency in a Single Question!

(Q.2) There are two gangsters, Murmu mobile & Pappu Pater they both fire at the target once then write its S. space.

Sol: various possibilities of this type

possibly even will hit

$$(\text{None will hit}) \cup (\text{Hit 1 & Missed 2}) \cup (\text{Missed 1 & Hit 2}) \cup (\text{Both hit}) = \Omega$$

$$(\bar{A} \cap \bar{B}) \cup (A \cap \bar{B}) \cup (A \cap B) \cup (\bar{A} \cap B) = \Omega$$

at least one will hit

$$S = \{\bar{A} \cap \bar{B}, A \cap \bar{B}, \bar{A} \cap B, A \cap B\} = \{E_1, E_2, E_3, E_4\}$$

$\Rightarrow E_1, E_2, E_3, E_4$ are mutually events. These are the individual element of S space.

$$P(E_1 \cup E_2 \cup E_3 \cup E_4) = 1$$

$\Rightarrow$ Ex 2:

Two person A & B fire at the target once with prob of their hitting is $\frac{4}{5}$ & $\frac{3}{4}$ respectively. find the Prob that:

(i) $P(A) = \frac{4}{5}$ $P(\bar{A}) = \frac{1}{5}$ $P(B) = \frac{3}{4}$ $P(\bar{B}) = \frac{1}{4}$ A & B are

(i) Find the Prob that both will hit?

ex) $P(A \cap B) = P(A) \cdot P(B) = \frac{4}{5} \times \frac{3}{4} = \frac{12}{20} = \frac{3}{5}$

(ii) Find the Prob that none will hit?

$$P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B}) = \frac{1}{5} \times \frac{1}{4} = \frac{1}{20}$$

(iii) Find the Prob that atleast one will hit =

$$= 1 - P(\text{none will hit}) = 1 - \frac{1}{20} = \frac{19}{20}$$

(iv) Find the Prob that target will be hit =

$$\text{atleast one will be hit} = \text{same} = \frac{19}{20}$$

(v) Find the Prob that either of them will hit!

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{4}{5} + \frac{3}{4} - \frac{3}{5} = \frac{19}{20}$$

Both
ques
will be
same.

(vi) Find the Prob A hit & B missed

$$P(A \cap \bar{B}) = P(A) \cdot P(\bar{B}) = \frac{4}{5} \times \frac{1}{4} = \frac{1}{5}$$

(vii) Find the Prob exactly 2 will hit

$$P[(A \cap \bar{B}) \cup (\bar{A} \cap B)]$$

$$P[E_1 \cup E_2] = P(E_1) + P(E_2) \quad \left. \begin{array}{l} E_1, E_2 \text{ are} \\ \text{ME} \end{array} \right\}$$

$$= P(A \cap \bar{B}) + P(\bar{A} \cap B)$$

$$= \frac{4}{5} \times \frac{1}{4} + \frac{1}{5} \times \frac{3}{4} = \frac{7}{20} \text{ or}$$

or

$$P(\text{only one will hit})$$

(viii) if only one person then find the probs that A hit & B missed:

(ix) original Prob = $P(S) = 1$

$$\text{Reduced Prob} = P(\text{condition}) = P(\text{only one person hit}) = \frac{7}{20} \text{ (using part 7)}$$

$$\text{fav Prob} = P(\text{which is required}) = P(A \text{ hit \& } B \text{ miss})$$

$$\text{condition Prob} = \frac{\text{fav}}{\text{P-Prob}} = \frac{1/5}{7/20} = \frac{1}{5} \text{ (using Part 6)}$$

$$= \frac{4}{7} \text{ or}$$

Lecture no. 04

Conceptual Question of Condition Prob

* Just Reduce the S-space according to condition

Q.) A coin is tossed twice & its outcome is known to be head the first the Prob of getting exactly 2H?

Sol) original S-space = $\{(HH), (HT), (TH), (TT), (HT), (TH), (TH), (TT)\} = S$

" Reduce S-space = $\{1^{st} \text{ outcome is head}\}$
 $= \{(HH), (HT), (TH), (HT)\} = 4$

fav elements = $\{\text{exactly 2H}\} = \{(HT), (TH)\} = 2$

" So, Condition Prob = $P\left[\frac{\text{Favable Out}}{1^{st} \text{ outcome is H}}\right] = \frac{\text{Favable}}{\text{Red S-space}} = \frac{2}{4}$
 $= \frac{1}{2} \text{ Ans}$

Ans) Req Prob = $P[(HT) \text{ or } (TH)]$

" $= 1 \times \frac{1}{2} \times \frac{1}{2} + 1 \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{2} \text{ Ans}$

Note) A coin is tossed twice then find the Prob of getting exactly 2H

" It is a question not based on Condition Prob
 " So Req Prob = $\frac{\text{fav}}{\text{total}} = \frac{2}{2^2} = \frac{1}{2} \text{ Ans}$

Q) A coin is tossed 6 times & ^{1st} two outcomes are H, H then find the prob occurring of tail in remaining 4 tosses?

Sol: App 3: Req prob = $P[HHTTTT] = 1 \times 1 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$
 $= \frac{1}{16}$ Ans

App 1: original S space $S = \{HHHHHH, (HHHTTT), (HHTHTT), (HTHTTT), (THTTTT), (TTHTTT), (TTTHTT), (TTTTHH)\}$

Reduced S space = $\{HHHHHH\} \dots \dots \dots \rightarrow$ Not using

App 2: Reduced S space = $\{(HH) \times 2 \times 2 \times 2 \times 2\}$
 $1 \times 1 \times 2 \times 2 \times 2 \times 2 = 16$ ways

fav cases = $\{(HHTTTT)\} = 1$

Hence Condition Prob = $\frac{\text{fav}}{\text{P.S. Space}} = \frac{1}{16}$ Ans

Note: ~~Here the condition was not there when a coin would have been = $\frac{1}{16} = \frac{1}{4}$~~

A coin is tossed 6 times then find the Prob of occurring tail in 3rd, 4th, 5th, 6th times

Req Prob = $P[\underline{S} \underline{H} \underline{T} \underline{T} \underline{T} \underline{T}]$

$1 \times 1 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$ Ans

study time

PAPER

Q1) Two dice are thrown simultaneously if there is a sum of the outcome is 7 then find the Prob that 4 appears atleast once!

Sol) Total die = $6^2 = 36$

Reduced S space = {sum of outcome is 7} = 6 pair

fav case = {4 appears atleast once} = {(3,4), (4,3)} = 2

$$\text{condition Prob} = \frac{\text{fav case}}{\text{d.s space}} = \frac{2}{36} = \frac{1}{18}$$

Q2) Two dice are thrown simultaneously then find the Prob that 4 appears atleast once

Sol) It is a same question w/o condition.

$$S = 6^2 = 36$$

$$\text{fav case} = \left\{ \begin{matrix} (1,4), (2,4), (3,4), (4,4), (5,4), (6,4) \\ (4,1), (4,2), (4,3), (4,5), (4,6) \end{matrix} \right\} = 11$$

$$\therefore \text{so req Prob} = \frac{11}{36} = \frac{11}{36}$$

Q3) Two integers are to be selected from integer 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 if there sum is even, find the Prob that both the selected integer are odd.

Sol) (a) $3/11$ (b) $3/5$ (c) $4/11$ (d) 0

App 2 Total ways of selecting two integers $11C_2 = 55$

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 $\begin{cases} \rightarrow 2, 4, 6, 8, 10 \text{ (even)} \\ \rightarrow 1, 3, 5, 7, 9, 11 \text{ (odd)} \end{cases}$

Reduce ways = {Those ^{sum} is even}

= Either (both are even) or (Both are odd)

$$= 5C_2 + 5C_2 = 10 + 10 = 20$$

fav. ways = {Both are odd}

$$= 5C_2 = 10$$

$$\text{Condition} = \frac{\text{fav.}}{\text{R.S. space}} = \frac{10}{20} = \frac{1}{2} \quad \text{A}$$

Note: Had the condition were not there, Ans would be 100

$$\text{Sol:} \quad \text{Req. Prob} = \frac{\text{fav.}}{\text{Total}} = \frac{5C_2}{11C_2} = \frac{10}{55} \quad \text{A}$$

App 10 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 $\begin{cases} \rightarrow 2, 4, 6, 8, 10 \\ \rightarrow 1, 3, 5, 7, 9, 11 \end{cases}$

=) original S. space = $\begin{pmatrix} (1,1), (1,2), (1,3), (1,4), \dots, (1,11) \\ (2,1), (2,2), (2,3), \dots, (2,11) \\ (3,1), \dots, (3,11) \\ \vdots \end{pmatrix}$

=) Reduced S. space = $\begin{pmatrix} (1,6), (1,7), (1,8), (1,9), (1,10), (1,11) \\ (2,5), (2,6), (2,7), (2,8), (2,9), (2,10), (2,11) \\ (3,4), (3,5), (3,6), (3,7), (3,8), (3,9), (3,10), (3,11) \\ \vdots \end{pmatrix}$

$$\Rightarrow \text{Condition Prob} = \frac{\text{fav.}}{\text{Req.}} = \frac{10}{20} = \frac{1}{2} \quad \text{A}$$

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- Q.1) A couple has 2 kids then find the Prob. of having exactly 2 Boys! - if elder child is Boy
- 2) one of the child is a Boy

i) $S = \{(0,0), (0,1), (1,0), (1,1)\} = 4$

$A = S \cap B = \{(0,0), (0,1)\} = 2$

$f_{\text{av}} = \{(0,0)\} = 1$

$= \frac{1}{2} \times 2$

ii) $A = S \cap B$

Req. Prob. = $P[0,0]$

$= 1 \times \frac{1}{2} = \frac{1}{2}$ Ans

iii) $S = \{(0,0), (0,1), (1,0), (1,1)\} = 4$

$A = S \cap B = \{(0,0), (0,1), (1,0)\} = 3$

$f_{\text{av}} = \{(0,0)\} = 1$

Cond'n. Prob. = $\frac{f_{\text{av}}}{A \cap B} = \frac{1}{3}$ Ans

- Q.2) Parcel are sending from sender S to receiver B sequentially through two Post office. The Prob. of losing an incoming parcel is $\frac{1}{5}$ by each P.O. independently of all other parcels.

Given that parcel was lost, then find the Prob. that was lost by 2nd P.O.

$$P(L) = \frac{1}{5} \quad P(W) = \frac{4}{5}$$

study time

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Sol: Original Prob = $P(S) = 1$

Reduced Prob = $P(\text{Parcel is lost})$
 $= P(\text{Lost by 1st Po}) \cup (\text{Lost by 2nd Po})$

$$= P[\text{lost by 1st Po}] + P[\text{lost by 2nd Po}]$$

$$= P[\text{lost by 1st Po}] + P[WL \text{ by 1st}] + P[\text{lost by 2nd}]$$

$$= \frac{1}{5} + \frac{4}{5} \times \frac{1}{5} = \frac{9}{25}$$

fav prob = {which is required}

$$= \{ \text{lost by 2nd } \}$$

$$= \frac{4}{5} \times \frac{1}{5} = \frac{4}{25}$$

Condition Prob = $\frac{\text{fav}}{\text{R-Event}} = \frac{4/25}{9/25} = \frac{4}{9}$

Q.1) Considered an unbalanced die numbered 1 to 6.
 The prob of an ^{odd} face is 50% the prob of an ^{even} face and Prob of any even number face is same.

The prob of an even face given that face value exceeds 3 is 0.75 then find the Prob that face value exceeds 3.

A and B are considering to apply for a job.
 The Prob that A apply for a job is $\frac{1}{2}$
 The Prob of that A apply for job given that B apply for a job is $\frac{1}{2}$ and

Lecture no, 0.5

Standard Result of condition Prob

- $P(A/B) = \frac{P(A \cap B)}{P(B)}$ = Prob of A when B has already occurred.
- $P(B/A) = \frac{P(B \cap A)}{P(A)}$ = Prob of B when A has already occurred.
- $P(A \cap B | C) = \frac{P(A \cap B \cap C)}{P(C)}$ = Prob of simultaneous occurrence of A & B when C has already occurred.

Note $\Rightarrow$ In case of Ind events, Cond has no significance
 i.e. if A & B are Ind then $\begin{cases} P(A/B) = P(A) \\ P(B/A) = P(B) \end{cases}$

Proof $\Rightarrow$ Let A & B are Ind then

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = P(A)$$

2) To check Independency of events, we have 3 methods

$\Rightarrow$ Method 1:-

using Def of Ind. event

Method 2:-

$\Leftrightarrow$

$\Rightarrow$ If Prob $P(A \cap B) = P(A) \cdot P(B)$ then A & B are Ind

Method 3:-

$\Rightarrow$ If $P(A/B) = P(A)$ then A and B are Ind

Q:) if $P(A) = \frac{1}{6}$, $P(B) = \frac{1}{12}$, $P(A|B) = \frac{1}{12}$ then find the Prob of these simultaneous occurrence!

Sol:) a) $\frac{1}{6}$ c) $\frac{1}{12}$ ✓ $P(A|B) \neq P(A)$
 b) $\frac{1}{8}$ d) 0 $A \& B$ are not ind

∴ we can not use

$$P(A \cap B) = P(A) \cdot P(B)$$

Q:) if Prob = $P(A) = 1$ & $P(B) = \frac{1}{2}$ then $\rightarrow P(A|B)$
 $\rightarrow P(B|A)$

Sol:) $P(A) = 1$ $P(A|B) = 1$
 $P(B|A) = \frac{1}{2}$ ✓

∴ A is sure event

A will definitely occur

A is ind from B (By m-1)

B is also ind from A.

Q:) An hydrolic structure has 4 gates which operates ind the Prob of failure of each gate is 0.2 given that gate 1 has failed then find the Prob that gate 2 & gate 3 also failed!

Sol:) $P(g_1) = P(g_2) = P(g_3) = P(g_4) = 0.2$ each

$$P(g_2 \cap g_3 | g_1) = P(g_2 \cap g_3) = 0.2 \cdot 0.2 = 0.04$$

study time



$$P(\text{odd}) = 90 + P(\text{even})$$

$$P(\text{odd}) = \frac{90}{100} + P(\text{even})$$

Now

$$P(2) + P(4) + P(6) = n(10)$$

$$P(\text{even} / < 3) = 0.75$$

$$P(\text{free} > 3) = ? ? ?$$

$$\Rightarrow P(\text{even} / < 3) = 0.75$$

$$= \frac{P(\text{even} \cap < 3)}{P(< 3)} = \frac{3}{4}$$

$$= \frac{P(4 \cup 6)}{P(< 3)} = \frac{2}{4}$$

$$= \frac{P(4) + P(6)}{P(< 3)} = \frac{2}{4}$$

$$= \frac{n + n}{P(< 3)} = \frac{2}{4} = P(< 3) = \frac{8n}{3} = \frac{8 \times 10}{3} = \frac{80}{3}$$

By common sense we can write

$$P(\text{odd}) + P(\text{even}) = 1$$

$$\frac{90}{100} P(E) + P(E) = 1$$

$$P(E) = \left[\frac{9}{10} + 1 \right] = 1$$

$$P(E) \left(\frac{19}{10} \right) = 1$$

$$P(\text{even}) = \frac{10}{19}$$

$$n + n + n = \frac{10}{19} = \left(n = \frac{10}{57} \right)$$

Q.1) A and B are considering to apply for a job. The prob that A apply for a job is $\frac{1}{4}$. The prob that B apply for a job given that A apply for a job is $\frac{1}{2}$ and the prob that B apply for a job given that A does not apply for a job is $\frac{1}{3}$. Find the prob that B does not apply for a job given that A does not apply for a job.

Sol: $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{3}$, $P(B|A) = \frac{1}{2}$, $P(B|\bar{A}) = \frac{1}{3}$

$P(\bar{A}|\bar{B}) = ?$

$P(\bar{A}|\bar{B}) = P(\bar{A}/\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{1 - P(A \cap B)}{1 - P(B)}$

$= \frac{1 - \{P(A) + P(B) - P(A \cap B)\}}{1 - P(B)}$

$\Rightarrow P(\bar{A}|\bar{B}) = \frac{1}{3} = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{1}{3} \Rightarrow P(\bar{A} \cap \bar{B}) = \frac{1}{3} \cdot P(\bar{B})$
 $= \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$

Now $P(A|B) = \frac{1}{2} = \frac{P(A \cap B)}{P(B)} = 2P(A \cap B) = P(A)$
 $P(A \cap B) = 2 \times \frac{1}{4} = \frac{1}{2}$

$P(B) = \frac{1}{3}$

$= \frac{1 - \{\frac{1}{4} + \frac{1}{3} - \frac{1}{2}\}}{1 - \frac{1}{3}} = \frac{4}{3}$

Note: $P(A \cap B) \neq P(A) \cdot P(B)$ A & B are not independent

ME & Exhaustive events:

if E_1, E_2, E_3 are three events associated with S. space S s.t. $E_i \cap E_j = \emptyset$ for every $i \neq j$
 $\& E_1 \cup E_2 \cup E_3 = S$

Then events are called ME & Exhaustive events.

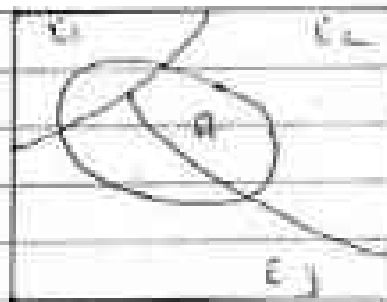
$$\therefore P(E_1 \cup E_2 \cup E_3) = P(S) = 1$$

$$\Rightarrow P(E_1) + P(E_2) + P(E_3) = 1$$

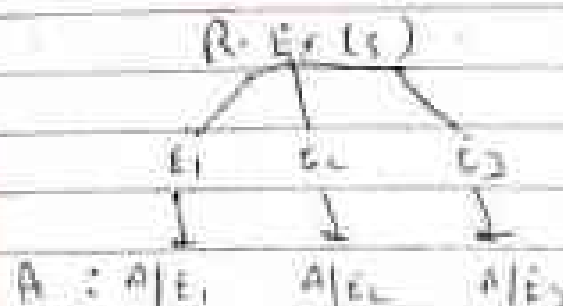
E_1, E_2, E_3 are exhaustive s.t.

Law of total Prob:

if E_1, E_2, E_3 are ME as well as exhaustive events associated with sample space S & A is an event which can occur with all E_1, E_2, E_3 then



$$P(A) = P(E_1) P(A|E_1) + P(E_2) P(A|E_2) + P(E_3) P(A|E_3)$$



Bayes's Theorem (Inverse Prob th) -

This theorem is useful to solve complex problem of conditional probability.

$i = 1, 2, \dots, n$

$\Rightarrow$

$$P(E_i | A) = \frac{P(E_i) \cdot P(A | E_i)}{P(A)} = \frac{P(E_i | A)}{P(A | E_i)}$$

$$P(E_i | A) = \frac{P(E_i) \cdot P(A | E_i)}{P(A)}$$

$$P(E_i | A) = \frac{P(E_i) \cdot P(A | E_i)}{P(A)}$$

Note: $\Rightarrow$ Look for Bayes theorem
Associated events are ME & exhaustive
i.e. $P(E_1) + P(E_2) + P(E_3) = 1$

$\Rightarrow$ whenever question, there is a feeling of cross check the given condition we run use Bayes's theorem.
and if there is no feeling of cross check anything then we use Law of total Prob

$\Rightarrow$ In Bayes's Th: What should be Assume
 $A = \{ \text{Assume A that even when A is not as we} \}$
In Law of total Prob. $A = \{ \text{Assume A that since which is required} \}$.

$\Rightarrow$ In both cases write A at the bottom of tree.

Q.) Consider a Computers are supplied to an organ according to following chart

Company	% of Computer supply	Prob of being defective
X	60%	0.01
Y	30%	0.02
Z	10%	0.03

- (i) Find the prob that supplied computer is defective
 (ii) Given that supplied computer is defective then find the prob that it was supplied by company Y!

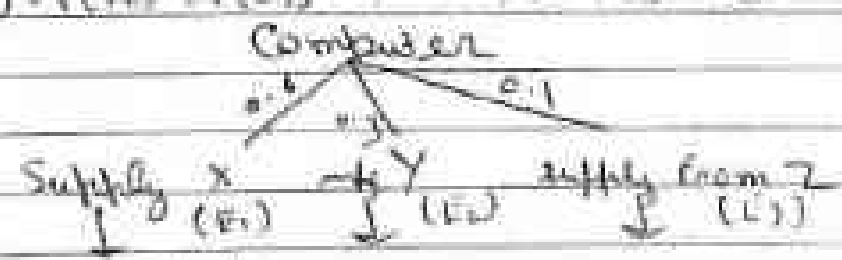
Sol (i) $A = \{\text{Supplied computer is defective}\}$

$E_1 = \{\text{Comp supplied from X}\}$

$E_2 = \{\text{" " " " Y}\}$

$E_3 = \{\text{" " " " Z}\}$

$P(E_1) + P(E_2) + P(E_3) = 1$ E_1, E_2, E_3 are MC & exhaustive



Supplied comp. is defective

A:	A/E_1	A/E_2	A/E_3
	0.01	0.02	0.03

(ii) $P(A) =$ using law of total prob

$$P(A) = P(0.6 \times 0.01 + 0.3 \times 0.02 + 0.1 \times 0.03)$$

$$= 0.015 \text{ Ans}$$

(ii) using Baye's Th:

$$P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(A)} = \frac{P(E_2) \cdot P(A|E_2)}{P(A)}$$

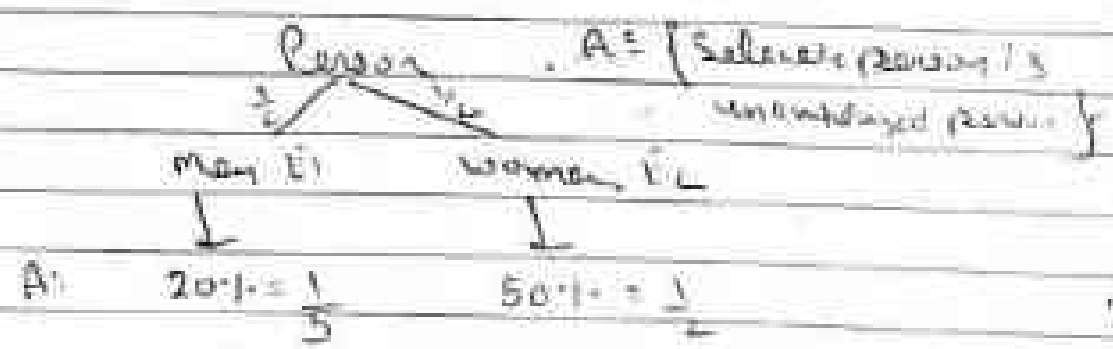
$$= \frac{P(\text{vertical path})}{P(\text{Horizontal path})} = \frac{0.3 \times 0.02}{0.015} = \frac{6}{15} \text{ Pf}$$

Similarly $P(E_1|A) = \frac{0.6 \times 0.01}{0.015} = \frac{6}{15} \text{ Pf}$

$$P(E_3|A) = \frac{0.1 \times 0.03}{0.015} = \frac{1}{15} \text{ Pf}$$

Q.) In a town there are equal no. of men & women if a in which 20% men & 50% women are employed A person is selected at random then find the Prob that selected person will be an unemployed person.

sol.)



$$P(A) = \frac{1}{2} \times \frac{1}{5} + \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{1}{10} + \frac{1}{4} = \frac{2+5}{20} = \frac{7}{20} = 35\%$$

→ Here E_1 & E_2 are ME & Exhaustive.

✓ Parcel rule.

A: {Parcel is lost}

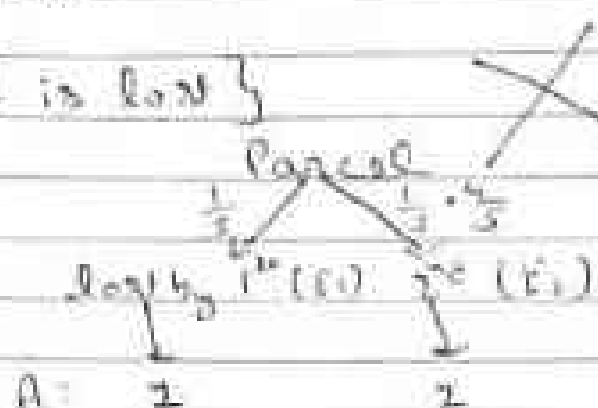


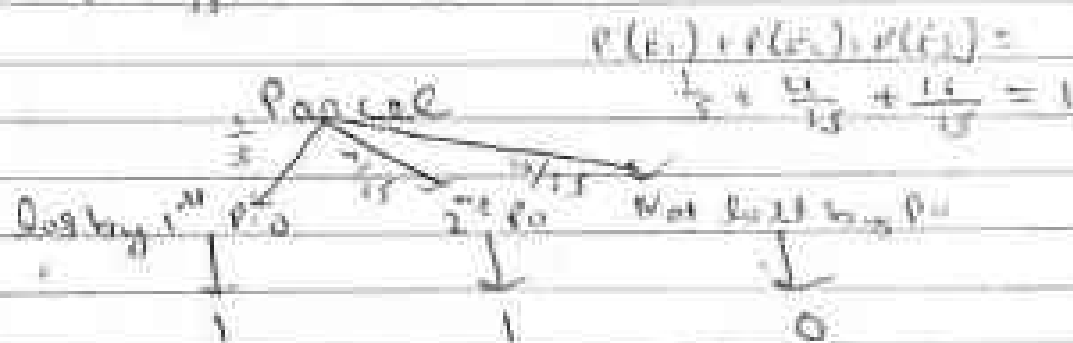
Diagram is wrong

$$\frac{1}{5} + \frac{4}{5} \neq 1$$

So not m.e.s.
not full set of mutually
exclusive events

$$P(\text{Prob}) = \frac{9/25}{\frac{1}{5} + \frac{4}{5} = 1} = \frac{9}{9}$$

⇒



$$P(E_1) + P(E_2) + P(E_3) = \frac{1}{5} + \frac{4}{15} + \frac{16}{15} = 1$$

$$P(A) = \frac{1}{5} \times 1 + \frac{4}{15} \times 1 + \frac{16}{15} \times 0 = \frac{9}{15}$$

$$P(E_1/A) = \frac{\frac{4}{15} \times 1}{\frac{9}{15}} = \frac{4}{9} \text{ or } \frac{4}{9} \text{ of}$$

1.1 Lecture no. 06

Note ⇒ if a given question 4 choices in which only one is correct then what is the prob of getting correct option if guess method is followed.

$$\text{Ans. Prob} = \frac{1}{4} \text{ or}$$

- Q. In a test an Examinee either Guess or Copy or know the answer of an objective having 4 choice each.
- The Prob of making guess is $\frac{1}{3}$ & copying is $\frac{1}{6}$.
 The Prob that his answer is correct given that he copied it is $\frac{1}{8}$ then find the Prob that he copy the answer if he answered correctly?

Ans. $A = \{\text{Examinee answered correctly}\}$

$E_1 = \text{Guess method}$

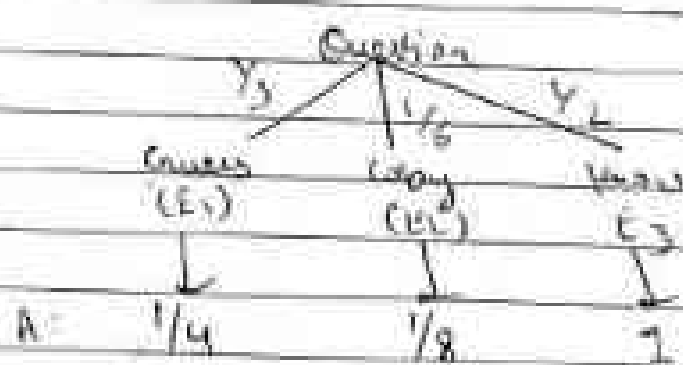
$$P(E_1) = \frac{1}{3}$$

$E_2 = \text{Copy method}$

$$P(E_2) = \frac{1}{6}$$

$E_3 = \text{Know method}$

$$P(E_3) = 1 - \left[\frac{1}{3} + \frac{1}{6} \right] = \frac{1}{2}$$



By definition

We can say these

E_1, E_2, E_3 are

ME as well as Exhaustive

$$P(A) = \frac{1}{3} \times \frac{1}{4} + \frac{1}{6} \times \frac{1}{8} + \frac{1}{2} \times 1$$

$$P(A) = \frac{29}{48} \rightarrow 29$$

$$P(E_2|A) = \frac{P(\text{v. Path})}{P(\text{H. Path})} = \frac{\frac{1}{6} \times \frac{1}{8}}{\frac{29}{48}} = \frac{1}{29} \quad \text{Ans}$$

Similarly

$$P(E_1|A) = \frac{\frac{1}{3} \times \frac{1}{4}}{\frac{29}{48}} = \frac{\frac{1}{12}}{\frac{29}{48}} = \frac{4}{29} \quad \text{Ans}$$

$$P(E_3|A) = \frac{1}{29} \quad \text{Ans}$$

⇒ out of 48 attempted question, 29 are correct & out of 29 correct answers, he copied only 1 while guessed 4 & he knew 24 ans.

2) if each question is of 1 marks w/o surmounting then marks obtained by student = $\frac{29}{48}$

" " " " " (ideally) = $\frac{24}{48}$

" " " " " (if correct answer is a guess) = $\frac{25}{48}$

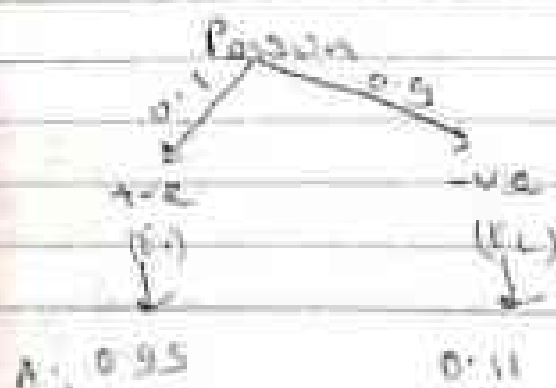
Q:) In a town 10% of the population is said to be "A new diagnostic kit arrive in the market this kit correctly identifies covid +ve individuals 95% of time and covid -ve individuals 99% of time.

A person is tested by the kit & is found to be +ve then find the prob that person is actually +ve

sol:) $A = \{\text{Person actually found +ve by kit}\}$

100 +ve person $\begin{cases} 95\% \text{ true result} \\ 5\% \text{ false result} \end{cases}$

100 -ve person $\begin{cases} 99\% \text{ true result} \\ 1\% \text{ false result} \end{cases}$



$$P(A) = (0.1 \times 0.95) + (0.9 \times 0.01) = 0.104$$

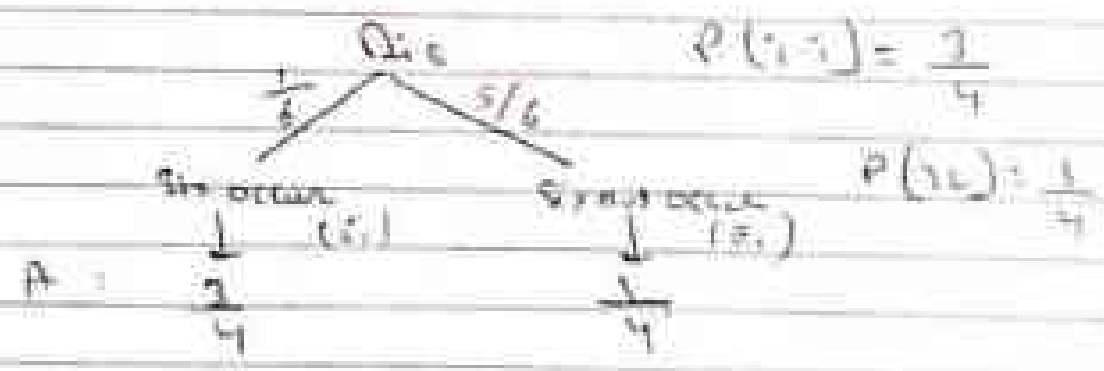
$P(\text{Actually true})$

$$P(E|A) = \frac{0.1 \times 0.95}{0.157} = \frac{95}{157}$$

$\Rightarrow$ out of 100 persons interviewed by Vit, 157 are declared true by Vit & out of 157, true is 95 persons. 95 are Actually true.

Q.3) A man is known to speak truth 3 out of 4 times he throws a die and reports that it is 6 then find the Prob that it is actually 6.

Sol.1) $A = \{\text{man reports that it is 6}\}$



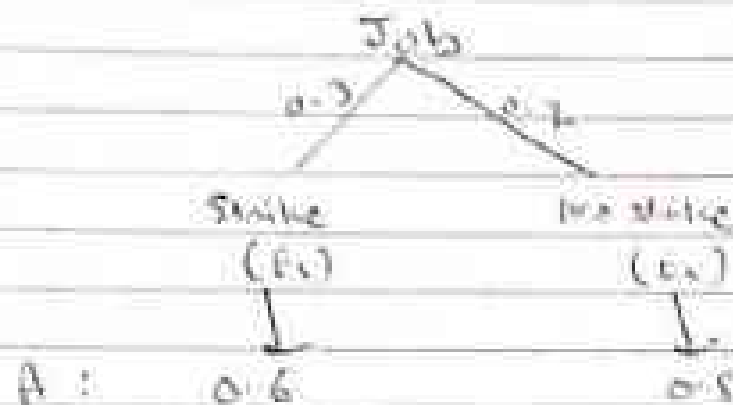
$$P(A) = \frac{1}{6} \times \frac{3}{4} + \frac{5}{6} \times \frac{1}{4} = \frac{8}{24}$$

$$P(E|A) = \frac{\frac{1}{6} \times \frac{3}{4}}{\frac{8}{24}} = \frac{3}{8}$$

$\Rightarrow$ Out of 24 Reports given by him, only 8 are correct reports showing that it is 6. Out of 8 reports, actually it is only 3 are correct.

Q.1) A person has undertaken a construction job. The prob that there will be strike is 0.3. The prob that construction job will be completed on time if there is strike is 0.6. If there is no strike 0.8 then find the prob that construction job will be completed on time.

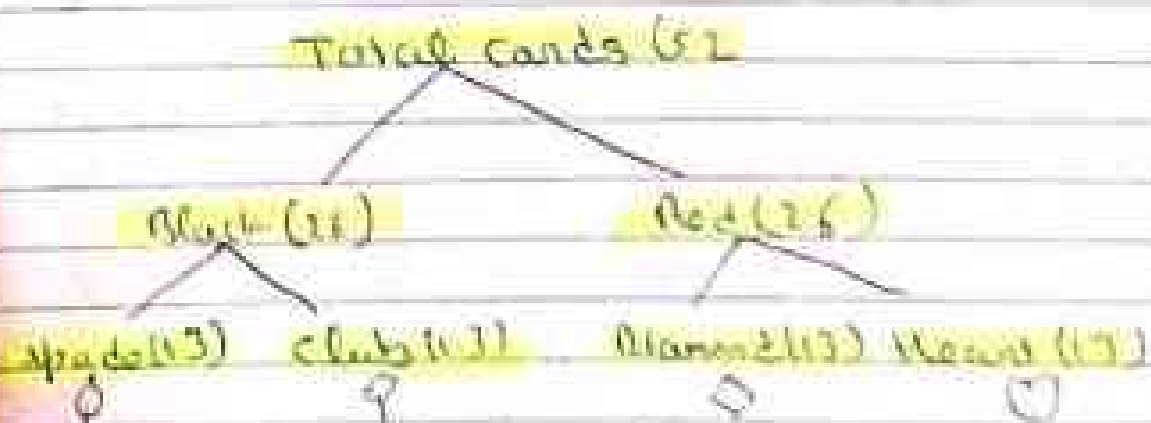
Solⁿ



$$P(A) = 0.3 \times 0.6 + 0.7 \times 0.8$$

$$= 0.18 + 0.56 = 0.74 \quad \text{Ans}$$

• # Concept of playing cards:-



Black card = 16 (K, Q, J, A)

Face card = 12 (K, Q, J)

Q:) From a pack of 52 cards, while shuffling four cards are accidentally dropped then find the prob that missing cards will be one from each suit?

Sol:) Total ways of dropping 4 cards: $52C_4$
 fav ways = missing cards will be 1 from each other
 $= 13C_1 \times 13C_1 \times 13C_1 \times 13C_1$

$$\text{Req Prob} = \frac{\text{fav}}{\text{Total}} = \frac{(13C_1)^4}{52C_4}$$

Q:) From a pack of 52 card while shuffling one card is lost then

(i) Find the Prob that lost card will be spade?

$$\frac{13C_1}{52C_1} = \frac{1}{4}$$

(ii) Find the Prob that lost card will not be spade

$$= \frac{39C_1}{52C_1} = \frac{39}{52} = \frac{3}{4}$$

Q:) From a pack of 52 card, while shuffling one card is dropped & then two cards are drawn at random then find the Prob that both the selected cards will be spade?

Case 1. study done

$$\text{Ex: } P(\text{lost card is a spade}) = \frac{{}^{13}C_1}{{}^{52}C_1} = \frac{13}{52} = \frac{1}{4}$$

$$P(\text{both the selected card are of spade}) = \frac{{}^{12}C_1}{{}^{51}C_2}$$

(from 12 spade & 39 other)

Case 2.)

$$P(\text{lost card is not spade}) = \frac{{}^{39}C_1}{{}^{52}C_1} = \frac{39}{52} = \frac{3}{4}$$

$$P(\text{both the selected card are spade}) = \frac{{}^{13}C_1}{{}^{51}C_2}$$

(from 13 spade & 38 other)

So

$$\text{Req prob will be} = P(\text{case 1}) + P(\text{case 2})$$

$$= P(\text{prob 1}) + P(\text{prob 2}) + P(\text{prob 1}) + P(\text{prob 2})$$

$$= \frac{1}{4} \times \frac{{}^{12}C_1}{{}^{51}C_2} + \frac{3}{4} \times \frac{{}^{13}C_1}{{}^{51}C_2}$$

$$= \underline{\underline{\frac{13}{51}}}$$

Concept of with & w/o Replacement -

Ex: 1) From the pack of 52 cards, 3 cards are drawn then find the number of ways if cards are drawn

$$1) \text{ Simultaneously} = {}^{52}C_3$$

$$2) \text{ one by one with Replacement} = {}^{52}C_1 \times {}^{52}C_1 \times {}^{52}C_1$$

$$3) \text{ one by one w/o } " = {}^{52}C_1 \times {}^{51}C_1 \times {}^{50}C_1$$

Date: _____

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Q-2 Hypergeometric Distribution method is not applicable

(i) There are 10 ^{cups} on a table in which 6 are defective & 4 are non def & we want to draw three cups one by one w/o replacement then find the prob that there will be exactly one defective.

Ans:

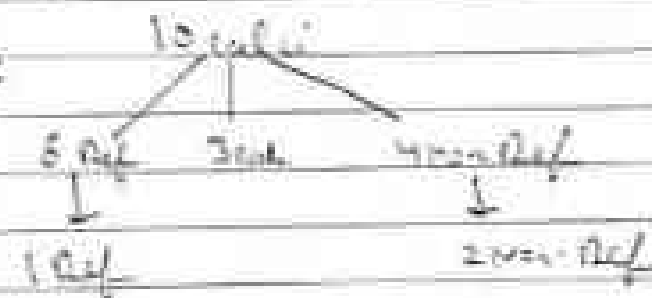
Req. Prob

$$= P[DNV \text{ or } NVN \text{ or } VNN]$$

$$= \left(\frac{6}{10} \cdot \frac{4}{9} \cdot \frac{3}{8} \right) + \left(\frac{4}{10} \cdot \frac{6}{9} \cdot \frac{3}{8} \right) + \left(\frac{4}{10} \cdot \frac{3}{9} \cdot \frac{6}{8} \right)$$

$$= 3 \times \left(\frac{6}{10} \cdot \frac{4}{9} \cdot \frac{3}{8} \right)$$

Q-3 Hypergeometric



$$\begin{aligned} \text{Req. Prob} &= \frac{\text{favor}}{\text{Total}} = \frac{6C_2 \times 4C_1}{10C_3} = \frac{6 \times 4 \times 3}{10 \times 9 \times 8} \\ &= \left(\frac{6 \times 4 \times 3}{10 \times 9 \times 8} \right) \times 3 \end{aligned}$$

(ii) also find the answer if we are draw cups one by one with replacement:

$$\begin{aligned} &\left(\frac{6}{10} \cdot \frac{4}{10} \cdot \frac{4}{10} \right) + \left(\frac{4}{10} \cdot \frac{6}{10} \cdot \frac{4}{10} \right) + \left(\frac{4}{10} \cdot \frac{4}{10} \cdot \frac{6}{10} \right) \\ &= 3 \times \left(\frac{6}{10} \cdot \frac{4}{10} \cdot \frac{4}{10} \right) \text{ Ans.} \end{aligned}$$

study time

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Chapter - 3

Date: 10/10/2020

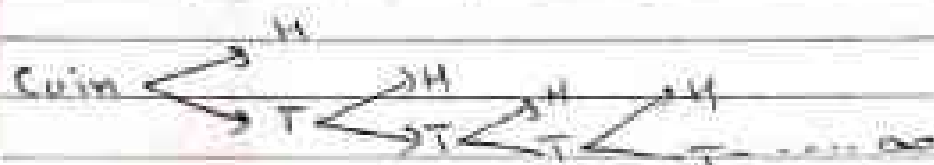
Statistics - 1

Tree diagram Based Question of Probability:-

Q1) A Note:- $S_{\infty} = a + ar + ar^2 + ar^3 + \dots \infty = \frac{a}{1-r}$

$$|r| < 1$$

Q2) A coin is tossed until head appears then find the Prob that Required no. of tosses will be odd?

sol) $S = \{H, TH, TTH, TTTH, TTTTH, \dots\}$ 

fav case = {Head is coming on odd tosses}
 = {H or TH or TTH or ...}

Req. prob = $P\{H \text{ or } TH \text{ or } TTH \text{ or } \dots\}$

$$\begin{aligned}
 P(\text{odd toss}) &= P(H) + P(TH) + P(TTH) + \dots \\
 &= \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \dots \\
 &= \frac{\frac{1}{2}}{1 - \left(\frac{1}{2}\right)} = \frac{2}{3} \text{ Ans.}
 \end{aligned}$$

Note) Both A & B can't win simultaneously so $P(A \cap B) = 0$
 2) $\frac{1}{6}$ and $\frac{5}{6}$ are MC.

Q.1) Two persons A & B play a game of dice in which alternately in which ~~one~~ anyone can win if 6 appear 1st time then find their respective chances of winning if A begins?

Ans.)

$$P(\text{winning}) = P(\text{A wins}) = \frac{1}{6} \text{ \& } P(\text{B wins}) = \frac{5}{6}$$

P(6 not come) = $\frac{5}{6}$



$$S = \left\{ \frac{1}{6}, \frac{5}{6}, \frac{25}{36}, \frac{125}{216}, \frac{625}{1296}, \dots \right\}$$

$$\begin{aligned} P(A \text{ wins}) &= P\left(\frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} + \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} + \dots\right) \\ &= \frac{1}{6} + \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^4 \cdot \frac{1}{6} + \dots \end{aligned}$$

$$= \frac{1}{6} \left[1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 + \dots \right]$$

$$\frac{1}{6} \left[\frac{1}{1 - \left(\frac{5}{6}\right)^2} \right] = \frac{6}{11}$$

$$P(B \text{ wins}) = 1 - \frac{6}{11} = \frac{5}{11}$$

Q.2)

3 friends A & B & C play a die in turn. If one get a six A wins the game then find their respective chances of winning?

$$P(A) = \frac{31}{51}$$

$$P(B) = \frac{30}{51} \quad P(C) = \frac{20}{51}$$

4. Random variable is a variable that takes numerical value depending on the outcome of R.E.

study time

Statistics

probability

Probability, Distribution

The table representing distribution of Prob is called probabilities distribution.

(eg.) A coin is tossed thrice then find the Prob dist of number of heads.

→ Random variable

sol: let $n = \{\text{no. of heads}\} = \{0, 1, 2, 3\}$

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

$$p_1 = P(n=0H) = P(TTT) = \frac{1}{8}$$

$$p_2 = P(n=1H) = P(HTT) = \frac{3}{8}$$

$$p_3 = P(n=2H) = \frac{3}{8}$$

$$p_4 = P(n=3H) = \frac{1}{8}$$

So Prob dist is

x	0	1	2	3
$P(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Random variable: When we are not sure about the outcome of an Exp then such type of Exp are called Random Exp & variable involved in R Exp is called R variable (for eg.)

Number of student, No. of vehicles, No. of books -

Discrete R variable

Height, weight etc

Continuous R variable

Counting Related variable

which has finite probability
R variable is in a certain range.

$N =$ No. of observations

only one

sample size

Random variable

D.R.V (x)

↓

Discrete Prob Dist

Binomial Dist. Poisson

↓

Prob mass func (PMF)

$$p_i \geq 0 \text{ \& } \sum p_i = 1$$

↓

1.) $\text{mean}(x) = E(x) = \sum p_i x_i$

2.) $\text{variance}(x) = E(x^2) - E^2(x)$

3.) $S.D(x) = \sqrt{\text{var}(x)}$

C.R.V (u)

↓

Continuous Prob Dist

normal / gamma / uniform

↓

Prob Density func (PDF)

$$f(u) \geq 0 \text{ \& } \int_{-\infty}^{\infty} f(u) du = 1$$

↓

1.) $\text{mean} = E(u) = \int u f(u) du$

2.) $\text{var}(u) = E(u^2) - E^2(u)$

3.) $S.D(u) = \sqrt{\text{var}(u)}$

1. V.S. $x = \{ \text{which is Required should be assumed as } x \}$

$(\bar{x}) / (M) / E(x)$

1.) Mean = / Central value / Average value / — Expected value

childhood method $\Rightarrow \bar{x} = \frac{\sum x}{N}$

Mature method $\Rightarrow E(x) = \frac{\sum p_i x_i}{\sum p_i} = \sum p_i x_i$

2.) $\therefore E(x) = \sum p_i x_i$ similarly $E(x^2) = \sum p_i x_i^2$
 $E(x^3) = \sum p_i x_i^3$

3.) variance (x) $\Rightarrow$ it measures the spread of dist. about its central value.

any one

any one

i.e. var = fluctuation = spread of data.

(m.s.D)

var(x) = variance is the average of squares of Dev from central value

$$\text{i.e. } \text{var}(x) = \frac{\sum (x - \bar{x})^2}{n} = \dots \dots \dots E(x^2) - (E(x))^2$$

$$\text{ii) } \text{S.D}(x) = \sqrt{\text{var}(x)}$$

$$\text{Note: } E(x^2) \neq (E(x))^2$$

Q:) A coin is tossed thrice then find the average number of head occur also find its variance & S.D.

$$\text{Sol: } X = \{\text{Number of heads}\} = \{0, 1, 2, 3\}$$

X:	0	1	2	3
P(X)	1/8	3/8	3/8	1/8

$$\begin{aligned} E(X) &= \sum p_i x_i = \sum p_1 x_1 + \sum p_2 x_2 + \sum p_3 x_3 + \sum p_4 x_4 \\ &= \frac{1}{8}(0) + \frac{3}{8}(1) + \frac{3}{8}(2) + \frac{1}{8}(3) \\ &= 1.5 = \frac{3}{2} \end{aligned}$$

i.e. Average number of head = 1.5

$$\begin{aligned} \text{ii) } E(x^2) &= \sum p_i x_i^2 = p_1 x_1^2 + p_2 x_2^2 + p_3 x_3^2 + p_4 x_4^2 \\ &= \frac{1}{8}(0)^2 + \frac{3}{8}(1)^2 + \frac{3}{8}(2)^2 + \frac{1}{8}(3)^2 \\ &= 3 \end{aligned}$$

Note

Question Based on Prob D.S. :-

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study time

10/10/2020

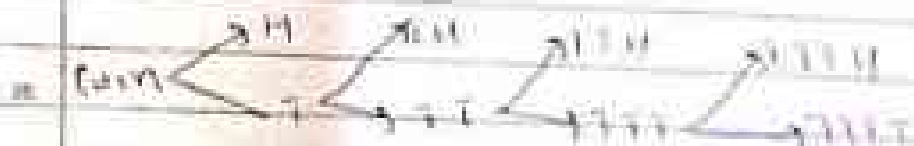
$$\begin{aligned} \text{var}(x) &= E(x^2) - (E(x))^2 \\ &= 3 - \left(\frac{3}{2}\right)^2 \\ &= \frac{3}{4} \text{ Ans.} \end{aligned}$$

$$\Rightarrow \text{S.D.}(x) = \sqrt{\text{var}(x)} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2} \text{ Ans.}$$

Lecture no. 02

Q:) A coin is tossed until head appears or tail appears 4 times in succession then find the average no. of tosses required.

Sol: / $x = \{\text{Number of tosses}\} = \{1, 2, 3, 4\}$
 $E(x) = S = \{H, TH, TTH, TTTH, TTTT\}$



x	1	2	3	4	$E(x) = 2$	😊
$P(x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$		

$$\Rightarrow P_1 = P(x=1 \text{ toss}) = P(H) = \frac{1}{2}$$

$$P_2 = P(x=2 \text{ toss}) = P(TH) = \frac{1}{4}$$

$$P_3 = P(x=3 \text{ toss}) = P(TTH) = \frac{1}{8}$$

$$P_4 = P(x=4 \text{ toss}) = P(TTTT) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

study time

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$$\begin{aligned}
 E(x) &= \sum p_i x_i = p_1 x_1 + p_2 x_2 + p_3 x_3 + p_4 x_4 \\
 &= \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(4) \\
 &= \frac{15}{8} \therefore = 1.87
 \end{aligned}$$

Ex-1) minimum tosses Req. = 1
 max " " " = 4
 Average " " " = 1.87

Q:) A die is thrown large no. of times then find the expected value of outcome!

Sol: $X = \{\text{outcome}\} = \{1, 2, 3, 4, 5, 6\}$

X :	1	2	3	4	5	6
$P(x)$:	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$\begin{aligned}
 \text{Now } E(x) &= \sum p_i x_i = \frac{1}{6} [1 + \frac{1}{6} \times 1 + \frac{1}{6} \times 2 + \frac{1}{6} \times 3 + \frac{1}{6} \times 4 + \frac{1}{6} \times 5 + \frac{1}{6} \times 6] \\
 &= \frac{1}{6} \times 6 = 1
 \end{aligned}$$

$$= \frac{1}{6} [1 + 2 + 3 + 4 + 5 + 6]$$

$$= \frac{1}{6} \times \frac{6 \times 7}{2} = 3.5$$

Q:) Each of the 9 words in the sentence
 "The quick brown fox jumps over the lazy
 dog" are written on a separate of piece of
 paper & these pieces are kept in a box
 if one piece is drawn at random then find
 the average length of the word drawn!

Sol: $X = \{\text{length of the word drawn}\} = \{3, 4, 5\}$

X	3	4	5
$P(X)$	$\frac{4}{9}$	$\frac{2}{9}$	$\frac{3}{9}$

$$\begin{aligned}
 E(X) &= \sum P_i X_i = P_1 X_1 + P_2 X_2 + P_3 X_3 \\
 &= \frac{4}{9}(3) + \frac{2}{9}(4) + \frac{3}{9}(5) \\
 &= \frac{25}{9} = 3.89
 \end{aligned}$$

minimum length of word drawn = 3
 Max " " " " = 5
 Average " " " " = 3.89

Q: In a game when a person is paid Rs 5 if he gets all heads or all tails, & he will have to pay Rs 3 if either one or two head occur, then find the expected amount wins or losses by him on an average per game.

(ii) if above game is played by 100 person 150 times each then find the expected amount wins or losses by game organizer!

Sol: Let $N_1 = 100$ person, for single person in game
 $N_2 = 150$ game, $X = \{\text{amount received}\} = \{5, -3\}$

$S = \{(HHH), (HHT), (HTH), (HTT), (TTH), (THT), (TTH), (TTT)\} = 8 \text{ samples}$

$$\begin{aligned}
 P(X = 5) &= P(\text{win}) = P(\{(HHH), (TTT)\}) \\
 &= \frac{2}{8} = \frac{1}{4}
 \end{aligned}$$

study time

Date: / / Page: /

$$P(x) = P(x = -3 \text{ Rs}) = P(\text{Loss}) = \frac{6}{8} = \frac{3}{4}$$

x	5	-3
$P(x)$	$\frac{1}{4}$	$\frac{3}{4}$

$$E(x) = \sum p_i x_i = p_1 x_1 + p_2 x_2$$

$$= \frac{1}{4}(5) + \frac{3}{4}(-3)$$

$$= -1 \text{ Rs}$$

Average amount Received by single person is - single game time

i.e. he will, 1 Rs on an Average per game

(ii) On an average single person in single game will lose = 1 Rs

" " " " " " 150 game " " = 150 Rs
 " " " 200 person in 150 game " " = 200 x 150 = 30000 Rs

i.e. game organizer will earn Rs 30000 in 150 game

Q.) In a game man wins 1 Rs for 6 and also loses one Rs for any other number when a fair die is thrown.

man decided to throw die thrice but to

quit as & when he gets six then find the expected amount wins or losses by game org.

(ii) if above game is played by 100 person 200 times each then find the expected amount wins or losses by game organizer.

study time

probability

Q.1) Let $N_1 = 108$ person
 $N_2 = 200$ person games

• for single person in single game

$$X = \{\text{Amount Received}\} = \{-1, 0, 1, 3\}$$



$$S = \{W \text{ or } LW \text{ or } LLW \text{ or } LLL\}$$

$$= \{1, 0 \text{ or } -1, -3\}$$

$$p_1 = P(X=1) = P(W) = 1/6$$

$$p_2 = P(X=0) = P(LW) = 5/6 \cdot 1/6$$

$$p_3 = P(X=-1) = P(LLW) = (5/6)^2 \cdot 1/6$$

$$p_4 = P(X=-3) = P(LLL) = (5/6)^3$$

$$P(X) = \begin{matrix} X: & 1 & 0 & -1 & -3 \\ p_i: & 1/6 & 5/36 & 25/216 & 125/216 \end{matrix} \quad \sum p_i = 1$$

$$E(X) = \sum p_i X_i = p_1 X_1 + p_2 X_2 + p_3 X_3 + p_4 X_4$$

$$= \frac{1}{6} (1) + \frac{5}{36} (0) + \frac{25}{216} (-1) + \frac{125}{216} (-3)$$

$$= -\frac{91}{54} \text{ Rs}$$

Study time

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Av amount losses by single pair in single game
 $= \frac{91}{54}$ Rs

" " " " " " " 200 game $= 200 \times \frac{91}{54}$

" " " " 108 game " 200 " $= 200 \times \frac{91}{54} \times 108$

$= 36400$ Rs.

- Q.) 2000 students are appearing in a test in which each student has to solve 150 MCQ's. Each correct answer gives 1 marks & each incorrect answer fetches -0.25 marks. If all the students have given all their answers randomly then find the sum total of these expected marks.

Sol.) $N_1 = 2000$ students
 $N_2 = 150$ Questions

For single student in single Question:

$X = \{\text{Marks obtained}\} = \{1, -1/4\}$

$P_1 = P(X = 1 \text{ mark}) = P(\text{correct}) = 1/4$ (randomly)

$P_2 = P(X = -1/4 \text{ mark}) = P(\text{wrong}) = 3/4$ " "

X	1	-1/4
$P(X)$	1/4	3/4

$$E(X) = P_1 \cdot x_1 + P_2 \cdot x_2$$

$$= \frac{1}{4} (1) + \frac{3}{4} (-\frac{1}{4})$$

$= \frac{1}{16}$ Marks

i.e. Av marks obtained by single student in single question

$$= \frac{1}{16} \left[\text{" " " " " 2000 " i.e. 150 ques.} \right] = \frac{1}{16} \times 2000 \times 150$$

$$= \frac{150000}{8} = 18750$$

Lecture no. 03

⇒ Measure of central Tendency →

∴ There are mean, median, mode.

maximum 24/05/2021

⇒ Mode: The data having highest frequency is called mode.

⇒ Median: After arranging the data either in increasing order or in decreasing order the middle most value is called median.

Case 1: if $N = \text{odd}$ $Md = \left(\frac{N+1}{2}\right)^{\text{th}}$ observation.Case 2: if $N = \text{even}$ $Md = \frac{\left(\frac{N}{2}\right)^{\text{th}} + \left(\frac{N}{2} + 1\right)^{\text{th}}}{2}$ ⇒ mean: $\bar{X} = \frac{\sum x}{N} = \frac{\text{Sum of observation}}{N}$

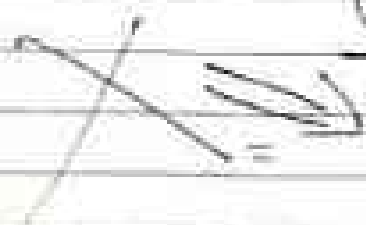
(eg) Find the mean, Md, & Mode of

2, 3, 3, 4, 4, 5, 6, 6, 6, 7, 8, 8, 8, 8, 8, 9, 9, 10, 10

Sol: ∴ mode = 8 ∵ it is repeating max no. of times

Median = $N = 20$ ~~$\frac{N+1}{2}$~~

$$Md = \frac{\left(\frac{20}{2}\right)^{\text{th}} + \left(\frac{20}{2} + 1\right)^{\text{th}}}{2} = \frac{10^{\text{th}} + 11^{\text{th}}}{2}$$



$$= \frac{21}{2} = 10.5 \quad \text{Blunder}$$

study time

20 marks

10 marks

10 marks

$$= \frac{6+7}{2} = \frac{13}{2} = 6.5$$

+ 2+3+3+0

$$\Rightarrow \text{New mean} = \frac{\sum x_i}{N} = \frac{2+2+3+3+4+5+4+3+5+8}{10}$$

$$= 6.5$$

Note: 1.) Mean ($\bar{x}$) / central value / Average ($\bar{x}$) = $\frac{\sum x_i}{N}$

also we know that Expected value $E(x) = \sum p_i x_i = \bar{x}$

i.e. $E(x) = \frac{\sum x_i}{N} = \sum p_i x_i$ similarly $E(x^2) = \frac{\sum x_i^2}{N} = \sum p_i x_i^2$

2.) Variance: $\text{var}(x) = E[(x - \bar{x})^2] = \frac{E(x - \bar{x})^2}{N} = \dots = E(x^2) - E(x)^2$

3.) S.D (σ) = $\sqrt{\text{var}(x)}$

4.) var & S.D ≥ 0 (True)

equality holds when all are equal

5.) var & S.D ≤ 1 By common sense. Always consistently

$\Rightarrow$ Covariance: $\rightarrow$ (To see in next page).

Q.) marks obtained by 100 student are showing in the table

	Marks obtained (x_i)	No. of student (f_i)
	25	20
$\sum x_i =$	30	20
	35	40
	40	20

$N = 100$

Then find the mean, median & mode of marks obtained

Study time 10 Marks To solve 50 Marks

25 15 15 ... 15, 30, 30, 30 ... 30, 35, 35, 35 ... 35

40 40 40 ... 40

20 Marks

$$\Sigma x = 25 \times 10 + 30 \times 20 + 35 \times 40 + 40 \times 20 = 3300$$

$$\Rightarrow \text{Avg marks} (\bar{x}) = \frac{\Sigma x}{N} = \frac{3300}{100} = 33 \text{ marks}$$

$$\Rightarrow \text{mode} = 35 \text{ marks}$$

$$\Rightarrow \because N=100 \therefore \text{md} = \frac{\left(\frac{N}{2}\right)^{\text{th}} + \left(\frac{N}{2} + 1\right)^{\text{th}}}{2}$$

$$= \frac{\frac{100}{2} + \left(\frac{100}{2} + 1\right)}{2} = \frac{50^{\text{th}} + 51^{\text{th}}}{2}$$

$$= \frac{35 + 35}{2} = \frac{70}{2} = 35 \text{ marks}$$

Ques:-) In class 10th we have studied that $\text{mode} = 3\text{md} - 2\text{mean}$ then why this result is not valid these questions.

Because this result is valid only when we have moderately skewed Distribution.

Measure of Dispersion:-
To these are variance, S.D. Covariance

$\Rightarrow$ Covariance: it measure simultaneous variance of two variable X & Y.

it is defined as $Cov(X, Y) = E\{(X - \bar{X})(Y - \bar{Y})\}$
 $= \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{N} = \frac{\sum (X_i Y_i) - \bar{X} \sum Y_i}{N} = \frac{\sum (X_i Y_i) - \bar{X} \sum Y_i}{N}$

Note: 1) if X & Y are two independent R.V then

$$Cov(X, Y) = 0$$

(An example) After the age of 20 yrs, $Cov(H, Age) = 0$
 $X \quad Y$

$$2) E(X) = \frac{\sum X_i}{N}, E(Y) = \frac{\sum Y_i}{N}, E(X^2) = \frac{\sum X_i^2}{N}, E(Y^2) = \frac{\sum Y_i^2}{N}$$

$$E(XY) = \frac{\sum (X_i Y_i)}{N}$$

⇒ Some more standard results:

Let X & Y are R.V & a, b, c are constants

$$1) E(ax + by + c) = aE(X) + bE(Y) + E(c)$$

$$= aE(X) + bE(Y) + c$$

$$2) Var(ax + b) = a^2 Var(X) + Var(b)$$

$$= a^2 Var(X) + 0$$

$$3) Var(ax + by) = a^2 Var(X) + b^2 Var(Y) + 2ab Cov(X, Y)$$

$$4) Var(X) = \frac{\sum (X_i - \bar{X})(X_i - \bar{X})}{N} = \frac{\sum (X_i - \bar{X})^2}{N} = Var(X)$$

∴ we know that $Cov(X, X) = E(X^2) - E(X) \cdot E(X)$

$$Cov(X, X) = E(X^2) - E(X) \cdot E(X)$$

$$= E(X^2) - E^2(X) = Var(X)$$

Q) if the difference b/w expected value of square of random and square of expected value is given as R then.

let X is the random variable

$$E(X^2) - (E(X))^2 = \text{var}(X)$$

Q:) Which one of the following batsman more consistent?

Batsman	AV	SD
V.K	54.21	4.22
R.S	51.45	5.23
S.H.Y	63.21	6.23
M.S.D	58.14	4.12

Measure of consistency

Consistency $\propto \frac{1}{SD}$

More consistency $\Rightarrow$ less S.D

(A) V.K

(B) R.S

(C) S.H.Y

(D) M.S.D ✓

Q:) if X & Y are two Ind Random variable then choose correct option

a. $E(XY) = E(X) \cdot E(Y)$ ✓

b. $\text{cov}(X, Y) = 0$ ✓

c. $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$ ✓

d. $E(X^2+Y^2) = E^2(X) \cdot E^2(Y)$ ✗

$$\text{var}(aX+bY) = a^2 \text{var}(X) + b^2 \text{var}(Y) + 2ab \text{cov}(X, Y)$$

$$\text{var}(X-Y) = 1^2 \text{var}(X) + (-1)^2 \text{var}(Y) + 2(1)(-1) \cdot 0$$

$$\text{var}(X-Y) = \text{var}(X) + \text{var}(Y)$$

Q:) if X_1 & X_2 are two zero mean Independent Random variable having variance $\frac{4}{9}$ & $\frac{1}{9}$ resp then find mean & var of $(2X_1 + 3X_2)$.

only case **Standard result using**

the binomial theorem

Ex. 1) A.T.E. $E(x_1) = E(x_2) = 0$

$$\text{var}(x_1) = \frac{1}{4}, \quad \text{var}(x_2) = \frac{1}{9}$$

Also given that x_1, x_2 are ind

i.e. $\text{cov}(x_1, x_2) = 0$

$$\begin{aligned} \text{Let } Y &= 2x_1 + 3x_2 \Rightarrow E(Y) = E(2x_1 + 3x_2) \\ &= 2E(x_1) + 3E(x_2) \\ \text{i.e. } &= 2(0) + 3(0) = 0 \end{aligned}$$

$$\begin{aligned} \text{var}(Y) &= \text{var}(2x_1 + 3x_2) \\ &= 4\text{var}(x_1) + 9\text{var}(x_2) + 2(2)(3)\text{cov}(x_1, x_2) \\ &= 4\left(\frac{1}{4}\right) + 9\left(\frac{1}{9}\right) + 0 = 1 + 1 = 2 \end{aligned}$$

a) (2, 1) b) (1, 1) c) (0, 1) d) (0, 0)

Binomial Theorem $\rightarrow$

Case 1: if $n = +ve$ integer then $(n+a)^n$

$$= \sum_{r=0}^n {}^nC_r n^{n-r} a^r$$

$$\begin{aligned} (n+a)^n &= {}^nC_0 n^n + {}^nC_1 n^{n-1} a + {}^nC_2 n^{n-2} a^2 + \dots \\ &\quad + \boxed{{}^nC_{n-1} n a^{n-1} + \dots} + {}^nC_n a^n \end{aligned}$$

Case 2: if $n = -ve$ integer or fraction then

$$(1+n)^n = 1 + \frac{n(n-1)}{2!} n^2 + \frac{n(n-1)(n-2)}{3!} n^3 + \dots$$

eg) $(1-n)^{-2}$

$$= 1 + 2n + 3n^2 + 4n^3 + 5n^4 + \dots$$

Learn by heart

$$\begin{aligned} \Rightarrow (1-n)^2 &= 1 - n + n^2 + n^2 \dots \dots \dots \text{sum of } 2 \text{ terms} \\ \Rightarrow (1-n)^3 &= 1 - n + n^2 + n^3 \dots \dots \dots \text{sum of } 3 \text{ terms} \\ \Rightarrow (1-n)^4 &= 1 - n + n^2 + n^3 + n^4 \dots \dots \dots \text{sum of } 4 \text{ terms} \end{aligned}$$

Bernoulli trials:

If any random experiment has only two possible outcomes (known as success & failure) then such type of R exp are called —

• Bernoulli trials

where p = Prob of success, q = Prob of failure

$$\therefore \boxed{q+p=1}$$

• Success = which is Required

(e.g.) A Die is thrown 6 is assumed as success then

$$P(\text{success}) = \frac{1}{6}, \quad P(\text{failure}) = \text{not getting } 6 = \frac{5}{6}$$

Geometric

Geometric Distribution: / vs Binomial Dis.

This distribution helps to find the number of trials required to get 1st success.

Def:

Let we have n trials with Prob of success = p & p (failure) = q & let x = (Number of trials required to get 1st success)

$$\text{then } x = \{1, 2, 3, 4, 5, \dots\}$$

Prob dist x : 1 2 3 4 5 6

$$P(x) = p, q^1p, q^2p, q^3p, q^4p, q^5p, \dots$$

1.) Hence $P(X=x) = P(x \text{ trials are required to get 1st success})$

$$P(X=x) = q^{x-1} p \quad \text{where } q+p=1$$

2.) $E(X) = \sum p_i x_i = p_1 x_1 + p_2 x_2 + p_3 x_3 + \dots$

$$= p(1) + q p(2) + q^2 p(3) + \dots$$

$$= p [1 + 2q + 3q^2 + 4q^3 + \dots]$$

$$= p [(1-q)^{-2}] = \frac{p}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p}$$

Similarly

$\Rightarrow \text{Var}(X) = E(X^2) - E^2(X) = \dots = \frac{q}{p^2}$

$\Rightarrow \text{SD}(X) = \frac{\sqrt{q}}{p}$

Q.) Suppose you are playing a game of Dard and the prob of getting success is 0.4 then find the prob that you will hit the Bull's Eye in 4th trial.

Sol.) $X = \{\text{No. of trials get 1st success}\}$

$p = P(\text{Bull's eye}) = 0.4$

$q = P(\text{Not getting Bull's eye}) = 0.6$

$\therefore P(X=4) = P(\text{success in 4th trial})$

$$= q^3 p$$

$$= (0.6)^3 (0.4)$$

$$= 0.0864 \quad \underline{\underline{A}}$$

$$n(4) = 1 \quad n=6$$

$$p(1) = 1 \quad p=2$$

study class

probability

Q:) Consider a company that produce on an average 3 defective bulbs out of 20 bulbs. Then find the prob that 1st def bulb will be found when 6th one is selected.

Sol:) Each has $\begin{cases} \rightarrow \text{Def Bulb} = \text{success} \\ \rightarrow \text{Non def Bulb} = \text{failure} \end{cases}$

$$p = P(\text{Def Bulb}) = \frac{3}{20} = \frac{1}{20}$$

$$q = P(\text{non def Bulb}) = \frac{17}{20}$$

$X = \{ \text{No. of trials Required to get 1st success} \}$

$$P(X=6) = P(\text{6 trials are req 1st Def bulbs})$$

$$= q^5 p = q \left(\frac{17}{20} \right)^5 \left(\frac{1}{20} \right) = 0.0386$$

Lecture no. 04

Binomial Dist:

e.g.) A coin is tossed 6 times then find the prob of getting exactly 2H?

App 2: Req prob = $\frac{\text{fav}}{\text{total}} = \frac{{}^6C_2}{2^6} = \frac{15}{64} = 0.22973$

App 1: Req prob = $P(\text{getting exactly 2H 2HT}) = {}^6C_2 \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)^4$

H = success

T = failure

$$= {}^n C_r p^r q^{n-r}$$

$$= {}^6 C_2 \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)^4 = \frac{15}{64}$$

study time

DATE: / /

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(eg.) A coin is tossed 6 times find the prob that only first two tosses are H

Ans: App 3 = $P(HHTTTT) = \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^4$

(eg.) A coin is tossed 6 times find the prob 1st two tosses are H, H

Ans: $P(HH \text{ some time}) = \left(\frac{1}{2}\right)^2 \times 1^4$

Necessary condition for B. Dist:-

There are four necessary condition.

- 1) No. of trials (B. Exp) are should be finite
i.e. $n = \text{finite}$
- 2) Each trial (B. Exp) should be Independent.
- 3) Each trial (B. Exp) has only two possible outcomes known as success & failure
i.e. (each trial must be of Bernoulli type)
- 4) The Prob of success for each trial (B. Exp) should be constant

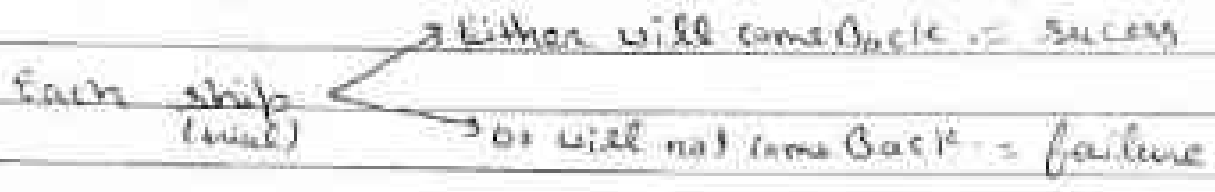
Special Trick:

When in a question we are not sure about the location of success, we can apply —
B. Dist.

Q: g:) 10 ships are going in an Atlantic ocean the find the Prob that exactly 3 will be come back if history suggest that out of 11000 ships only 1000 come back?

Ans:) $n = 10 \approx \text{finite}$

Each ship in Ind



Ex prob = $P(\text{success for each ship}) = \frac{1000}{11000} = \frac{10}{11}$

$X = \{\text{Number of ship coming back}\}$ $\begin{cases} p = \frac{10}{11} \\ q = \frac{1}{11} \end{cases} \quad \begin{matrix} n = 10 \\ r = 3 \end{matrix}$

$P(X=3) = {}^{10}C_3 \left(\frac{10}{11}\right)^3 \left(\frac{1}{11}\right)^7 =$

Definition:

Let X is Discrete Random Variable s.t. it's prob mass func is

defined as $P(X = x \text{ times}) = {}^n C_x p^x q^{n-x}$ where $p+q=1$

X is called Binomial Random variable. n.t.b
It is denoted as $X \sim B(n, p)$

Note-1

Parameters / statistical attribute $\rightarrow$
That minimum info which is necessary to apply standard result is called parameter

2.) $X = \{\text{which is require}\}$ success

3.) Prob Dist: $X: 0, 1, 2, 3, 4, \dots, n-1, n$
 $P(X): p_0, p_1, p_2, p_3, p_4, p_5, \dots, p_{n-1}, p_n$

where $\sum p_i = 1$



Q) In a Box 10% items are Defective. if 10 items are choosen at random then find the Prob. of getting exactly 2 Defective item?

$X = \{ \text{Number of defective item} \}$ success

$n = 10$

$r = 2$

$p = P(\text{Def item}) = 10\% = 0.1 \approx \text{constant}$

$q = P(\text{Not Def item}) = 0.9$

$P(X = r \text{ success}) = {}^n C_r p^r q^{n-r}$

$P(X = 2 \text{ Def item}) = {}^{10} C_2 (0.1)^2 (0.9)^8$

$= 0.1937$

Q) Out of 1000 families with 5 children each how many would you expect

- (i) Having exactly 2 ^{Boys} ~~Boys~~
- (ii) Having either 2 ^G ~~sons~~ or 3 ^B ~~daughters~~
- (iii) Having all girls.

- (a) 312 (b) 615 (c) 31 (d) 500

Let $N = 1000$ families

For Single family:

(i) $X = \{ \text{Number of Boys} \}$ success

$n = 5$

$r = 2$

$p = \frac{1}{2}$

$q = \frac{1}{2}$

$= {}^n C_r p^r q^{n-r}$

$= {}^5 C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3$

$P(X = 2) = \frac{10}{32} = \frac{0.3125}{1} = \frac{312.5}{1000}$

(ii) $x = \{\text{No. of Boys}\} = \{0, 1, 2, 3, 4, 5\}$ is binomial

$$\begin{aligned} P(x=2B, 3B) &= P(x=2B) + P(x=3B) \\ &= {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 + {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 \\ &= \frac{10}{32} = \frac{0.125}{1} = \frac{125}{1000} \end{aligned}$$

i.e. out of 1000 families no. of families having either 2B or 3B = 125 f.

(iii) $x = \{\text{No. of girls}\} = \{0, \text{success}\}$

$$\begin{aligned} {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 &= \frac{1}{32} = \frac{2.5\%}{1} = \frac{25}{1000} \\ &= \frac{0.0071}{1} = \frac{71}{1000} \text{ f.} \end{aligned}$$

Q:) In a Bike race, The Prob of a motorist being killed in an accident is $\frac{1}{2400}$ then find the prob that in a race having 200 motorist

(i) There will be no fatal accident
(ii) " " " atleast ^{one} fatal accident.

Sol:) $x = \{\text{No. of fatal accident}\}$ is binomial.

$$n=200, p = \frac{1}{2400}, q = \frac{2399}{2400}$$

$$P(x = x \text{ success}) = {}^nC_x p^x q^{n-x}$$

$$\begin{aligned} P(x=0 \text{ f.}) &= {}^{200}C_0 \left(\frac{1}{2400}\right)^0 \left(\frac{2399}{2400}\right)^{200} \\ &= 0.92 \end{aligned}$$

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Out of 100 Bacc of above type, number of race when no one will be killed = 92

$$(ii) \quad 1 - P(\text{no one will be killed}) \\ 1 - 0.92 = 0.08 = \frac{8}{100}$$

$$4) \text{ Mean } (x) = E(x) = \sum p_i x_i = p_1 x_1 + p_2 x_2 + \dots + p_n x_n \\ = \dots \dots \dots np, \text{ (Mean)}$$

$$5) \text{ var } (x) = E(x^2) - E^2(x) = \sum p_i x_i^2 - (np)^2 \dots npq$$

$$6) \text{ S.D. } (\sigma) = \sqrt{\text{var}(x)}$$

e.g.) In case of Binomial dist | Mean > var

$$7) \text{ Representation of complete B. Dist} = \sum_{x=0}^n {}^n C_x p^x q^{n-x}$$

$$\sum p_i = \sum_{x=0}^n ({}^n C_x p^x q^{n-x}) = (q+p)^n = 1^n = 1$$

e.g.) Mean & var of B. Dist are 4 & 12 respectively then find complete B. Dist.

$$\text{e.g.) } \begin{cases} \text{mean}(np) = 4 \\ \text{var}(npq) = 12 \end{cases} \left\{ \begin{array}{l} \text{i.e. Question is wrong} \\ \therefore \text{Mean} > \text{var} \end{array} \right\} \times$$

- Mean & var of B. D are 12 & 4 respectively then find complete B. Dist

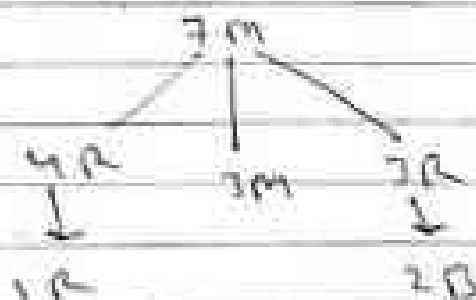
$np = 12$	$q + p = 1$	$\Rightarrow 12(q) = 4$	$(q+p)^n$
$npq = 4$		$q = \frac{1}{3}$	$= \left(\frac{1}{3} + \frac{2}{3}\right)^n$
$n = 18$		$p = \frac{2}{3}$	

Analysis of hypergeometric & Binomial Dist.

- if we are performing B. Exp one by one w/o replacement then we can also use the concept of hypergeometric Dist.
- if we are performing B. Exp one by one with Replacement then we can also use the concept of Binomial Distribution.

(e.g.) A box contains 4R & 3B marbles & we want to draw 3 marbles one by one w/o replace then find of drawing 1R & 2B marbles.

sol:) m-1 (making cases) already do that
m-2 (using hypergeometric Dist.)



$$\therefore \text{So req. Prob} = \frac{{}^4C_1 \times {}^3C_2}{{}^7C_3} = \frac{4 \times 3}{7 \times 6 \times 5} = \frac{4 \times 3 \times 2}{7 \times 6 \times 5} \times 3$$

(ii) Also find the ans if marble are drawn one by one with replacement

sol:) m-1 (By making cases)

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Req (m-2) using B-Dist

$X = \{\text{Number of Red marbles}\}$, success

$$p = P(RM) = \frac{4}{7}, \quad q = \frac{3}{7}, \quad n = 3$$

$$P(X = \text{success}) = {}^n C_r p^r q^{n-r}$$

$$\begin{aligned} P(X = 1 RM) &= {}^3 C_1 \left(\frac{4}{7}\right)^1 \left(\frac{3}{7}\right)^2 \\ &= 3 \times \frac{4}{7} \times \frac{3}{7} \times \frac{3}{7} \end{aligned}$$

Q:) There are 10 calci on a table in which 6 are Defective & 4 are non def & we want to draw 3 calci one by one w/o Replacement then find the Prob that there will be exactly one Defective?

sol:) m-1 using case m-2 using hypergeometric

$$= \left(\frac{6 \times 4 \times 3}{10 \times 9 \times 8} \right) \times 3$$

(ii) also find the answer if we are drawing calci one by one with replacement.

m-1 making case
m-2 using B-Dist

$$P(X = 10) = {}^3 C_1 \left(\frac{6}{10}\right)^1 \left(\frac{4}{10}\right)^2$$

$X = \{\text{No. of Def}\}$

$$= 3 \times \frac{6}{10} \times \frac{4}{10} \times \frac{4}{10} \text{ Ans}$$

$$p = \frac{6}{10}, \quad q = \frac{4}{10}, \quad n = 3$$

$$P(X = \text{success}) = {}^n C_r p^r q^{n-r}$$

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

Euler's constant

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Poisson Distribution:

It is a particular case of Binomial Dist under following restrictions.

- (i) $n \rightarrow \infty$ (very large)
- (ii) $p \rightarrow 0$ (very small)
- (iii) $np \rightarrow \lambda$ (i.e. constant)

$\Rightarrow$ Trick:

Whenever we are not sure about it but we can find its average value (λ) then we can apply Poisson Dist.

Def: Let X is D.R.V such that its p.m.f is defined as

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

then X is called Poisson random variable with parameter λ

It is denoted as $X \sim P(\lambda)$

Note: 1) $X = \{\text{which is required}\}$ success

$$2) \text{mean}(X) = E(X) = \sum x \cdot p(x) = 0 \cdot p_0 + 1 \cdot p_1 + 2 \cdot p_2 + \dots$$

$$3) \text{Prob Dist: } X: \begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & \dots \end{array}$$

$$P(X): \begin{array}{cccccc} p_0 & p_1 & p_2 & p_3 & p_4 & \dots \end{array}$$

where $\sum p_i = 1$

Q:) An observer counts on an average 240 veh/hr on a specific highway location then find the prob that no vehicle will arrive in a 30 seconds time interval?

Sol: $x = \{\text{No. of vehicles arrive in 30 sec}\}$ Poisson

$$\begin{aligned}\lambda &= \text{Average no. of vehicle} = 240 \text{ veh/hr} \\ &= 4 \text{ veh/min} \\ &= 2 \text{ veh/30 sec}\end{aligned}$$

$$\begin{aligned}P(x = 0 \text{ veh}) &= \frac{e^{-\lambda} \cdot \lambda^x}{x!} = P(x = 0 \text{ veh}) \\ &= \frac{e^{-\lambda} \cdot \lambda^0}{0!} = e^{-\lambda} \\ &= e^{-2} = 0.135\end{aligned}$$

out of 1000 intervals of 30 sec each, number of times when no vehicle will arrive = 135 times

Q:) Traffic office imposes on an average 5 no of penalties per day on traffic violations then find the prob that office will impose fewer than 3 penalties on a particular day?

Sol:) $x = \{\text{number of penalties on particular day}\}$ Poisson

Average no of penalties (λ) = 5 Pen/day

$$\begin{aligned}P(x = 1 \text{ veh}) &= \frac{e^{-\lambda} \cdot \lambda^x}{x!} = P(x < 3) = P(x \leq 2) = P(x = 0) \\ &+ P(x = 1) + P(x = 2)\end{aligned}$$

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$$= e^{-\lambda} \left[\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} \right]$$

$$= e^{-5} \left(1 + 5 + \frac{25}{2} \right) = \frac{37}{2e^5}$$

Q.1) A car hire firm has two cars which it hires out day by day. The no. of demands for a car follows poisson Distribution with mean 1.5. then find the no. of days in a year when car is not used?

Sol:1) $N = 365$ days

for single day: $x = \{ \text{No. of demands on day} \}$

Average demands (λ) = 1.5 per day

$$P(x = x_{\text{success}}) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \Rightarrow P(x = 0 \text{ demand}) = \frac{e^{-\lambda} \cdot \lambda^0}{0!}$$

$$= e^{-\lambda} = e^{-1.5} = 0.2231$$

i.e. the prob that on a single day car is not used = $e^{-1.5}$

hence

no. of days in a year when car is not used
 $= 365 \times e^{-1.5} = 81$ days.

Q.2) Wireless sets are manufactured with 25 solder joints each. On an average one joint in 500 is defective. then find the no. of wireless sets to be free from defective joints in a consignment of 10000 sets.

Sol:2) let $N = 10000$ sets

In single W-set: $x = \{ \text{Number of defective making one W-set} \}$

Average no. of defective rounds (λ) = np

$$p = P(\text{success}) = P(\text{Def. round}) = \frac{1}{500}$$

$n = 1.5$ rounds

$$np = 1.5 \times \frac{1}{500} = \frac{1}{333} = (\lambda)$$

$$P(x = \text{success}) = \frac{e^{-\lambda} \lambda^x}{x!} = e$$

$$P(x = \text{success}) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda} = 0.9312$$

$$\frac{0.9312}{1} \rightarrow \frac{9312}{10000}$$

out of 10000 v. uts. no. of v. uts. to be free from def. rounds = 9312

m-l Using Binomial Dist $\Rightarrow N = 10000$ v. uts.
for single v. ut, $x = \{ \text{no. of Def. rounds} \}$

$$n = 1.5 \quad \text{so } P(x = \text{success}) = {}^n C_x p^x e^{-(n-p)}$$

$$n = \frac{1}{500}$$

$$15 C_0 \left(\frac{1}{500} \right)^0 \left(\frac{499}{500} \right)^{1.5}$$

$$= 1 - \frac{499}{500}$$

$$= 0.9511$$

Special note:

Question Based on B. Dist can also be solved by using Poisson Dist

i.e. B. Dist $\Rightarrow$ P. Dist $\left\{ \begin{array}{l} \text{we use it if } n \text{ is small} \\ \text{we use B. Dist} \\ \text{if } n \text{ is large we} \\ \text{and converse is not true. Poisson Dist.} \end{array} \right.$
(n, p given) $\Rightarrow$ (λ, np)

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Q: if $X \sim \text{Poisson}$ is P.R.V s.t. $P(X=1) = P(X=2)$ then find variance of X

a) 0 b) 1 c) 2 d) $\sqrt{2}$

$$\frac{e^{-\lambda} \lambda^1}{1!} = \frac{e^{-\lambda} \lambda^2}{2!}$$

$$\lambda = \lambda^2$$

$$2\lambda = \lambda^2$$

$$2\lambda - \lambda^2 = 0$$

$$\lambda(2 - \lambda) = 0$$

$$\lambda = 0, \lambda = 2$$

But for Poisson Dist
Mean = variance = λ
So
 $\text{Var}(X) = \lambda = 2$ ✓

Recurrence Relation:

The relation b/w $P(X=x+1)$ & $P(X=x)$ is called R.R.

⇒ R.R for Binomial Dist →

We know that

$$P(X=x \text{ success}) = {}^n C_x p^x q^{n-x}$$

$$P(X=x+1 \text{ success}) = {}^n C_{x+1} p^{x+1} q^{n-(x+1)}$$

So $\frac{P(X=x+1)}{P(X=x)} = \frac{?}{?}$

$$\frac{P(X=x+1)}{P(X=x)} = \frac{{}^n C_{x+1} p^{x+1} q^{n-x-1}}{{}^n C_x p^x q^{n-x}}$$

$$\frac{P(X=x+1)}{P(X=x)} = \frac{\frac{n!}{x!(x+1)!} \cdot \frac{n!}{(n-x)! (n-x-1)!} \cdot p^{x+1} q^{n-x-1}}{\frac{n!}{x!(n-x)!} \cdot p^x q^{n-x}}$$

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$$= \frac{x!(n-x)!}{(x+1)!(n-x-1)!} \cdot \frac{p}{2}$$

$$= \frac{x!(n-x)(n-x-1)!}{(x+1)x!(n-x-1)!} \cdot \frac{p}{2}$$

$$\frac{P(n=x+1)}{P(n=x)} = \frac{n-x}{x+1} \cdot \frac{p}{2}$$

$$P(n=x+1) = \frac{n-x}{x+1} \cdot \frac{p}{2} \cdot P(n=x)$$

$\Rightarrow$ R.R for Poisson Dist $\Rightarrow$

we know that

$$P(x=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$= P(x=x+1) = \frac{e^{-\lambda} \cdot \lambda^{x+1}}{(x+1)!}$$

$$\text{So } \frac{P(x=x+1)}{P(x=x)} = \frac{\frac{e^{-\lambda} \cdot \lambda^{x+1}}{(x+1)!}}{\frac{e^{-\lambda} \cdot \lambda^x}{x!}}$$

$$= \frac{x! \lambda^x \cdot \lambda}{(x+1)! \cdot \lambda}$$

$$= \frac{x! \lambda}{(x+1)x!} = \frac{\lambda}{(x+1)}$$

$$P(1) = \frac{\lambda}{2+1} P(2)$$

$$P(2) = \frac{\lambda}{1+1} P(1)$$

$$P(3) = \frac{\lambda}{2+1} P(2) \dots$$

$$P(x=x+1) = \left(\frac{\lambda}{x+1} \right) P(x=x)$$

Lecture no. 06

⇒ Basic of Integration:-

⇒ Even function:-

if $f(-x) = f(x)$ then $f(x)$ is called function
& graph of an even func is symmetrical about Y axis. i.e Y axis is like a mirror

⇒ Odd function:-

if $f(-x) = -f(x)$ then $f(x)$ is called odd function
& graph of an odd function is symmetrical about origin

i.e Image I $\longleftrightarrow$ III & II $\longleftrightarrow$ IV

⇒ Neither even nor odd function:-

if $f(-x) \neq f(x)$ then $f(x)$ is called NEVO funct
 $\neq -f(x)$

i.e graph is neither symmetrical about Y axis
nor about origin.

(eg) check the nature of the following function:-

1) $y = f(x) = x^2$

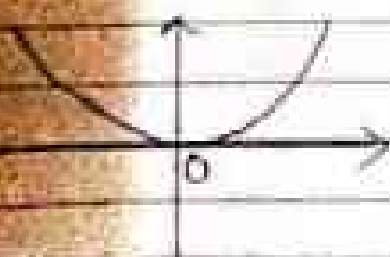
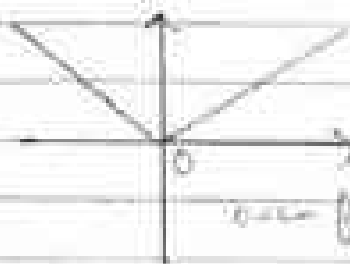
1.)

2) $y = f(x) = x^3$

3) $y = f(x) = |x|$

4) $y = f(x) = \sin x$

5) $y = f(x) = \cos x$



$y = f(x) = x^2$
even function

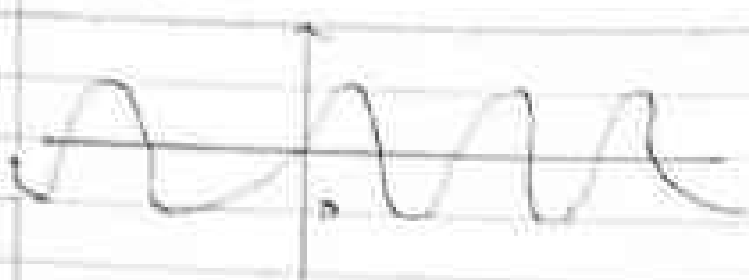
2.)



odd function
 $y = f(x) = x^3$

$$\left[\log(x) = -\infty \quad \log(1) = 0 \quad \log(x) = +\infty \right] \rightarrow \log \text{ is not odd or even}$$

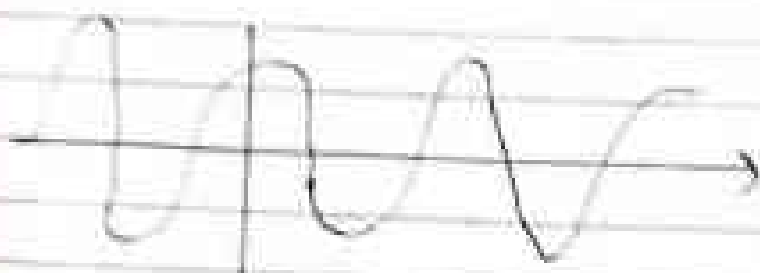
4)



$$y = f(x) = \sin(x)$$

odd function

5)



$$y = f(x) = \cos(x)$$

even function

6)

$$y = f(x) = e^x$$



$$f(-x) = e^{-x}$$

$$= \frac{1}{e^x} = \frac{1}{f(x)}$$

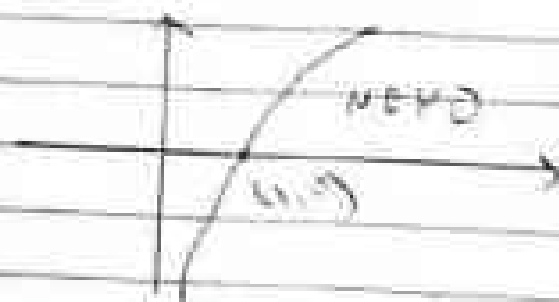
$$f(-x) \neq f(x)$$

$$f \neq f(x)$$

so this is not even

7)

$$y = f(x) = \log(x)$$



$$f(-x) = \log(-x)$$

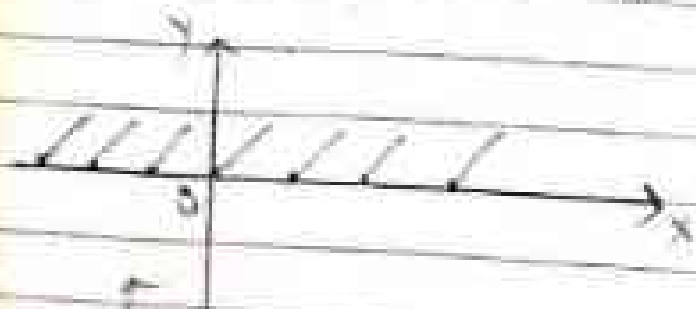
= Not Defined

$$f(-x) \neq f(x)$$

$$f \neq f(x)$$

so this is not even function

8)



NEED function

$$f(x) = x - [x]$$

9)

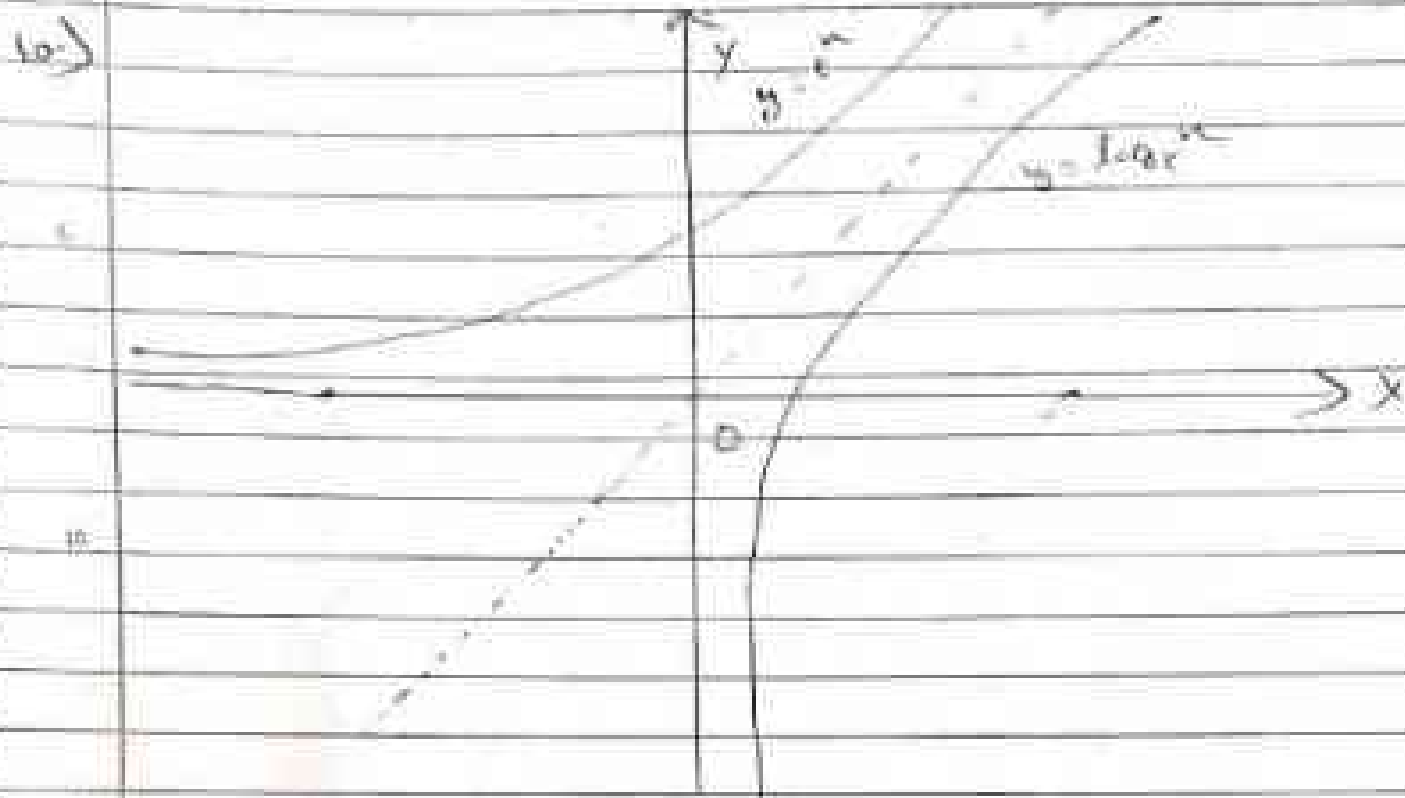


$$e^{-\infty} = 0, e^0 = 1, e^{\infty} = \infty \rightarrow \text{Osgood's Lemma}$$

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$\Rightarrow y = e^x$ & $y = \ln(x)$ are inverse.

$\Rightarrow$ Special formula:

$$\Rightarrow \int_{-x}^x f(t) dt = \begin{cases} 2 \int_0^x f(t) dt, & \text{if } f(t) \text{ is even function} \\ 0, & \text{if } f(t) \text{ is an odd function.} \end{cases}$$

(e.g.) $\int_{-\pi/3}^{\pi/3} \sin^3 x dx = 0$ $f(x) = \sin^3 x$ is an odd function

$$\begin{aligned} f(-x) &= \sin^3(-x) \\ &= -(\sin x)^3 = -\sin^3 x = -f(x) \end{aligned}$$

(e.g.) $\int_{-\pi/4}^{\pi/4} \log \left(\frac{2+\sin x}{2-\sin x} \right) dx = 0$

$$\begin{aligned} \because f(x) &= \log \left(\frac{2+\sin x}{2-\sin x} \right) \text{ is an odd function.} \\ f(-x) &= \log \left(\frac{2+\sin(-x)}{2-\sin(-x)} \right) \\ &= \log \left(\frac{2-\sin x}{2+\sin x} \right) = \log \left(\frac{2+\sin x}{2-\sin x} \right)^{-1} = -f(x). \end{aligned}$$

Note: Definite integration is the presentation of area b/w

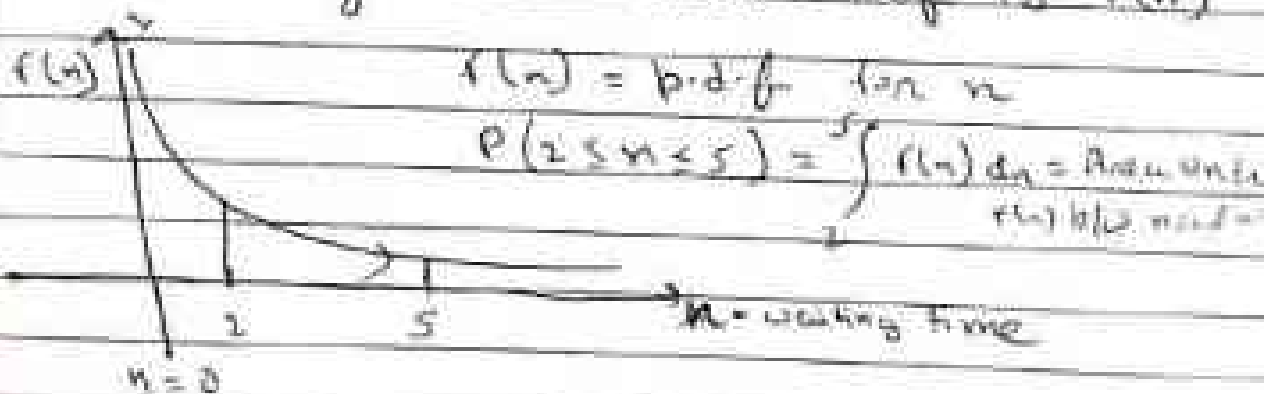
$$\int_a^b f(x) dx = \left[\phi(x) \right]_a^b = \phi(b) - \phi(a) = \text{Area}$$

= Area under $f(x)$ b/w $x=a$ & $x=b$ with x axis

Continuous Random Variable (C.R.V = x)

if a random variable has infinite possibilities in a given range then it is called C.R.V
for e.g. height, weight, time, etc.

Let x = waiting time then its p.d.f is $f(x)$



$\Rightarrow$ p.d.f = $f(x) \cong$ Prob at x {have patience}

$\Rightarrow$ let x is C.R.V, & $f(x)$ is its p.d.f then we have following results

1) $f(x) \geq 0$ i.e. graph of p.d.f always lies above x -axis

2) $\int_{-\infty}^{\infty} f(x) dx = 1$
or

$P(-\infty < x < \infty) = 1 = \text{Total area under } f(x) \text{ w.r.t } x$

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for all x
 $f(x) \geq 0$
 $\int_{-\infty}^{\infty} f(x) dx = 1$

$$3) P(a < x < b) = \int_a^b f(x) dx = \text{Area under } f(x) \text{ b/w } x=a \text{ \& } x=b$$

$$\text{or } P(a \leq x \leq b) = \dots$$

(for C.R.V)

4) if in a class we have 100 student then prob of choosing any random = $\frac{1}{100}$

=> (for C.R.V)

A ————— B Here we have ∞ possibilities for it

$\therefore P(\text{choosing any random})$
 on this $\text{rod} = \frac{1}{\infty} = 0$

i.e. At any particular point, prob will be meaning less so we will try to calculate the prob of Continuous random variable in a certain interval as follows

$$\text{i.e. } P(a < x < b) = \int_a^b f(x) dx$$

Q:) if $f(x) = K e^{-2|x|}$ & $x \in \mathbb{R}^1$ is density funct for C.R.V x then find $K=?$ p.d.f

sol:) $\therefore f(x)$ is p.d.f for x hence

total area under $f(x) = 1$

$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad = 2 \int_0^{\infty} f(x) dx = 1$$

$$= 2 \int_0^{\infty} K e^{-2|x|} dx = 1 \quad = 2K \int_0^{\infty} e^{-2|x|} dx = 1$$

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$$2K \int_0^{\infty} \frac{e^{-\alpha(x)} dx}{e^{-\alpha(x)}} = 1$$

$$= 2K \left(\frac{e^{-\alpha(x)}}{-\alpha} \right)_0^{\infty} = 1$$

$$-\frac{2K}{\alpha} [e^{-\infty} - e^0] = 1$$

$$-\frac{2K}{\alpha} (0 - 1) = 1$$

$$\boxed{K = \frac{\alpha}{2}}$$

Q:) if $\alpha = 2$ in previous question then evaluate $P(-1 \leq x \leq 1) = ?$

Sol:) $\because K = \frac{\alpha}{2} = \frac{2}{2} = 1$

Hence p.d.f is $f(x) = K e^{-\alpha(x)} = \frac{e^{-2(x)}}{e} = e^{-2(x)}$

$P(-1 \leq x \leq 1) = \text{Area under } f(x) \text{ b/w } -1 \text{ \& } 1$

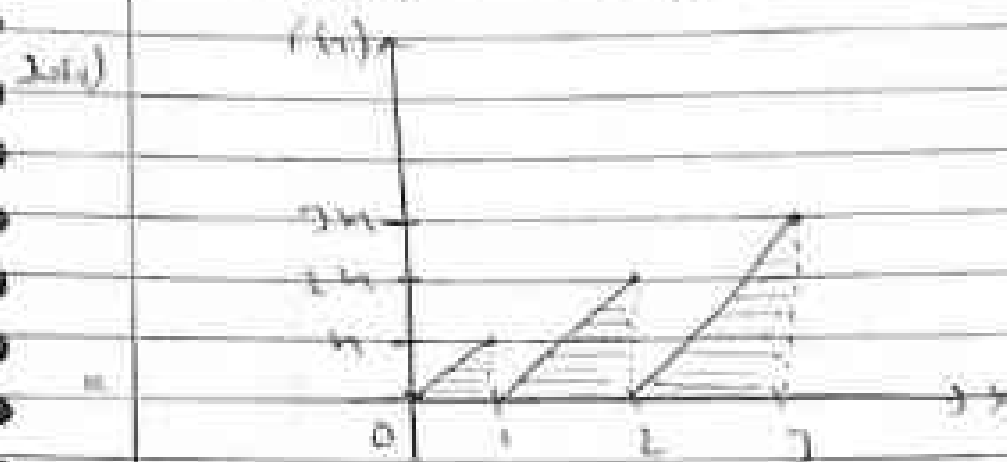
$$= \int_{-1}^1 f(x) dx$$

$$= 2 \int_0^1 f(x) dx = 2 \int_0^1 e^{-2(x)} dx$$

$$= 2 \int_0^1 e^{-2(x)} dx = 2 \left(\frac{e^{-2(x)}}{-2} \right)_0^1$$

$$= - \left(e^{-2} - e^0 \right) = 1 - \frac{1}{e^2} = 0.8647$$

Q.) Find n if $f(n)$ is valid p.d.f for R.V n as shown in diagram:



Ans.) We know that area under $f(n) = 1$

$$\text{Area of } (\Delta_1 + \Delta_2 + \Delta_3) = 1$$

$$\frac{1}{2} (1 \times n + 1 \times 2n + 1 \times 3n) = 1$$

$$\frac{6n}{2} = 1 \quad \boxed{n = \frac{1}{3}}$$

Q. 5.) Mean $(n) = E(n) = \int_{-\infty}^{\infty} n f(n) dn$

2^{nd} moment $= E(n^2) = \int_{-\infty}^{\infty} n^2 f(n) dn$

3^{rd} moment $= E(n^3) = \int_{-\infty}^{\infty} n^3 f(n) dn$

$$= E(n) = \int_{-\infty}^{\infty} |n| f(n) dn$$

In all these cases $f(n)$ is the p.d.f of n only.

Now in general if $f(n)$ is the p.d.f of n and $g(n)$ is another function of n then

$$E\{g(n)\} = \int_{-\infty}^{\infty} g(n) f(n) dn$$

$$6.) \text{var}(x) = E(x^2) - [E(x)]^2$$

$$7.) \text{S.D.}(x) = \sqrt{\text{var}(x)}$$

Q:) If $f(x) = \begin{cases} e^{-x} & , x \geq 0 \\ 0 & , x < 0 \end{cases}$ is p.d.f for x and $g(x) = e^{-x/4}$ is another function of x then find $E\{g(x)\} =$

$$\begin{aligned} \text{sol:)} \quad E\{g(x)\} &= \int_{-\infty}^{\infty} g(x) \cdot f(x) dx = \int_{-\infty}^0 g(x) \cdot 0 dx + \int_0^{\infty} g(x) (e^{-x}) dx \\ &= 0 + \int_0^{\infty} e^{-x/4} \cdot e^{-x} dx = \int_0^{\infty} e^{-5x/4} dx \\ &= \left(\frac{e^{-5x/4}}{-5/4} \right)_0^{\infty} \end{aligned}$$

$$= -4 \left[e^{-\infty} - e^0 \right] = -4(0-1) = 4 \quad \text{Ans.}$$

Q Note:) Draw the graph of $f(x) = \begin{cases} 1+x & -1 \leq x \leq 0 \\ 1-x & 0 \leq x \leq 1 \end{cases}$

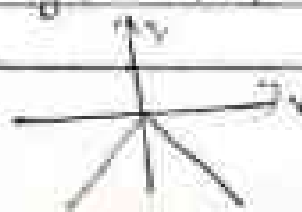
$$y = f(x) = \begin{cases} 1+x & -1 \leq x \leq 0 \\ 1-x & 0 \leq x \leq 1 \end{cases}$$

$$= 1 - |x| ; -1 \leq x \leq 1$$

$$y = |x|$$



$$y = -|x|$$



$$y = -|x| + 1$$



study time

for the C.R.V.

PROBABILITY DENSITY

Q: if x is C.R.V. & its p.d.f is $f(x) = \begin{cases} 1-x & 0 \leq x \leq 1 \\ 1-x & 0 < x < 1 \\ 1-x & 0 \leq x \leq 1 \end{cases}$
 find, mean, 2nd moment, variance & S.D
 of x

sol: Mean(x) = $E(x) = \int_{-\infty}^{\infty} x f(x) dx$

$$= \int_{-\infty}^0 x f(x) dx + \int_0^1 x f(x) dx + \int_1^{\infty} x f(x) dx$$

$$= \int_{-\infty}^0 x f(x) dx \quad \left\{ \begin{array}{l} \text{Here } f(x) \text{ is even} \\ \text{function} \end{array} \right.$$

$\Rightarrow$ $= 0$ (zero) $\rightarrow$ mean

$\Rightarrow$ 2nd moment = $E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$

$$= \int_{-\infty}^0 x^2 f(x) dx + \int_0^1 x^2 f(x) dx + \int_1^{\infty} x^2 f(x) dx$$

$$= 2 \int_0^1 x^2 (1-x) dx$$

$$= 2 \left(\frac{x^3}{3} - \frac{x^4}{4} \right)_0^1$$

$$= \frac{1}{6} \text{ unit}$$

$$\Rightarrow \text{var}(x) = E(x^2) - (E(x))^2$$

$$= \frac{1}{6} - (0)^2$$

$$= \frac{1}{6}$$

$$\text{S.D} = \sqrt{1/6}$$

$$= \frac{1}{\sqrt{6}} \text{ unit}$$

study time { again watch this question carefully.
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Q.1) if $f(n) = \begin{cases} kn+1 & 0 \leq n \leq 4 \\ 0 & \text{otherwise} \end{cases}$ is p.d.f for n then k = ?

sol.1) $f(n)$ is p.d.f so $a = -3/8$
 $b = 8/3$
 $c = 16/3$
 $d = f(n)$ is not p.d.f

$$\int_0^4 f(n) dn = 1$$

$$\int_0^4 (kn+1) dn = 1$$

$$\left(\frac{k n^2}{2} + n \right)_0^4 = 1$$

$$\left\{ k \left(\frac{16}{2} \right) + 4 \right\} - \left\{ k(0) + 0 \right\} = 1$$

$$8k + 4 = 1$$

$$\boxed{k = -3/8} \quad \text{A}$$

But due to option (D) we will cross check graph of $f(n)$ lies below x axis which is not possible.

$f(x) \geq 0 \text{ and } 0 \leq F(x) \leq 1$

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Lecture no. 07

⇒ C.O.F (Cumulative Density Function)

Let x is continuous random variable such that $f(x)$ is it's p.d.f then it's C.O.f is denoted as $F(x)$ and is defined as

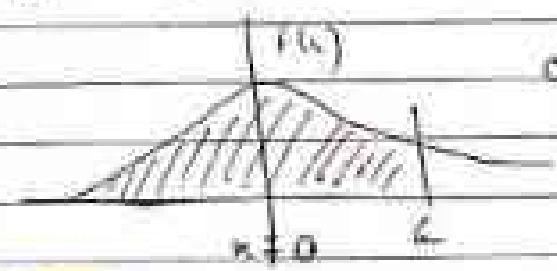
$$F(x) = \int_{-\infty}^x f(t) dt$$

$f(-\infty) = \int_{-\infty}^{-\infty} f(x) dx = 0$
 $f(\infty) = \int_{-\infty}^{\infty} f(x) dx = 1$

i.e C.O.F at $x=a$ is $F(a) = \int_{-\infty}^a f(x) dx =$

sum of all the probs from starting point upto a .

e.g.) Let x is C.R.V its p.d.f is shown in the following diagram.



area under $f(x)$ b/w $-\infty$ to a

① Notes
 $\Rightarrow F(x) = \int_{-\infty}^x f(t) dt$

$\Rightarrow F(x) = \int_{-\infty}^x f(t) dt$



$P(a < x < b) = \int_a^b f(x) dx$

i.e $f(x) = F(x) - F(a) \Rightarrow \text{p.d.f} = F(b) - F(a)$

2) $f(x) \geq 0$ Graph of p.d.f always above x axis but it is not the representation of Prob $\because f(x)$ may be greater than 1

if integrate p.d.f then come c.d.f.

1.2

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3.) For $0 \leq f(x) \leq 1$ graph of c.d.f always lies above x axis & it is the area under of Prob & $f(x) \geq 1$.

Q.1) if $f(x) = e^{-2|x|}$ is p.d.f for C.R.V. then find

Q.1) $F(-3)$?

Q.2) $F(0)$?

Q.3) $F(5)$?

Sol.1) $F(-3) = \int_{-\infty}^{-3} f(x) dx = \int_{-\infty}^{-3} e^{-2|x|} dx$



$= \int_{-\infty}^{-3} e^{-2(-x)} dx = \int_{-\infty}^{-3} e^{2x} dx = \left(\frac{e^{2x}}{2} \right)_{-\infty}^{-3}$

$= \frac{1}{2} (e^{-6} - e^{-\infty}) = \left(\frac{1}{2e^6} \right)$

4.) Let x is D.R.V. s.t. its Prob Dist is

x :	0	1	2	3
$P(x)$:	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

find c.d.f at $n=2$.

$$\sum_{i=0}^2 p_i = p_0 + p_1 + p_2$$
$$= \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}$$

5.) if x is C.R.V. s.t. its p.d.f is $f(x)$ then

c.d.f at $n=a$ is

$$F(a) = \int_{-\infty}^a f(x) dx$$

6.) $F(x) = \int_{-\infty}^x f(x) dx$

$$f(x) = \frac{d}{dx} F(x)$$

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(ii) $F(0)$:

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^0 \frac{-1(x)}{x} dx + \int_0^{\infty} \frac{1(x)}{x} dx \\ &= \int_{-\infty}^0 \frac{-1(x)}{x} dx + \int_0^{\infty} \frac{1(x)}{x} dx \\ &= \left(\frac{e^{-2x}}{2} \right)_{-\infty}^0 = \frac{1}{2} (e^0 - e^{-\infty}) = \left(\frac{1}{2} \right) \end{aligned}$$

(iii) $F(5)$:

$$\begin{aligned} \int_{-\infty}^5 f(x) dx &= \int_{-\infty}^0 \frac{-1(x)}{x} dx + \int_0^5 \frac{1(x)}{x} dx \\ &= \int_{-\infty}^0 \frac{-1(x)}{x} dx + \int_0^5 \frac{1(x)}{x} dx \\ &= \frac{1}{2} \left[\left(\frac{e^{-1x}}{-1} \right)_{-\infty}^0 \right] = \frac{1}{2} \left[\left(\frac{e^{-10}}{-1} - \frac{e^{-\infty}}{-1} \right) \right] \\ &= \left(1 - \frac{1}{2e^{10}} \right) \end{aligned}$$

Q: if $f(x) = e^{-1|x|}$ is p.d.f of x then find $F(x)$ for $x > 0$

Ans: Given $x > 0$ so $F(x) = \int_{-\infty}^x f(x) dx = \int_{-\infty}^x e^{-1|x|} dx$

$$\begin{aligned} &= \int_{-\infty}^0 e^{-1|x|} dx + \int_0^x e^{-1|x|} dx \\ &= \int_{-\infty}^0 e^{-1x} dx + \int_0^x e^{-1x} dx = \frac{1}{2} - \frac{1}{2} (e^{-2x} - e^0) \end{aligned}$$

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$$\frac{1}{2} = \frac{1}{2} \left(\frac{1}{e^{1-x}} - 1 \right) = \left(1 - \frac{1}{2e^{1-x}} \right) dx$$

Q.2) if $f(x) = e^{-2|x|}$ is p.d.f for R.V. x then find C.d.f for $x < 0$

$$F(x) = \int_{-\infty}^x f(x) dx = \int_{-\infty}^x e^{-2|x|} dx$$

$$= \int_{-\infty}^x f(x) = \int_{-\infty}^x e^{-2x} = \left(\frac{e^{-2x}}{-2} \right)_{-\infty}^x$$

$$= \frac{1}{2} (e^{2x} - e^{-\infty}) = \frac{1}{2} (e^{2x} - 0) = \frac{1}{2} e^{2x}$$

Q.3) if $f(x) = e^{-|x|}$ is p.d.f for x then find C.d.f for $x \geq 0$

$$\begin{aligned} \text{Sol.} \quad F(x) = F(x) &= \int_{-\infty}^x f(x) dx = \int_{-\infty}^x e^{-|x|} dx \\ &= \int_{-\infty}^x e^{-x} = \left(\frac{e^{-x}}{-1} \right)_{-\infty}^x \\ &= \frac{1}{2} (1 - e^{-x}) = \frac{1}{2} \text{ Ans} \end{aligned}$$

Notes

1) if $f(x) = e^{-|x|}$ is p.d.f then C.d.f $F(x)$ is given as

$$F(x) = \begin{cases} \frac{1}{2} e^{2x} & , \quad x < 0 \\ 1/2 & , \quad x = 0 \\ 1 - \frac{1}{2} e^{-2x} & , \quad x > 0 \end{cases}$$

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$$\Rightarrow \text{Evaluate } f(-1) = \frac{1}{20^{-1}} \quad \text{Ans}$$

$$f(1) = 1 - \frac{1}{20^{1/2}} \quad \text{Ans}$$

$$(ii) \quad \frac{d}{dn} \text{ Evaluate } \frac{d}{dn} f(n) =$$

$$= \begin{cases} \frac{1}{2} e^{2n} & n < 0 \\ 0 - \frac{1}{2} (-2e^{2n}) & n > 0 \end{cases}$$

$$\begin{cases} e^{2n} & n < 0 \\ e^{2n} & n > 0 \end{cases} = e^{2|n|}, \quad \text{Ans}$$

$$\therefore \left(\frac{d}{dn} f(n) \right) = f(n) \quad \& \quad \frac{d}{dn} f(n) = 0$$

Q: if $f(n) = a e^{b|n|}$ is p.d.f of C.R.V then find C.D.f for $n > 0$

$$\text{Ans: } F(n) = \int_{-\infty}^n f(n) dn = \int_{-\infty}^0 a e^{b|n|} dn + \int_0^n a e^{b|n|} dn$$

$$= a \int_{-\infty}^0 e^{2n} dn + a \int_0^n e^{2n} dn$$

$$= a \left(\frac{e^{2n}}{2} \right)_{-\infty}^0 + a \left(\frac{e^{2n}}{2} \right)_0^n$$

$$= \frac{a}{2} (1 - 0) + \frac{a}{2} (e^{2n} - 1)$$

$$= \frac{a}{2} (2 - 1) \quad \text{Ans}$$

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Q.1) if $f(n) = \begin{cases} n e^{-n\lambda} & n > 0 \\ 0 & \text{otherwise} \end{cases}$ is the density function of C.R.V. n then find its c.d.f. for any n .

$$\begin{aligned} \text{Sol: (i) c.d.f. is } F(n) &= \int_{-\infty}^n f(n) dn = \int_{-\infty}^0 (0) dn + \int_0^n n e^{-n\lambda} \\ &= 0 + \int_0^n n e^{-n\lambda} \\ &= 0 + n \left(\frac{e^{-n\lambda}}{-\lambda} \right) \Big|_0^n \\ &= - \left[e^{-n\lambda} - 1 \right] = (1 - e^{-n\lambda}) \end{aligned}$$

Q.2) (ii) Also find Prob of $(n < 5) = ?$

$$\begin{aligned} P(-\infty < n < 5) &= \int_{-\infty}^5 f(n) dn = \int_{-\infty}^0 0 dn + \int_0^5 n e^{-n\lambda} \\ &= 0 + \int_0^5 n e^{-n\lambda} dn = 1 - e^{-5\lambda} \end{aligned}$$

(M-L)

$$P(n < 5) = P(-\infty < n < 5) = F(5) = (1 - e^{-5\lambda}) \text{ in above question}$$

(iii) Evaluate $P(2 \leq n \leq 5) = ? = F(5) - F(2)$

$$\begin{aligned} P(a \leq n \leq b) &= F(b) - F(a) = (1 - e^{-5\lambda}) - (1 - e^{-2\lambda}) \\ &= \left(\frac{1 - e^{-5\lambda}}{1 - e^{-2\lambda}} \right) \end{aligned}$$

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Q:) if $f(x) = \begin{cases} 1/8, & 0 \leq x \leq 2 \\ 3/8, & 2 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$ is p.d.f then find it's c.d.f?

Sol:) Case 1: if $x < 0$ then $F(x) = \int_{-\infty}^x f(t) dt$

$$F(x) = \int_{-\infty}^x 0 dt = 0$$

Case 2: if $0 \leq x \leq 2$ then $F(x) = \int_{-\infty}^x f(t) dt$

$$= \int_{-\infty}^0 f(t) dt + \int_0^x (1/8) dt = 0 + \frac{1}{8}(x-0) = \frac{x}{8}$$

$$= 0 \times \frac{1}{8} \times \frac{2}{8} \times \frac{3}{8} \times \frac{1}{8}$$

Case 3: if $2 \leq x \leq 4$ then $F(x) = \int_{-\infty}^x f(t) dt$

$$= \int_{-\infty}^0 f(t) dt + \int_0^2 f(t) dt + \int_2^x f(t) dt$$

$$0 + \frac{1}{8} + \frac{3}{8}(x-2) = \frac{2}{8} + \frac{3x}{8} - \frac{6}{8}$$

$$= \left(\frac{3x}{8} - \frac{1}{2} \right) \quad \text{or} \quad \frac{3x}{8} - \frac{1}{2}$$

Case 4:) if $x > 4$

c.d.f is $F(x) = \int_{-\infty}^x f(t) dt$

$$= \int_{-\infty}^0 0 dt + \int_0^2 1/8 dt + \int_2^4 3/8 dt + \int_4^x 0 dt$$

$$0 + \frac{1}{8}(2-0) + \frac{3}{8}(4-2) + 0 = 1 \quad \checkmark$$

Lecture no. 08

Exponential Distribution:

Whenever in a question, we have a feeling of waiting time then we can use E.D.

E.g.: Arrival of customer in a Queue is gov by Poisson Distribution

" while their waiting time in a Queue is governed by Exponential Distribution.

Def:

Let t is C.R.V s.t it's p.d.f is define as

$$f(t) = \begin{cases} \lambda e^{-\lambda t}, & t > 0 \\ 0, & t \leq 0 \end{cases} = (MEPO)$$

t is called Exponential variable with parameter λ i.e. $t \sim E(\lambda)$

Verification: 1) Total area under $f(t)$ is

$$1.) = \int_{-\infty}^{\infty} f(t) dt = \int_{-\infty}^0 (0) dt + \int_0^{\infty} (\lambda e^{-\lambda t}) dt$$

$$= 0 + \lambda \left(\frac{e^{-\lambda t}}{-\lambda} \right)_0^{\infty} = - (e^{-\infty} - e^0) = - (0 - 1) = 1$$

$$2.) \text{Mean}(t) = E(t) = \int_{-\infty}^{\infty} t f(t) dt = \int_{-\infty}^0 (0) dt + \int_0^{\infty} (t \lambda e^{-\lambda t}) dt$$

$$= 0 + \frac{1}{\lambda}$$

$$= \frac{1}{\lambda}$$

study time

(learn)

probability

2) i.e. $\text{Mean}(t) = E(t) = \frac{1}{\mu} = \text{Average waiting time}$

3.) $\text{var}(t) = E(t^2) - E^2(t) = \int_0^{\infty} t^2 f(t) dt - \left(\frac{1}{\mu}\right)^2$
 $= \int_0^{\infty} t^2 (\mu e^{-\mu t}) dt = \frac{1}{\mu^2}$

$= \dots = \frac{1}{\mu^2}$ d (learn)

4.) S.D (t) = $\sqrt{\text{var}(t)} = \frac{1}{\mu}$ d (learn)

5.) In Poisson dist, mean = variance
 & in exponential dist, mean = S.D

5.) $t = \{\text{waiting time}\}$
 $= \{\text{which is required}\}$

6.) $P(t_1 < t < t_2) = \int_{t_1}^{t_2} f(t) dt = \text{area under } f(t) \text{ b/w } t_1 \text{ \& } t_2$

$\underline{m-2} = F(t_2) - F(t_1)$ where $F(t)$ is C.D.f

7.) C.D.f of E.R.V. $f(t) = \begin{cases} \mu e^{-\mu t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$

$F(t) = \int_{-\infty}^t f(t) dt = \int_{-\infty}^0 (0) dt + \int_0^t (\mu e^{-\mu t}) dt = \dots$
 $= \boxed{1 - e^{-\mu t}}$

8.) If the average length of phone call at public telephone booth and is of 10 mins & when you are about to start your call, someone arrives immediately ahead of you and starts calling (with your permission) then find the Prob that you will have to wait b/w 10 & 20 mins?

Q.1.) $t = \{\text{waiting time}\} = \{\text{length of phone call}\}$

Average waiting time = 10 min $\Rightarrow \frac{1}{\mu} \Rightarrow \mu = \frac{1}{10} \text{ min}^{-1}$

$$\text{L.p.d.f } f(t) = \begin{cases} \mu e^{-\mu t} & , t \geq 0 \\ 0 & , t < 0 \end{cases}$$

So,

$$P(10 < t < 20) = \int_{10}^{20} f(t) dt = \int_{10}^{20} \mu e^{-\mu t} dt$$

$$= \mu \left[\frac{e^{-\mu t}}{-\mu} \right]_{10}^{20} = - \left(e^{-20\mu} - e^{-10\mu} \right)$$

$$= \frac{1}{e^{20\mu}} - \frac{1}{e^{10\mu}} = \frac{1}{e^2} - \frac{1}{e} = 0.232$$

$$= \frac{232}{1000}$$

i.e. out of 1000 persons facing same problem,
No. of person who will have to wait 10 to 20 min = 232 persons

M-2. $P(10 < t < 20) = F(20) - F(10) = (1 - e^{-20\mu}) - (1 - e^{-10\mu})$

$$= e^{-10\mu} - e^{-20\mu}$$

$$= \frac{1}{e} - \frac{1}{e^2} = 0.234$$

Q.2) Consider a company that produces ceiling fan with Parameter $0.0003 (\text{hr})^{-1}$ then find the % of fans that will provide more than 10000 hrs service.

study time

Date: _____
Page: _____Let Company produce $N = 1000$ fans

for single fan

 t = (service hrs of the fan)

$$\lambda = 0.0003 (\text{hr})^{-1}$$

$$\text{p.d.f } f(t) = \begin{cases} \lambda e^{-\lambda t} & , t \geq 0 \\ 0 & , t < 0 \end{cases}$$

$$P(t > 10000) = P(10000 < t < \infty) = \int_{10000}^{\infty} f(t) dt$$

$$= \int_{10000}^{\infty} \lambda e^{-\lambda t} dt$$

$$= \lambda \left[\frac{e^{-\lambda t}}{-\lambda} \right]_{10000}^{\infty} = - \left[0 - e^{-10000\lambda} \right]$$

$$= e^{-3} = 0.049$$

$$P(t > 10000) = \frac{0.049}{1} = \frac{4.9}{100} \approx \frac{5}{100}$$

i.e. $\frac{5}{100} = 1\%$ Fans will provide more than 10000 service.

Q:) if X is E.R.V with parameter λ and it's C.O.f is given as

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & , x \geq 0 \\ 0 & , x < 0 \end{cases} \text{ then evaluate}$$

$$P\left(\frac{x > 2}{x > 1}\right) = ?$$

$$\text{Q:) } m-1 \text{ v. } n \text{ that } P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned}
 \Rightarrow P\left(\frac{n \geq 2}{n \geq 1}\right) &= \frac{P\{n \geq 2 \cap n \geq 1\}}{P(n \geq 1)} = \frac{P(n \geq 2)}{P(n \geq 1)} \\
 &= \frac{1 - P(0 \leq n \leq 1)}{1 - P(0 \leq n \leq 0)} \\
 &= \frac{1 - P(2)}{1 - P(1)} = \frac{1 - (1 - e^{-2})}{1 - (1 - e^{-1})} = \frac{e^{-2}}{e^{-1}} \\
 &= \frac{1}{e}
 \end{aligned}$$

M-L C.O.F, $F(n) = \begin{cases} 1 - e^{-n}, & n \geq 0 \\ 0, & n < 0 \end{cases}$

$$f(n) = \frac{d}{dn} F(n) = \begin{cases} e^{-n}, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$$\begin{aligned}
 P\left(\frac{n \geq 2}{n \geq 1}\right) &= \frac{P(n \geq 2 \cap n \geq 1)}{P(n \geq 1)} = \frac{P(n \geq 2)}{P(n \geq 1)} \\
 &= \frac{\int_2^{\infty} f(n) dn}{\int_1^{\infty} f(n) dn} = \frac{\left(\frac{e^{-n}}{-1}\right)_2^{\infty}}{\left(\frac{e^{-n}}{-1}\right)_1^{\infty}} \\
 &= \frac{(0 - e^{-2})}{(0 - e^{-1})} = \frac{1}{e}
 \end{aligned}$$

9.) Traffic is moving at the Rate of 360 veh/hr on a specific highway location. if the arrival of the vehicle at the location follows Poisson distribution then find the Prob that GAP b/w two successive veh lies b/w 6 to 10 seconds?

Exd.) $t = \{\text{Gap b/w two successive vehicles}\}$

Av. no. of vehicle coming at junction $(\lambda) = 360$

$$\lambda = 360 \text{ vehicle/hr} = 6 \text{ veh/min} = \frac{1 \text{ veh}}{10}$$

i.e. A.V time gap b/w two successive vehicle
 $= 10 \text{ sec} = \frac{1}{\lambda} = \lambda = \frac{1}{10}$

$$\therefore \text{p.d.f } f(t) = \begin{cases} \lambda e^{-\lambda t} & , t \geq 0 \text{ where } \lambda = \frac{1}{10} \text{ per sec} \\ 0 & , t < 0 \end{cases}$$

$$P(6 < t < 10) = \int_6^{10} \lambda e^{-\lambda t} dt = \lambda \left[\frac{e^{-\lambda t}}{-\lambda} \right]_6^{10}$$

$$= - \left[e^{-10 \times 4} - e^{-6 \times 4} \right] = \frac{1}{e^{10}} - \frac{1}{e^6}$$

$$= 0.18$$

Note: Inter arrival time follows Exponential dist.

Uniform Distribution:

If Probabilities are Distributed uniformly for given value of n then you should follow 'U-Dist'.

Ex) if a die is thrown once then its sample points outcome are

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$s: 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6$$

$$p(s): \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6}$$

i.e. for $a \leq x \leq b$ $P(x) = \frac{1}{b-a}$ const $\Rightarrow x$ is U.R.V of Discrete type

e.g.) let x is Random point of stick of length l

$0 \leq x \leq l$ i.e. $x \in (0, l)$

$$P(\text{choosing any Random point } x) = \frac{1}{l}$$

i.e. $P(x) = \frac{1}{l} = \text{const}$ so x is U.R.V is $(0, l)$

Def:

let x is define in $[a, b]$ & x is C.R.V & it's p.d.f is defined as given as

$$f(x) = \begin{cases} \frac{1}{b-a} & , a \leq x \leq b \\ 0 & , \text{otherwise} \end{cases}$$

then x is called U.R.V

$$\Rightarrow \text{mean of } (x) = E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_a^b x \frac{1}{b-a} dx$$

$$= \frac{a+b}{2} \text{ (Average) } dx$$

$$\Rightarrow \text{variance}(x) = E(x^2) - E^2(x) = \int_a^b x^2 f(x) dx - \left(\frac{a+b}{2}\right)^2$$

$$= \dots = \frac{(b-a)^2}{12}$$

$$\Rightarrow S.D.(x) = \frac{b-a}{\sqrt{12}} \text{ p. form}$$

$$\Rightarrow P(a \leq x \leq b) = \int_a^b f(x) dx = \text{Area under } f(x) \text{ b/w } a \text{ \& } b$$

$$= \frac{b-a}{b-a}$$

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Q.) Suppose you break a stick of unit length at a uniformly chosen random point then find Expected

- (i) length of broken stick
- (ii) length of shorter stick.
- (iii) length of longer stick.

Ans.) (i) $X = \{\text{length of broken stick}\}$

So

$$X \in (0, 1)$$


$$\text{p.d.f } f(x) = \begin{cases} \frac{1}{1-0} & , 0 \leq x \leq 1 \\ 0 & , \text{other} \end{cases}$$

ii.

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^1 x(1) dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2} = 0.5 \text{ m}$$

(ii) $Y = \{\text{length of shorter stick}\}$



$$\text{p.d.f } Y \in (0, \frac{1}{2}) \Rightarrow f(y) = \begin{cases} \frac{1}{\frac{1}{2}-0} & , 0 \leq y \leq \frac{1}{2} \\ 0 & , \text{other} \end{cases}$$

$$E(Y) = \int_{-\infty}^{\infty} y f(y) dy$$

$$= \int_0^{\frac{1}{2}} y(2) dy = 2 \left[\frac{y^2}{2} \right]_0^{\frac{1}{2}} = \frac{1}{4} \text{ m}$$

Q7.) $n = \{\text{length of longer stick}\}$



$$n \in \left(\frac{1}{2}, 1\right) \Rightarrow f(n) = \begin{cases} \frac{1}{1-n} & \frac{1}{2} \leq n \leq 1 \\ 0 & \text{other} \end{cases}$$

$$E(n) = \int_{-\infty}^{\infty} n \cdot f(n) = \int_{\frac{1}{2}}^1 n(n) \, dn$$

$$= \left(\frac{n^2}{2}\right)_{\frac{1}{2}}^1 = 1 - \frac{1}{4} = \frac{3}{4} \quad \text{A}$$

Q8.) if x & y are two ^{ind} i.i.d. R.V. defined in $[-1, 1]$ then evaluate $P[\max(x, y) \leq \frac{1}{2}] = ?$

Sol.) $\because$ x is V.R.V. in $[-1, 1]$ so p.d.f. is

$$f(x) = \begin{cases} \frac{1}{1-(-1)} & -1 \leq x \leq 1 \\ 0 & \text{other} \end{cases}$$

$\because$ y is also V.R.V. in $[-1, 1]$ so p.d.f. is

$$f(y) = \begin{cases} \frac{1}{1-(-1)} & -1 \leq y \leq 1 \\ 0 & \text{other} \end{cases}$$

$$P(\max(x, y) \leq \frac{1}{2}) = P(-1 \leq (x, y) \leq \frac{1}{2})$$

$$= P(-1 \leq x \leq \frac{1}{2}) \cdot P(-1 \leq y \leq \frac{1}{2}) \quad (\text{indep.})$$

$$= \int_{-1}^{\frac{1}{2}} f(x) \, dx \times \int_{-1}^{\frac{1}{2}} f(y) \, dy$$

$$= \int_{-1}^{\frac{1}{2}} f(x) \, dx \times \int_{-1}^{\frac{1}{2}} f(y) \, dy$$

$$= \frac{1}{4} \left(\frac{2}{2} \right) \left(\frac{2}{2} \right) = \frac{9}{16} \text{ Ans}$$

Lecture no. 09

Q:) if p is the length of the position vector of any random point with in the rectangle formed by $(0,0), (1,0), (1,1), (0,1)$ then find $E\{p^2\}$ = ?

Ans:) let $P(x,y)$ be any random point in the rectangle



then P.V of $P(x,y) = \vec{OP} = x\hat{i} + y\hat{j}$

$$\text{so } p = |\vec{OP}| = \sqrt{x^2 + y^2}$$

$$\text{i.e. } p^2 = x^2 + y^2 \Rightarrow E\{p^2\} = E\{x^2 + y^2\} = E\{x^2\} + E\{y^2\}$$

$$= \int_{-\infty}^{\infty} x^2 f(x) dx + \int_{-\infty}^{\infty} y^2 f(y) dy \quad \text{--- (1)}$$

$\because 0 \leq x \leq 1$ & x is U.R.V so p.d.f is $f(x)$

$$f(x) = \begin{cases} \frac{1}{1-0} & 0 \leq x \leq 1 \\ 0 & \text{other} \end{cases}$$

$0 \leq y \leq 1$ & y is U.R.V so p.d.f is $f(y) = \begin{cases} \frac{1}{1-0} & 0 \leq y \leq 1 \\ 0 & \text{other} \end{cases}$

$$\text{Now by (1) } E\{p^2\} = \int_{-\infty}^{\infty} x^2 f(x) dx + \int_{-\infty}^{\infty} y^2 f(y) dy$$

$$= \int_0^1 x^2 (1) dx + \int_0^1 y^2 (1) dy$$

$$= \left(\frac{x^3}{3} \right)_0^1 + \left(\frac{y^3}{3} \right)_0^1 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \text{ Ans}$$

Q.1) At a certain traffic junction the cycle of traffic lights is 2 min of green (vehicle does not stop) + 3 min of Red (vehicle stop). If the arrival time of the vehicle at the junction follows uniform distribution over 5 min cycle then find the expected waiting time of the vehicle at the junction!

Sol:1) Let $x = \{\text{Arrival time of vehicle at the junction}\}$

$$0 \leq x \leq 5 \text{ so p.d.f is } f(x) = \begin{cases} \frac{1}{5-0}, & 0 \leq x \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{waiting time} = g(x) = \begin{cases} 0, & 0 \leq x \leq 2 \\ 5-x, & 2 < x \leq 5 \end{cases}$$

$$E\{g(x)\} = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx = \int_0^5 g(x) \left(\frac{1}{5}\right) dx$$

$$= \int_0^2 (0) \frac{1}{5} dx + \int_2^5 (5-x) \frac{1}{5} dx$$

$$0 + \int_2^5 (5-x) \frac{1}{5} dx = \frac{1}{5} \left(5x - \frac{x^2}{2} \right)_2^5$$

$$= \dots = 0.5 \text{ min}$$

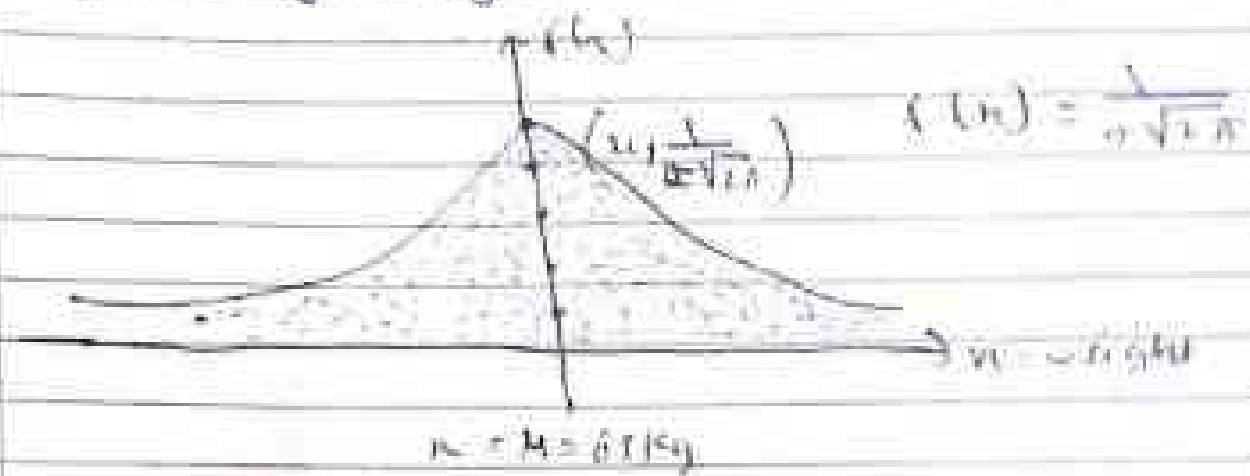
$$\text{Expected waiting time} = 0.5 \times 60 = 30 \text{ sec}$$

Normal Dist (Gaussian Dist) / Bell curve
(100% they will ask Ques from this topic)

$\Rightarrow$ Whenever random variable x has a tendency to accumulate about x 's average value

(H) then x is called normal random variable and the distribution formed is called Normal Dist / Gaussian Dist / Bell curve.

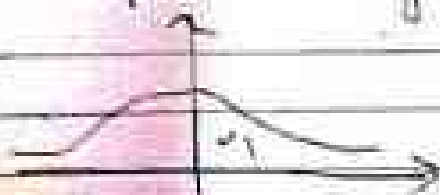
(g) Height, weight, Age, Marks distribution, Marks, etc.



$\Rightarrow$ This is **NEN0** function

Notes:

- 1.) N. curve is a symmetrical curve with symmetry about mean $(x = u)$
- 2.) N. curve is a bell shaped curve where variance is represented by width of the bell.



i.e. more flat curve is more variance.

3.) N. curve is a unimodal curve i.e. Mean = Median = Mode

4.) Highest point of normal curve occurs at $x = u$
 and highest ^{value} $y = f(u) = \frac{1}{\sigma\sqrt{2\pi}}$
 * point = x
 value = y

Def:)

Let x is C.V. s.t its p.d.f is given as

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{--- (1)}$$

then x is called normal random variable with parameter μ & σ^2 It is denoted as $x \sim N(\mu, \sigma^2)$

Note:)

1.) $x = \{\text{which is Required}\}$

$$2.) \text{Mean}(x) = \int_{-\infty}^{\infty} x \cdot f(x) dx = \dots$$

$$\dots = \mu$$

$$3.) \text{variance}(x) = E(x^2) - E^2(x) = \int_{-\infty}^{\infty} x^2 f(x) dx - (\mu)^2$$

$$\dots = \sigma^2$$

$$\Rightarrow \text{Total area under } f(x) = \int_{-\infty}^{\infty} f(x) dx = \dots$$

$$\text{--- using given facts ---} = 1$$

$$\Rightarrow P(a \leq x \leq b) = \int_a^b f(x) dx = \text{Not easy to integrate}$$

M-L = Area under $f(x)$ b/w $x = a$ & $x = b$
 = Still not easy to find area

Hence we have to develop new concept to find above prob.

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Note: if x is N.B.V s.t. $x \sim N(\mu, \sigma^2)$ & $z = \frac{x - \mu}{\sigma}$
then find mean & variance of z ?

$$\text{Sol: } \mu_z = E(z) = E\left[\frac{x - \mu}{\sigma}\right] = \frac{1}{\sigma} [E(x) - E(\mu)]$$

$$= \frac{1}{\sigma} [\mu - \mu] = 0$$

&

$$\sigma_z^2 = \text{var}(z) = \text{var}\left(\frac{x - \mu}{\sigma}\right) = \frac{1}{\sigma^2} [\text{var}(x) - \text{var}(\mu)]$$

$$= \frac{1}{\sigma^2} [\sigma^2 - 0] = 1$$

$$\therefore x \sim N \quad \mu_z = 0, \sigma_z = 1, \sigma_z^2 = 1$$

Def: Standard Normal variable (z) $\Rightarrow$

if $x \sim N(\mu, \sigma^2)$ & $z = \frac{x - \mu}{\sigma}$ then mean (μ_z)
& $\text{var}(z) = 1$ & in that case z is called
standard normal variable with p.d.f. is defined as

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2}$$

$\mu = 0, \sigma = 1$
 $z = \left(\frac{x - \mu}{\sigma}\right)$

$$= \boxed{f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}} \Rightarrow \text{Even function}$$

$$\therefore z \sim N(0, 1)$$

$\Rightarrow$ Standard normal variable free from any parameter

$\Rightarrow$ Special conclusion:

$$P(a \leq n \leq b) = \text{Area b/w } a \text{ \& } b = P(z_1 \leq z \leq z_2)$$

Area b/w z_1 \& z_2 = use normal table



$$z = \frac{n - \mu}{\sigma}$$

At $n = \mu$, $z = \frac{\mu - \mu}{\sigma} = 0$

At $n = a$, $z_1 = \frac{a - \mu}{\sigma}$

At $n = b$, $z_2 = \frac{b - \mu}{\sigma}$

1) N. Table starts from $z=0$ and is defined for the value of z

$$\begin{aligned} \text{eg } P(0 \leq z \leq 1.28) &= \int_0^{1.28} f(z) dz = \int_0^{1.28} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2\sigma^2}} dz \\ &= 0.4 \quad \left\{ \text{use table} \right\} \end{aligned}$$

2) Flow chart of solving Question:

Step 1) First find n = {which is Required}

Step 2) Convert n into z by using $z = \frac{n - \mu}{\sigma}$

Step 3) Use concept of symmetry

Step 4) Use normal table

⇒ **Concepts of Symmetry =**

1) $P(-\infty < Z < \infty) = \text{Total area under } f(z) = 1$

2) $P(-\infty < Z < 0) = P(0 < Z < \infty) = \frac{1}{2}$
 (left side area) (right side area)

3) $P(-1 \leq Z \leq 1) = 2P(0 \leq Z \leq 1)$
 $= 2(\text{use normal table})$
 $= 2(0.3413)$

4) $P(-1 \leq Z \leq 2) = P(-1 \leq Z \leq 0) + P(0 \leq Z \leq 2)$
 $= P(0 \leq Z \leq 1) + P(0 \leq Z \leq 1.7)$
 $= (0.3413) + (0.4435)$

5) $P(-\infty < Z \leq 3) = P(-\infty < Z < 0) + P(0 \leq Z \leq 3)$
 $= \frac{1}{2} + (0.392)$

6) $P(Z > 2.58) = P(0 < Z < \infty) - P(0 \leq Z \leq 2.58)$
 $= \frac{1}{2} - (0.55)$



Q) 2400 students are appearing in an exam in which distribution of marks are assumed to be normal with mean 70 marks and S.D 6.25 marks. then find the no. of students getting marks b/w 20 and 40. It is given that area under the N curve b/w $Z=0$ & $Z=1.6$ is 0.4452.

study class

probability

Sol:) let $N = 2000$ student

for single student: $X = \{\text{marks obtained}\}$

$$\mu = 30, \quad \sigma = 6.25, \quad Z = \frac{X - \mu}{\sigma}$$

$Z_1 = \frac{20 - 30}{6.25} = -1.6$

$Z_2 = \frac{40 - 30}{6.25} = 1.6$

$$P(20 < X < 40) = P(-1.6 < Z < 1.6)$$

$$= 2 \left(0.5 - 2 < 1.6 \right)$$

$$= 2 (0.4491) = 0.8982 = \frac{898.2}{1000} = \frac{1796.4}{2000}$$

$$= \frac{1796}{2000}$$

∴ out of 2000 students no. of student getting b/w 20 & 40 = 1796

Q:) In a school of 1000 students average height of student is 68.22 inches and variance is 8 inches² then find the no. of students having height over 6 feet tall if we have a confidence of 36.461 within the limit $Z = 0$ to 1.5

Sol:) let $N = 1000$ student for a single student
 $X = \{\text{Height}\}$

$$\mu = 68.22, \quad \sigma = \sqrt{10.8}, \quad Z = \frac{X - \mu}{\sigma} = \frac{72 - 68.22}{\sqrt{10.8}} = 1.15$$

$$P(\text{Height} \geq 6 \text{ feet}) = P(X > 72) = P(Z > 1.15)$$

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$$= \frac{1}{2} - P(0 \leq Z \leq 1.15)$$

$$\frac{1}{2} - 0.3746 = \underline{0.1254}$$

$$= \frac{125.4}{1000} \approx 1.25 \text{ standard}$$

Q.1) if x is Gaussian variable having mean 1 & variance 4 & let y is another Gaussian variable with mean -1 & variance a unknown also we have $P(x \leq -1) = P(y \geq 2)$ then find S.D of y

Q.2) if x is zero mean unit variance Gaussian variable then find $E(|x|)$

Q.3) $x \sim (1, 4)$ $y \sim (-1, \sigma^2)$ & we have to find σ_y

$$\text{ATQ } P(x \leq -1) = P(y \geq 2)$$

$$Z_x = \frac{x - \mu_x}{\sigma_x} = \frac{-1 - 1}{2} = -1$$

$$Z_y = \frac{y - \mu_y}{\sigma_y} = \frac{2 - (-1)}{\sigma_y} = \frac{3}{\sigma_y}$$

$$P(Z_x \leq -1) = P(Z_y \geq \frac{3}{\sigma_y})$$

$$P(Z_x \geq 1) = P(Z_y \geq \frac{3}{\sigma_y})$$

$$1 = \frac{3}{\sigma_y} \Rightarrow \boxed{\sigma_y = 3} \text{ Ans}$$

Q: 3 if u & v are two independent normal variables gaussian variable having variance $\frac{1}{4}$ & $\frac{1}{9}$ resp then $P(3v \geq 2u) = ?$

Lecture no. 10

Sol: 2: $\mu_n = 0$, $\sigma_n = 1$

" $u \sim N(\mu, \sigma^2)$

or

$u \sim N(0, 1)$

is S.V.V and no need to convert it into S.V.V its p.d.f is $f(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$

(i) $E(u) = \text{mean}(u) = 0$ (given)

(ii) $E(g(u)) = \int_{-\infty}^{\infty} g(u) f(u) du$

or $E(|u|) = \int_{-\infty}^{\infty} |u| \cdot \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \right) du$

" $= 2 \int_0^{\infty} \frac{1}{\sqrt{2\pi}} |u| \cdot e^{-\frac{u^2}{2}} du$

$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} u e^{-\frac{u^2}{2}} du$

$= \text{Put } \frac{u^2}{2} = t$
 $u du = dt$

$$E(|u|) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-t} dt$$

$$= \sqrt{\frac{2}{\pi}} \left(\frac{e^{-t}}{-1} \right)_0^{\infty}$$

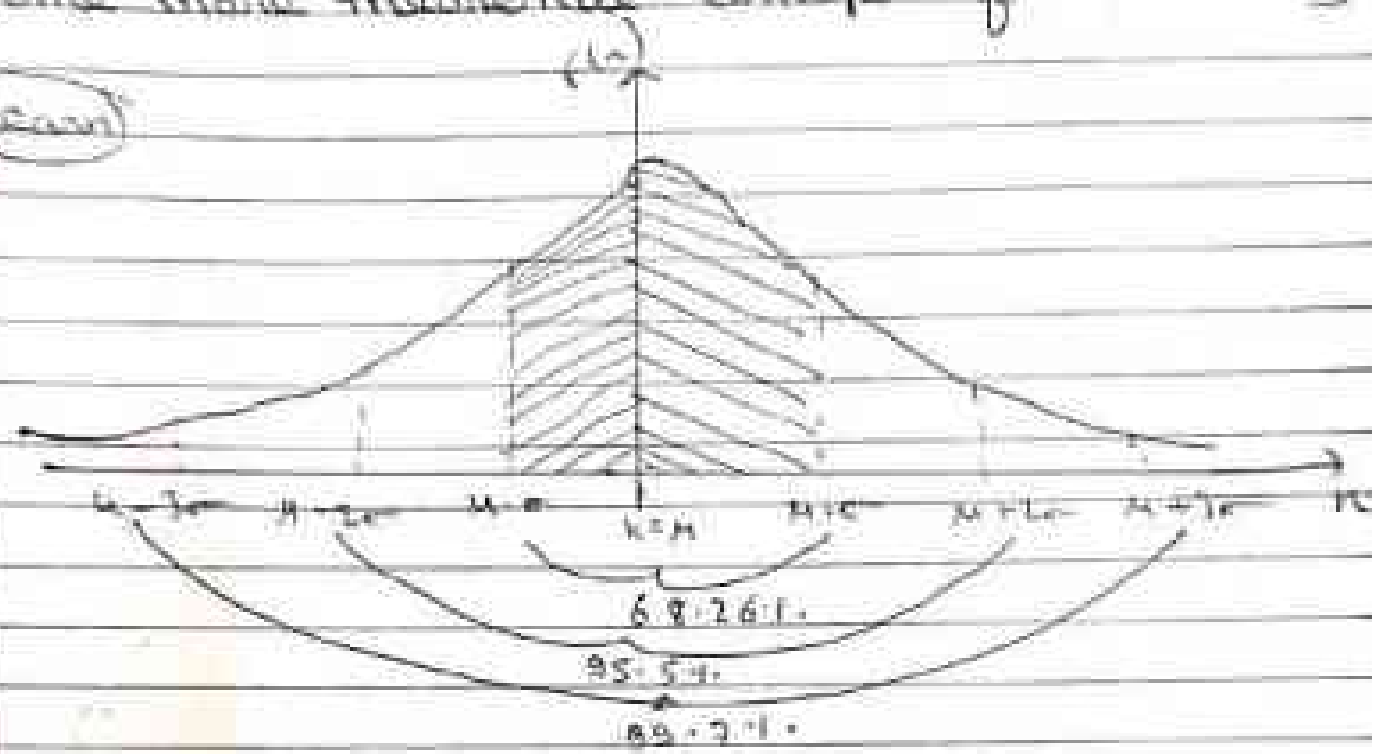
$$= \sqrt{\frac{2}{\pi}} (0 - (-1)) = \sqrt{\frac{2}{\pi}} = 0.798$$

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Some more theoretical concept of N. Dist $\rightarrow$

(Learn)



$$P(\mu - \sigma \leq x \leq \mu + \sigma) = 68.26\% = 0.6826$$

$$P(\mu - 2\sigma \leq x \leq \mu + 2\sigma) = 95.54\% = 0.955$$

$$P(\mu - 3\sigma \leq x \leq \mu + 3\sigma) = 99.74\% = 0.997$$

Note: $P(\mu - \sigma \leq x \leq \mu + \sigma) = 0.6826$ } $\mu = 0, \sigma = 1$
 1.) $P(\mu - 2\sigma \leq x \leq \mu + 2\sigma) = 0.955$ } $x = 2$
 $P(\mu - 3\sigma \leq x \leq \mu + 3\sigma) = 0.997$

$$P(-1 \leq z \leq 1) = 0.6826$$

$$P(-2 \leq z \leq 2) = 0.955$$

$$P(-3 \leq z \leq 3) = 0.997$$

2) Evaluate $P(0 \leq z \leq 1) =$ where notation have their usual meaning.

W.K. that

$$P(-1 \leq z \leq 1) = 0.6826$$

$$= 2P(0 \leq z \leq 1) = 0.6826$$

$$P(0 \leq z \leq 1) = \frac{0.6826}{2} = 0.3413$$

Q:) if $x \sim N\{102, 27^2\}$ then evaluate Prob
 $P(90 \leq x \leq 102) = ?$

- a.) 68.1. $x \sim N\{\mu, \sigma^2\} \begin{cases} \mu = 102 \\ \sigma = 27 \end{cases}$
 b.) 34.1.
 c.) 50.1. $\mu - \sigma = 102 - 27 = 75$
 d.) 16.7.1. ✓ $\mu + \sigma = 102 + 27 = 129$

i.e area b/w 75 & 129 = 68.26%
 = 0.68

Area b/w 75 & 102 = 34.13%
 i.e



Area b/w 90 & 102 $\approx 17.1\%$

= i.e Best possible choice is option (d).

Normal Sum theorem:

if x_1 & x_2 are Ind, N.B.V with mean μ_1, μ_2 & var σ_1^2, σ_2^2 respectively then the var $x = c_1 x_1 + c_2 x_2$ (c_1 & c_2 are const) is also N.B.V with mean $(c_1 \mu_1 + c_2 \mu_2)$ & var $(c_1^2 \sigma_1^2 + c_2^2 \sigma_2^2)$
 i.e

$x_1 \sim N\{\mu_1, \sigma_1^2\}$ then $x = c_1 x_1 + c_2 x_2$
 $x_2 \sim N\{\mu_2, \sigma_2^2\}$

x is also N.B.V defn as
 $x \sim N\{(c_1 \mu_1 + c_2 \mu_2), (c_1^2 \sigma_1^2 + c_2^2 \sigma_2^2)\}$
 Here x_1 & x_2 are Ind.

e.g:) if $x_1 \sim N\{2, 9\}$, $x_2 \sim N\{3, 16\}$ & x_1 & x_2 are Ind then find mean & variance of $x = 2x_1 - 5x_2$.

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30.1) $\frac{n-1}{r-1} \text{ D.F. } N.S.T$

$$u_1 \sim N\{2, 3^2\} \begin{cases} \mu_1 = 2 \\ \sigma_1 = 3 \end{cases}$$

$$u_2 \sim N\{3, 4^2\} \begin{cases} \mu_2 = 3 \\ \sigma_2 = 4 \end{cases}$$

$$\begin{aligned} E(u) &= \mu_u = 2(\mu_1) - 5(\mu_2) \\ &= 4 - 15 = -11 \end{aligned}$$

$$\begin{aligned} \text{var}(u) &= 2^2(\sigma_1^2) + 5^2(\sigma_2^2) \\ &= 4 \times 3^2 + 25 \times 16 = 436 \end{aligned}$$

m-2 using properties of mean & variance

$$\begin{aligned} E(u) &= E(2u_1 - 5u_2) \\ &= 2E(u_1) - 5E(u_2) \\ &= 2 \times 2 - 5 \times 3 = 4 - 15 = -11 \end{aligned}$$

$$\begin{aligned} \text{var}(u) &= \text{var}(2u_1 - 5u_2) \\ &= 4\text{var}(u_1) + 25\text{var}(u_2) + 2(2)(-5)\text{cov}(u_1, u_2) \\ &= 4(9) + 25(16) = 436 \quad \text{or} \end{aligned}$$

$$\begin{aligned} u &\sim N\left\{0, \frac{1}{2}\right\} & P(3u \geq 1u) &= \\ v &\sim N\left\{0, \frac{1}{3}\right\} & P(3v - 2u \geq 0) &= \\ \text{cov}(u, v) &= 0 & = P(u \geq 0) &= \\ & & = P(2u \geq 0) &= \\ & & = \frac{1}{2} \quad \text{or} & \end{aligned}$$

$$\boxed{u = 3v - 2u}$$

$$\begin{aligned} \mu_n = E(n) &= E(3v - 2u) \\ &= 3E(v) - 2E(u) \\ &= 3(0) - 2(0) = 0 \end{aligned}$$

$$\begin{aligned} \text{var}(n) &= \text{var}(3v - 2u) \\ &= 9\text{var}(v) + 4\text{var}(u) + 2(3)(-2)\text{cov}(u, v) \\ &= 9\left(\frac{1}{9}\right) + 4\left(\frac{1}{4}\right) + 0 = 1 + 1 = 2 \end{aligned}$$

$$\text{i.e. } \sigma_n = 2$$

$$Z_n = \frac{n - \mu_n}{\sigma_n} = \frac{0 - 0}{\sqrt{2}} = 0$$

9.) Average marks obtained by students in math portion of SAT exam are 519 & variance 5184 while in verbal portion, Av marks obtained by student are 474 & variance 6368. We are selecting two student at random then find $P(n > y) = ?$ where n = marks obtained by 1st student in math portion.
 y = Marks obtained by 2 student in verbal & we know that area under the N curve upto 0.5 is 0.6915?

$$\text{Sol:)} \quad E(0.5) = 0.6915$$

$$\begin{aligned} P(n > y) &= P(n - y > 0) = P(u > 0) \\ &= P\left(Z > \frac{0}{\sqrt{2}}\right) \end{aligned}$$

$$n \sim N[519, 5184]$$

$$y \sim N[474, 6368]$$

$$n \text{ \& } y \text{ are i.i.d.}$$

$$= P\left(\frac{1}{\sqrt{2}}\right) = 0.6915$$

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let $u = n - y$

$$\mu_u = E(u) = E(n - y) = (n) - (n/4) = 55$$

$$\begin{aligned} \text{var}(u) &= \text{var}(n - y) = \text{var}(n) + \text{var}(y) + 0 \\ &= 5 + 32 + 6768 = 12100 \end{aligned}$$

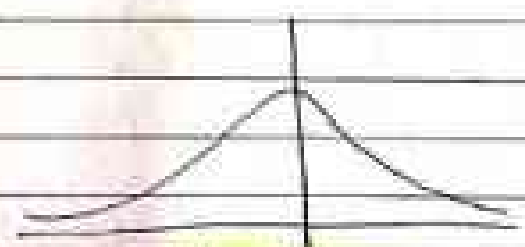
$$\text{so S.D.}(u) = \sqrt{12100} = 110$$

or

$$u \sim N\{55, 12100\}$$

$$Z_u = \frac{u - \mu_u}{\sigma_u} = \frac{0 - 55}{110} = -\frac{1}{2}$$

Some more info for N-curve $\rightarrow$
(Skewness in N-curve)



$m_e < m_d < m_o$

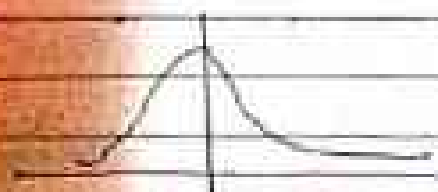
(Positively skewed)



$m_o < m_d < m_e$

(Negatively skewed)

$m_e < m_d < m_o$



$m_o < m_d < m_e$

(Symmetric)

{ See in slide }

negatively skewed = when Tail is coming more to the left than it is called -ve skewed
positively skewed = " " " " " "

Lecture no. 11

Curve fitting :-

Q1) Find the line of Best fit for the follow Data.

X	1	2	3	4	5
Y	3	5	7	9	11

Sol:1) M-1 By observation

$y = 2x + 1$ will be the line of best fit

M-2	x	y	xy	x ²
	1	3	3	1
	2	5	10	4
	3	7	21	9
	4	9	36	16
	5	11	55	25

$$\begin{aligned}\sum x &= 15 \\ \sum y &= 35 \\ \sum xy &= 125 \\ \sum x^2 &= 55\end{aligned}$$

$$n = 5$$

Case 1) Line of Best fit:

let us consider a set of values given as

x_1	x_2	x_3	...	x_{n-1}	x_n
y_1	y_2	y_3	...	y_{n-1}	y_n

let line of Best fit is $y = a + bx$ where a & b can be obtained by solving

$$\left. \begin{aligned} \sum y &= na + b \sum x \\ \sum xy &= a \sum x + b \sum x^2 \end{aligned} \right\} \begin{array}{l} \text{these two eq. are} \\ \text{called Normal eq.} \\ \text{for the line of} \\ \text{Best fit.} \end{array}$$

study class

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Normal equations are:

$$\sum y = na + b \sum x \quad \Rightarrow \quad 35 = 5a + b15$$

$$\sum xy = a \sum x + b \sum x^2 \quad \Rightarrow \quad 125 = a15 + b55$$

$$5a + 15b = 35$$

$$15a + 55b = 125$$

$$a = 1, b = 2$$

Hence the line of Best fit $y = a + bx$

or

$$y = 1 + 2x$$

Q: Find the line of Best fit for

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

m-i By observation $\rightarrow$ Not possible

$$y = 7.28 + 0.928x$$

x	y	xy	x ²
1	9	9	1
2	8	16	4
3	10	30	9
4	12	48	16
5	11	55	25
6	13	78	36
7	14	98	49
Σ	77	374	140

Let the line of Best fit is $y = a + bx$

n-ry are:

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

$$= 77 = 7a + b28$$

$$374 = a28 + b140$$

$$a = 7.28 \text{ \& } b = 0.915$$

Hence $y = a + bx$

$$= 7.28 + 0.928x$$

Parabola of best fit:

Consider the set of Data given as

(Not)

x_1	x_2	x_3	...	x_n
y_1	y_2	y_3		y_n

Let the parabola that fits above data is as follows

$$y = a + bx + cx^2$$



Normal equations are

$$\sum y = na + b \sum x + c \sum x^2$$

1

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

after solving these 3 eq. we can find a, b, c

Hence parabola of best fit can be obtained

Correlation:

=> Covariance: it measures the simultaneous variation of two random variable x & y

=> Correlation: it measure the simultaneous variation of two random variable x & y

$$Cov(x, y) = E(xy) - E(x) \cdot E(y)$$

$$\text{Correlation coeff } r = \frac{Cov(x, y)}{\sigma_x \sigma_y}$$

$$-\infty < Cov(x, y) < \infty$$

$$-1 \leq r \leq 1$$

study time

Correlation

Note) if $r = 1$ then x & y are perfectly correlated in positive sense.

2) if $r = -1$ " " " " " " " " -ve sense.

3) if $r = 0$ " " " " " " " " Not related / independent.

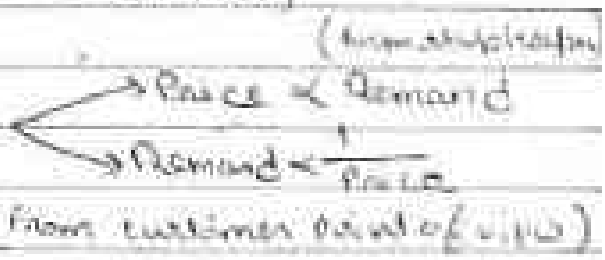
Covariance	Correlation
1) it measures the direction of relation.	1) it measures the direction as well as strength of relation.
2) $-\infty < \text{cov}(x, y) < +\infty$	2) $-1 \leq r \leq 1$
3) $\text{cov}(x, y) = E(xy) - E(x) \cdot E(y)$	3) $r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$
4) x & y are ind then $\text{cov}(x, y) = 0$	4) if x & y are ind $r = 0$
5) in different unit cov is also diff.	5) in diff unit cor coeff is same.

⇒ E.g of covariance: weight (y)



Regression:

If supply is constant, then



Regression

⇒

Correlation

Regression

1.) it measure the direction as well as strength of relation. 1.) It measure the value of dependent var 0.5 to 1.0 is

2.) It has nothing to do with cause & effect relationship. 2.) It measure the cause & effect relationship.

$$3.) r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

$$\text{Effect} = (\text{Cause}) \begin{cases} y = f(x) \\ x = f(y) \end{cases}$$

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4.) Some time there may be Non-sense correlation.

3.) R. line y on $x: y = a + b_1x$
 R. line x on $y: x = a + b_2y$ / $y = y$
 where $b_2b_1 = \frac{r^2}{r^2}$

$$b_2b_1 = r^2$$

& there are R. coeff.

4.) Here we do not have any non-sense regression line.

Q.) Find the correlation coeff, regression coeff R. line y on x & R. line x on y for

X	1	2	3	4	5
Y	3	5	7	9	11

Sol:)

X	Y	XY	X ²	Y ²
1	3	3	1	9
2	5	10	4	25
3	7	21	9	49
4	9	36	16	81
5	11	55	25	121

$$\sum x = 15$$

$$\sum y = 35$$

$$\sum xy = 115$$

$$\sum x^2 = 55$$

$$\sum y^2 = 285$$

study time

$$n = 5$$

Study Time

$$E(n) = \bar{n} = \frac{\sum n}{n} = \frac{15}{5} = 3$$

$$\bar{y} = E(y) = \frac{\sum y}{n} = \frac{35}{5} = 7$$

$$\overline{n^2} = E(n^2) = \frac{\sum n^2}{n} = \frac{55}{5} = 11$$

$$\overline{y^2} = E(y^2) = \frac{\sum y^2}{n} = \frac{285}{5} = 57$$

$$\overline{ny} = E(ny) = \frac{\sum ny}{n} = \frac{125}{5} = 25$$

$$\begin{aligned} \text{Cov}(n, y) &= E(ny) - E(n) \cdot E(y) \\ &= 25 - (3)(7) \\ &= 25 - 21 \\ &= 4 \end{aligned}$$

$$\begin{aligned} \text{var}(n) &= E(n^2) - E^2(n) & \text{var}(y) &= E(y^2) - E^2(y) \\ &= 11 - (3)^2 & &= 57 - 49 \\ &= 2 & &= 8 \end{aligned}$$

$$\sigma_n = \sqrt{2} \quad , \quad \sigma_y = \sqrt{8} = 2\sqrt{2}$$

$$(i) \text{ Correlation coeff } (r) = \frac{\text{Cov}(n, y)}{\sigma_n \sigma_y} = \frac{4}{\sqrt{2} \cdot 2\sqrt{2}} = 1$$

i.e. perfectly correlated in positive sense

$$(ii) \text{ B. coeff } \begin{cases} b_{yn} = r \frac{\sigma_y}{\sigma_n} = 1 \cdot \frac{2\sqrt{2}}{\sqrt{2}} = 2 \\ b_{ny} = r \frac{\sigma_n}{\sigma_y} = 1 \cdot \frac{\sqrt{2}}{2\sqrt{2}} = \frac{1}{2} \end{cases}$$

$$\begin{aligned} (iii) \text{ R. line } y \text{ on } n & \\ &= y - \bar{y} = b_{yn}(n - \bar{n}) \\ &= y - 7 = 2(n - 3) \Rightarrow \boxed{y = 2n + 1} \end{aligned}$$

$$\begin{aligned} (iv) \text{ R. line } n \text{ on } y &= \\ &= n - \bar{n} = b_{ny}(y - \bar{y}) \end{aligned}$$

Q:) if $\bar{x} = 10$, $\bar{y} = 90$, $\sigma_x = 3$, $\sigma_y = 12$ & $r = 0.8$
then find R. line y on x and x on y ?

Sol: R. line y on $x =$

$$b_{yx} = r \cdot \frac{\sigma_y}{\sigma_x} = 0.8 \times \frac{12}{3} = 3.2$$

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 90 = 3.2(x - 10)$$

$$y = 3.2x - 32 + 90$$

$$\boxed{y = 3.2x + 58} \quad \text{Ans}$$

R. line x on $y =$

$$b_{xy} = r \cdot \frac{\sigma_x}{\sigma_y} = 0.8 \times \frac{3}{12} = 0.2$$

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 10 = 0.2(y - 90) \Rightarrow x = 0.2y - \frac{1}{5}(90) + 10$$

$$x = 0.2y - 45 + 10$$

$$\boxed{x = 0.2y - 35} \quad \text{Ans}$$

Q:) Find r , b_{yx} , b_{xy} , R. line x on y &
R. line y on x for the data

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

Sol:) =

⇒ Analysis:

1) Intersection points of two regression line is not the same but $(\bar{x}, \bar{y})$

$$\left. \begin{array}{l} y = 2x + 1 \\ x = \frac{1}{2}y + \frac{1}{2} \end{array} \right\} \Rightarrow (x, y) = (1, 7) = (\bar{x}, \bar{y})$$

2) Correlation coefficient is the geometric mean of R. coffs

$$\text{i.e. } r = \pm \sqrt{b_{yx} \cdot b_{xy}}$$

3) r , b_{yx} , b_{xy} are of same sign (+)

i.e. either all are +ve or all are -ve

⇒

4) if $b_{yx} > 1$ then $b_{xy} < 1$ but converse need not be true
 ≠ i.e. both may be < 1 also

⇒ Geometric mean: (GM)

for two +ve no. a & b : $GM = \sqrt{ab}$

" " -ve " " " " : $GM = \sqrt{ab}$

" one +ve & one -ve no. : $GM = \textcircled{N.O.}$

5) Angle b/w two regression line:-

$$\tan \theta = \left(\frac{1 - r^2}{r^2} \right) \cdot \frac{\sigma_{yx} \sigma_{xy}}{\sigma_{xx} \sigma_{yy}} = \theta$$

eg) if two R. line are $\perp$ then correlation coeff will be

$$\theta = \frac{\pi}{2}$$

$$\tan \theta = \tan \frac{\pi}{2} = \infty$$

$$\infty = \left(\frac{1 - r^2}{r^2} \right) \frac{\sigma_{yx} \sigma_{xy}}{\sigma_{xx} \sigma_{yy}}$$

$$r^2 = \left(\frac{1 - r^2}{\infty} \right) \cdot \frac{\sigma_{yx} \sigma_{xy}}{\sigma_{xx} \sigma_{yy}}$$

$$r^2 = 0 \Rightarrow \boxed{r = 0}$$

✓

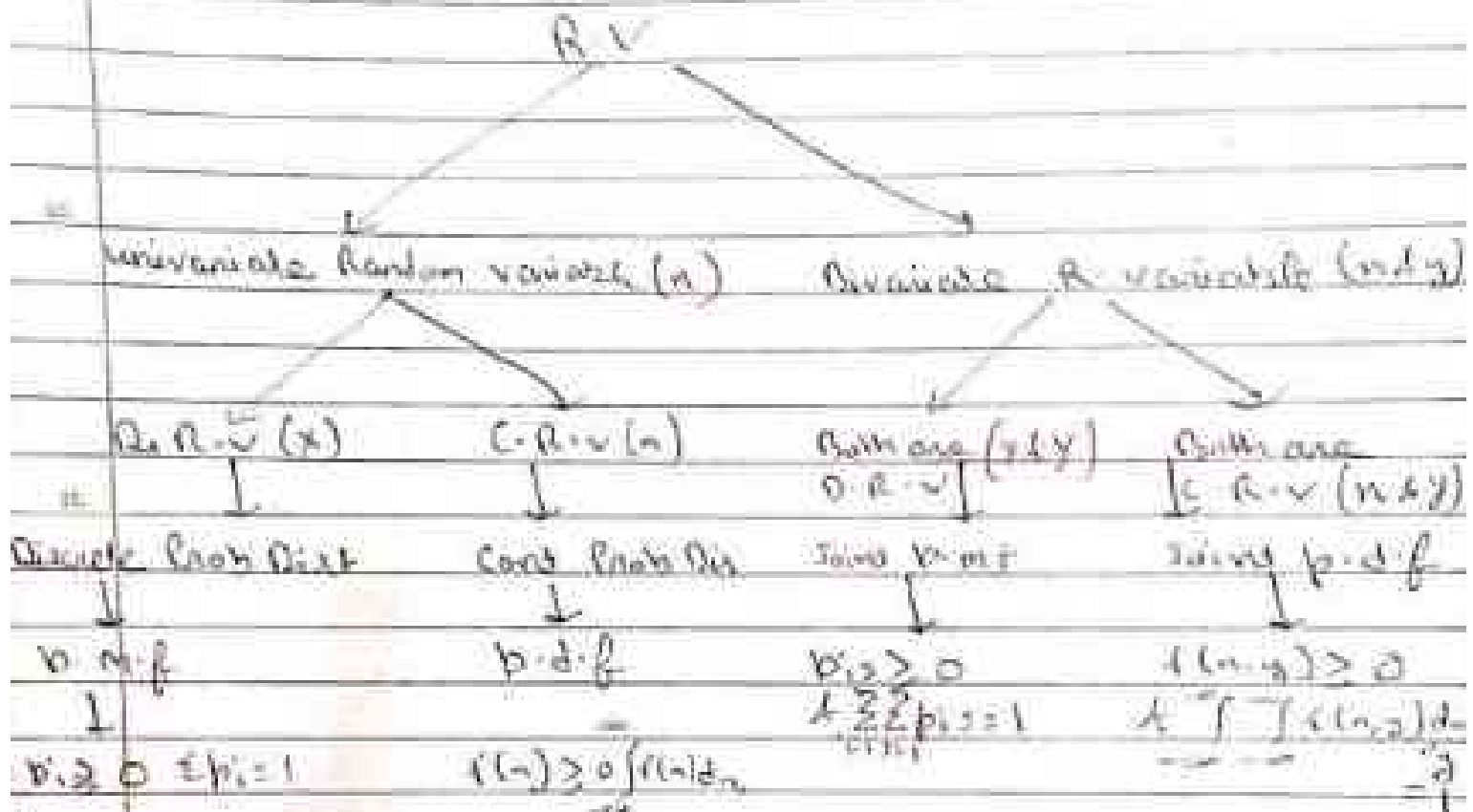
study time



Chapter - 04

Statistics - 02

Lecture, no, 01



Joint p.m.f

- 1) if n & y are two random variables then their joint p.m.f is defined as

$$P(n = n_i, y = y_j) = P(n_i, y_j) = p_{ij} \begin{cases} p_{ij} \geq 0 \\ \sum_{i,j} p_{ij} = 1 \end{cases}$$

- 2) Individual Prob. m.f of x and y are called marginal density function (m.d.f) & these are given as:

$x: n_1, n_2, n_3, \dots, n_m$	$y: y_1, y_2, \dots, y_n$
$P(x): p_1, p_2, p_3, \dots, p_m$	$P(y): p_{y1}, p_{y2}, \dots, p_{yn}$

Q.) A fair coin is tossed twice. if x denote the number heads on the 1st toss and Y denote the total number of heads then write it's

- (i) Joint p.m.f
- (ii) Also find marginal distribution of x and y
- (iii) Evaluate $P(x=1, y \geq 1)$, $P(x \geq 0, y \geq 2)$, $P(0 \leq x \leq y)$
- (iv) Conditional distribution of x when $y=1$
- (v) Conditional distribution of Y when $x=1$
- (vi) $P(x=y)$
- (vii) $P(1 \leq y \leq 3)$
- (viii) $P(y=2)$

$$x = \{\text{No. of heads on 1st toss}\} = \{0, 1\}$$

$$Y = \{\text{Total no. of heads}\} = \{0, 1, 2, 3\}$$

$$S = \{(HH), (HT), (TH), (TT), (THH), (THT), (TTH), (TTT)\} = 8 \text{ samples}$$

$$x = \{\text{No. of heads on 1st toss}\} = \{0, 1\}$$

iii)	$x =$	0	1	} this is called m.f of x
	$P(x) =$	$\frac{4}{8}$	$\frac{4}{8}$	
		8	8	

$$Y: \{\text{Total no. of heads}\} = \{0, 1, 2, 3\}$$

	$Y:$	0	1	2	3	} this is called m.f of y
	$P(y) =$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	
		8	8	8	8	

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$X \backslash Y$	0	1	2	3	$P(X)$
0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0	$\frac{4}{8}$
1	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{4}{8}$
$P(Y)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1

$$P(X=0, Y=0) = P(0,0) = P(\text{No. of heads on 1st coin} = 0 \\ \text{ \& total no. of heads} = 0) \\ = P(TT) = \frac{1}{8}$$

$$P(0,1) = P(X=0, Y=1) = P(\text{No. of heads on 1st coin} = 0 \\ \text{ \& total no. of heads} = 1) \\ = P(THT, THT) = \frac{2}{8}$$

$$P(0,2) = P(X=0, Y=2) = P(\text{No. of heads on 1st coin} = 0 \\ \text{ \& total no. of heads} = 2) \\ = P(THTT) = \frac{1}{8}$$

$$P(0,3) = P(X=0, Y=3) = P(\text{No. of heads on 1st coin} = 0 \\ \text{ \& total no. of heads} = 3)$$

Similarly for Y .

$$(iii) P(X=1, Y \geq 1) = P(X=1, Y \in \{2, 1\}) = P(1,0) + P(1,1) \\ = \frac{1}{8} + \frac{2}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$$

$$(iv) P(X \geq 0, Y \geq 2) = P(X=0, 1, 2, Y=2, 3) \\ = P[(0,2) (0,3), (1,2) (1,3)] \\ = \frac{1}{8} + 0 + \frac{2}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$$

only one

$$P(0,0), P(0,1), P(0,2), P(0,3), P(1,0), P(1,1), P(1,2), P(1,3)$$

$$c) P(x=0, y=3) = P[(0,1), (1,1)]$$

$$= P[(0,3), (1,3)]$$

$$= 0 + \frac{1}{8} = \frac{1}{8} \quad \underline{A}$$

$$(vi) P(x=3) = P[(1,0), (2,1), (1,1)] = \frac{1}{8} + \frac{1}{8} + \frac{2}{8} = \frac{4}{8}$$

$$(vii) P(1 \leq y \leq 3) =$$

$$m-1 \text{ (Using m.d.f of } x) = P(1 \leq y \leq 3) = P(y=1, 2, 3)$$

$$= \frac{7}{8} + \frac{3}{8} + \frac{1}{8} = \frac{7}{8}$$

$$= m-2 \text{ (Using joint table):}$$

$$P(1 \leq y \leq 3) = P[(1,0), (1,1), (1,2), (1,3)]$$

$$= P[(0,1), (0,2), (0,3), (1,1), (1,2), (1,3)]$$

$$= \frac{2}{8} + \frac{1}{8} + 0 + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$$

$$= \frac{7}{8} \quad \underline{A}$$

$$(viii) P(y=3)$$

$$m-1 \text{ (Using m.d.f of } y) = \frac{1}{8}$$

$$m-2 \text{ (Using joint table)}$$

$$= P(x=0, y=3)$$

$$= P[(0,3), (1,3)]$$

$$= 0 + \frac{1}{8}$$

$$= \frac{1}{8} \quad \underline{A}$$

3.) Conditional distribution of X when $Y = y_j$ is given can be obtained by using,

$$P\left[\frac{X=n}{Y=y_j}\right] = \frac{P[n_i, y_j]}{P(Y=y_j)}$$

$$P\left(\frac{X=n}{Y=y_j}\right) :$$

$X :$	n_1	n_2	n_3	...	n_m
$P\left(\frac{X=n}{Y=y_j}\right) :$					

4.) Conditional distribution of Y when $X = n_i$ is given can be obtained by using,

$$P\left[\frac{Y=y}{X=n_i}\right] = \frac{P[n_i, y]}{P(X=n_i)}$$

5.) if $P[X=n_i, Y=y_j] = P(X=n_i) \cdot P(Y=y_j) \forall i, j$ then X & Y is called Ind random variable

$$(iv) P\left(\frac{X=n}{Y=1}\right) = \frac{P(n, 1)}{P(Y=1)} \begin{cases} \rightarrow P(X=0/Y=1) = \frac{P(0, 1)}{P(Y=1)} = \frac{1/8}{1/8} = 1 \\ \rightarrow P(X=1/Y=1) = \frac{P(1, 1)}{P(Y=1)} = \frac{1/8}{1/8} = 1 \end{cases}$$

$X :$	0	1
$P(X=n/Y=1) :$	$\frac{2}{3}$	$\frac{1}{3}$

$$(v) P\left(\frac{Y=y}{X=1}\right) = \frac{P(1, y)}{P(X=1)}$$

$$P\left(\frac{Y=0}{X=1}\right) = \frac{P(1, 0)}{P(X=1)} = \frac{0}{1/8} = 0$$

$$P\{y=1/x=1\} = \frac{P\{1,1\}}{P\{x=1\}} = \frac{1/8}{1/2} = \frac{1}{4}$$

$$P\{y=2/x=1\} = \frac{P\{1,2\}}{P\{x=1\}} = \frac{1/8}{1/2} = \frac{1}{4}$$

$$P\{y=3/x=1\} = \frac{P\{1,3\}}{P\{x=1\}} = \frac{1/8}{1/2} = \frac{1}{4}$$

iii

So joint conditional distribution for y when $x=1$ is given as:

$$P\{y=y/x=1\} :$$

$y = y :$	0	1	2	3
	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

9) Four coins are tossed, let x denotes number of heads and y denotes (number of head - number of tails). Then write the joint pdf for such type of random experiment.

- (i) Evaluate $P(-2 \leq y \leq 4)$
- (ii) $P(x \leq 2, y \geq 2)$
- (iii) $P(x \geq 3, -2 \leq y \leq 4)$
- (iv) Conditional distribution of x given the value of y
- (v) " " " " y given the value of x
- (vi) marginal distribution of x
- (vii) marginal distribution of y
- (viii) $P(x=y) = ?$ (ix) $P(x \leq 1) = ?$

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$S = \{ \underbrace{(HHHH)}_4, \underbrace{(HHHT)}_4, \underbrace{(HHTH)}_4, \underbrace{(HTHH)}_4, \underbrace{(THHH)}_4, \underbrace{(HHTT)}_4, \underbrace{(HTTT)}_4, \underbrace{(TTT)}_3 \} = 2^4 = 16$ possible outcomes
So 16 Quadruples

$X = \{\text{Number of heads}\} = \{0, 1, 2, 3, 4\}$

$X:$	0	1	2	3	4
$P(X):$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

it is marginal density of X

$Y = \{\text{Number of heads} - \text{Number of tails}\}$

$= \{-4, -2, 0, 2, 4\}$

$Y:$	-4	-2	0	2	4
$P(Y):$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

it is m.d.f of Y

$\Rightarrow$ Joint table is as follows:

$X \backslash Y$	-4	-2	0	2	4	$P(Y)$
0	$\frac{1}{16}$	0	0	0	0	$\frac{1}{16}$
1	0	$\frac{4}{16}$	0	0	0	$\frac{4}{16}$
2	0	0	$\frac{6}{16}$	0	0	$\frac{6}{16}$
3	0	0	0	$\frac{4}{16}$	0	$\frac{4}{16}$
4	0	0	0	0	$\frac{1}{16}$	$\frac{1}{16}$
$P(X)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$	"1"

$$P(X=0, Y=-4) = P(TTTT) = \frac{1}{16}$$

$$P(X=1, Y=-2) = P(HTTT) = \frac{4}{16}$$

$$P(X=2, Y=0) = \frac{6}{16}$$

$$P(X=3, Y=2) = P(HTHT) = \frac{4}{16}$$

$$\begin{aligned}
 \text{(i)} \quad P(-2 \leq y \leq 4) &= P(y = -2, 0, 2) \\
 &= \frac{4}{16} + \frac{6}{16} + \frac{4}{16} = \frac{14}{16} = \frac{7}{8}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad P(x \leq 2, y \geq 2) &= P(x = 0, 1, 2, y = 2, 4) \\
 &= P[(0, 2), (0, 4), (1, 2), (1, 4), (2, 2), (2, 4)] \\
 &= 0 + 0 + 0 + 0 + 0 + 0 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad P(x \geq 3, -2 \leq y \leq 4) &= P(x = 3, 4, y = 0, 2, 4) \\
 &= P[(3, 0), (3, 2), (3, 4), (4, 0), (4, 2), (4, 4)] \\
 &= 0 + \frac{4}{16} + 0 + 0 + 0 + \frac{1}{16} \\
 &= \frac{5}{16}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad P(x \leq 2) &= P(0, 1, 2) \\
 &= \frac{1}{16} + \frac{4}{16} + \frac{6}{16} = \frac{11}{16}
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad P(x = y) &= P[(0, 0), (2, 2), (4, 4)] \\
 &= P(0 + 0 + \frac{1}{16}) = \frac{1}{16}
 \end{aligned}$$

(vi) Conditional dist of x when $y = 0$

$$P\left[\frac{x=y}{y=0}\right] = \frac{P(x, 0)}{P(y=0)} = \frac{P(x, 0)}{\left(\frac{6}{16}\right)}$$

study time

PRACTICE

	2	0	1	2	3	4
$P\left[\frac{X=Y}{Y=0}\right] = \frac{P(0,0)}{4/16}$	$\frac{1}{4}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$

$X=Y:$	0	1	2	3	4
$P\left[\frac{X=Y}{Y=0}\right] =$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$
$=$	0	0	1	0	0

(v) Conditional distribution of Y given that $X=0$

$$P\left[\frac{Y=Y}{X=0}\right] = \frac{P(0,Y)}{P(X=0)} = \frac{P(0,Y)}{4/16}$$

$$Y=Y \quad -4 \quad -2 \quad 0 \quad 2 \quad 4$$

$P\left[\frac{Y=Y}{X=0}\right] = \frac{P(0,-4)}{4/16}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{4/16}{4/16}$	$\frac{0}{4/16}$
--	------------------	------------------	------------------	---------------------	------------------

$$Y=Y = -4 \quad -2 \quad 0 \quad 2 \quad 4$$

$P\left[\frac{Y=Y}{X=0}\right] =$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{0}{4/16}$	$\frac{4/16}{4/16}$	$\frac{0}{4/16}$
$=$	0	0	0	1	0

Q] Two Dice are rolled simultaneously. Let X denote the sum of two faces and Y the absolute value of their difference. Find joint p.m.f for X and Y .

(i) Also evaluate $P(X=Y)$

(ii) $P(X=7 \cap Y \geq 2)$

(iii) $P[(X+Y=9) \cap (X-Y=1)]:$

(iv) Find marginal distribution of X and Y .

(v) Find conditional distribution of X given $Y=1$

Lecture no- 01

Sol:1) $X = \{\text{Sum of two faces}\} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

x	2	3	4	5	6	7	8	9	10	11	12
$P(X)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$Y = \{\text{absolute value of throw}\} = \{0, 1, 2, 3, 4, 5\}$

y	0	1	2	3	4	5
$P(Y)$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$

$\Rightarrow$ Joint P.m.f table.

$X \backslash Y$	0	1	2	3	4	5	$P(X)$
2	$\frac{1}{36}$	0	0	0	0	0	$\frac{1}{36}$
3	0	$\frac{2}{36}$	0	0	0	0	$\frac{2}{36}$
4	$\frac{1}{36}$	0	$\frac{2}{36}$	0	0	0	$\frac{2}{36}$
5	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	0	$\frac{2}{36}$
6	$\frac{1}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	$\frac{4}{36}$
7	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	$\frac{5}{36}$
8	$\frac{1}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{6}{36}$
9	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	$\frac{5}{36}$
10	$\frac{1}{36}$	0	$\frac{2}{36}$	0	0	0	$\frac{4}{36}$
11	0	$\frac{2}{36}$	0	0	0	0	$\frac{3}{36}$
12	$\frac{1}{36}$	0	0	0	0	0	$\frac{1}{36}$
$P(Y)$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$	1

$$\begin{aligned}
 (i) \quad P(X=Y) &= (2,2) + (3,3) + (4,4) + (5,5) \\
 &= 0 + 0 + 0 + 0 \\
 &= 0
 \end{aligned}$$

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2) $P(X=7 \cap Y \geq 3) = P(X=7 \text{ and } Y=3 \text{ or } 4 \text{ or } 5)$

$P[(7,3) (7,4) (7,5)]$

$= \frac{2}{36} + 0 + \frac{2}{36} = \frac{4}{36} = \frac{1}{9}$ Ans

3) $P\{(X+Y=4) \cap (X-Y=2)\}$

$X+Y=4 \quad (X,Y) = (3,1)$

$X-Y=2$

$2X=6 \quad = \frac{2}{36} = \frac{1}{18}$ Ans

$X=3$

$3+Y=4$

$Y=1$

4) $P\left[\frac{X=4}{Y=2}\right] = \frac{P(X=4)}{P(Y=2)} = \frac{P(X=4)}{8/36}$

$P(1,2) = 0$

$P(2,2) = 0$

$P(3,2) = 1/4$

$P(4,2) = 0$

$P(5,2) = 1/4$

$P(6,2) = 0$

$P(7,2) = 1/4$

$P(8,2) = 0$

$P(9,2) = 1/4$

$P(10,2) = 0$

$P(11,2) = 0$

$P(12,2) = 0$

X \ Y	2	3	4	5	6	7	8	9	10	11	12
$P\left(\frac{X=Y}{X+Y}\right)$	0	0	1/4	0	1/4	0	1/4	0	1/4	0	0

Q.) Joint P.M.F for random variable x and y are given in the table. find $P[1 \leq x \leq 3, y \geq 1]$

$x \backslash y$	0	1	2	$P(x)$
0	$1/18$	$1/6$	$1/6$	$2/18$
1	$1/9$	$1/18$	$1/9$	$3/18$
2	$1/6$	$1/6$	$1/18$	$7/18$
$P(y)$	$2/18$	$2/18$	$2/18$	1

Sol.) $P[1 \leq x \leq 3, y \geq 1]$

$$P(1, 2 \leq x, y \geq 1, 2)$$

$$= P(1, 1) + P(1, 2) + P(2, 1) + P(2, 2)$$

$$= \frac{1}{18} + \frac{1}{9} + \frac{1}{6} + \frac{1}{18} = \frac{1+2+3+1}{18} = \frac{7}{18}$$

Q.) Joint p.d.f of two discrete random variable x and y is given as $f(x, y) = c(2x + y)$ where $0 \leq x \leq 2$ & $0 \leq y \leq 3$ then.

Sol.) $x = 0, 1, 2$

$y = 0, 1, 2, 3$

$x \backslash y$	0	1	2	3	$P(x)$
0	0	c	$2c$	$3c$	$6c$
1	$2c$	$3c$	$4c$	$5c$	$14c$
2	$4c$	$5c$	$6c$	$7c$	$21c$
$P(y)$	$6c$	$9c$	$12c$	$15c$	$42c$

(i) Find c

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$$\sum_{i,j} p_{ij} = 1$$

$$42c = 1$$

$$c = \frac{1}{42}$$

$$(ii) P(X=1, Y=1)$$

$$= P(1,1)$$

$$= 5c$$

$$= \frac{5 \times 1}{42} = \frac{5}{42}$$

$$3) P(X \geq 1, Y \leq 2) =$$

$$= P(X=1, 2, Y=0, 1, 2)$$

$$P(1,0) + P(1,1) + P(1,2) + P(2,0) + P(2,1) + P(2,2)$$

$$= 2c + 3c + 4c + 4c + 5c + 6c$$

$$= 24c$$

$$= \frac{24 \times 1}{42} = \frac{24}{42} = \frac{4}{7}$$

Q:) X and Y are two random variable having the joint density function

$$f(x,y) = \frac{1}{27} (2x+y)$$

where x and y can only be integer 0, 1 or 2
find

(i) marginal distribution of x and y

(ii) Conditional distribution of y given x=1

Sol:)

X \ Y	0	1	2	P(x)
0	0	1/12	2/12	3/12
1	2/12	3/12	4/12	9/12
2	4/12	5/12	6/12	15/12
P(y)	6/12	9/12	12/12	1

(i)

X: 0 1 2
P(x): 3/12 9/12 15/12 } is is m.d.f of x.

Y: 0 1 2
P(y): 6/12 9/12 12/12 } m.d.f of y

(ii)

$$P\left(\frac{y=x}{n=1}\right) = \frac{P(x=y=2)}{n=1} = \frac{P(1,2)}{n=1}$$

$$\frac{P(1,0)}{9/12} = \frac{2/12}{9/12} = \frac{2}{9}$$

$$\frac{P(1,1)}{9/12} = \frac{3/12}{9/12} = \frac{3}{9}$$

$$\frac{P(1,2)}{9/12} = \frac{4/12}{9/12} = \frac{4}{9}$$

Y: 0 1 2
P(y=x) $\frac{2}{9}$ $\frac{3}{9}$ $\frac{4}{9}$ } cond prob of stud.

Q.)

Let X & Y be two random variable with joint distribution given below.

X \ Y	0	1	2
0	1/12	3/12	2
1	1/12	3	1/12
2	3/12	1/12	1/12

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if a & b are constant then evaluate $P(x=1, y \geq 0)$

(a) $1/2$

(b) $1/4$

(c) $1/3$

(d) $2/3$

sol) $\sum_{i=1}^n \sum_{j=0}^n b_{ij} = 1$

$$\frac{1}{12} + \frac{3}{12} + a + \frac{2}{12} + b + \frac{1}{12} + \frac{3}{12} + \frac{1}{12} + \frac{1}{12} = 1$$

$$a + b + \frac{16}{12} = 1$$

But we know that

a & b can't be $-ve$

these are probabilities.

$$a + b = 0$$

$$\boxed{a = -b}$$

So only possibility for a & b will be

$$a = b = 0$$

$$P(x=1, y \geq 0) = P(x=1, y=0, 1, 2)$$

$$P(1,0) + P(1,1) + P(1,2)$$

$$= \frac{3}{12} + 0 + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$$

$\therefore$ (c)

Q) If X is uniformly distributed over $\{1, 2, 3, 4, 5, 6\}$ & represent number of throws of fair die. Let Y denote the number of times 2 appear in X throws which of the following option is correct.

(Divine 21 Jan)

$$a) P(Y=1/n=2) = \frac{1}{4}$$

$$b) P(Y=2/n=2) = 5/6$$

$$c) P(Y=5/n=6) = 5/6$$

$$d) P(Y=6/n=5) = 5/6$$

So, $P(X=1) = P(X=2) = P(X=3) = P(X=4) = P(X=5) = P(X=6) = \frac{1}{6}$
 let

$$Y = \text{no. of times 2 appears} \quad \text{success} \rightarrow p = P(1) = \frac{1}{6}$$

$$q = P(2) = \frac{5}{6}$$

$$P(X = x \text{ success}) = {}^n C_x p^x q^{n-x}$$

$$a) P(X=2/Y=2) = P(\text{no. 2 occurring twice in two throws}) = {}^2 C_2 p^2 q^{2-2} = \left(\frac{1}{6}\right)^2$$

$$b) P(X=2/X=4) = P(\text{no. 2 occurring twice in 4 throws})$$

$$= {}^4 C_2 p^2 q^{4-2} = \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^2$$

$$= \frac{25}{36}$$

$$c) P(Y=5/X=6) = P(\text{no. 2 occurring 5 times in 6 throws})$$

$$= {}^6 C_5 p^5 q^{6-5} = 6 \times \left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)$$

$$= \frac{5}{6}$$

$$d) P(X=1/n=5) = P(\text{no. of 2 occurring twice in 5 throws})$$

$$\text{Event} = \emptyset$$

Q.) If X & Y are two Discrete Random variables show in the table

$X \backslash Y$	0	1	$P(X)$
0	$1/4$	$1/2$	$3/4$
1	$1/8$	$1/8$	$1/4$
$P(Y)$	$3/8$	$1/8$	1

then find $E\{Y/n=1\} = 1$

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$x:$	0	1
$P(x):$	$3/4$	$1/4$

$y:$	0	1
$P(y):$	$3/4$	$1/4$

$$P\left(\frac{y=y}{n=1}\right) = \frac{P(1,y)}{P(n=1)} = \frac{P(1,1)}{2} =$$

$$\frac{P(1,0)}{n=1} = \frac{1/8}{1/4} = \frac{1}{2}$$

$y:$	0	1
$P(y=y/n=1)$	$\frac{1}{2}$	$\frac{1}{2}$

$$\frac{P(1,1)}{P(n=1)} = \frac{1/8}{1/4} = \frac{1}{2}$$

w.k that Expected $(x) = \sum p_i x_i = \sum x_i p_i$

$$E\left\{\frac{y}{n=1}\right\} = \sum y \cdot P(y=y/n=1)$$

$$= 0 \times \frac{1}{2} + 1 \times \frac{1}{2} = 0.5 \text{ or } 1/2$$

Lecture no. 03

⇒ Double Integral working Rule:-

Q1) $\int_1^3 \int_0^n (n^2 y) dy dn = ?$

$$\int_1^3 \int_0^n n^2 y dy dn = \int_1^3 \left[\int_0^n y dy \right] (n^2) = \int_1^3 \left(\frac{y^2}{2} \right)_0^n (n^2) dn$$

$$= \frac{1}{2} \int_1^3 n^3 dn (n=0) =$$

$$Q2) \int_{-2}^2 \int_{-\sqrt{y}}^{\sqrt{y}} (1) dndy = ? = \frac{1}{2} \left(\frac{n^2}{2} \right)_1^2 = \frac{1}{2} \left[\frac{2^2}{2} - 1 \right]$$

$$\int_{y=0}^4 \int_{n=0}^{\sqrt{y}} (1) dndy = 14.2.2$$

$$\int_0^1 \left\{ \int_{\sqrt{y}}^1 (1) dx \right\} \cdot \left(\frac{1}{2} dy \right)$$

$$\begin{aligned} \int_0^1 \left(\frac{x}{\sqrt{y}} \right) dy &= \int_0^1 \frac{1}{2} (y^1 - \sqrt{y}) dy \\ &= \left[\frac{y^2}{2} - \frac{y^{3/2}}{3/2} \right]_0^1 \\ &= \frac{1}{2} - \frac{2}{3} = \frac{1}{6} \end{aligned}$$

$\Rightarrow$ **Case 1:** if y has variable limits we should first integrate dy keeping x constant & in this case strip is vertical.

$\Rightarrow$ **Case 2:** if x has variable limit, we should first integrate dx , keeping y constant & in this case strip is horizontal.

$$Q: I = \int_0^3 \int_1^2 e^{(m+xy)} dx dy = ?$$

$$= \int_0^3 \int_1^2 (e^m \cdot e^{xy}) dx dy \begin{cases} m-1 \\ m-2 \\ m-3 \end{cases}$$

$$I = \int_0^3 \left(\int_1^2 e^m dx \right) (e^{xy} dy)$$

$$= \int_0^3 \left(e^m \right)_1^2 (e^{xy} dy)$$

$$= \int_0^3 (e^2 - e) (e^{xy} dy) = (e^2 - e) (e^y)_1^2 = (e^2 - e) (e^2 - 1) e^0$$

$$= (e^2 - 1) (e^2 - e)$$

$$= (e^2 - 1) (e^2 - e)$$

See in lecture 10/11

⇒ **Case 1:** if both x & y have variable limits then it is a variable question.

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⇒ **Case 2:** if both the limits are constant we can integrate any order keeping other variable constant. But it is advisable to integrate in that order which is given in question.

⇒ **Case 3:** if both the limits are constants and integral is an explicit function of x & y we can integrate separately.

⇒ **Explicit function:** if it is possible to separate dependent & independent variable then func. is called explicit function & it is denoted as $y = f(x)$.

⇒ **Implicit function:** if it is not possible to separate dependent & independent variable then func. is called implicit func. & it is denoted as $f(x, y) = c$.

(e.g.) $x^3 + y^3 + 3xy^2 = 4x$
 $y^3(1+2x) = 4x - x^3$
 $y^3 = \frac{4x - x^3}{1+2x}$
 $y = \left(\frac{4x - x^3}{1+2x} \right)^{1/3}$ i.e. Explicit.

(e.g.) $x^3 + y^3 + 3xy^2 = 1$
 $f(x, y) = c$ i.e. Implicit.

(e.g.) $xy = xy$
 $y = xy = x$
 $y(1-x) = x$
 $y = \frac{x}{1-x}$ i.e. Explicit.

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$$\text{Q.1) } I = \int_0^{\pi/2} \int_0^{\pi/2} \sin(u+y) \, du \, dy$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} \sin(u+y) \, du \, dy$$

$$\text{Ans. } = \int_0^{\pi/2} \left\{ \int_0^{\pi/2} \sin(u+y) \, du \right\} \cdot dy$$

$$= \int_0^{\pi/2} \left\{ -\cos(u+y) \right\}_0^{\pi/2} dy = \int_0^{\pi/2} \left\{ -\cos\left(\frac{\pi}{2}+y\right) + \cos(y) \right\} dy$$

$$= \int_0^{\pi/2} (\sin y + \cos y) dy = (\cos y + \sin y) \Big|_0^{\pi/2}$$

$$= -1 + 1 = 0$$

$$\text{Q.2) } I = \int_0^2 \int_0^{\pi/2} (u^2 y) \, dy \, du$$

Ans. The concept of an explicit form is applicable only when limits are constants.

i.e. this is question based on case 1.

$$I = \int_0^2 \int_0^{\pi/2} (u^2 y) \, dy \, du$$

$$= \int_0^2 y \, dy = \int_0^2 \left(\frac{y^2}{2} \right) (u^2 \, du)$$

$$= \frac{1}{2} \int_0^2 (u^4 - 0) (u^2 \, du) =$$

$$\frac{1}{2} \int_0^2 u^4 \, du = \frac{1}{2} \left(\frac{u^5}{5} \right)_0^2 = \frac{32}{10} = \frac{16}{5}$$



study time

$$Q:) \int_0^2 \int_0^{y_1} e^{n^2} \cdot dy \, dn = ?$$

$$\text{Sol.}) \int_0^2 \left\{ \int_0^{y_1} 1 \, dy \right\} \cdot e^{n^2} \, dn$$

$$= \int_0^2 \left(y \right)_0^{y_1} \cdot e^{n^2} \, dn$$

$$= \int_0^2 \left(\frac{n}{2} - 0 \right) \cdot e^{n^2} \, dn = \frac{1}{2} \int_0^2 n e^{n^2} \, dn$$

$$= \text{Put } n^2 = t \Rightarrow \begin{cases} \text{At } n=0, t=0 \\ \text{At } n=2, t=4 \end{cases}$$

$$n \, dn = dt/2$$

$$= \frac{1}{2} \int_0^4 e^t \cdot \frac{dt}{2} = \frac{1}{4} \left(e^t \right)_0^4 = \frac{1}{4} (e^4 - e^0)$$

$$= \frac{1}{4} (e^4 - 1) \, \text{Ans}$$

$$Q:) I = \int_0^1 \int_{2y}^2 e^{n^2} \, dn \, dy = ? \quad (\text{Do your self})$$

(Patience)

Application of Multiple Integrals

$$\text{Area} = \iint (1) \, dn \, dy = \iint (1) \, dy \, dn$$

(using n-Var) (using y-Var)

1) Area under $f(n)$ b/w $n=a$ to b & x -axis $\Rightarrow$

$$\text{Volume} = \iiint (1) \, dn \, dy \, dz = \iint dz \, dn \, dy = \iint f(n,y) \, dn \, dy$$

$$\text{Mass} = \iiint \rho \, dn \, dy \, dz \quad \text{where } \rho = \rho(n,y,z) = \text{Density}$$

Density $\times$ volume = mass

Q.1) Find the intersecting point of the curve $y^2 = x$ & $y = x$

Sol.

$$y^2 = x$$

$$y = x$$

$$x^2 = x$$

$$x^2 - x = 0 \Rightarrow x = 0 \text{ or } x = 1$$

$$x^2 - x = 0$$

$$\text{at } x = 1, y = 1$$

$$x(x-1) = 0 \Rightarrow x = 0 \text{ or } x = 1$$

i.e. $(0,0)$ & $(1,1)$ are two intersection points

Q.2) Find the area bounded by $y^2 = x$ & $y = x$ using both type of strips

Sol. Case 1 using vertical strip

$$\text{Area} = \int \int (1) dy dx$$

$$= \int_0^1 \int_0^x dy dx$$

$$= \frac{1}{6} \text{ sq. units}$$

$$\text{Case 2: (using } y\text{-strip)} = \frac{1}{6} \text{ sq. units}$$

Q.3) Find the area bounded by $y^2 = x$ & $x^2 = y$ using both type of strips

Sol.

$$y^2 = x$$

$$(x^2)^2 = x$$

$$x^4 - x = 0$$

$$x(x^3 - 1) = 0$$

$$x = 0 \text{ or } x = 1$$

Case 1: using vertical strip

$$\text{Area} = \int \int (1) dy dx$$

$$= \frac{1}{6} \text{ sq. units}$$

$$x = 0 \text{ or } x = 1$$

Case 2: using horizontal strip

$$\text{Area} = \int \int (1) dx dy$$

$$= \frac{1}{6} \text{ sq. units}$$

$$x^2 + y^2 = y^2 \Rightarrow x = y \quad y^2 = x^2$$

$$x^2 + y^2 = x^2 \Rightarrow y = x \quad x^2 = y^2$$

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Q: If $f(x, y)$ is a function of two variables i.e. $z = f(x, y)$ and we have given two constraints $x^2 + y^2 = a^2$ then volume under $f(x, y)$ will be

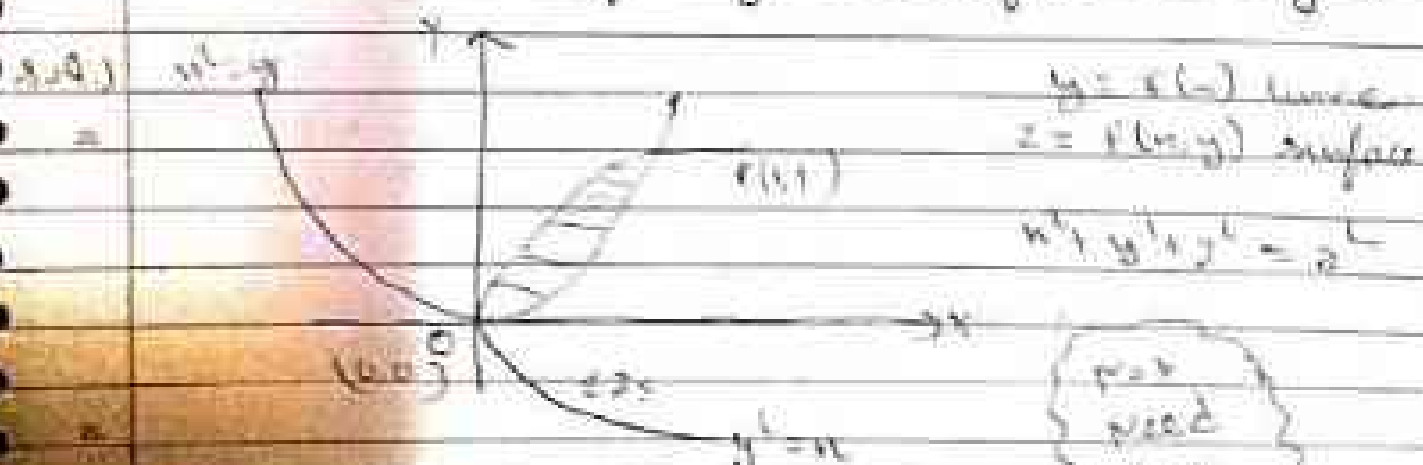
(a) $\int_0^{2\pi} \int_0^a f(x, y) dx dy$ ✓

(b) $\int_0^{2\pi} \int_0^a f(x, y) dy dx$

(c) $\int_0^{2\pi} \int_0^a f(x, y) dx dy$ ✓

(d) $\int_0^{2\pi} \int_0^a f(x, y) dy dx$

Q: Find the volume of solid standing on the area common to the parabola $y^2 = x$ & $x^2 = y$ as base and cut off by the surface $z = 12 - 4x - y^2$



Req. vol = $\iiint (1) dx dy dz$

not need our answer

$\int_0^1 \int_0^x \int_0^{12-4x-y^2} 1 dz dy dx = \frac{584}{140}$ ✓

Lecture no. 04

Q:) Evaluate $\iint n^2 dy dn$ where O is the region in 1st quadrant bounded by $ny=16$, $y=0$, $y=n$, $n=8$.

Sol:) $\iint n^2 dy dn = \int \int_{onc} n^2 dy dn + \int \int_{n=0c} n^2 dy dn$

$$= \int_0^4 \int_0^n n^2 dy dn + \int_4^8 \int_0^{16/n} n^2 dy dn$$

$$= \dots \dots \dots$$

$$\text{II} \quad 448 \quad 2$$

(ii) Also find the area bounded by $ny=16$, $y=0$, $y=n$, $n=8$?

Sol:) Area: $\iint (1) dy dn = \int \int_{onc} (1) dy dn + \int \int_{n=0c} (1) dy dn$

$$= \int_0^4 \int_0^n (1) dy dn + \int_4^8 \int_0^{16/n} (1) dy dn$$

$$= 8 + 16 \ln 2$$

Change of order of integration:

- if vertical strip is given then to change the order of integration we should make horizontal strip & put the limits accordingly.

Q:) $\int_0^1 \int_{2y}^2 e^{xy} dx dy$

$$=$$

Lecture no. 05

$\Rightarrow$ Joint p.d.f $\Rightarrow$ if x & y are two continuous random variable then their joint p.d.f is denoted as $f(x, y)$ & it's condition are

(i) $f(x, y) \geq 0 \Rightarrow$ surface $z = f(x, y)$ lies above xy plane

(ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1 \Rightarrow$ total volume under $f(x, y)$ & above xy plane is of unit.

Note: i) $P(a \leq x \leq b, c \leq y \leq d)$

$$= \int_{y=c}^d \int_{x=a}^b f(x, y) dx dy$$

2) if x & y are two Ind R.V then $f(x, y) = f(x) \cdot f(y)$

3) M.d.f of x is given as $f(x) = \int_{-\infty}^{\infty} f(x, y) dy$

& M.d.f of y is given as $f(y) = \int_{-\infty}^{\infty} f(x, y) dx$

• Make vertical strip

• Make horizontal strip

4) Condition Prob. of $w = a$ when $y = c$ is given as $P\left(\frac{x=a}{y=c}\right)$

Conditional density funct of x for given y is $f(x/y) = \frac{f(x, y)}{f(y)}$

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Conditional density function of y for given x is:

$$f(y|x) = \frac{f(x,y)}{f(x)}$$

Q: if x & y are two ind uniform random variables defined in $[-1, 1]$ then find $P[\max(x, y) \leq \frac{1}{2}]$

sol: (n-1) (using graph)



Total area = $(2)(2) = 4$

fav area = shaded area = $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}$

$$P[\max(x, y) \leq \frac{1}{2}] = P[-1 \leq (x, y) \leq \frac{1}{2}]$$

$$= \frac{\text{fav area}}{\text{total area}} = \frac{1/4}{4} = \frac{1}{16} \text{ Ans}$$

Using concept of U.R.V:-

$x \in [-1, 1]$ & x is U.R.V so p.d.f is

$$f(x) = \begin{cases} \frac{1}{1-(-1)} & -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$y \in [-1, 1]$ & y is also U.R.V so p.d.f is

$$f(y) = \begin{cases} \frac{1}{1-(-1)} & -1 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

study time



$$P(\max(x, y) < \frac{1}{2}) = P\left[-1 \leq (x, y) < \frac{1}{2}\right]$$

$$= P\left[-1 \leq x < \frac{1}{2}\right] \cdot P\left[-1 \leq y < \frac{1}{2}\right]$$

$$= \int_{-1}^{\frac{1}{2}} f(x) dx \cdot \int_{-1}^{\frac{1}{2}} f(y) dy$$

$$= \int_{-1}^{\frac{1}{2}} \frac{1}{4} dx \cdot \int_{-1}^{\frac{1}{2}} \frac{1}{4} dy$$

$$= \frac{1}{4} \left(\frac{1}{2} + 1\right) \cdot \frac{1}{4} \left(\frac{1}{2} + 1\right) = \frac{1}{4} \times \frac{3}{2} \times \frac{1}{4} \times \frac{3}{2} = \frac{9}{16}$$

m-1

Using the concept of joint p.d.f

$\therefore x, y$ are ind R.V so their joint p.d.f is given as $f(x, y) = f(x) \cdot f(y)$

$$= \frac{1}{4} \cdot \frac{1}{4} = \begin{cases} \frac{1}{16} & \text{if } -1 \leq x < \frac{1}{2} \text{ and } -1 \leq y < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

Now

$$P(\max(x, y) < \frac{1}{2}) = P\left\{-1 \leq (x, y) < \frac{1}{2}\right\}$$

$$= \int_{-1}^{\frac{1}{2}} \int_{-1}^{\frac{1}{2}} f(x, y) dx dy$$

$$= \frac{1}{16} \int_{-1}^{\frac{1}{2}} \int_{-1}^{\frac{1}{2}} 1 dx dy$$

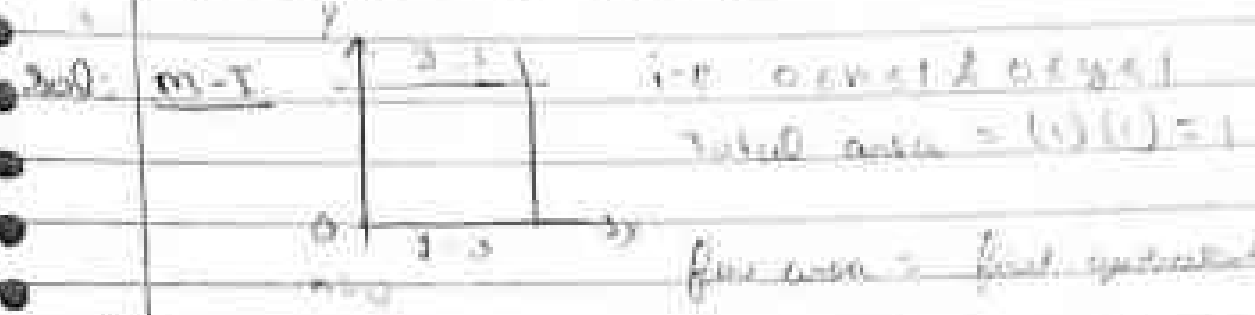
$$= \frac{1}{16} \int_{-1}^{\frac{1}{2}} \left(\frac{1}{2} + 1\right) dy$$

$$= \frac{1}{16} \left(\frac{1}{2} + 1\right) \left(\frac{1}{2} + 1\right) = \frac{9}{16}$$

study time

Probability

Q.) Two points are randomly selected at random in an interval to determine the prob that sum of their squares is less than 1.



$$= \frac{1}{4} (\pi \cdot 1^2) = \frac{\pi}{4} (1) = \frac{\pi}{4}$$

So $P(x^2 + y^2 < 1) = \frac{\text{fav area}}{\text{interval}} = \frac{\pi/4}{1} = \frac{\pi}{4}$

Ex: $x \in [0, 1]$ & x is U.R.V so pdf is

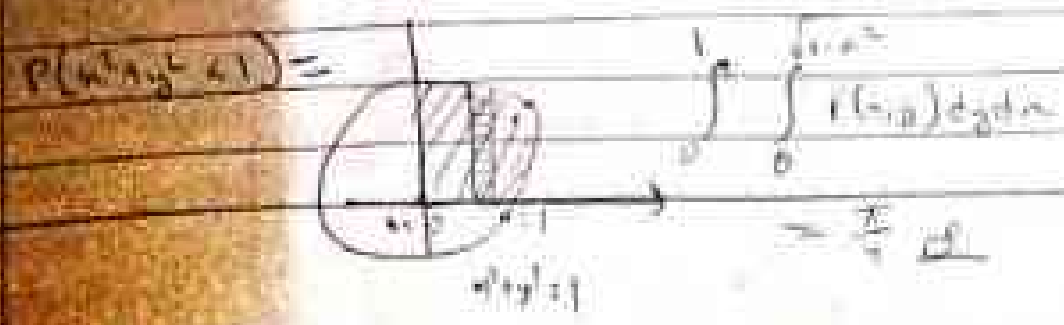
$$f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{other} \end{cases} \quad \text{(Probability density)}$$

$y \in [0, 1]$ & y is U.R.V so pdf is $f(y) = \begin{cases} 1 & 0 \leq y \leq 1 \\ 0 & \text{other} \end{cases}$

Ex: Selection of n x, y with in the interval $[0, 1]$ are of independent nature
So we can say that P.V x, y are ind

$\Rightarrow$ Joint p. $P(x^2 + y^2 < 1) = P(0 \leq x < 1) \cdot P(0 \leq y < 1)$

Joint pdf is $f(x, y) = f(x) \cdot f(y)$
 $= \begin{cases} 1 & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{other} \end{cases}$



Q.1 The joint p.d.f of 2D random variable (x, y) is given as:

$$f(x, y) = \begin{cases} 2, & 0 < x < 1, 0 < y < x \\ 0 & \text{other} \end{cases}$$

- (i) verify that it is valid p.d.f
- (ii) Find marginal density function of x & y
- (iii) Check for the independency of x and y
- (iv) Find the conditional density function of y for given $x = x$
- (v) Find the conditional density function of x for given $y = y$.

Ans. Region of integration is



$$(i) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$$

$$= \int_0^1 \int_0^x (2) dy dx = 2 \int_0^1 \left\{ \int_0^x (1) dy \right\} dx$$

$$= 2 \int_0^1 (x) dx = 2 \int_0^1 (1/2) dx$$

$$= 2 \left[\frac{x^2}{2} \right]_0^1 = 2 \left[\frac{1}{2} - 0 \right]$$

$$(ii) \text{m.o.f of } x: f(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad \text{--- (1)}$$

study time

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$$\int_0^1 f(x) dx = 2n, \quad 0 \leq n \leq 1$$

$$f(y) = \int_{n=y}^1 f(x, y) dx = \int_y^1 f(x) dx = 2(1-y), \quad 0 \leq y \leq 1$$

i.e. m.d.f of x is $f(x) = 2n, \quad 0 \leq n \leq 1$

ii. A m.d.f is y $f(y) = 2(1-y), \quad 0 \leq y \leq 1$

$$(iii) f(x) \cdot f(y) = (2n) \cdot 2(1-y) = 4n(1-y) \neq f(x, y)$$

$$\therefore f(x, y) \neq f(x) \cdot f(y)$$

i.e. $f(x, y)$ is not independent.

(iv) Conditional density function of y for given x is

$$f(y/x) = \frac{f(x, y)}{f(x)} = \frac{2}{2n} = \frac{\text{Joint p.d.f.}}{\text{m.d.f. of } x}$$

$$= \frac{1}{n}, \quad 0 \leq n \leq 1$$

(v) Conditional density function of x for given y is

$$f(x/y) = \frac{f(x, y)}{f(y)} = \frac{2}{2(1-y)} = \frac{\text{Joint p.d.f.}}{\text{m.d.f. of } y}$$

$$= \frac{1}{(1-y)}, \quad 0 \leq y \leq 1$$

Q. Joint p.d.f of x and y is given as

$$f(x, y) = 4xye^{-(x+y)}, \quad x \geq 0, y \geq 0$$

(i) verify that it is valid p.d.f

(ii) Find marginal p.d.f for x and y

(iii) Check the independency of x & y

(iv) Find conditional density of x for given y

study done

Note: Evaluate $\int_0^{\infty} 2x e^{-x^2} dx = \int_0^{\infty} 2y e^{-y^2} dy = 1$

Sol: verification $\int_0^{\infty} \int_0^{\infty} f(x, y) dx dy$

$$= \int_0^{\infty} \int_0^{\infty} (2xy e^{-(x^2+y^2)}) dx dy$$

$$= \int_0^{\infty} \int_0^{\infty} (2x e^{-x^2}) \cdot (2y e^{-y^2}) dx dy$$

$$= \int_0^{\infty} 2x e^{-x^2} dx \int_0^{\infty} 2y e^{-y^2} dy$$

$$= 1 \quad \text{Hence verified.}$$

(ii) m.d.f of x :

$$f(x) = \int_0^{\infty} f(x, y) dy =$$

$$= \int_0^{\infty} f(x, y) dy = \int_0^{\infty} 4xy e^{-(x^2+y^2)} dy = 2x e^{-x^2} \int_0^{\infty} 2y e^{-y^2} dy$$

$$f(x) = 2x e^{-x^2}, \quad 0 < x < \infty$$

m.d.f of y :

$$f(y) = \int_0^{\infty} f(x, y) dx =$$

$$= \int_0^{\infty} 4xy e^{-(x^2+y^2)} dx = 2y e^{-y^2} \int_0^{\infty} 2x e^{-x^2} dx$$

$$f(y) = 2y e^{-y^2}, \quad 0 < y < \infty$$

o.e.y.e

(iii) Check the independency

$$f(x) \cdot f(y) = f(x, y) = 2n e^{-nx} \cdot 2y e^{-y} = 4nxy e^{-(x+y)}$$

So x and y are independent.

(iv) Conditional density function of x for y given

$$f(x/y) = \frac{f(x, y)}{f(y)} = \frac{4nxy e^{-(x+y)}}{2y e^{-y}} = 2n e^{-x}$$

Here we observed that whenever x and y are ind then $f(x/y) = f(x)$
Similarly $f(y/x) = f(y)$.

$$\Rightarrow P[n < a/y < c] = \frac{P[n < a \cap y < c]}{P[y < c]}$$

Q.) if Joint P.d.f for x and y is given as $f(x, y) =$

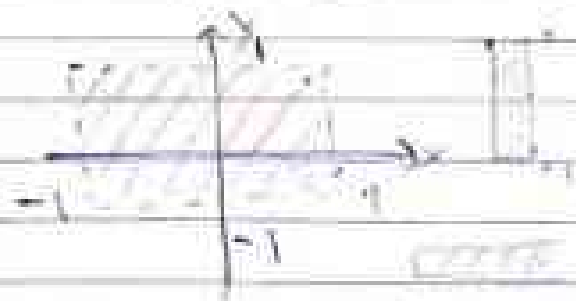
$$\begin{cases} \frac{1}{4}(1+xy), & |x| < 1 \\ & |y| < 1 \\ 0 & \text{other} \end{cases}$$

Show that x and y are not independent.

$$\Rightarrow |x| < 1 \Rightarrow -1 < x < 1$$

$$|y| < 1 \Rightarrow -1 < y < 1$$

This shaded Region is
Region of integration.



$$\text{M.d.f of } x: f(x) = \int_{-1}^1 f(x, y) dy$$

$$= \int_{-1}^1 \frac{1}{4}(1+xy) dy = \frac{1}{4} \int_{-1}^1 1 dy + x \int_{-1}^1 y dy$$

$$= \frac{1}{4} [1 - (-1)] + 0 = \frac{1}{4} (2) = \frac{1}{2} \quad -1 < x < 1$$

$$E(x) = \int x f(x) dx$$

study time

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$$\begin{aligned} \text{O.d.f of } y: \int f(x, y) dx &= \int \frac{1}{4} (1+xy) dx \\ &= \int \frac{1}{4} (1+xy) dx = \frac{1}{4} \int 1 dx + y \int \frac{x}{4} dx \\ &= \frac{1}{4} (2) + y \left(\frac{x^2}{8} \right) \\ &= \frac{1}{2} + \frac{y}{8} (1-1) = \frac{1}{2} + \frac{y}{8} (1-1) \\ &= \frac{1}{2} + y \end{aligned}$$

$$\Rightarrow \text{Since } f(x) \cdot f(y) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \neq f(x, y)$$

$\Rightarrow$ x & y are not independent.

$$\Rightarrow E(x/y) = \int x$$

Lecture no. 06

$$\begin{aligned} \Rightarrow \text{Conditional expectation: } & \left\{ \begin{aligned} E(x/y) &= \int x f(x/y) dx \\ E(y/x) &= \int y f(y/x) dy \end{aligned} \right. \end{aligned}$$

Q) if x & y are two random variable & $f(y/x)$ is the conditional expectation of x for given y then evaluate $E[E(x/y)] = ?$

- a) $E(x) / f(y)$ w.k that $f(x/y) = \frac{f(x, y)}{f(y)}$ (1)
 b) $E(x/y)$
 c) $E(x)$ ✓ $\because E(x/y) = \int x f(x/y) dx$
 d) $E(y)$

$$= \phi(y) \text{ let.}$$

$$\begin{aligned} \text{Now } E[E(x/y)] &= E[\phi(y)] \\ &= \int \phi(y) \cdot f(y) dy \end{aligned}$$

= Do your self.

study time



Q.1) if joint p.d.f is

$$f(n, y) = \begin{cases} \frac{e^{-y}}{y} e^{-ny/y}, & 0 < n < \infty \text{ and } 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

then find $E(n)$ for given y i.e. $E(n|y=y) = ?$

Sol.1) $E(n|y) = \int_{-\infty}^{\infty} n \cdot f(n|y) dn$

" m.d.f of y is $= \int_{-\infty}^{\infty} f(n, y) dn = \int_0^{\infty} \left(\frac{e^{-y}}{y} e^{-ny/y} \right) dn$

$$= \frac{e^{-y}}{y} \int_0^{\infty} e^{-ny/y} dn$$

$$= \frac{e^{-y}}{y} \left(\frac{e^{-ny/y}}{-1/y} \right)_0^{\infty} = -e^{-y} \cdot (-1) = e^{-y}$$

Now $f(n|y) = \frac{f(n, y)}{f(y)} = \frac{\left(\frac{e^{-y}}{y} e^{-ny/y} \right)}{e^{-y}} = \frac{1}{y} e^{-n/y}$

So

Expected value of n given y is $E(n|y) = \int_{-\infty}^{\infty} n \cdot f(n|y) dn$

i.e. $E(n|y) = \int_0^{\infty} n \cdot \left(\frac{1}{y} e^{-n/y} \right) dn = \frac{1}{y} \int_0^{\infty} n e^{-n/y} dn$

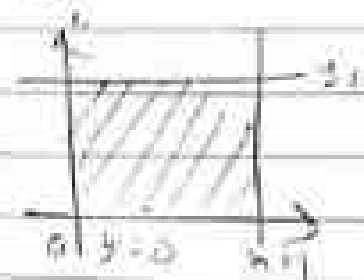
$$= \dots = y$$

Q.2) Suppose that two continuous Random variables (n, y) has joint p.d.f given as

$$f(n, y) = \begin{cases} 6n^2 y & 0 < n < 1, 0 < y < 1 \\ 0 & \text{other} \end{cases}$$

(i) verify that $\int_0^1 \int_0^1 f(x,y) dx dy = 1$

$$= \int_0^1 \int_0^1 (6x^2y) dx dy$$



$$= \int_0^1 6x^2 dx \int_0^1 y dy$$

$$= 6 \left(\frac{x^3}{3} \right)_0^1 \left(\frac{y^2}{2} \right)_0^1$$

$$= \frac{6}{3 \cdot 2} (1) (1) = 1 \quad \text{Hence verified.}$$

(vi) m.d.f of x is $f(x) = \int_{-\infty}^{\infty} f(x,y) dy$

$$= \int_0^1 6x^2y dy$$

$$= 6x^2 \left(\frac{y^2}{2} \right)_0^1 = 3x^2 = f(x)$$

m.d.f of y is $f(y) = \int_{-\infty}^{\infty} f(x,y) dx$

$$= \int_0^1 6x^2y dx = 6y \left(\frac{x^3}{3} \right)_0^1$$

$$= 2y = f(y)$$

$$f(x)f(y) = (3x^2)(2y) = 6x^2y = f(x,y) \quad \text{as } x \text{ \& } y \text{ are ind.}$$

(ii) $P(0 < x < 1, \frac{1}{2} < y < 2) =$

$$\int_{1/2}^2 \int_0^1 f(x,y) dx dy$$



$$y = \frac{1}{4} + 1$$

$$n = 0, 1/4$$

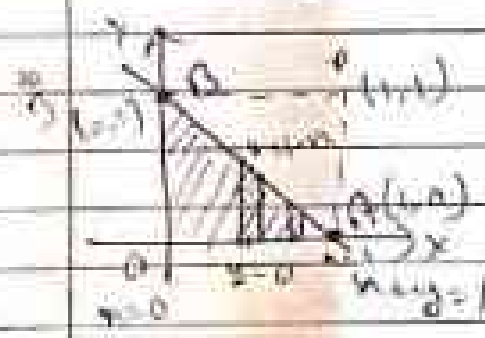
$$\int_{x=1/4}^1 \int_{n=0}^{1/4} (6n^2y) dy dx =$$

$$6 \int_0^{1/4} n^2 dn \int_{1/4}^1 y = 6 \left(\frac{n^3}{3} \right)_0^{1/4} \left(\frac{y^2}{2} \right)_{1/4}^1$$

$$= 6 \left(\left(\frac{1}{4} \right)^3 \right) \cdot \left(\frac{1^2 - (1/4)^2}{2} \right)$$

$$= \frac{3}{8} \text{ or } 0.375$$

(ii) $P(n+y \leq 1) = ?$ $\int_{n=0}^1 \int_{y=0}^{1-n} f(n,y) dy dn$



$$= \int_0^1 \int_0^{1-n} (6n^2y) dy dn$$

$$= \int_0^1 \left(\int_0^{1-n} y dy \right) 6n^2 dn$$

$$= \int_0^1 \left(\frac{y^2}{2} \right)_0^{1-n} 6n^2 dn$$

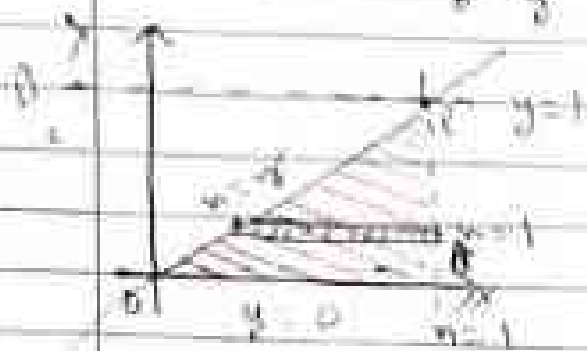
$$= \int_0^1 \frac{(1-n)^2}{2} \cdot 6n^2 dn$$

$$= 3 \int_0^1 (1+n^2-2n) n^2 dn$$

$$= 3 \int_0^1 (n^3 + n^4 - 2n^3) dn = 3 \left(\frac{n^4}{4} + \frac{n^5}{5} - \frac{2n^4}{4} \right)$$

$$= \frac{1}{10} \text{ or } 0.1$$

$$(iv) P(x > y) = ? \quad \int_0^1 \int_y^1 f(x, y) dx dy$$



$$= \int_0^1 \int_y^1 2xy dx dy$$

$$= \int_0^1 \left(\int_y^1 2x dx \right) (y dy)$$

$$= \int_0^1 2 \left(\frac{x^2}{2} \right) (y dy)$$

$$= \int_0^1 (1 - y^2) \cdot y dy$$

$$= 2 \int_0^1 (y - y^3) dy = 2 \left(\frac{y^2}{2} - \frac{y^4}{4} \right)_0^1$$

$$= 2 \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{1}{2}$$

$$(v) P(x < 1 | y < 2) = ? \quad = \frac{\int_0^1 f(x, y) dx}{\int_0^2 f(x, y) dy}$$

$$= P(x < 1) \quad \because x \text{ and } y \text{ are ind.}$$

$$= P(0 < x < 1) = \int_0^1 f(x) dx$$

$$= \int_0^1 (2x^2) dx \quad \because \text{marginal of } x \text{ is } f(x) = 2x^2$$

$$= 1$$

$$\text{m.u.} \quad \frac{P(x < 1, y < 2)}{P(y < 2)} = \frac{P(x < 1, y < 1)}{P(y < 1)}$$

$$= \frac{\int_0^1 \int_0^1 f(x, y) dx dy}{\int_0^1 \int_0^1 f(x, y) dx dy} = \frac{\int_0^1 \int_0^1 2xy dx dy}{\int_0^1 \int_0^1 2xy dx dy} = 1$$

Q) Also evaluate Prob $P(n < 1 / y < 1/2)$ using $n=2$

$$\begin{aligned} \text{Sol.) } \frac{P(n < 1 \cap y < 1/2)}{P(y < 1/2)} &= \frac{\int_0^{1/2} \int_0^y n^2 y \, dn \, dy}{\int_0^{1/2} \int_0^y 2y \, dn \, dy} \\ &= \frac{\int_0^{1/2} y^3 \, dy}{\int_0^{1/2} y^2 \, dy} \\ &= \frac{\frac{1}{4} \cdot \frac{1}{4}}{\frac{1}{3} \cdot \frac{1}{8}} = 1 \end{aligned}$$

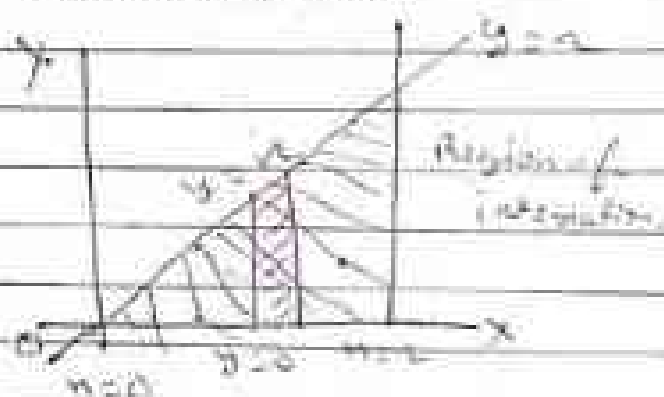
Q) if $f(n, y) = \begin{cases} cn^2 y, & 0 \leq n \leq 2, \text{ and } 0 \leq y \leq n \\ 0 & \text{otherwise} \end{cases}$ is

Joint p.d.f for x and y then evaluate

(i) $P(Y_1 < n < 1) = ?$

(ii) $P(y \geq 1) = ?$

(iii) $P(Y_1 < n < 1, y \geq 1) = ?$



Sol.) $\int_0^2 \int_0^n f(n, y) \, dn \, dy = 1$

$$\int_0^2 \int_0^n (cn^2 y) \, dy \, dn = 1$$

$$\int_0^2 \left[cn \left(\frac{y^2}{2} \right) \right]_0^n \, dn = 1$$

$$\frac{c}{2} \int_0^2 (n^2) \cdot n \, dn = 1$$

$$\frac{c}{2} \left(\frac{n^4}{4} \right)_0^2 = 1 \Rightarrow \frac{c}{2} \times \frac{16}{4} = \frac{16c}{8} = 1$$

$$\frac{16c}{8} = 1 \Rightarrow c = \frac{1}{2}$$

Now joint pdf is $(x, y) = \begin{cases} \frac{1}{4}xy & 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$

(i) m.f of x is $(x) = \int_0^1 f(x, y) dy$

$$\begin{aligned} \left| \begin{array}{c} \text{Area} \\ \text{under } f(x, y) \end{array} \right| &= \int_0^1 (x_1 y) dy \\ &= \frac{x}{2} \left(\frac{y^2}{2} \right)_0^1 = \frac{x^2}{4} \quad 0 \leq x \leq 1 \end{aligned}$$

m.f of y is $(y) = \int_0^1 f(x, y) dx$

$$\begin{aligned} \left| \begin{array}{c} \text{Area} \\ \text{under } f(x, y) \end{array} \right| &= \int_0^1 (x_1 y) dx \\ &= \frac{y}{2} \left(\frac{x^2}{2} \right)_0^1 = \frac{y}{4} (1 - 0) \\ &= \frac{y}{4} (1 - 0) = y - \frac{y^2}{2} \end{aligned}$$

$$(ii) P\left(\frac{1}{2} < x < 1\right) = \int_{1/2}^1 f(x) dx = \int_{1/2}^1 \frac{x^2}{4} dx$$

$$= \frac{1}{4} \left(\frac{x^3}{3} \right)_{1/2}^1 = \frac{1}{12} \left(1 - \frac{1}{8} \right)$$

$$= \frac{1}{12} \left(\frac{18-1}{18} \right) = \frac{1}{18} = \frac{15}{270}$$

$$m-2 \quad P(x < 1) = P[x < 1, 0 < y < 1]$$

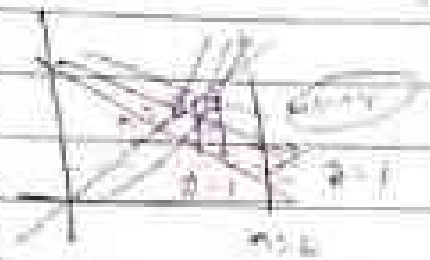
$$= \int_{1/2}^1 \int_0^1 f(x, y) dy dx$$

$$= \int_{1/2}^1 \int_0^1 (x_1 y) dy dx = \frac{15}{270}$$

study time

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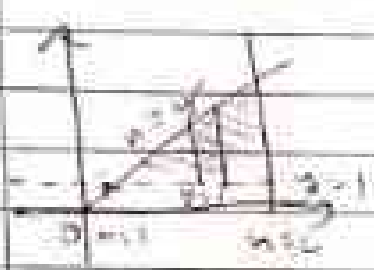
(ii) $P(y \geq 1) = \int_1^2 f(y) dy = \int_1^2 (y - \frac{y^2}{4}) dy$



$$= \frac{1}{4} \left(\frac{y^2}{2} - \frac{y^3}{3} \right) \Big|_1^2$$

$$= \left(\frac{2^2 - 1}{2} \right) - \left(\frac{16 - 1}{12} \right) = \frac{2}{2} - \frac{15}{12} = \frac{6}{12}$$

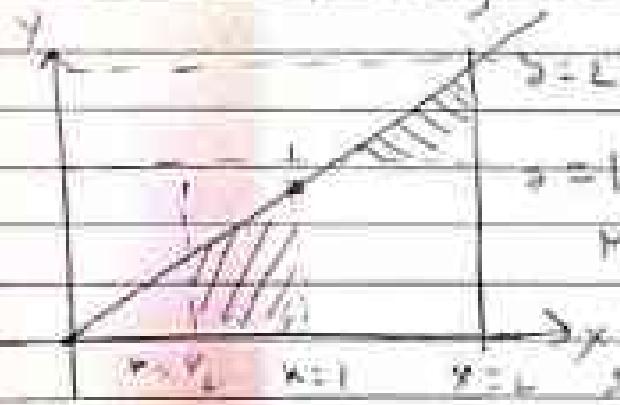
M-L $P(y \geq 1) = P(0 <= x <= 2, y \geq 1) = (12 - 15) \times 10^{-2} = 0.07$



$$\int_1^2 \int_0^2 f(x, y) dx dy = \int_1^2 \int_0^2 (y - \frac{y^2}{4}) dx dy$$

$$= \dots = \frac{1}{12}$$

(iii) $P(y_L <= x <= 1, y \geq 1) = \int_{y_L}^1 \int_1^2 f(x, y) dx dy$



No common region so $f(x, y) = 0$

$\Rightarrow P(y_L <= x <= 1, y \geq 1) = 0$

Q.2) if Joint p.d.f for x and y is given as $f(x, y) =$

$\{ \dots \}$

Lecture no. 078) Joint C.D.f $\Rightarrow$ if $f(x, y)$ is joint p.d.f & $F(x, y)$ is joint C.D.f then

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) du dv \quad \Rightarrow \text{general case}$$

A C.D.f at $x=a, y=b$ is $F(a, b) = \int_{-\infty}^a \int_{-\infty}^b f(u, v) du dv$
 $= \text{C.D.f at } (a, b)$

$$9) f(x, y) = \frac{\partial^2}{\partial x \partial y} (F(x, y))$$

10) if x is uniformly random variable b/w 0 & 1 and consider $y = x^2$ be another random variable then p.d.f of y is?

a) $f(y) = \sqrt{y}; 0 \leq y \leq 1$

b) $f(y) = -2\sqrt{y}; 0 \leq y \leq 1$

c) $f(y) = 1/2\sqrt{y}; "$

d) $f(y) = y^2, "$

Sol. In we will try to find C.D.f of y at $y=a$

" x is U.R.V in $(0, 1)$ so p.d.f of x is

$$f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{other} \end{cases} = 1$$

" $y = x^2$ & $x \in (0, 1)$ so y also lies b/w 0 & 1 but not necessarily uniform.

study time



∴ Now first we will try to calculate c.d.f of y at a where $0 < a < 1$

$$F_y(a) = P(y \leq a) = P(x^2 \leq a) = P(\sqrt{a} \leq x \leq \sqrt{a})$$

$$\begin{aligned} \text{definition of c.d.f} &= P(-\sqrt{a} \leq x < 0) + P(0 \leq x \leq \sqrt{a}) \\ \text{since } x &= 0 + P(0 \leq x \leq \sqrt{a}) \end{aligned}$$

$$= 0 + \int_0^{\sqrt{a}} f(x) dx$$

$$F_y(a) = \int_0^{\sqrt{a}} 1 dx = \sqrt{a}, \quad 0 < a < 1$$

∵ a is any random point in $(0, 1)$ so we can take $a = y$

$$\text{Hence C.D.F } \boxed{F(y) = \sqrt{y}}, \quad y \in (0, 1)$$

So p.d.f of y is $f(y) = \frac{d}{dy} F(y)$

$$= \frac{d}{dy} (\sqrt{y}) = \frac{1}{2\sqrt{y}}, \quad 0 < y < 1$$

Q1) if X is Exponential λ and $Y = X^2$, what is the p.d.f of Y ?

$$(a) f(y) = \frac{1}{\lambda} e^{-\lambda \sqrt{y}}, \quad y > 0$$

$$(b) f(y) = \lambda y^{2\sqrt{y}}, \quad y > 0$$

$$(c) f(y) = \frac{1}{\sqrt{y}} e^{-\frac{1}{\lambda} \sqrt{y}}, \quad y > 0$$

$$(d) f(y) = \lambda e^{-\lambda \sqrt{y}}, \quad y > 0$$

If X is E.R.V with parameter λ so it's p.d.f is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & \text{other} \end{cases}$$

$\therefore$ since $x = \sqrt{y} = n^2$ as y also has b/w region

• C.D.f of y (at $y=a$) is

$$F_y(a) = P(y \leq a) = P(n^2 \leq a)$$

$$= P(-\sqrt{a} \leq n \leq \sqrt{a})$$

$$= 0 + P(0 \leq n \leq \sqrt{a})$$

$$\int_0^{\sqrt{a}} f(n) dn = \int_0^{\sqrt{a}} \lambda e^{-\lambda n} dn$$

$$= \lambda \left(\frac{e^{-\lambda n}}{-\lambda} \right)_0^{\sqrt{a}} = - \left(e^{-\lambda \sqrt{a}} - 1 \right)$$

$$f_y(a) = 1 - e^{-\lambda \sqrt{a}}, \quad a > 0$$

$\therefore$ a is any random point so we can take $a=y$
 So C.D.f of y is $\boxed{F(y) = 1 - e^{-\lambda \sqrt{y}}}$

• L.p.d.f is

$$f(y) = \frac{d}{dy} F(y) = \frac{d}{dy} [1 - e^{-\lambda \sqrt{y}}]$$

$$= 0 - e^{-\lambda \sqrt{y}} \left(-\lambda \frac{1}{2\sqrt{y}} \right)$$

$$\boxed{f(y) = \frac{\lambda}{2\sqrt{y}} e^{-\lambda \sqrt{y}}} \quad y > 0$$

Q:) What is the ^{density func} distribution of Y if $y = -\frac{1}{2} \log(1-x)$ and x is uniform $(0, 1)$?

- a) $y = e^{-2x}$
- b) $y = \frac{1}{2} e^{-2x/2}$
- c) $y = 2 e^{-2x}$
- d) $y = 4 e^{-4x}$

sol:) $\because$ x is U.R.V in $(0, 1)$ so pdf of x is

$$f(x) = \begin{cases} \frac{1}{1-0} & , 0 < x < 1 \\ 0 & \text{other} \end{cases} = 1$$

$$\because y = -\frac{1}{2} \log(1-x)$$

$$-2y = \log(1-x)$$

$$\log(1-x) = -2y$$

$$1-x = e^{-2y}$$

$$-x = e^{-2y} - 1$$

$$x = 1 - e^{-2y}$$

Now try to find C.D.f of y as $(y=a)$ which is given as

$$F_y(a) = P(y \leq a)$$

$$F_y(a) = P\left[-\frac{1}{2} \log(1-x) \leq a\right]$$

$$= P(\log(1-x) \geq -2a)$$

$$= P[(1-x) \geq e^{-2a}]$$

$$= P[-x \geq e^{-2a} - 1]$$

$$= P[x \leq 1 - e^{-2a}]$$

study time

$$F_y(a) = P\left[X \leq 1 - e^{-2a}\right] = \int_0^{1-e^{-2a}} f(x) dx$$

$$= \int_0^{1-e^{-2a}} 2e^{-2x} dx$$

$$= 1 - e^{-2a}$$

So

$$F_y(a) = 1 - e^{-2a}, \quad 0 < a < 1$$

a is any random point so taking $a = y$

$$\boxed{F(y) = 1 - e^{-2y}} \quad 0 < y < 1$$

So v.d.f is

$$f(y) = \frac{d}{dy} F(y) = \frac{d}{dy} (1 - e^{-2y})$$

$$= (-e^{-2y})(-2) = 2e^{-2y}$$

$$f(y) = 2e^{-2y} \quad 0 < y < 1 \quad \text{Ans}$$

9.) What is the joint p.d.f of x and y if their joint distribution function is given as

$$F(x, y) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-(x+y)}; & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

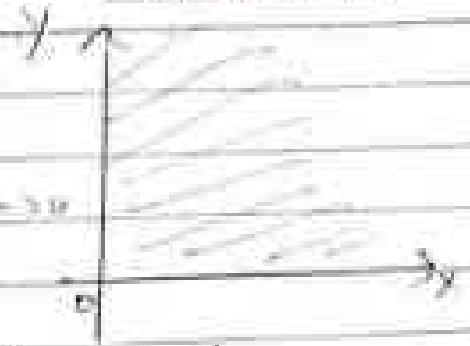
- (i) check the independencies of x and y
- (ii) Find the marginal densities of x and y
- (iii) Evaluate $P(x \leq 1 \cap y \leq 1)$
- (iv) Find $P(x+y \leq 1)$
- (v) Also find density function $f_n(x/y)$?
- (vi) Also find the density function of n for given y !
- (vii) Also find the expected value of n for given y !

study time

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(i) Joint C.D.f is

$$F(n, y) = \begin{cases} 1 - e^{-n} - e^{-y} + e^{-(n+y)} & n > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

So joint p.d.f is $f(n, y) = \frac{\partial^2}{\partial n \partial y} (F(n, y))$

$$= \frac{\partial}{\partial n} \left[\frac{\partial}{\partial y} (1 - e^{-n} - e^{-y} + e^{-(n+y)}) \right]$$

$$= \frac{\partial}{\partial n} [0 - 0 + e^{-y} + e^{-n}(-e^{-y})]$$

$$= (0 - 0 + 0 + e^{-n}(-e^{-y})) = -e^{-(n+y)} \quad n > 0, y > 0$$

i.e. p.d.f is $f(n, y) = e^{-n} \cdot e^{-y}, n > 0, y > 0$ (ii) m.d.f of n is $f(n) = \int_{-\infty}^{\infty} f(n, y) dy$ 

$$= \int_0^{\infty} e^{-n} e^{-y} dy$$

$$= e^{-n} \left[\frac{e^{-y}}{-1} \right]_0^{\infty} = -e^{-n} (e^{-\infty} - e^0)$$

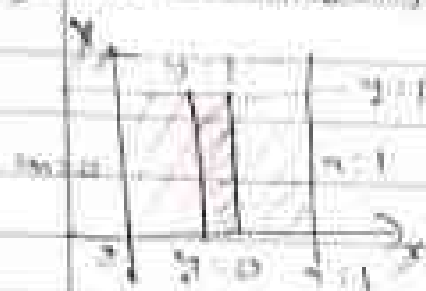
$$f(n) = -e^{-n} (0 - 1) = e^{-n}$$

m.d.f of y is $f(y) = \int_{-\infty}^{\infty} f(n, y) dn$

$$f(y) = e^{-y} \quad n > 0, y > 0$$

(iii) $f(n) \cdot f(y) = f(n, y)$ so this is the independence of n and y .

$$(iii) P(n \leq 1, n \leq 1) = \int_0^1 \int_0^1 f(n, y) dy dx$$



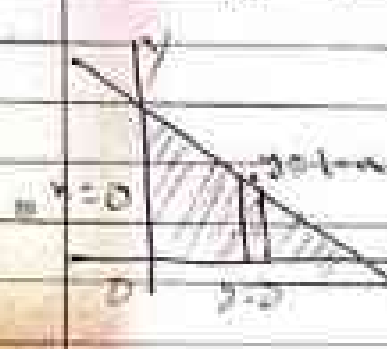
$$= \int_0^1 \int_0^1 e^{-x} \cdot e^{-y} dx dy$$

$$= \int_0^1 e^{-x} dx \cdot \int_0^1 e^{-y} dy$$

$$= \left(\frac{e^{-x}}{-1} \right)_0^1 \cdot \left(\frac{e^{-y}}{-1} \right)_0^1 = + (e^{-1} - 1) (e^{-1} - 1)$$

$$= \left(1 - \frac{1}{e} \right)^2 \quad \underline{\underline{A}}$$

$$(iv) P(n+y \leq 1) = \int \int f(n, y) dy dx$$



$$= \int_0^1 \int_0^{1-n} e^{-x} e^{-y} dy dx$$

$$= \int_0^1 e^{-x} \left(\int_0^{1-n} e^{-y} dy \right) dx$$

$$= \int_0^1 e^{-x} \left(\frac{e^{-y}}{-1} \right)_0^{1-n} dx$$

$$= \int_0^1 e^{-x} (e^{-n+1} - e^{-1}) dx$$

$$= \int_0^1 (e^{-x} e^{-n+1} - e^{-x} e^{-1}) dx$$

$$= \left(\frac{e^{-x} e^{-n+1}}{-1} - \frac{e^{-x} e^{-1}}{-1} \right)_0^1 = \left(\left(\frac{e^{-1}}{-1} - \frac{1}{e} \right) \right) - \left(\frac{1}{-1} - 0 \right)$$

$$= 1 - \frac{2}{e} \quad \underline{\underline{A}}$$

Learn integration of product formula

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(vi) Density function of n for given y is

$$f(n/y) = \frac{f(n,y)}{f(y)} = \frac{e^{-ny} y^n}{e^{-y}} = e^{-n(y-1)} \quad n \geq 0$$

(vii) Expected value of n for given y is

$$E(n/y) = \int_0^{\infty} n \cdot f(n/y) dn$$

$$\therefore E(n/y) = \int_0^{\infty} n \cdot (e^{-n(y-1)}) dn$$

$$= \left[n \cdot \frac{e^{-n(y-1)}}{-(y-1)} - \int_0^{\infty} 1 \cdot \frac{e^{-n(y-1)}}{-(y-1)} dn \right]_0^{\infty}$$

$$= \left(-n e^{-n(y-1)} - e^{-n(y-1)} \right)_0^{\infty} = (- (0 - 0) - (-1))$$

$$= 1 \text{ Ans.}$$

Now

(v) Here n will be treated as new random variable & we will try to find its C.O.f. so first we will try to find its C.O.f. let $n = z$ then C.O.f of z at a is given as

$$F_z(a) = P(Z \leq a) = P\left(\frac{n}{y} \leq a\right) = P(n \leq ay)$$

$$= \int_0^{ay} \int_0^{\infty} e^{-ny} y^n dn dy$$

$$= \int_0^{\infty} \left[\int_0^{ay} e^{-ny} dn \right] y^n dy$$

$$\therefore F_z(a) = P(Z \leq a) = \int_0^{\infty} \left(1 - e^{-nay} \right) y^n dy$$

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$$\int_0^{\infty} (e^{-ay} - \frac{e^{-(a+1)y}}{a+1}) dy$$

$$= \left[\frac{e^{-ay}}{-1} - \frac{e^{-(a+1)y}}{-(a+1)} \right]_0^{\infty} = \left[(0-0) - \left(-1 + \frac{1}{a+1} \right) \right]$$

$$= 1 - \frac{1}{a+1}, \quad a > 0$$

$\therefore$ as a is random so we can take $a = 2$ that is C.D.f is $F(z) =$

$$F(z) = 1 - \frac{1}{z+1}, \quad z > 0$$

So, p.d.f of z is $f(z) = \frac{d}{dz} F(z) = \frac{1}{(z+1)^2}$ $z > 0$
(where $z = \frac{x}{y}$)

i.e.

C.D.f of $\frac{x}{y}$ is $F_{\frac{x}{y}}(a) = 1 - \frac{1}{a+1}, \quad a > 0$

h

p.d.f of $\left(\frac{x}{y}\right)$ is $f_{\frac{x}{y}}(a) = \frac{1}{(a+1)^2}, \quad a > 0$

Lecture no. 08

Sampling ^(universe)

⇒ Population: The group of individual under consideration is called population.

• Parameter: Those numerical quantities from which we can understand population are called parameter.
 (e.g. ⇒ Mean μ , S.D. σ are the parameter, Variance σ^2)

• Statistic ⇒ Those numerical quantities from which we can understand sample.
 (e.g. Mean $\bar{x}$, S.D. s , Variance s^2)

⇒ Sample: Any subset of population is called sample.

⇒ Sampling: With the help of statistics we will try to understand parameter & hence can conclude the information for population. This process is called sampling.

⇒ Proportion: The ratio of successful items & total items is called proportion.

$$i.e. \text{ Proportion} = \frac{\text{Number of success}}{\text{Total items}}$$

$$L + 1 - P$$

$$\Rightarrow \text{Sample proportion of success } (\bar{p}) = \frac{\text{No. of success}}{\text{Sample size}} = \frac{n}{n}$$

$$\Rightarrow \text{Population " " " } (p) = \frac{\text{No. of success}}{\text{Population size}} = \frac{X}{N}$$

study time

Q.7

Probability of boys in India = $\frac{7500}{15000} = \frac{1}{2} = 0.5$

in a family 5 kids there are 2 boys. then population of boys

$$= \frac{2}{5} \neq 0.5$$

then boy is assumed as success

⇒ Standard Error :-

It is the S.D. of sample mean or sample proportion.

1) S.E. of $(\bar{x}) = \text{S.D. of } \bar{x} = \frac{\sigma}{\sqrt{n}}$

2) S.E. of $(\bar{p}) = \sqrt{\frac{p\bar{p}}{n}}$

⇒ Most Probable Limits →

a) $\bar{x} - 3 \text{ S.E.}(\bar{x}) \leq \text{Population Mean}(\mu) \leq \bar{x} + 3 \text{ S.E.}(\bar{x})$

b) $\bar{p} - 3 \text{ S.E.}(\bar{p}) \leq \text{Population Proportion}(\mu_p) \leq \bar{p} + 3 \text{ S.E.}(\bar{p})$

Q.8) A sample of 400 members has mean mean 4.0 if the population is normal with S.D 2.6 and its mean is unknown then find the most probable limits for population mean.

Sol.) $n = 400, \bar{x} = 4, \sigma = 2.6$

S.E. of $\bar{x} = \frac{\sigma}{\sqrt{n}} = \frac{2.6}{\sqrt{400}} = 0.13$ → $3 \text{ S.E.}(\bar{x})$
 $= 3 \times 0.13 = 0.39$

Hence most probable limits of μ will be as follow

$$\bar{x} - 3 \text{ S.E.}(\bar{x}) \leq \mu \leq \bar{x} + 3 \text{ S.E.}(\bar{x})$$

study time

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$$4 - 2 \times 0.39 \leq \mu \leq 4 + 2 \times 0.39$$

$$= 4 - 0.78 \leq \mu \leq 4 + 0.78$$

$$= 3.21 \leq \mu \leq 4.78 \quad \text{So, it is concluded that the audit is correct.}$$

Q: In a town 350 out of 600 person were found to be vegetarian on the basis of this data can we say that majority of population in the town is vegetarian?

Sol: $n = \{\text{No. of veggie person}\}$, success

$n = 600$ $x = 350$ veggie person so sample proportion of success $(\bar{p}) = \frac{x}{n} = \frac{350}{600} = 0.58$

& sample proportion of failure $(\bar{q}) = 1 - \bar{p} = 0.42$

$$S.E. \text{ of } (\bar{p}) = \sqrt{\frac{\bar{p}\bar{q}}{n}} = \sqrt{\frac{0.58 \times 0.42}{600}} = 0.02$$

$$3SE(\bar{p}) = 0.06$$

most probable limit of population proportion

$$\bar{p} - 3SE(\bar{p}) \leq p_0 \leq \bar{p} + 3SE(\bar{p})$$

$$0.58 - 0.06 \leq p_0 \leq 0.58 + 0.06$$

$$0.52 \leq p_0 \leq 0.64$$

i.e. proportion of veggie person in population lies b/w 0.52 & 0.64

Hence 52% of veggie person = 52% & 64%

Ans: Yes

Q.) A die was thrown 9000 times and 1 or 6 obtained 3120 times can we say that the die is unbiased? (fair die).

sol.) $n = 9000$ $x = 3120$ (number of times 1 or 6 obtained) $\rightarrow 14000$
 Sample proportion of success ($\hat{p}$) = $\frac{x}{n}$

$$= \frac{3120}{9000} = 0.3467 = 0.347$$

$$\text{failure } (\hat{q}) = 1 - \hat{p} = 0.653$$

$$\text{S.E. of } \hat{p} = \sqrt{\frac{\hat{p}\hat{q}}{n}} = \sqrt{\frac{0.347 \times 0.65}{9000}} = 0.005$$

$\therefore$

$$3SE(\hat{p}) = 3 \times 0.005 = 0.015$$

Most probable value of population proportion (p_0) =

$$\hat{p} - 3SE(\hat{p}) \leq p_0 \leq \hat{p} + 3SE(\hat{p})$$

$$0.347 - 0.015 \leq p_0 \leq 0.347 + 0.015$$

$$0.332 \leq p_0 \leq 0.362$$

(a) we assume that the die is unbiased (fair) then $p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = \frac{1}{6}$
 $\therefore p(1 \text{ or } 6) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} = 0.333$

$\therefore$ population proportion of success

$\therefore$ exact value of population proportion lies in b/w most probable

study time

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Q1) In previous question, if success is occurring 3240 times then prove that die is biased. (info 20)

Q2) A coin was tossed 400 times and head turned up 210 times. Discuss whether coin is unbiased or not.

sol.) $n = 400$, $x = \{\text{no. of heads}\} = 210$

Sample proportion of head $\hat{p} = \frac{x}{n} = \frac{210}{400} = 0.525$

" " " " Tail $q = 1 - \hat{p} = 0.475$

S.E. of $\hat{p} = \sqrt{\frac{\hat{p}q}{n}} = \sqrt{\frac{0.525 \times 0.475}{400}} = 0.035$

M.P.L.

$$\hat{p} - 3SE(\hat{p}) \leq p_0 \leq \hat{p} + 3SE(\hat{p})$$

$$= 0.525 - 0.035 \times 3 \leq p_0 \leq 0.525 + 0.035 \times 3$$

$$0.415 \leq p_0 \leq 0.600$$

If coin was unbiased then $P(\text{getting head}) = \frac{1}{2}$
= proportion of success in population

$$\text{i.e. } p_0 = 0.500$$

Since value of p_0 lies within M.P.L. so coin was unbiased.

⇒ Special note →

Population variance = $\frac{\sum (x_i - \mu)^2}{N}$, N = population size

Sample variance = $\frac{\sum (x_i - \bar{x})^2}{n-1}$, n = sample size

Lecture no. 09

Central Limit Theorem:

⇒ let us have a population of large size N not necessarily normal but let us assume that its Average = μ & $S.D = \sigma$

($n \geq 30$)

let us consider some sample of this population as follows:

$S_1 = \{x_1, x_2, x_3, \dots, x_n\} \Rightarrow \bar{S}_1$

$S_2 = \{y_1, y_2, y_3, \dots, y_n\} \Rightarrow \bar{S}_2$

$S_3 = \{z_1, z_2, z_3, \dots, z_n\} \Rightarrow \bar{S}_3$

$\vdots$

$S_{1000} = \{x_1, x_2, x_3, \dots, x_n\} \Rightarrow \bar{S}_{1000}$

→ samples

$\rightarrow \frac{\bar{S}_1 + \bar{S}_2 + \bar{S}_3 + \dots + \bar{S}_{1000}}{1000}$

i.e. Average of sample mean = population mean

$\therefore \text{var} \left\{ \frac{\bar{S}_1 + \bar{S}_2 + \bar{S}_3 + \dots + \bar{S}_n}{n} \right\} = \frac{\sigma^2}{n}$

⇒ i.e. $\bar{S}_i \sim N \left\{ \mu, \frac{\sigma^2}{n} \right\}$ This is called C.L.T

⇒ Distribution formed by sample means is Normal distributed.

2.) Average of Distribution formed by sample ^{mean} $\bar{x}$ is almost equal to population mean.

3.) Variance of Distribution formed by sample mean $= \frac{\sigma^2}{n}$

$\Rightarrow$ Necessary condition for C.L.T.

- 1.) Sample members should be selected at random
- 2.) " " must be independent
- 3.) Population members should be distributed identically
- 4.) $30 \leq$ sample size $< 10\%$ of population size

Q.2) C.L.T does not hold in which case:

- (a) Sample size ≥ 30 & population is normal
- (b) " " ≥ 30 & " is not normal
- (c) " " < 30 & " is normal
- (d) " " < 30 & " is not normal. $\times$

Q.3) A set of sample have been collected from large data and the sample mean values are 12.8, 10.9, 11.4, 14.2, 12.5, 13.6, 15, 9, 12.8 then find population mean (μ)

Ans) By C.L.T we know that

Population mean = Average of sample space

$$= \frac{12.8 + 10.9 + 11.4 + 14.2 + 12.5 + 13.6 + 15 + 9 + 12.8}{9}$$

$$\mu = 12.44$$

study time

15/11/2023

Q:) In a survey of Lucknow city, it was reported that Average age of mobile user is of 30 yrs with S.D 12 yrs, which is the mean and S.D of mobile users in a sample of 100 persons.

sol:) $\mu = 30, \sigma = 12, n = 100$

So Sample mean = population mean = 30 yrs

$$\text{Sample variance} = \frac{\sigma^2}{n} = \frac{144}{100} = 1.44$$

$$\therefore \text{S.D} = 1.2$$

(ii) Also find the probability that Average age of mobile user in a sample of 100 persons is less than 28 yrs. it is the given that area under normal curve from $z=0$ to 1.6 is 0.4452

sol:) Here Here Sample mean ($\bar{x}$) is normally Distributed with mean 30 & var = 1.44

$$\text{i.e. } \bar{x} \sim N\{30, 1.44\} \quad (0, 1)$$

$$\text{i.e. } \mu_{\bar{x}} = 30 \text{ \& } \sigma_{\bar{x}} = 1.2$$

$$\text{Let } Y = \{\text{Average age of mobile users}\}$$

$$P(Y < 28) = P(Z_Y < -1.67) = P(Z_Y > 1.67)$$

$$Z_Y = \frac{Y - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{1}{2} = P(0 < Z < 1.67)$$

$$= \frac{-2}{1.2} = -1.67, \quad \frac{1}{2} = 0.4452$$

$$= 0.054 = 5.4\% \text{ (Ans)}$$

study time (minutes)

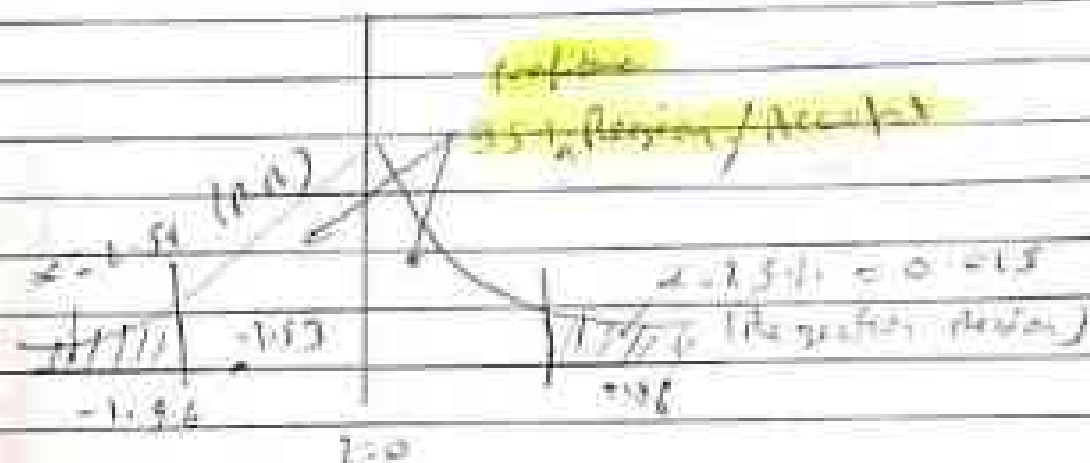
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Q.) A sample of 400 members has a mean = 4 where sample is taken from normal population with unknown mean and standard deviation 2.6. Can we say that population mean is 4.2 with 5% level of significance? It is given that for two tailed test $z = 1.96$ for $\alpha = 0.025$.

(YES)

Sol.) Sample size (n) = 400

$$\bar{x} = 4 \quad \mu = ? \quad \sigma = 2.6$$

Null Hypothesis $H_0: \mu = 4.2$ Alternative Hypothesis $H_1: \mu \neq 4.2$ i.e. $\mu < 4.2$ or $\mu > 4.2$

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{4 - 4.2}{2.6 / \sqrt{400}} = \frac{-0.2}{0.13} = -1.57$$

i.e. z lies in confidence region so accept H_0 & reject H_1 .

Hypothesis Testing:-

- Hypothesis is on the basis of sample information we make certain assumption for population & these assumption are called hypothesis.
- Null Hypothesis: it is a kind of statement in which we assume that H_0 : "There is no significant difference b/w sample attribute & population attribute."
(e.g.)

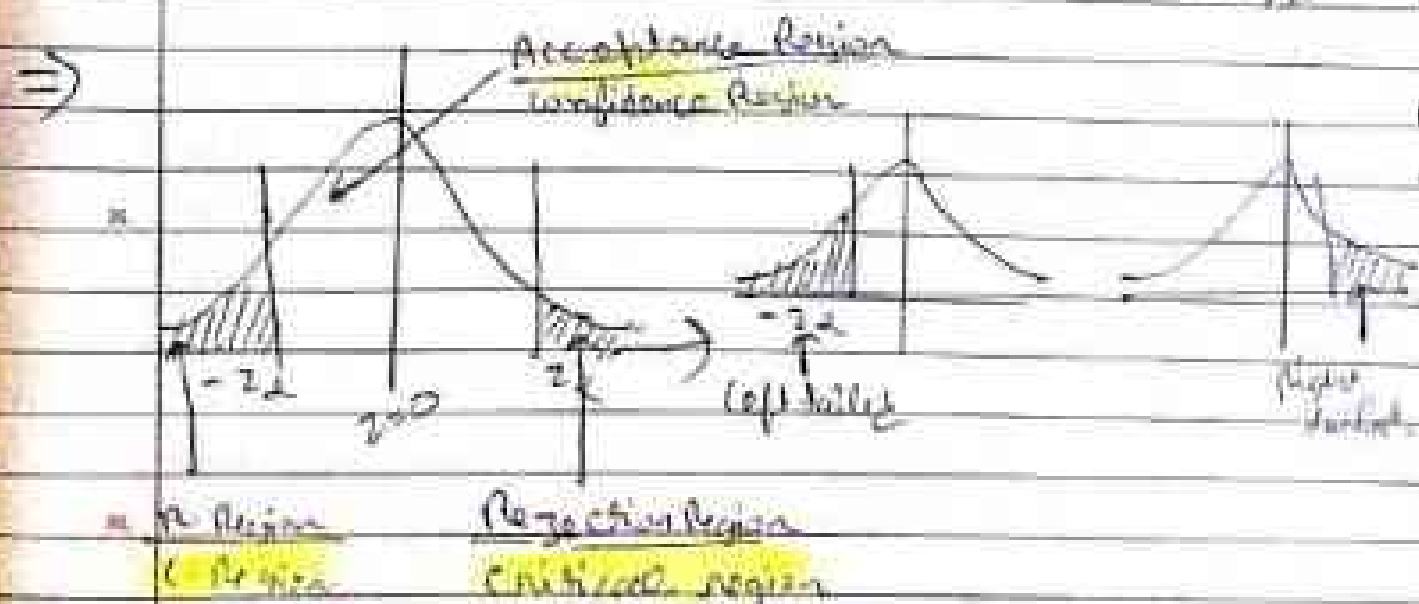
$$H = H_0$$

- Alternative hypothesis:-

H_1 : "There is a significant b/w sample attr & population attribute"

$$H_1: \mu \neq \mu_0 \text{ or } H_1: \mu > \mu_0 \text{ or } H_1: \mu < \mu_0$$

Two tailed test Right tailed test Left tailed test



⇒ we can follow following three type of test to check hypothesis

1) Z-test (parametric test)

2) t-test → small sample test ($n < 30$) (small variance)

3) chi-square test → large sample test (generally $n > 30$) (large variance)

⇒ With the help of Z-test we can solve following type of questions:

- Type-1

" Significance of population mean:

$$H_0: \mu = \mu_0 \quad \& \quad H_1: \mu \neq \mu_0$$

i.e.

there is no significant difference b/w sample mean & population mean.

$$Z = \frac{\bar{x} - \mu_0}{SE(\bar{x})} = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

if Z lies in the confidence region we will accept H_0 otherwise we reject H_0

- Type-2:

Significance of difference b/w two population mean

$$H_0: \mu_1 = \mu_2 \quad \& \quad H_1: \mu_1 \neq \mu_2$$

$$Z = \frac{\bar{x} - \bar{y}}{SE(\bar{x} - \bar{y})} = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

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Type-3:

Significance of population proportion:

$$H_0: \bar{p} = p_0 \quad \& \quad H_1: \bar{p} \neq p_0$$

i.e. there is no significant diff b/w sample prop ($\bar{p}$) & pop prop (p_0)

$$z = \frac{\bar{p} - p_0}{\sigma} = \frac{\bar{p} - p_0}{\sqrt{p_0 q_0}} = \frac{\bar{p} - p_0}{\sqrt{p_0 (1 - p_0)}}$$

$$= \frac{\left(\frac{x}{n}\right) - p_0}{\frac{1}{n} \sqrt{p_0 q_0}} = \boxed{\frac{\bar{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}}$$

Type-4:

Significance of difference b/w two population proportions

$$H_0: p_1 = p_2 \quad \& \quad H_1: p_1 \neq p_2$$

i.e. There is no significant difference b/w two population proportions.

$$\hat{p}_1 = \frac{n_1}{n_1}, \quad \hat{p}_2 = \frac{n_2}{n_2}, \quad \bar{p} = \frac{n_1 + n_2}{n_1 + n_2}$$

where $\bar{p}$ = common proportion of two sample

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sigma(\hat{p}_1 - \hat{p}_2)} = \boxed{\frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\bar{p} q}{n_1} + \frac{\bar{p} q}{n_2}}}}$$

lecture no. 10 :-

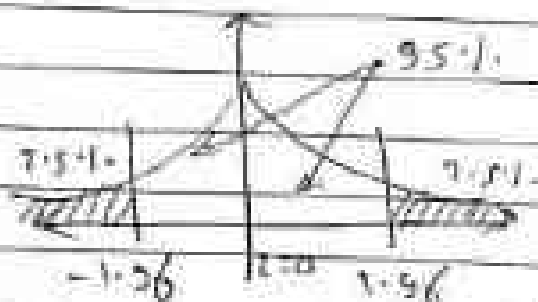
⇒ Type 1:

• we have to check significance of sample mean

• $H_0: \mu = \mu_0$ (No difference b/w S.M & P.M)• $H_1: \mu \neq \mu_0$ (Significant difference)

$$Z = \frac{\bar{x} - \mu_0}{SE(\bar{x})} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

• if Z lies in A.R so Accept H_0
 " " " " R.R so Reject H_0

sol:) Sample size (n) = 400, $\bar{x} = 4$, $\mu_0 = ?$ • $\sigma = 2.6$ 

• $H_0 = \mu = 4.2$ i.e. there is no significant difference between $\bar{x}$ & μ_0

$H_1 = \mu \neq 4.2$ i.e. there is significant difference

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{4 - 4.2}{2.6 / \sqrt{400}} = \frac{-0.2}{2.6}$$

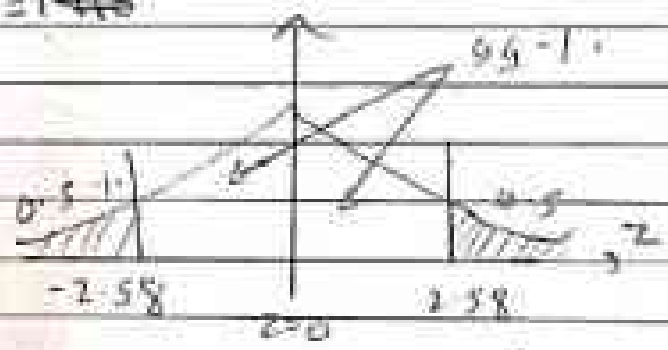
$$= \frac{-4}{2.6} = -1.53$$

Z lies in acceptance region so H_0 is Accepted & Reject H_1 i.e. $\mu = 4.2$ ✓

Q:) While calculating the average monthly income of a family in a town sample of 81 families was taken. The mean income and standard deviations of these 81 families were found to be 4108 Rs and 24 Rs respectively show that the assumption "average" income of family in a town is 4100 Rs is not reasonable for 1% level of significant if $Z_c = 2.58$

(ii) Also find the most probable limits for every income

Sol:) Sample size $(n) = 81$, $\bar{x} = 4108$, $\sigma = 24$
 $\mu_0 = 4100$



$H_0 : \mu_0 = 4100$ (There is no significant b/w $\bar{x}$ & μ_0)
 $H_1 : \mu_0 \neq 4100$ (There is significant b/w $\bar{x}$ & μ_0)

$z = \text{statistic is}$

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$$z = \frac{4108 - 4100}{24 / \sqrt{81}} = 3.15$$

(ii) $SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} = 2.67$

$$3 SE(\bar{x}) = 8.01 = 8.01$$

$$= \bar{x} - 3 SE(\bar{x}) \leq \mu_0 \leq \bar{x} + 3 SE(\bar{x})$$

$$= 4108 - 8.01 \leq \mu_0 \leq 4108 + 8.01$$

$$= \frac{8.01}{24} = \frac{4108 \pm 8.01}{24}$$

z lies in the rejection region so reject H_0
 Accept H_1
 i.e. $\mu_0 \neq 4100$

⇒ Type - 2:

(iii) Also find Probable limit for 99% Region

$$= \bar{x} - 2.58 \text{SE}(\bar{x}) \leq \mu_0 \leq \bar{x} + 2.58 \text{SE}(\bar{x})$$

$$= 4103 - 2.58(2.62) \leq \mu_0 \leq 4103 + 2.58(2.62)$$

$$= \boxed{4101.12 \leq \mu_0 \leq 4104.88}$$

i.e. why we have concluded that $\mu_0 \neq 4100$

⇒ Type - 2:

" We can check the significance of difference b/w two sample means.

H_0 : There is no S. Diff b/w $\bar{x}_1$ & $\bar{x}_2$
i.e. $H_0: \bar{x}_1 = \bar{x}_2$

H_1 : There is S. Diff b/w $\bar{x}_1$ & $\bar{x}_2$
i.e. $H_1: \bar{x}_1 \neq \bar{x}_2$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\text{SE}(\bar{x}_1 - \bar{x}_2)} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Q.) The mean of two sample of 1000 and 2000 members are 67.5 and 68.0 inches respectively. Can the sample be regarded as drawn from the same population of standard deviation 2.5 inches. Take $\alpha = 0.05$
i.e. $Z_{\alpha/2}(0.05) = 1.96$

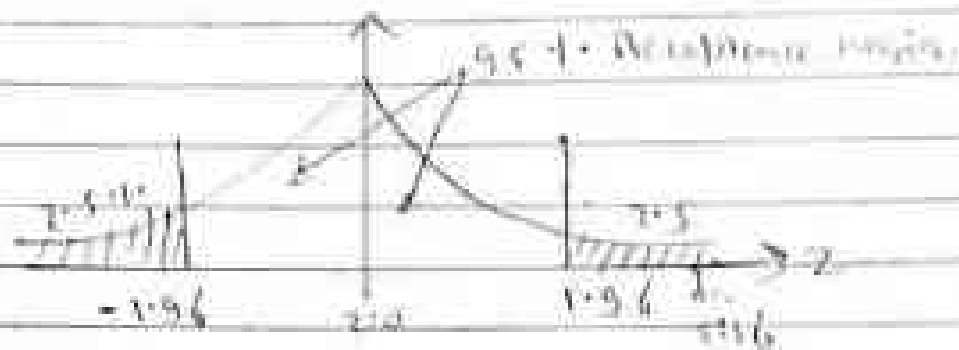
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Sol: $n_1 = 1000$, $n_2 = 2000$, $\sigma^2 = 2.5$

$\bar{x} = 67.5$, $\bar{y} = 68.0$



$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

$$Z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}}$$

$$= Z = \frac{67.5 - 68.0}{\sqrt{\frac{(2.5)^2}{1000} + \frac{(2.5)^2}{2000}}} = -5.16$$

$\therefore Z$ lies in rejection region $\therefore$ Reject H_0 & Accept H_1

i.e. 20 sample are not drawn from same population

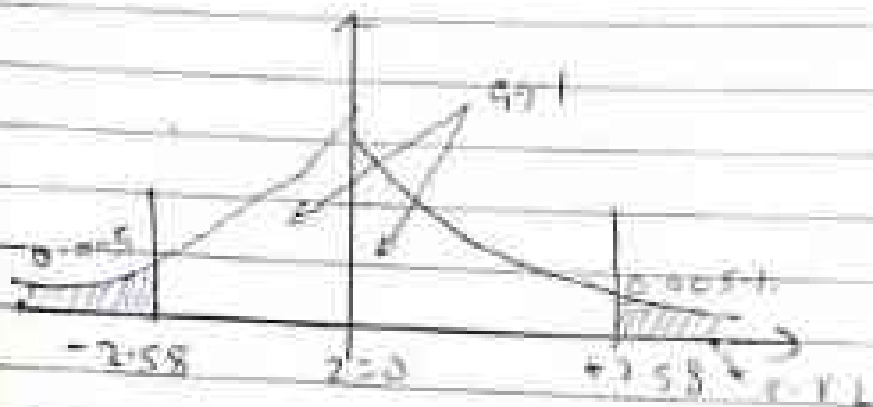
Q: If means of two samples (of same size 400) are 250 and 220 with their standard deviation 40 and 55 respectively then test $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$ at 1% level of significance i.e. $Z_{\alpha}(0.01) = 2.58$

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Sol: $n_1 = n_2 = 400$, $\bar{n} = 250$, $\bar{y} = 210$

$\sigma_1 = 40$, $\sigma_2 = 55$



$$Z = \frac{\bar{n} - \bar{y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{250 - 210}{\sqrt{\frac{1600}{400} + \frac{3025}{400}}} = 9.8219$$

$\therefore Z$ lies in critical region $\therefore$ reject H_0
 Accept H_1
 $i.e. H_1 \neq H_2$

$\Rightarrow$ Type-3:

We can justify the significance of a population proportion with the help of sample proportion.

$H_0: p = p_0$

$H_1: p \neq p_0$

$$Z = \frac{n - \mu_p}{\sigma_p} = \frac{n - E(n)}{SD(n)} = \frac{n - np_0}{\sqrt{np_0(1-p_0)}}$$

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$$= n \left(\frac{n}{n} - p_0 \right)$$

$$n \sqrt{\frac{p_0 q_0}{n}}$$

$$= \frac{\bar{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$$

Q.) A coin was tossed 400 times and head tossed up 210 times discuss whether coin is unbiased or not?

Sol.) $n = \{\text{Number of heads}\}$ success.

$$n = 210, \quad N = 400$$

Sample Proportion of head $\bar{p} = \frac{n}{N} = \frac{210}{400} = 0.52$

" population proportion of head $= p_0 = \frac{x}{n} = \frac{1}{2}$

$$p_0 = \frac{1}{2} = 0.5 \quad \& \quad q_0 = 0.5$$

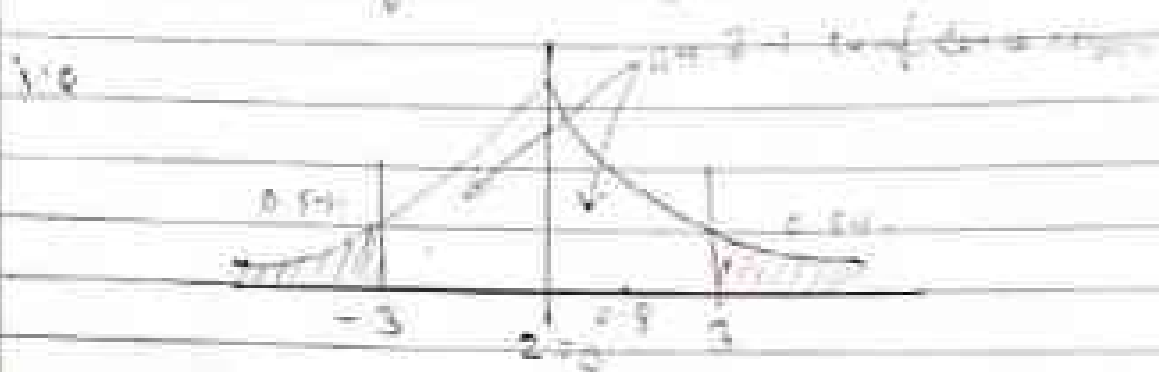
$$H_0: \bar{p} = p_0 \quad \& \quad H_1: \bar{p} \neq p_0$$

$$Z = \frac{\bar{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.52 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{400}}} = \frac{0.02}{0.05}$$

$$= \frac{4}{5}$$

$$= 0.8$$

∴ No level of significance is given so we will take confidence level as 95% i.e.



∴ 2 lies in confidence level so Accept H_0 & Reject H_1

i.e. $\bar{p} = p_0 = \frac{1}{2}$ Yes

So coin is unbiased.

Q.) A Die was thrown 9000 times and 1 or 6 was obtained 3240 times can we say that die was unbiased for 1% level of significance?
 $Z_{\alpha}(1\%) = 2.58$

Sol.) $n = 3240$, $n = 9000$ so sample proportion of success

$$\tilde{p} = \frac{n}{n} = \frac{3240}{9000} = 0.36$$

for population, we know that $P(1) = P(6) = \frac{1}{6}$

(a) die is unbiased then $P(1) = P(6) = \frac{1}{6}$ so population proportion of success $p = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

$$p = 0.33$$

$$1 - p = 0.67$$

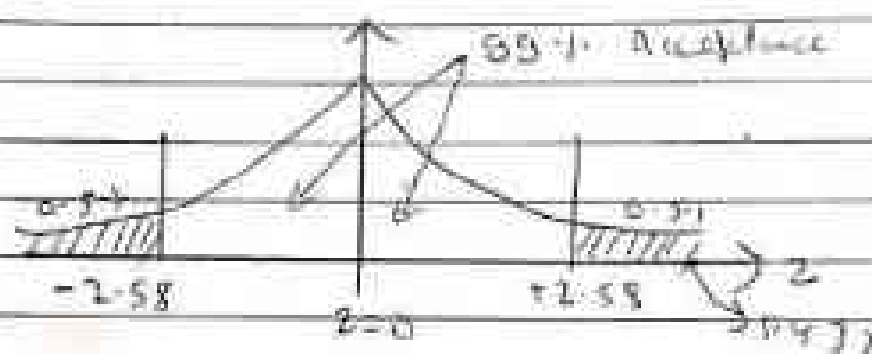
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H_0 : There is no significant difference b/w $\bar{p}$ & p_0
i.e. $\bar{p}$ is unbiased
OR

$H_0: \bar{p} = p_0$

H_1 : " " significant " " " " Biased
i.e. $H_1: \bar{p} \neq p_0$



$$Z = \frac{\bar{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.38 - 0.33}{\sqrt{\frac{0.33 \times 0.67}{5000}}}$$

$$= 6.052 \approx 6.05$$

$\therefore$ Value of Z statistic lies in critical region
so reject H_0 & accept H_1 i.e. $\bar{p} \neq p_0$

$\therefore$ Die is certainly Biased.

Q) In a town, 350 out of 600 person were found to be vegetarian. On the basis of this data can we say that majority of population in the town is vegetarian at 5% level of significance
Given that for right tailed $Z_{\alpha}(0.05) = 1.645$

only one

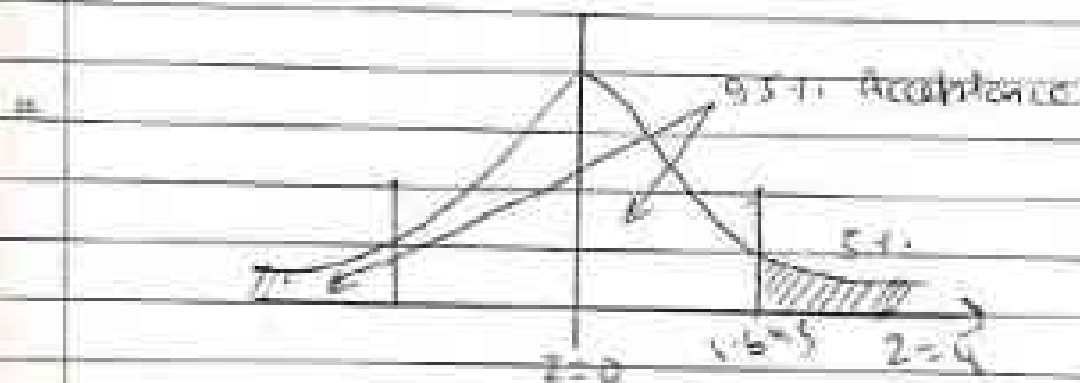
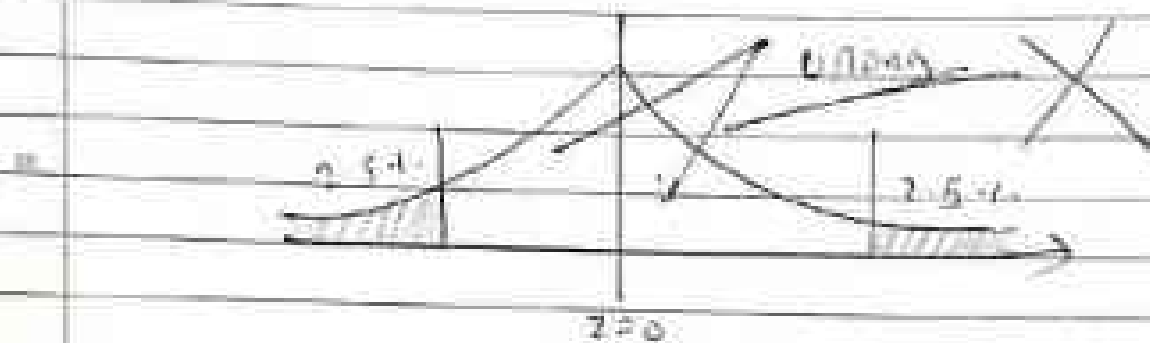
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sol: $n = 350$, $n = 600$

So sample proportion of veggie persons

$$\hat{p} = \frac{x}{n} = \frac{404}{600} = 0.673 = \frac{350}{600} \cdot (1)$$



let us assume that population is a two-sided test

let us assume that equal no. of veggie & non veggie person

i.e. there is no significant diff. b/w proportion of veggie & non-vegie person

i.e.

$$H_0: \hat{p} = p_0 = \frac{1}{2} \quad \text{No. of veggie person} = \text{No. of non-vegie person.}$$

$$H_1: p_0 > \frac{1}{2}$$

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$H_1: p > \frac{1}{2} \Rightarrow$ No. of vegetarians > no. of non-vegetarians

Now $p = \frac{1}{2}$, $q = \frac{1}{2}$, $n = 600$, $\hat{p} = 0.58$

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.58 - 0.50}{\sqrt{\frac{0.5 \times 0.5}{600}}}$$

$$\approx 4.02$$

$\therefore Z$ lies in rejection region $\therefore$ reject H_0 and H_1

i.e. $p > \frac{1}{2} \Rightarrow$ No. of vegetarians > no. of non-vegetarians

= majority of persons in a town are vegetarians.

$\Rightarrow$ Type-IV

Significance of difference b/w two population proportion

$$H_0: p_1 = p_2 \quad \text{vs} \quad H_1: p_1 \neq p_2$$

$$p_1 = \frac{n_1}{n_1 + n_2}, \quad p_2 = \frac{n_2}{n_1 + n_2}, \quad \bar{p} = \frac{n_1 + n_2}{n_1 + n_2}, \quad \bar{q} = 1 - \bar{p}$$

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}\hat{q}}{n_1} + \frac{\hat{p}\hat{q}}{n_2}}} = \frac{\hat{p}_1 - \hat{p}_2}{SE(\hat{p}_1 - \hat{p}_2)}$$

Q.3) There are two samples of 1000 person from City A and 800 person from city B. 400 person from each sample are found to be consumers of wheat. Can we say that number of wheat consumers from city A and city B differs significantly in desired 1% of confidence (i.e. 99%)?

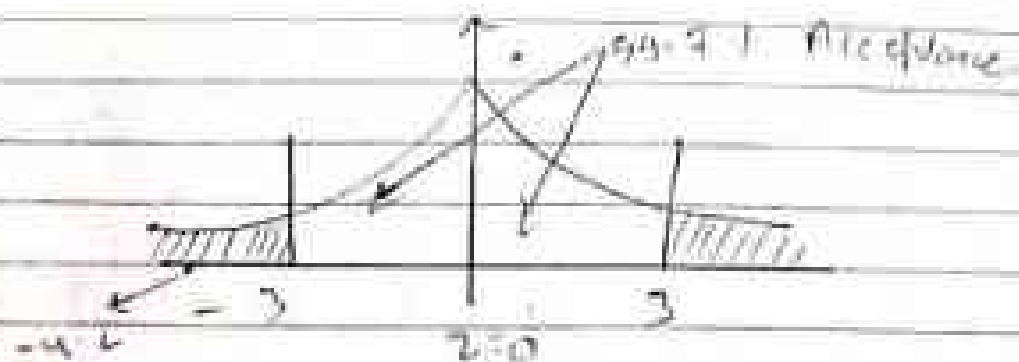
Sol: $n_1 = 1000$, $n_2 = 800$, $n = \text{consumers of wheat}$
 $n_1 = 400$, $n_2 = 400$

$$\bar{p}_1 = \frac{n_1}{n_1} = \frac{400}{1000} = 0.4$$

$$\bar{p}_2 = \frac{n_2}{n_2} = \frac{400}{800} = 0.5$$

Common proportion $\bar{p} = \frac{n_1 + n_2}{n_1 + n_2} = \frac{400 + 400}{1000 + 800}$
 $= \frac{800}{1800} = \frac{4}{9} = 0.44$

$$\bar{q} = 1 - \bar{p} = 0.56$$



∴ There is no significant diff b/w wheat consumer City A + City B

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$$i.e. H_0: p_1 = p_2$$

 $H_1: \text{There is significant}$

i.e.

$$H_1: p_1 \neq p_2$$

$$Z = \frac{\bar{p}_1 - \bar{p}_2}{\sqrt{\frac{\bar{p}\bar{q}}{n_1} + \frac{\bar{p}\bar{q}}{n_2}}} = \frac{0.4 - 0.5}{\sqrt{0.44 \times 0.56 \left\{ \frac{1}{100} + \frac{1}{100} \right\}}}$$

$$= -4.2$$

$\therefore Z$ lies in rejection region so Reject H_0 & Accept H_1

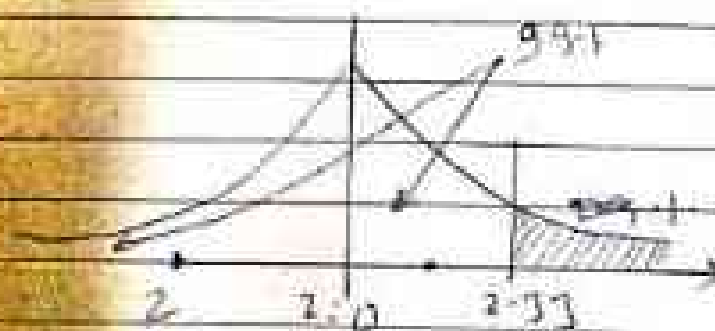
$$i.e. p_1 \neq p_2$$

So

Ans = Yes

Q.2.

(9i) A machine produced 20 defective articles in a lot of 400 and after overhauling it produced 20 defective articles in a lot of 300. Has the machine improved at $\alpha = 0.01$
 For right tailed $Z_{\alpha}(0.01) = 2.33$



n_1 = Number of Def articles producing by machine
 n_2 = Number of defective articles before overhauling
 $\bar{p}_1$ = 0.05
 $\bar{p}_2$ = 0.04

study time



Lecture no. 11

1- Distribution & student's t-test (Small Sample test i.e. for $n < 30$)

- Sample variance (s^2) = $\frac{\sum (x_i - \bar{x})^2}{n-1}$

- Population var (σ^2) = $\frac{\sum (x_i - \bar{x})^2}{n}$



- Degree of freedom = $DF(V) = \text{No. of observation} - \text{No. of condition imposed}$

or

or

V : Number of variable that have freedom to vary independently.

e.g. $n=10$ s.t. $\sum x_i = 80$

So we have freedom to choose $x_1, x_2, x_3, \dots, x_{n-1}$
 $\therefore x_n = 80 - (x_1 + x_2 + \dots + x_{n-1}) =$

i.e. no freedom to choose x_n , so degree of freedom
 $(V) = 9 \quad (10-1) = 9$

- $n=26, \mu, \sigma^2$ then $V = 26 - 2 = 24$

Note:

1) For one sample t-test, $[DF = n-1]$ & for two sample t-test $[DF = n-2]$ ($n_1 + n_2 - 2$)

2) In large DF, Distribution convert into Normal Distribution

⇒ Type-2 focus

Significance of population mean

$$H_0: \mu = \mu_0, \quad H_1: \mu \neq \mu_0$$

$$t = \frac{\bar{x} - \mu_0}{\text{SE}(\bar{x})} = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

Standard Error

Sample mean difference

Where S = Sample S.D

$$S^2 = \text{Sample variance} = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Q.) A machine produces washers of thickness 10mm. A sample of 10 washers has an average thickness of 9.52 mm with S.D of 0.6 mm. Find out 't'.

Sol.) $\mu_0 = 10\text{mm}, \quad \bar{x} = 9.52, \quad n = 10, \quad S = 0.6\text{ cm}$

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}} = \frac{9.52 - 10}{0.6/\sqrt{10}} = -2.58$$

Q.) A random sample of 16 values from a normal population has mean of 41.5 cm and sum of squares of deviation from mean is 135 cm². Can we say that the population mean is 43.5 cm? with 5% level of significance.

(i) Also find the 95% and 95% confidence for μ . It is given that $t_{15}(0.05) = 2.131$ and $t_{15}(0.01) = 2.94$

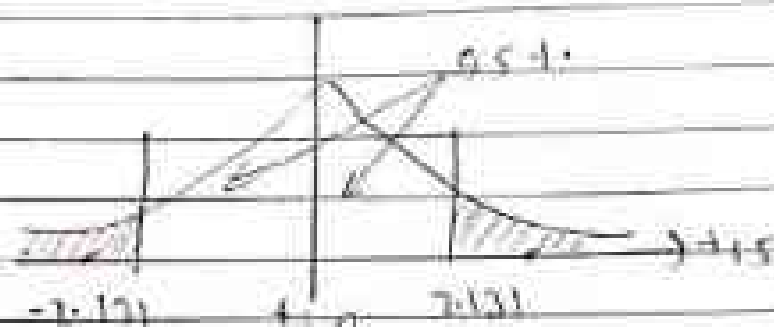
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sol: $n = 16$, $\bar{n} = 41.5$, $\sum (n_i - \bar{n})^2 = 135$, $\mu = ?$

$\mu_0 = 43.5$ (we have to check)

D.F.M = $n-1 = 16-1 = 15$ & $t_{15}(0.05) = 2.131$



Sample variance (s^2) = $\frac{\sum (n_i - \bar{n})^2}{n-1} = \frac{135}{16-1} = 9$

so $S = 3$

$H_0: \mu_0 = 43.5$

$H_1: \mu_0 \neq 43.5$

$t = \frac{\bar{n} - \mu_0}{s/\sqrt{n}} = \frac{41.5 - 43.5}{3/\sqrt{16}} = -2.66$

$\therefore t$ lies in rejection region so reject H_0 & accept H_1 i.e. $\mu_0 \neq 43.5$ for 95% C. Region.

$\Rightarrow$ Also check the validity of H_0 & H_1 for 99% C. Region.

$t = \frac{\bar{n} - \mu_0}{s/\sqrt{n}} = -2.66$ & At 9% significance level $t_{15}(0.01) = 2.602$

t lies in acceptance region accept H_0 for 99% of cases so $\mu_0 = 43.5$ for 99% C. Region

$$S.V(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{2}{\sqrt{16}} = \frac{1}{2} = 0.5$$

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15555555

ii) C.I. for 95%

$$\bar{x} \pm t_{\alpha/2} (0.05) SE(\bar{x}) \leq \mu_0 \leq \bar{x} + t_{\alpha/2} (0.05) SE(\bar{x})$$

$$= 41.5 \pm 2.131 \cdot 0.75 \leq \mu_0 \leq 41.5 + 2.131 \cdot 0.75$$

$$= \boxed{39.5 \leq \mu_0 \leq 43.0}$$

$\Rightarrow$ C.I. for 99%

$$t_{\alpha/2} (0.01) = 2.54 \text{ \& } SE(\bar{x}) = 0.75$$

$$= \bar{x} \pm t_{\alpha/2} (0.01) \cdot SE(\bar{x}) \leq \mu_0 \leq \bar{x} + t_{\alpha/2} (0.01) SE(\bar{x})$$

$$= 41.5 \pm 2.54 \cdot (0.75) \leq \mu_0 \leq 41.5 + 2.54 \cdot (0.75)$$

$$= \boxed{39.29 \leq \mu_0 \leq 43.70}$$

Q:) Average height of 10 student in a school is observed as 67 inches with sum of the square of deviations from central value is 88. can we say that average height of student in a school is 65 inches. it is given that, value of t at 5% level of significance with 9 degree of freedom is 2.62

sol:)

$$n = 10$$

$$D.F(V) = 10 - 1 = 9$$

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$t_9 (0.05) = 2.62$$

$$S^2 = \frac{88}{9}$$

$$\sum (x_i - \bar{x})^2 = 88$$

$$\bar{x} = 67$$

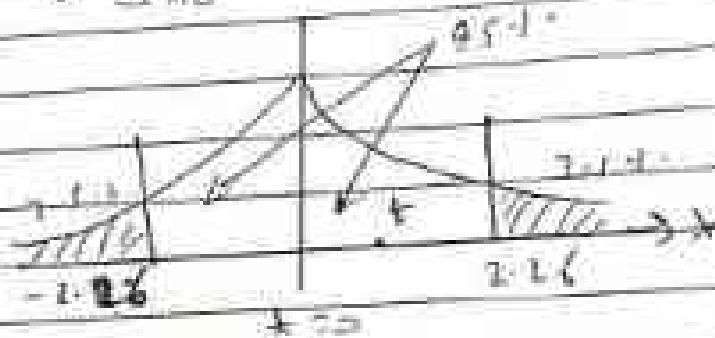
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$$H_0: \mu_0 = 65$$

$$H_1: \mu \neq 65$$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}}$$

$$= \frac{67 - 65}{\sqrt{\frac{55}{5710}}} = 2 = 2.024$$



$\therefore t$ lies in Acceptance region so Accept H_0
 $\therefore \mu = 65$ M

Q.1) if the mean and variance of sample of 10 observation are 66 and 14.67 respectively then find the 95% confidence limits for population mean (μ) given that upper 2.5% points of t distribution with 9 degree of freedom is 2.26

$$\text{Sol: } n=10, \bar{x}=66, s^2=14.67$$

$$\text{DF}(n) = 9, t_{9}(2.5\%) = 2.26$$



$$SE = \frac{s}{\sqrt{n}} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{14.57}{10}} = 1.21$$

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→) M.P.L. for 95%:

$$= \bar{x} - t_{\alpha/2}(0.05) (SE(\bar{x})) \leq \mu_0 \leq \bar{x} + t_{\alpha/2}(0.05) (SE(\bar{x}))$$

$$= 66 - 2.26 \times 1.21 \leq \mu_0 \leq 66 + 2.26 \times 1.21$$

$$= \boxed{63.26 \leq \mu_0 \leq 68.73}$$

" Type - 2:

Significance of Difference b/w two population mean

$$H_0: \mu_1 = \mu_2, \quad H_1: \mu_1 \neq \mu_2$$

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}}$$

$$S_1^2 = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1}$$

$$S_2^2 = \frac{\sum (y_i - \bar{y})^2}{n_2 - 1}$$

Covariance variance

$$S^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2} = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Q1) The means of two random sample of size 9 and 7 are 196.42 and 198.82 respectively variances are 26.94 and 18.73 can we say that the sample are drawn from same normal population?

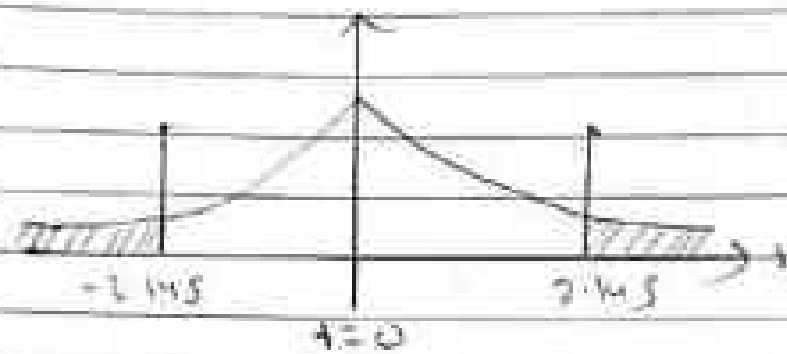
$$t_{19}(0.05) = 2.145$$

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2.1.1) $n_1 = 9$, $n_2 = 7$
 $DF(v) = n_1 + n_2 - 2$
 $= 9 + 7 - 2 = 14$



$\bar{x} = 196.42$, $\bar{y} = 198.82$

$S_1^2 = 26.94$, $S_2^2 = 18.77$

Common variance $s^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$

$s^2 = \frac{(9-1) \times 26.94 + (7-1) \times 18.77}{9+7-2}$

$s^2 = 23.42$

$H_0: \mu_1 = \mu_2 \Rightarrow$ Yes samples are drawn from same population.

$H_1: \mu_1 \neq \mu_2 \Rightarrow$ Samples are not drawn from "

$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}} = \frac{196.42 - 198.82}{\sqrt{\frac{23.42}{9} + \frac{23.42}{7}}}$

$= -0.58$

t falls in acceptance region so H_0 is accepted

Ques:1) Justify that for large D.F. χ^2 curve converts into normal curve

#. Lecture no. 12

Chi-Square Distribution
(large sample test i.e. for $n > 50$)

$$u \sim N(\mu, \sigma^2) \quad (10 \text{ p.v.})$$

$$z \sim N(0, 1) \quad (10 \text{ p.v.})$$

where

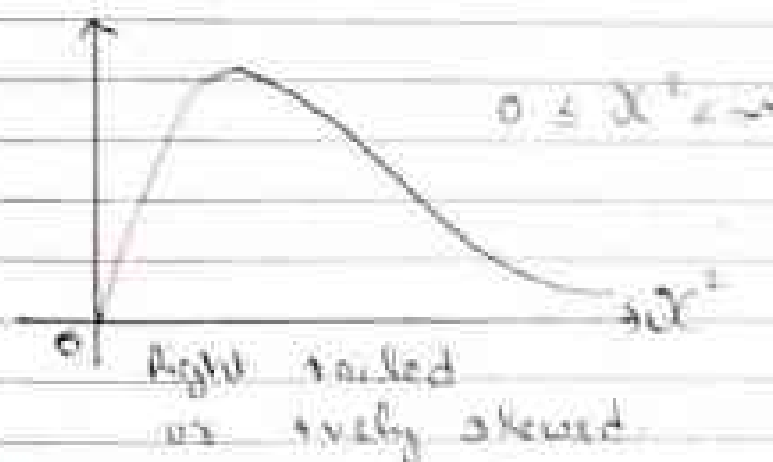
$$z = \frac{u - \mu}{\sigma}$$

• $z_i^2 = \left(\frac{u_i - \mu}{\sigma} \right)^2$ is called chi-square vari

i.e. $z_1^2, z_2^2, z_3^2, \dots, z_n^2$ are called chi square variable &

the distribution formed $z_1^2, z_2^2, z_3^2, \dots, z_n^2$ is called chi-sq Dist

$$\text{i.e. } \chi^2 = \sum_{i=1}^n (z_i^2) = \sum_{i=1}^n \left(\frac{u_i - \mu}{\sigma} \right)^2$$



• $D.F. = n - 1$

i.e. for large D.F. χ^2 distribution converts into N Dist

study time

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with the help of chi square test, we can solve following type of question

= Type - 1 Justification of population variance $\rightarrow$

$$H_0: \sigma^2 = \sigma_0^2$$

$$H_1: \sigma^2 > \sigma_0^2$$

$$= \text{Sample variance } (s^2) = \frac{\sum (x_i - \bar{x})^2}{n}$$

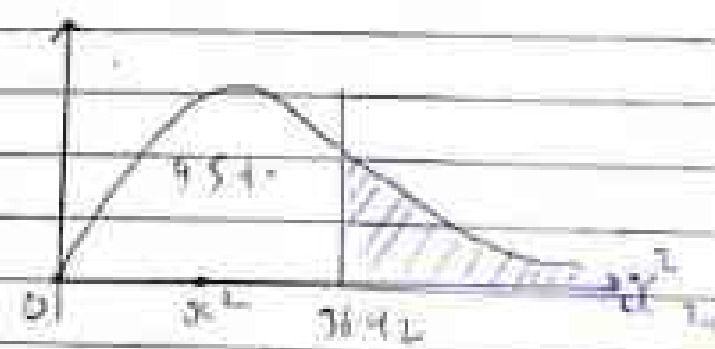
$\therefore$ Sample size is not very small ($n > 50$)

$$\chi^2 = \frac{\sum (x_i - \bar{x})^2}{s^2} = \boxed{\frac{ns^2}{\sigma^2}}$$

if value of χ^2 statistic lies in Accept H_0
Accept H_0

Q: A random sample of size 25 has sample standard deviation = 9. Test the hypothesis that population S.D is 10.5
Given $\chi^2_{24}(0.05) = 36.42$

sol: $n = 25$, $s = 9$, $H_0: \sigma = 10.5$, $H_1: \sigma \neq 10.5$
 $D.F = 25 - 1 = 24$, $\chi^2_{24}(5\%) = 36.42$



$$\chi^2 = \frac{ns^2}{\sigma^2} = \frac{25 \times 9^2}{(10.5)^2} = 19.30$$

value of χ^2 is in acceptance region $\therefore$ we accept H_0

Q) A random sample of size 20 has variance 25. if the population standard deviation is 8 calculate χ^2

(ii) Also calculate the hypothesis that $\sigma = 8$ for 95% confidence region given $\chi^2_{19}(0.05) = 30.19$

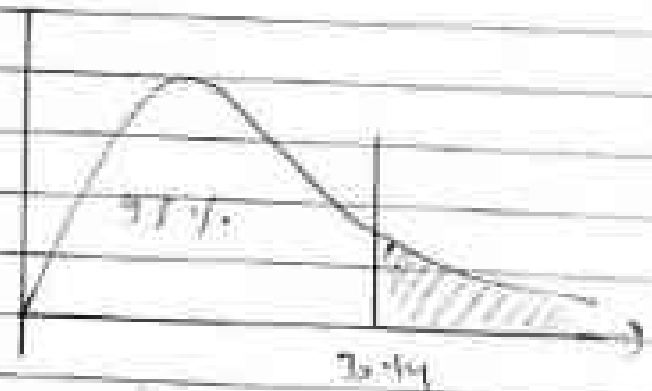
sol:) $n = 20, S^2 = 25, \sigma = 8$

$$\chi^2 = \frac{nS^2}{\sigma^2} = \frac{20 \times 25}{8^2} = 78.125$$

(ii) $df = 20 - 1$
 $= 19$

Ho: $\sigma = 8$

$H_1: \sigma \neq 8$



$\therefore$ value of χ^2 lies

in 5% C. Region $\therefore$ accept Ho & reject H_1

=) Special note:

Sample value = Experimental value: Exact value
= observed value (O)

population value = Theoretical value = Expected value (E)

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⇒

Type-2:

Significance of goodness of fit =

H_0 : There is no difference b/w O_i & E_i i.e.
 $O_i = E_i$

H_1 : There is significant ^{difference} b/w O_i & E_i i.e.

$$O_i \neq E_i$$

Here we have one condition i.e. $\sum O_i = \sum E_i$

So $D.F. (v) = n - 1$

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Q:) The following table shows the number of car accidents that occurred during the week. Test whether the accidents are uniformly distributed over the week.

Given $\chi^2_{0.05}(2.05) = 12.592$

Day	Mon	Tue	Wed	Thurs	Fri	Sat	Sun	Total
No. of accidents	14	18	12	11	15	19	14	93
								$\sum O_i$

sol.) Let accidents are uniformly distributed over the week.

& For χ^2 distribution we know that, $\sum E_i = \sum O_i = 93$

$$\text{So } E_1 = E_2 = E_3 = E_4 = E_5 = E_6 = E_7 = \frac{93}{7} = 13.2857$$

H_0 : There is no significant difference b/w O_i & E_i i.e. $O_i = E_i$

i.e. accident are uniformly distributed over the week.

H_1 : There is significant b/w O_i & E_i i.e. $O_i \neq E_i$

i.e. accident are not uniformly distributed over the week.

Days	O_i	E_i	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
M	14	14	0	0/14
T	18	14	16	16/14
W	12	14	4	4/14
Th	11	14	9	9/14
F	15	14	1	1/14
S	14	14	0	0/14
S	14	14	0	0/14

$$\text{i.e. } \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{0 + 16 + 4 + 9 + 1 + 0 + 0}{14} = \frac{30}{14}$$

$$= 2.14$$

$$D.F. (v) = n - 1 = 7 - 1 = 6$$



χ^2 falls in Acceptance region so Accept H_0 & Reject H_1 , so accident are $\propto$ over the week.

study time



Q:) A die is thrown 276 times and frequencies for various outcome are $f(1) = 40$, $f(2) = 32$, $f(3) = 29$, $f(4) = 59$, $f(5) = 57$, $f(6) = 59$. Test whether die is biased or not
it is given that $\chi^2_{(0.05)} = 11.09$

sol:) let us assume that die is unbiased then
 $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$
∴ Expected frequency of various
the outcome are as follows:

$$f(1) = \frac{276}{6}, f(2) = \frac{276}{6} = f(3) = f(4) = f(5) = f(6) = 46$$

and observed frequency are given as

$$\begin{matrix} f(1) & f(2) & f(3) & f(4) & f(5) & f(6) \\ 40 & 32 & 29 & 59 & 57 & 59 \end{matrix}$$

Outcome	O_i	E_i	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
1	40	46	36	36/46
2	32	46	196	196/46
3	29	46	289	289/46
4	59	46	169	169/46
5	57	46	121	121/46
6	59	46	169	169/46

$$276 \quad 276$$

$$\chi^2_{(n)} = 6 - 1 = 5$$

$$\begin{aligned} \chi^2 &= \frac{36 + 196 + 289 + 169 + 121 + 169}{46} \\ &= 21.70 \end{aligned}$$

study time

PRACTISE NOW 10

H_0 : There is no difference b/w O_i & E_i
i.e. $O_i = E_i$

OR

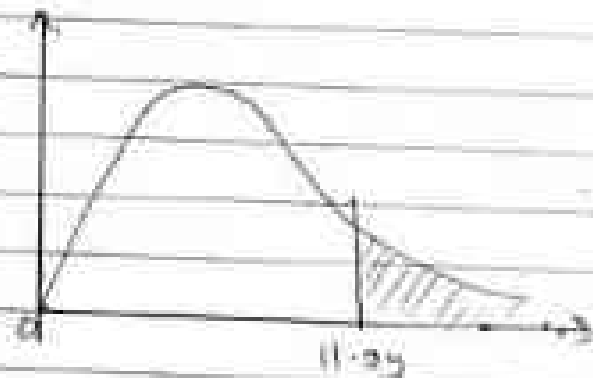
" Die is unbiased.

H_1 : There is significant difference b/w O_i & E_i
i.e. $O_i \neq E_i$

OR

" Die is biased.

$$\therefore \chi^2_{(0.05/5-1)} = 11.05$$



" χ^2 lies in rejection region i.e. reject H_0 & Accept H_1 .

Difference b/w O_i & E_i is highly significant. Hence die is biased.

Q.10. A car manufacture company produces four different color cars in which 2881 are white, 313 Grey, 287 Red and 112 Black. while at the beginning they had decided that they will produce white, grey, red & Black cars in the ratio 9:3:3:1 respectively. if $\chi^2_{(0.05)} = 7.815$ establish the correspondence b/w Theory and experiment. Does the experimental result support the theory?

⇒ Type - 3:

Significance of Independence of attributes

Ho: There is no relationship b/w two attributes i.e. both are Independent.

H1: There is a relationship b/w two attributes i.e. both are dependent.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Now information will be given in the form of matrix of order $R \times C$ then Degrees of freedom (V) = (R-1)(C-1)

Where

$$E_i = \frac{R_i \times C_j}{\text{Total}}$$

Row total $\times$ Col total

A:) What are the expected frequency of 2×2 contingency table given table

(i)

a	b	= a+b
c	d	= c+d

$$T = a+b+c+d$$

$$E(a) = \frac{(a+b)(a+c)}{a+b+c+d}$$

$$E(c) = \frac{(a+c)(a+b)}{a+b+c+d}$$

$$E(b) = \frac{(a+b)(b+d)}{a+b+c+d}$$

$$E(d) = \frac{(c+d)(b+d)}{a+b+c+d}$$

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(ii)

2	10	12
6	6	12
1	1	2

$$E(1) = \frac{12 \times 8}{24}$$

$$= DF = (r-1)(c-1)$$

$$E(10) = \frac{12 \times 16}{24}$$

$$= \frac{(1-1)(2-1)}{1} = 0$$

$$E(6) = \frac{12 \times 8}{24}$$

$$E(6) = \frac{12 \times 16}{24}$$

It is above table

Only one non value has freedom to vary and

Q) In an engineering college of Delhi, the I.O and economic condition of home of 1000 students are given in table.

I.O \ E (economic)	High	Low	Total
Rich	100	300	400
Poor	350	250	600
Total	450	550	1000

It is 2x2 contingency table

can we conclude any association b/w economic condition of home and I.O of the student.

$$\text{Here } \chi^2_{(0.05)} = 3.84$$

Ho: There is no association b/w E & I.O i.e. both are independent

study time

Hypothesis Testing

H_1 : There is association b/w E & A i.e. both are dependent

$$D.F. (v) = (2-1)(2-1) = 1 \quad \& \quad \chi^2_1 = 3.84$$

$$E(100) = \frac{400 \times 450}{1000} = 180$$

$$E(300) = \frac{400 \times 550}{1000} = 220$$

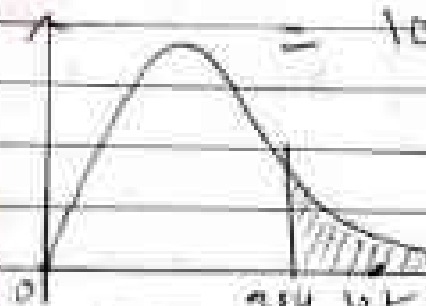
$$E(350) = \frac{600 \times 450}{1000} = 270$$

$$E(250) = \frac{600 \times 550}{1000} = 330$$

Info	O_i	E_i	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
High IQ	100	180	6400	6400/180
Low IQ	300	220	1400	1400/220
Rich	350	270	6400	6400/270
Poor	250	330	6400	6400/330

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$= 107.7$$



i.e. Reject H_0 & accept H_1 .

$\therefore$ There is an association b/w economic condition & IQ.

Q.) From the following data, find out whether there is any relationship b/w sex and preference of color:

Color \ Sex	Males	Females	Total
Red	10	40	50
White	70	30	100
Green	30	20	50
Total	110	90	200

it is 3×2 contingency table ^{Of} differences = $(3-1)(2-1)$
and $\chi^2 = 5.991$

sol.) H_0 : There is no relationship b/w gender and preference of colour
i.e. there are independent

H_1 : There is a relationship b/w gender and preference of colour
i.e. there are dependent

$$E(50) = \frac{50 \times 110}{200} = 27.5$$

$$E(40) = \frac{50 \times 90}{200} = 22.5$$

$$E(70) = \frac{100 \times 110}{200} = 55$$

$$E(30) = \frac{100 \times 90}{200} = 45$$

$$E(20) = \frac{50 \times 110}{200} = 27.5$$

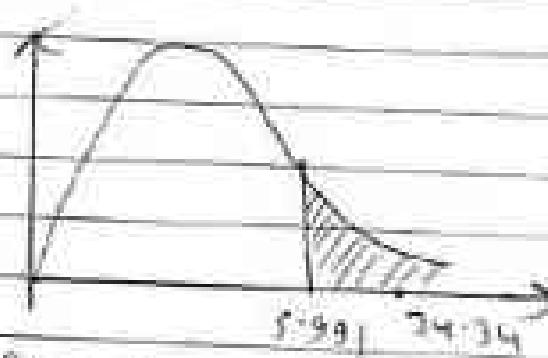
$$E(10) = \frac{50 \times 90}{200} = 22.5$$

study done

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Info	O_i	E_i	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
M & R	10	21.5		
F & R	40	21.5		
M & U	70	55		
F & U	30	45		
M & G	30	21.5		
F & G	20	21.5		

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 34.34$$



$\therefore$ value of χ^2 lies in critical region so reject H_0 & Accept H_1

There is relationship b/w gender & preference of colour.

Q. In a survey of 200 boys of which 75 were intelligent and remaining were unintelligent out of intelligent boys 40 had educated father while out of — unintelligent boys 85 had uneducated father. Can we conclude that educated father have intelligent boys given that $\chi^2_{(0.05)} = 3.841$.

(Table is not given)

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Sol:)

Boys Education of	Intelligent	Unintelligent	Total
Educated Father	40	40	80
Uneducated Father	35	95	130
	75	135	210

• Table is 2×2

$$D.F. (v) = (R-1)(C-1) \\ = (2-1)(2-1) = 1$$

• H_0 : There is No relationship b/w education of father & Intelligent of boys.

i.e.

both are Independent.

• H_1 : There is a relationship b/w education of father and intelligent of boys

i.e.

both are Dependent.

$$E(40) = \frac{80 \times 75}{210} = 30$$

$$E(40) = \frac{80 \times 135}{210} = 50$$

$$E(35) = \frac{130 \times 75}{210} = 45$$

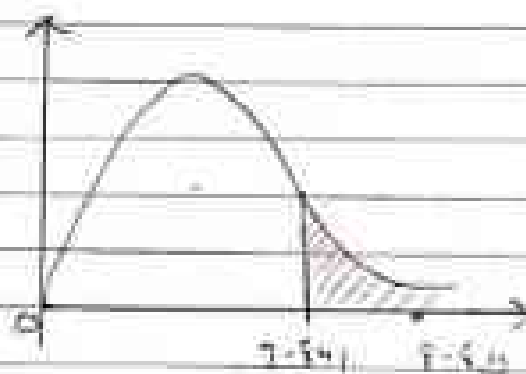
$$E(95) = \frac{130 \times 135}{210} = 80$$

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Info	O_i	E_i	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
100% F	40	70	100	1.43
75% F	40	50	100	2.00
50% F	35	45	100	2.22
25% F	85	75	100	1.33

$$\chi^2 = 9.99$$



i.e. Reject H_0 & accept H_1
or

educated further have intelligent boys.

⇒ Degree of freedom

Using χ^2 distribution, D.F. for various d.f.

1) Binomial Dist. D.F. = $(n-1)$ [$\{E_i = E_i\}$]

2) Poisson Dist. D.F. = $(n-1)$ [$\{E_i = E_i\}$]

3) Normal Dist. D.F. = $(n-1)$ [$\{E_i = E_i \& \mu, \sigma\}$]

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analysis

Chapter 10

Q: The number of screws declared fit or unfit by three inspectors X, Y and Z is shown in the following table.

Inspector	X	Y	Z	Total
Fit Screw	50	47	56	153
Unfit Screw	5	14	8	27
Total	55	61	64	180

$$\chi^2 = 4.945$$

Given that $\chi^2 = 0.05 = 3.84$ and $\chi^2_{(1, 95)} = 3.84$
 test the hypothesis that proportion of screws declare unfit by three inspectors are same.

Q: H_0 : There is no difference b/w proportion of unfit screws declared

H_1 : There is a significant diff b/w ---

$$DF = (2-1)(3-1) = 1 \times 2 = 2$$

Info	O_i	E_i	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
	50	46.75	10.5625	0.2259
Fit	47	51.85	4.8525	0.09353
Unfit	56	54.4	1.61	0.02941
X	5	8.25	3.251	0.3939
Y	14	9.15	4.851	0.5300
Z	8	9.6	1.61	0.1666

$$E_i = \frac{(50 + 47 + 56)(50 + 5)}{180} = \frac{153 \times 55}{180} = \frac{8415}{180} = 46.75$$

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$$E_2(47) = \frac{153 \times 61}{180} = 51.25$$

$$E_3(56) = \frac{153 \times 64}{180} = \frac{9792}{180} = 54.4$$

$$E_4(5) = \frac{37 \times 55}{180} = 11.25$$

$$E_5(10) = \frac{27 \times 61}{180} = 9.15$$

$$E_6(8) = \frac{24 \times 64}{180} = 8.6$$

$$X^L = \frac{(0 \dots 0 \dots)^L}{E_i} = 4.25 \text{ M}$$

Date _____

Linear Algebra - 01

01/10/2025

Lecture no: 01

Determinants:

(Represent any Number)

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = (a)(d) - (b)(c)$$

eg $|A| = \begin{vmatrix} 2 & -3 \\ 4 & -1 \end{vmatrix} = (-2) - (-12) = 10$

o.g $|A| = \begin{vmatrix} 3+4i & -i \\ i & 3-4i \end{vmatrix} = 3^2 - (4i)^2 - (-i^4) = 9 + 16 - 1 = 24$

=> Sign Convention: $\begin{vmatrix} + & - \\ - & + \end{vmatrix}, \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}, \begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix}$

sign of $(i,5) = (-1)^{i+j}$

eg Sign of $a_{23} = (-1)^{2+3} = -ve$, sign

Sign of $a_{42} = (-1)^{4+2} = +ve$

eg $\begin{vmatrix} 1 & 3 & 6 & 2 \\ 1 & 5 & 0 & 8 \\ 1 & 2 & 4 & \end{vmatrix} = - (1) \begin{vmatrix} 3 & 2 \\ 2 & 4 \end{vmatrix} + 5 \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} - (2) \begin{vmatrix} 1 & 3 \\ 1 & 2 \end{vmatrix}$

$$= - (12 - 4) + 5 (4 - 2) - 2$$

$$= -12 + 4 + 20 - 2$$

$$= 10$$

$$= 14 \text{ Ans.}$$

Spred

⇒ Sarrus (Sarrus Method)



$$|A| = (2 \times 0 + (-4) \times (-1) + (-1) \times 0 \times 2) - ((-1) \times 0 \times 2 + 2 \times (-1) \times 0 + (-4) \times 1 \times 2)$$

$$= 16 - 2 = 14$$

e.g.)

1	2	3
2	1	-2
1	-4	1

$$|A| = (1 \times (-4) + (-14) + (-2 \times 1 \times 2)) - (2 \times 1 \times 1 + 1 \times 2 \times 2 + (-4) \times 1 \times 2)$$

$$= -42$$

e.g.)

1	5	25
1	4	16
1	2	4

$$|A| = (5 \times 4) - (4 \times 2) - (2 \times 5)$$

$$= -6$$

⇒ We know that in class 12th to find det

1.)

1	a	a ²
1	b	b ²
1	c	c ²

$$= (a-b)(b-c)(c-a)$$

2.)

1	a	a ²
1	b	b ²
1	c	c ²

$$= (a-b)(b-c)(c-a)(a+b+c)$$

⇒ Det of order 4x4

e.g.

0	1	2	3
1	0	3	0
2	3	0	1
3	0	1	2

$$|A| = - (1 \times 3 \times 1 \times 1) + (2 \times 0 \times 1 \times 2) + (3 \times 0 \times 1 \times 2) + (0 \times 1 \times 2 \times 3) = -1$$

$$= -1$$

0	1	2	3
1	0	3	0
2	3	0	1
3	0	1	2

Del

$$= - \left[-(-4) \{4-1\} + 0-1 \{1-0\} + 0-3 \{0-1 \{4-3\} \} - 3 \right]$$

$$= - \left[-2-1 \right] - 3 \left[-1-27 \right] = 88$$

$\Rightarrow$ Shortcut (Elementary operation)

E- Row operation

E- Column operation

- | | |
|-----------------------------------|------------------------------------|
| i) $R_i \leftrightarrow R_j$ | ii) $C_i \leftrightarrow C_j$ |
| ii) $R_i \rightarrow KR_i$ | iii) $C_i \rightarrow KC_i$ |
| iii) $R_i \rightarrow R_i + KR_j$ | iiii) $C_i \rightarrow C_i + KC_j$ |

(Where K is any constant or any scalar)

$$R_2 \rightarrow R_2 - 5R_1 \quad (1^{st} \text{ type E-operation})$$

$$R_2 \rightarrow 5R_2 - R_1 \quad (\text{Not E-operation})$$

Note: (also valid for column)

- 1) if we interchange any two rows in a matrix value of its determinant change by -1 times
 - 2) if we multiply any row by constant K then value of det. is also by K times
 - 3) Only 1st E-operation does not affect the value of Determinant
- If we are free to apply 1st E-operation in a matrix while solving the det.

Sand

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* In a single step only use either Row operation or column operation at a time not both.

9:) The det value of the matrix

$$\begin{vmatrix} 13 & 2 & 1 & 3 \\ 31 & 4 & 5 & 6 \\ 26 & 3 & 7 & 4 \\ 10 & 1 & 3 & 2 \end{vmatrix} \text{ is } \underline{\hspace{2cm}}$$

Sol:) $C_2 \rightarrow C_2 - C_4$

$$\begin{vmatrix} 13 & -1 & 1 & 3 \\ 31 & -2 & 5 & 6 \\ 26 & -1 & 7 & 4 \\ 10 & -1 & 3 & 2 \end{vmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\begin{vmatrix} 13 & -1 & 1 & 3 \\ 5 & 0 & 3 & 0 \\ 13 & 0 & 6 & 1 \\ -3 & 0 & 2 & -1 \end{vmatrix}$$

$$-(-1) \begin{vmatrix} 5 & 3 & 0 \\ 13 & 6 & 1 \\ -3 & 0 & -1 \end{vmatrix} = 5(-6-4) - 3(-17+7) + 0$$

$$= -40 + 30 = -10$$

eg

$$|A| = \begin{vmatrix} 11 & 12 & 13 & 14 \\ 21 & 22 & 23 & 24 \\ 31 & 32 & 33 & 34 \\ 41 & 42 & 43 & 44 \end{vmatrix}$$

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$$\text{sol:)} \quad R_4 \rightarrow R_4 - R_3, \quad R_3 \rightarrow R_3 - R_2$$

$$\begin{vmatrix} 11 & 12 & 13 & 14 \\ 21 & 22 & 23 & 24 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \end{vmatrix}$$

$$R_4 \rightarrow R_4 - R_3$$

$$\begin{vmatrix} 11 & 12 & 13 & 14 \\ 21 & 22 & 23 & 24 \\ 10 & 10 & 10 & 10 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

$$= 0 \quad \underline{\underline{A}}$$

Properties of Determinant: (Both suitable)

1) if in a matrix, Any two rows (or any two columns) are identical then its $\text{Det} = 0$

2) if in a mat. All the elements in any Row - are all zero $= 0$ (then its value is)

3) $|A \ B \ C| = |A| \cdot |B| \cdot |C|$

4) $|A+B| \neq |A| + |B|$ $|A+B| \neq |A| + |B|$ $|A+B| \neq |A| + |B|$

5) $|A^T| = |A|$, $m \times n$

6) $|A^T| = |A|$

7) $|A^{-1}| = \frac{1}{|A|}$

8) $|A^{-1}| = \frac{1}{|A|}$ $\therefore \det(A^{-1}) = \frac{1}{\det A}$ Solved

where $A^{-1} = \frac{1}{|A|} \text{adj} A$ Date:

Q:) if $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ then $|A^3| = ?$

Sol:) $(3)(3) - (2)(2) = 5$

$$|A^3| = |A|^3 = 5^3 = 125$$

Q:) if $A = \begin{bmatrix} x & 2 \\ 2 & x \end{bmatrix}$ s.t. $|A^3| = 125$

So $x = ?$

a) 3 ✓

b) -3

c) 0

d) None

$$|A^3| = 125$$

$$|A|^3 = 125$$

$$|A| = 5$$

$$x^2 - 4 = 5$$

$$x^2 = 9$$

$$x = \pm 3$$

Q:) If $A = \begin{bmatrix} 2 & a & -b & 0 \\ a & 1 & -4 & 2 \\ 3 & 1 & 4 & -1 \\ 0 & 2 & 1 & 4 \end{bmatrix}$ then $\det |A^T A^{-1}| = ?$

Sol:) $|A^T| \cdot |A^{-1}|$

$$= |A| \cdot \frac{1}{|A|}$$

$$= 1$$

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Q: if A and B are matrices of det 1 then

(a) Det of $A+B$ is 2

(b) Det of $A+B$ is 1

(c) Det of $A+B$ is 0

(d) Nothing can be concluded about the determinant of $A+B$

Ans: $|A| = 1, |B| = 1$

Now we know that

$$|A+B| \leq |A| + |B|$$

$$\leq 1 + 1$$

$$\leq 2 \quad \text{i.e. } |A+B| = 2 \text{ or } |A+B|$$

Q(9). Max number of terms in the general expansion of $|A|_{n \times n} = n!$

(a) n

(b) n^2

(c) $2n$

(d) $n!$ ✓

* Total no. of terms in matrix $|A|_{n \times n} = n!$

(10). Area of Δ formed by $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is =

$$\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \rightarrow \text{det}$$

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e.g. Area of Δ formed by $(2, 7)$ $(3, 6)$ $(4, 5)$ will be

$$\frac{1}{2} \begin{vmatrix} 2 & 7 & 1 \\ 3 & 6 & 1 \\ 4 & 5 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 7/2 & 1/2 \\ 3 & 3 & 1/2 \\ 4 & 5 & 1/2 \end{vmatrix}$$

$$= \dots \dots \dots 0$$

i.e. these points are collinear.

e.g. Area of Δ formed by $(1, 0)$ $(2, 2)$ $(4, 2)$

$$= \frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ 2 & 2 & 1 \\ 4 & 2 & 1 \end{vmatrix} = \begin{vmatrix} 1/2 & 0 & 1/2 \\ 1 & 1 & 1/2 \\ 2 & 3/2 & 1/2 \end{vmatrix}$$

$$= -\frac{3}{2}$$

$\therefore$ area can't be negative so Ans = 1.5

$$(11.) \quad \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} a' & b' & c' \\ d' & e' & f' \\ g' & h' & i' \end{vmatrix} + \begin{vmatrix} a & b & c \\ d & e' & f' \\ g & h & i \end{vmatrix} + \begin{vmatrix} a & b & c \\ d & e & f \\ g & h' & i' \end{vmatrix}$$

$$(12.) \quad \begin{vmatrix} a+l & b & c \\ d+m & e & f \\ g+n & h & i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} l & b & c \\ m & e & f \\ n & h & i \end{vmatrix}$$

$$\begin{vmatrix} a+l & d+m & g+n \\ c & e & h \\ f & h & i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} l & m & n \\ c & e & f \\ g & h & i \end{vmatrix}$$

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$$\text{eg.) } \begin{vmatrix} 1 & -3 & 0 \\ 2 & -1 & 2 \\ 2 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 1 & -3 & 0 \\ 2 & -1 & 2 \\ 2 & 0 & 3 \end{vmatrix} + \begin{vmatrix} 1 & -3 & 0 \\ 3 & -1 & 2 \\ 2 & 0 & 3 \end{vmatrix}$$

$$= - - - - - = -17 \quad \text{✓}$$

Lecture no. 02

Q.) Which one of the following does not equal

$$\begin{vmatrix} 1 & n & n^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$$(a) \begin{vmatrix} 1 & n(n+1) & n+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$$

$$(b) \begin{vmatrix} 1 & n+1 & n^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$$

$$(c) \begin{vmatrix} 0 & n-y & n^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$$

$$(d) \begin{vmatrix} 2 & n+y & n^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$$

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Q:) if $\Delta = \begin{vmatrix} n & n^2 & n^3 \\ 1 & 2 & 3n \\ 0 & 2 & 5n \end{vmatrix}$ then $\frac{d\Delta}{dn}$

Sol:) $\frac{d\Delta}{dn} = \begin{vmatrix} 1 & 2n & 3n^2 \\ 1 & 2 & 3n \\ 0 & 2 & 5n \end{vmatrix} - \begin{vmatrix} n & n^2 & n^3 \\ 0 & 0 & 3 \\ 0 & 2 & 5n \end{vmatrix} + \begin{vmatrix} n & n^2 & n^3 \\ 1 & 2 & 3n \\ 0 & 0 & 5 \end{vmatrix}$

$$= 1(10n - 6n) - 1(10n^2 - 6n^2) + 0($$

$$+ +(-3)(2n) + 5(2n - n^2)$$

$$= 4n - 4n^2 + 6n + 10n - 5n^2$$

$$= -9n^2 + 8n$$

Q:) if $f(n) = \begin{vmatrix} n & n^2 & n^3 \\ 1 & 2 & 3n \\ 0 & 2 & 5n \end{vmatrix}$ s.t. $f'(n) = -1$ then $n = ?$

Sol:) $f'(n) = -9n^2 + 8n$

a) 1 ✓

b) -1

c) $-1/9$ ✓

d) 0

$$-9n^2 + 8n = -1$$

$$9n^2 - 8n = +1$$

$$9n^2 - 8n - 1 = 0$$

$$(9n+1)(n-1) = 0$$

$n \in \mathbb{R}$ $n = -\frac{1}{9}$ or

$\Rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \frac{-(a+b+c)}{2} [a^2 + b^2 + c^2 - ab - bc - ca]$

or

$$= \frac{-(a^3 + b^3 + c^3 - 3abc)}{2}$$

or

$$= \frac{1}{2} (a+b+c) [(a-b)^2 + (b-c)^2 + (c-a)^2]$$

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1.3) Special property of Det:

$$\text{let } A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & 1 \\ 1 & 2 & -1 \end{bmatrix} \quad 4 \quad O = \begin{bmatrix} +8 & -4 & +12 \\ 16 & 0 & 4 \\ 4 & 8 & -4 \end{bmatrix} = 4A$$

$$|A| = 15, \quad |O| = 960 = 4^3 \times 15 = (4 \times 15) = 4^3 |A|$$

m-2

$$|O| = \begin{vmatrix} 8 & -4 & 12 \\ 16 & 0 & 4 \\ 4 & 8 & -4 \end{vmatrix} = 4 \begin{vmatrix} 2 & -1 & 3 \\ 4 & 0 & 1 \\ 1 & 2 & -1 \end{vmatrix} = 4^3 |A|$$

i.e. in general if $A_{n \times n}$ & k is any scalar then
 $[kA] = k \cdot [A]$ in matrix
 $|kA| = k^n |A|$ in determinant.

e.g. if $|A|_{3 \times 3} = 6$ then

$$(i) |2A| = ? \quad \frac{1}{|2A|} = \frac{1}{2^3 \times 6} = \frac{1}{48}$$

$$(ii) |2A^{-1}| = ? \\ = 2^3 |A^{-1}| = 2^3 \cdot \frac{1}{|A|} = \frac{8^3}{6} = \frac{4}{3}$$

e.g. if $|A|_{3 \times 3} = -1$, $|O|_{3 \times 3} = 4$ then $|5AO| = ?$

$$\begin{aligned} \therefore a) & 0 & \because A_{3 \times 3} \text{ & } O_{3 \times 3} & = (AO)_{3 \times 3} \\ b) & -4 & \therefore & & \\ c) & -20 & & & \\ d) & -500 & & & \end{aligned}$$

$$|5AO| = 5^3 |AO| \\ = 125 |A| |O| \\ = 125 (-1) (4) = -500$$

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14.) If A is $n \times n$ then evaluate $|A \cdot \text{adj} A|$, $|\text{adj} A|$, $|\text{cof} A|$,
 $|A^{-1}| = ?$

Sol: We have already seen that

$$(i) |A \cdot \text{adj} A| = |A|^n$$

$$(ii) |\text{adj} A| = |A| \cdot |\text{adj} A| = |A|^n$$

$$|\text{adj} A| = \frac{|A|^n}{|A|} = |A|^{(n-1)}$$

$$(iii) |\text{cof} A| = |(\text{cof} A)^T| = |\text{adj} A| = |A|^{(n-1)}$$

We know that

Det of any M^{-1} = Det of it's transpose

$$(iv) A^{-1} = \frac{\text{adj} A}{|A|} = \frac{\text{Matrix}}{\text{number}}$$

$$|A^{-1}| = \left| \frac{\text{adj} A}{|A|} \right| = \frac{1}{|A|^n} |\text{adj} A|$$

$$= \frac{|A|^{n-1}}{|A|^n} = \frac{1}{|A|}$$

$$\therefore \frac{\text{Det}(A^{-1})}{\text{Det}(A)} = 1$$

15.) If A is $n \times n$ then $|\underbrace{\text{adj} \text{adj} \text{adj} \dots \text{adj} A}_{n \text{ times}}| = |A|$

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e.g: $|adj A| = |A|^{n-1}$, $|adj adj A| = |A|^{(n-1)^2}$

Q: if $|A|, g = 5$ then $|adj adj adj A| = ?$

Sol: a) 5^7 Here $n=4, r=3$
 b) 5^9
 c) 5^{14}
 d) 0
 $|A|^{(n-1)^3} = 5^{(4-1)^3} = 5^3$

Note: $(a^b)^c = a^{bc}$
 $(a^b)^c = a^{b \times c}$

MATRIX

Matrix is the rectangular arrangement of numbers

$$A = [a_{ij}]_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} \begin{matrix} 1 \leq i \leq m \\ 1 \leq j \leq n \\ \text{m. line} = \text{row} \\ \text{n. line} = \text{col} \end{matrix}$$

Q(1) Square mat: if no. of rows = no. of columns.

Q(2) if in a matrix diagonal exist then it must be square matrix.

$$A = [a_{ij}]_{n \times n} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

Q(3) if in a matrix corresponding element exist for every element

Dnc

- 1) for diag elements, $i = j$ $\forall i, j$
- 2) for upper diag elements, $i < j$ $\forall i, j$
- 3) for lower diag elements, $i > j$ $\forall i, j$
- 4) for off diag elements, $i \neq j$ $\forall i, j$
- 5) Corresponding elements are a_{ij} & a_{ji}

$\Rightarrow$ Trace (A) = Sum of diag elements

$$\text{Tr}(A) = \sum_{i=1}^n (a_{ii})$$

Some special matrices:

$$\begin{bmatrix} 2 & -4 & 0 & 3 \\ 0 & -1 & 2 & 1 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix} = \text{UTM} \quad \text{Tr} = 8$$

$$\text{Det} = -24$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 4 & 0 & 0 \\ 3 & 2 & 0 & 0 \\ -1 & 2 & 1 & 2 \end{bmatrix} = \text{UTM} \quad \text{Tr} = 7$$

$$\text{Det} = 0$$

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} = \text{Diag mat} \quad \text{Tr} = 4$$

$$\text{Det} = 0$$

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$$\begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

= Scalar mat

$$\text{Tr} = 20$$

$$\text{Det} = 625$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

= Identity mat/Unit mat. $\text{Tr} = 4$

$$\text{Det} = 1$$

* To find Det of U.T.M., L.T.M., Diag mat, Scalar mat & Identity mat we can multiply diagonal elements i.e.

Det = Product of Diag elements

* if $A = [a_{ij}]_{n \times n}$ s.t. $a_{ij} = 0 \forall i > j$ then A is U.T.M.
 (Lower diagonal elem = 0)

Q:) Evaluate $|A| =$ $\begin{bmatrix} 0 & 1 & 2 & 0 \\ -1 & 0 & 4 & 0 \\ -2 & -4 & 0 & 6 \\ 0 & 0 & -6 & 0 \end{bmatrix} = ?$

a.) 0

b.) 4

c.) 16

d.) 36

M-J $|A| = 0 \times 0 \times 0 \times 0 = 0$ (Sad face) X

$\therefore$ A is neither U.T.M., nor L.T.M. & not a diag element so we can not use the shortcut method defined in previous slide.

So we will find $|A|$ by conventional approach
 Spud

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M-2 Expanding along C_1

$$|A| = -6 \begin{vmatrix} 0 & 1 & 2 \\ 0 & 0 & 4 \\ 0 & 0 & -6 \end{vmatrix} = -6 [(-6)(0-1)]$$

$$= 36 \underline{A}$$

Lecture no. 03

$\Rightarrow$ Singular Matrix:

if $|A| = 0$ Then A is called singular mat.

$\Rightarrow$ Non-Singular Matrix:

if $|A| \neq 0$ then A is called non-singular mat.

$\Rightarrow$ Invertible matrix:

if A^{-1} exist then A is called Invertible Mat.

Def:

$$A^{-1} = \frac{\text{adj } A}{|A|} = \frac{(C_{ij} A)^T}{|A|}$$

A necessary condition for a mat to be invertible is $|A| \neq 0$.

* if A & B are two matrices s.t. $\boxed{AB = BA = I}$

Inverse of $A = B$ i.e. $A^{-1} = B$

Inverse of $B = A$ i.e. $B^{-1} = A$

i.e.
Both are inverses of each other.

Eg.) if M is a 2x2 mat $MY = I$ then

$$M^{-1} = Y$$

$$(M^{-1})^{-1} = I$$

$$(M^{-1})^{-1} = I$$

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Ex: 1) ① $m^4 = I$
 $m \cdot m^3 = I \Rightarrow m^{-1} = m^3$

② $m^4 = I$
 $m^2 \cdot m^2 = I \Rightarrow (m^2)^{-1} = m^2$

③ $m^4 = I$
 $m^3 \cdot m = I \Rightarrow (m^3)^{-1} = m$

* Q: if M is a matrix s.t. $M \neq I$, $M^2 \neq I$, $M^3 \neq I$, But $M^4 = I$
 then $M^{-1} = ?$

a.) m^{4k+1}

b.) m^{4k+2}

c.) m^{4k+3}

d.) m^{4k} $k \in \mathbb{N}$

Given $m^4 = I$

$m^8 = m^4 \cdot m^4 = I \cdot I = I^2 = I$

$m^{12} = m^4 \cdot m^4 \cdot m^4 = I \cdot I \cdot I = I^3 = I$

similarly

$m^6 = I$, $m^{10} = I$, ..., $m^{4k} = I$

$m^4 = I$

$m \cdot m^3 = I \Rightarrow m^{-1} = m^3 = m^3 \cdot I = m^3 \cdot m^4$
 $= m^{4k+3}$ $\square$

Q: if A is a matrix s.t. $A^2 = I$ then $A^{-1} = ?$

a.) I

b.) A

c.) A^2

d.) O

$A^2 = I$

$A \cdot A = I$

— (1)

$\Rightarrow \boxed{A^{-1} = A}$

or $A A^{-1} = I$ — (2)

i.e.

A is inverse of itself.

A is self invertible.

Spad

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* Shortest method of finding Inverse of 2×2 matrix

$$\text{if } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ then } A^{-1} = \frac{ad - bc}{|A|}$$

$$= \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Q.9:) $A = \begin{bmatrix} (3+2i) & -i \\ i & (3-2i) \end{bmatrix}$ then $A^{-1} = ?$

sol:) $A^{-1} = \frac{ad - bc}{|A|}$

$$= \frac{1}{12} \begin{bmatrix} (3-2i) & i \\ -i & (3+2i) \end{bmatrix}$$

Q:) if $A = \begin{bmatrix} 2 & -0.1 \\ 0 & 3 \end{bmatrix}$ & $A^{-1} = \begin{bmatrix} 0.5 & a \\ 0 & b \end{bmatrix}$ then

$a+b = ?$

sol:) $|A| = 6$ i.e. A^{-1} exist

$$A^{-1} = \frac{1}{6} \begin{bmatrix} 3 & 0.1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0.5 & 1/60 \\ 0 & 1/3 \end{bmatrix}$$

$a = \frac{1}{60}$, $b = \frac{1}{3}$ so $a+b = \frac{7}{20}$ $\underline{D}$

Date

* Shortcut method of finding Inverse of 3×3 mat

e.g. $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 0 & 4 \\ -3 & 1 & 2 \end{bmatrix}$ then $A^{-1} = ?$

3×3 $= \frac{\text{adj } A}{|A|} = \frac{(\text{cof } A)^T}{|A|}$

sol: $|A| = 34$ ($\neq 0$)

So A is non-singular $\Rightarrow A^{-1}$ exist

$\begin{array}{c|cccc} 1 & -2 & 3 & 1 & -2 \\ 2 & 0 & 4 & 2 & 0 \\ -3 & 1 & 2 & -3 & 1 \\ 1 & -2 & 3 & 1 & -2 \\ 2 & 0 & 4 & 2 & 0 \end{array}$

$= \frac{1}{34} \begin{bmatrix} -4 & 7 & -8 \\ -16 & 11 & 2 \\ 2 & 5 & 4 \end{bmatrix}$

$\Rightarrow$ Top row of $A^{-1} = \begin{bmatrix} -4 & 7 & -8 \end{bmatrix}$ X (wrong)

$= \begin{bmatrix} -4/34 & 7/34 & -8/34 \end{bmatrix}$ $\checkmark$ (correct)

$\Rightarrow \text{cof } A = (\text{adj } A)^T = \begin{bmatrix} -4 & -16 & 2 \\ 7 & 11 & 5 \\ -8 & 2 & 4 \end{bmatrix}$

Q:) if $A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & -1 \\ 2 & 3 & 2 \end{bmatrix}$ Then top row of A^{-1} is

a.) 5, 6, 4 $|A| = 1 \neq 0$ A^{-1} exist.

b.) 5, -3, 1

c.) 2, 0, -1 $A^{-1} = \frac{\text{adj } A}{|A|} = \frac{1}{1} \begin{bmatrix} 5 & -3 & 1 \\ \dots & \dots & \dots \end{bmatrix}$

d.) 2, -1, 1

$\begin{array}{c|cccc} 1 & 0 & -1 & 1 & 0 \\ 2 & 1 & -1 & 2 & 1 \\ 2 & 3 & 2 & 2 & 3 \\ 1 & 0 & -1 & 1 & 0 \\ 2 & 1 & -1 & 2 & 1 \end{array}$

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Q:) if $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 4 & 5 \end{bmatrix}$ then $A^{-1} = ?$ also find $\text{adj } A$

sol:) $|A| = 0$ i.e. A is singular so A^{-1} does not exist.

$\begin{array}{cc|cc} 1 & 2 & 3 & 1 & 2 \\ 4 & 5 & 6 & 4 & 5 \\ 3 & 4 & 5 & 3 & 4 \end{array}$
 $\text{adj } A = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$

Q:) if $A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 3 & 1 \\ 0 & 5 & -2 \end{bmatrix}$ and $\text{Adj } A = \begin{bmatrix} -11 & -9 \\ 4 & -2 \\ 10 & 16 \end{bmatrix}$

then $K = ?$

a) -5

b) 3

c) -3

d) 5

Inverse of matrices of 4×4 , 5×5 , 6×6 -
Not in a syllabus using Conventional Approach

Procedure: Write $A = I \cdot A$
 $\downarrow \quad \downarrow \quad \downarrow$
 $I = B \cdot A$ use E-operations
 $A^{-1} = B$ Ans.

Date

Addition & Subtraction in a matrix $\Rightarrow$
 is possible only when they are of same size

e.g.:
 $A = \begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & 2 \end{bmatrix}_{2 \times 3}$ $B = \begin{bmatrix} -1 & 2 & 3 \\ 1 & -2 & 0 \end{bmatrix}_{2 \times 3}$ $C = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 4 \\ 1 & 3 & 2 \end{bmatrix}_{3 \times 3}$

$A+B$ = defined But $A+C$ = NO & $A+C$ = NO

$A+B = \begin{bmatrix} 1 & 5 & 2 \\ 1 & -1 & 2 \end{bmatrix} = B+A$

$A-B = \begin{bmatrix} 3 & 1 & -4 \\ -1 & 3 & 2 \end{bmatrix} \neq B-A = \begin{bmatrix} -3 & -1 & 4 \\ 1 & -3 & -2 \end{bmatrix}$

i.e. matrix addition is commutative law.
 matrix subtraction is not " " .

Matrix Multiplication:

Is defined when no. of columns in 1st mat =
 no. of rows in 2nd mat

e.g.: Consider A and B then AB = (Defined)
 BA = (NO)

e.g. $A = \begin{bmatrix} 1 & 2 & -4 \end{bmatrix}_{1 \times 3}$ & $B = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}_{3 \times 1}$ then AB = ? & BA

$AB = \begin{bmatrix} -8 \end{bmatrix}_{1 \times 1} = -8$

$\Rightarrow [1 \times 2 + 2 \times 1 + (-4) \times 3]_{1 \times 1}$

Date

$$BA = \begin{bmatrix} 2 & 4 & -8 \\ 1 & 2 & -4 \\ 3 & 6 & -12 \end{bmatrix}$$

Q.) The value of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \times \begin{bmatrix} 3 & 2 & 4 \end{bmatrix} \times \begin{bmatrix} 4 \\ 6 \\ 7 \end{bmatrix}$ equals

a.) $\begin{bmatrix} 52 \\ -104 \\ 156 \end{bmatrix}$

b.) $\begin{bmatrix} 52, -104, 156 \end{bmatrix}$

c.) $\begin{bmatrix} 52 \\ 104 \\ 156 \end{bmatrix}$

d.) None of these.

e.g.) $A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 1 & 2 \end{bmatrix}$ 2×3 & $B = \begin{bmatrix} 2 & -1 & 3 \\ -1 & 0 & 4 \\ 1 & 2 & -2 \end{bmatrix}$ 3×3 then $AB = ?$
 1×1 & 1×1

Sol.) $\begin{bmatrix} 1 & -2 & 3 \\ 4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 \\ -1 & 0 & 4 \\ 1 & 2 & -2 \end{bmatrix}$

See in the slide.)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

Date

⇒ Number of time, symbol of multiplication is used to find $AO = 18$

⇒ Number of time, symbol of Addition is used to find $AO = 12$ time.

Shortcut:

Consider $A_{m \times n}$ & $B_{n \times p}$ then to find $(AO)_{m \times p}$

→ No. of multiplication Required = $m \cdot n \cdot p$

→ No. of addition Req = $m(n-1)p$.

eg) if $A_{2 \times 3}$ & $B_{3 \times 3}$ then find $AO \rightarrow$ no. of multi
no. of addition

Sol: $m=2, n=3, p=3$

$$\text{no. of multi} = 2 \times 3 \times 3 = 18$$

$$\text{no. of addition} = 2(3-1)3 = 12$$

eg) if $A_{3 \times 4}$ & $B_{4 \times 3}$ then find $AO \rightarrow$ no. of multi
no. of addition

$m=3, n=4, p=3$

$$\text{no. of multi} = 36 = (3 \times 4 \times 3) =$$

$$\text{no. of add} = 24 = (3(4-1)3) =$$

BA

$$m=4 \quad \text{no. of multi} = 4 \times 3 \times 4 = 48$$

$$n=3 \quad \text{no. of addition} = 4 \times (3-1) \times 4 = 16 \times 2 = 32$$

$$p=4$$

na 2000 question

Date

Q: Consider $A_{2 \times 3}$, $B_{3 \times 4}$ & $C_{4 \times 2}$ then find the minimum number of multiplication & Addition that will be required to find matrix product (ABC)

- a.) 40, 28 we can find by using $\begin{cases} \text{Case 1 } (AB)C \\ \text{Case 2 } A(BC) \end{cases}$
 b.) 26, 26
 c.) 40, 24

d.) 48, 28 Case: I $A_{2 \times 3}$, $B_{3 \times 4} = (AB)_{2 \times 4}$

$$m=2, n=3, p=4 \quad = \quad m=2 \times n=4$$

Now

$$A = 16$$

$$(AB)_{2 \times 4} \cdot C_{4 \times 2} = (ABC)_{2 \times 2}$$

$$m=2 \quad = \quad m=16$$

$$n=4 \quad = \quad A = 2(4-1)2 = 1$$

$$p=2$$

$$\text{mul} = 40 \quad \& \quad \text{Addition} = 28$$

Case 2:

$$A(BC)$$

$$B_{3 \times 4} \cdot C_{4 \times 2} = (BC)_{3 \times 2}$$

$$m=3, n=4, p=2 \quad = \quad m=3 \times 4 \times 2 = 24$$

$$A = 3 \times (4-1)2 = 18$$

$$A_{2 \times 3} (BC)_{3 \times 2} = A(BC)_{2 \times 2}$$

$$m=2 \quad n=3 \quad p=2$$

$$= m \cdot n \cdot p = 12$$

$$n \cdot (p-1) \cdot 2 = 8$$

$$\text{mul} = 26 \quad \& \quad \text{Addition} = 26$$

Date

Commutative Law

$$\bullet A+B = B+A$$

$$\bullet AB \neq BA$$

Associative Law

$$\bullet A+(B+C) = (A+B)+C$$

$$\bullet A(BC) = (AB)C$$

i.e. Matrix multiplication is associative but not commutative in general.

e.g.) $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 0 & 1 \end{bmatrix}_{2 \times 3}$; $B = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix}_{3 \times 2}$

$$AB = \begin{bmatrix} 3 & -3 \\ 3 & 2 \end{bmatrix}_{2 \times 2}$$

Here $AB \neq BA$ in general

$$BA = \begin{bmatrix} 4 & -4 & 7 \\ 5 & -2 & 5 \\ 0 & 2 & 3 \end{bmatrix}_{3 \times 2}$$

But $\text{Tr}(AB) = \text{Tr}(BA)$

In general all

Q:) If A & B are two matrices of same order then which of the following is true?

$$a.) (A+B)^2 = A^2 + 2AB + B^2 \quad \times$$

$$b.) (A-B)^2 = A^2 - 2AB + B^2 \quad \times$$

$$c.) (A+B)^2 + (A-B)^2 = 2A^2 + 2B^2 \quad \checkmark$$

$$d.) (A+B)(A-B) = A^2 - B^2 \quad \times$$

$$a.) (A+B)^2 = (A+B)(A+B) \\ = A^2 + AB + BA + B^2$$

$$b.) (A-B)^2 = (A-B)(A-B) \\ = A^2 - AB - BA + B^2$$

$$c.) (A+B)(A-B) = A^2 - AB + BA - B^2$$

Lecture no. 04

* In Matrix Algebra, concept of Division is not defined. b/w two matrices.

In number system $ab=c \Rightarrow b=\frac{c}{a}$, $a \neq 0$
 While in matrix algebra $ab=c \xrightarrow{a} b=\frac{c}{a}$ (Always)

Note: If A is non-singular matrix s.t. $AB=C$

$$AB = C$$

$$A^{-1}(AB) = A^{-1}C$$

$$(A^{-1}A)B = A^{-1}C$$

$$IB = A^{-1}C$$

$$B = A^{-1}C$$

* In matrix algebra "Cancellation Law holds only when matrix is non-singular"

e.g.) if A is a non-sing mat s.t.

$AB=A$ ~~$AB=A$~~ = then $B=I$ (True)

e.g.) if $AB=A$ then $B=I$ (False)

e.g.) if $XY=Y$ & $YX=X$ then $X^2+Y^2=?$

Case 2: if X & Y are non-singular

a) $X+Y$

b) $2I$

c) 0

d) I

X^{-1} & Y^{-1} exist

So

$XY=Y \Rightarrow \boxed{X=I}$

$YX=X \Rightarrow \boxed{Y=I}$

$X^2+Y^2 = I^2+I^2 = 2I$

X^2+Y^2

I^2+I^2

$2I$

Spinal

Date:

Case II: if X & Y are singular

$$XY = Y \quad \text{--- (1)} \quad YX = X \quad \text{--- (2)}$$

Now

$$\begin{aligned} X^2 + Y^2 &= X \cdot X + Y \cdot Y \\ &= X(YX) + Y(XY) \\ &= (XY)X + (YX)Y \\ &= (Y)X + (X)Y \\ &= X + Y \quad \text{--- (3)} \end{aligned}$$

Q: if X and Y are two singular matrices such that $XY = Y$ and $YX = X$ then $X^2 + Y^2 =$?

$$\begin{aligned} \text{a) } X+Y & \quad X^2+Y^2 = XX+YY \\ \text{b) } XY & \quad = X \cdot X + Y \cdot Y \\ \text{c) } YX & \quad = X(YX) + Y(XY) \\ \text{d) } 2(X+Y) & \quad = (XY)X + (YX)Y \\ & \quad = (Y)X + (X)Y \\ & \quad = X + Y \end{aligned}$$

Conjugate of complex no. $\Rightarrow$

$$\begin{aligned} Z &= x+iy & \bar{Z}_1 &= 2-3i = \bar{Z}_1 = 2+3i \\ \bar{Z} &= x-iy & \bar{Z}_2 &= -5 = \bar{Z}_2 = -5 \\ & & \bar{Z}_3 &= 4i = \bar{Z}_3 = -4i \end{aligned}$$

Transjugate matrix $\Rightarrow$

(Transpose conjugate of mat)

if A is given mat then $A^{\theta} = (\bar{A}^T)$ or $(\bar{A})^T$

$$\text{eg:)} \quad A = \begin{bmatrix} 2 & 4+i & -3i \\ 2-3i & 0 & 4 \end{bmatrix} \quad \text{then } A^{\theta} =$$

Real no. of conjugate is always conjugate.

Date

$$A^T = \begin{bmatrix} 2 & 2-i \\ 4+i & 0 \\ -7i & 4 \end{bmatrix}, \quad \bar{A} = \begin{bmatrix} 2 & 4-i & 7i \\ 2+i & 0 & 4 \end{bmatrix}$$

$$A^{\Theta} = (\bar{A}^T) = \begin{bmatrix} 2 & 2+i \\ 4-i & 0 \\ 7i & 4 \end{bmatrix}$$

⇒ Real matrices

If all the element in a mat are real numbers then A is called real matrices.

If $\bar{A} = A$ or $(A^{\Theta} = A^T)$ then A is Real mat.

⇒ Complex matrices

If atleast one element in a mat is complex no. then mat is called Complex matrices.

e.g.) $A = \begin{bmatrix} 2 & 4 & -1 \\ 2 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 4 & -1 \\ 1 & 2 & 4i \end{bmatrix}$

$=$ Real mat $\quad \quad \quad =$ Complex mat.

If $\bar{A} \neq A$ then A is complex mat.

↳ Eigenvalue

* The Concept of trace, Det and Inverse, are defined for Square matrices.

While Concept of Transpose, Conjugate, Transpose Rank, are also defined for Rectangular matrices.

Some Confusions =

$$1) (A+B+C)^T = A^T + B^T + C^T$$

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$$2.) (A+B+C)^0 = A^0 + B^0 + C^0$$

$$3.) \ln(A+B+C) = \ln(A) + \ln(B) + \ln(C)$$

$$4.) |A+B+C| \leq |A| + |B| + |C|$$

$$5.) (A+B+C)^{-1} = \frac{\text{adj}(A+B+C)}{|A+B+C|}$$

$$6.) AB \neq BA \text{ but } \ln(AB) = \ln(BA)$$

$$7.) |A \cdot B \cdot C| = |A| \cdot |B| \cdot |C|$$

8.) Reversal Laws:

$$(i) (ABC)^T = C^T B^T A^T$$

$$(ii) (ABC)^0 = C^0 B^0 A^0$$

$$(iii) (ABC)^{-1} = C^{-1} B^{-1} A^{-1}$$

Q:) if A, B, C, D are non-singular matrices of same order then if $ABCD = I$ then B^{-1} & $D^{-1} = ?$

sol:) m-1 =) $B = \frac{A^{-1} C^{-1} D^{-1}}{A^{-1} C^{-1} D^{-1}} = I A^{-1} C^{-1} D^{-1} = A^{-1} C^{-1} D^{-1}$ X wrong

m-2 =) $ABCD = I$

$$[A^{-1} C^{-1} D^{-1} \neq A^{-1} D^{-1} C^{-1}]$$

$$A^{-1} (ABCD) D^{-1} = A^{-1} I D^{-1}$$

Super Bounder

$$A^{-1} A B C D D^{-1} = A^{-1} D^{-1}$$

$$BC = A^{-1} D^{-1}$$

$$(BC) C^{-1} = A^{-1} D^{-1} C^{-1} \rightarrow C$$

$$B = A^{-1} D^{-1} C^{-1} \text{ Ans}$$

$$A^{-1} D^{-1} C^{-1}$$

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$$\begin{aligned} \text{(ii)} \quad B^{-1} &= (A^{-1} \ O^{-1} \ C^{-1})^{-1} \\ &= (C^{-1})^{-1} \cdot (O^{-1})^{-1} \cdot (A^{-1})^{-1} \quad (\text{By R. law}) \\ &= COA \quad \text{Ans.} \end{aligned}$$

Some special type of matrices

eg:)

a	b	-c
b	d	e
-c	e	h

a	b	-c
-b	0	e
c	-e	0

a	4+i	2i
4-i	0	-5
-2i	5	3

Symmetric Mat

$$\therefore (A^T = A)$$

Skew-Symmetric

(Anti-symmetric)

$$(A^T = -A)$$

Hermitian

$$(A^H = A)$$

2i	4+i	2i
-4+i	0	-5
2i	5	-3i

Skew Hermitian matrices.

$$(A^H = -A)$$

Note:)

1.) Symm Mat:

Is symmetrical about leading diagonal
i.e corresponding element are same & $A^T = A$

2.) Skew-symm mat:

Diagonal elements are all zero and corresponding element are of opposite sign & $(A^T = -A)$

3.) Hermitian mat:

Diagonal element are either zero or purely Real
& corresponding element are conjugate of each other
& $A^H = A$

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4) Show Hermitian mat.

Diagonal elements are either zero or purely imaginary number and corresponding elements are -ve conjugate of each other $[A^0 = -A]$

* if A is s.t. A is skew sym then $|A| =$

$$\begin{cases} 0, & n = \text{odd} \\ \text{perfect square}, & n = \text{even} \end{cases}$$

i.e.

Det of skew sym of odd order = 0

e.g.) $|A| = \begin{vmatrix} 0 & -1 & -2 & 3 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & -1 & 0 & 4 & 1 \\ -3 & 2 & -4 & 0 & 2 \\ -4 & 0 & -1 & -2 & 0 \end{vmatrix}$ then $|A| = ?$

5x5

sol: This is skew-sym so $|A| = 0$ (by $n = \text{odd}$)

e.g.) $|A| = \begin{vmatrix} 0 & 1 & 2 & 0 \\ -1 & 0 & 4 & 0 \\ -2 & -4 & 0 & 6 \\ 0 & 0 & -6 & 0 \end{vmatrix}$ then $|A| = ?$

4x4

$|A| = - - - - - 36 = (6)^2 = \text{perfect square.}$

e.g.) if $A = \begin{bmatrix} 0 & - & - & - \\ -0 & - & - & - \\ 0 & -0 & - & - \\ - & - & - & 0 \end{bmatrix}$ is any skew sym mat then

$|A| = ?$ a) 3
b) 12
c) 81
d) 215

Date

eg: if $A = \begin{bmatrix} 0 & - & - & - \\ - & 0 & - & - \\ - & - & 0 & - \\ - & - & - & 0 \end{bmatrix}$ is any skew symm mat
 then $|A| = ?$
 a.) 3
 b.) 0
 c.) 81
 d.) C.N.D.

$\therefore 0$ is a perfect square matrix.

eg: if $A = \begin{bmatrix} 0 & -1 & -2 & -3 \\ 1 & 0 & -1 & -2 \\ 2 & 1 & 0 & -1 \\ 3 & 2 & 1 & 0 \end{bmatrix}$ then find $|A| = ?$
 (H.W.)
 $4 \times 4 = 16$

Q: if $A = [a_{ij}]_{n \times n}$ s.t. $a_{ij} = i^2 - j^2 \forall i, j$ then
 for $n \geq 3$, $n = \text{odd}$ $A^{-1} = ?$

sol: possible value of $n = 3, 5, 7, 9, \dots$

$n=3$ $A = [a_{ij}]_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 0 & -3 & -8 \\ 3 & 0 & -5 \\ 8 & 5 & 0 \end{bmatrix}$

Skew-symmetric mat

So, $|A| = 0$ So A^{-1} does not exist.

(ii) Sum of all the element in skew-sym
 $A_{n \times n} = 0$

Q: if $A = [a_{ij}]_{n \times n}$ s.t. $\forall i, j$; $a_{ij} = -a_{ji}$ then the
 value of $\sum_{i=1}^n \sum_{j=1}^n (a_{ij}) = ?$

Skew

Date

Corresponding element are of opposite sign
 i.e. Now for diagonal element i

We can take $i = j$

$$\text{i.e. } a_{ii} = -a_{ii}$$

$$a_{ii} = -a_{ii}$$

$$2a_{ii} = 0$$

$$a_{ii} = 0$$

$$\boxed{\text{Diag element} = 0}$$

i.e. A is skew symmetric

$$\text{So } A_{nn} = 0$$

$\Rightarrow$ Every Sq. mat^A can be expressed as the sum of
 symm & skew-symm mat.

By Common sense:

$$A = \left(\frac{A + A^T}{2} \right) + \left(\frac{A - A^T}{2} \right)$$

$$\boxed{A = P + Q}$$

$$P = \frac{A + A^T}{2} = \text{symm mat}$$

$$Q = \frac{A - A^T}{2} = \text{skew-symm mat}$$

Sq. mat = symm + skew symm

$$\text{where } P^T = \left(\frac{A + A^T}{2} \right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A^T + A}{2} = P$$

i.e. P is symm

$$Q^T = \left(\frac{A - A^T}{2} \right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = - \left(\frac{A - A^T}{2} \right)$$

$$= -Q$$

i.e. Q is skew-symmetric

⇒ Every sq. mat can be expressed as the sum of Hermitian & skew Hermitian mat as follows
Consider sq. mat $A_{n \times n}$ then by common sense we can write.

$$A = \left(\frac{A + A^H}{2} \right) + \left(\frac{A - A^H}{2} \right)$$

Q:) if A and B are two symmetric matrices. Then consider the following statements.

- (i) $A+B$ is symmetric $(A+B)^T = A^T + B^T = (A+B)$
- (ii) AB is symmetric $\times (AB)^T = B^T A^T = BA \neq AB$
- (iii) $AB+BA$ is symmetric $(AB+BA)^T = (AB)^T + (BA)^T$
 $= B^T A^T + A^T B^T$
 $= BA + AB = AB+BA$

Q:) if A & B are two Hermitian Matrices then $(AB-BA)$ will be?

Given:

- a.) Symm. $A^H = A$ & $B^H = B$
- b.) Skew-Symm. $\therefore$ Now
- c.) Hermitian $(AB-BA)^H = (AB)^H - (BA)^H$
 $= B^H A^H - A^H B^H$
 $= BA - AB$
 $= -(AB-BA)$
- d.) Skew Hermitian $\checkmark$

⇒ Idempotent mat:

if $A^2 = A$ then is called Idempotent matrices.
 & for idempotent mat, $A^2 = A^3 = A^4 = \dots = A$
 i.e if A is an Idempotent mat then $|A| = 0$ or 1

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⇒ Involuntary mat:

if $A^2 = I$ then A is called involuntary mat:

* For involuntary mat $A^2 = A^4 = A^6 = I$

i.e if A is an involuntary mat then $|A| = 1 \text{ or } -1$

⇒ Nilpotent mat:

if $A^k = 0$ then A is called nilpotent matrices of power k .

Eg

if A is s.t $A^2 \neq 0$, $A^3 \neq 0$ But $A^4 = 0$ then $A^5 = A^6 = 0$.

So, A is nilpotent matrix power 4

* if $A^2 = A$ then $|A^2| = |A|$

$$|A|^2 = |A| = 0$$

$$|A|(|A| - 1) = 0$$

$$|A| = 0 \text{ or } 1$$

* if $A^2 = I$ then $|A^2| = |I|$

$$|A^2| = 1$$

$$|A| = 1 \text{ or } -1$$

* if $A^k = 0$ then $|A^k| = |0|$

$$|A|^k = 0$$

$$|A| = 0$$

Ex:) if $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$ then A is

that

Sol:)

$$A^2 = A \cdot A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

∴ $A^2 = 0$ so A is nilpotent mat of power 2

Spiral

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Q:) $A = \begin{bmatrix} 6 & 0 \\ -4 & -6 \end{bmatrix}$ then A is? Nullpotent of index 2

sol:) $(a=2, b=3)$

Q:) if A is an Idempotent mat then $(I-A)$ will be

$$A^2 = A$$

a.) Null mat. $M-I$ $A \cdot A = A$ $\times$ wrong step

b.) Identity. $A = I$

c.) Involuntary. $\therefore (I-A) = I-I = 0$

d.) Idempotent. $\therefore$ for an idempotent $(I-A)^2 = 0$ or I
i.e we are not sure that mat A given in

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question is not regular not possible to a

$$M-2 \quad A^2 = A \quad - (1) \text{ given}$$

Now

$$(I-A)^2 = (I-A)(I-A)$$

$$= I^2 - IA - AI + A^2$$

$$= I - A - A + A$$

$$= (I-A)$$

So $(I-A)$ is an Idempotent mat.

Q:) if T is idempotent then $T^k = T$ for

a.) $k=2$

if T is idempotent

b.) All integer k

$$T^2 = T$$

c.) All positive integer $k \geq 2$

$$T^3 = T$$

d.) All of the above.

$$T^4 = T$$

$$T^5 = T$$

Orthogonal matrix:

if $AA^T = I$ or $(A^T = A^{-1})$ then A is called orthogonal matrix.

⇒ Unitary mat:

if $AA^H = I$ (or $A^{-1} = A^H$) then A is called unitary mat.

* Unitary mat of real no. is orthogonal also
let A is unitary mat formed by real no. then it is called Real mat also.

$$AA^H = I \quad - (1) \text{ (unitary)}$$

$$A^H = A^T \quad - (2) \text{ (real no.)}$$

Using (1) & (2) we can conclude that $AA^T = I$ which is condition of orthogonal mat.

* if A is an orthogonal mat then $|A| = 1$ or -1

Q.) check the nature of following matrices

$$1.) A = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$2.) B = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$$

sol:) let us calculate AA^T .

$$A \cdot A^T = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

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So, A is orthogonal mat.

Note: $\omega^3 = 1, 1 + \omega + \omega^2 = 0, \bar{\omega} = \omega^2, \bar{\omega^2} = \omega$

2.)

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}, \quad \bar{B} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \bar{\omega} & \bar{\omega^2} \\ 1 & \bar{\omega^2} & \bar{\omega} \end{bmatrix}$$

$$= \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$$

Now

$$-AB = (\bar{A})^T$$

$$= \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$$

$$\text{Now } AA^T = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix} \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = I$$

$$\text{So, } AA^T = I$$

$\therefore A$ is unitary matrices.

Now

Date

Q:) if $A = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix}$ is orthogonal matrix then $(AA')^{-1} = ?$

a) I_2 b) I_3 c) I_4 d) I

Now

$$AA' = I \quad (\text{given})$$

$$(AA')^{-1} = (I)^{-1} = I = I_{4 \times 4} = I_4$$

Q:) if $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ then A is -

a) orthogonal

b) involutory

c) unitary

d) Hermitian

Now

$$A^2 = A \cdot A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = I$$

$$[A^2 = I]$$

i.e. A is involutory also

Now

$$AA^T = A \cdot A = A^2 = I$$

So

$AA^T = I$ i.e. unitary also

P.O.D

Date

Vectors & their properties:

ordered pair $= (n_1, n_2) = n_1\hat{i} + n_2\hat{j} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} = 2 \text{ dim vector}$
(Point in 2D)

ordered triplet $= (n_1, n_2, n_3) = n_1\hat{i} + n_2\hat{j} + n_3\hat{k} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = 3 \text{ dim vector}$

ordered quadruple $= (n_1, n_2, n_3, n_4) = ? = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \end{bmatrix} = 4 \text{ dim vector}$

ordered n -tuple $= \dots = ? = \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_n \end{bmatrix} = n \text{ dim vector}$

$\Rightarrow$ ordered n -tuple: Any ordered set of n numbers is called ordered n -tuple.

Generally it is represented in the form of column matrix. (But we can also represent it in the form of Row mat).

$$x = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ \vdots \\ n_n \end{bmatrix}$$

$$\text{or } x = [n_1, n_2, n_3, \dots, n_n]$$

i.e. 3×4  has 3 row vectors & 4 column vectors

$\Rightarrow$ that pass Consider $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ & $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$ then:

Note: 1) Three vector are orthogonal, they are pairwise orthogonal if

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1) Dot Product = $\boxed{x \cdot y = x^T y} = (n_1 y_1 + n_2 y_2 + \dots + n_n y_n)$

Note: Try to calculate $x \cdot y = x_1 y_1 + y_1 x_1 = N \cdot 0$

2) Norm of vector = $\boxed{\|x\| = \sqrt{x^T x}} = \sqrt{n_1^2 + n_2^2 + \dots + n_n^2}$
(length)

Note: $x \cdot y = y \cdot x = (n_1 y_1 + n_2 y_2 + \dots + n_n y_n)$
i.e. dot product is commutative

3) Normalized vector = if $\boxed{\|x\| = 1}$ then x is called normalized vector
(Unit vector)

4) Orthogonal vector = if $x \cdot y = 0$ then x & y are called orthogonal vector.

5) orthonormal vector =
if $\boxed{x \cdot y = 0 \text{ and } \|x\| = 1, \|y\| = 1}$ then vector are orthonormal vector.

Q:) check the nature of following vectors give in following set.

$$\{(1, 2, 1)^T, (2, 1, -4)^T, (3, -2, 1)^T\}$$

Sol:) $x_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}$, $x_3 = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$

$$\begin{aligned} x_1 \cdot x_2 &= 1 \times 2 + 2 \times 1 + 1 \times -4 = 0 \\ x_2 \cdot x_3 &= 2 \times 3 + 1 \times -2 + -4 \times 1 = 0 \\ x_3 \cdot x_1 &= 3 \times 1 + (-1) \times 2 + 1 \times 1 = 0 \end{aligned} \quad \left\{ \begin{array}{l} n_1 y_1 + n_2 y_2 + n_3 y_3 \end{array} \right.$$

i.e. x_1, x_2, x_3 are orthogonal. $\Rightarrow$ L.I also.

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$$\left. \begin{aligned} \|x_1\| &= \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \\ \|x_2\| &= \sqrt{2^2 + 1^2 + (-4)^2} = \sqrt{18} \\ \|x_3\| &= \sqrt{3^2 + (-1)^2 + 1} = \sqrt{11} \end{aligned} \right\} \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

∴ These vectors are not of unit norm so they are not orthonormal.

Two Characteristic properties:-

1) Column vectors of an orthogonal matrix are orthonormal vectors.

e.g.) $A = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$ is an orthogonal matrix. $(AA^T = I)$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $c_1 \quad \quad c_2 \quad \quad c_3$

∴ $c_1 \cdot c_2 = c_2 \cdot c_3 = c_3 \cdot c_1 = 0$
 & $\|c_1\| = \|c_2\| = \|c_3\| = 1$ Hence verified.

2) if vectors are orthogonal then these are L.I. also [True]

3) $A = \begin{bmatrix} 1/9 & -4/9 & 8/9 \\ 5/9 & 4/9 & 1/9 \\ 4/9 & -7/9 & 8/9 \end{bmatrix}$ is an orthogonal mat then $\alpha + \beta = ?$

2) 0 ✓
 3) 4
 4) -4
 1) 8

∴ $c_1 \cdot c_2 = 0$

$$\frac{1}{81} [1 \times (-4) + 8 \times 4 + (-7) \times 4] = 0$$

$$28 - 7\alpha = 0$$

$$\alpha = 4$$

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$$C_2 \cdot C_3 = 0$$

$$\frac{1}{81} [-32 + 4 - 7\beta] = 0$$

$$-28 - 7\beta = 0$$

$$\boxed{\beta = -4}$$

$$\boxed{\lambda + \beta = 4 - 4 = 0}$$

Q.) The vector $(1, 1, 1)$ & $(1, a, a^2)$ (where $a = \frac{-1 + i\sqrt{3}}{2}$) are

a) L.S ✓ $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $y = \begin{bmatrix} 1 \\ a \\ a^2 \end{bmatrix} = \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}$

b) L.O

c) orthogonal ✓

d) orthonormal ✗

$$x \cdot y = 1 \times 1 + 1 \times \omega + 1 \times \omega^2$$

$$= 1 + \omega + \omega^2 = 0 \quad \Rightarrow \text{a}$$

i.e. x & y are orthogonal $\Rightarrow$ L.S also

Now $\|x\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$ ☹️

& $\|y\| = \sqrt{1 + \omega^2 + (\omega^2)^2} = \sqrt{1 + \omega^2 + \omega^4} = \sqrt{1 + \omega^2 + \omega} = 0$

* Max number of different elements that are required to construct a square mat $A_{n \times n} = \frac{n(n+1)}{2}$

e.g. $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}_{2 \times 2}$ i.e. max element req. = 3 = $\frac{2(2+1)}{2}$

e.g. $A = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}_{3 \times 3}$ i.e. max element req. = 6 = $\frac{3(3+1)}{2}$

e.g. $A = \begin{bmatrix} a & b & c & d \\ b & e & f & g \\ c & f & h & i \\ d & g & i & j \end{bmatrix}_{4 \times 4}$ i.e. max element req. = 10 = $\frac{4(4+1)}{2}$
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* Max number of 1's entries that are required to determine system mat $A_{n \times n} = \frac{n(n-1)}{2}$

Total elements in any general mat of $n \times n = n^2$



5 10 13 20 27

Note: E J O T Y

Lecture no: 06

Rank of matrix:

e.g 1: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -2 \\ 1 & -4 & 1 \end{bmatrix}$ 3×3 $\therefore |A| = -42$
 $\therefore |A| \neq 0$ or A is non-singular $\Rightarrow P(A) = \text{order of } A = 3$

e.g 2: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 4 & 5 \end{bmatrix}$ 3×3 $\therefore |A| = 0$
 $\therefore A$ is singular $\therefore P(A) \neq 3$

Consider a submatrix of the type

$$A_1 = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \text{ (By deleting } P_3 \text{ \& } C_3)$$

$|A_1| = -3$ $\therefore A_1$ is non-singular

$P(A) = \text{order of } A_1 = \text{Two}$

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eg 3: $A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 4 & -6 \\ -1 & 2 & 7 \end{bmatrix}$ $\therefore |A| = 0$
 3×3 i.e. A is singular

So rank $P(A) \neq 3$

Consider the submatrix $A_1 = \begin{bmatrix} 4 & -6 \\ 2 & 7 \end{bmatrix}$ 2×2

 $\therefore |A_1| = \dots = 40$ i.e. A_1 is non-singular $P(A) = \text{two}$

eg 4: $A = \begin{bmatrix} 1 & 2 & 2^2 & 2^3 & 2^4 \\ 1 & 2 & 2^2 & 2^3 & 2^4 \\ 1 & 2 & 2^2 & 2^3 & 2^4 \\ 1 & 2 & 2^2 & 2^3 & 2^4 \end{bmatrix}$ 4×5

$= \begin{bmatrix} 1 & 2 & 4 & 8 & 16 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 2 & 4 & 8 & 16 \end{bmatrix}$ 4×5

$P(A) \neq 5$ So, we are not able to find any submatrix of order 5×5

 $P(A) \neq 4$ or 3 or 2

$\therefore$ all the submatrices 4×4 , 3×3 , 2×2 are singular.

So,

$P(A) = 1$

$\Rightarrow$ Submatrix: By deleting some rows or some columns or both the matrix obtained is called submatrix.

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Def of rank: It is the order of non-singular submatrix of highest order that can exist in a given mat.

$\Rightarrow$ if $P(A_{7 \times 7}) = 4$ then $\left\{ \begin{array}{l} \text{at least one non-sing submat} \\ \text{of order } 4 \times 4 \end{array} \right.$
Every square submat of order
5x5, 6x6 are singular.

e.g.: $A = \begin{bmatrix} 2 & -3 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ or diag (2, -3, 0, 4, 1)
 the $P(A) = ?$

Sol: $|A| = 0$ i.e A is singular $\therefore P(A) \neq 5$

Consider a submatrix $A_1 = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ By del(n)

$$|A_1| = -24$$

i.e A_1 is non-singular

$\therefore P(A) = \text{order of } A_1 = \text{Four.}$

e.g.: $A = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$ $P(A) \neq 7$
 only submatrix order
 7×7 ONE
 $P(A) \neq 6, 5, 4$ all
 the sub matrix of
 6×7 order $6 \times 6, 5 \times 6, 4 \times 6$
 are singular

Most of the submatrices of order 3×3 are also singular.

But $\exists$ a submatrices of the type $A_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

By deleting $(R_2, R_4, R_5, C_4, C_5, C_6)$.

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$\therefore |A| = -6 \text{ i.e. } \neq 0$ so $|A|$ is non-singular

So $P(A) = \text{order of } A = \text{three}$

* E-operation do not alter the rank of a matrix
i.e. we are free to apply all three E-operations while calculating Rank."

That's why Rank is an Invariant property of matrices.

$\Rightarrow$ Equivalent matrix: ~~one or more~~
matrix obtained by applying E-operation are
equivalent to each other.

Q.) Equivalent matrices have same rank.
But they may have different diff determinant.

(e.g.) $A = \begin{bmatrix} 11 & 12 & 13 & 14 \\ 21 & 22 & 23 & 24 \\ 31 & 32 & 33 & 34 \\ 41 & 42 & 43 & 44 \end{bmatrix}$ then $P(A) = 1$

$R_4 \rightarrow R_4 - R_3$
 $R_2 \rightarrow R_2 - R_1$

11	12	13	14
10	10	10	10
10	10	10	10
10	10	10	10

$R_3 \rightarrow R_3 - R_2$

11	12	13	14
11	22	23	24
10	10	10	10
10	10	10	10

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$$R_2 \rightarrow R_2 - R_1 \quad \begin{bmatrix} 10 & 11 & 12 & 13 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \end{bmatrix} \quad 3 \times 3$$

$$\begin{array}{l} R_4 \rightarrow R_4 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array} \quad \begin{bmatrix} 10 & 11 & 12 & 13 \\ 10 & 10 & 10 & 10 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad 3 \times 3$$

$$P(A) \neq 4 \text{ or } 3$$

all 4×4 & 3×3
submatrices are
singular.

$$A_1 = \begin{bmatrix} 11 & 12 \\ 10 & 10 \end{bmatrix} \quad \therefore |A_1| = -10 \neq 0$$

i.e. Non-singular submatrices of 2×2 exist
So $P(A) = 2$.

Q:) if $A = [a_{ij}]_{n \times n}$ s.t. $a_{ij} = i \cdot j \quad \forall i + j$ then
for $n \geq 4$, $P(A) = ?$

$$\text{let } n=4, \quad A = [a_{ij}]_{4 \times 4} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \\ 4 & 8 & 12 & 16 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$R_4 \rightarrow R_4 - 4R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow P(A) = \text{one}$$

4×4

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Q:) The rank $(n+1) \times (n+1)$ mat A will be

$$A = \begin{bmatrix} 1 & a^1 & a^2 & \dots & a^n \\ 1 & a^2 & a^3 & \dots & a^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a^n & a^n & \dots & a^n \end{bmatrix} \quad (n+1) \times (n+1)$$

a.) $n+1$

b.) n

c.) 0

d.) 1

Q:) if $A = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & -1 \end{bmatrix}$ then $\rho(A) = ?$

What rank
Question

sol:) $R_2 \leftrightarrow R_3, R_4 \leftrightarrow R_5$

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_5 \rightarrow R_5 + (R_1 + R_4)$$

$$= \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = 5 \therefore |A| = 0$$

⇒ Dikrha ke Kisi bhi matrix ko Echelon form me convert kar sakte hai. 50

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$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

⇒ (Non-sing U.T.M) $\Rightarrow$

$$= A_1 = |A_1| = 1$$

$m=2$

$$P(A_1) = 4$$

$$A \sim A_1$$

Using Echelon form
In slide.

$$\text{So } P(A) = 4 \text{ Any.}$$

⇒ e.g of Echelon form:

$$\begin{bmatrix} 2 & 4 & -2 & 3 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

$$P(A) = 4$$

$$\begin{bmatrix} 2 & -1 & 3 & 1 \\ 0 & 3 & -1 & 2 \\ 0 & 0 & 2 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$P(A) = 3$$

$$\begin{bmatrix} 3 & 1 & 2 & 4 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$P(A) = 3$$

$$\begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & -5 & 2 & 1 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

$$P(A) = 3$$

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

✗

$$\begin{bmatrix} 2 & 4 & 1 & 3 \\ 0 & 7 & 5 & -1 \\ 0 & 0 & 2 & 5 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

✗

A Echelon form (Triangular form)

Any mat $A_{m \times n}$ is said to be in E-chelon form if

1) No. of zeroes before the 1st non-zero element in a row should be in an increasing order in the subsequent rows.

2) Every zero row if exist should occur at the bottom of mat.

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Note: 1) $\rho(\text{Echelon form}) = \text{Number of non-zero rows}$

2) Any mat can be converted into E-form by using E-operations

3) It is advisable to apply only E-operations while converting given matrices into an E-form (as per our syllabus)

Flow chart of converting given mat into an E-form

1) make a_{11} unity (Not compulsory but advisable)

2) make all the element of C_1 (that lies below a_{11}) zero by using E-row operation.

3) make a_{22} unity (Not compulsory but advisable)

4) make the element of C_2 (that lies below a_{22}) zero by using elementary operation.

5) make a_{33} unity & so on

Note:

Take care, In E-form, $a_{ii} = 1$

Q: $A = \begin{bmatrix} 2 & -1 & 4 & 2 \\ 1 & 3 & 0 & -2 \\ -3 & -2 & -4 & 0 \\ 0 & 2 & 1 & 3 \end{bmatrix}_{4 \times 4}$

a.) 1

b.) 2

c.) 3

d.) 4

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Sol:) $R_1 \leftrightarrow R_2$

1	3	0	-2
2	-1	4	2
-3	-2	-4	0
0	2	1	3

$R_2 - 2R_1$
 $R_3 + 3R_1$

1	3	0	-2
0	-7	4	6
0	7	-4	-6
0	2	1	3

$R_3 = R_3 + R_2$

1	3	0	-2
0	-7	4	6
0	0	0	0
0	2	1	3

$R_1 \leftrightarrow R_2 + R_4$

1	3	0	-2
0	-1	7	15
0	0	0	0
0	2	1	3

$R_4 \rightarrow R_4 + 2R_2$

1	3	0	-2
0	-1	7	15
0	0	0	0
0	0	15	33

$\rightarrow R_4 \leftrightarrow R_3$

1	3	0	-2
0	-1	7	15
0	0	15	33
0	0	0	0

$= A_1$

No. of non-zero rows = 3
So

$$\rho(A) = 3$$

Q:) The rank of the matrix

1	4	8	7
0	0	3	0
4	2	3	1
3	12	24	2

a.) 3

b.) 1

c.) 2

d.) 4.

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Sol:)

$$\begin{aligned} R_3 &= 4R_1 \\ R_4 &= 3R_1 \end{aligned}$$

$$\begin{bmatrix} 1 & 4 & 8 & 7 \\ 0 & 0 & 3 & 0 \\ 0 & -14 & -13 & -22 \\ 0 & 0 & 0 & -19 \end{bmatrix}$$

$$\rightarrow R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 4 & 8 & 7 \\ 0 & -14 & -13 & -22 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & -19 \end{bmatrix} = A, \quad 4 \times 4$$

$$\rho(A) = (\text{non-zero V.I.M})_{4 \times 4} = 4$$

$$\therefore \rho(A) = 4$$

Q.) Applying elementary transform to a matrix its rank

- increase
- decrease
- does not change ✓
- None of the above.

Q.) The rank of the matrix

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 3 & 6 & 9 & 12 \\ 0 & 5 & 10 & 15 & 20 \end{bmatrix}$$

$$a.) \quad 0 \quad R_2 \rightarrow R_2 - 3R_1$$

$$b.) \quad 1 \quad R_3 \rightarrow R_3 - 5R_1$$

$$c.) \quad 2$$

$$d.) \quad 3 \quad \boxed{\rho(A) = \text{one}}$$

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad 3 \times 5$$

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Lecture no. 07

Q:) $A = \begin{bmatrix} 2 & & & & & \\ & -1 & & & & \\ & & 0 & & & \\ & & & 3 & & \\ & & & & 4 & \\ & & & & & 0 \end{bmatrix}$ or $A = \text{diag}(2, -1, 0, 3, 4, 0)$
 then $P(A) = ?$

Sol:) $A = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3} \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$R_4 \leftrightarrow R_5$ so $|A| \neq 0$ for 4×4 so $P(A) = 4$.
 No. of non-zero.

Properties of Rank.

1) $P(A)_{\max} \leq \min\{m, n\}$

e.g. $P(A_{6 \times 4}) \leq 4$

2) $P(A)_{\min} \geq 1$

i.e. $P(A)_{\min} \geq 1 \leq P(A)_{\max} \leq \min\{m, n\}$

3) $P(\text{Null mat}) = 0$

4) if $A_{n \times n}$ s.t. A is non-singular $P(A) = n$.
 & $P(A^{-1}) = n$

i.e. $P(A) = P(A^{-1})$

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e.g. $A_{5 \times 5}$ s.t. $|A| \neq 0$ so $P(A) = 5$

$$5.) P(A) = P(A^T) = P(A^0) = P(AA^T) = P(AA^0)$$

X i.e. $P(A) = P(\text{orthogonal mat})$ are same (obviously) (P(AA^0))

6.) if A & B are two matrices s.t. AB is defined then $P(AB) \leq \min\{P(A), P(B)\}$

e.g.: $P(A_{6 \times 7}) = 5$ & $P(B_{7 \times 5}) = 3$ then $P(AB)_{6 \times 5} \leq 3$

i.e. Rank of the product can never exceeds their individual Rank.

7.) $P(A+B) = P(A) + P(B)$ (Not always true)

i.e.

$P(A+B) \leq P(A) + P(B)$ (True) { See in slide }

8.) if $A = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$ then $P(A)$ = 4

4×4

$P(A) = 4$ ✓

9.1.) $|A| = - - - - = 5$

i.e. A is not-singular.

9.2.) if $A = \begin{bmatrix} M & -1 & 0 & 0 \\ 0 & M & -1 & 0 \\ 0 & 0 & M & -1 \\ -6 & 11 & -6 & 1 \end{bmatrix}$ s.t. $P(A) = 3$ ✓

4×4

then no. of different value of M satisfying it will be Spiral

sol:) A is singular $|A| = 0$

$$\begin{vmatrix} \lambda & -1 & 0 & 0 \\ 0 & \lambda & -1 & 0 \\ 0 & 0 & \lambda & -1 \\ -6 & 11 & -6 & 1 \end{vmatrix} = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$(\lambda - 1)(\lambda - 2)(\lambda - 3) = 0$$

$$\lambda = 1, \lambda = 2, \lambda = 3$$

a.) 0

b.) 1

c.) 2

d.) 3 ✓

e.) all (b), (c), (d)

Q:) if $P(A_{m \times n}) = n$ $P(B_{n \times p}) = p$ then $P(AB) = ?$
 $n \leq m$ $p \leq n$

a.) m

b.) n

c.) p ✓

d.) n.p.

$$\boxed{p \leq m \leq m} \quad \text{so } P(AB)_{n \times p} \leq p$$

Q:) If $A = \begin{bmatrix} p & q \\ x & y \end{bmatrix}$, $B = \begin{bmatrix} p^2 + q^2 & 2p + 2q \\ p + 2q & 2^2 + 3^2 \end{bmatrix}$ s.t
 $P(A) = N$ then $P(B) = ?$

$n-1$ let us try to calculate $AA^T = B$

a.) N ✓

b.) $N-1$

c.) $2N$

d.) N^2

i.e. $AA^T = B$, Now we know that
 $= P(B) = P(AA^T) = P(A) = N$

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$$m-2 \cdot |A| = p^2 - q^2 \quad \& \quad |B| = \dots \dots \dots - (p^2 - q^2)^2$$

i.e. $|A|$ & $|B|$ will be simultaneous zero expressions

if

$$|A| = 0 \Rightarrow |B| = 0 \quad \text{so } N = 1 \quad \text{so } P(A) = N = 1 \quad \text{Ans.}$$

$$P(B) = N. \text{ Ans.}$$

$$\text{if } |A| \neq 0 \Rightarrow |B| \neq 0, \quad \text{so } P(A) = N = 2 = P(B) \quad \text{Ans.}$$

Case 2 $\Rightarrow (m-3)$

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}, \quad B = \begin{bmatrix} p^2 + q^2 & p^2 + s^2 \\ pr + qs & rp + sq \end{bmatrix}$$

Case 1: let $p = s = 4, q = r = 0$

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 16 & 0 \\ 0 & 16 \end{bmatrix}$$

$$P(A) = 2 = N \quad P(B) = 2 = N$$

Case 2: $p = 4, q = r = s = 0$

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix}$$

$$P(A) = 1 = N \quad P(B) = 1 = N$$

$$\boxed{P(A) = P(B) = N} \text{ always.}$$

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Q:) let $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ be two matrices. Then the rank of $P+Q$ is = ?

sol:) $P+Q = \begin{bmatrix} 0 & -1 & -2 \\ 8 & 9 & 10 \\ 8 & 8 & 8 \end{bmatrix}$

$|P+Q| = 0$ i.e. it is singular

So $P(P+Q) \neq 0$

$\Rightarrow P(P+Q) = 2$ or

$\Rightarrow 8$

$P(\text{Row mat}) = P(A_{n \times n}) = 1 \Rightarrow P(\text{Row} \times \text{column}) = P(AO)$

$P(\text{column mat}) = P(OA) = 1$

$P(\text{column} \times \text{row}) = P(OA)_{n \times n}$

$1 \times 1 = 1$ or 0

e.g:) if $A = \begin{bmatrix} 2 & -1 & 3 \end{bmatrix}_{1 \times 3}$, $O = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}_{3 \times 1}$ then $P(AO) = 1$
 $\& P(OA) = 1$

sol:) $AO = \begin{bmatrix} 8 & -2 & 3 \end{bmatrix}_{1 \times 1}$
 $= [9] = P(AO)_{1 \times 1} = 1$

$OA = \begin{bmatrix} 8 & -4 & 12 \\ 4 & -2 & 6 \\ 2 & -1 & 3 \end{bmatrix}_{3 \times 3}$ it is not a row mat
 so $P(OA) = 1$

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Q:) if $x = [x_1, x_2, x_3, \dots, x_n]$ is an ordered n -tuple non-zero vector s.t. $xx' = v$ then $P(v) = ?$

- a) n $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$ = column mat
 b) $n-1$
 c) 1
 d) n^2

$$x' = [x_1, x_2, x_3, \dots, x_n] = \text{Row mat}$$

$$v = (xx') = (\text{column} \times \text{Row})_{n \times n}$$

$$P(v) = P(xx')_{n \times n} = 1$$

Q:) The rank of matrix

$$\begin{bmatrix} 0 & i & -i \\ -i & 0 & i \\ i & -i & 0 \end{bmatrix}$$

is

sol:) skew symmetric $|A| = 0$

so $P(A) \neq 3$

$$A_1 = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \quad |A_1| \neq 0 \text{ so } P \text{ exist}$$

So A_1 is non-singular

$$P(A) = 2$$

Q:) The rank of 4×4 skew-symmetric matrix

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ -1 & 0 & 1 & 0 \end{bmatrix}$$

is

sol: $|A| = 4$ i.e. A is non-singular of 4×4

$$P(A) = 4$$

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$$\Theta: \begin{bmatrix} 0 & -1 & -2 & -3 \\ 1 & 0 & -1 & -2 \\ 2 & 1 & 0 & -1 \\ 3 & 2 & 1 & 0 \end{bmatrix}_{4 \times 4}$$

sol: $|A| = 0 = (-1)^1 = \rho(A) \neq 4$
 $\rho(A) = 2$

* $\rho(\text{skew-symm mat}) \geq 2$

i.e. $\rho(\text{skew-symm cannot be 1})$

$\therefore$ Skew-symm mat of order $(\times 1)$ (one)

e.g. check the nature of vector $\begin{bmatrix} 2 \\ -1 \\ 0 \\ 3 \end{bmatrix}$ $\begin{bmatrix} -1 \\ 2 \\ 3 \\ 1 \end{bmatrix}$ $\begin{bmatrix} -6 \\ 6 \\ 6 \\ 4 \end{bmatrix}$

$\Rightarrow$ By observation

$$x_1 - x_2 = \begin{bmatrix} 2 \\ -3 \\ -3 \\ 2 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} -6 \\ 6 \\ 6 \\ 4 \end{bmatrix} = -\frac{1}{2} x_3$$

$$2x_1 - 2x_2 + x_3 = 0$$

Linear relationship exist b/w
 x_1, x_2, x_3 and I.O. ρ .

M-2 w/o using observation

Linear Dependent & Linearly Independent vectors

L.D vectors: Vectors are called L.D if $\exists$ **Linear Relation** b/w them.

L.I vectors: " " " " L.I if there are **ONE** **any linear relationship** b/w them.

Linear Combination of vectors:

Let $x_1, x_2, x_3, \dots, x_r$ are the given vectors
 $k_1, k_2, k_3, \dots, k_r$ are the scalars (const)

then the relationship of the type

$$k_1 x_1 + k_2 x_2 + k_3 x_3 + \dots + k_r x_r = 0 \quad \text{--- (1)}$$

is called **Linear combination of vectors.**

Special note: $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \end{bmatrix}$ then $x^T = \begin{bmatrix} & & & \end{bmatrix} = k \cdot 0$
 So relationship their
 Linear will never exist.

$\Rightarrow$ **Method of checking the nature of vectors:**

Consider the given vectors are $x_1, x_2, x_3, \dots, x_r$

then construct a matrix A as follow

$$A = [x_1 \ x_2 \ x_3 \ \dots \ x_r] \neq \text{Row mat.}$$

m-1 (Generally method) (always applicable).

(i) if $\rho(A) = \text{No. of vectors} \Rightarrow$ **L.I**

(ii) if $\rho(A) < \text{ " " " } \Rightarrow$ **L.D**

m-2 (Tricky method) (applicable only when A is

(i) if $|A| \neq 0 \Rightarrow$ **L.I** $\Rightarrow$ **sq. mat)**

(ii) if $|A| = 0 \Rightarrow$ **L.D.**

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e.g.) check 1

$$A = [x_1 \ x_2 \ x_3] = \begin{bmatrix} 2 & -1 & -6 \\ -1 & 2 & 6 \\ 0 & 3 & 6 \\ 3 & 1 & -4 \end{bmatrix}_{4 \times 3}$$

$$R_1 \leftrightarrow R_2 \quad \begin{bmatrix} -1 & 2 & 6 \\ 2 & -1 & -6 \\ 0 & 3 & 6 \\ 3 & 1 & -4 \end{bmatrix} \quad \begin{array}{l} R_2 \leftrightarrow R_1 \\ R_4 + 3R_1 \end{array}$$

$$\begin{bmatrix} -1 & 2 & 6 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - \frac{2}{3}R_2 \end{array} \quad \begin{bmatrix} -1 & 2 & 6 \\ 0 & 3 & 6 \\ 0 & 3 & 6 \\ 0 & 7 & 14 \end{bmatrix}$$

$$P(A) = 2 < 3$$

So

vectors are L.D.

9:) the vectors $(1 \ 2 \ -1)$, $(2 \ 3 \ 4)$, $(0 \ 1 \ 2)$, $(4 \ -3 \ 2)$

Ans: --- are linearly independent

$$\text{Sol:)} \quad x_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \quad x_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \quad x_4 = \begin{bmatrix} 4 \\ -3 \\ 2 \end{bmatrix}$$

$$A = [x_1 \ x_2 \ x_3 \ x_4] = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 2 & 3 & 1 & -3 \\ -1 & 4 & 2 & 2 \end{bmatrix}_{3 \times 4}$$

$$P(A) \leq 3 \quad (\text{By property})$$

$$P(A) < 4 \quad \text{definitely by common sense}$$

$$P(A) < \text{No. of vectors} \quad \text{vectors are Linearly Independent (L.D.)}$$

Date: _____

sol:	$x_1 =$	1	$x_2 =$	4	$x_3 =$	2
		2		5		1
		3		6		9

$$A \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 1 \\ 3 & 6 & 9 \end{bmatrix}_{3 \times 3}$$

$ A = 0$	1	4	2	$\lambda = 8$	
	2	5	λ		$= 0$
	3	6	9		

So, for $\lambda \neq 8$ - " LI i.e. $\lambda \in \mathbb{R} - \{8\}$

sol: $x_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ $x_2 = \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}$ $x_3 = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$

y_1, y_2, y_3 are orthogonal so there are L_i also

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Note: if there are two vector x_1 & x_2 then we need to use 6. method or 7. method, only use observation method.

i.e. consider x_1 & x_2 are given vectors
if $x_1 = Kx_2$ for any K the vector are L.I.

* if $x_1 \neq Kx_2$ (i.e. $K = 0$ or ∞) the " " L.I.

e.g. $x_1 = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} -4 \\ 6 \\ -2 \end{bmatrix}$

$$\therefore +2x_1 = -x_2$$

$$2x_1 + x_2 = 0$$

e.g.

$$x_1 = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} -4 \\ 6 \\ 0 \end{bmatrix}$$

$$\Rightarrow x_1 \neq Kx_2 \text{ for any non-zero } K$$

i.e. $K = 0$ or $\infty \Rightarrow$ L.I.

Lecture no. 8

e.g. $A = \begin{bmatrix} 1 & 2 & 3 & 3 & 6 \\ 2 & 11 & -2 & 3 & 1 \\ 1 & 4 & 1 & -3 & 2 \end{bmatrix}$

3x5

e.g. $\rho(A) = 3 \rightarrow A$ will have at most 3 L.I. vectors

choosing

System of equation

Non-homogeneous system
($AX=B$)

Homogeneous system
($AX=0$)

e.g

$$\begin{aligned} 2x - y + 4z &= 0 \\ x - 2 - 3z &= 2 \\ 3x + 1y - z &= 0 \\ -4x + 4y + 1z &= 0 \end{aligned}$$

$2x - y + 4z = 0$	$=$	$\begin{bmatrix} 2 & -1 & 4 \end{bmatrix}$	$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$	$=$	$\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$
$x - 2 - 3z = 2$	$=$	$\begin{bmatrix} 1 & 0 & -3 \end{bmatrix}$	$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$	$=$	$\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$
$3x + 1y - z = 0$	$=$	$\begin{bmatrix} 3 & 2 & -1 \end{bmatrix}$	$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$	$=$	$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
$-4x + 4y + 1z = 0$	$=$	$\begin{bmatrix} -4 & 4 & 1 \end{bmatrix}$	$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$	$=$	$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$4 \times 3 \quad 3 \times 1 \quad 4 \times 1$

$$A_{4 \times 3} X_{3 \times 1} = B_{4 \times 1}$$

$$[A:B] = \left[\begin{array}{ccc|c} 2 & -1 & 4 & 0 \\ 1 & 0 & -3 & 2 \\ 3 & 2 & -1 & 0 \\ -4 & 4 & 1 & 0 \end{array} \right]$$

= Augmented matrices

Consider

Coefficient mat

Constant mat

$$A_{m \times n} X_{n \times 1} = B_{m \times 1}$$

$$[A:B] = \text{Aug}$$

No. of equatⁿ

No. of variable

solution of system
variable matrices

Unknown vector.

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

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- 2.) if $m > n$ then system is overdetermined. (i.e.)
 if $m = n$ then system is equally determined.
 if $m < n$ then system is underdetermined. (i.e.)

3.) Nature of solutions $\rightarrow$

e.g. $2x + 3y = 5$
 $x - 3y = 4$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1/3 \end{bmatrix} = \text{unique sol exist}$$

e.g. $2x + 3y = 5$
 $4x + 6y = 10$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 4/3 \end{bmatrix}$$

= multiple vector exist
 Infinite sol exist

e.g. $2x + 3y = 5$
 $4x + 6y = 9$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \\ \end{bmatrix} = \text{No vector exist}$$

= No solution exist

Mathematically both the equations are contradicting each other.

Method of solving Non-homog system $\rightarrow$

Consider $Ax = b$, $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$

1.) Rank method (always applicable)

(i.e. for $m > n$, $m = n$, $m < n$)

- 1) if $\rho(A) = \rho(A : b) = \text{No. of variables} \Rightarrow \text{unique sol}$
- 2) if $\rho(A) = \rho(A : b) < \text{No. of variables} \Rightarrow \infty \text{ sol}$

Another condition for consistency: $\rightarrow$
 ρ must be L.O on columns of A

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c) $\rho(A) \neq \rho(A:B) = \text{No solution}$

2) Matrix method
 (applicable only $m=n$)

if $|A| \neq 0 \Rightarrow$ unique sol

if $|A| = 0 \Rightarrow \infty$ sol, $(\text{adj } A)B = 0$

if $|A| = 0 \Rightarrow$ No sol, $(\text{adj } A)B \neq 0$

Note: if A is $n \times n$ i.e. we have equally determined system.

$$\text{Then } AX = B \quad \text{--- (1)}$$

$$A^{-1}(AX) = A^{-1}B$$

$$X = A^{-1}B$$

$$X = \frac{(\text{adj } A)}{|A|} \cdot B$$

$$X = \frac{(\text{adj } A)B}{|A|}$$

If $|A| = 0$, $(\text{adj } A)B \neq 0$ then $X = \text{Something} = \text{N.O}$

$\Rightarrow$ Consistent system:

System is called consistent if $\exists$ solution
 (whether unique or ∞ sol)

$\Rightarrow$ Inconsistent system:

System is called inconsistent if we have no solution

* Necessary Condition for a system $AX = B$ to be consistent is

$$\rho(A) = \rho(A:B)$$

At least one solution = consistency

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Q:) Write the condition for the existence of at least one solution of an unknown vector in the system $PY = Q$

sol:) $P(P) = P(P:Q)$

Q:) The system of equation:

$$2x + y = 5$$

$$x - 3y = -1$$

$$3x + 4y = K$$

is consistent when K is

a.) $[A:B] = \begin{bmatrix} 2 & 1 & 5 \\ 1 & -3 & -1 \\ 3 & 4 & K \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2}$

b.) $\begin{bmatrix} 1 & -3 & -1 \\ 2 & 1 & 5 \\ 3 & 4 & K \end{bmatrix} \xrightarrow{\begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix}} \begin{bmatrix} 1 & -3 & -1 \\ 0 & 7 & 7 \\ 0 & 13 & K+3 \end{bmatrix}$

$R_2 \rightarrow \frac{1}{7} R_2$ $\begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 1 \\ 0 & 13 & K+3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 13R_2} \begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & K-10 \end{bmatrix}$

$= P(A) = 2$ for consistency
 $P(A:B) = 2$

$$K-10 = 0$$

$$K = 10$$

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$$\begin{array}{rcl}
 m-2 & 2x + 3y = 5 & \text{--- (1)} \\
 & x - 3y = -1 & \text{--- (2)} \\
 & 3x + 4y = k & \text{--- (3)}
 \end{array} \quad \left. \vphantom{\begin{array}{rcl} 2x + 3y = 5 \\ x - 3y = -1 \end{array}} \right\} x=2, y=1$$

$$3(2) + 4(1) = k$$

$$\boxed{k=10}$$

Q:) The solution of the system:

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + 5z = 14$$

$$a = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$b = \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}$$

$$c = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$d = \begin{bmatrix} -4 \\ 0 \\ -2 \end{bmatrix}$$

m-1 = using options Ans is C.

m-2 = without using options.

solve above three equation simultaneously as discuss in class 8th → Do your self.

$$\begin{array}{c|ccc|c}
 m-2 & [A:D] & = & \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & 5 & 14 \end{bmatrix} & \begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \end{array} & \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & 4 & 8 \end{bmatrix} & \begin{array}{l} R_3 - R_2 \\ R_2 \times 2 \end{array} & \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}
 \end{array}$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$P(A) = 3 = P(A:D) = \text{No. of variables}$$

$$P(A) = P(A:D) = \text{No. of variables}$$

So, unique sol exist.

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(ii) Procedure of finding solution

$$Ax = B$$

1	1	1	x	=	6
1	2	3	y	=	10
1	2	5	z	=	14

$$\approx \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 4 \end{bmatrix}$$

$x + y + z$	=	6
$y + 2z$	=	4
$2z$	=	4

$$= x + y + z = 6$$

$$y + 2z = 4$$

$$2z = 4$$

$$\approx \boxed{I = 2}, \boxed{y + 2z = 4 = 0}, \boxed{x + 0 + 2 = 6}$$

$$\boxed{x = 4}$$

$$\text{So } x = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

=> This method is called Rank method, Echelon form method, gauss elimination method, or Backward substitution method.

9.) Find the value of λ & μ for which following system has

∞ sol

No. sol

Unique sol

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

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Q.) Consider the system of eq $\begin{bmatrix} 2 & 1 & -4 \\ 4 & 3 & -12 \\ 1 & 2 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$

noticed that 2nd & 3rd column of coeff mat are L.D then for how many value of λ does the above system has ∞ solution.

a.) 1
b.) 2
c.) 3
d.) $\frac{1}{5}$

$$\begin{bmatrix} 1 & 2 & -8 & | & 7 \\ 0 & 0 & 0 & | & 5\lambda - 1 \\ 0 & 0 & 0 & | & 5\lambda - 1 \end{bmatrix}$$

$$= \frac{5\lambda - 1}{2} = 0$$

$$\rho(A) = \rho(A:B) < \text{No. of variables}$$

$$5\lambda - 1 = 0$$

$$5\lambda = 1$$

$$\lambda = \frac{1}{5}$$

Q.) if A is s.t $A^T = I$ then $Ax = 0$ has

a.) unique sol

$n-1$

$$A^T = I$$

b.) ∞ sol

$$|A^T| = |I|$$

c.) No sol

$$|A|^2 = 1$$

d.) more than one but

$$|A| = \pm 1$$

finite no of sol.

$$\text{i.e. } |A| \neq 0$$

By matrix method

unique sol exists

M-2 $A^2 = I \Rightarrow A = A^{-1}$
 i.e. A^{-1} exist $\Rightarrow |A| \neq 0$
 ||

Dy matrix method
 unique sol exist.

M-3 Given $A^2 = I$
 $A^2 = I$ (given)

Now $Ax = 0$

$A(Ax) = A \cdot 0$

$A^2 x = A \cdot 0 \Rightarrow Ix = 0$

$Ix = [A \cdot 0]_{n \times 1}$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

unique vector exist

So,

unique solution exist.

Lecture no. 09

Q:) if $A_{3 \times 4}$ matrix s.t. $Ax = 0$ is an inconsistent system then highest possible Rank of A will be!

sol:) a.) 0 M-1) $A_{3 \times 4}$ $[A:B]_{3 \times 5}$

b.) 1

c.) 2

d.) 3

* for inconsistency $P(A) \neq P(A:B)$

$\therefore \max P(A:B)_{3 \times 5} \leq 3$

$\therefore \max P(A) = 2$

$P(A) < P(A:B)$

M-2 for $P(A) < P(A:B)$ we have following cases:

$P(A) = 1, P(A:B) = 2$

$P(A) = 1, P(A:B) = 3$

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Q.) Solve

$$3x + 3y + 2z = 1$$

$$x + 2y = 4$$

$$10y + 3z = -2$$

$$2x - 3y - z = 5$$

a) No solution

b) ∞ "

c) unique " ✓

d) finite no. of solution

Q.) The nature of the sol of following system

$$1.) \quad x_1 - 2x_2 + 4x_3 = 5 \quad \Rightarrow \text{No solution}$$

$$2x_1 - 4x_2 + 8x_3 = 7$$

$$2.) \quad x_1 - 2x_2 + 4x_3 = 5 \quad \Rightarrow \text{sol.}$$

$$2x_1 - 4x_2 + 8x_3 = 10$$

Undetermined system can not have unique sol.

(Note:)

Undetermined non-homog system if consistent

always consist infinite sol. $P(A) = P(A:0)$ In case of overdetermined system we may have ∞ type of answer

$$\text{if } P(A) < P(A:0) \Rightarrow \text{No sol}$$

$$\text{if } P(A) = P(A:0) \Rightarrow \begin{cases} \text{unique sol.} \\ \text{or } \infty \text{ sol.} \end{cases}$$

Q:) Given a system of equation

$$x + 2y + 2z = b_1$$

$$5x + y + 3z = b_2$$

which of the following is true regarding its sol:

{ see in slide }

sol:)

Q:) Find the nature of sol of

$$x + y + z = 1$$

$$2x + y + 4z = 1$$

$$4x + y + 10z = 1$$

$$A:0 = \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 2 & 1 & 4 & | & 1 \\ 4 & 1 & 10 & | & 1 \end{bmatrix}$$

a) No sol

b) Unique sol

c) ∞ sol ✓

d) Finite no. of ind sol. ✓

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & -1 & 2 & | & -1 \\ 0 & -3 & 6 & | & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & -1 & 2 & | & -1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\rho(A) = 2 = \rho(A:0) < 3$$

Homogeneous System: ($Ax=0$)

eg: $x - 4y + 2z = 0$

$$3x + 1y - 2z = 0$$

$$-4x - y + 4z = 0$$

$$7x + 1y + 3z = 0$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

is the sol of this system (T)

& this solution Always exist that's why

Homogeneous system never inconsistent or
homog system Always consistent

Initial soln variable = 0

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Non trivial soln solution $\Rightarrow$ It always exist for

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Another reason of consistency:-

$P(A) = P(A:0)$ always homogeneous

homogeneous (unique sol not possible)

Homogeneous system always consist ∞ solutions

$\downarrow$
(No solution not possible)

$\Rightarrow$ Method of solving Homog system:

$$(A_{m \times n} X_{n \times 1} = 0_{m \times 1})$$

Rank method (always)

matrix method

($m > n, m = n, m < n$)

(only for $m = n$)

$\Rightarrow$ if $P(A) = \text{no. of variable} = \text{unique}$

(i) if $|A| \neq 0 \Rightarrow$ unique

$\Rightarrow$ if $P(A) < \text{no. of variable} = \infty$ sol

(ii) if $|A| = 0 \Rightarrow \infty$ sol

$\Rightarrow$ In this chapter

(i) unique sol $\cong$ trivial sol $\cong$ zero sol always exist

(ii) ∞ sol $\cong$ Non-trivial sol also exist $\cong$ Non-zero sol also exist

(iii) No solution $\neq$ zero solution

Q. Find k for which

a.) $2/3$

$$(3k-8)x + 3y + 3z = 0$$

b.) $1/3$

$$3x + (3k-8)y + 3z = 0$$

(i) $4/3$

$$2x + 3y + (3k-8)z = 0$$

c.) $1/3$

has non trivial sol.

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$$\text{sol:)} \quad A = \begin{vmatrix} 3k-8 & 3 & 3 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix}$$

For non-trivial sol ($\Rightarrow$ sol) : $|A| = 0$

$$\begin{vmatrix} 3k-8 & 3 & 3 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix} = 0$$

$$C_1 \rightarrow C_1 + (C_2 + C_3)$$

$$\begin{vmatrix} 3k-2 & 3 & 3 \\ 3k-2 & 3k-8 & 3 \\ 3k-2 & 3 & 3k-8 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 3 & 3 \\ 3k-2 & 3k-8 & 3 \\ 1 & 3 & 3k-8 \end{vmatrix} = 0 \rightarrow (3k-2) \begin{vmatrix} 1 & 3 & 3 \\ 0 & 3k-11 & 0 \\ 0 & 0 & 3k-11 \end{vmatrix}$$

$$(3k-2)(3k-11) = 0$$

$$k = \frac{2}{3}, \frac{11}{3}, \frac{11}{3}$$

Q.) For the system to have infinite sol, which of the following is/are true

$$pn + qy + rz = 0$$

$$qn + ry + pz = 0$$

$$rn + py + qz = 0$$

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- a.) $b=q=r$
 b.) $b+q+r=0$
 c.) $b^3+q^3+r^3-3bqr=0$
 d.) None.

$\Rightarrow$ Consider $A_{m \times n} X_{n \times 1} = B_{m \times 1}$ then which one is false

- a.) if $m > n$, $B \neq 0$, $\rho(A) < \rho(A:B)$ then system has no solution. (T)
 b.) if $m = n$, $B = 0$, $|A| \neq 0$ then system has only trivial sol. (T)
 c.) if $m = 3$, $n = 5$, $B = 0$ then system has also non zero sol. (T)
 d.) if $m = 5$, $n = 3$, $B = 0$ s.t. $\rho(A) = 3$ then system has only zero sol. (T)
 e.) if $m = n$, $B = 0$, then system has sol. (T)
 f.) if $m > n$, $B = 0$ then system has multiple sol.
 g.) if $m > n$, $B = 0$, $\rho(A) < n$ then system has multiple sol. (T)

3.) The value of λ for which the system of eqs

$$x+y+z=0$$

$$y+2z=0$$

$$\lambda x + z = 0$$

has more than one solution

$$|A| = 0$$

No. of independent solution = nullity of A

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$$\text{nullity of } A = n - P(A)$$

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a.) -1 ✓

b.) 0

c.) $\frac{1}{2}$

d.) 1

Q:) if A is $m \times n$ then $Ax = 0$ has _____ sol.

a.) No sol.

$$A \times n \times 1 = 0 \times 1$$

b.) unique sol.

c.) multiple sol. ✓

Understanding as this comes from

d.) Data is inadequate.

Lecture no. 10

Analysis:-

Consider $xy + z = 0 \Rightarrow \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = [0] \Rightarrow A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$

Q:) How many LI sol are there for $xy + z = 0$ at a time?

sol:) two.

Nullity of A / Dimension of null space.

⇒ Shortcut: No. of LI solutions of $Ax = 0$

$$\text{No. of columns} - P(A)$$

• Rank nullity theorem

$$N(A) = N(I) - P(A)$$

⇒ Null space / Solution space / solution set :-

Set formed by various solutions of homogeneous system $Ax = 0$ is called Null space.

↳ Dimension of null space = counting of LI sol of $Ax = 0$
Nullity of A .

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(Range space)

$\Rightarrow$ Column space: if $A_{m \times n}$ then set formed by all the linear combinations of columns of A .

• Dimension of column space & row space = $\rho(A)$

$\Rightarrow$ Row space: if $A_{m \times n}$ then set formed by all the linear combinations of rows of A .

e.g. Find the nullity of $x+y+z=0$

Here $A_{1 \times 3} \Rightarrow \rho(A) = 1$ so $N(A) = \text{No. of column} - \rho(A)$
 $= 3 - 1 = 2$

i.e. has two L.I. solutions

e.g. if all the solutions of $Ax=0$ are scalar multiples of $\begin{bmatrix} 2 \\ -1 \\ 3 \\ 4 \end{bmatrix}$ then $\rho(A) = 1$

Sol: $\Rightarrow$ At Θ eq. (1) has only one L.I. sol.

So,

Nullity of A $N(A) = 1$ Hence by Rank-nullity theorem
 $N(A) = \text{No. of column} - \rho(A)$
 $1 = 5 - \rho(A) \Rightarrow \rho(A) = 4$

Eigen - Value and Eigen Vectors:

$\Rightarrow$ Eigen value can be zero but eigen vector is not zero

eigen vector is always non-zero vector

Def:)

Consider Sq mat $A_{n \times n}$ then non-zero vector x is called eigen vector corresponding to eigenvalue λ (Real/complex) if we are able to find a relation of the type

$$Ax = \lambda x$$

$\lambda = \text{Eigen value}$
 $x = \text{Eigen vector}$

LHS is the multi of two matrices = (Tough to do)
 RHS is the scalar multi in a mat. (Easy to do)

$\Rightarrow$ Here we are considering Homogeneous system as follows

$$Ax = \lambda x \Rightarrow Ax - \lambda x = 0 \Rightarrow (A - \lambda I)x = 0$$

Hence this system will satisfy all the property of homogeneous system.

$\Rightarrow$ Characteristic Equation of A

$$\begin{aligned}
 Ax &= \lambda x \\
 (A - \lambda I)x &= 0 \quad - (1) \\
 mx &= 0 \quad - (2)
 \end{aligned}$$

non-zero eigenvector

Non zero solution

$\Rightarrow$ solution

$$P(m) < n \quad \text{or} \quad |m| = 0$$

$$P(A - \lambda I) < n \quad \text{or} \quad |A - \lambda I| = 0$$

Date

i.e. Necessary condition for the existence of non-zero eigen vector is

$$P(A - \lambda I) < n \quad \text{or} \quad |A - \lambda I| = 0 \quad \text{--- (2)}$$

Equation (2) is called characteristic eq. of A
 & Roots of this equation i.e.
 values of λ are called Eigen values / Eigen roots /
 char. value / char. Roots /
 latent roots / special roots

e.g.:) Find the E-value of

$$1.) \quad A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Sol:)} \quad A - \lambda I = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{bmatrix} \Rightarrow \text{char. mat}$$

Now char eq. is $|A - \lambda I| = 0$

$$= \begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0 \quad \text{char. eq.}$$

$$\dots \dots \dots (\lambda - 3)(\lambda - 1)^2 = 0$$

$$\lambda = 3, 1, 1$$

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$$2.) A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$\Rightarrow$ char eq is $|A - \lambda I| = 0$

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0-\lambda & -1 \\ 1 & 0-\lambda \end{bmatrix} = 0$$

$$\lambda^2 - (1)(-1) = 0$$

$$\lambda^2 + 1 = 0 \rightarrow \text{char eq}$$

$$= \boxed{\lambda = i, -i}$$

$$3.) A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} = 0 \text{ char mat}$$

$$|A - \lambda I| = 0 \quad \begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} = 0$$

$$= \lambda(\lambda-3)(\lambda-15) = 0 \rightarrow \text{char eq}$$

$$\boxed{\lambda = 0, 3, 15} \text{ are eigen values.}$$

Spiral

Date

Q:) Find the E. value of $A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ -1 & 2 & 0 & 1 \\ -1 & 0 & 2 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix}_{4 \times 4}$

sol:) C. eq. is $|A - \lambda I| = 0$

$$\begin{vmatrix} 0-\lambda & 0 & 1 & 1 \\ -1 & 2-\lambda & 0 & 1 \\ -1 & 0 & 2-\lambda & 1 \\ 1 & 0 & -1 & 0-\lambda \end{vmatrix} = 0$$

Note:

1.) Sum of E. value = $\text{Tr}(A) = 4$

2.) Product of E. value = $0 = |A|$

3.) $\lambda = 0 \Leftrightarrow |A| = 0$

$$\lambda(\lambda-2)(\lambda-1)^2 = 0$$

4.) Total no. of E. value = order

$$\lambda = 0, 2, 1, 1$$

Properties of ^{eigen} λ value $\rightarrow$

Let $A_{n \times n}$ having eigen value $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$

1.) Total number of E. value of $A =$ order of A
(whether different or Repeated).

2.) Sum of E. value = Trace (A)
 $\lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_n = \text{Tr}(A)$

3.) Product of E. value = Det (A)
 $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 \cdot \dots \cdot \lambda_n = |A|$

4.) Zero is an E. value of A iff A is singular.
i.e. $\lambda = 0 \Leftrightarrow |A| = 0$

Signed

- 5.) Number of non-zero E-value of $A \in P(A)$ only if $P(A, \lambda) = 0$ then always less than $\text{rank } A$.
- 6.) if sum of all the elementⁱⁿ of each row (or each column) is unique constant K then the constant K will be one of the E-value of A .
- 7.) Don't use E-operation in a given mat while calculating E-value.
But we can apply 3rd E-operation in its C-rq.
i.e. $|A - \lambda I| = 0$.
- 8.) E-value of U.T.M., L.T.M., Diag mat, scalar mat, Identity mat are just the diagonal elements.
- 9.) if λ is an eigen value of A then to find eigen value of any algebraic expression formed by A , we can replace A with λ in that expression.
- 10.) let $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ are E-value of A then
 (i) E-value of A^T are $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ i.e. same as
 (ii) E-value of A^m are $\lambda_1^m, \lambda_2^m, \lambda_3^m, \dots, \lambda_n^m$ ($m \in \mathbb{N}$)
 (iii) E-value of A^{-1} are $1/\lambda_1, 1/\lambda_2, 1/\lambda_3, \dots, 1/\lambda_n$ ($A \neq 0$)
 (iv) E-value of $(\text{adj } A)$ are $\frac{|A|}{\lambda_1}, \frac{|A|}{\lambda_2}, \frac{|A|}{\lambda_3}, \dots, \frac{|A|}{\lambda_n}$.
 But take care $|A| \neq 0$
 if $|A| = 0$ then to find E-value of $(\text{adj } A)$, use Special convention Approach.

Date

$$Q: \text{) } A = \begin{bmatrix} -2 & -1 & 3 & 4 \\ 0 & 1 & 3 & -4 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

then.

$$(i) \text{ E. value of } A \text{ are } = -2, 1, 4, 3$$

$$(ii) \text{ " " } A^T \text{ " } = -2, 1, 4, 3$$

$$(iii) \text{ " " } A^3 \text{ " } = -8, 1, 64, 27$$

$$(iv) \text{ " " } A^{-1} \text{ are } = 1/-2, 1, 1/4, 1/3$$

$$(v) \text{ E. value of adj } A, \text{ are } = -24/-2, -24, -24/4, -24/3$$

$$Q: \text{) } A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then.

$$(i) \text{ E. value of } A \text{ are } = 1, -2, 0, 4$$

$$(ii) \text{ " " } A^T \text{ are } = 1, -2, 0, 4$$

$$(iii) \text{ " " } A^2 \text{ are } = 1, 2, 0, 16$$

$$(iv) \text{ " " } A^{-1} \text{ are } = 1, \cancel{1/2}, \cancel{0}, \cancel{1/4}$$

(Blunder)

$$A^{-1} = O.M.E.$$

Spiral

Date ONE

Q. $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ then E-value of A^{-1} and
 (i) " of $\text{adj} A$ are?

sol: E-value of A are = 0, 7, 15. (already calculated)

$$\therefore |A| = (0)(7)(15) = 0$$

it means $A^{-1} = \text{O.N.E.}$

$$\begin{array}{ccccc} 8 & -6 & 2 & 8 & -6 \\ -6 & 7 & -4 & -6 & 7 \\ 2 & -4 & 3 & 2 & -4 \\ 8 & -6 & 2 & 8 & -6 \\ -6 & 7 & -4 & -6 & 7 \end{array} \quad \text{adj} A = \begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 20 \\ 10 & 20 & 20 \end{bmatrix}$$

$$= |\text{adj} A| = 0$$

$$\therefore \lambda = 0$$

Now calculator E-value = 0, 0, 4.5

Lecture no. 11 :-

Q: if E-value of $A = \begin{bmatrix} 2 & 3 \\ n & 8 \end{bmatrix}$ are 4 & 8 then
 $n+y = ?$

$$\text{sol: } \lambda_1 + \lambda_2 = \text{tr}(A) \quad \& \quad \lambda_1 \cdot \lambda_2 = |A|$$

$$4 + 8 = 2 + y$$

$$\boxed{y = 10}$$

$$(4) \cdot (8) = 2y - 3n$$

$$32 = 20 - 3n$$

$$\boxed{n = -4}$$

$$n + y = 10 - 4 = 6$$

Q: if trace of $A_{2 \times 2}$ are -2 & -35 resp
 then $\lambda_1 + \lambda_2 =$

sol:

a) 12, b) -12, c) 2, d) -2

Date

Q:) Consider a 2×2 square matrix

$$A = \begin{bmatrix} \sigma & \lambda \\ \omega & \sigma \end{bmatrix}$$

where λ is unknown. if the eigen value of the matrix A are $(\sigma + i\omega)$ and $(\sigma - i\omega)$ then λ is equal to

w.v. that

a.) $+i\omega$

$$\lambda_1 \cdot \lambda_2 = |A|$$

b.) $-i\omega$

$$(\sigma + i\omega)(\sigma - i\omega) = \sigma^2 - \omega^2$$

c.) $+i\omega$

$$\sigma^2 - \cancel{\sigma^2} - \cancel{\sigma^2} = \sigma^2 - \omega^2$$

d.) $-i\omega$

$$- \cancel{\sigma^2} \omega^2 = -\omega^2$$

$$\cancel{\sigma^2} \omega^2 = \omega^2$$

$$\lambda = i\omega = i\omega = (-i\omega)$$

Q:) if P and Q are two sq. mat of same order s.t. $PQ = QP, P^{-1} = Q$ then Q is an eigen value of

a.) P But not Q

$$PQ = QP = I$$

$$P^{-1} = Q$$

b.) Q But not P

$$Q^{-1} = P$$

c.) Both P & Q

$$i.e. \text{ Both } P^{-1} \text{ \& } Q^{-1} \text{ exist}$$

d.) Neither P nor Q

$$P \neq Q \text{ are non singular}$$

$$i.e. |P| \neq 0 \text{ \& } |Q| \neq 0$$

so By prop 4.

Q:) if one of the E. value of $A =$

$$\begin{bmatrix} 40 & -16 & -24 \\ -11 & 30 & -19 \\ 26 & 24 & -50 \end{bmatrix}$$

is λ then other E. values will be

a.) $\lambda - 20$

By prop 6 one eigen value is $\lambda = 20$

b.) $20 - \lambda$

c.) $20 + \lambda$

d.) 0

202:) $|A| = ?$ $C_1 \rightarrow C_1 + C_2 + C_3$

$$\begin{vmatrix} 0 & -16 & -24 \\ 0 & 30 & -19 \\ 0 & 24 & -53 \end{vmatrix} = 0$$

$|A| = 0$ using prop 4 $\lambda = 0$

Now

$$\lambda_1 + \lambda_2 + \lambda_3 = \text{Tr}(A) = 20$$

$$0 + \lambda + \lambda_2 = 20$$

$$\therefore \boxed{\lambda_2 = 20 - \lambda}$$

Q:) one of the E. value of $A =$

$$\begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix}$$

a) 5

b) 0

c) 10 ✓

d) -1

$$= 10 \quad 10 \quad 10 \quad 10$$

m-1 using prop ② one of the E. value is $\lambda = 10$

$\Rightarrow$ **Equivalent** matrices have same **Rank**

But may have different **Determinant** as well as different **Eigen values**.

Q:) the E. value of $A =$

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{vmatrix}$$

one

By prop ② One E. value

is $\lambda = 4$,

$$4 \quad 4 \quad 4 \quad 4$$

Now $\rho(A) = \text{one}$.

i.e. A has almost one non-zero eigen value which is $\lambda = 4$

So

other E-values will be $0, 0, 0$

Hence

$$\lambda = 4, 0, 0, 0$$

Q.) if $A_{2 \times 2}$ s.t. $a_{11} = a_{12} = a_{21} = 1$ & $a_{22} = -1$ then E-value of A^{19} are

a.) $\pm \sqrt{2}$ let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

b.) ± 2

c.) ± 1

d.) $\pm 512\sqrt{2}$ $\lambda_1 + \lambda_2 = \text{tr}(A) = 0$ — (1)

$\lambda_1 \cdot \lambda_2 = |A| = -2$ — (2)

$\Rightarrow \lambda_1 = \sqrt{2}, \lambda_2 = -\sqrt{2}$

C. eq. is $|A - \lambda I| = 0$

$\lambda^2 - 2 = 0$

$\lambda = \pm \sqrt{2}$

So E-values of A are $\sqrt{2}$ & $-\sqrt{2}$

A^{19} are $(\sqrt{2})^{19}$ & $(-\sqrt{2})^{19}$

$= (\sqrt{2})^{18} \sqrt{2} \quad (-\sqrt{2})^{18} \sqrt{2}$

E-value of A^{19} are $= 2^9 \sqrt{2}, -2^9 \sqrt{2}$

$= 512 \sqrt{2}, -512 \sqrt{2}$

$= \pm 512 \sqrt{2}$

$$\omega = \frac{-1 + i\sqrt{3}}{2}, \quad \omega^2 = \frac{-1 - i\sqrt{3}}{2}, \quad \omega = \omega, \quad \omega^2 = \omega$$

$$\omega^3 = 1, \quad 1 + \omega + \omega^2 = 0$$

91

Date

Q:) $A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & -1+i\sqrt{3}/2 & 0 \\ 3 & 4 & -1-i\sqrt{3}/2 \end{bmatrix}$ the $\ln(A^{102}) = ?$

sol:) $A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & \omega & 0 \\ 3 & 4 & \omega^2 \end{bmatrix}$ E. values of A are $\lambda = 1, \omega, \omega^2$

E. values of $A^{102} = 1^{102}, \omega^{102}, (\omega^2)^{102}$

$$= 1, -\omega^{102}, \omega^{204}$$

$$= 1, (\omega^3)^{34}, (\omega^3)^{68}$$

$$= 1, 1, 1, = 1, 1, 1$$

i.e. E.V of A^{102} are $= 1, 1, 1$

So

$$\ln(A^{102}) = -1 + 1 + 1 = 3$$

Q:) if $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ & $B = A^2 - 2A + 3I$ then find the product of the eigen values of B

Q:) 81 By prop (6), so eigen value is $\lambda_1 = 6$

Q:) 30

$$\ln(A) = 8$$

$$\lambda_1 + \lambda_2 = 8$$

$$6 + \lambda = 8 \Rightarrow \lambda = 2$$

$$\boxed{\lambda_2 = 2}$$

i.e. Eigen values of A are 2 & 6

Now using prop (6) $\rightarrow \lambda_B = 2^2 - 2(2) + 3(1) = 1$

$$B = A^2 - 2A + 3I$$

$$\lambda_A = 1^2 - 2(1) + 3(2) = 27$$

So Req. product $= 27 \times 27 = 81$ ✓

Spiral

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⇒ Shortcut method to write C. eq. of 2×2 mat A =

$$\lambda^2 - (\text{tr } A)\lambda + |A| = 0$$

e.g.: $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \Rightarrow \lambda^2 - 8\lambda + 12 = 0$
 $\Rightarrow \lambda = 6 \text{ \& } 2$

e.g.: $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \Rightarrow \lambda^2 - \lambda(0) - 2 = 0$
 $\Rightarrow \lambda^2 - 2 = 0$
 $\Rightarrow \lambda = \pm \sqrt{2}$

e.g. $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \Rightarrow \lambda^2 - 0 + 1 = 0$
 $\Rightarrow \lambda^2 + 1 = 0$
 $\Rightarrow \lambda = \pm i$

Q: if eigen value of $x_1 x_2$ are -2 & -3 then trace of $(x+2)^{-1} (x+5)$ will be

a) -4 ✓ $\text{ATQ } \Rightarrow x_1 x_2 \begin{cases} \lambda x = -2 \\ \lambda x = -3 \end{cases}$

b) 4

c) -5

d) 12

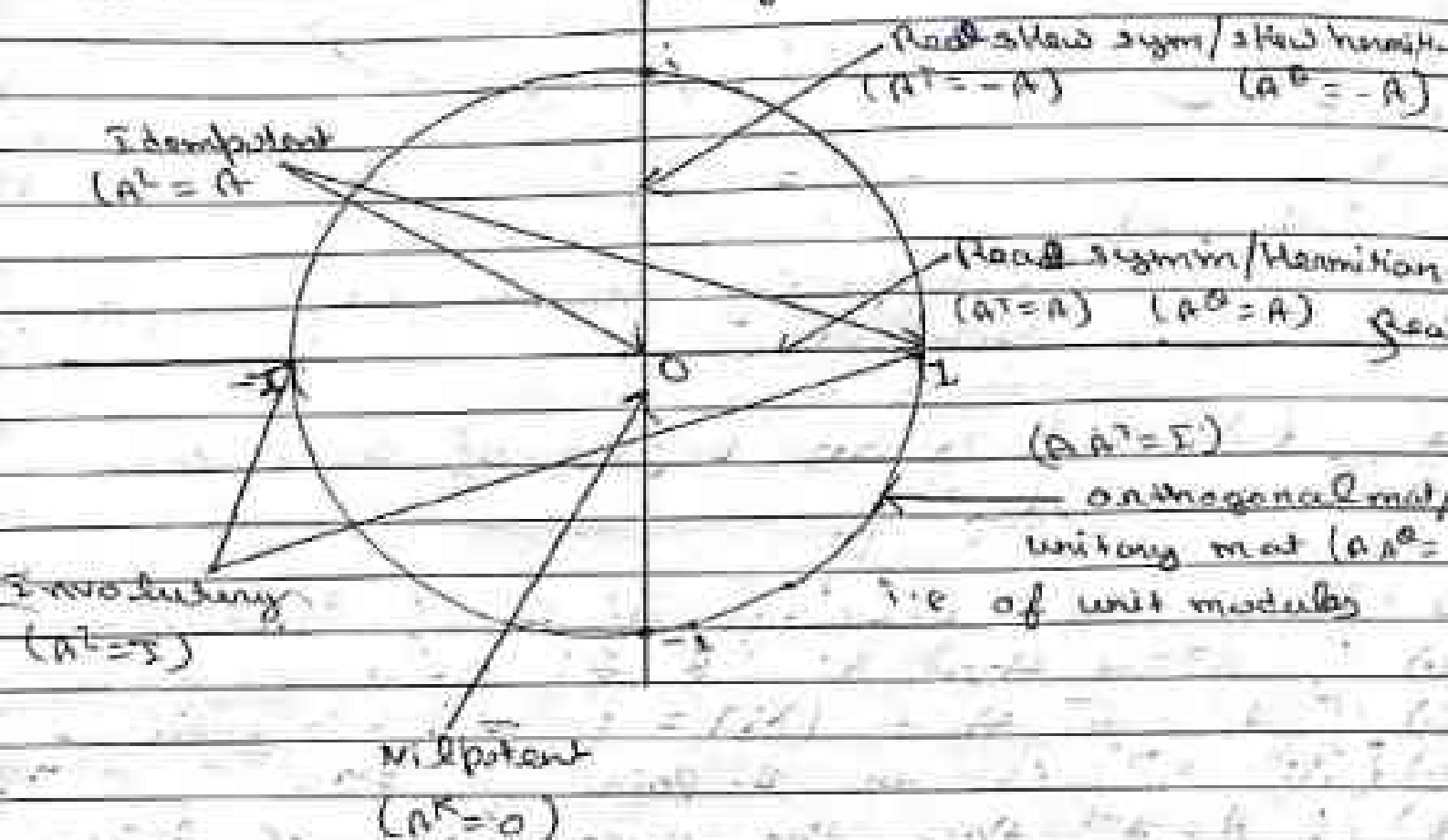
Q) $O = (x+2)^{-1} (x+5)$ $\begin{cases} \lambda_0 = (-2+1)^{-1} (-2+5) \\ = (-1)^{-1} (3) = -3 \\ \lambda_0 = (-3+1)^{-1} (-3+5) \\ = (-2)^{-1} (2) = -1 \end{cases}$

$\text{Tr}(O) = -3 + (-1) = -4$ ✓

Date

Shortcut method to learn various theorems based on E value.

Consider a unit circle centered at origin i.e. $x^2 + y^2 = 1$



Q.) if $A = \begin{bmatrix} - & - & - & - \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{bmatrix}_{n \times n}$ is an O-mat then $(A A^T)^{-1} =$

(ii) which of the following can not be the E value of A

a) $-1/2 + i\sqrt{1}/2$

b) -1

c) $\sqrt{1}/2 + i/2$

d) $1/4 + i\sqrt{3}/4$ ✓

Date

Q:) if E values of $A = \begin{bmatrix} 2 & 5+i & -3 \\ x & -1 & 4 \\ -3 & 4 & 3 \end{bmatrix}$ are all real? No. then is?

- a.) $5+i$
 b.) $5-i$ ✓
 c.) 0
 d.) Both a & b.
- Real eigen values $\Rightarrow$ $\left\{ \begin{array}{l} \text{Either A is real} \\ \text{or A is Hermitian} \end{array} \right.$
- $a_{12} = \text{complex}$

Corresponding elements are conjugate of each other

* if $\{\lambda_i, x_i\}$ is an E-pair of A then which is false?

- a.) $\exists$ Non zero E-vector of A if $P(A-\lambda_i) < nT$
 b.) if $A^T = A$ then $\lambda_i \in \mathbb{R} \forall i$. A is Hermitian
 c.) if $A^{-1} = A^T$ then $|\lambda_i| = 1 \forall i$ A is orthogonal
 d.) $\{\lambda_i^m, x_i^m\}$ is an E-pair of A^m $\leftarrow$ if m is an integer
 e.) if $A = A^{-1}$ then the Eigen value of A is 1

Lecture no. 12

mat A & A^m have "Same" E-vectors.

Q:) The eigen value of $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ are.

- a.) Purely imaginary
 b.) zero
 c.) Real ✓
 d.) None of the above.
- Skew symmetric $\rightarrow$ (Purely imag)
 Hermitian $\rightarrow$ (Purely Real)

Cayley - Hamilton Theorem (C.H.T)

Every sq. mat satisfy its own equation.

i.e. we can replace $\lambda \rightarrow A$ in C equation

Consider A nx n then its C. eq. is $|A - \lambda I| = 0$

i.e. $\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_{n-1} \lambda + a_n = 0$

By C.H.T $\lambda \rightarrow A$

$$\therefore A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_{n-1} A + a_n I = 0$$

Application:

Using this theorem, we can find $\text{Tr}(A)$, $|A|$ and A^{-1} as follows.

1) $\text{Tr}(A) = -a_1$

2) $|A| = (-1)^n a_n$

3) $A^{-1} = \dots$ will be discussed in question.

$\Rightarrow$ Take care coefficient of A^n should be unity.

Q:) Using C.H.T for $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ evaluate $\text{Tr}(A)$, $|A|$, $A^{-1} = ?$

sol:) C. eq. is $|A - \lambda I| = 0$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

By C.H.T, $\lambda \rightarrow A$

$$A^3 - 5A^2 + 7A - 3I = 0 \quad \text{--- (1)}$$

$$1.) \operatorname{Tr}(A) = -(\alpha) = -(-5) = 5$$

$$2.) |A| = (-1)^3 \alpha_3 = -(-3) = 3$$

$$3.) A^{-1} = |A|^{-1} \neq 0 \Rightarrow A^{-1} \text{ exist}$$

now multiply by A^{-1} in (1)

$$A^{-1}(A^3 - 5A^2 + 7A - 3I) = A^{-1} \cdot 0$$

$$A^2 - 5A + 7I - 3A^{-1} =$$

$$(-3A^{-1}) = -(A^2 - 5A + 7I)$$

$$\boxed{A^{-1} = \frac{1}{3}(A^2 - 5A + 7I)}$$

Q:) Using C.H.T, evaluate A^{-1} for $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Ans: solve.

Q:) For $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ evaluate $B = A^3 - 5A^2 + 7A^4 - 3A^5 + A$
 $- 5A^3 + 7A^2 - 4A + 2I$

Sol: By C.H.T.

$$\boxed{A^3 - 5A^2 + 7A - 3I} = 0$$

$$0 = A^3 - 5A^2 + 7A^4 - 3A^5 + A^4 - 5A^3 + 7A^2 - 4A + 2I$$

$$A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) - A + 2I$$

$$= A^5(0) + A(0) - A + 2I$$

$$0 = -A + 2I \quad \checkmark$$

Date

Q:) $P_{3 \times 3}$ s.t. it's C equation is $a(\lambda) = |\lambda I - P| = \lambda^3 - \lambda^2 + \lambda + 1 = 0$ then find $\text{Trace}(P)$, $|P|$, P^{-1} and $|O| = 1$
 where $O = P^8 + P^7 + P^6 + P^5 + P^4 + P^3 + P^2 + P + 4I$

sol:) $f(\lambda) = 0$ C eq of P is

$$|P - \lambda I| = 0$$

$$(-1)^3 |\lambda I - P| = 0$$

$$|\lambda I - P| = 0$$

$$\lambda^3 + \lambda^2 + \lambda + 1 = 0$$

on comparison with,

$$(\lambda^3 + a_1 \lambda^2 + a_2 \lambda + a_3) = 0$$

$$\text{Tr}(P) = -a_1 = -(1) = -1$$

$$|P| = (-1)^3 (a_3) = -(1) = -1$$

(Replace $\lambda \rightarrow P$)

$$P^{-1}(P^3 + P^2 + P + I) = P^{-1} \cdot 0$$

$$P^2 + P + I + P^{-1} = 0$$

$$P^{-1} = -(P^2 + P + I) \quad \text{--- (2)}$$

$$P^3 + P^2 + P + I = 0 \quad \text{--- (1)}$$

$$O = P^8 + P^7 + P^6 + P^5 + P^4 + P^3 + P^2 + P + 4I$$

$$P^5(P^3 + P^2 + P + I) + P(P^3 + P^2 + P + I) + 4I$$

$$P^5(0) + P(0) + 4I$$

$$O_{3 \times 3} = 4I_{3 \times 3} \Rightarrow |O| = |4I_{3 \times 3}| = 4^3 |I_{3 \times 3}|$$

$$= 64 \times 1$$

$$= 64$$

Spiral

Date

Q:) if E-value of $A_{n \times n}$ are ± 1 & $\pm i$ then $A^4 = ?$

- a.) A $A \neq I, \lambda = 1, -1, i, -i$
 b.) O So
 c.) I_2 C.E. is $(\lambda-1)(\lambda+1)(\lambda-i)(\lambda+i) = 0$
 d.) I_4 $(\lambda^2-1)(\lambda^2+1) = 0$
 $\lambda^4 - 1 = 0$
 $A^4 - I = 0$
 $A^4 = I = I_4 = I_{n \times n}$

Q:) if $A = \begin{bmatrix} -1 & -2 \\ 0 & 3 \end{bmatrix}$ then which is true.

- a.) $A^2 - 2A - 3I$ ✗ C.E. is $\lambda^2 - 2\lambda + (-1) = 0$
 b.) $(A-3I)(A+I) = 0$ ✓ $\lambda^2 - 2\lambda + (-1) = 0$
 c.) $A - 2I - 3A^{-1}$ ✗ $\lambda^2 - 2\lambda - 3 = 0$
 d.) $A^2 - 3A - 2I$ By C.H.T.

$$A^2 - 2A - 3 = 0$$

$\swarrow$ $\searrow$
 multiply by A^{-1} $A - 2I - 3A^{-1} = 0$

Factor

$$(A-3I)(A+I) = 0$$

Q:) if C.E. of $A_{n \times n}$ is $2\lambda^4 - 6\lambda^3 + 4\lambda^2 - 8\lambda + 11 = 0$ then $|A| = ?$

sol:) C.E. is $\lambda^4 - 3\lambda^3 + 2\lambda^2 - 4\lambda + 6 = 0$
 $a_1 = -3, a_2 = 2, a_3 = -4, a_4 = 6$

$$|A| = (-1)^4 a_4 = 6 \quad \text{Ans}$$

Date

Q: Constant term in the eq of $A =$

0	1	2	3
1	0	2	0
2	3	0	1
3	0	1	2

Sol: In the constant term $= a_{44} = +|A|$
 $= \dots = 88$ Ans.

Note (i) Constant term in the C.E of $A_{n \times n}$
 $= a_n = \begin{cases} -|A| & n = \text{odd} \\ |A| & n = \text{even} \end{cases}$

2) Shortcut method of finding C. eq. of $A_{2 \times 2}$
 mat: $\rightarrow$

C. eq. is $|A - \lambda I| = 0$

$$\lambda^2 + a_1 \lambda + a_2 = 0$$

$$\lambda^2 - (-a_1) \lambda + ((-1)^2 a_2) = 0$$

$$\boxed{\lambda^2 - (tr A) \lambda + (|A|) = 0}$$

Q: In matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ if $(a+d) = (ad-bc) = 1$ then

 $A^3 =$

C. eq.

$$\lambda^2 - (tr A) \lambda + |A|$$

$$\lambda^2 - (1) \lambda + 1 = 0$$

or C.H.T

$$A^2 - A + I = 0$$

$$\boxed{A^2 = A - I} \quad \text{--- (1)}$$

$$A^3 = A \cdot A^2$$

$$A^3 = A(A - I)$$

$$A^3 = A^2 - A$$

$$A^3 = (A - I) - A$$

$$= A - I - A$$

$$= -I \text{ Answer}$$

lecture no. 13# Eigen vector:if $Ax = \lambda x$ or $(A - \lambda I)x = 0$ $\lambda \rightarrow$ E. value, $x \rightarrow$ Eigen vector

eg $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \begin{cases} \lambda_1 = 2 & , & x_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\ \lambda_2 = 6 & , & x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{cases}$

$$Ax_1 = 2x_1, \quad Ax_2 = 6x_2$$

Now our aim to ^{know} find the procedure of finding E. vectors!

Q:) Find the E. value & E. vectors of $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$

sol:) C. eq. is $\lambda^2 - (\text{tr}(A))\lambda + |A| = 0$

$$\lambda^2 - 8\lambda + 12 = 0$$

$$\lambda = 2 \text{ \& } 6$$

E. vector for $\lambda = 2$:

Consider $Ax = \lambda x$

$$(A - \lambda I)x = 0$$

$$(A - 2I)x = 0$$

$$\begin{bmatrix} 4-2 & 2 \\ 2 & 4-2 \end{bmatrix} x = 0$$

$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} x = 0$$

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$$A_2 - A_1 \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2w_1 + 2w_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left. \begin{aligned} 2w_1 + 2w_2 &= 0 \\ 0 &= 0 \end{aligned} \right\}$$

i.e.

$$w_1 = -w_2$$

let $w_1 = k$, $w_2 = -k$

So

$$x = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = k \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

i.e.

$$\text{for } \lambda = 4, \quad x = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

⇒ Eigen vector, $\lambda = 6$:

$$Ax = \lambda x$$

$$(A - \lambda I)x = 0$$

$$(A - 6I)x = 0$$

$$\begin{bmatrix} 4-6 & 2 \\ 2 & 4-6 \end{bmatrix} x = 0$$

$$\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} x = 0$$

$$R_2 \rightarrow R_2 + R_1 \Rightarrow \begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} x = 0$$

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$$\begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -2x_1 + 2x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left. \begin{array}{l} -2x_1 + 2x_2 = 0 \\ 0 = 0 \end{array} \right\} \quad \boxed{x_1 = x_2} = k \text{ (let)}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} k \\ k \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

i.e.

$$\text{for } \lambda = 6, \quad x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Q.) Find the eigenvalues and eigen vectors of
 $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$?

Ans.) A is U.T.M so $\lambda = 3, 2, 2$ are the eigenvalues.
E-vectors ($\lambda = 3$) Consider $Bx = \lambda x$ or $(A - \lambda I)x = 0$
 $(A - 3I)x = 0$

$$\begin{bmatrix} 2-3 & 1 & 0 \\ 0 & 2-3 & 0 \\ 0 & 0 & 3-3 \end{bmatrix} x = 0$$

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} x = 0$$

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$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -x_1 + x_2 \\ -x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -x_1 + x_2 &= 0 \\ -x_2 &= 0 \\ 0 &= 0 \end{aligned}$$

$$\begin{aligned} x_1 &= x_2 = 0 \\ x_2 &= 0 \end{aligned}$$

Let $x_3 = k$ then $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ k \end{bmatrix} = k \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

∴ for $\lambda = 3$, $x = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

So, null

∴ for $\lambda = 3$, $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Q:) Find the E. values & E. vectors of $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

sol:) ∵ A is U.T.M so $\lambda = 2, 2, 3$

E. vector for $\lambda = 2$

$$Ax = \lambda x$$

$$(A - \lambda I)x = 0$$

$$(A - 2I)x = 0$$

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} x = 0$$

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$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -x_1 + x_3 \\ -x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left. \begin{array}{l} -x_1 + x_3 = 0 \\ -x_2 = 0 \end{array} \right\} \begin{array}{l} x_1 = x_3 = k \\ x_2 = 0 \end{array}$$

$$x_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} k \\ 0 \\ k \end{bmatrix} = k \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

So,

eigen vector for $\lambda = 1$, $x = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

eigen vector for $\lambda = 2$

$$Ax = \lambda x$$

$$(A - \lambda I)x = 0$$

$$(A - 2I)x = 0$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_2 = R_1$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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$$\begin{bmatrix} n_3 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$n_3 = 0, \text{ let } n_1 = k_1, \quad n_2 = k_2$$

$$X = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} k_1 \\ k_2 \\ 0 \end{bmatrix} = \begin{bmatrix} k_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ k_2 \\ 0 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

i.e. $\lambda = 2 \rightarrow X_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \lambda = 2, 2 \rightarrow (L)^2$
 $X_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

Properties of eigen vector:-

1.) If X is an eigen vector for λ then (kX) is also an E-vector for λ . ($k \neq 0$)

2.) E-vectors corresponding to different E-value of symm mat are orthogonal & hence they are LI also

Ex.) if A is 3×3 s.t. $A^T = A$ and a, b are the E-val & corresponding E-vector are $\begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$ & $\begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$ then for $a \neq b$ the value of $n_1 r_1 + n_2 r_2 + n_3 r_3 = 0$

Ex.) By above property, $X \cdot X = 0 \Rightarrow n_1 r_1 + n_2 r_2 + n_3 r_3 = 0$
 So

$$\begin{aligned} \text{Am of } 3 &= 1 & \text{Am of } 2 &= 2 \\ \text{Gm of } 3 &= 1 & \text{Gm of } 2 &= 1 \end{aligned}$$

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3.) Shortcut method of finding E vectors:
(if options are given)
"Should be unique in all comparisons there is not be any invalid result".

4.) For Different E values, corresponding E vectors also E.T

5.) if E value repeats (then headache will start) then corresponding E vectors may be L.T or may be L.O.

6.) Model Matrix: Matrix formed by E vector is called model mat

Let $A_{3 \times 3}$ is the given mat & x_1, x_2, x_3 are the E vectors then model mat is $P = [x_1, x_2, x_3]$

Q:) Find Eigen values & Eigen vectors of A

$$= \begin{bmatrix} 0 & 0 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

& Hence find Am, Gm

Sol:) A is U.T.M so $\lambda = 0, 0, 0$, Hence Am of $\lambda = 0$ three

Eigen vector for $\lambda = 0$ $Ax = \lambda x$ $(A - \lambda I)x = 0$ $(A - 0I)x = 0$ $Ax = 0$	M-L Gm $G_m(\lambda = 0)$ $= \text{order} - \rho(A - 0I)$ $= 3 - \rho(A)$ $= 3 - 1 = \text{two H}$
---	--

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$$\begin{bmatrix} 0 & 0 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 5x_3 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$5x_3 = 0 \Rightarrow x_3 = 0, \quad x_1 = k_1, \quad x_2 = k_2$$

$$Y = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} k_1 \\ k_2 \\ 0 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

i.e.

for $\lambda = 0$ we have

$$Y_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$Y_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Hence G.M. of $\lambda = 0 = \text{two}$. $\square$

7.) Algebraic multiplicity (A.M.)

Number of times particular eigenvalue repeat is called its A.M.

8.) Geometric multiplicity (G.M.)

Number of L.I. E. vector (or particular E. value λ) is called its G.M.

9.) Shortcut to finding G.M. (for λ):

$$\text{G.M. of } \lambda = \text{order} - P(A - \lambda I)$$

Let λ is the repeated E. value then G.M. of $\lambda =$

$$\text{order} - P(A - \lambda I)$$

no. of L.I. E. vectors of λ

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Note: For Non-depended E-value $\rightarrow$ $AM = 1$
 $\rightarrow$ $GM = 1$

Q.) one of the E-vector of $A = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 5 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 1 \end{bmatrix}$ will be

a.) $\begin{bmatrix} 1 \\ -1 \\ 3 \\ 0 \end{bmatrix}$ b.) $\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$

(ii) Also find the corresponding E-value.

c.) $\begin{bmatrix} 2 \\ 3 \\ 0 \\ -1 \end{bmatrix}$ d.) $\begin{bmatrix} 1 \\ -1 \\ 2 \\ 3 \end{bmatrix}$

Sol.) Taking options

$$Ax = \lambda x$$

$$\begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 5 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 3 \\ 0 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ -1 \\ 3 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 5 \\ 5 \\ 6 \\ 9 \end{bmatrix} = \begin{bmatrix} \lambda \\ -\lambda \\ 3\lambda \\ 0 \end{bmatrix} \Rightarrow \lambda = 5, \lambda = -5/4$$

$\lambda = 2$ (Not unique) $\lambda = 0$ (considered)

c.) cannot be E-vector

Taking option b.

$$Ax = \lambda x$$

$$\begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 5 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\begin{bmatrix} 5-\lambda & 0 & 0 & 0 \\ 0 & 5-\lambda & 5 & 0 \\ 0 & 0 & 2-\lambda & 1 \\ 0 & 0 & 0 & 1-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\lambda = 5$
 $\lambda = 5$
 $0 = 0$
 $0 = 0$

} unique
 } Same.
 } valid.

(ii) Corresponding Eigen value is 5
 $\therefore (Ax = 5x)$

Q:) The E-vectors of $A = \begin{bmatrix} 0 & 0 & \alpha \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\alpha \neq 0$ is/are

- a) $(1, 0, 0)'$ ✓
 b) $(0, 0, 1)'$
 c) $(0, \alpha, 0)'$ ✓
 d) $(0, 0, \alpha)'$

Taking option a.

$$Ax = \lambda x$$

$$\begin{bmatrix} 0 & 0 & \alpha \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & \alpha \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda x = 0, \alpha = 0, 0 = 0$$

$$\alpha \neq 0, \lambda = 0$$

valid

unique
Same

one value valid.

Special

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Taking option (b)

$$Ax = \lambda x$$

$$\begin{bmatrix} 0 & 0 & x \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \lambda \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

option b is
not eigen
vector!

$$\begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \lambda \end{bmatrix}$$

$x=0$, $0=0$, $\lambda=0$
invalid, valid, unique

Taking option (c) ✓

Taking option (d) ✗

(H.O.)

Q:) For a matrix $A = \begin{bmatrix} 3 & -2 & 2 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ one of the eigen value is equal to -2 which of the following is an eigen vector

a.) $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ b.) $\begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$ c.) $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ d.) $\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$ ✓

Q2.) Taking options.

Q:) The vector $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ is an eigen vector of $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -4 & 0 \end{bmatrix}$ then corresponding eigen value of A is

a.) 1 b.) 2 c.) 5, d.) -1

Spinal

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Sol:) $AX = \begin{bmatrix} -2 & 2 & -1 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} -2 & 2 & -1 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ -3 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

⇒ one pair of L.I. E. vectors of $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ will be $-5X$

a.) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ ∴ there are L.O

$AX_1 = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$

b.) $\begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ ∴ there are L.O

c.) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ ∴ there are L.I. But 2nd is not an E. vector

d.) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ 4 \end{bmatrix}$ ∴ there are L.I. as well as E. vector ✓

Q:) Find the E. value & E. vectors of $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$

if one pair of L.I. E. vectors of $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ is

$\begin{bmatrix} 1 \\ a \end{bmatrix}$ & $\begin{bmatrix} 1 \\ b \end{bmatrix}$ then $a+b = 1-0.5 = 0.5$

one pair of L.I. E. vectors for $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ will be.

Sol:) a.) $\begin{bmatrix} 1 & 0 \end{bmatrix}, \begin{bmatrix} -2 & 0 \end{bmatrix}$

b.) $\begin{bmatrix} 2 & 1 \end{bmatrix}, \begin{bmatrix} -2 & -1 \end{bmatrix}$

c.) $\begin{bmatrix} 4 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 3 \end{bmatrix}$

d.) $\begin{bmatrix} 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \end{bmatrix}$

P.T.O

Date

Some confusion (most imp)

- 1.) Number of L.I. Row vectors of $A = \rho(A)$
- 2.) No. of L.I. column vectors of $A = \rho(A)$
- 3.) No. of L.I. solutions of homogeneous system $(Ax=0)$ will be = $\left[\text{No. of columns} - \rho(A) \right]$ (nullity)
- 4.) No. of L.I. Eigen vectors for repeated eigen value $(\lambda) = \left[\text{order} - \rho(A - \lambda I) \right]$ (dim of λ)

Lecture no. 14

Similar Matrices:

Two matrices A & B are said to be similar then exist a relationship of the type $P^{-1}AP = B$ where P is any invertible mat.

Note:

- 1.) Similar matrices have same trace $\nearrow$ same eigen values
 det same char eq, submatrices will have same Rank while equivalent matrices have same Rank & other properties may be diff

2.) Similarity relation b/w matrices is an equivalence relation.

i.e. Reflexive, symmetric as well as transitive.

Date

e.g. verify the process of diagonalisation for

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

Sol: we know that $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \rightarrow \lambda = 6, \lambda_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
 $\rightarrow \lambda = 2, \lambda_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

So, modal matrix = $[x_1 x_2]$

$$= \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

Consider modal mat $P = [x_1 x_2] = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$ &

Let, $D = \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix} = \text{Diag mat}$

Since $m = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow m^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ s.d

$$P^{-1} = \frac{1}{(-2)} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

Now, $P^{-1}AP =$

$$\frac{1}{-2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix} = D \quad P^{-1}AP = D$$

Use: Now calculating $PDP^{-1} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$

$$= \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} = A$$

Spiral

Diagonalisation:

A is called Diagonalisation if it is similar to Diagonal mat.

i.e. $\exists P$ s.t. $P^{-1}AP = D$ & this process is called Diagonalisation

Consider $A_{3 \times 3}$ & let it's E-value are $\lambda_1, \lambda_2, \lambda_3$ & Eigen vector are x_1, x_2, x_3

then, ^{mat}mat is a $P = [x_1 \ x_2 \ x_3]$ & let $D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$ & let $|P| \neq 0$

let us try to calculate

$$P^{-1}AP = [P^{-1}] [A] [P] = \text{---} \text{---} \text{---} D$$

i.e. $P^{-1}AP = \text{Diag mat}$

this is very special property of mat & this property is called Diagonalisation mat.

Note

1.) $P^{-1}AP = D$ i.e. by calculating $PDP^{-1} = A$

$$P(P^{-1}AP)P^{-1} = PDP^{-1}$$

$$IAS = PDP^{-1}$$

$$A = PDP^{-1}$$

2.) N^o cond for Diagonalisation

$$\boxed{\text{Number of L.I. E-vectors} = \text{order of } A}$$

i.e. ^{mat}mat must be non-sing or $|P| \neq 0$

3.) if E-vectors are L.O then ^{mat}mat become singular & Diagonalisation is not possible.

$$Ax = \lambda x$$

$$P^{-1}AP = D \Rightarrow \lambda$$

Q: if $A_{2 \times 2}$ s.t. $A = \begin{bmatrix} -1 & 2 \\ 1 & 2 \end{bmatrix}$ & $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ then $A = ?$

a.) $\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$ $\lambda = 2 \Rightarrow \begin{bmatrix} -1 & 2 \\ 1 & 2 \end{bmatrix}$ & $\lambda_2 = 6, x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

b.) $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ $0 = \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix}$ & $P = [x_1, x_2] = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$

c.) $\begin{bmatrix} -2 & 4 \\ 2 & 6 \end{bmatrix}$ $P^{-1} = (-1) \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$

Now using the concept of diagonal

d.) $\begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix}$ $P^{-1}AP = D \Rightarrow A = PDP^{-1}$
 $A = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

Q: if $A_{2 \times 2}$ s.t. $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $A \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ then find $A = ?$

a.) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ b.) $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ c.) $\begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$ d.) $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$

$$\lambda_1 = 1, y_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \lambda_2 = 2, y_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, P = [y_1, y_2] = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, P^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$P^{-1}AP = D \Rightarrow A = PDP^{-1} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

Spiral

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Q:) The number of LI eigen vectors of the mat

$$A = \begin{bmatrix} 2 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Is

Doubt

a.) 1

b.) 2

c.) 3

d.) 4

$$\text{Tr} A = 10$$

$$|A| = -24$$

A has 4 different E values

$\Rightarrow$ " " 4 LI eigen vector = order 30 Diag

L-U Decomposition:

If we want to factorize any square mat into the product of LTM & UTM then we have following method.

1.) Doolittle Method

$$A = (\text{unit LTM}) (UTM)$$

$$A_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix} \begin{bmatrix} d & e & f \\ 0 & g & h \\ 0 & 0 & i \end{bmatrix}$$

2.) Crout method

$$A = (LTM) (\text{unit UTM})$$

$$\begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix} \begin{bmatrix} 1 & g & h \\ 0 & 1 & i \\ 0 & 0 & 1 \end{bmatrix}$$

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3) Cholesky method

$$A = L L^T$$

or

$$A = L L^T$$

$$L = L \cdot T \cdot M$$

Q: If $A = \begin{bmatrix} 1 & 2 \\ 2 & 13 \end{bmatrix}$ is to be decomposed into the product of $L \cdot T \cdot M$ & $U \cdot T \cdot M$ then properly decomposed L & U matrices are

M-1 Using Doolittle method.

Write $A = (\text{unit } L \cdot T \cdot M) U \cdot T \cdot M$

$$A = \begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix} \begin{bmatrix} b & c \\ 0 & d \end{bmatrix} \quad \text{--- (1)}$$

$$\begin{pmatrix} 1 & 2 \\ 2 & 13 \end{pmatrix} = \begin{bmatrix} b & c \\ ab & ac+d \end{bmatrix}$$

$$b=1, \quad c=2, \quad ab=2$$

$$ac+d=13 \quad \therefore \quad a=2 \quad \& \quad d=9$$

$$\text{So } A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 9 \end{bmatrix} = \text{(a)}$$

~~$P \cdot T \cdot M$~~

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m-2 Crout method:

$$A = (L \cdot T \cdot M) \text{ (unit d.r.m)}$$

$$A = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix} \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix} \quad \text{--- ①}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 13 \end{bmatrix} = \begin{bmatrix} a & ad \\ b & bd+c \end{bmatrix}$$

$$a=1, \quad b=2, \quad \underset{d=2}{ad=2}, \quad \underset{c=9}{bd+c=13}$$

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad \text{--- ②} = A$$

m-3 Cholesky method:

$$A = LL^T \text{ or } LL^T$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 13 \end{bmatrix} = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix} \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \quad \text{--- ①}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 13 \end{bmatrix} = \begin{bmatrix} a^2 & ab \\ ab & b^2+c^2 \end{bmatrix}$$

$$a^2 = 1, \quad ab = 2, \quad b^2+c^2 = 13$$

$$a=1, \quad b=2, \quad c=3$$

Ans ①

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{--- ②}$$

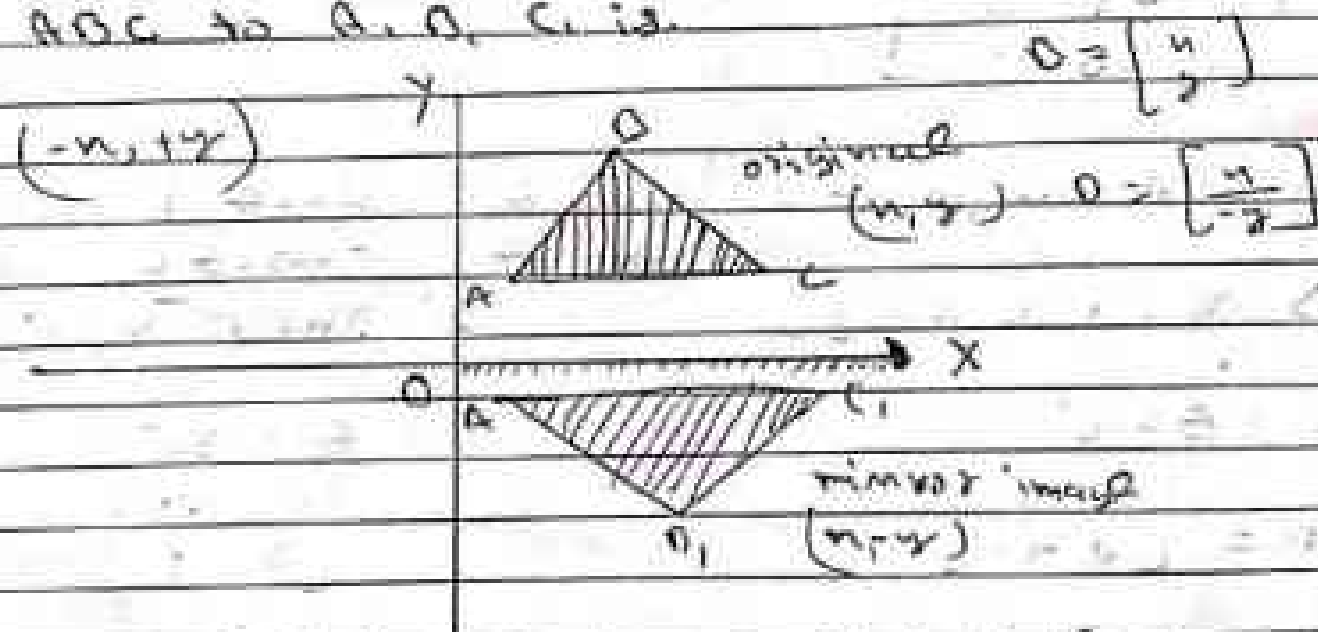
Date

Q:) In the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 \\ 9 & 5 \end{bmatrix}$ if the diagonal element of U are both 1 then the lower diagonal entry L_{21} of L is

sol:) $L_{21} = 5$

Reflection matrix:

Q:) The figure shows a shape ABC and its mirror image A₁B₁C₁ across the horizontal axis (x-axis). The coordinate transform matrix that maps ABC to A₁B₁C₁ is.



a.) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} n \\ y \end{bmatrix} = \begin{bmatrix} y \\ n \end{bmatrix}$ c.) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} n \\ y \end{bmatrix} = \begin{bmatrix} -n \\ y \end{bmatrix}$

b.) $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} n \\ y \end{bmatrix} = \begin{bmatrix} y \\ -n \end{bmatrix}$ d.) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} n \\ y \end{bmatrix} = \begin{bmatrix} n \\ -y \end{bmatrix}$

option (d) is given mirror image about y axis.

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Q.) The transformation matrix to find mirror image of any point $P \begin{bmatrix} x \\ y \end{bmatrix}$ in $x-y$ plane about the line $y=x$

a.) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\begin{matrix} \text{not} \\ y=x \end{matrix}$

b.) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ✓

c.) $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

d.) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

⇒ with angle $\theta = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$

1.) About x axis

2.) About y axis

$\theta = 0^\circ$

R.M. = $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\theta = 90^\circ$

R.M. = $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

3.) About $y=x$

4.) About $y=-x$

$\theta = 45^\circ$

R.M. = $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$\theta = -45^\circ$

R.M. = $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

Special case: About origin

R.M. = $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

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Rotation Matrix

Q:) if the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is to be rotated by 45° in an ACW sense then what are the coordinates of new vector?

- a.) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ b.) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ c.) $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$ d.) $\begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix}$ ✓



$|\vec{OB}| = |\vec{OA}| = \sqrt{2} \Rightarrow \vec{OB} = \begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix}$

M-2

$$R_{ACW} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Here $\theta = 45^\circ$ so

$$\begin{aligned} \vec{OB} = \begin{bmatrix} x' \\ y' \end{bmatrix} &= R_{ACW} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix} \quad \text{Ans.} \end{aligned}$$

P.T.O.

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Note: if vector is rotated by angle θ in CW sense then replace $\theta \rightarrow -\theta$ in Rot mat.

$$\text{i.e. } R = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$= R \begin{bmatrix} x \\ y \end{bmatrix}$$

$\Rightarrow$ Rotation mat in 2D in ACW sense by an angle θ .

Case - 1 (if vector is rotating)

1.) About X axis

2.) About Y axis

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$R_y = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

3.) About Z axis

$$R_z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow$ if coordinate system is rotating $\rightarrow$ (in ACW sense) take transpose of case I.

1.) Rotation matrix are orthogonal matrix

2.) $\det(\text{Rotation mat}) = 1 + 2\cos\theta$

2

Spiral

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Q: If the vector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is to be rotated by 30° in the ACW about x-axis then find new vector.

Ans.

$$\begin{bmatrix} 1 \\ \sqrt{3}/2 \\ \sqrt{3}/2 \end{bmatrix}$$

✓

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Linear Algebra - II

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lecture no. 01# Quadratic Form

=> General observation:

Consider a vector $x = \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{3 \times 1}$ & let $A = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}_{3 \times 3}$ let us calculate: $x^T A x = \begin{bmatrix} - & - & - \end{bmatrix}_{1 \times 3} \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}_{3 \times 3} \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{3 \times 1}$
 $= \begin{bmatrix} - & - & - \end{bmatrix}_{1 \times 3} \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{3 \times 1}$ $Q(x) = \begin{bmatrix} - \end{bmatrix}_{1 \times 1} = \text{Scalar quantity.}$ Again considering $x = \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{n \times 1}$, $A = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}_{n \times n}$ then $x^T A x = \begin{bmatrix} - & - & - \end{bmatrix}_{1 \times n} \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}_{n \times n} \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{n \times 1}$
 $= \begin{bmatrix} - & - & - \end{bmatrix}_{1 \times n} \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{n \times 1}$ $Q(x) = \begin{bmatrix} - \end{bmatrix}_{1 \times 1} = \text{Scalar quantity.}$ Later on this concept of converting x into a value is possible when A is a symm mat.

Quadratic Form: "It is a process of assigning value to any column mat"

& it is denoted as $Q(x)$ and is define as $Q(x) = x^T A x = \begin{bmatrix} - \end{bmatrix}_{1 \times 1} = \text{constant / scalar value}$ where A is called matrix associated with Quadratic Form

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Note:

1.) Here A is always a symmetric mat.2.) if $A = I$ then $Q(x) = x^T A x = x^T I x = x^T x = \|x\|^2$
i.e.if A is an I then $Q(x) = (\text{Norm } x)^2$
or

$$Q(x) = \|x\|^2$$

where $\|x\| = \sqrt{x^T x}$

$$= x_1^2 + x_2^2 + \dots + x_n^2 = \|x\|^2$$

e.g. if $A = \begin{bmatrix} 7 & 4 \\ 4 & 5 \end{bmatrix}_{2 \times 2}$ then find quad form of
 $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1}$

$$\begin{aligned} \text{Sol:)} \quad Q(x) &= x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix}_{1 \times 2} \begin{bmatrix} 7 & 4 \\ 4 & 5 \end{bmatrix}_{2 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1} \\ &= \begin{bmatrix} (7x_1 + 4x_2) & (4x_1 + 5x_2) \end{bmatrix}_{1 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1} \\ &= \begin{bmatrix} 7x_1^2 + 4x_2x_1 + 4x_1x_2 + 5x_2^2 \end{bmatrix} \\ &= 7x_1^2 + 8x_1x_2 + 5x_2^2 \end{aligned}$$

(ii) if $x = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ then $Q(x) =$

$$\begin{aligned} &= 7(3)^2 + 8(3)(-1) + 5(-1)^2 \\ &= 63 - 24 + 5 = 44 \text{ Ans} \end{aligned}$$

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 $\Rightarrow$ Type - 1

Q:) Find the Quadratic forms of

1.) $A = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$

2.) $A = \begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix}$

Sol:) let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ then

$$Q(x) = x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & -2x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= (x_1^2 - 2x_2^2) \quad \text{Ans}$$

$$Q(x) = x_1^2 - 2x_2^2$$

2.)

$$Q(x) = x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} (3x_1 - x_2) & (-x_1 + 4x_2) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= 3x_1^2 - x_2^2$$

$$= (3x_1^2 - x_1 x_2 - x_1 x_2 + 4x_2^2) \quad \text{Ans}$$

$$= 3x_1^2 - 2x_1 x_2 + 4x_2^2$$

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Q:) Find the quadratic forms

$$1.) A = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 4 & 3 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\text{Sol: let } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ then}$$

$$Q(x) = x^T A x = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 2 & 0 & -1 \\ 0 & 4 & 3 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= \begin{bmatrix} 2x_1^2 - x_1x_3 & 4x_2^2 + 3x_2x_3 & -x_1x_3 + 3x_2x_3 + x_3^2 \end{bmatrix}$$

$$= \begin{bmatrix} 2x_1^2 - x_1x_3 + 4x_2^2 + 3x_2x_3 + (-x_1x_3 + 3x_2x_3 + x_3^2) \end{bmatrix}$$

$$= \boxed{2x_1^2 + 4x_2^2 + x_3^2 - 2x_1x_3 + 6x_2x_3} \quad \text{A}$$

$$2.) \begin{bmatrix} 1 & -3 & 0 & 1 \\ -3 & 0 & 2 & 4 \\ 0 & 2 & 3 & 5 \\ 1 & 4 & 5 & -3 \end{bmatrix}$$

$$Q(x) = x^T A x = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 & 1 \\ -3 & 0 & 2 & 4 \\ 0 & 2 & 3 & 5 \\ 1 & 4 & 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$= x_1^2 + 3x_2^2 - 3x_3^2 + 6x_1x_2 + 2x_1x_3 + 4x_1x_4 + 10x_2x_3 + 10x_2x_4 + 10x_3x_4 - 3x_4^2$$

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Q:) if $A = \begin{bmatrix} 2 & -1 & 3 & 4 \\ -1 & 1 & 5 & 0 \\ 3 & 5 & 3 & 1.5 \\ 4 & 0 & 1.5 & -4 \end{bmatrix}$ & $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ then $Q(x) =$

sol:) $Q(x) = x^T A x =$

$$2x_1^2 + x_2^2 + 3x_3^2 - 4x_4^2 - 2x_1x_2 + 6x_1x_3 + 8x_1x_4 + 10x_2x_3 + 3x_3x_4$$

Type - 2

Q:) Find the symmetric matrix that is associated with the following Quadratic form

1) $x_1^2 - 5x_1x_2 + 3x_2^2$

sol: $Q(x) = 1 \cdot x_1^2 - \frac{5}{2}x_1x_2 - \frac{5}{2}x_2x_1 + 3x_2^2$

$$= x^T A x$$

$$= \begin{bmatrix} 1 & -5/2 \\ -5/2 & 3 \end{bmatrix}$$

= Symm

2) $x_1^2 + 4x_2^2 - 2x_1x_2 + 5x_3x_4$

sol:) $Q(x) = x^T A x$

$$= 1 \cdot x_1^2 + 0x_1^2 + 4x_2^2 + 0x_3^2 + 2x_1x_2 - 2x_1x_2 + 0x_1x_3 + 0x_1x_4 + 0x_2x_3 + 5x_2x_4 + 0x_3x_4$$

where $A =$

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 4 & 0 & 5/2 \\ 0 & 0 & 4 & 0 \\ 0 & 5/2 & 0 & 0 \end{bmatrix}$$

= Symm matrix

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3.) $x^2 + 2y^2 - 3z^2 + 4xy - 5yz + 3zx$

sol: $Q(x) = x^2 + 2y^2 - 3z^2 + 4xy - 5yz + 3zx$

$$= x_1^2 + 2x_2^2 - 3x_3^2 + 4x_1x_2 - 5x_2x_3 + 3x_3x_1 = ?$$

$$= 1 \cdot x_1^2 + 2x_2^2 - 3x_3^2 + 4x_1x_2 - 5x_2x_3 + 3x_3x_1$$

$$\begin{bmatrix} 1 & 2 & 3/2 \\ 2 & 2 & -5/2 \\ 3/2 & -5/2 & -3 \end{bmatrix} = \text{sym}$$

Q:) if Quad form of x is given as $Q(x) = a^2 - 2b^2 + 3c^2 + d^2 - ab + 4bc - 2cd + 2ad$ then find symm mat associated with above Q form

sol:) $Q(x) = x_1^2 - 2x_2^2 + 3x_3^2 + x_4^2 + x_1x_2 + 4x_2x_3 - 2x_3x_4 + 2x_1x_4$

$$A = \begin{bmatrix} 1 & 1/2 & 0 & 1 \\ -1/2 & -2 & 2 & 0 \\ 0 & 2 & 3 & -1 \\ 1 & 0 & -1 & 1 \end{bmatrix} = \text{symm mat.}$$

Type - 3

Q: if quadratic form of any vector x is given as:

1.) $x_1^2 - 4x_2^2 + 3x_1x_2$ then evaluate $Q \begin{bmatrix} 1 \\ -3 \end{bmatrix} = ?$

sol:) $(1)^2 - 4(-3)^2 + 3(1)(-3)$

$$= 1 - 36 - 9 = -44 \text{ A}$$

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2) $x^2 + 4y^2 - z^2 + 2zx + 4xy$ then evaluate Q

$$\text{sol: } = (1)^2 + 4(2)^2 - (-3)^2 + 2(-3)(1) + 4(1)(2)$$

$$= 1 + 16 - 9 - 6 + 8$$

$$= 10 \quad \text{Ans.}$$

Relation b/w Quadratic form & Norm of x :

$$\text{let } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{then } \|x\| = \sqrt{x^T x} = \sqrt{(x_1^2 + x_2^2 + x_3^2)}$$

$$\text{i.e. } x^T x = \|x\|^2$$

let us take a symm mat of the type

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{say}$$

& we will try to calculate $Q(x) = ?$

$$Q(x) = x^T A x = x^T I x = x^T x = \|x\|^2$$

Conclusion: if $A = I$ then $Q(x) = \|x\|^2$

$$\text{e.g. if } A = I \text{ \& } x = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} \text{ then } Q(x) = ?$$

$$= 3^2 + (-1)^2 + 1^2 = 14 \quad \text{Ans.}$$

Maximum and minimum value of quadratic form $Q(x)$ (when $\|x\| = 1$)

let $A = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}_{3 \times 3}$ = symm mat $\begin{cases} \lambda_1, \lambda_2, \lambda_3 \text{ are } \lambda_i \\ \text{w } N_i = \frac{x_i}{\|x_i\|} \end{cases}$

let $\lambda_{\min} = \lambda_1$ & $\lambda_{\max} = \lambda_3$
Then min value of $Q(x)$ will be $= \lambda_{\min}$.
max value of $Q(x) = \lambda_{\max}$

i.e. $\lambda_{\min} \leq Q(x) \leq \lambda_{\max}$

& these min and max values of quadratic form occur at unit eigen vector of $\lambda_{\min}$ & $\lambda_{\max}$ respectively

(e.g.) let $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ & $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be any vector s.t. $\|x\| = 1$ then min & max value of $Q(x)$ for $x =$

sol: $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \begin{cases} \lambda_1 = 4, & x_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\ \lambda_2 = 6, & x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{cases} \quad N_1 = \frac{x_1}{\|x_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$N_2 = \frac{x_2}{\|x_2\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$Q(x) = x^T A x = 4x_1^2 + 4x_2^2$

Now $Q(N_1) = N_1^T A N_1 = 4 \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}^T \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = 4 \left(\frac{1}{2} \right) + 4 \left(\frac{-1}{2} \right) \left(\frac{1}{2} \right) = 2 - 2 = 0$
 $\lambda_{\min} = 0$

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Again

$$Q(u) = u^T A u = 0 \begin{bmatrix} 1/u_1 \\ 1/u_2 \end{bmatrix} = 4 \times \frac{1}{u_1} + 4 \times \frac{1}{u_2} + \frac{1}{u_1} + 4 \times \frac{1}{u_2} \\ = 2 + 2 + 2 = 6 = \lambda_{\max}$$

i.e.

$$2 \leq Q(x) \leq 6$$

Now let us take another random vector say

$$x = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}, \|x\| = 1$$

$$Q(x) = 4x_1^2 + 4x_2^2 + 4x_1^2$$

$$= 4 \times \left(\frac{3}{5}\right)^2 + 4 \times \frac{3}{5} \times \frac{4}{5} + 4 \times \left(\frac{4}{5}\right)^2$$

$$= \frac{36 + 48 + 64}{25} = \frac{148}{25} = 5.92 \text{ Hence verified}$$

Type - 4

Q:) For the matrix $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$, the max and min value of $x^T A x$ will be subject to the constraint $x^T x = 1 \Rightarrow \|x\| = 1$ (given)

Sol:) Already discussed:

Q:) For the matrix $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$, max & min value of $x^T A x$ lies on the interval (provided unit)

a) $[1, 6]$ ✓

b) $[-5, 5]$

c) $[1, 7]$ ✓

d) $[2, 2]$

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Q:) Min & max values of $\phi(x)$ on $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ lies in b/w

A is v. m. $\lambda = 1, 2$

min max

a.) $(-1, 1)$

b.) $(1, 5)$

c.) $(1, 2)$

d.) $(2, 5)$

e.) (None) ✓

wrong Data given in Question.

$A = \text{No 1 Symm}$

Relation b/w Quadratic form and eigen values

$\phi(x) > 0$ +ve definite. if All the E values are +ve

$\phi(x) \geq 0$ semi +ve definite. " " " " " " -ve

$\phi(x) < 0$ -ve definite

$\phi(x) \leq 0$ semi -ve definite

$\phi(x) > 0$ & < 0 ind. definite. if all the E values are +ve as well as -ve.

Positive definite mat $\Rightarrow$ if all the E values are +ve then A is called +ve definite

Negative definite mat $\Rightarrow$ if all the E values are -ve then A is called -ve definite

Indefinite mat $\Rightarrow$ if E values are +ve as well as -ve then A is Indefinite mat

Semi +ve Definite $\Rightarrow$ if E values are 0 or +ve then A is called +ve Definite mat.

Type 5:

Q:) The nature of $\phi(x) = 3x_1^2 + 2x_2^2 + x_3^2 + 4x_1x_2 + 4x_1x_3$ will be?

a.) +ve definite

b.) -ve definite

c.) Indefinite ✓

d.) None of these

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Sol:)

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 1 \end{bmatrix}_{3 \times 3}$$

$\lambda_1 + \lambda_2 + \lambda_3 = 6$ (+ve)
 $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 = -10$ (-ve)

Case-1: $\Delta = +ve \Rightarrow$ All 3 can't be -ve

$|\Delta| = -ve \Rightarrow$

- one +ve & two -ve ✓
- two negative one +ve (N.P.) $\therefore |\Delta| =$
- all three -ve (N.P.) $\therefore \Delta = +ve$

i.e. A has -ve as well as +ve Δ value $\therefore$
 $Q(x) = \text{Indefinite.}$

M-1

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 1 \end{bmatrix}_{3 \times 3}$$

C.O.F. is $|A - \lambda I| = 0$

$$(\lambda + 1)(\lambda - 5)(\lambda - 1) = 0$$

$$\lambda = -1, 3, 5$$

i.e. $Q(x) = \text{Indefinite.}$

Q:) Let $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ which of the following statement is true regarding the quadratic form $Q(x) = x^T A x$?

Transfer

- $Q(x)$ is positive definite.
- $Q(x)$ is negative definite.
- $Q(x)$ is indefinite.
- $Q(x)$ is positive, semi-definite.

M-1 $\lambda_1 + \lambda_2 + \lambda_3 = 8$

$$\lambda_1 \cdot \lambda_2 \cdot \lambda_3 = 8$$

all three are +ve
 two negative & 1 +ve

Spiral

$\lambda_1 + \lambda_2 + \lambda_3 = 8$

All positive ✓ Date
 two +ve, one -ve ✗
 one +ve, two -ve ✓

⇒ Not feasible method this time.

M-2

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}_{3 \times 3}$$

C. eq. is $|A - \lambda I| = 0$

$$\begin{vmatrix} (3-\lambda) & 1 & 2 \\ 1 & (2-\lambda) & 1 \\ 2 & 1 & (3-\lambda) \end{vmatrix} = 0 \quad \dots \quad \boxed{\lambda^3 - \text{tr}(A)\lambda^2 + (\dots)\lambda + (-|A|) = 0}$$

$$(3-\lambda) [(2-\lambda)(3-\lambda) - 1] - 1 [(3-\lambda) - 2] + 2 [1 - 2(1-\lambda)]$$

$$= \dots \dots \dots (\lambda - 1) (\text{and E.g. in } \lambda) = 0 =$$

$$\lambda = 1 \quad 4 - \lambda = \text{+ve, +ve}$$

So

A is +ve definite.

$$M_{11} = \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 6 - 1 = 5$$

⇒ 3x3 should be:

$$M_{22} = \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} = 9 - 4 = 5$$

$$\lambda^3 + (-\text{tr}(A))\lambda^2 + (\dots)\lambda + (-|A|) = 0$$

$$M_{33} = \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} = 6 - 1 = 5$$

$$= \lambda^3 + (-8)\lambda^2 + (15)\lambda + (-8) = 0$$

$$5 + 5 + 5 = (15)$$

~~Not possible~~

Date

Type 6

Q.1) if $Q(x) = 2x_1^2 - 2x_1x_2 + 2x_2^2$ then write it's another Quadratic form which is free cross product term.

a.) $2y_1^2 + 2y_2^2$ $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = I$ $\lambda = 1$
 b.) $y_1^2 + 3y_2^2$ $\lambda = 1$ & $\text{Tr}(A) = 4$
 c.) $y_1^2 + 2y_2^2$
 d.) $-y_1^2 + 3y_2^2$ $\lambda = 3$

i.e. eigen value of I & J

Consider $O = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$; $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ & $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$Q(y) = y^T O y$ (using principle) $= \lambda y_1^2 + \lambda y_2^2$

$= \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$= y_1^2 + 3y_2^2$

where

$Q(x) = x^T A x = y^T O y = Q(y)$
 (put $x = Py$)

Principle Axes Theorem:

Consider $A = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$ $n \times n$ symm mat

& let $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ then $Q(x) = x^T A x$.

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let $A \begin{cases} \lambda_1, x_1 \\ \lambda_2, x_2 \\ \lambda_3, x_3 \end{cases} \Rightarrow \begin{cases} N_1 = \frac{x_1}{\|x_1\|} \\ N_2 = \frac{x_2}{\|x_2\|} \\ N_3 = x_3 \end{cases}$ i.e. unit e-vector

 $\|x_3\| = 1$

More useful matrix

So A is symmetric & $\lambda_1, \lambda_2, \lambda_3$ are different (x_1, x_2, x_3) are orthogonal

So considering orthonormal matrix $P = [N_1 \ N_2 \ N_3]$ i.e. $P^{-1} = P^T$

let $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$, $D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$ let $x = PY$

if we put $x = PY$ then $Q(x) = x^T A x = \dots = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$

$\Rightarrow$ Note: $x = PY \Rightarrow Y = P^{-1}x$

Q:) The orthogonal Quadratic form of $Q(x) = 4x_1^2 + 4x_2^2 + x_3^2 - 2x_1x_2$ will be?

a.) $-3x_1^2 + 2x_2^2 + 4x_3^2$ in $\mathbb{R}^3$ $=$ $A = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

b.) $6x_1^2 + 5x_2^2 - 2x_3^2$ in $\mathbb{R}^3$, $|A| \neq 15$

c.) $2x_1^2 + 5x_2^2 + 1x_3^2$ in $\mathbb{R}^3$, $|A| = 15$

d.) $x_1^2 - x_2^2 + 6x_3^2$ in $\mathbb{R}^3$, $\text{in} = \lambda_1 + \lambda_2 + \lambda_3 = 9$
 $|A| = \lambda_1 \lambda_2 \lambda_3 = 15$

Don't worry, Now we will use the Concept of S.V.D which will be discussed after the understanding some basic information.

$\Rightarrow$ Singular values of $A \rightarrow$

sq. roots of ϵ value of $(A^T A)$ and called singular values of A .

if $A_{m \times n}$ then $(A^T A)_{n \times n}$ will have ϵ values of $A^T A$ are $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$.

then singular values of A are $\sqrt{\lambda_1}, \sqrt{\lambda_2}, \sqrt{\lambda_3}, \dots, \sqrt{\lambda_n}$.

Singular values A are $\sigma_1, \sigma_2, \sigma_3, \dots, \sigma_n$ where

$$\sigma_i = \sqrt{\lambda_i} \quad \forall i$$

e.g.: let $A_{2 \times 3}$ then $(A^T)_{3 \times 2}$ $\begin{cases} (A^T A)_{2 \times 2} \\ (A A^T)_{3 \times 3} \end{cases}$

let $\lambda_1, \lambda_2, \lambda_3$ are the ϵ values of $(A^T A)$ then $\sigma_1, \sigma_2, \sigma_3$ are the singular values of A where

$$\sigma_1 = \sqrt{\lambda_1}, \quad \sigma_2 = \sqrt{\lambda_2}, \quad \sigma_3 = \sqrt{\lambda_3}$$

$$\Rightarrow \sigma_1^2 + \sigma_2^2 + \sigma_3^2 = \lambda_1 + \lambda_2 + \lambda_3 = \text{Tr}(A^T A)$$

Q.) Find the singular values of $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$

$$\text{Sol:)} \quad A^T = \begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix}_{3 \times 2} = A^T A = \begin{bmatrix} 80 & 100 & 40 \\ 100 & 130 & 140 \\ 40 & 140 & 200 \end{bmatrix}_{3 \times 3}$$

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$$A A^T = \begin{bmatrix} 33 & 381 \\ 81 & 117 \end{bmatrix}_{2 \times 2} = \lambda^2 - 450\lambda + 32400 = 0$$

$$\lambda = 360, 90$$

$$\text{So } A^T A = \begin{bmatrix} 80 & 100 & 40 \\ 100 & 170 & 140 \\ 40 & 140 & 200 \end{bmatrix}_{3 \times 3} \Rightarrow P(A^T A) = 2 \Rightarrow |A^T A| = 0$$

$$\lambda = 0$$

= Both are symmetric
 $\Rightarrow \text{Tr}(A^T A) = \text{Tr}(A A^T)$

$$P(A) = P(A A^T) = P(A^T A) = 2$$

We know that

Non-zero E-value of $A^T A$ & $A A^T$ are same so
 Instead of calculating E-value of $(A^T A)_{3 \times 3}$ we
 will try to find the E-value $(A A^T)_{2 \times 2}$

i.e.

E-value of $A A^T$ are 360 & 90

So E-value of $(A^T A)$ are 360, 90, 0

Hence singular value of A are $\sqrt{360}, \sqrt{90}, \sqrt{0}$

$$\text{i.e. } \sigma_1 = \sqrt{360}, \sigma_2 = \sqrt{90}, \sigma_3 = \sqrt{0}$$

Q.) Find the singular value of $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$

$$\text{Sol: } A^T A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}_{2 \times 2} = 45$$

By property of eigen value $\lambda = 45$

Now $\text{Tr}(A^T A) = 50$

$$\lambda_1 + \lambda_2 = 50$$

$$45 + \lambda_2 = 50$$

$$\lambda_1 = 5$$

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E-values of $A^T A$ are 45 & 5

So

Singular values of A are $\sqrt{45}$ & $\sqrt{5}$

i.e.

$\sigma_1 = \sqrt{45}$, $\sigma_2 = \sqrt{5}$ are the singular values of

Note:

1.) E-values of $A^T A = 45$ & 5

2.) " " " $A A^T = 45$ & 5

3.) Rank = E-values of $A = 3$ & 5

Singular values of $A = \sqrt{45}$ & $\sqrt{5}$

3.) A is sq. mat (sq. observation)

$$|A| = \lambda_1 \cdot \lambda_2 = \sigma_1 \cdot \sigma_2$$

A:) Given a 3×3 matrix E with $SVD = 4 \epsilon V^T$ if E has exactly one zero singular value what can be said about determinant of E ?

a.) The det of E is zero

b.) The det of E is the product of its non-zero S

c.) The det of E is equal to sum of its singular val

d.) $\det(V) = 1$

sol: $E_{3 \times 3}$ i.e. sq. mat $= |E| = \text{Product of i. values}$
 $= \text{Product of singular val}$
 $= \sigma_1 \cdot \sigma_2 \cdot (0)$

$$= 0$$

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Q: Consider a matrix A where $A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$ the matrix $A^T A$ is:

Which of the following is correct about the eigen value of $A^T A$?

- (a) They are the sq of the singular values of A
- b.) They are the sum of the square of the singular value of A
- c.) They are the absolute value of singular value of A
- d.) They are one less than the singular values of A .

sol: Sing value of $A = \text{square root of E-value of } A^T A$
 $(\text{Sing value of } A)^2 = \text{E-value of } A^T A$

2.) Which of the following is correct about the sum of the Eigen values of $A^T A$?

$= \lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_n$
 $= \sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \dots + \sigma_n^2$
 $\Rightarrow$ Sum of the sq of the sing values

$\Rightarrow$ (b)

Q: Let A be a 2×2 matrix given by:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

The singular (SVD) of A is given by $A = U \Sigma V^T$ where U & V are orthogonal matrices and Σ is a diagonal matrix.

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matrix with non-negative entries. If the singular values of A are σ_1 and σ_2 where $\sigma_1 \geq \sigma_2 \geq 0$. What is the sum of the square of singular values of A ?

a.) 5.69

b.) 6.33

c.) 30

d.) 5.

Sol.) $\sigma_1^2 + \sigma_2^2 = \lambda_1 + \lambda_2 = \text{Tr}(A^T A)$
 $\text{Tr}(A) \quad \text{Tr}(A^T A)$

$$A^T A = \begin{pmatrix} 10 & 14 \\ 14 & 20 \end{pmatrix} = 30 = \text{Tr}(A^T A)$$

Q.) Consider a 3×3 matrix B given by

$$B = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

The singular (SVD) of B is given by $B = U \Sigma V^T$, where Σ is a diagonal mat with singular values on the diagonal. What is the sum of square of the singular value of B .

$$(B^T B)$$

a.) 29

b.) 30

c.) 31

d.) 32

$$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 = \lambda_1 + \lambda_2 + \lambda_3$$

$$= 16 + 9 + 4 = 29$$

$$= \text{Tr}(B^T B)$$

Sp. Note:

E-values of $B^T B$ are 16, 9, 4

So singular values of B are $\sqrt{16}$, $\sqrt{9}$, $\sqrt{4}$

i.e.

4, 3, 2 which are the eigen values of B

$\Rightarrow$ if A is symm mat then singular value of A = Eigen value of A (in sq. mat).

$\Rightarrow$ Note:

1) if $A_{m \times n}$ be any rectangular mat then $(A^T A)$ & $(A A^T)$ both are symmetric.

2) Non zero E value of $A^T A$ and $A A^T$ are same that why $P(A A^T) = P(A^T A)$.

3) Also we have $\text{Tr}(A A^T) = \text{Tr}(A^T A)$:

in fact $AB \neq BA$ But $\text{Tr}(AB) = \text{Tr}(BA)$

4) Let E-value of $A^T A$ is λ_i then singular values, A is $\sigma_i = \sqrt{\lambda_i}$ & i

5) if A is square mat of $n \times n$ then

$$|A| = \lambda_1 \cdot \lambda_2 \cdot \lambda_3 \cdots \lambda_n = \sigma_1 \cdot \sigma_2 \cdot \sigma_3 \cdots \sigma_n$$

E-value of A

7) if $A_{n \times n}$ & A is symmetric then
singular value of A = Eigen values of A

8) if $A_{m \times n}$ then $(A^T A)_{n \times n}$

Sum of the square of sing value of A = Sum of eigen values

$$A^T A = \text{Tr}(A^T A)$$

Date

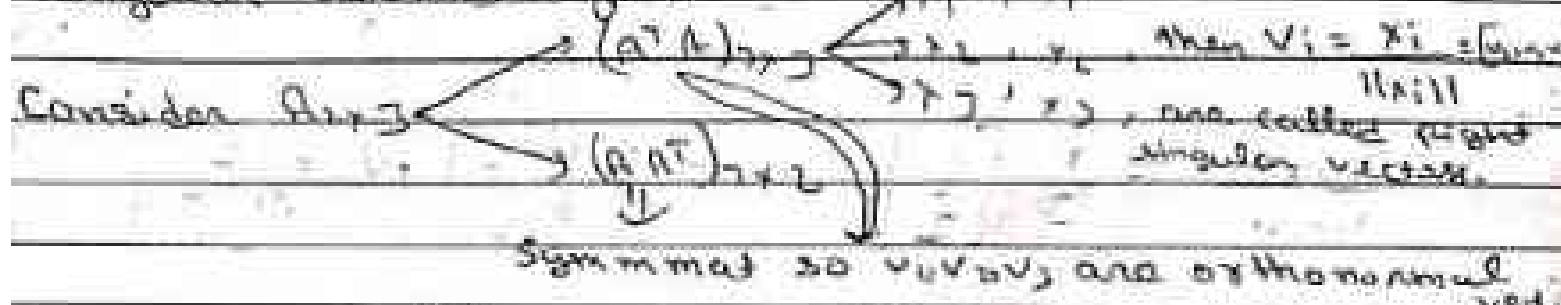
i.e.

$$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \dots + \sigma_n^2 = \lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_n = \text{Tr}(A^T A)$$

Right and left singular vector of A.

if A is $m \times n$ then unit E -vector of $A^T A$ are called right singular vectors of A.

Also E unit E -vectors of AA^T are called left singular vectors of A.

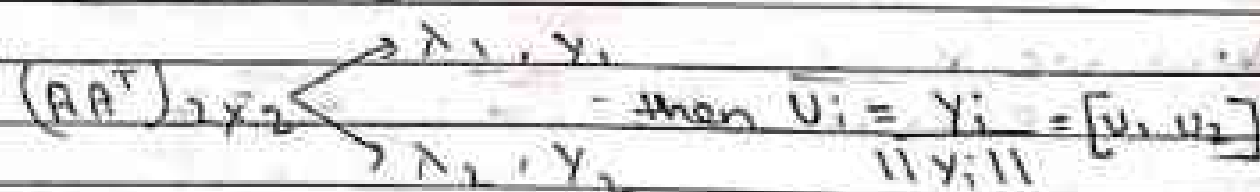


$$\text{i.e. } v_i \cdot v_j = v_i^T v_j = 0 \quad \forall i \neq j$$

$$\& v_i \cdot v_i = v_i^T v_i = 1$$

In short $v_i^T v_j = 0, v_i^T v_i = 1$

Similarly:



$\therefore (AA^T)$ is symm so $u_1 \& u_2$ are also orthonormal vectors.

$$\text{i.e. } u_i \cdot u_j = u_i^T u_j = 0 \quad \& u_i \cdot u_i = u_i^T u_i = 1 \quad \forall i \neq j$$

In short

Right singular vectors $\cdot V = [v_1, v_2, v_3]$

left " " " $U = [u_1, u_2, u_3]$

$\Rightarrow$ Relationship b/w v_i & $u_i \Rightarrow$

$$Av_i = \sigma_i u_i$$

$$\text{i.e. } Av_1 = \sigma_1 u_1$$

$$Av_2 = \sigma_2 u_2 \text{ Special}$$

Date:

Q:) Find the S.V.D of $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -4 \end{bmatrix}$

Sol:) $A^T A = \begin{bmatrix} 80 & 140 & 40 \\ 100 & 170 & 140 \\ 40 & 140 & 200 \end{bmatrix}$, $AA^T = \begin{bmatrix} 332 & 81 \\ 81 & 117 \end{bmatrix}$

On $A^T A \rightarrow \begin{cases} \lambda_1 = 360, \gamma_1 = [1 \ 2 \ 2] \\ \lambda_2 = 90, \gamma_2 = [-2 \ -1 \ 2] \\ \lambda_3 = 0, \gamma_3 = [2 \ -2 \ 1] \end{cases}$

So,
 $v_1 = \frac{\gamma_1}{\|\gamma_1\|} = \frac{1}{3} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, $v_2 = \frac{1}{3} \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$, $v_3 = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$

So
 set of right singular vectors
 $V = [v_1 \ v_2 \ v_3] = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

Now we have

$$\sigma_1 = \sqrt{360}, \sigma_2 = \sqrt{90}, \sigma_3 = 0$$

Now $Av_1 = \sigma_1 v_1 \Rightarrow v_1 = \frac{1}{\sigma_1} Av_1$

$$v_1 = \frac{1}{\sqrt{360}} \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}_{3 \times 1}$$

$$= \frac{1}{3\sqrt{360}} \begin{bmatrix} 54 \\ 18 \end{bmatrix}_{2 \times 1} = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$v_2 = \frac{1}{\sigma_2} Av_2 = \frac{1}{\sqrt{90}} \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -4 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$$

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$$= \frac{1}{3\sqrt{90}} \begin{bmatrix} 9 \\ -27 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

i.e. for $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$, R.Sing vector $V = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}$

sing values of A are $\sqrt{360}$, $\sqrt{90}$, 0

Left singular vector $U = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix}$

Now constructing a ^{diag} ~~mat~~ mat with the help of non-zero singular value.

$$D = \begin{bmatrix} \sqrt{360} & 0 \\ 0 & \sqrt{90} \end{bmatrix} \quad \text{Taking one } \& \text{ let special mat}$$

$$\Sigma = \begin{bmatrix} \sqrt{360} & 0 & 0 \\ 0 & \sqrt{90} & 0 \end{bmatrix}$$

Now calculating $U \Sigma V^T = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} \sqrt{360} & 0 & 0 \\ 0 & \sqrt{90} & 0 \end{bmatrix}$

$$= \dots = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} = A \quad \begin{matrix} 2 \times 3 \\ 3 \times 3 \end{matrix}$$

$$\Rightarrow U \Sigma V^T = A$$

Special matrix Involved in S.V.D $\rightarrow$

Let $\sigma_1, \sigma_2, \sigma_3$ are the non-zero singular values then

$$D = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} \quad \& \quad \Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$$

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SVD $\Rightarrow$ $A = U \Sigma V^T$ is called S.V.D of A.

$$A_{m \times n} = U_{m \times m} \Sigma_{m \times n} V_{n \times n}^T \text{ is S.V.D in proper order}$$

* Σ is special mat, include 0, which is formed by non-zero sing value of A.

Q:) obtain the S.V.D of $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$

Sol:) $A^T A = \begin{bmatrix} 9 & 32 \\ 0 & 25 \end{bmatrix} \quad \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix} \Rightarrow$

i.e. one E-value is 45 & $\lambda_2 = 5$

so other E-value is $\lambda = 5$

i.e. $A^T A = \lambda_1 = 45 \quad \lambda_2 = 5$

$$y_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad y_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Right Singular vectors $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \& \quad v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Hence $\sigma_1 = \sqrt{45} \quad \& \quad \sigma_2 = \sqrt{5}$

Right Singular vectors $V = [v_1 \ v_2] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

$Av_1 = \sigma_1 v_1 \Rightarrow u_1 = \frac{1}{\sigma_1} A v_1$

$$= \frac{1}{\sqrt{45}} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{45}} \begin{bmatrix} 3 \\ 9 \end{bmatrix}$$

$$u_1 = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Now

$$Av_2 = \sigma_2 v_2 = v_2 = \frac{1}{\sigma_2} Av_2$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

i.e.

$$v_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

So left singular vectors $U = [v_1, v_2]$

$$= \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}, \quad U = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}, \quad V = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$(\sigma_1 = \sqrt{45}, \sigma_2 = 5) \quad D = \begin{bmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

So

$$\text{S.V.D of } A \text{ is } = U \Sigma V^T = I$$

$$= \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

Q:) if A is 2×2 $M = A^T A$ and $N = A A^T$ and also we have

$$M \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 45 \\ 45 \end{bmatrix}, \quad M \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 5 \end{bmatrix}, \quad N \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 45 \\ 135 \end{bmatrix},$$

$$N \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -15 \\ 5 \end{bmatrix} \text{ then find } A = ?$$

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$$\text{ex:)} \quad m \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 45 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad m \begin{bmatrix} -1 \\ 1 \end{bmatrix} = 5 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$N = \begin{bmatrix} 1 \\ 3 \end{bmatrix} = 45 \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad N \begin{bmatrix} -3 \\ 1 \end{bmatrix} = 5 \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$\lambda_1 = 45$$

$$\lambda_2 = 5$$

$$\lambda_1 = 45$$

$$\lambda_1 = 5$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$y_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$V = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$$

$$\sigma_1 = \sqrt{45}, \quad \sigma_2 = \sqrt{5}$$

$$D = \begin{bmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} M$$

$$A = U \Sigma V^T = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

Q:) Given a 5×5 matrix B with SVD $U \Sigma V^T$ which of the following statement is correct regarding the relationship b/w B and its SVD components

a) See in the slide.

$$B = U \Sigma V^T$$

$$\rho(B) = \rho(U \Sigma V^T)$$

$$\rho(B) = \rho(\Sigma) = \rho(V^T B U) = \rho(B)$$

$$\rho(B) = \rho(\Sigma)$$

Q:) Let A be a 4×3 matrix. The singular value decomposition (SVD) of A is given by $A = U \Sigma V^T$. If U is a 4×4 orthogonal matrix, Σ is a 4×3 diagonal matrix and V is a 3×3 orthogonal matrix, which of the following statements is true.

- (a) Σ has exactly 3 non-zero singular values.
 (b) The columns of U form an orthogonal basis for $\mathbb{R}^4$.
 (c) The columns of V form an orthogonal basis for $\mathbb{R}^4$.
 (d) The rows of U form an orthogonal basis for $\mathbb{R}^3$.

sol $A_{4 \times 3}$ $\begin{cases} (A^T A)_{3 \times 3} \rightarrow V = [v_1, v_2, v_3]_{3 \times 3} \\ (A A^T)_{4 \times 4} \rightarrow U = [u_1, u_2, u_3, u_4]_{4 \times 4} \end{cases}$

$$A_{4 \times 3} = U_{4 \times 4} \Sigma_{4 \times 3} V^T_{3 \times 3}$$

=

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \sigma_1 \\ 0 \end{bmatrix}$$

Lecture no. 03

Vector space =

Any set of vectors V is called V space if it satisfy following property

⇒ Property:

- 1.) Let $x, y \in V \Rightarrow x + y \in V$ i.e. closure property of vector addition.
- 2.) If $x + y = y + x$ i.e. vector addition is commutative property holds.
- 3.) $\forall x, y, z \in V \Rightarrow (x + y) + z = x + (y + z)$ i.e. associative prop of V add holds.
- 4.) $\forall x \in V \exists 0 \in V$ s.t. $x + 0 = x$ i.e. additive identity exist.
- 5.) $\forall x \in V \exists -x \in V$ s.t. $x + (-x) = 0$ i.e. additive inverse exist.
- 6.) $\forall x \in V \exists$ scalar c s.t. $c \cdot x \in V$ i.e. closure prop for scalar multiplication holds.
- 7.) $\forall x \in V \exists$ scalar λ & μ s.t. $\lambda(\mu x) = (\lambda\mu)x$ i.e. associative prop for scalar multiplication holds.
- 8.) $\forall x \in V \exists$ (Scalar 1) s.t. $1 \cdot x \in V$ i.e. multiplicative identity exist.

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9.) $\forall x, y \in V \exists \text{ scalar } \lambda \text{ s.t. } \lambda(x+y) = \lambda x + \lambda y$

i.e. distributive prop holds.

10.) $\forall x \in V \exists \text{ scalar } \lambda \text{ s.t. } (\lambda + \mu)x = \lambda x + \mu x$

i.e. distributive prop holds.

$\Rightarrow$ Any set of vectors 'V' satisfying all 10 properties is called V-space.

Ex: 1) $V = \mathbb{R}^2 \Rightarrow 2-D \text{ vector space } x = \begin{bmatrix} x \\ y \end{bmatrix}$

$V = \mathbb{R}^3 \Rightarrow 3-D \quad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$V = \mathbb{R}^n \Rightarrow n-D \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

$V = \mathbb{R}^n \Rightarrow n-D \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

$\Rightarrow$ Verification

Consider $V = \mathbb{R}^2 \left\{ \begin{bmatrix} x \\ y \end{bmatrix}; x, y \in \mathbb{R} \right\}$

1.) let $x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, y = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ then $x+y = \begin{bmatrix} 1 \\ 5 \end{bmatrix} \in \mathbb{R}^2$

i.e. closure prop for vector addition holds.

2.) $x+y = \begin{bmatrix} 1 \\ 5 \end{bmatrix} = y+x$ i.e. commutative prop for v-add holds.

3.) let $z = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ then $(x+y)+z = \begin{bmatrix} 1 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$

$x+(y+z) = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$ i.e. associative prop holds.

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4.) $\because 0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{R}^2$ & $x + 0 = x$ i.e. Additive identity exist

5.) $x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $-x = \begin{bmatrix} -2 \\ -3 \end{bmatrix} \in \mathbb{R}^2$ & $x + (-x) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0$

i.e. additive inverse exist.

6.) let $c = 5$, $x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ then $cx = 5x = \begin{bmatrix} 10 \\ 15 \end{bmatrix} \in \mathbb{R}^2$

Closure prop for scalar multi holds:

7.) let $\lambda = 3$, $\mu = -5$ then $\lambda(\mu x) = 3(-5x) = 3 \begin{bmatrix} -10 \\ -15 \end{bmatrix} = \begin{bmatrix} -30 \\ -45 \end{bmatrix} \in \mathbb{R}^2$ & $(\lambda\mu)x = (3(-5))x = -15 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -30 \\ -45 \end{bmatrix} \in \mathbb{R}^2$

i.e. $\lambda(\mu x) = (\lambda\mu)x$ so Associative prop for vector addition scalar multiplication hold.

8.) let $c = 1 \Rightarrow cx = 1 \cdot x = 1 \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \in \mathbb{R}^2$ i.e. multi identity exist.

9.) let $\lambda = 3$, $x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $y = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ then $\lambda(x+y) = 3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 15 \end{bmatrix}$ &

$\lambda x + \lambda y = 3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \end{bmatrix} + \begin{bmatrix} -3 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ 15 \end{bmatrix}$

10.) $\lambda = 3$, $\mu = -5$, $x = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \rightarrow (\lambda + \mu)x = -2x$
 $\begin{bmatrix} 2 \\ 3 \end{bmatrix} \rightarrow (\lambda x + \mu x) = 3x - 5x = -2x$

i.e. both types of distributive prop hold.

Definition of the
subspace.

Date

Subspace:

Let V is any vector space & W is any subset of V i.e. $W \subset V$

if W is itself a vector space then W is called subspace of V .

or

if W satisfy all the properties of V space then W is called subspace of V space (V).

Show that:-

It is In order to check W is subspace or no only check following two condition during exam.

$$1) \forall x, y \in W \Rightarrow x + y \in W$$

$$2) \forall x, y \in W \Rightarrow \exists \text{ scalar } c \text{ s.t. } cx \in W$$

$$\Rightarrow \forall x, y \in W \Rightarrow c(x+y) \in W \text{ (both property)}.$$

Q:) Let S_1, S_2, S_3 be sets of real-valued function defined as

$$S_1 = \{f: f(x) = 0\}$$

$$S_2 = \{g: g(n) = n+1 \text{ for all } n \in \mathbb{R}\}$$

$$S_3 = \{h: h(n) = c \text{ for some constant } c \in \mathbb{R}\}$$

which of the following statements is correct?

i) S_1 is a VS & S_2 is not a VS

ii) S_2 is a VS & S_3 is not a VS

iii) S_1 & S_3 are VS but S_2 is not a VS

iv) S_1 & S_2 are VS but S_3 is not a VS.

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$$\text{ex: 1) } S_1 = \{f(x) : f(3) = 0\}$$

$$\text{let } f, g \in S_1$$

$$f(3) = 0 \text{ \& } g(3) = 0$$

Now

$$\begin{aligned} (f+g)(3) &= f(3) + g(3) \\ &= 0 + 0 = 0 \end{aligned}$$

i.e. $f+g \in S_1$ closure prop holds

Now

$$(cf)(3) = c f(3) = c(0) = 0$$

i.e.

$$cf \in S_1$$

$\forall f \in S_1, \exists \text{ scalar } c \text{ s.t. } cf \in S_1$

Hence

S_1 is a v-space.

$$\Rightarrow S_2 = \{h(x) : h(x) = c\}$$

let $h(x), g(x) \in S_2$ then $h(x) = c, g(x) = c$

Now

$$(h+g)(x) = h(x) + g(x) = c + c = 2c = c \cdot 1 \Rightarrow (h+g) \in S_2$$

i.e. $\forall h, g \in S_2 \Rightarrow h+g \in S_2 \Rightarrow$ closure holds.

let $h(x) \in S_2$ then $h(x) = c$

$$\text{let } \lambda \text{ is any scalar then } (\lambda h)(x) = \lambda [h(x)] = \lambda(c) = c$$

$$\Rightarrow (\lambda h) \in S_2$$

i.e.

$\forall h \in S_2, \exists \text{ scalar } \lambda \text{ s.t.}$

$$\lambda h \in S_2$$

i.e. closure prop holds

Hence S_2 is also a vector space.

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$$S_2 = \{g(n) : g(n) = n+1\}$$

let

$$g_1(n) + g_2(n) \in S_2 \text{ then } g_1(n) = n+1, g_2(n) = n+1$$

Now

$$(g_1 + g_2)(n) = g_1(n) + g_2(n) = n+1 + n+1 = 2n+2 \neq n+1$$

i.e.

$$g_1 + g_2 \notin S_2$$

Closure property for v-addition not hold So S_2 is not a vector space.

Q.) Let M_n denote the vector space of all $n \times n$ real mat. Consider the following subset of M_n .

- 1.) $W_1 = \{A \in M_n \mid A^2 = I\}$, where I is the identity
- 2.) $W_2 = \{A \in M_n \mid \text{rank}(A) = 1\}$
- 3.) $W_3 = \{A \in M_n \mid \text{trace}(A) = 0\}$
- 4.) $W_4 = \{A+B \mid A \in M_n, B \text{ is a fixed mat in } M_n\}$

Which of the following statement is correct.

- a.) W_1 is a linear subspace of M_n
- b.) W_2 is a linear subspace of M_n
- c.) W_3 is a linear subspace of M_n ✓
- d.) W_4 is a linear subspace of M_n only if B is a zero matrix.

$$\text{Sol: } W_1 = \{A \in M_{n \times n} : A^2 = I\}$$

$$\text{let } A, B \in W_1 \text{ then } A^2 = I, B^2 = I$$

$$(A+B)^2 = A^2 + B^2 + AB + BA = I + I + AB + BA \neq I$$

i.e.

$A+B$ is not necessarily involutory mat.

$$A+B \notin W_1$$

i.e W_1 is not closed under vector addition.
Hence W_1 is not a vector space.

$$2.) W_2 = \{A \in M_n : p(A) = 1\}$$

Let $A, O \in W_2$ then $p(A) = 1, p(O) = 1$

Now

$$(A+O) \leq p(A) + p(O) \quad \text{i.e. } p(A+O) \leq 1+1 = 2$$

$$p(A+O) \leq 2$$

it is not necessary that $p(A+O) = 1 \Rightarrow A+O \notin W_2$

$$3.) W_3 = \{A \in M_n : \text{Tr}(A) = 0\}$$

Let $A, O \in W_3$ then $\text{Tr}(A) = 0, \text{Tr}(O) = 0$

Now

$$(A+O) = \text{Tr}(A) + \text{Tr}(O) = 0 + 0 = 0$$

$$\therefore (A+O) \in W_3 \quad \text{closure prop't holds}$$

Again

$$c \cdot A \quad \text{Tr}(c \cdot A) = c \cdot \text{Tr}(A) = c \cdot 0 = 0$$

i.e. $(c \cdot A) \in W_3$ closure property holds.

Hence W_3 is a subspace of M_n .

$$4.) W_4 = \{A+O : B \text{ is a fixed mat}\}$$

We have two possibilities for B either $B=0$
if we take $B \neq 0$ then or $B \neq 0$

$A+O \neq A$ additive identity $O \notin W_4$

Hence W_4 is not a vector space.

if we take $B=0$ then

$$A+O = A \quad \text{i.e. additive identity exist}$$

Hence W_4 can be a vector space.

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Spanning in vector space:-

let W is a subspace of any V space

let $S \subseteq W$

i.e S is any subset of W

Linear span $\rightarrow$ let S is any set of vectors

then set of all (linear combination) of vectors in

S is called linear span of S

and it is denoted by $L(S)$

let $S = \{x_1, x_2, x_3, x_4\}$ then all the linear combination of these vectors can be written as $K_1x_1 + K_2x_2 + \dots$ where $K_i \in \mathbb{R}$

then

$$L(S) = \{ (K_1x_1 + K_2x_2 + K_3x_3 + K_4x_4) : K_i \in \mathbb{R} \} = L(S)$$

Spanning in vector space:

let W is any subspace & S is any subset of W

then W is called spanned by S if $W = L(S)$

S spans W i.e $L(S) = W$

i.e generates

Any random vector of W can be written as linear combination of vectors in S

for ex) Consider $W = \mathbb{R}^2$ & $S = \{x_1, x_2\} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

then $W = L(S)$

i.e W is spanned by vectors x_1 & x_2

i.e Any random vector in $\mathbb{R}^2$ can be generated with the linear comb of x_1 & x_2

Basis of vector space:

Let W is any subspace & S is any given set of vectors.

then S is called Basis for W if

- 1) S spans W
- 2) S contains L.I. set of vectors.

For e.g. Basis for $W = \mathbb{R}^2$ is $S_1 = \{x_1, x_2\} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

$\therefore L\{S_1\} = \mathbb{R}^2$ & S_1 contains L.I. vectors (By standard

~~method~~ Basis
Again, for $W = \mathbb{R}^3$ is $S_2 = \{x_1, x_2, x_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

$\therefore L\{S_2\} = \mathbb{R}^3$ & S_2 contains L.I. vectors (By Tachy method)

$\therefore |S_2| \neq 0$

Dimension of vector space:

Number of vectors in it's basis.

For e.g.: $\dim(\mathbb{R}^2) = 2$ $\therefore$ Basis for $\mathbb{R}^2 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

$\dim(\mathbb{R}^3) = 3$ $\therefore$ Basis for $\mathbb{R}^3 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

Let W is subspace of $\mathbb{R}^3$ then $\dim W =$ Not necessarily it will depend on the basis of W (will be discussed later)

If V is any vector space & it's Basis is $= \{e_1, e_2, e_3, \dots, e_n\}$ then $\dim(V) = \text{No. of vectors in Basis} = n$.

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Q. Consider the set of column vectors defined by
 $x = \{(n_1, n_2, n_3)^T : n_1 + n_2 + n_3 = 0\}$ then
 $S = \{(1, -1, 0)^T, (0, -1, 1)^T\}$ is basis for subspace

sol. 1) $X \subset \mathbb{R}^3$ so X is subspace of $\mathbb{R}^3$ & defined as
 $X = \left\{ \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} : n_1 + n_2 + n_3 = 0 \right\}$

Now
 $S = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\} = \{x_1, x_2\}$
 S will span X .
 will be basis for X if $\begin{cases} \text{S contains 2 vectors} \end{cases}$

Clearly x_1 & x_2 are L.I. by observation hence 2nd cond holds.

Now
 let us take any random vector $y = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$ then
 $y = 3x_1 - x_2$ so y can be generated by x_1 & x_2 Hence S span X .

$\Rightarrow S$ is basis for subspace X .

YES Q. & $\dim(X) = 2$.

P. P.

Lecture no. 04

Properties of Basis & Dimension:

1) Basis of any vector space is not unique.

2) if $\dim(\text{v. space}) = n$ then Any set of n l.i. vectors forms basis.

e.g. $V = \mathbb{R}$ that $\dim(\mathbb{R}^3) = 3 \Rightarrow$ Any set of 3 l.i. vectors form basis for $\mathbb{R}^3$.

Explanation:

for the vector space $V = \mathbb{R}^3$ Standard Basis is $S_1 = \{e_1, e_2, e_3\}$

$$S_1 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

& its another take

$$S_2 = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right\} \text{ then}$$

S_2 is also basis for $\mathbb{R}^3$

(e.g.) Consider the vector space $V = \mathbb{R}^4$ then its standard basis is $S_1 = \{e_1, e_2, e_3, e_4\}$

$$= \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} = \dim(\mathbb{R}^4) = 4.$$

its another basis are as follows:

$$S_2 = \left\{ \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\} = \{x_1, x_2, x_3, x_4\}$$

Spanned

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$$\therefore |S_2| = \dots \dots \dots 88 \text{ i.e. } |S_2| \neq 0$$

i.e. $|S_2| \neq 0 \Rightarrow x_1, x_2, x_3, x_4$ are L.I. and hence S_2 form basis for R^4 .

$$\Rightarrow S_3 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix} = \{y_1, y_2, y_3, y_4\}$$

$$\therefore |S_3| = \dots \dots \dots = 5 \text{ i.e. } |S_3| \neq 0$$

$\Rightarrow y_1, y_2, y_3, y_4$ are L.I.

Hence

S_3 is also Basis for R^4

It goes on $\dots \dots \dots$

Q:) it is given that $x_1, x_2, \dots, x_m$ are m non-zero orthogonal vectors. The dimension of the vector space spanned by the $2m$ vectors $x_1, x_2, \dots, x_m, -x_1, -x_2, \dots, -x_m$ is

a) $2m$

b) $m+1$

c) m ✓

d) dependent on the choice of $x_1, x_2, \dots, x_m$

$$\text{Sol:)} \quad S = \{x_1, x_2, \dots, x_m, -x_1, -x_2, \dots, -x_m\}$$

$$\text{A.T.Q.}, L(S) = V$$

W.K. that orthogonal vectors are L.I. also
i.e. out of $2m$ vectors only $x_1, x_2, x_3, \dots, x_m$ are L.I.
So Basis = $\{x_1, x_2, x_3, \dots, x_m\} = \dim(V) = m$.

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Q.) if $v_1, v_2, \dots, v_6$ are six vectors in $\mathbb{R}^4$, which one of the following statement is false.

a.) if $\{v_1, v_2, v_5, v_6\}$ spans $\mathbb{R}^4$, then it forms a basis for $\mathbb{R}^4$.

b.) These vectors are not LI

c.) it is not necessary that these vectors span $\mathbb{R}^4$

d.) Any four these vectors form a basis for $\mathbb{R}^4$

Q.) $P = \begin{bmatrix} -1 & 0 \\ -1 \\ 3 \end{bmatrix}^T$, $Q = \begin{bmatrix} -2 \\ -5 \\ 9 \end{bmatrix}^T$ and $R = \begin{bmatrix} 2 \\ -2 \\ 12 \end{bmatrix}^T$ are three

vectors. An orthogonal set of vectors having a span that contain P, Q, R is

a.) $\begin{bmatrix} -6 \\ 2 \\ 6 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}$ b.) $\begin{bmatrix} -4 \\ 2 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 5 \\ 7 \\ -11 \end{bmatrix}$, $\begin{bmatrix} 8 \\ 2 \\ -3 \end{bmatrix}$ $x \cdot y \neq 0$

c.) $\begin{bmatrix} 6 \\ 2 \\ -1 \end{bmatrix}$, $\begin{bmatrix} -3 \\ 2 \\ -2 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 9 \\ -4 \end{bmatrix}$ d.) $\begin{bmatrix} 4 \\ 2 \\ 11 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 5 \\ 2 \\ 4 \end{bmatrix}$

1) $x_1 \cdot x_2 = 0$ i.e. orthogonal

2) $x \cdot x_2 \neq 0$ not orthogonal

3) $x_1 \cdot y_1 \neq 0$ not orthogonal

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$$m=2 \quad x_1 = \begin{bmatrix} -6 \\ -3 \\ 6 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix}$$

Let $S = \{x_1, x_2\}$ then $P, Q, R \in L[S]$

$$\therefore P = x_1 - x_2 \text{ so } P \in L[S]$$

$$\therefore Q = x_1 + x_2 \text{ so } Q \in L[S]$$

$$\therefore R = x_1 + 2x_2 \text{ so } R \in L[S]$$

Hence S spans P, Q, R contain orthogonal vectors also P, Q, R so correct option is (A.)

$\Rightarrow$ The following vector is L.O upon the solution to the previous problem.

$$\begin{array}{ll} \text{a.) } \begin{bmatrix} 8 \\ 9 \\ 3 \end{bmatrix} & \text{b.) } \begin{bmatrix} -2 \\ -13 \\ 30 \end{bmatrix} \end{array} \quad \begin{array}{ll} \text{c.) } \begin{bmatrix} 4 \\ 4 \\ 5 \end{bmatrix} & \text{d.) } \begin{bmatrix} 13 \\ 2 \\ -3 \end{bmatrix} \end{array}$$

$$\text{a.) } A = [x_1, x_2, x_3] = \begin{bmatrix} -6 & 4 & 8 \\ -3 & -2 & 9 \\ 6 & 3 & 3 \end{bmatrix} \quad \therefore |A| \neq 0 \quad \text{i.e. L.I.}$$

$$\text{b.) } B = [x_1, x_2, x_3] = \begin{bmatrix} -6 & 4 & -2 \\ -3 & -2 & -12 \\ 6 & 3 & 30 \end{bmatrix} \quad \therefore |B| \neq 0 \quad \text{i.e. } x_1, x_2, x_3 \text{ are L.O.}$$

$$\text{c.) } C = [x_1, x_2, x_3] = \dots \quad |C| \neq 0 \quad \text{(L.I.)}$$

$$\text{d.) } D = [x_1, x_2, x_3] = \dots \quad |D| \neq 0 \quad \text{(L.I.)}$$

So only option b is correct.

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Partition Matrix (Block Matrix)

⇒ Application: Find Det & E-value of

$$A = \begin{bmatrix} 2 & 4 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 3 & 2 \end{bmatrix}_{4 \times 4}$$

M-1 Using Conventional approach

Do your self.

$$\& |A - \lambda I| = 0 \quad \lambda = \text{Do your self}$$

M-2

$$A = \begin{bmatrix} \boxed{2 \ 4} & 0 & 0 \\ \boxed{0 \ -3} & 0 & 0 \\ 0 & 0 & \boxed{1 \ -2} \\ 0 & 0 & \boxed{3 \ 2} \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix}$$

$$= |A| = |A_1| \cdot |A_2| = (-6) \cdot (8) = -48$$

$$\text{E-value of } A_1 = \begin{bmatrix} 2 & 4 \\ 0 & -3 \end{bmatrix} \text{ are } \lambda = 2, -3$$

$$\text{" " } A_2 = \begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix} \text{ are } \lambda^2 - 3\lambda + 8 = 0$$

$$\lambda = \frac{3 \pm \sqrt{9-32}}{2} = \frac{3 \pm \sqrt{-23}}{2}$$

$$= 1.5 \pm i\sqrt{23} \quad \text{M}$$

$$\text{eg } \begin{bmatrix} 1 & 2 & -3 & 0 & 1 \\ 2 & 0 & 1 & 2 & -1 \\ 3 & 1 & 2 & -1 & 0 \\ 1 & 2 & 1 & 3 & 4 \end{bmatrix}_{5 \times 5} ; \begin{bmatrix} 1 & 2 & -3 & 0 & 1 \\ 2 & 0 & 1 & 2 & -1 \\ 3 & 1 & 2 & -1 & 0 \\ 1 & 2 & 1 & 3 & 4 \end{bmatrix}_{4 \times 4} ; \begin{bmatrix} 2 & -1 & 3 & 4 & 0 \\ 1 & 2 & 0 & -1 & 2 \\ 3 & -1 & 2 & 1 & 1 \\ 0 & 1 & 3 & 1 & 3 \\ 1 & 2 & 0 & -1 & 2 \end{bmatrix}$$

Special 2

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$$A = \begin{bmatrix} 1 & 2 & -3 & 0 & 1 \\ 2 & 0 & 1 & 2 & -1 \\ 3 & 1 & 2 & -1 & 0 \\ 1 & 2 & 1 & 3 & 4 \end{bmatrix} \quad 4 \times 5$$

$$B = \begin{bmatrix} 1 & 2 & -3 & 0 & 1 \\ 2 & 0 & 1 & 2 & -1 \\ 3 & 1 & 2 & -1 & 0 \\ 1 & 2 & 1 & 3 & 4 \end{bmatrix} \quad 4 \times 5$$

$$C = \begin{bmatrix} 2 & -1 & 3 & 4 & 0 \\ 1 & 2 & 0 & -1 & 2 \\ 3 & -1 & 2 & 1 & 4 \\ 0 & 1 & 3 & 1 & 3 \\ 1 & 2 & -1 & 2 & 1 \end{bmatrix} \quad 5 \times 5$$

$\Rightarrow$ These matrices are called partition matrices

Def: With the help of horizontal & vertical lines we can convert given matrices into a partition matrix.

Partition
or

Square Block matrices

$$\text{eg. } \begin{bmatrix} 1 & -2 & 4 & 3 \\ 2 & 0 & 1 & 4 \\ 0 & 1 & -1 & 7 \\ 3 & 1 & 0 & 5 \end{bmatrix} \quad 4 \times 4$$

$$\text{eg. } \begin{bmatrix} 1 & 3 & 1 & 6 & -2 \\ 0 & -1 & 5 & 7 & 0 \\ 0 & 0 & 2 & 4 & 1 \\ 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 & -3 \end{bmatrix} \quad 5 \times 5$$

$$\text{e.g. } \begin{bmatrix} 1 & -2 & 4 & 1 & 3 & 2 \\ 2 & 4 & -1 & 0 & 1 & 4 \\ 1 & 1 & 2 & -1 & 0 & 3 \\ -2 & 3 & 4 & 1 & 2 & 7 \\ 7 & 3 & 4 & -1 & 2 & 3 \\ 0 & 7 & 5 & 2 & -1 & 4 \end{bmatrix} \quad 6 \times 6$$

$$\text{e.g. } \begin{bmatrix} 1 & 2 & 4 & -5 & 0 & 7 \\ 2 & 4 & 1 & -1 & 2 & 1 \\ 0 & 7 & 5 & 1 & 2 & 3 \\ 4 & 1 & 0 & 2 & -1 & 4 \\ 1 & 2 & -1 & 4 & 2 & 1 \\ 7 & 2 & 4 & -1 & 2 & 9 \end{bmatrix} \quad 6 \times 6$$

Ans: There are two necessary conditions.

- 1.) No. of horizontal line = No. of vertical line
- 2.) Diagonal blocks must be square matrices.
(not necessarily of same order)

Block Diagonal matrix $\rightarrow$

$$A = \begin{bmatrix} A_1 & 0 & 0 & \dots & 0 \\ 0 & A_2 & 0 & \dots & 0 \\ 0 & 0 & A_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & A_n \end{bmatrix} n \times n$$

- 1.) if block exist only at diag positions.
- 2.) All the off diag blocks must be null matrix

e.g: $A = \begin{bmatrix} 4 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 4 & 2 \end{bmatrix} 4 \times 4$ $B = \begin{bmatrix} 4 & -2 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 7 & 1 \\ 0 & 0 & 0 & 0 & -5 & 2 \end{bmatrix} 6 \times 6$

$|A| = (8)(2) = 16$

$C = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 8 & 0 \\ 0 & 0 & 3 & 2 & 7 & 0 \\ 0 & 0 & 0 & 4 & -6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 8 \end{bmatrix} 6 \times 6$ $= \begin{bmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ 0 & 0 & A_3 \end{bmatrix} = 1140$

$|C| = (-4)(-12)(8) = 384$

$|C| = (-2)(136)(8) = -2176$

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⇒ Application:

$$|A| = |A_1| \cdot |A_2| \cdot |A_3| \cdots |A_n|$$

= Product of Dets of Diagonal blocks

Block U.T.M

If all the blocks that lies below the Diagonal blocks are null matrices then these matrices are called block U.T.M.

$$\text{eg } A = \begin{pmatrix} 1 & 3 & 1 & 6 & -2 \\ 0 & -1 & 5 & 7 & 0 \\ 0 & 0 & 2 & 4 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} A_1 & A_2 & A_3 \\ 0 & A_4 & A_5 \\ 0 & 0 & A_6 \end{pmatrix}$$

$$|A| = (-1)(-4)(-3) = -12$$

Block L.T.M:

If all the matrices that lies above the diag blocks are null matrices then mat are called Block L.T.M.

$$\text{eg } A = \begin{pmatrix} 1 & 3 & 0 & 0 & 0 \\ 5 & -1 & 0 & 0 & 0 \\ 0 & 7 & 2 & 4 & 0 \\ 3 & 6 & 1 & -1 & 0 \\ -2 & 4 & 1 & 3 & -3 \end{pmatrix} = \begin{pmatrix} A_1 & 0 & 0 \\ A_2 & A_3 & 0 \\ A_4 & A_5 & A_6 \end{pmatrix}$$

$$|A| = (-16)(-6)(-3) = -288$$

⇒ Application:

$$|A| = |A_1| \cdot |A_4| \cdot |A_6| \text{ in U.T.M}$$

= Product of det of Diag blocks

$$|A| =$$

Product of det of Diag blocks.

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Q:) if $A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ -1 & 5 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & -1 & 3 \end{bmatrix}$ then $|A| = ?$

sol:) $|A| = \cancel{(5)} \cancel{(-2)} (5+2) (6+0) = (7)(6) = 42$

Q:) $A = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ How to find E. vals of A?

sol:) Simply by using the concept of partition matrix

$$= A_1 = \begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix} \Rightarrow \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 3\lambda + 15 = 0$$

$$\Rightarrow \lambda(\lambda - 5) - 3(\lambda - 5) = 0$$

$$A_2 = [1]_{1 \times 1} \quad (\lambda - 5)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 5, \lambda = 3, \lambda = 1$$

Q:) The number of L.I. eigen vectors of the matrix

$$\begin{bmatrix} 2 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$
 is

a.) 1 $m-1$ c. eq. $|A - \lambda I| = 0$

b.) 2 $\lambda = 3, 4, 1, 1$

c.) 3 all the eigen value are different $\Rightarrow$

d.) 4. Eigen vectors are also L.I.

$$\lambda = 3, 4, \frac{3 \pm \sqrt{17}}{2}$$

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$$M-L \quad \textcircled{0MAH} = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix}$$

$$= A_1 = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$$

$$= \lambda = 3, 4$$

$$\text{E-value of } A_1 = \lambda^2 - \text{tr}(A_1)\lambda + |A_1| = 0$$

$$= \lambda^2 - 3\lambda + (-2) = 0$$

$$\lambda = \frac{3 \pm \sqrt{9+8}}{2} = \frac{3 \pm \sqrt{17}}{2} \quad \text{Ans}$$

Q:) Given a 3×3 block matrix

$$A = \begin{bmatrix} A_{11} & 0 & 0 \\ 0 & A_{22} & A_{23} \\ 0 & 0 & A_{33} \end{bmatrix}_{3 \times 3} \quad |A| = |A_{11}| \cdot |A_{22}| \cdot |A_{33}|$$

where A_{11}, A_{22}, A_{33} are 3×3 mat. of A is upper triangular & invertible, which of the following must be correct.

- a.) A_{11}, A_{22} & A_{33} must be invertible. ✓
- b.) A_{11}, A_{22}, A_{33} & A_{23} must be invertible.
- c.) $\det(A_{11}) = \det(A_{22}) = 0$
- d.) $A_{11}, A_{23}; A_{11}, A_{22}, A_{33}$ must be invertible.

Q:) Consider the block diagonal mat A given

$$A = \begin{bmatrix} \textcircled{3} & \textcircled{1} & 0 & 0 & 0 \\ \textcircled{1} & \textcircled{2} & 0 & 0 & 0 \\ 0 & 0 & \textcircled{4} & \textcircled{2} & 0 \\ 0 & 0 & \textcircled{0} & \textcircled{3} & 0 \\ 0 & 0 & 0 & 0 & \textcircled{5} \end{bmatrix} \quad \text{tell what is } A^{-1}?$$

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Sol: $\begin{bmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ 0 & 0 & A_3 \end{bmatrix} \Rightarrow A^{-1} = \begin{bmatrix} A_1^{-1} & 0 & 0 \\ 0 & A_2^{-1} & 0 \\ 0 & 0 & A_3^{-1} \end{bmatrix}$

a.) $\begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & -2 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}$ $A_1 = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$
 $A_1^{-1} = \frac{1}{5} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$

b.) $\begin{bmatrix} 2/3 & -1/5 & 0 & 0 & 0 \\ -1/5 & 2/3 & 0 & 0 & 0 \\ 0 & 0 & 3 & -2 & 0 \\ 0 & 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}$ Ans (c) is correct

c.) $\begin{bmatrix} 1/5 & 1/5 & 0 & 0 & 0 \\ 1/5 & 2/5 & 0 & 0 & 0 \\ 0 & 0 & 1/4 & -1/6 & 0 \\ 0 & 0 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 1/5 \end{bmatrix}$ $A_2 = \begin{bmatrix} 4 & 2 \\ 0 & 3 \end{bmatrix}$
 $A_2^{-1} = \frac{1}{12} \begin{bmatrix} 3 & -1 \\ 0 & 4 \end{bmatrix}$

d.) A is invertible $\begin{bmatrix} 1/2 & -1/6 \\ 0 & 2/3 \end{bmatrix}$

Q: if $A = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 2 & 5 \\ 3 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$ then eval AB using included part

Sol: $\begin{bmatrix} O_{12} & O_{21} \\ O_{12} & A_{11} \end{bmatrix} \begin{bmatrix} I_{23} & I_{11} \\ O_{13} & O_{11} \end{bmatrix}$

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$$A \cdot O = \begin{bmatrix} P_{11} & Q_{11} \\ 0_{12} & R_{11} \end{bmatrix} \begin{bmatrix} S_{12} & T_{11} \\ 0_{13} & U_{11} \end{bmatrix} = \begin{bmatrix} (P_{11}S_{12} + Q_{11}0_{13}) \\ (0_{12}S_{12} + R_{11}0_{13}) \end{bmatrix}$$

$$= \begin{bmatrix} P_{11}T_{11} + Q_{11}U_{11} \\ 0_{11}T_{11} + R_{11}U_{11} \end{bmatrix}$$

$$= \begin{bmatrix} - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{bmatrix}_{3 \times 4} = A$$

Lecture no. 05

Projection matrix:

> General points:

Consider $A = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n \times 1}$, $A^T = \begin{bmatrix} \text{---} & \text{---} & \text{---} \end{bmatrix}_{1 \times n}$ over

$$M = A^T A \rightarrow \begin{bmatrix} \text{---} & \text{---} & \text{---} \end{bmatrix}_{1 \times n} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n \times 1} = \begin{bmatrix} \text{---} \end{bmatrix}_{1 \times 1} = \text{Scalar}$$

$$N = A A^T \rightarrow \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n \times 1} \begin{bmatrix} \text{---} & \text{---} & \text{---} \end{bmatrix}_{1 \times n} = \begin{bmatrix} \text{---} & \text{---} & \text{---} \end{bmatrix}_{n \times n} = \text{Symm mat}$$

$$N^T = (A A^T)^T = (A^T)^T \cdot A^T = A A^T = N \quad \text{i.e. Symm}$$

Systematic way:

$$1) \because \hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{|\vec{a}|} \vec{a} = |\vec{a}| \cdot \hat{a}$$

Any vector = (it's magnitude) $\times$ (unit vector in it's dir)

$$2) \text{ let } A_{n \times 1} \text{ then } A_{n \times 1}^T \begin{cases} (A^T A)_{1 \times 1} = \text{symm} = \text{scalar} \\ (A A^T)_{n \times n} = \text{Symm mat} \end{cases}$$

$$3) \text{ let } A_{n \times 1} \text{ \& } B_{n \times 1} \begin{cases} (A^T B) = \text{scalar Quantity} \\ (A \cdot B)^T = A^T B = \text{scalar Quantity} \\ \|A\| = \sqrt{A^T A} = \|A\|^2 = A^T A \end{cases}$$

4) if $A \cdot B = 0$ then $A \& B$ are called orthogonal vectors. Spiral

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⇒ Projection of vector:-



if we want to find projection of $\vec{B}$ onto the vector $\vec{A}$ then process will (along) be as follows.

Now Any vector = (it's magnitude) (unit vector in direction of)

$$\vec{b} = (om) (\hat{A})$$

$$= |\vec{B}| \cos \theta \frac{\vec{A}}{|\vec{A}|}$$

$$\Delta OMB, \frac{om}{OB} = \cos \theta$$

$$om = |\vec{B}| \cos \theta$$

$$om = |\vec{om}| |\hat{om}|$$

where $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$

$$\vec{b} = |\vec{B}| \cos \theta \frac{\vec{A}}{|\vec{A}|}$$

$$= |\vec{B}| \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} \left(\frac{\vec{A}}{|\vec{A}|} \right)$$

$$= \left(\frac{\vec{A} \cdot \vec{B}}{|\vec{A}|^2} \right) \vec{A} = \left(\frac{\vec{A} \cdot \vec{B}}{|\vec{A}|^2} \right) \vec{A}$$

$$\vec{b} = \left(\frac{\vec{A}^T \vec{B}}{\vec{A}^T \vec{A}} \right) \vec{A} = (\lambda) \vec{A}$$

i.e vector $\vec{B}$ & $\vec{A}$ are collinear

Q:) Find the projection of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ onto the vector $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

sol:) Here $B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $A = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$$A^T B = \begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \end{bmatrix}_{1 \times 1} = -1$$

$$A^T A = \begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \end{bmatrix}_{1 \times 1} = 5$$

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$$\text{So, } \vec{b} = \begin{pmatrix} A^T O \\ A^T A \end{pmatrix} \vec{A} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} \vec{A} = \frac{-1}{5} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} 2/5 \\ -1/5 \end{bmatrix}$$

M-2:

We have show that

$$\vec{b} = \begin{pmatrix} A^T O \\ A^T A \end{pmatrix} \vec{A} = \vec{A} \begin{pmatrix} A^T O \\ A^T A \end{pmatrix} = \begin{pmatrix} A A^T \\ A^T A \end{pmatrix} O$$

Proj. vect. $\vec{b} = P \vec{A}$ where $P = \frac{A A^T}{A^T A}$ = Projection mat.

Conclusion: Projection along a line:Projection of vector $\vec{B}$ along a line $\vec{A}$ is given as

$\vec{b} = P \vec{A}$ where $P = \frac{A A^T}{A^T A}$ = Projection mat.

M-2 in Question

$$A^T A = \begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \end{bmatrix}_{1 \times 1} = 5$$

$$A \cdot A^T = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \begin{bmatrix} -2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}_{2 \times 2}$$

So

Projection mat $P = \frac{A A^T}{A^T A}$

Proj. Mat. = $\frac{1}{5} \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$ & Proj. vect. $\vec{b} = P \vec{A}$

$$\vec{b} = \frac{1}{5} \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2/5 \\ -1/5 \end{bmatrix}$$

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Q:) What multiple of $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ will be closest to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

a.) $-1 = \begin{bmatrix} 2/5 \\ -1/5 \end{bmatrix} = -\frac{1}{5} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = -\frac{1}{5} \text{ (A)}$

b.) $2/5$

c.) $-1/5$ ✓

d.) $-2/5$

Q:) Find the point on the line $x+y=0$ which is closest to the point $(1,1)$

Sol:) Vector along the line $x+y=0$ is $A = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

* Given point $D = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ so closest point = proj. vect.

$\vec{p} = \begin{bmatrix} 2/5 \\ -1/5 \end{bmatrix}$

a.) $(-2, 1)$

b.) $(2, -1)$

c.) $(2/5, -1/5)$ ✓

d.) $(-2/5, 1/5)$

Q:) Find a matrix that projects every point in 2D plane onto the line $x+y=0$. Also find the project of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ onto this line.

Sol:) Vector along the line $x+y=0$ is $A = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

Proj. mat = $P = \frac{AA^T}{A^T A} = \frac{1}{5} \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$ already calculated

* $\vec{p} = PD = \begin{bmatrix} 2/5 \\ -1/5 \end{bmatrix}$ already calculated

$\det(A) = 1$, $|A| = 0$, $\rho(A) = 1$, $\lambda = 0 \text{ or } 1$, Diagonalizable Special
 N.S. of L_2 (vector) = order $P = \text{symm}$, idempotent.

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Q: The projection of $B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ onto the line $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ will be.

Sol:)

$$P = \frac{A A^T}{A^T A} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \text{projection matrix}$$

So

$$A A^T = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+1+1 \end{bmatrix} = 3.$$

$$\vec{P} = P B = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$$

Q:) What multiple of $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is closest to the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$?

$$\vec{P} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Sol:)

a.) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ ✗

b.) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

c.) 2 ✓

d.) 6

Q:) The point along the vector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ which is closest to the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ will be?

Sol:)

a.) (1 1 1) b.) 2 c.) (2 2 2) d.) (1 2 3)

Spiral

Q:) What multiple of $A(111)$ is closest to the point $O(244)$

a.) 3

$$A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, A^T = [1 \ 1 \ 1]$$

b.) 10

c.) $10/3$ ✓

d.) $3/10$

$$A^T A = 3$$

$$A A^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$P = \frac{A A^T}{A^T A} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\vec{b} = P\vec{O} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 10/3 \\ 10/3 \\ 10/3 \end{bmatrix} = \frac{10}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Note:

$$\text{Tr}(P) = 1$$

$$|P| \neq 0$$

$P =$ Symm as well as Idempotent

E-value $(\lambda) = 0, 0, 1$ gm of '0' = two

$P(A) = 1$ (one) gm of '1' = one

No. of LI e-vectors = three

Hence $P =$ Diagonalizable.

Q:) Find the projection matrix that projects every point in $\mathbb{R}^3$ onto the line of intersection of the planes $x+y+z=0$ & $x-z=0$ & hence find the projection of $O(123)$?

a:) Solving 1 & 2 $x+y+z=0$ & $x-z=0$ $\Rightarrow 2x+y=0$

$$= \boxed{2x+y+0z=0} \quad \text{--- (3)}$$

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Any vector along line (1) is $A = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1+4+0 \end{bmatrix} = 5$$

$$A A^T = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

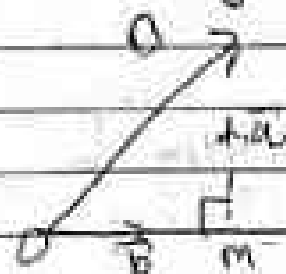
So projection mat is

$$\vec{P} \cdot P = \frac{A A^T}{A^T A} = \frac{1}{5} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{Projection matrix}$$

$$\vec{P} = P \vec{O} = \frac{1}{5} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} -3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} -3/5 \\ 6/5 \\ 0 \end{bmatrix} = \text{proj vector}$$

orthogonal complement of $B \rightarrow$



$\vec{P}$ = Projection vector

$\vec{Q}$ = orthogonal complement of $\vec{P}$

By vector law, $\vec{P} + \vec{Q} = \vec{B}$

$$\Rightarrow \boxed{\vec{Q} = \vec{B} - \vec{P}} \rightarrow \text{Day no. 1}$$

or

$$P \vec{B} + \vec{Q} \vec{B} = \vec{B}$$

$$(P + \vec{Q}) \vec{B} = \vec{B} \Rightarrow \boxed{P + \vec{Q} = I} \rightarrow \text{Day no. 2}$$

where $\vec{Q}$ = projection matrix on orthogonal complement of $\vec{P}$.

Date

What is the nature of PQ ?

Sol: $PQ = P(I - P) = PI - P^2 = P - P = 0$

(Hence Ans will be Null mat)

Q:) Find the projection matrix P onto the line through $A = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ & hence find the projection of $B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ onto A .

(ii) Also find the matrix Q that projects onto the line $L^\perp$ to A & hence find orthogonal projection of B onto A .

Sol: $P = \frac{AA^T}{A^T A} = \frac{1}{10} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$AA^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{2 \times 1} \begin{bmatrix} 1 & 1 \end{bmatrix}_{1 \times 2} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}_{2 \times 2}$

$A^T A = \begin{bmatrix} 1 & 1 \end{bmatrix}_{1 \times 2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{2 \times 1} = 1 + 1 = 2$

we want $\vec{b} = P\vec{a} = \frac{1}{10} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2/10 \\ 2/10 \end{bmatrix}$

(ii) w.k that

$P + Q = I$

$Q = I - P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{1}{10} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 9/10 & -1/10 \\ -1/10 & 9/10 \end{bmatrix}$

orthogonal projection matrix

Spiral

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1. orthogonal complement of A is

$$\vec{q} = \vec{a} - \vec{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 4/10 \\ 11/10 \end{bmatrix} = \begin{bmatrix} 6/10 \\ -1/10 \end{bmatrix}$$

↓
orthogonal complement of

Projection matrix (along a subspace spanned by set of vectors)

Q:) The projection of $O = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$ onto the column space of $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix}$ will be?

Ans:) $C(A) =$ set of all linear combinations of columns of A where $A = [x_1 \ x_2]$

$$= \{ (c_1 x_1 + c_2 x_2) : c_1, c_2 \in \text{Scalar}, x_1, x_2 \text{ are col of } A \}$$

Here x_1, x_2 lies in 2D as well as they are LI also (By observation)

So,

$C(A)$ is nothing but 2D space or $C(A) = \mathbb{R}^2$

$$A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}, |A^T A| = 1$$

$$(A^T A)^{-1} = \frac{1}{1} \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix}$$

Standard form of projection mat onto the subspace spanned by A is given as:

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$$P = A(A^T A)^{-1} A^T$$

$$= \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 13+10 & -5-4 \\ 13+15 & -5-6 \\ 0-0 & 0-0 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 23 & -9 \\ 28 & -11 \\ 0 & 0 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 13+18 & 23+27 & 0 \\ 28+16 & 28+13 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

i.e. Projection mat $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

& Projected vector is $\vec{p} = P\vec{\alpha} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}$

$\Rightarrow \text{Tr}(P) = 2$
 $|P| = 0$
 $\rho(P) = 2$

$P^T = P = \text{symm}$ | (m. of '0' =) one

$P^2 = P \Rightarrow$ Idempotent | (m. of '1' =) two

Hence no. of L.I. E. vectors for $A = \text{three} = \text{rank}$
 $\therefore A$ is Diagonalizable.

Date

$\Rightarrow$ let x_1, x_2, x_3 are given vectors & we want to find the projection of $\vec{b}$ (onto the subspace) - spanned by x_1, x_2, x_3 then process will be follows

Consider a mat $A = [x_1, x_2, x_3]$ then

$$\text{Projection Mat} = A(A^T A)^{-1} A^T$$

$$\text{projection vector} = \boxed{\vec{p} = P\vec{b}}$$

Properties of Projection mat:

1.) Trace of projection Mat onto the line = 1 (ub)

2.) Trace of projection mat onto the subspace is not necessarily 1

3.) Det of projection Mat is always zero except identity mat

doesn't matter whether on the line or onto the

4.) Projection mat is also an Idempotent matrix if we take the projection of $\vec{b}$ onto A again then it will be vector itself.

$$P(\vec{b}) = P(P\vec{b}) = P\vec{b}$$

$$\Rightarrow P^2 \vec{b} = P\vec{b} \Rightarrow P^2 = P$$



5.) $\therefore P$ is an idempotent mat

$$\text{i.e. } P^2 = P$$

$$\lambda^2 = \lambda \Rightarrow \lambda(\lambda - 1) = 0$$

$$\text{or } \lambda = 0 \text{ or } 1$$

i.e. value of projection mat either 0 or 1 or both.

Spinal

Lecture no. 06

Q:) Given a vector $v = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and a subspace W spanned by $w = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ what is the projection matrix P that projects any vector onto W , and what is the projection of v onto W ?

a.) $P = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$, Proj of $v = \begin{pmatrix} 2.5 \\ 2.5 \end{pmatrix}$ ✓

b.) $P = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$, " " " " = $\begin{pmatrix} 1.5 \\ 1.5 \end{pmatrix}$

c.) $P = \begin{pmatrix} 0.5 & 2.5 \\ 1.5 & 1.5 \end{pmatrix}$, " " " " = $\begin{pmatrix} 1.5 \\ 1.5 \end{pmatrix}$ = Proj = 2.5 / 1

d.) $P = \begin{pmatrix} 0.5 & 0.15 \\ 1.5 & 1.15 \end{pmatrix}$, " " " " = $\begin{pmatrix} 0.5 \\ 1.5 \end{pmatrix}$ = Proj = 1 / 1

Let $A = W = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow P = \frac{WW^T}{W^T W}$

$W^T W = \frac{1}{2} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Here vector $B = v$

$P = PB = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 2.5 \end{bmatrix}$

→ A, B, C & D are not symmetric

Q:) In a 2-dim space R^2 , consider the subspace W spanned by the vector $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Let P be the projection matrix onto W . Which of the following vector is the image of $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ under projection P .

- a.) $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ b.) $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$ c.) $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ d.) $\begin{bmatrix} 7/2 \\ 7/2 \end{bmatrix}$

$$A = W \quad B = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad P = \frac{AA^T}{A^T A} = \frac{WW^T}{W^T W} = \frac{v_1 v_1^T}{v_1^T v_1}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\vec{B} = P \vec{B} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 7/2 \\ 7/2 \end{bmatrix}$$

Q:) For a 2×2 Proj mat P that projects onto the line $y=x$. What is the matrix P .

- a.) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ b.) $\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ c.) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ d.) $\begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$

$y=x \Rightarrow y-x=0$
so

$$A = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ x \end{bmatrix} \quad \text{i.e. } A = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$P = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

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Q:) Which of the following matrices represent a projection onto the line L with direction vector $d = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

a.) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ in (x) b.) $\begin{bmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{bmatrix}$ c.) $\begin{bmatrix} 1/5 & 2/5 \\ 2/5 & 4/5 \end{bmatrix}$ ✓

d.) $\begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}$ in (x) $P = \frac{AA^T}{A^T A} = \frac{dd^T}{d^T d} = \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

⇒ Q:) if $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ then find projection of B

onto $A^T A$? Also find the orthogonal vector (F) onto the col of A . Complement

sol: $A^T A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$(A^T A)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

$P = A(A^T A)^{-1}A^T$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$= \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

Date:

$$\text{i.e. let } A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \text{ \& } B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, P = \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\vec{b} = PB = \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$\Rightarrow$ if $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ \& $B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ then split B into $\vec{b} + \vec{q}$ where $\vec{b}$ is in the column space of A \& $\vec{q}$ is orthogonal to that space.

$\Rightarrow$ we know that $\vec{b} + \vec{q} = \vec{B}$ where $\vec{q}$ = orthogonal complement of $\vec{b}$

$$\vec{q} = B - \vec{b} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 \\ 2/3 \\ -1/3 \end{bmatrix}$$

$\Rightarrow$ Also find proj mat for orthogonal complement

sol:

We know that

$$P + Q = I$$

$$Q = I - P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 & 1/3 & -1/3 \\ 1/3 & 1/3 & -1/3 \\ 1/3 & -1/3 & 1/3 \end{bmatrix}$$

Spiral

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Q:) if V is a subspace spanned by $(1, 1, 0, 1)$ & $(0, 0, 1, 0)$ then find

- 1.) The projection matrix P onto V
- 2.) The vector in V which is closest to the vector $B = (0, 1, 0, -1)$.

Sol:) (i) $x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}$

$x_1 \cdot B = 0$ & $x_2 \cdot B = 0$
i.e. B is $\perp$ to columns of A so $\begin{bmatrix} \vec{p} \\ \vec{0} \end{bmatrix} = \vec{0}$ $\checkmark$

(ii) Do it by conventional method. $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ $A^T A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$
 $(A^T A)^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ $P = A(A^T A)^{-1} A^T$ $\checkmark$

Q:) Find the proj mat & proj of $B(b_1, b_2, b_3)$ onto column space of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -2 \\ 1 & -4 & 1 \end{bmatrix}$ By 7th property

Sol:) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -2 \\ 1 & -4 & 1 \end{bmatrix}$ $|A| = -42 \neq 0$ i.e. A^{-1} exists.
By property (7) $\text{proj} = I$
 $\text{proj} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\checkmark$

Q:) Find the proj of $B(1, 3, 4)$ onto $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$
Sol:) By observation
 $B = x_1 + 3x_2$ $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

i.e. B is LD on columns of A
So By property (9)
 $\vec{p} = B = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$ $\checkmark$

M-2 Consider $M = \{x_1, x_2, B\}$
 $= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 7 \end{bmatrix}$ $|M| = 0$
So x_1, x_2, x_3, B are L.D. $\checkmark$ Special

6.) Projection mat always symmetric.

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7.) if A is sq. mat s.t A^{-1} exist then proj mat
 $= P = I$ & $\vec{p} = \vec{b}$
i.e vector itself.

Proof $P = A(A^T A)^{-1} A^T$

$$= A(A^{-1}(A^T)^{-1})A^T = (AA^{-1})(A^T)^{-1}A^T = I \cdot I = I$$

$\vec{p} = P \cdot \vec{b} = I \vec{b} = \vec{b}$ i.e vector itself. Hence proved.

8.) if B is orthogonal to the columns of A then

$P =$ No idea

$\vec{p} = \vec{b}$.

9.) if B is L.D on columns of A then $P =$ No idea
 $\vec{p} = \vec{b}$ itself.

Q.) Find the projection of $B(3, 2, 1)$ onto the

column of $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & -4 \end{bmatrix}$

$x_1 \cdot x_2$

a.) $\vec{b}$ $\therefore x_1 \cdot 0 = 0, x_2 \cdot 0 = 0$

b.) $\vec{b}$ ✓ i.e.

c.) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

B is orthogonal to the column of A
Hence by prop (8)

d.) $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$\vec{p} = 0$

option B is correct.

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Q:) Consider the vector space $\mathbb{R}^3$ and the subspace W spanned by vector $u_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ & $u_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ what is the projection matrix P that projects any vector onto W and what is the projection of $v = \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix}$ onto W .

a.) $P = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$, projection of $v = \begin{bmatrix} 5 \\ 4 \\ 3 \end{bmatrix}$ $|P| \neq 0$
 $\neq 0$ $\Rightarrow$ not sym

b.) $P = \frac{1}{3} \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$, " " " = $\frac{1}{3} \begin{bmatrix} 5 \\ 4 \\ 3 \end{bmatrix}$ $|P| \neq 0$
 $\neq 0$ $\Rightarrow$ not sym

c.) $P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$, " " " = $\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ $|P| \neq 0$
 $\neq 0$ $\Rightarrow$ not sym

d.) None of the above.

Q:) Given 3×3 matrix P that projects onto the plane spanned by the vectors $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ what will be the norm of P $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ H.W.

a.) 1
 b.) $\sqrt{2}$
 c.) $\sqrt{3}$
 d.) $\sqrt{3}$ ✓
 $\|P\| = 1$
 $\sqrt{\frac{4}{3} + \frac{4}{3} + \frac{1}{3}}$
 $= \sqrt{\frac{24}{3}} = 2\sqrt{2}$ ✓

$P = A(A^T A)^{-1} A^T$

formula

Date _____

Q:) For a $m \times n$ matrix P that projects onto a subspace spanned by $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^2$, what will be $\det(P)$?

a.) 0 ✓ By property.

b.) 0.5

c.) 1

d.) 2

Q:) if P is a 2×2 $m \times n$ mat such that $P \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ and $P \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$, what is $\det(P)$?

sol:) $\Rightarrow$ we know that $|P| = 0$ always except identity mat & for identity mat $|P| = 1$.

a.) 1.5

b.) 0.15

c.) 0.5

d.) No valid $m \times n$ matrix P . ✓

Q:) Given a $m \times n$ mat P in $\mathbb{R}^n$ how can determine the dimension of the subspace onto which P projects?

a.) it is the no. of non-zero col in P

b.) it is the rank of P . ✓

c.) it is the no. of zero eigen values of P

d.) it is the trace of P .

Q. Rank of $m \times n$ mat = Dim of subspace formed by all A

$$P(P) = \dim(\text{col space of } A)$$

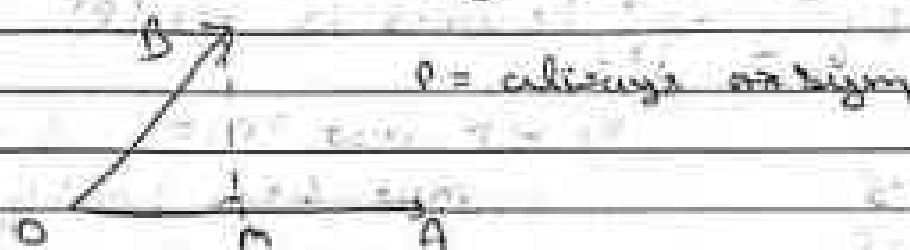
Date:

Q:) Let P be a proj mat. which of the following properties does not necessarily hold?

- a.) P is idempotent.
- b.) The eigen value of P are either 0 or 1.
- c.) P is always diagonalizable.
- d.) P is always symmetric ✓
- e.) All are correct.

Ans:

By default we take orthogonal projection.



Ans:

Sometimes projection may be of following type



This is not orthogonal type of projection

$\Rightarrow P = \text{Not Necessarily symmetric}$

Date

Date _____

Calculus

DPP-2

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Date

Lecture no. 01

Function & Graph.

Type of function

Algebraic
function

- 1.) Polynomial funct.
- 2.) Rational funct.
- 3.) Fractional funct.
- 4.) Piecewise funct.

- a.) mod funct
- b.) Signum funct
- c.) G.I.F
- d.) L.I.F
- e.) F.P.F

Transcendental
function

- 1.) Exponential funct
- 2.) Log funct
- 3.) Trigonometric funct
- 4.) Inverse trig funct

- ⇒ G.I.F = Greatest Integer funct (Floor function)
⇒ L.I.F = Least Integer funct (Ceiling function)
⇒ F.P.F = Fractional Part funct.

1.) Polynomial:

It's domain is $(-\infty, \infty)$ & Degree = 0, 1, 2, 3, 4, 5, ...
It's definition is same at all points in the domain of $y = f(x)$.

Date

e.g.:

$$y = k \text{ (constant poly)} = \text{degree} = 0$$

$$y = ax + b \text{ (linear poly)} = \text{deg} = 1$$

$$y = ax^2 + bx + c \text{ (quad poly)} = \text{deg} = 2$$

$$y = ax^3 + bx^2 + cx + d \text{ (cubic poly)} = \text{deg} = 3$$

Basic graphs:-

1) Eq. of X axis $y = 0$

2) Eq. of line $\parallel$ to X axis $y = k$ (Horizontal line)

3) Eq. of Y axis $x = 0$

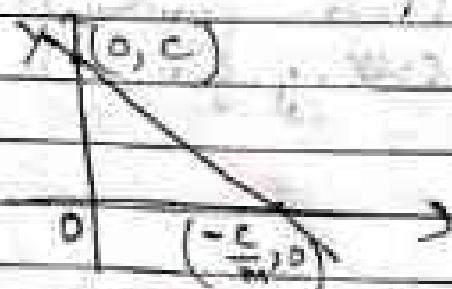
4) Eq. of line $\parallel$ to Y axis $x = k$ (Vertical line).

5) Eq. of line passing through origin is $y = mx$
when $m = \text{slope of line} = \tan \theta$

6) Slope Intercept form.

$$y = mx + c$$

$c =$ Intercept cuts on Y axis.



in obtuse angle

$\tan \theta$ is negative

So $m = \tan \theta$

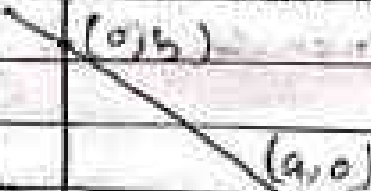
$$= -ve.$$

7) Intercept form

$$\frac{x}{a} + \frac{y}{b} = 1$$

$a = x$ intercepts

$b = y$ intercepts

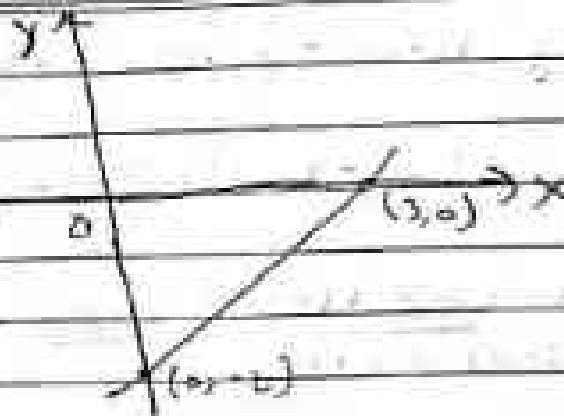


e.g.) $2x - 3y = 6$

or

$$\frac{2x}{6} - \frac{3y}{6} = 1$$

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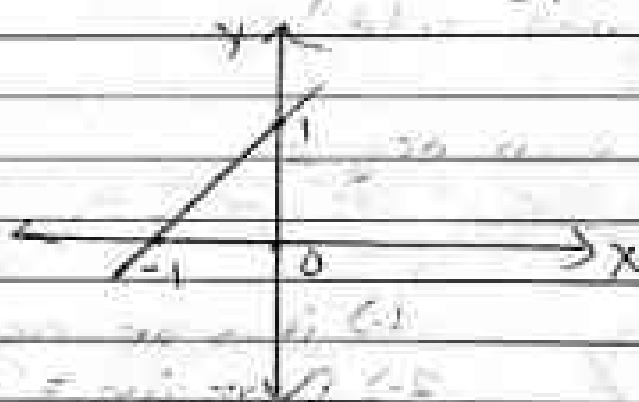
Q:) Draw the graph of $y = \begin{cases} 1+x & -1 \leq x \leq 0 \\ 1-x & 0 \leq x \leq 1 \end{cases}$

Sol:) Case 1:

if $-1 \leq x \leq 0$

$$y = 1+x$$

$$-x + y = 1 \quad \Rightarrow \quad \frac{-x}{1} + \frac{y}{1} = 1$$



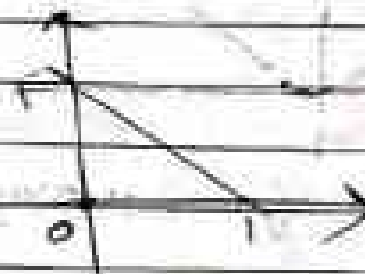
Case 2:

if $0 \leq x \leq 1$

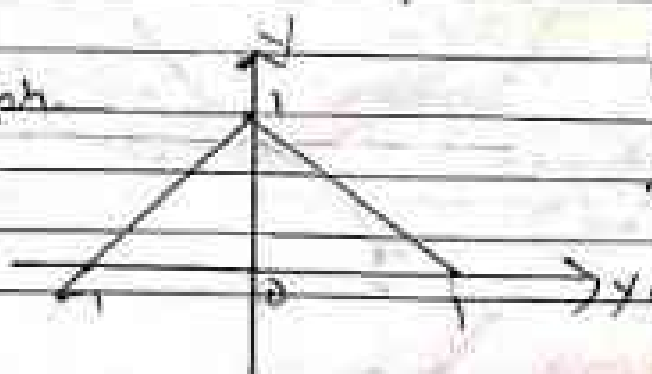
$$y = 1-x$$

$$x + y = 1$$

$$\Rightarrow \frac{x}{1} + \frac{y}{1} = 1$$



Final graph



Since $f(-x) = f(x)$
Even function

Date

$\Rightarrow$ **Even funct:** if $f(-x) = f(x) \Rightarrow f(x)$ is called an even

$\Rightarrow$ **odd funct:** if $f(-x) = -f(x) \Rightarrow f(x)$ is called an odd fu

$\Rightarrow$ **NEVO funct:** if $f(-x) \neq f(x)$ } then $f(x)$ is called NEVO
 $\wedge f(-x) \neq -f(x)$

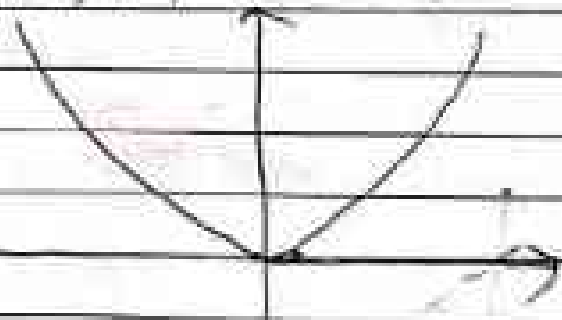
$\Rightarrow$ if graph of $f(x)$ is symmetrical about Y axis
 $f(x)$ is an even function.

$\Rightarrow$ if graph of $f(x)$ is symmetrical about origin
 then $f(x)$ is an odd function.
 (I $\longleftrightarrow$ III & II $\longleftrightarrow$ IV)

$\Rightarrow$ NEVO (neither even nor odd).

* Polynomial of even degree

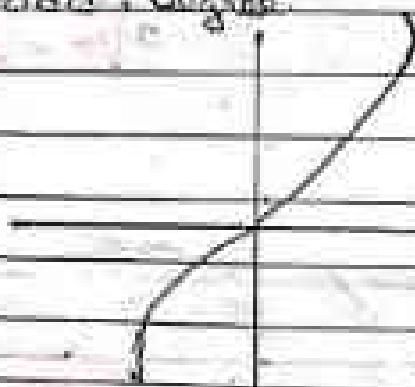
$$y = x^2, x^4, x^6, \dots$$



- 1.) it is an even function
- 2.) Domain = $(-\infty, \infty)$
- 3.) Range = $[0, \infty)$

* Polynomial of odd degree

$$y = x^3, x^5, x^7$$

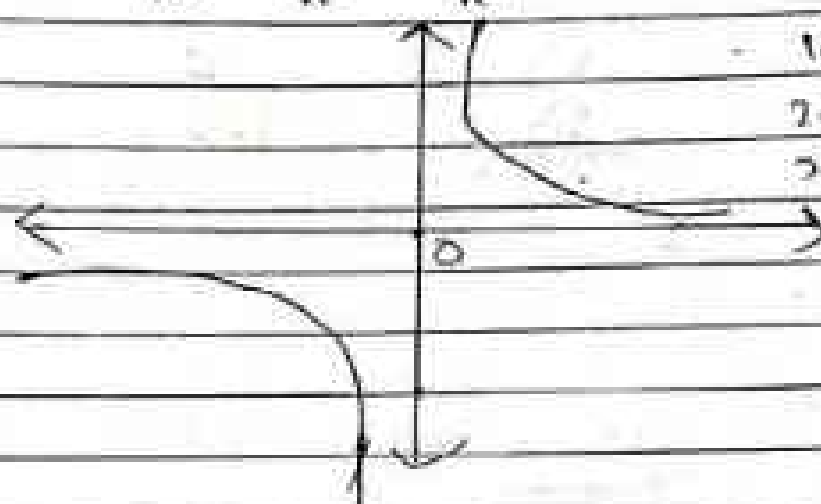


- 1.) it is odd fu
- 2.) Domain = $(-\infty, \infty)$
- 3.) Range = $(-\infty, \infty)$

Date

* Rational function (Type 1)

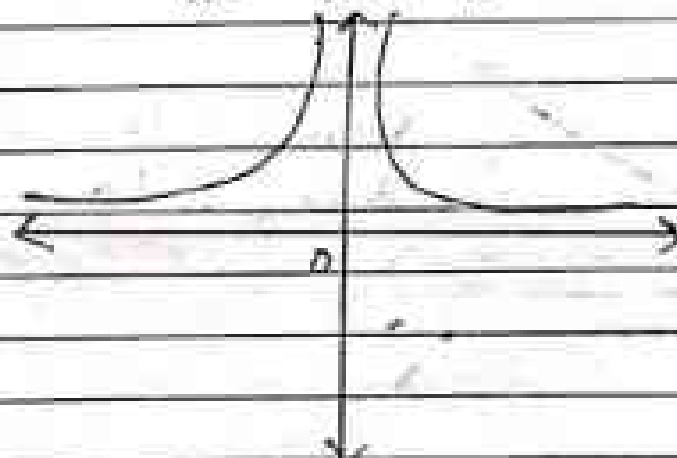
$$y = \frac{1}{n}, \frac{1}{n^2}, \frac{1}{n^3}, \dots$$



- 1.) odd function
- 2.) Domain: $\mathbb{R} - \{0\}$
- 3.) Range: $\mathbb{R} - \{0\}$

* Type 2:

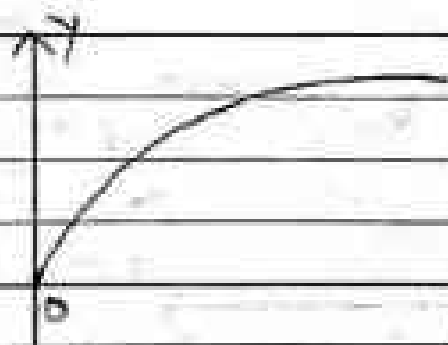
$$y = \frac{1}{n^2}, \frac{1}{n^4}, \frac{1}{n^6}, \dots$$



- 1.) even function
- 2.) Domain: $\mathbb{R} - \{0\}$
- 3.) Range: $(0, \infty)$
i.e. $y > 0$

* Irrational function (Type 1)

$$y = \sqrt[n]{n}, \sqrt[n]{4}, \sqrt[n]{6}, \dots \text{ or } y = \sqrt{n}$$



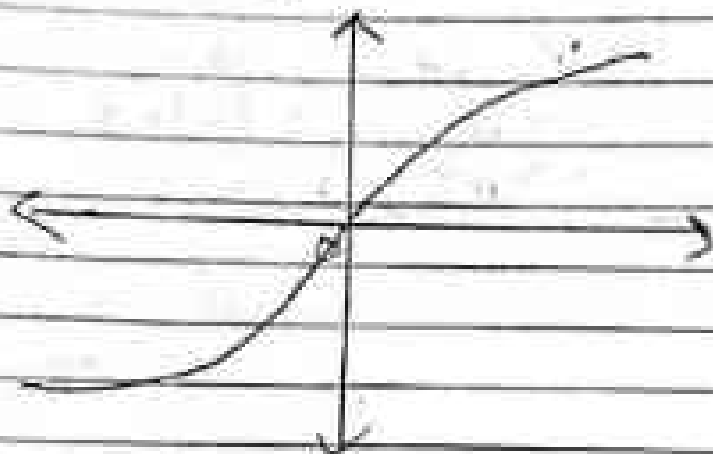
- 1.) $\mathbb{N} \in \mathbb{N}$
- 2.) Domain: $[0, \infty)$
i.e. $n \geq 0$
- 3.) Range: $[0, \infty)$
i.e. $y \geq 0$

$$\frac{1}{\infty} = \text{Something} = 0$$

Date

• Type 2

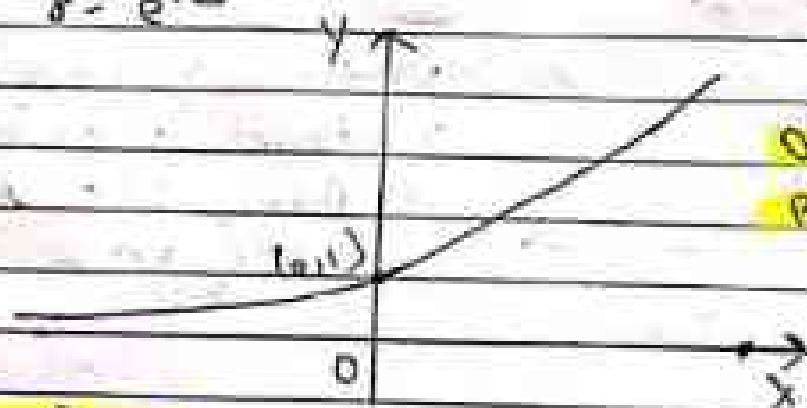
$$y = \frac{y_1}{x} + \frac{y_2}{x} + \frac{y_3}{x} + \dots$$



- 1.) odd function
- 2.) Domain: $(-\infty, \infty)$
- 3.) Range: $(-\infty, \infty)$

⇒ Exponential function:

$$y = e^x$$

Domain: $(-\infty, \infty)$ Range: $(0, \infty)$ i.e. $x > 0$

$$y(0) = e^0 = 1$$

$$y(-\infty) = e^{-\infty} = 0$$

$$y(\infty) = e^{\infty} = +\infty$$

⇒ log function:

$$y = \log e^x \text{ or } y = \ln x$$



$$y(1) = \log e^1 = 0$$

$$y(0) = \log e^0 = -\infty$$

$$y(\infty) = \log e^{\infty} = +\infty$$

$$y(-\infty) = \log e^{-\infty} = \text{N.D.}$$

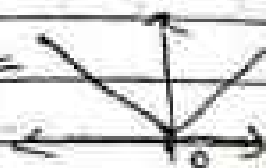
Domain: $(0, \infty)$ Range: $(-\infty, \infty)$

NEED function

Date

 $\Rightarrow$ Modulus function:

$$y = |x| = \sqrt{x^2} = \begin{cases} -x, & x < 0 \\ +x, & x \geq 0 \end{cases} = \max\{x, -x\}$$

Domain: $(-\infty, \infty)$ Range: $(0, \infty)$ i.e. Mod is always +ve

Even function:

e.g. 1:) $|3| = +3, |-3| = -(-3) = +3$

e.g. 2:) $|3| = \max\{3, -3\} = +3$
 $|-3| = \max\{-3, -(-3)\} = \max\{-3, +3\} = 3$

e.g. 3:) $|3| = \sqrt{(3)^2} = \sqrt{9} = +3$
 $|-3| = \sqrt{(-3)^2} = \sqrt{9} = +3$

"Sq. root of real no. is always taken as +ve"

 $\Rightarrow$ Piecewise function:

if funct is defined by multiple sub functions
 s.t. Domain of each sub function is different the
 function is called piecewise function.

Ex e.g.:

Mod funct, signum funct, h.i.f, l.i.f
 fractional part funct

P. 1. 1. 0

Date

Q:) The number of solution $n^2 - 5|n| + 6 = 0$ is andsol:) Case I if $n < 0$

$$n^2 - 5(-n) + 6 = 0$$

$$n^2 + 5n + 6 = 0$$

$$(n+2)(n+3) = 0$$

$$\boxed{n = -2 \text{ \& } n = -3}$$

Case II if $n > 0$

$$n^2 - 5(n) + 6 = 0$$

$$n^2 - 5n + 6$$

$$(n-2)(n-3) = 0$$

$$\boxed{n = 2 \text{ \& } n = 3}$$

Four solutions

Q:) The no. of sol of $n^2 + 5|n| + 6 = 0$ is and
(common sense).sol:) a) 0 ✓ $m-1$ (ive) 0 + (ive 0) + (ive 0) = 0

b) 2 sum of three +ve quantity cannot be zero

c) 3 so we have no solution

d) 4 i.e

No value of n satisfying above eqn

m-2 Do yourself

Lecture no. 02

$$n^2 + 5|n| + 6 = 0 \quad \text{--- (1)}$$

Case 1: $n < 0$ Case 2: $n > 0$

$$n^2 + 5(-n) + 6 = 0$$

$$n^2 + 5(n) + 6 = 0$$

$$n^2 - 5n + 6 = 0$$

$$n^2 + 5n + 6 = 0$$

$$(n-2)(n-3) = 0$$

$$(n+2)(n+3) = 0$$

$$\boxed{n = 2, \& 3}$$

$$\boxed{n = -2, \& -3}$$

Contradiction

Contradiction

value is take -ve so
 result come -ve but
 result come +ve

} So we have no sol
 i.e no value of n satisfying
 or (C)

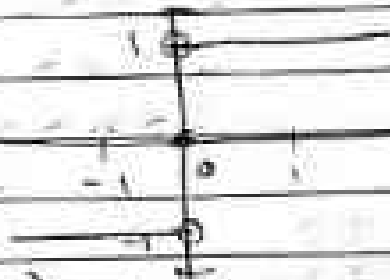
Spiral

Date _____

 $\Rightarrow$ Signum function:

$$\text{Sgn}(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$= \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ +1 & x > 0 \end{cases}$$

Domain = $(-\infty, \infty)$ Range = $\{-1, 0, 1\}$

odd function.

$\text{Sgn}(0.1) = 1$

$\text{Sgn}(0.9) = 1$

$\text{Sgn}(1.5) = 1$

$\text{Sgn}(2.7) = 1$

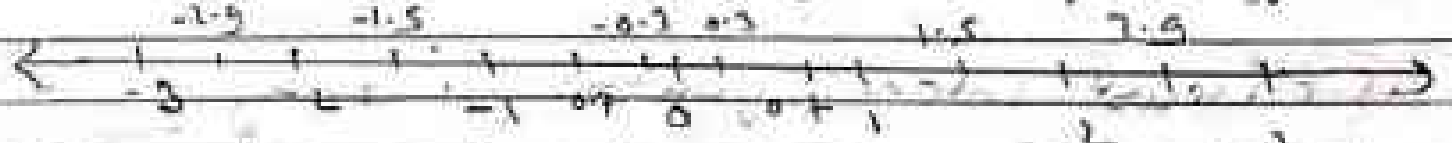
$\text{Sgn}(-0.2) = -1$

$\text{Sgn}(-0.9) = -1$

$\text{Sgn}(-1.5) = -1$

$\text{Sgn}(-2.7) = -1$

$\text{Sgn}(0) = 0$

 $\Rightarrow$ G.I.F (Greatest integer function) / Floor function

$\lfloor 0.3 \rfloor = 0, \lfloor 0.7 \rfloor = 0, \lfloor 1.5 \rfloor = 1, \lfloor 2.9 \rfloor = 2$

$\lfloor -0.3 \rfloor = -1, \lfloor -0.7 \rfloor = -1, \lfloor -1.5 \rfloor = -2, \lfloor -2.9 \rfloor = -3$

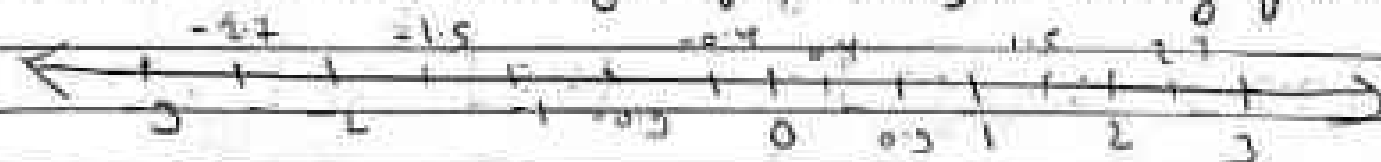
$\lfloor 2.1 \rfloor = 2, \lfloor 2 \rfloor = 2, \lfloor 0 \rfloor = 0$

$\lfloor -2.1 \rfloor = -3, \lfloor -2 \rfloor = -2, \lfloor 3 \rfloor = 3$

Date _____

Def:

$$y = [x] = \begin{cases} -2 & -2 \leq x < -1 \\ -1 & -1 \leq x < 0 \\ 0 & 0 \leq x < 1 \\ 1 & 1 \leq x < 2 \\ 2 & 2 \leq x < 3 \\ 3 & 3 \leq x < 4 \end{cases}$$

Domain: $(-\infty, \infty)$ Range: $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ = set of integers
NEED function (Neither even nor odd). $\Rightarrow$ L.I.F (Least Integer function) / (Ceiling function)

$$[0.4] = 1, [0.9] = 1, [1.5] = 2, [2.7] = 3$$

$$[-0.4] = 0, [-0.9] = 0, [-1.5] = -2, [-2.2] = -3$$

$$[0] = 0, [1] = 1, [2] = 2, [-1] = -1, [-2] = -2$$

Spiral

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Def:

$$y = |x|$$

$$\begin{cases} -1 \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ \vdots \end{cases}$$

$$-2 < x \leq -1$$

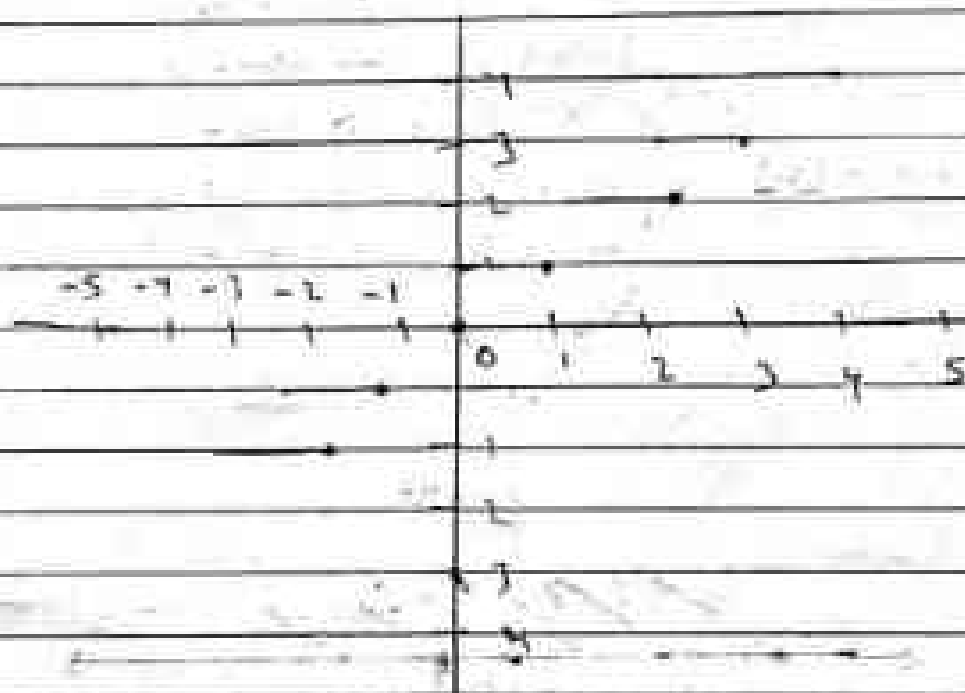
$$-1 < x \leq 0$$

$$0 < x \leq 1$$

$$1 < x \leq 2$$

$$2 < x \leq 3$$

$$3 < x \leq 4$$

Domain: $= (-\infty, \infty)$ Range: $= \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\} = \mathbb{Z}$ {w. fix.

MEMO function.

⇒ Fractional part function:

$$y = \{x\} = x - [x]$$

$$\text{e.g.) } \{2.3\} = 0.3, \quad \{1.5\} = 0.5, \quad \{0.7\} = 0.7 \dots$$

$$\{-2.3\} = (-2.3) - [-2.3] = -2.3 - (-3) = -2.3 + 3 = 0.7$$

$$\{-1.5\} = (-1.5 + 2) = 0.5$$

Date

$$f(-0.7) = -0.7 + 1 = 0.3$$

$$f(-3.2) = -3.2 + 4 = 0.8 \text{ \& so on } \dots$$

And

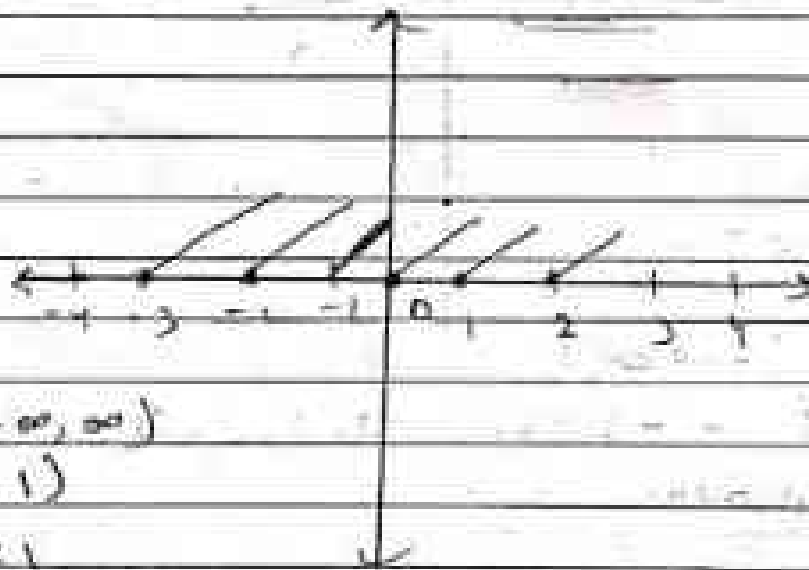
$$f(0) = 0, f(1) = 0, f(2) = 0$$

$$f(-1) = 0, f(3) = 0, f(4) = 0$$

$$\text{Def:)} \quad y = f(x) = \{x\} = x - [x]$$

$$y = \{x\} = x - [x]$$

$$\begin{cases} x+2, & -2 \leq x < -1 \\ x+1, & -1 \leq x < 0 \\ x, & 0 \leq x < 1 \\ x-1, & 1 \leq x < 2 \\ x-2, & 2 \leq x < 3 \end{cases}$$



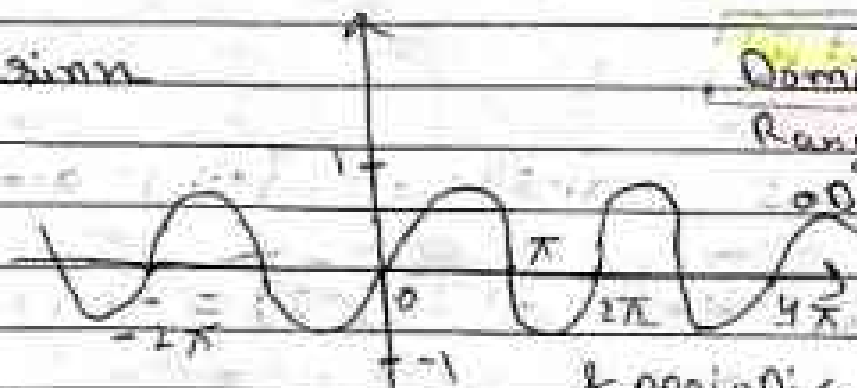
$$\text{Domain} = (-\infty, \infty)$$

$$\text{Range} = [0, 1)$$

$$\text{i.e. } 0 \leq y < 1$$

NGND function

$$* y = \sin x$$



$$\text{Domain} = (-\infty, \infty)$$

$$\text{Range} = [-1, 1]$$

odd function

& periodic function with
Period = 2π Spiral

Date

$$f(-n) = \sin n$$

$$f(-n) = \sin(-n)$$

$$= -\sin n$$

$$= -f(n)$$

So $f(n)$ is an odd function.

$$* y = \cos n$$



Domain: $(-\infty, \infty)$

Range: $[-1, 1]$

Even function & periodic with period $= 2\pi$

$\Rightarrow$ Periodic function:

$f(n)$ is called periodic function of period T if

$$f(n+T) = f(n)$$

e.g. $\sin(n+2\pi) = \sin n$ so period of $\sin n$ is 2π

e.g. $\tan(n+\pi) = \tan n$ i.e. period is $\tan n \pi$

$$f(n) = |\sin n| \quad T = \pi \text{ (periodic)}$$

$$f(n) = \sin |n| \quad T = \text{Non-periodic (even function)}$$

~~Q. 4. 0~~

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$\Rightarrow$ Domain of $y = f(x) \Rightarrow$ Set of permissible value of x is called Domain.

$\Rightarrow$ Range of $y = f(x) \Rightarrow$ Set of permissible value of y is called range.

i.e.

Restriction imposed on Inputs (x) is called Domain
 & " " " " outputs (y) " " Range

$\Rightarrow$ Domain: Set of permissible value of x is called Domain of $y = f(x)$ & there is no shortcut method to find Domain of given funct, knowledge it can be calculated only by using common sense or by previous knowledge.

e.g Find the Domain of the following funct:

1.) $y = f(x) = \frac{1}{x^2 - 5x + 6}$

Sol: $y = \frac{1}{(x-2)(x-3)}$ $\text{Dom} = \mathbb{R} - \{2, 3\}$

2.) $y = \sqrt{-x^2 + 5x - 6}$
 $= \sqrt{-(x-2)(x-3)}$

w.k. that $-(x-2)(x-3) \geq 0$
 $(x-2)(x-3) \leq 0$
 $2 \leq x \leq 3$

$\therefore$ Domain $[2, 3]$ $\mathbb{R}$

Spiral

Common rules:

$$1.) \frac{1}{f(n)} ; f(n) \neq 0$$

$$2.) \sqrt{f(n)} ; f(n) \geq 0$$

$$3.) \frac{1}{\sqrt{f(n)}} ; f(n) > 0$$

$$4.) \log_e(f(n)) ; f(n) > 0$$

$$5.) \sin(f(n)) ; -\infty < f(n) < \infty$$

$$Q:) y = f(n) = \log(n-3)$$

$$\text{sol:)} \text{ w.k that, } (n-3) > 0$$

$$n > 3$$

$$\text{So Domain} = (3, \infty)$$

$$Q:) y = f(n) = \sin(n-1)$$

$$\text{w.k that } \sin(n-1) < \infty$$

$$-\infty < n < \infty$$

$$\text{Domain} = (-\infty, \infty)$$

$$Q:) y = \sin^{-1}(n-3)$$

$$\text{w.k that we can write, } \sin y = (n-3)$$

$$\text{w.k that}$$

$$-1 \leq \sin y \leq 1$$

$$-1 \leq (n-3) \leq 1$$

$$2 \leq n \leq 4$$

$$\text{So Domain} = [2, 4]$$

$$Q.1) y = n^n$$

$$= e^{\log_e(n^n)} = (n \log n)$$

Not indeterminate

So Domain = $(0, \infty)$.

i.e. in above funt -ve inputs are not allowed.

Lecture no. 03⇒ Some useful Result:

$$1.) |n-a| < l \Rightarrow a-l < n < a+l$$

e.g.

$$|n| < l \Rightarrow -l < n < l$$

$$2.) |n-a| > l \Rightarrow n < a-l \text{ or } n > a+l$$

e.g.

$$|n| > l \Rightarrow n < -l \text{ or } n > l$$

$$3.) \text{ if } a < b \text{ and } (n-a)(n-b) < 0 \\ \Rightarrow a < n < b$$

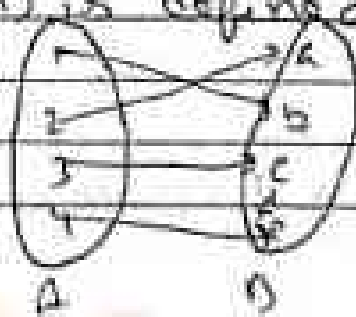
$$4.) \text{ if } a < b \text{ and } (n-a)(n-b) > 0 \\ \Rightarrow n < a \text{ or } n > b$$

Some more Concept:-Let $f: A \rightarrow B$ i.e. $f(n)$ is defined from A to B then:

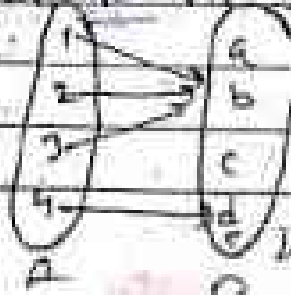
eg ①

one-one

into



2.)



many

one/

into

3.)

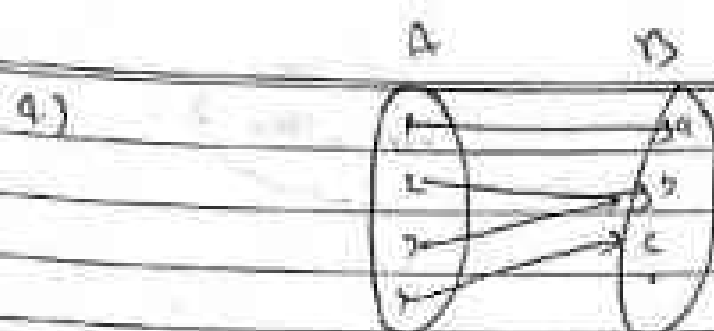


many

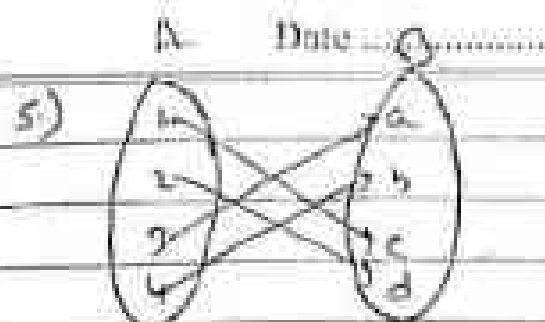
one/

into

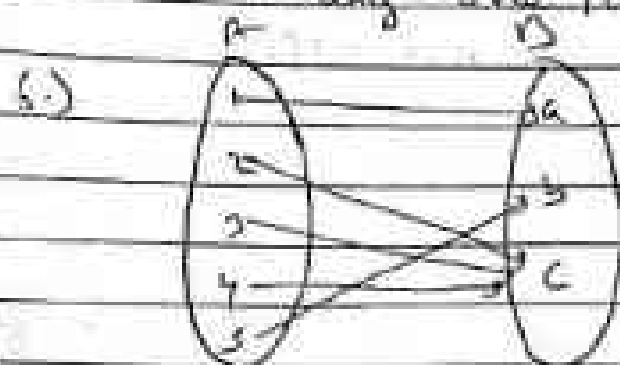
B- C₂-familiën:



many - ang / and o



One - one / One



many-one / onto.

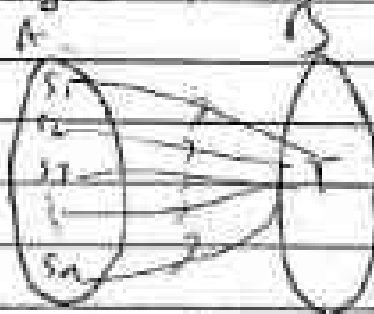
1.) Range = {a, b, c, d}, Range $\in$ codomain. Hence inv.

2.) Range = $\{b, d\}$, $\therefore$ "C" = 26, "E" = 13, "A" = 1

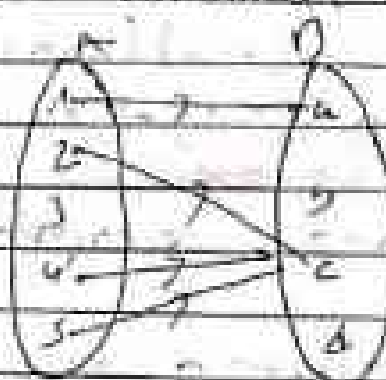
2.) Range = $\{a, b, c\}$ $13 \in C$ " = 2" "

5.) Range = $\{4, 5, 6\}$. Range = Codomain. Hence odd.

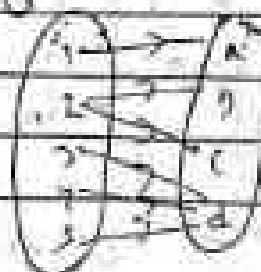
6.) Range = $\{a, b, c\}$ Range = Cardinal Number



Many one / into



Not a
function
to condition
not satisfied



Not a
function
uniqueness property held.
Special

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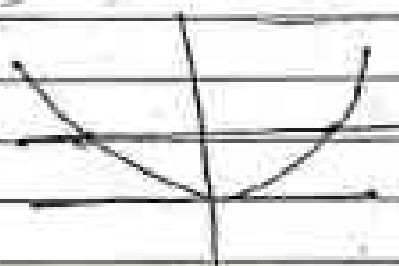
Note:)

- 1.) if different element have different image then f is called one-one
- 2.) if different element have same image then f is many one function.

Shortcut:

If any random line $\parallel^x$ to x axis cuts the graph at one point then f is called one-one otherwise many one

Let $y = x^2$ is many-one.



* one-one mapping $\approx$ Injective mapping.

* onto mapping $\approx$ Surjective mapping.

* one-one / onto mapping $\approx$ Bijective mapping.

e.g.) if $y = f(x) = x^2 ; \mathbb{R} \rightarrow \mathbb{R}$ then find range
 $= [0, \infty] = \mathbb{R}^+$ Domain $\mathbb{R}$ Co-domain $\mathbb{R}$ [$\mathbb{R} \subset \mathbb{C} \subset \mathbb{R}$ = Int.]

if $y = f(x) = x^2 ; \mathbb{R} \rightarrow \mathbb{R}^+$ then find it's nature
 Range $= \mathbb{R}^+$ Hence onto

$\Rightarrow y = f(x)$ & $y = f^{-1}(x)$
 one-one / onto one-one / onto

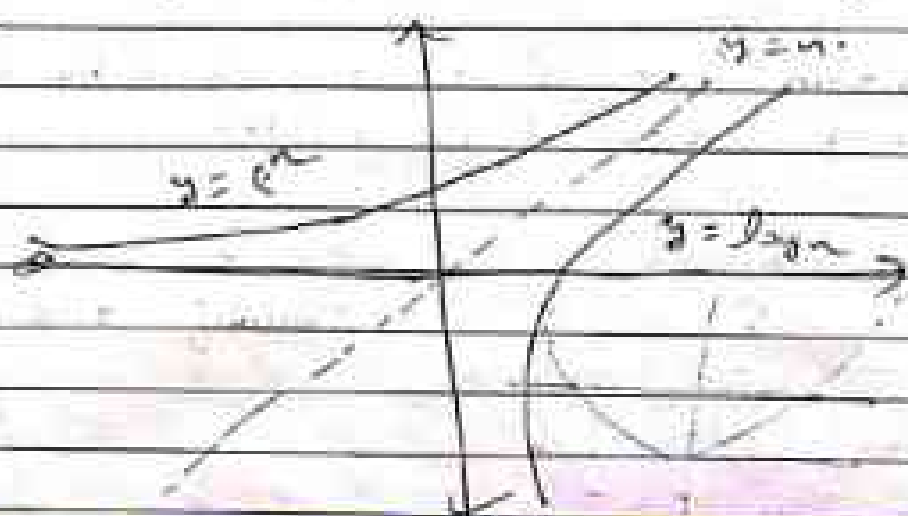
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Inverse funct:

if $y = f(x)$ & $y = g(x)$ are inverse funct of each other then they are symmetrical about the line $y = x$.

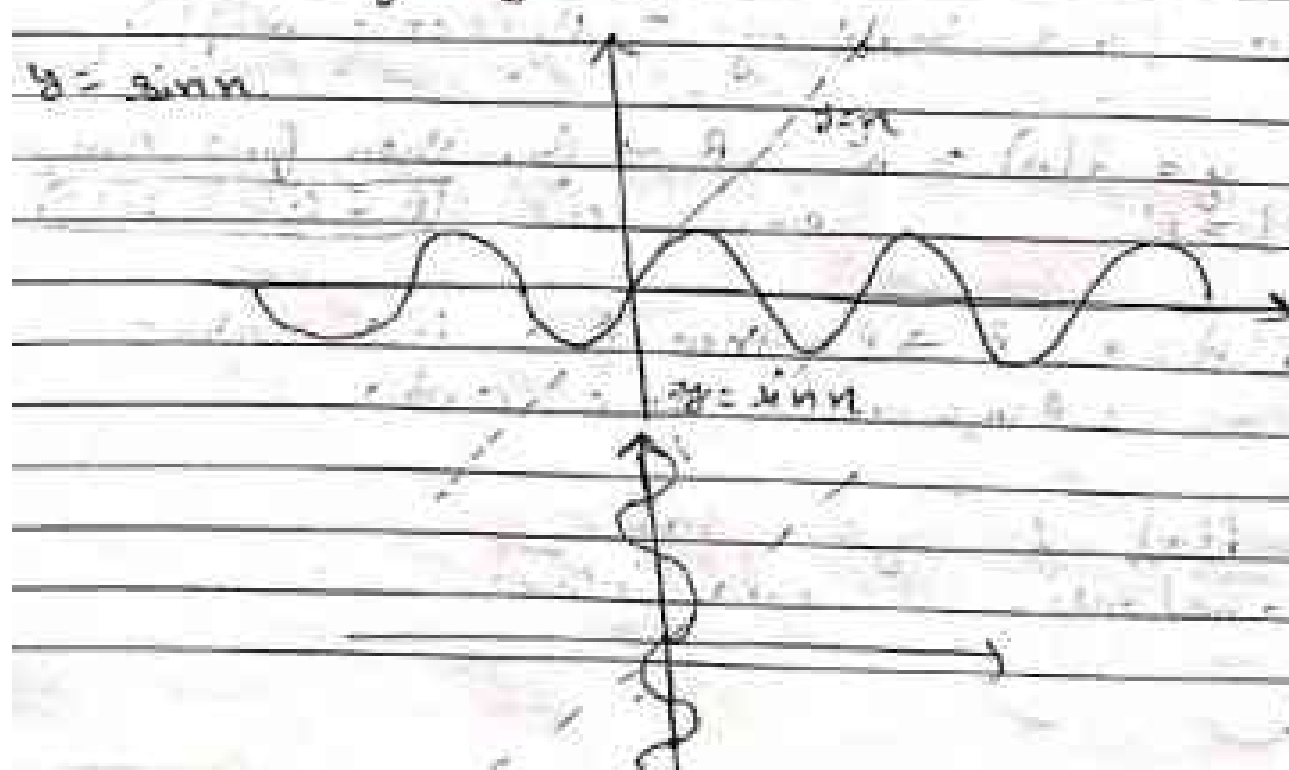
i.e you will behave like a mirror for $f(x)$ & $g(x)$

e.g $y = e^x$ & $y = \log_e x$ are inverse funct of each other.



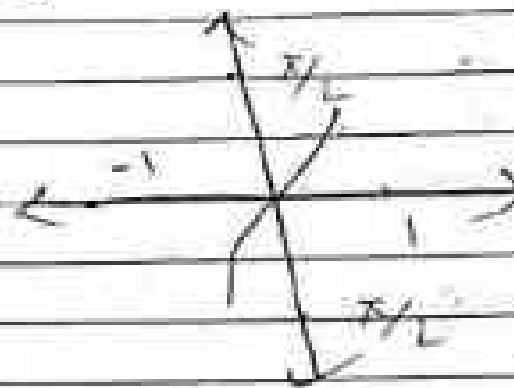
Note: if $f(x)$ is one-one/onto then only $f^{-1}(x)$ will exist & $f^{-1}(x)$ will also be one-one/onto.

* Graph of $y = \sin^{-1} x$



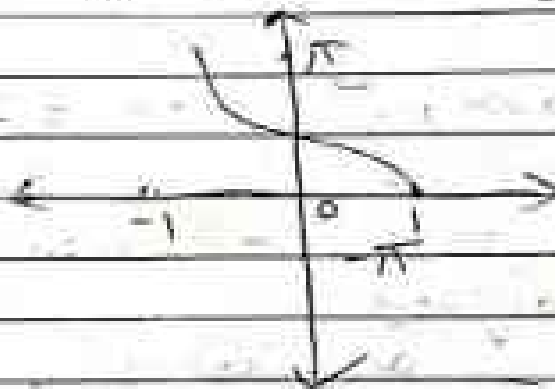
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So graph of $\sin^{-1} x$ is



Domain: $[-1, 1]$
 Range: $[-\pi/2, \pi/2]$
 odd function.

* Graph of $y = \cos^{-1} x$.



Domain: $[-1, 1]$
 Range: $[0, \pi]$
 NE NO function.

* Graph of $y = \tan^{-1} x$.



Domain: $(-\infty, \infty)$
 Range: $(-\pi/2, \pi/2)$
 It is an odd function.

Basic of Limits

$\Rightarrow$ Infinity: Two $\parallel$ line never intersect
 OR

Two $\parallel$ line intersect at ∞

1) Not a very large no.

2) ∞ is not unique.

Date

$$3.) \frac{\text{Something}}{\infty} = 0 \quad (\text{Assumption})$$

$$\frac{\text{Something}}{0} = \infty \quad (\text{fact})$$

$$4.) \infty + \infty = \infty = \text{N.D.}$$

$$5.) \infty \times \infty = \infty = \text{N.D.}$$

$$6.) \frac{\infty}{\infty} = \infty = \text{N.D.}$$

$$7.) 0^0, \infty^0, \frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, 1^\infty, \infty - \infty = \text{Indeterminate}$$

Note: if final ans is ∞ or $-\infty$ then we can say that it is Not defined.

But if final ans is $\frac{0}{0}$ then we say that it is Indeterminate form.

$\Rightarrow$ Properties of logs:

$$1.) \log_a n = b \Rightarrow n = a^b, \quad \log_a a = 1$$

$$2.) \log_a n > \log_a y \Rightarrow n > y, \quad \log_a \infty = \infty$$

$$3.) \log_a (n \cdot y) = \log_a n + \log_a y, \quad \log_a 1 = 0$$

$$4.) \log_a \left(\frac{n}{y} \right) = \log_a n - \log_a y$$

$$5.) \log_a (n^y) = y \log_a n$$

$$6.) \log_y n = \frac{\log_a n}{\log_a y}$$

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$$7.) \log_y y = \frac{1}{\log_y y}$$

$$8.) \log_a b(x) = \frac{1}{b} \log_a x$$

$$9.) a = e^{\log_e a}$$

$$10.) a^n = e^{\log_e(a^n)} = e^{n(\log_e a)}$$

$$11.) a^{\log_e b} = b^{\log_e a}$$

⇒ Some useful result:

$$(i) a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$$

$$(ii) a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$$

$$(iii) (a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$(iv) (a-b)^3 = a^3 - b^3 - 3ab(a-b)$$

$$(v) a^2 - b^2 = (a-b)(a+b)$$

Indeterminate forms:

There are exactly 7 indeterminate forms for e.g.

$$\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, \infty - \infty, 0^0, 1^\infty, \infty^0$$

"If any mathematical expression has multiple answer then that expression is said to be Indeterminate form"

$$0^0 = 1$$

$$0^\infty = 0$$

$$\frac{\infty}{\infty} = \infty$$

L-Hospital Rule $\rightarrow$ This rule provides a unique answer out of ∞ answers & this rule is applicable only for $\frac{0}{0}$ and $\frac{\infty}{\infty}$ form.

Rule:

Separately differentiate N^x & D^x , equal number of times, until we have free from $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.

(e.g.) $\lim_{n \rightarrow \infty} \left(\frac{n^3}{e^n} \right) = \frac{\infty}{\infty}$

$$= \lim_{n \rightarrow \infty} \left(\frac{3n^2}{e^n} \right) = \frac{\infty}{\infty}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{6n}{e^n} \right) = \frac{\infty}{\infty}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{6}{e^n} \right) = \frac{6}{\infty} = \frac{6}{\infty} = 0$$

Note: limit exist means "There exist unique, constant value that must be free from n ".

Lecture no. 04

Neighbourhood of real no. ' a ' $\rightarrow$ means we are considering an open interval of length $2h$, with centered at point ' a '.

i.e. nbhd of $a = (a-h, a+h)$ where $h > 0$ & h depends on our choice.

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Note: if n lies in the Nbd of $a \Rightarrow$

$$n \in (a-h, a+h) \text{ or}$$

$$a-h < n < a+h$$

or

$$|n-a| < h$$

or

$$n \rightarrow a$$

or

n is about ' a '.

e.g.) if $f(n) = \frac{n^2-9}{n-3}$ then $f(3) = ?$ DNE

$$D = A - \{3\}$$

$$\rightarrow \text{LHL} = 6$$

$$\rightarrow \text{RHL} = 6$$

$$\rightarrow \lim_{n \rightarrow 3} f(n) = 6$$

$$n \rightarrow 3$$

i.e. Approx value of $f(n)$ in the Nbd of 3

e.g.) $f(n) = \frac{|n-4|}{n-4}$ then $f(4) = ?$ DNE i.e. Exact value

$$D = A - \{4\}$$

$$\rightarrow \text{LHL} = -1$$

$$\rightarrow \text{RHL} = +1$$

$$\rightarrow \lim_{n \rightarrow 4} f(n) = \text{DNE}$$

$$f(n) = \begin{cases} \frac{-(n-4)}{n-4}, & n < 4 \end{cases}$$

$$\begin{cases} \frac{(n-4)}{n-4} & n > 4 \end{cases}$$

$$f(n) = \begin{cases} -1 & n < 4 \\ 1 & n > 4 \end{cases} \quad \text{i.e. } f(3.999) = -1$$

$$f(4.001) = 1$$

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(eg:) $f(n) = 1$ then $f(5) = ONE$

Don = A - {5}

$\nearrow$ LHL = ONE
 $\nearrow$ AHL = ONE
 $\nearrow$ $\lim_{n \rightarrow 5} f(n) = ONE$

$\Rightarrow$ method of solving question:

M-1 By Direct substitution. (Best method)

M-2 By factorisation.

M-3 By Rationalisation

M-4 By using I.M.O form concept

M-5 By using standard result.

M-6 Using Common sense.

$\Rightarrow$ Sp. formula

$$1.) \sum_{n=1}^N n = 1 + 2 + 3 + \dots + N$$

$$= \frac{N(N+1)}{2}$$

$$2.) \sum_{n=1}^N n^2 = 1^2 + 2^2 + 3^2 + \dots + N^2$$

$$= \frac{N(N+1)(2N+1)}{6}$$

$$3.) \sum_{n=1}^N n^3 = 1^3 + 2^3 + 3^3 + \dots + N^3$$

$$= \left(\frac{N(N+1)}{2} \right)^2$$

Date

$$4) \sum_{n=1}^N (a) = \underbrace{a + a + a + \dots + a}_{N \text{ times}} = N \cdot a.$$

Def of limit $\rightarrow \boxed{\lim_{n \rightarrow a} f(n) = l} \Rightarrow$

whenever n lies in the Nbd of 'a'
 $f(n)$ lies in the Nbd of l .

Continuity: $f(n)$ is said to be continuous at $n=a$ if

$$\boxed{\lim_{n \rightarrow a} f(n) = f(a)} \Rightarrow \text{App value} = \text{Exact value}$$

$\Rightarrow$ whenever n lies in the Nbd of a , $f(n)$ lies in the Nbd of $f(a)$.

i.e. $\boxed{LHL = RHL = f(a)}$

Ex: 1)

Check the nature of following funct:

1.) $f(n) = \frac{\sin n}{n}$

Dom = $\mathbb{R} - \{0\}$

Discontinuous at $n=0$

2.) $f(n) = \begin{cases} \frac{\sin n}{n} & n \neq 0 \\ 1 & n = 0 \end{cases}$

Dom = $\mathbb{R} \text{ i.e. } (-\infty, \infty)$

Continuous at $n=0$

3.) $f(n) = \begin{cases} \frac{\sin n}{n} & n \neq 0 \\ 2 & n = 0 \end{cases}$

Approx value = $\lim_{n \rightarrow 0} f(n) = 1$

Exact value = $f(0) = 2$ (given)

$\therefore$ App value $\neq$ Exact value $\therefore$ Discontinuous at $n=0$. Spiral

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S.p note:) To check Discontinuity of $f(x)$ at $x=a$ we have 3 methods:

M-1 if Exact value = DNE then $f(x)$ is Discontinuous

M-2 if Approx value = DNE then " " "

M-3 if Both exist But not equal then $f(x)$ is Discontinuous.

i.e $\lim_{x \rightarrow a} f(x) \neq f(a) \Rightarrow f(x)$ is Discontinuous at $x=a$

Q:1) $\lim_{x \rightarrow 2} \left(\frac{x^2 - 4}{x + 3} \right) = ?$

Sol:)

$$= \frac{2^2 - 4}{2 + 3} = \frac{4 - 4}{5} = \frac{0}{5} = 0$$

Unique, constant value
exists.

Q:2) $\lim_{n \rightarrow \infty} \left(\frac{1 + 2 + 3 + 4 + \dots + n}{n^2} \right) = ?$

Sol:)

$$= \lim_{n \rightarrow \infty} \left[\frac{n(n+1)}{2} \cdot \frac{1}{n^2} \right] = \lim_{n \rightarrow \infty} \frac{n^2 \left(1 + \frac{1}{n} \right)}{2n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2} \left(1 + \frac{1}{n} \right) = \frac{1}{2} \left[1 + \frac{1}{\infty} \right] = \frac{1}{2} + 0 = \frac{1}{2}$$

3) $\lim_{n \rightarrow \infty} \left[\frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n^3} \right] = ?$

Date

Sol: $\lim_{n \rightarrow \infty} \frac{n(n+1)(2n+1)}{6n^3}$

$$= \lim_{n \rightarrow \infty} \frac{n^3(1+\frac{1}{n})(2+\frac{1}{n})}{6n^3}$$

$$= \lim_{n \rightarrow \infty} \frac{(1+\frac{1}{n})(2+\frac{1}{n})}{6} = \frac{(1+0)(2+0)}{6}$$

$$= \frac{2}{6} = \frac{1}{3} \quad \square$$

Q: $\lim_{n \rightarrow \infty} \frac{1-2+3-4+5-6+\dots+(2n-1)-2n}{\sqrt{n^2+1}+\sqrt{n^2-1}}$

Sol: $\lim_{n \rightarrow \infty} \frac{(1-2)+(2-4)+(5-6)+\dots+(2n-1)-2n}{n\sqrt{1+\frac{1}{n^2}}+n\sqrt{1-\frac{1}{n^2}}}$

$$= \lim_{n \rightarrow \infty} \frac{(-1)+(-1)+(+)(-1)+\dots+(-1)}{n\left\{\sqrt{1+\frac{1}{n^2}}+\sqrt{1-\frac{1}{n^2}}\right\}}$$

$$= \lim_{n \rightarrow \infty} \frac{-n}{n\sqrt{1+0}+\sqrt{1+0}} = \frac{-1}{1+1} = -\frac{1}{2} \quad \square$$

a) $\frac{1}{2}$

b) $-\frac{1}{2}$ ✓ = R

c) 1

d) D.N.E

Date

Q:) $\lim_{n \rightarrow 0} \left(\frac{1}{n} \right) = \frac{1}{0}$ i.e. limit DNE $= \infty$ or $-\infty$ i.e.

Q:) $\lim_{n \rightarrow \infty} \left(\frac{1}{n} \right) = \frac{1}{\infty} = 0$ i.e. unique So limit.

Q:) $\lim_{n \rightarrow 0} \left(\frac{1}{n^2} \right) = \frac{1}{0^+} = +\infty$ But it is not unique so limit DNE.

Q:) $\lim_{n \rightarrow \infty} \left(\frac{1}{n^2} \right) = \frac{1}{\infty^2} = \frac{1}{\infty} = 0$ i.e. unique So limit exist.

Type (2) Factorization

Q:) $\lim_{n \rightarrow 2} \left(\frac{n^2 - 5n + 6}{n^2 - 4} \right) = \frac{0}{0}$ form

$\frac{0}{0}$
= using L-Hospital's Rule

$= \lim_{n \rightarrow 2} \left(\frac{2n - 5}{2n} \right) = \left(\frac{4 - 5}{4} \right) = \frac{-1}{4} = -\frac{1}{4}$

M-2 By using factorization

$\lim_{n \rightarrow 2} \left(\frac{n^2 - 3n - 4n + 6}{(n^2 - 2)(n+1)} \right) = \frac{(n-6)(n-3)}{(n-2)(n+1)}$

$= \lim_{n \rightarrow 2} \left(\frac{n-3}{n+1} \right) = \frac{-1}{4} = -\frac{1}{4}$

Date

$$Q:) \lim_{y \rightarrow 1} \left(\frac{y^2 - 1}{y - 1} \right) = ?$$

$$\text{Sol:)} \lim_{y \rightarrow 1} \frac{\cancel{(y-1)} (y^2 + 1 + y)}{\cancel{(y-1)}}$$

$$= 1^2 + 1 + 1 = 3 \quad \text{Ans}$$

$$Q:) \lim_{y \rightarrow 1} \left(\frac{y^4 - 1}{y^3 - 1} \right) = ?$$

$$\text{Put } y = x^{1/2}$$

$$= \text{When } y \rightarrow 1, \quad x \rightarrow 1$$

$$= \lim_{x \rightarrow 1} \left(\frac{x^2 - 1}{x^3 - 1} \right) = \frac{\cancel{(x-1)} (x^2 + 1 + x)}{\cancel{(x-1)} (x+1) (x^2 + 1)}$$

$$= \lim_{x \rightarrow 1} \left[\frac{(x^2 + 1 + x)}{(x+1)(x^2 + 1)} \right] = \frac{1+1+1}{(1+1)(1+1)} = \frac{3}{4}$$

$$= \frac{3}{4} \quad \text{Ans}$$

$$Q:) \lim_{n \rightarrow 1} \left(\frac{n^4 - 1}{n^5 - 1} \right) = ? \quad \frac{0}{0}$$

$$= \frac{\frac{1}{4} n^{\frac{1}{4}-1}}{\frac{1}{5} n^{\frac{1}{5}-1}} = 0 \quad = \lim_{n \rightarrow 1} \frac{5}{4} \left(\frac{n^{\frac{1}{4}}}{n^{\frac{1}{5}}} \right)$$

$$\frac{1}{5} n^{\frac{1}{5}-1} = 0 \quad = \frac{5}{4} \quad \text{Ans}$$

$$\text{n.r.) } n = y^{20} \quad \text{put}$$

Do your self.

Special

Date

Type 3 (Rationalization)

$$Q: \lim_{n \rightarrow \infty} (\sqrt{n^2+n+1} - \sqrt{n^2+1}) = ?$$

Sol: $(\infty - \infty)$ form.

$$= \lim_{n \rightarrow \infty} (\sqrt{n^2+n+1} - \sqrt{n^2+1}) \left[\frac{\sqrt{n^2+n+1} + \sqrt{n^2+1}}{\sqrt{n^2+n+1} + \sqrt{n^2+1}} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{(n^2+n+1) - (n^2+1)}{\sqrt{n^2+n+1} + \sqrt{n^2+1}} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{\cancel{n^2} + \cancel{n} + 1 - \cancel{n^2} - 1}{\sqrt{1 + \frac{1}{n} + \frac{1}{n^2}} + \sqrt{1 + \frac{1}{n^2}}} \right]$$

$$= \frac{1}{\sqrt{1+0} + \sqrt{1+0}} = \frac{1}{1+1} = \frac{1}{2} \quad \text{Ans}$$

$$Q: \lim_{t \rightarrow \infty} (\sqrt{t^2+1} - t) = ? \quad = \frac{1}{2} \quad \text{Ans}$$

Sol: $(\infty - \infty)$ form.

$$Q: \lim_{n \rightarrow 4} \left[\frac{3 - \sqrt{5+n}}{1 - \sqrt{5-n}} \right] = ?$$

a.1 $\sqrt{2}$ b.1 $-1/2$ c.1 $-1/3$

d.1 3

$$\left[\frac{3 - \sqrt{5+n}}{1 - \sqrt{5-n}} \right] \times \left[\frac{1 + \sqrt{5-n}}{1 + \sqrt{5-n}} \right] = \frac{(3 - \sqrt{5+n})(1 + \sqrt{5-n})}{1 - (5-n)}$$

Spiral

Date

~~$$\lim_{n \rightarrow 4} (n-4)$$~~

Sol:)

$$\lim_{n \rightarrow 4} \frac{3 - \sqrt{5+n}}{1 - \sqrt{5-n}} \times \frac{1 + \sqrt{5-n}}{1 + \sqrt{5-n}} \times \frac{3 + \sqrt{5+n}}{3 + \sqrt{5+n}}$$

$$= \lim_{n \rightarrow 4} \frac{9 - (5+n)}{1 - (5-n)} \times \frac{1 + \sqrt{5-n}}{3 + \sqrt{5-n}}$$

$$= \lim_{n \rightarrow 4} \frac{4-n}{n-4} \times \frac{1 + \sqrt{5-n}}{3 + \sqrt{5-n}} = -1 \times \frac{1+1}{3+\sqrt{3}} = -\left(\frac{2}{6}\right) = -\frac{1}{3}$$

Type IV (IND form concept)

$\frac{0}{0}, \frac{\infty}{\infty}$ } Directly use L-Hospital Rule

$0 \times \infty, \infty - \infty$ } Use L-Hospital Rule after convert them into one of the 1st two forms

$0^0, \infty^0, 1^\infty$ } use log concept.

Note:)

$$\frac{d}{dn} (n^a) = a n^{a-1}$$

$$a.) \frac{d}{dn} (n^n) = n^n (1 + \log n)$$

$$\frac{d}{dn} (a^n) = a^n \log_e a$$

$$\frac{d}{dn} (a^a) = 0$$

Date

$$Q:) \lim_{n \rightarrow a} \left(\frac{n^n - a^a}{n^a - a^n} \right) = ? \quad \frac{0}{0} \text{ form}$$

$$\text{Sol:)} \lim_{n \rightarrow a} \left[\frac{n^n (1 + \log_e n) - 0}{a n^{a-1} - a^n \log_e a} \right]$$

$$= \frac{a^a (1 + \log_e a)}{a a^{a-1} - a^a \log_e a} = \frac{1 + \log_e a}{1 - \log_e a} = \text{unique constant}$$

i.e. limit exist.

$$Q:) \lim_{n \rightarrow \infty} \left(\frac{n^n}{e^n} \right) = ? \quad \frac{\infty}{\infty} \text{ form}$$

Sol:)} Applying L-Hospital rule n-times.

$$= \lim_{n \rightarrow \infty} \frac{D^n(n^n)}{D^n(e^n)} = \lim_{n \rightarrow \infty} \frac{n!}{e^n} = \frac{n!}{e^n}$$

$$= \frac{n!}{\infty} = 0$$

Note: $D^n(n^n) = n!$, $D^{n+1}(n^n) = 0$

$$\bullet D(n) = 1 = 1!$$

$$\bullet D^2(n^2) = D(2n) = 2! = 2!$$

$$\bullet D^3(n^3) = D^2(3n^2) = D(3 \times 2n) = 3 \times 2 \times 1 = 3!$$

$$\bullet D^4(n^4) = 4!$$

In general $D^n(n^n) = n!$ & $D^{n+1}(n^n) = 0$

Date

Lecture no. 05

$$Q: \lim_{n \rightarrow 0} \left[\frac{\log(1+n)}{n} \right] = \frac{0}{0} \text{ form}$$

$$\text{sol:} \lim_{n \rightarrow 0} \left[\frac{1/1+n}{1} \right] = \frac{1/1+0}{1} = 1 \text{ Ans}$$

$$Q: \lim_{n \rightarrow 3} (3-n) \tan\left(\frac{\pi n}{6}\right) = ? \quad 0 \times \infty \text{ form}$$

$$\text{sol:} \lim_{n \rightarrow 3} \frac{3-n}{\cot\left(\frac{\pi n}{6}\right)} = \frac{0}{0} \text{ form}$$

$$= \lim_{n \rightarrow 3} \left[\frac{-1}{-\csc^2\left(\frac{\pi n}{6}\right) \cdot \frac{\pi}{6}} \right] = \lim_{n \rightarrow 3} \left[\frac{6}{\pi} \sin^2\left(\frac{\pi n}{6}\right) \right]$$

$$= \frac{6}{\pi} \cdot \sin^2 \frac{\pi}{2} = \frac{6}{\pi} \cdot (1)^2 = \frac{6}{\pi} \text{ Ans}$$

$$Q: \lim_{n \rightarrow 0} [n \cdot \log(1/n)] = ? \quad 0 \times \infty = \text{form}$$

$$\text{sol:} \lim_{n \rightarrow 0} \frac{\log(1/n)}{\frac{1}{n}} = \frac{\infty}{\infty} \text{ form}$$

Applying L. Hospital rule

$$\lim_{n \rightarrow 0} \frac{(1/1/n)^{-1/n}}{-1/n^2} = \lim_{n \rightarrow 0} \frac{-1/n}{-1/n^2} =$$

$$\lim_{n \rightarrow 0} (1/n) = 0 \quad \text{i.e. limit exists.}$$

Date

$$Q: \lim_{n \rightarrow 0} (\tan n \cdot \log n) = ?$$

 $0 \times (-\infty)$ form

Sol:)

$$\lim_{n \rightarrow 0} \frac{\log n}{\cot n} = \frac{\infty}{-\infty} \text{ form}$$

$$= \lim_{n \rightarrow 0} \frac{\frac{1}{n}}{-\operatorname{cosec}^2 n} = \lim_{n \rightarrow 0} \frac{-\sin^2 n}{n} =$$

$$= -\lim_{n \rightarrow 0} \left(\frac{\sin n}{n} \right) \times \lim_{n \rightarrow 0} (\sin n) = -1 \times 0$$

$$= 0 \quad \text{Ans.}$$

$$Q: \lim_{n \rightarrow 0} (n^n) = ? \quad 0^0 \text{ form} \quad \text{Ans} = 1$$

Sol:)

$$Q: \lim_{n \rightarrow 0} \left(\frac{1}{n} \right)^n = ? \quad \infty^\infty \text{ form}$$

Sol:)

$$\text{let } K = \lim_{n \rightarrow 0} \left(\frac{1}{n} \right)^n$$

$$\log_e K = \lim_{n \rightarrow 0} \log \left(\frac{1}{n} \right)^n$$

$$= \lim_{n \rightarrow 0} \left[n \log \left(\frac{1}{n} \right) \right] \quad \left\{ \begin{array}{l} \text{already solved} \\ \text{in previous ques} \end{array} \right.$$

$$= 0 \times \infty \text{ form}$$

$$= 0$$

i.e.

$$\log_e K = 0 \Rightarrow K = e^0 = 1 \quad \square$$

Spiral

$$e^0 = 1$$

$$e^\infty = \infty$$

$$e^{-\infty} = 0$$

Date

$$\text{Q:)} \lim_{n \rightarrow \infty} (n)^{\frac{1}{n}} = ?$$

Ans = 1

$$\text{Q:)} \lim_{n \rightarrow \infty} (1+n^2)^{\frac{1}{n}} e^{-n} = ? \quad \infty^0 \text{ (form.)}$$

Sol: let

$$K = \lim_{n \rightarrow \infty} (1+n^2)^{\frac{1}{n}} e^{-n}$$

$$\log_e K = \lim_{n \rightarrow \infty} \log (1+n^2)^{\frac{1}{n}} e^{-n}$$

$$= \lim_{n \rightarrow \infty} e^{-n} \cdot \log (1+n^2) = 0 \times \infty \text{ (form.)}$$

$$= \lim_{n \rightarrow \infty} \left[\frac{\log (1+n^2)}{e^n} \right] = \frac{0}{\infty}$$

Apply 1- Hospital Rule

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{1+n^2} \right) (0+2n) = \left[\frac{2n}{1+n^2} \right]$$

$$= \frac{2n}{(1+n^2) e^n} = \lim_{n \rightarrow \infty} \frac{2}{(1+n^2) (0) + e^n (2n)}$$

$$= \frac{2}{1+\infty + \infty} = \frac{2}{\infty + \infty} = 0$$

$$\log_e K = 0$$

$$K = e^0 = K = 1 \quad \text{Ans}$$

$$(ab)^{\frac{1}{n}} = a^{\frac{1}{n}} \cdot b^{\frac{1}{n}}$$

Date

e.g.: $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$

e.g.: $\lim_{n \rightarrow \infty} (1 - \frac{1}{n})^n = \lim_{n \rightarrow \infty} \left[(1 - \frac{1}{n}) (1 + \frac{1}{n}) \right]^{\frac{n}{2}}$
 $= \lim_{n \rightarrow \infty} (1 - \frac{1}{n})^{\frac{n}{2}} (1 + \frac{1}{n})^{\frac{n}{2}}$
 $= e^{-\frac{1}{2}} \cdot e^{\frac{1}{2}} = e^0 = 1$ Ans

Q:) $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^{n+p} = ?$, p finite = ?

a.) e^x
 b.) e^x ✓
 c.) e^p
 d.) 1

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n \cdot \left(1 + \frac{x}{n}\right)^p$$

$$= \left(1 + \frac{x}{n}\right)^n \left(1 + \frac{x}{n}\right)^p$$

$$= e^x \cdot (1+0)^p$$

$$= e^x \cdot 1^p$$

$$= e^x (1) = e^x$$
 B

Some more standard Result:

1.) $\lim_{n \rightarrow 0} \left(\frac{\sin n}{n} \right) = \lim_{n \rightarrow 0} \left(\frac{n}{\sin n} \right) = \lim_{n \rightarrow 0} \left(\frac{\sin^{-1} n}{n} \right) =$

$$\lim_{n \rightarrow 0} \left(\frac{\tan n}{n} \right) = \lim_{n \rightarrow a} \left(\frac{\sin(n-a)}{(n-a)} \right) = 1$$

2.) $\lim_{n \rightarrow 0} \left(\frac{a^n - 1}{n} \right) = \log_e a$

Date

e.g.) $\lim_{n \rightarrow 0} \left(\frac{e^n - 1}{n} \right) = 1$ Ans.

3.) $\lim_{n \rightarrow 0} \left[\frac{\log(1+n)}{n} \right] = 1$

4.) $\lim_{n \rightarrow 0} \left(\frac{1 - \cos n}{n^2} \right) = \frac{1}{2}$

* All these result are valid when $n \rightarrow 0$ can be evaluated with the help of L-Hospital's

Q.) $\lim_{a \rightarrow 0} \frac{x^a - 1}{a} = 1$

sol.) loge x

Q.) $\lim_{n \rightarrow 0} \left(\frac{1 - \cos 3n}{n^2} \right) = ?$

sol.)

$$\lim_{3n \rightarrow 0} \left[\frac{1 - \cos 3n}{(3n)^2} \right] \times 9 \quad \because \quad n \rightarrow 0 \quad 3n \rightarrow 0$$

$$= \frac{1}{2} \times 9 = \frac{9}{2} \text{ (Ans.)}$$

Q.) $\lim_{n \rightarrow 0} \left(\frac{\log n}{n} \right) = \infty$ or DNE

Q.) $\lim_{n \rightarrow \infty} \left(\frac{\log n}{n} \right) = 0$ Ans.

Q.) $\lim_{n \rightarrow \infty} \left(\frac{\log n}{n} \right) = 0$ Ans.

Date

$$Q:) \lim_{n \rightarrow 0} \left[n \cdot \sin \frac{1}{n} \right] = ?$$

$$Sol:) \rightarrow 0 \times \sin \infty = 0 \times (\text{No. b/w } -1 \& 1) = 0 \quad \text{Ans}$$

$$\lim_{n \rightarrow 0} \left[\frac{\sin \frac{1}{n}}{\frac{1}{n}} \right] = \text{put } n = \frac{1}{y} \Rightarrow n \rightarrow 0 \Rightarrow y \rightarrow \infty$$

$$\lim_{y \rightarrow \infty} \left(\frac{\sin y}{\frac{1}{y}} \right) = 0 \quad \text{Ans}$$

$$Q:) \lim_{n \rightarrow 0} \left[\sin \left(\frac{1}{n} \right) \right] = ?$$

$$Ans:) \sin \infty = \text{Any no. b/w } -1 \& 1 = \text{Not unique} \Rightarrow \text{limit DNE.}$$

$$Q:) \lim_{n \rightarrow \infty} \left(\frac{3n - \sin n}{2n + 5 \cos n} \right) = ?$$

$$a.) 3 \quad \frac{\infty - \sin \infty}{\infty + \cos \infty} = \frac{\infty}{\infty}$$

$$b.) 1.5 \quad \frac{\infty - \sin \infty}{\infty + \cos \infty} = \frac{\infty}{\infty}$$

$$c.) 1$$

$$d.) 0 \quad \lim_{n \rightarrow \infty} \left[\frac{3 - \cos n}{2 - 5 \sin n} \right] = \frac{3 - \cos \infty}{2 - 5 \sin \infty}$$

$$e.) \text{DNE} \quad \frac{3 - \cos \infty}{2 - 5 \sin \infty} = \frac{3 - (\text{No. b/w } -1 \& 1)}{2 - (\text{No. b/w } -1 \& 1)} = \text{give up by this method.}$$

m-2

$$\lim_{n \rightarrow \infty} \frac{n \left(3 - \frac{\sin n}{n} \right)}{2 + 5 \frac{\cos n}{n}} = \frac{3 - \sin y/n}{2 - 5 \cos y/n} = \frac{3 - 0}{2 - 0} = \frac{3}{2}$$

Next See slide. Spiral

Date _____

Q:) if $-2 - \frac{n^2}{3} < \frac{n \cdot \sin n}{1 - \cos n} < 2$ then evaluate $\lim_{n \rightarrow 0} \left(\frac{n \sin n}{1 - \cos n} \right)$

Sol:)

$$\lim_{n \rightarrow 0} \left[\frac{n^2 \cdot \frac{\sin n}{n}}{(1 - \cos n)} \right] = \lim_{n \rightarrow 0} \left[\frac{\sin n / n}{(1 - \cos n) / n^2} \right]$$

$$= \frac{1}{\frac{1}{2}} = 2 \text{ Ans}$$

M-2 Using L-Hospital Rule

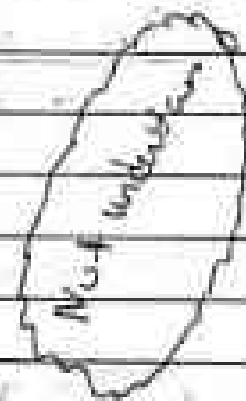
$$\lim_{n \rightarrow 0} \left(\frac{n \sin n}{1 - \cos n} \right) = \frac{0}{0} = \dots = 2 \text{ Ans}$$

M-3 Using Sandwich Theorem:

if $f(n) < g(n) < h(n)$ then

$$\lim_{n \rightarrow a} f(n) < \lim_{n \rightarrow a} g(n) < \lim_{n \rightarrow a} h(n)$$

$$\text{Q:) } \lim_{n \rightarrow 0} \frac{\sin(n^5)}{\sin^3(2n)} = \frac{0}{0}$$



Sol:) Using Standard result

$$\lim_{n \rightarrow 0} \left\{ \left(\frac{\sin(n^5)}{\sin(2n)} \right)^3 \cdot \frac{n^5}{8n^3} \right\}$$

$$\therefore 1 \times 1^3 \times \lim_{n \rightarrow 0} \frac{n^2}{8} = 1 \times 1 \times 0 = 0 \text{ Ans.}$$

Date

$$Q:) \lim_{n \rightarrow 0} \left(\frac{e^{n^2} - \cos 3n}{2n^2} \right) = ?$$

$$Q:) \lim_{n \rightarrow 0} \frac{(4^n - 1)^3}{\sin\left(\frac{n}{4}\right) \cdot \log\left(1 + \frac{n^2}{3}\right)} = ?$$

Lecture no. 06

Sol:) Using standard result:

$$\lim_{n \rightarrow 0} \left[\frac{e^{n^2} - 1 + 1 - \cos 3n}{2n^2} \right]$$

$$\lim_{n \rightarrow 0} \left(\frac{e^{n^2} - 1}{2n^2} \right) + \lim_{n \rightarrow 0} \left(\frac{1 - \cos 3n}{2n^2} \right)$$

$$= \frac{1}{2} \left(\lim_{n \rightarrow 0} \left(\frac{e^{n^2} - 1}{n^2} \right) \right) + \frac{1}{2} \lim_{3n \rightarrow 0} \left(\frac{1 - \cos 3n}{(3n)^2} \right) \times 9$$

$$= \frac{1}{2} \times 1 + \frac{1}{2} \times \frac{1}{2} \times 9 = \frac{1}{2} + \frac{9}{4} = \frac{11}{4} \quad \square$$

$$\text{Sol: 2} \quad \lim_{n \rightarrow 0} \left(\frac{4^n - 1}{n} \right)^3 \cdot (n^3)$$

$$\left[\frac{\frac{\sin \frac{n}{4}}{\frac{n}{4}} \cdot \frac{\log\left(1 + \frac{n^2}{3}\right)}{\frac{n^2}{3}}}{\frac{n}{4}} \right] \left(\frac{n^2}{12} \right)$$

$$= \frac{(\log e 4)^3 \times 12}{1 \times 1} = 12(\log e 4)^3 \quad \square$$

Continuity:

Q:) Find K for which $f(x)$ is continuous at $x=0$

$$1.) f(x) = \begin{cases} \frac{e^{4x} - \cos 2x}{2x^2} & , x \neq 0 \\ K & , x = 0 \end{cases}$$

sol:) At $x=0$, Exact value $= f(0) = K$ & In the value abd of $x=0$

Approx value =

$$\lim_{x \rightarrow 0} f(x) = \frac{11}{4} \quad \left[\begin{array}{l} \text{Already} \\ \text{solved} \end{array} \right]$$

Now for continuity at $x=0$

$$\boxed{\text{Exact value} = \text{Approx value}}$$

$$\boxed{K = \frac{11}{4}} \quad \text{Ans}$$

$$2.) f(x) = \begin{cases} \frac{(4^x - 1)^3}{\left(\sin\left(\frac{\pi}{2}\right) \cdot \log\left(1 + \frac{x^2}{5}\right) \right)} & , x \neq 0 \\ K & , x = 0 \end{cases}$$

$$\text{sol:)} \quad \text{Approx value} = \lim_{x \rightarrow 0} = 12 (\log 4)^3$$

So Exact value = Approx value

$$\boxed{K = 12 (\log 4)^3} \quad \text{Ans}$$

Date

$$3.) \quad f(x) = \begin{cases} x^k \cdot \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Sol: At $x=0$, Exact value = $f(0) = 0$ (given)

When $x \rightarrow 0$, App value = $\lim_{x \rightarrow 0} f(x)$

$$= \lim_{x \rightarrow 0} \left[x^k \cdot \sin\frac{1}{x} \right]$$

$$= 0^k \times \sin\infty = \{ \text{Any no b/w } -1 \& 1 \} \times 0^k$$

$$\lim_{x \rightarrow 0} f(x) = 0^{k-1} \cdot 0 \quad (\text{Any no b/w } -1 \& 1)$$

$$= 0^{k-1} \cdot [0 \times \text{Any no b/w } -1 \& 1]$$

$$= 0^{k-1} \cdot [0] = 0^k$$

$$0^k = \begin{cases} \infty, & k < 0 \\ \text{IND}, & k = 0 \\ 0 \text{ (prior)}, & k > 0 \end{cases}$$

i.e. for $k > 0$ approx value = exact value = 0

So our answer is $k > 0$ Ans

Q:) Check the continuity of the following function

$$1.) \quad f(x) = \begin{cases} x \sin\frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} \quad (\text{cont})$$

Sol: In previous question

we have taken $k=1$ (i.e. $k > 0$)

So this fnc is also continuous

Spiral

Date

$$2) f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

Sol: Exact value at $x=0 = f(0) = 1$

$$\text{App value} = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) = 1$$

$\therefore$ Exact value = App value

So

$f(x)$ is cont at $x=0$

Q.) Find K of for which $f(x)$ is continuous at $x=0$

$$1) f(x) = \begin{cases} \frac{\log(1+3x) - \log(1-2x)}{x}, & x \neq 0 \\ K, & x = 0 \end{cases}$$

a.) 1

b.) 0

c.) -1

d.) 5 ✓

Sol:

Exact value = $f(0) = K$

$$\text{App value} = \lim_{x \rightarrow 0} \left\{ \frac{\log(1+3x)}{x} - \frac{\log(1-2x)}{x} \right\}$$

$$= \frac{1 \times 3 - (-1) \times 2}{1}$$

$$= 3 + 2 = 5$$

Approx value = Exact value = 5

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$$2.) f(n) = \begin{cases} \sin \frac{1}{n} & , n \neq 0 \\ k & , n = 0 \end{cases}$$

a.) 0

b.) 1

c.) $(-\frac{1}{2}, \frac{1}{2})$ (No. b/w -1 & 1)

d.) No such value of k ✓

Sol.) Exact value $f(0) = k$

Appx. value $= \lim_{n \rightarrow 0} \sin\left(\frac{1}{n}\right) = \sin \infty$

$=$ Any no. b/w -1 & $1 = \text{ONE}$

So $f(n)$ is discontinuous by method - 2

So

No such value of k exist.Q:) The value of a and b for which the function

$$f(n) = \begin{cases} 2n+1 & \text{if } n \leq 1 \\ an^2+b & \text{if } 1 < n < 3 \\ 5n+2a & \text{if } n \geq 3 \end{cases}$$
 is continuous every

where

a.) $a=2, b=1$

$n=1$ & $n=3$

b.) $a=1, b=2$

and the problem existing

c.) $a=3, b=2$

d.) $a=2, b=3$

It is given that $f(n)$ is continuous every where it will be cont at $n=2$ & 3 also

At $n=1$ L.H.L = R.H.L $= (2(1)+1) = a(1)+b$ - (1)

$n=3$ L.H.L = R.H.L $= a(3)^2+b = 5(3)+2a$ - (2)

Now solving eq (1) & (2)

$a=2, b=1$

Spiral

Mathematical Def of differentiability:

$f(x)$ is said to be diff at $x=a$ if $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exist.

& value of this limit is called $f'(a)$

i.e

$$f'(a) = \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} \right)$$

$\rightarrow$ L.H.D. $\rightarrow$ R.H.D.
if L.H.D. = R.H.D. then

when L.H.D = R.H.D then $f(x)$ is said to be differentiable at $x=a$

Slope of chord AB: $\frac{f(x) - f(a)}{x - a}$

$\therefore x$ lies in the Nbd of a i.e $x \rightarrow a$

then $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \text{slope of tangent} = f'(a)$

$\Rightarrow \frac{dy}{dx} = f'(x) = \tan \theta = \text{slope of tangent at any point.}$

$\Rightarrow$ When L.H.D = R.H.D then $f(x)$ is said to be differentiable at $x=a$.

$$\text{L.H.D} = \lim_{h \rightarrow 0} \left(\frac{f(a-h) - f(a)}{-h} \right)$$

$$\text{R.H.D} = \lim_{h \rightarrow 0} \left(\frac{f(a+h) - f(a)}{h} \right)$$

Note:

1) Continuity is the necessary condition for diff. But it is not sufficient condition.

2) N. Cond is $\longrightarrow$ Continuity

Sufficient cond is $\longrightarrow \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} \right)$ must exist. Sufficient

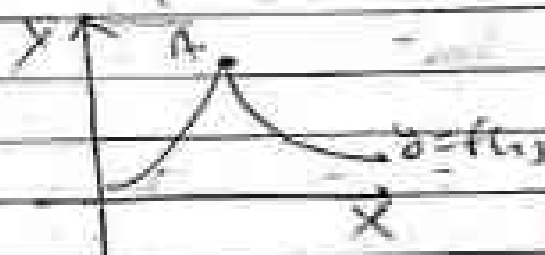
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$\Rightarrow$ Differentiability $\Rightarrow$ Continuity (True)

$\Rightarrow$ Continuity $\Rightarrow$ Differentiability (False)

3.) At sharp point, $f(x)$ is said to be non-differentiable.

At A, $f(x)$ is non Diff, it is not possible to find unique tangent at A.



e.g.) At $x=0$, $f(x) = |x|$ is Continuous but Non-differentiable.

4.) Shortcut method of finding L.H.L and R.H.L w.k that

$$LHL = \lim_{x \rightarrow a^-} f(x) \quad \& \quad RHL = \lim_{x \rightarrow a^+} f(x)$$

similarly

$$\begin{aligned} LHO \text{ of } f(x) &= LHL \text{ of } f'(x) \\ &= \lim_{x \rightarrow a^-} f'(x) \end{aligned}$$

$$\begin{aligned} RHO \text{ of } f(x) &= RHL \text{ of } f'(x) \\ &= \lim_{x \rightarrow a^+} f'(x) \end{aligned}$$

P.T.O

Q.) Find the derivative of $y = |x|$?

$$\text{Sol:)} \quad \frac{d}{dx} |x| = \frac{d}{dx} (\sqrt{x^2}) = \frac{1}{2\sqrt{x^2}} \cdot \frac{d}{dx} (x^2)$$

$$= \frac{1}{2\sqrt{x^2}} \cdot 2x = \frac{x}{\sqrt{x^2}} = \left(\frac{x}{|x|} \right) \quad \text{---}$$

i.e.

$$\frac{d}{dx} |x| = \frac{x}{|x|} = \frac{x|x|}{|x|^2} = \frac{x|x|}{x^2} = \left(\frac{|x|}{x} \right) \quad \text{---}$$

So,

$$\frac{d}{dx} (|x|) = \frac{|x|}{x} = \begin{cases} -1 & x < 0 \\ 0NE & x = 0 \\ +1 & x > 0 \end{cases}$$

$$\text{i.e. } f'(0) = 0NE$$

$$f'(1) = f'(2) = f'(3) = f'(4) = \dots = 1$$

$$f'(-0.5) = f'(-1) = f'(-2.3) = \dots = -1$$

$$\text{L.H.D at } (x=0) = -1 \quad \therefore \text{L.H.D} \neq \text{R.H.D}$$

$$\text{R.H.D at } (x=0) = +1 \quad f(x) = |x| \text{ is not diff at } x=0$$

Q.) At $x=0$, $f(x) = |x|^3$ is

a.) once diff but not twice

b.) twice " " " " thrice ✓

c.) thrice diff

d.) Not Differentiable

$$f(x) = |x|^3 = \begin{cases} -x^3 & , x < 0 \\ x^3 & , x > 0 \end{cases}$$

$$= f'(x) = \begin{cases} -3x^2 & , x < 0 \\ 3x^2 & , x > 0 \end{cases} \Rightarrow f''(x) = \begin{cases} -6x & , x < 0 \\ 6x & , x > 0 \end{cases}$$

$$\text{L.H.D} = \text{R.H.D} = 0 \quad \checkmark$$

$$\text{L.H.D} = \text{R.H.D} = 0 \quad \checkmark$$

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$$f''(x) = \begin{cases} -6 & , x < 0 \\ 6 & , x > 0 \end{cases}$$

$$= \frac{L.H.O \neq R.H.O}{-6 \quad 6} \quad \times$$

Q:) if $f(x) = \begin{cases} e^x & , x < 1 \\ ax^2 + bx + \ln x & , x \geq 1 \end{cases}$ is differentiable

everywhere then find a & b !

sol:) $\because f(x)$ is diffn everywhere so it must be continuous at every point.

Continuity at $x=1$

$$f(x) = \begin{cases} e^x & , x < 1 \\ ax^2 + bx + \ln x & , x \geq 1 \end{cases}$$

$$= L.H.L = R.H.L = \text{Exact value}$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$$

$$e^1 = a + b = a + b \quad \text{--- (1)}$$

=) Differentiability at $x=1$

$$f'(x) = \begin{cases} e^x & , x < 1 \\ 2ax + b + \frac{1}{x} & , x \geq 1 \end{cases}$$

$$L.H.O = R.H.O$$

$$\lim_{x \rightarrow 1^-} f'(x) = \lim_{x \rightarrow 1^+} f'(x)$$

$$e^1 = 2a + b + 1 \quad \text{--- (2)}$$

So

eq (1) & (2) is

$$\boxed{a = -1, b = e + 1}$$

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Q:) A real function

$$f(x) = \begin{cases} ax^2 + \beta x, & \text{for } x < 0 \\ ax^3 + \beta x^2 + 5 \sin x, & x \geq 0 \end{cases}$$

if $f(x)$ is twice differentiable then

a.) $a=1, \beta=0$

$$f'(x) = \begin{cases} 2ax + \beta & x < 0 \\ 3ax^2 + 2\beta x + 5 \cos x & x \geq 0 \end{cases}$$

b.) $a=1, \beta=5$

c.) $a=5, \beta=10$

d.) $a=5, \beta=5$

$$f''(x) = \begin{cases} 2a & x < 0 \\ 6ax + 2\beta - 5 \sin x & x \geq 0 \end{cases}$$

L.H.O. = R.H.O. = (1)

$\beta = 5$ — (1)

$2a = 2\beta$ — (2)

So, $\boxed{a=5, \beta=5}$

Q:) At $x=1$, $f(x) = |\log x|$ issol:) Q-1 (w/d using graph) $\rightarrow$

$$f(x) = |\log x| = \begin{cases} -\log x, & 0 < x < 1 \\ \log x, & x \geq 1 \end{cases}$$

$$f'(x) = \begin{cases} -1/x, & 0 < x < 1 \\ 1/x, & x \geq 1 \end{cases}$$

To check continuity at $x=1$

L.H.L = R.H.L = $f(1)$

$-\log 1 = \log 1 = \log 1$

$0 = 0 = 0$ i.e. verified

L.H.L = $f(1^-) = -\log 1 = 0$

or R.H.L = $f(1^+) = \log 1 = 0$

Exact value = $f(1) = 0$

So at $x=1$, $f(x) = |\log x|$ is continuous

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To check different at $n=1$

$$\cancel{L.H.O} = \cancel{R.H.O}$$

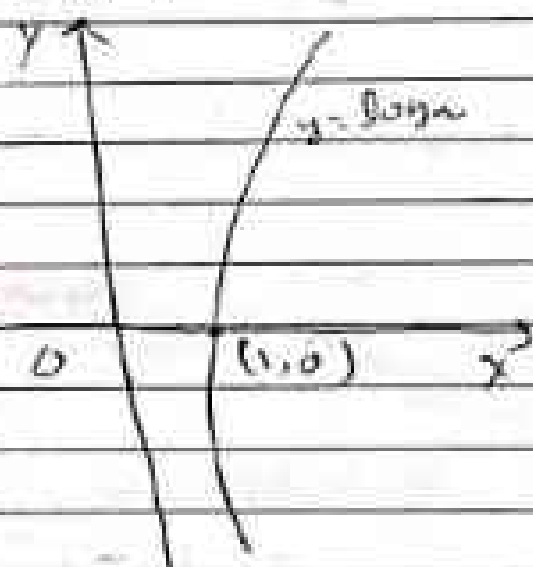
$$L.H.O \text{ at } n=1 = L.H.L \text{ of } f'(n) = -1$$

$$R.H.O \text{ at } n=1 = R.H.L \text{ of } f'(n) = 1$$

 $\therefore L.H.O \neq R.H.O$ So

 $f(n) = |\log n|$ is not diff at $n=1$
 $n=1$

Using graph :-



Take mirror image about
x axis of that portion
of graph which lies
below x axis.



So,
Can't but not
differentiable.

 $n=1$ is sharp point

H.W.

Date

Q:) Check the differentiability of $f(x) = x|x|$?

Sol:) $f(x) = x|x| = \begin{cases} -x^2, & x < 0 \\ x^2, & x > 0 \end{cases}$

$f'(x) = \begin{cases} -2x, & x < 0 \\ 2x, & x > 0 \end{cases}$

At $x = 0$

L.H.L = $f(0^-) = 0$ i.e. continuous at $x = 0$

R.H.L = $f(0^+) = 0$

Exact value = $f(0) = 0$

$\Rightarrow$ L.H.O of $f(x) = \text{L.H.L of } f'(x) = f'(0^-) = 0$

R.H.O of $f(x) = \text{R.H.L of } f'(x) = f'(0^+) = 0$

i.e. $\therefore \text{L.H.O} = \text{R.H.O}$

So Diff at $x = 0$ $\therefore$

Lecture no. 07

$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots = \frac{e^x - e^{-x}}{2}$

$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots = \frac{e^x + e^{-x}}{2}$

$\frac{d}{dx} \sinh x = \cosh x = \frac{e^x + e^{-x}}{2}$

$\frac{d}{dx} \cosh x = \sinh x = \frac{e^x - e^{-x}}{2}$

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

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⇒ Increasing power of n then this is Maclaurin series.

Taylor & Maclaurin Series :-

(Maclaurin Series) →

Let $f(x)$ is continuous, func defined in a Domain 0 & it's all derivative exist then $f(x)$ can be expanded in the Nbd of 0 ($x \rightarrow 0$) as follows:

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

Note:)

1.) This series provides Approx value of $f(x)$ in the Nbd of 0

2.) Coeff of x^n in the M.S. exp = $\frac{f^{(n)}(0)}{n!}$

3.) In this series we are getting Increasing power of x .

⇒ Some standard Result of Maclaurin series:
All these Result are valid only when $x \rightarrow 0$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$$

$$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \dots$$

$$a^x = 1 + x(\ln a) + \frac{x^2}{2!} (\ln a)^2 + \frac{x^3}{3!} (\ln a)^3 + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

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$$3.) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots = \frac{e^x - e^{-x}}{2}$$

$$4.) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots = \frac{e^x + e^{-x}}{2}$$

$$5.) \tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots$$

 $\Rightarrow$ Note:

$$1.) \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$2.) \coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

Q.) Which of the following series consist only odd power of x in its Taylor series expansion about $x=0$?

$$a.) \sin(x^2) = x^2 - \frac{(x^2)^3}{3!} + \frac{(x^2)^5}{5!} - \dots$$

$$b.) \cos(x^2) =$$

$$c.) \sin(x^3) = x^3 - \frac{(x^3)^3}{3!} + \frac{(x^3)^5}{5!} - \dots$$

$$d.) \cos(x^3) =$$

Q.) Expand $\log(1+x)$ in the increasing powers of x .

$$sol.) f(x) = \log(1+x), \quad f'(x) = \frac{1}{1+x}, \quad f''(x) = \frac{-1}{(1+x)^2}, \quad f'''(x) = \frac{2}{(1+x)^3}$$

$$f(0) = \log(1+0) \quad f'(0) = \frac{1}{1+0} \quad f''(0) = \frac{-1}{(1+0)^2} \quad f'''(0) = \frac{2}{(1+0)^3}$$

$$= 0 \quad = 1 \quad = -1 \quad \text{Expand} = 2$$

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Now Maclaurin series expansion of $f(x)$ in the Mtd of $x=0$ is

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$\log(1+x) = 0 + x(1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(2)$$

$$\log(1+x) = x - \frac{x^2}{2!} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Hence proved.

Q:) The Taylor series expansion of $(3\sin x + 2\cos x)$

a) $2 + 3x - \frac{x^2}{2} - \frac{x^3}{2} + \dots$ ✓

b) $2 - 3x + \frac{x^2}{2} - \frac{x^3}{2} + \dots$

c) $2 + 3x + \frac{x^2}{2} + \frac{x^3}{2} + \dots$

d) $2 - 3x - \frac{x^2}{2} + \frac{x^3}{2} + \dots$

Now

$$\text{Sol:} \quad 3 \left[x - \frac{x^2}{2!} + \frac{x^3}{3!} - \dots \right] + 2 \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right]$$

$$= 2 + 3x - \frac{x^2}{2} - \frac{x^3}{2} + \dots$$

Taylor series:

if we want to expand our function $f(x)$ in the Mtd of 'a' then expansion is called T.S expansion & it is defined as follows;

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots$$

Note:

1.) This series will provide Approx value of $f(x)$ in the Nbd of a

2.) coeff of $(x-a)^n = \frac{f^n(a)}{n!}$

3.) Here we are getting increasing power of $(x-a)$

4.) To find Approx value of $f(x)$ is linear form, neglect 2nd & higher degree term.

5.) " " " " " " Quadratic form then neglect 3rd & higher degree term.

Q.) Expand $e^x + \sin x$ in the increasing power of $(x-\pi) = ?$

or

Find the power series expansion of $e^x + \sin x$ when $|x-\pi| < \epsilon$

or

Find the coeff of $(x-\pi)^2$ in the power series expansion of $(e^x + \sin x)$

or

Find the linear approximation of $(e^x + \sin x)$ when $x \rightarrow \pi$

or

" " Quad " of $(e^x + \sin x)$ about π

sol:) $f(x) = e^x + \sin x$, $f'(x) = e^x + \cos x$, $f''(x) = e^x - \sin x$
 $f(\pi) = e^\pi + 0$, $f'(\pi) = e^\pi - 1$, $f''(\pi) = e^\pi - 0$

So T.S expansion

Date

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots$$

$$e^x + \sin x = (e^\pi) + (x-\pi)(e^\pi) + \frac{(x-\pi)^2}{2!} (e^\pi) + \dots$$

Ans. of part (i) & (ii) is

$$\text{(iii)} = \text{coeff of } (x-\pi)^2 = \frac{e^\pi}{2}$$

(iv) Linear Approximation

$$= e^\pi + (x-\pi)(e^\pi - 1) + \text{Neglect}$$

(v) Quad Approximation

$$= e^\pi + (x-\pi)(e^\pi - 1) + \frac{(x-\pi)^2}{2!} (e^\pi) + \text{Neglect}$$

Q. Linear Approximation of e^{-x} about 2 will be

a) $(x-2)e^{-x}$ Sol: $f(x) = e^{-x}$ About $x=2$ ($x \rightarrow 2$)

b) $(3-x)/e^2$ Ans: $f(x) = e^{-x}$ $f'(x) = -e^{-x}$

c) $(3-x)e^2$ $f(x) = e^{-x}$ $f'(x) = -e^{-x}$

d) $(2-x)e^{-2}$ - By expansion is $f(x) = f(2) + (x-2)f'(2)$

$$e^{-x} = e^{-2} + (x-2)(-e^{-2})$$

$$= \left(\frac{3-x}{e^2} \right) \text{ d.}$$

Q. Expand $\log x$ in the nbd of 1 & hence eval $\log e(1.1)^2$

Ans: $f(x) = \log x$, $f'(x) = \frac{1}{x}$, $f''(x) = -\frac{1}{x^2}$, $f'''(x) = \frac{2}{x^3}$

$f(1) = \log 1$, $f'(1) = \frac{1}{1}$, $f''(1) = -\frac{1}{1^2}$, $f'''(1) = \frac{2}{1^3}$ - - -
 $= 0$, 1 , -1 , 2 Signs

Date: _____

Now T.S expansion $f(x)$ in the Mbd of $x=1$ is given as,

$$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1) + \dots$$

$$\log_e x = (0) + (x-1)(1) + \frac{(x-1)^2}{2!}(-1) + \frac{(x-1)^3}{3!}(1) + \dots$$

$$\log_e x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \dots$$

(ii part)

put $x=1.1$ $\therefore$ (it lies in the Mbd of 1)

$$\log(1.1) = (1.1-1) - \frac{(1.1-1)^2}{2} + \frac{(1.1-1)^3}{3} - \dots$$

$$= (0.1) - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} - \dots$$

$$= 0.0953 + \text{Neglect} \approx 0.0953 \text{ Ans}$$

Q:) The function $f(x) = \sin x$ can be approximated about $\frac{\pi}{6}$ as follows:

$$a.) x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad x \rightarrow \frac{\pi}{6}$$

$$x - \frac{\pi}{6} \rightarrow 0$$

b.) $Y_1 = \text{exact value.}$

But in $f(x) = \sin x$ a

when is not 0 i.e. $x \neq \frac{\pi}{6}$

$$c.) \left(x - \frac{\pi}{6}\right) - \frac{\left(x - \frac{\pi}{6}\right)^3}{3!} + \frac{\left(x - \frac{\pi}{6}\right)^5}{5!} - \dots$$

So we can not use the series

d.) None. ✓

$$f(x) = f\left(\frac{\pi}{6}\right) + \left(x - \frac{\pi}{6}\right)f'\left(\frac{\pi}{6}\right) + \frac{\left(x - \frac{\pi}{6}\right)^2}{2!}f''\left(\frac{\pi}{6}\right) + \dots$$

$$\sin x = \dots \quad (c)$$

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Q.) The function $f(x) = \sin\left(x - \frac{\pi}{6}\right)$ can be approximated about $\frac{\pi}{6}$ as follows:

a.) $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$

b.) y_2

c.) $\left(x - \frac{\pi}{6}\right) - \frac{\left(x - \frac{\pi}{6}\right)^3}{3!} + \frac{\left(x - \frac{\pi}{6}\right)^5}{5!} - \dots$

d.) None.

Sol.) $x \rightarrow \frac{\pi}{6}$, $\left(x - \frac{\pi}{6}\right) \rightarrow 0$

or Angle $\rightarrow 0$

So we can use Standard result of M.S

Q.) In the neighbourhood of $x=1$, $f(x)$ has the power series expansion of the type $f(x) = 1 + (1-x) + (1-x)^2 + (1-x)^3 + \dots \infty$ then find $f(x)$ will be?

a.) $1/x-1$

b.) $1/1-x$

c.) $1/x$ ✓

d.) $x^{-1}/x+1$

Sol.) W.K. that $S_{\infty} = a + ar + ar^2 + ar^3 + \dots = a$

if we take or

$a=1$ then $= 1 + r + r^2 + r^3 + \dots = \frac{1}{1-r}$

$f(x) = \frac{1}{1-(1-x)} = \frac{1}{x} = \frac{1}{x}$

Lecture no. 08# Mean Value Theorem:1.) Slope of line passing through (x_1, y_1) & (x_2, y_2)

$$\text{is } m = \frac{y_2 - y_1}{x_2 - x_1}$$

2.) In general, slope of tangent at any random point x is

$$m = \tan \theta = f'(x)$$

3.) Slope of horizontal line is $m = f'(x) = \tan 0 = 0$

4.) Elementary funct: All polynomial funct, Exp funct, Log funct, Trig funct, & Inverse funct are called E. functions.

5.) All E. funct are continuous as well as Differentiable in their Domain.

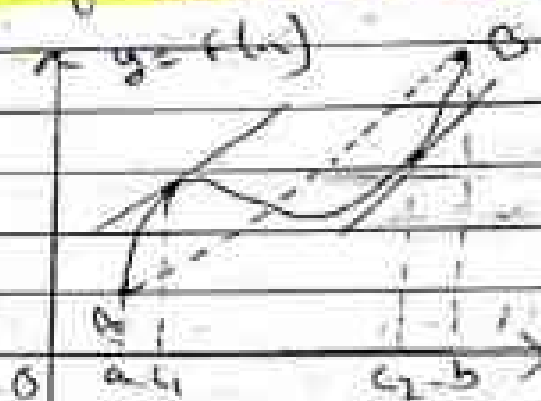
Lagrange's M.V.ThLet $f(x)$ is defined in $[a, b]$ s.t(i) $f(x)$ is cont in $[a, b]$ (ii) $f(x)$ is Diff in (a, b) Then $\exists$ at least one c in b/w a & b for whichtangent at c is $\parallel^r$ to chord.

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i.e. slope of tangent = slope of AB (chord)

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Note: Converse of L.M.V. Theorem is not necessarily True



A (a, f(a)), B (b, f(b))

Sub note:

$$\text{if } [f(x) \text{ is cont \& diff}] \Rightarrow \left[\exists c \in (a, b) \text{ s.t. } \frac{f(b) - f(a)}{b - a} = f'(c) \right]$$

Q:) Consider $f(x) = (x-1)^2 + 2$ then verify L.M.V.T. is $[0, 1]$

or

Q:) Consider $f(x) = (x-1)^2 + 2$ then find coordinate of that point on the curve where tangent is $\parallel$ to the chord joining $(0, 3)$ & $(1, 2)$.

or

Q:) if $f(x) = (x-1)^2 + 2$ then find eq. of tangent which is $\parallel$ to the chord joining $(0, 3)$ & $(1, 2)$.

or

Q:) For the funct $y = (x-1)^2 + 2$, At least at one point in the interval $[0, 1]$, the value of $\frac{dy}{dx} = ?$

Date:

Sol: $y = f(x) = (x-1)^2 + 2$ $\begin{cases} \text{at } x=0, f(0) = 3 \\ \text{at } x=1, f(1) = 2 \end{cases}$

Ans 1:)

$$A(a, f(a)) = (0, 3)$$

$$O(b, f(b)) = (1, 2)$$

$\therefore$ it is only of degree '2' hence caught as well as diff. So

By L.M.V.Th, $\frac{f(b) - f(a)}{b - a} = f'(c)$

$$= \frac{f(1) - f(0)}{1 - 0} = \left[2(x-1) \right]_{x=c}$$

$$= \frac{2 - 3}{1 - 0} = 2(c-1)$$

$$= -1 = 2c - 2$$

$$= -1 + 2 = 2c$$

$$= 2c = 1$$

$$c = \frac{1}{2}$$

Hence verified $\square$

Note: $f(x) = (x-1)^2 + 2$

$$f'(x) = 2(x-1) + 0$$

So $f'(c) = 2(c-1) + 0$

Ans 2 $f(c) = (c-1)^2 + 2$

$$f\left(\frac{1}{2}\right) = \left(\frac{1}{2} - 1\right)^2 + 2 = \frac{9}{4}$$

Hence reqd point = $(c, f(c)) = \left(\frac{1}{2}, \frac{9}{4}\right)$

$$m = f'(c)$$

Date

Ans 3: slope of tangent at $(x=c) = f'(c)$.

$$m = [x'(c-1)]$$

So

$$\text{Slope at } c = \frac{1}{2} \text{ is } m = 2 \left(\frac{1}{2} - 1 \right) = -1$$

1st

eq of tangent at $C \left(\frac{1}{2}, \frac{3}{4} \right)$ having slope

$$m = -1 \text{ is } y - y_1 = m(x - x_1)$$

$$= y - \frac{3}{4} = (-1) \left(x - \frac{1}{2} \right)$$

$$= \boxed{4x + 4y = 1}$$

Ans 4:

At least at one point in $[0, 1]$ the value of $\left(\frac{dy}{dx} \right)_{x=c} = f'(c) = \text{slope of } AB = -1$ ✓

$$\text{Slope of chord } (AB) = \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(1) - f(0)}{1 - 0} = \frac{2 - 3}{1 - 0} = -1$$

$$= \text{slope of tangent} = \left(\frac{dy}{dx} \right)_{x=c}$$

Q:) for the funct $y = 5x^2 + 10x$ At least at one point in $(1, 2)$, $\frac{dy}{dx}$ is exactly 1

$$\text{sol:)} \quad 5x^2 + 10x \quad \begin{cases} \text{at } 1 = 15 \\ \text{at } 2 = 40 \end{cases}$$

$$= \text{slope of chord} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(2) - f(1)}{2 - 1}$$

$$= \frac{40 - 15}{1} = +25 \text{ ✓}$$

$$\boxed{m = +25}$$

Spiral

Date _____

Q:) For the function $f(x) = |x|$, L.M. value is not applicable in?

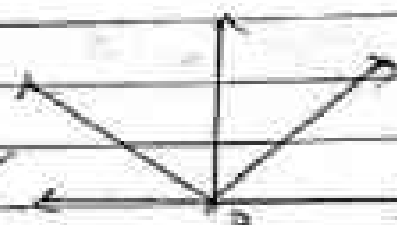
$$\text{Dom} = (-\infty, \infty)$$

a.) $1 \leq x \leq 3$

b.) $x < -1$ or $x > 1$

c.) $0 < x < 1$

d.) $-2 < x < 2$ ✓



At $x=0$, $f(x)$ is not diff

So we can not apply

L.M.Th in that interval in which we have $x=0$

Q:) verify L.M.Th for $f(x) = x^3$ in $[-1, 1]$ & hence evaluate $c=1$

$$\text{Sol:)} \quad f'(c) = \frac{f(1) - f(-1)}{1 - (-1)} = \text{DNE}$$

$$f'(x) = \frac{1}{3} x^{\frac{1}{3}-1} = \frac{1}{3x^{2/3}}$$

∴ By common sense, it is obvious that at $x=0$, $f(x)$ is not differentiable.

Hence

L.M.V.T Not applicable.

Note: $(1)^3 = 1$, $0, 0, 0$

$(-1)^3 = -1$, $0, -0, -0$

$f'(c) = \frac{1}{3c^{2/3}}$ =, Now forcefully try to satisfy L.M.

$$f'(c) = \frac{f(1) - f(-1)}{1 - (-1)}$$

Date

$$= \frac{1}{3\sqrt{3}} = \frac{(1)^{3/2} - (-1)^{3/2}}{1 - (-1)}$$

$$\frac{1}{3\sqrt{3}} = \frac{1 - (-1)}{2} = \frac{2}{2} = 1$$

$$c^{3/2} = \frac{1}{3} \Rightarrow c = \left(\frac{1}{3}\right)^{2/3}$$

$$c = \frac{1}{3\sqrt{3}}$$

• it is possible to find C but it does not imply that $f(x)$ is Cont & Diff both.

i.e. Converse need not be true.

Rolle's M.V. Theorem:

Let $f(x)$ is defined in $[a, b]$ s.t.

(i) $f(x)$ is Continuous in $[a, b]$

(ii) $f(x)$ is Diff. in (a, b)

(iii) $f(a) = f(b)$

then $\exists$ at least one point c in b/w a & b for which

tangent is horizontal

or tangent is $\parallel$ to x -axis

$$f'(c) = 0$$



Date

Q:) Verify R. Th for $f(x) = \sin^4 x + \cos^4 x$ is $[0, \pi/2]$ & Hence find $C = ?$

a.) 0 it is a combination of E. funct so cannot
b.) $\pi/2$ as well as diff.

c.) $\pi/4$ ✓

d.) $\pi/6$ Also we have $\boxed{f(0) = 1 = f(\pi/2)}$ i.e.

all three necessary condition are satisfied of Rolle's Theorem over

$\exists$ at least n b/w 0 & $\pi/2$ for which $\boxed{f'(c) = 0}$

$$f(x) = \sin^4 x + \cos^4 x$$

$$f'(x) = 4\sin^3 x \cos x + 4\cos^3 x \sin x$$

$$\begin{aligned} f'(x) &= 4\sin^3 x \cos x + 4\cos^3 x \sin x \\ &= 4\sin^2 x \cos^2 x (\sin^2 x - \cos^2 x) \\ &= -2(\sin 2x) \cdot 2\sin x \cos x (\cos 2x) \\ &= -\sin 4x \quad \text{B.C.} \end{aligned}$$

$$f'(c) = -\sin 4c$$

$$-\sin 4c = 0 \quad \text{or} \quad \sin 4c = \sin n\pi$$

$$\boxed{c = \frac{n\pi}{4}}, \quad n \in \mathbb{Z}$$

$$C = \dots, -\frac{\pi}{4}, -\frac{\pi}{4}, 0, \frac{\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}, \dots$$

$$\boxed{C = \pi/4}$$

Date

Q:) Find the eq. of tangent at the point on the curve $f(x) = \sin^4 x + \cos^4 x$ is $[0, \pi/2]$ whose tangent is $\perp$ to x axis?
 & Find coordinates of that point?

- a.) $(\frac{1}{2}, \frac{\pi}{4})$ Slope of tangent $(m) = f'(x) = 0$
 b.) $(\frac{\pi}{4}, \frac{1}{2})$ ✓ i.e
 c.) $(\frac{\pi}{2}, \frac{1}{2})$ eq. of tangent at $(c, f(c))$ which is
 d.) $(\frac{\pi}{4}, 0)$ to x axis

$$y - f(c) = m(x - c)$$

$$'' = 0(x - c)$$

$$'' = 0$$

$$\boxed{y = f(c)}$$

$$c = f\left(\frac{\pi}{4}\right) = (\sin^4 x + \cos^4 x) = \dots = \frac{1}{2}$$

i.e

$$\text{eq. of tangent is } y = \frac{1}{2}$$

$$\text{Coordinates} = (c, f(c)) = \left(\frac{\pi}{4}, \frac{1}{2}\right)$$

Cauchy M.V. Th $\rightarrow$

Let $f(x)$ & $g(x)$ are two funct defined in $[a, b]$ &

- (i) Both $f(x)$ & $g(x)$ are continuous in $[a, b]$
 (ii) " " " " " diff in (a, b)
 (iii) $g'(x) \neq 0 \forall x \in (a, b)$

Then $\exists$ at least one c in b/w a & b for which

$$\boxed{\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}}$$

Date

Ratio of diff of $f(x)$ & $g(x)$ at $a+h = ?$
 Ratio of slope of tangent of $f(x)$ & $g(x)$ at c .

Q:) if $f(x) = x^2$ & $g(x) = x^3$ defined in $[1, 2]$ then verify C.M.V.Th and hence evaluate $C = ?$

a.) 1.5 $\because$ Both $f(x)$ & $g(x)$ are poly function.

b.) 9/14 Hence Cont as well as diff.

c.) 17/9

d.) 14/9 $\checkmark$ $f(x) = x^2$, $g(x) = x^3$
 $f'(x) = 2x$, $g'(x) = 3x^2$

$$\because g'(x) \neq 0 \forall x \in (1, 2)$$

$$\text{So by C.M.V.Th; } \frac{f(2) - f(1)}{g(2) - g(1)} = \frac{f'(c)}{g'(c)}$$

$$= \frac{4 - 1}{8 - 1} = \frac{3C}{3C^2} = \boxed{C = 14/9} \quad \checkmark$$

$C = 14/9$ lies in b/w 1 & 2

Hence C.M.V.T is verified.

Q:) if $f(x) = \log_e x$ & $g(x) = \log_e \left(\frac{1}{x}\right)$ defined in $[1, 2]$ then By C.M.Th then $C = ?$

$f(x) = \log x$, $g(x) = -\log x$ (Both are Transcendental)

a.) 1.0

b.) 1.25

c.) 1.5

d.) Any value b/w 1 & 2 $\checkmark$ $f(1) = 0$ $g(1) = 0$
 $f(2) = \log 2$ $g(2) = -\log 2$
 $f'(x) = \frac{1}{x}$ $g'(x) = -\frac{1}{x}$

Date

$$\Rightarrow f'(c) = \frac{1}{c} \quad , \quad g'(c) = -\frac{1}{c}$$

$$\therefore g'(x) \neq 0 \quad \forall x \in (1, 2)$$

By C.M.V Th Then,

$$= \frac{f(2) - f(1)}{g(2) - g(1)} = \frac{f'(c)}{g'(c)}$$

$$= \frac{\log 2 - 0}{-\log 2 - 0} = \frac{1/c}{-1/c} = -1 = 1 \quad (\text{identity})$$

$\Rightarrow c$ can take any value 1 & 2

Lecture no. 09

Lagrange's mean value Theorem for Integral
Average height

if $f(x)$ is continuous func in $[a, b]$ then $\exists c$ in (a, b) s.t.

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx$$

Q:) The mean value of $f(x) = 4x^2 + 7x + 1$ b/w -1 & 2 is

$$a) \frac{7}{2} \quad f(c) = \frac{1}{2+1} \int_{-1}^2 (4x^2 + 7x + 1) dx$$

$$b) \frac{9}{2} \quad = \frac{1}{3} \left[\frac{4x^3}{3} + \frac{7x^2}{2} + x \right]_{-1}^2 = \dots = \frac{13}{2}$$

$$c) 14 \quad d) 17/2 \quad \checkmark$$

Spiral

Q:) The point on the curve $y = \frac{4}{x^2}$ b/w $x = -4$ to $x = 2$ where we can find mean valueth of the function will be!

a) $(-2\sqrt{2}, -0.5)$ $f(c) = \frac{1}{-2+4} \int_{-4}^{-2} \left(\frac{4}{u^2} \right) du$

b) $(-2\sqrt{2}, 0.5)$ ✓ $= \frac{4}{2} \int_{-4}^{-2} (u^{-2}) du$

c) $(2\sqrt{2}, -0.5)$

d) $(2\sqrt{2}, 0.5)$ $= \frac{4}{2} \left[\frac{u^{-2+1}}{-2+1} \right]_{-4}^{-2}$
 $= 2 \left[\frac{u^{-1}}{-1} \right]_{-4}^{-2} = -2 \left[-\frac{1}{2} + \frac{1}{4} \right]$

$= 0.5$

$\therefore f(c) = 0.5$

$\frac{4}{c^2} = \frac{1}{2} \Rightarrow c^2 = 8 \Rightarrow c = \pm 2\sqrt{2}$

In L.A.R.M.V. The c lies in b/w $x = -4$ to $x = 2$ so, $c = -2\sqrt{2}$

Q:) Find the average value of $f(x) = x^2$ b/w 1 to 4

Sol:) $f(c) = \frac{1}{4-1} \int_1^4 f(x) dx = \frac{1}{3} \int_1^4 (x^2) dx = \frac{63}{3} = 21$

$f(c) = 21 \Rightarrow c^2 = 21 \Rightarrow c = \pm \sqrt{21}$

(ii) Also find the coordinate of that point!

a) $\pm \sqrt{7}$

b) $(\sqrt{7}, 7)$ ✓ $f(c) = 7 \Rightarrow (c, f(c))$

c) $(\sqrt{5}, 7)$ $c^2 = 7$

d) $(2.5, 7)$ $c = \pm \sqrt{7} \Rightarrow (\sqrt{7}, 7)$

Derivatives.

Date

General points:

Funct $\begin{cases} \rightarrow \text{In the form of curve (in 2D)} : y = f(x) \\ \rightarrow \text{In the form of surface (in 3D)} : z = f(x, y) \end{cases}$
 or $f(x, y, z) = C$

* **Explicit Curve**: $y = f(x)$, $z = f(x, y)$

* **Implicit curve**: $f(x, y) = C$, $f(x, y, z) = C$

$\Rightarrow$ if we can do separate then it is called Explicit

1) Explicit funct:

if it is possible to **separate** Dependent & independent variable then funct is called Explicit.

For e.g.: $x^3 + y^3 + 3xy^2 = 4x^2$

$$(1+3x)y^3 = 4x^2 - x^3 \Rightarrow \left(\frac{4x^2 - x^3}{1+3x} \right)^{1/3} = y$$

2) Implicit funct:

if it is **not possible** to **separate** Dep & Ind variable then funct is called implicit funct.

For e.g.: $x^3 + y^3 + 3xy = 1$

$f(x, y) = C$ i.e. implicit.

Types of Question:

- 1.) Based on ordinary Derivatives (exist in case of curve $y = f(x)$)
- 2.) " " Partial Derivatives (exist in case of surface $z = f(x, y)$)

3) Based on Total derivative (if $z \rightarrow (x, y) \rightarrow t$ (alone))

4) " " chain Rule of partial derivative
(if $z \rightarrow (x, y) \rightarrow (x, y)$)

5) Based on Jacobian (if $(u, v) \rightarrow (x, y)$)

6) Based on Euler Theorem (if $U = U(x, y)$ is homogeneous function n dy then use E. Theorem).

$\Rightarrow$ ordinary derivative.

Power formula = $\frac{d}{dn} (n^a) = an^{a-1}$

Similarly

$$\frac{d^2}{dn^2} (n^a) = n^{a-2}$$

$$d^{n+1} (n^a) = 0$$

2) $\frac{d}{dn} a^n = a^n \log_e a \Rightarrow$ Exponential formula.

$$\frac{d}{dn} (e^n) = e^n$$

Q.1) if $y = n^a + a^n + a^a + n^n$ then $dy/dn = ?$

Sol: $\frac{dy}{dn} = a n^{a-1} + a^n \log_e a + \cancel{a^a \log_e a} + n^n (1 + \log_e n)$

Q.2) $y = \log_e n + \log_n n + \log_e a + \log_n a + \log_e a$ then $\frac{dy}{dn} = ?$

Sol: $y = \log_e n + \frac{\log_e n}{\log_e a} + \log_e a + 1 + 1$

= See in slide

Date

$$3.) \frac{d}{dn} (n^n) = n^n (1 + \log_e n)$$

$$4.) \frac{d}{dn} (\log_e n) = \frac{1}{n}$$

$$\Rightarrow \frac{d^2}{dn^2} (\log_e n) = \frac{d}{dn} \left(\frac{1}{n} \right) = -\frac{1}{n^2}$$

{chain rule}

$$\Rightarrow \frac{d}{dn} \left(\frac{1}{\log_e n} \right) = \frac{-1}{(\log_e n)^2} \cdot \frac{d}{dn} (\log_e n) = \frac{-1}{(\log_e n)^2} \left(\frac{1}{n} \right)$$

$$Q: y = \log_{10} 10 + \log_e n^n + \log_{10} n^n + \log_e n^{10} \text{ then } \frac{dy}{dn} = ?$$

$$\text{Sol:} = 1 + 1 + \frac{\log_e n^n}{\log_e 10} + \frac{\log_e 10}{\log_e n^n}$$

$$= 0 + 0 + \frac{1}{\log_{10} 10} \left(\frac{1}{n} \right) + \log_e 10 \frac{d}{dn} \left(\frac{1}{\log_e n^n} \right)$$

$$= \frac{1}{n \log_{10} 10} + \log_e 10 \left[\frac{-1}{(\log_e n^n)^2} \cdot \frac{1}{n} \right] \text{ Ans.}$$

Q: let $f(n) = e^{-|n|}$, where n is real. The value of $\frac{df}{dn}$ at $n=1$ is

$$f(n) = e^{-|n|} = \begin{cases} e^{-(-n)} & , n < 0 \\ e^{-(n)} & , n > 0 \end{cases} = \begin{cases} e^n & , n < 0 \\ e^{-n} & , n > 0 \end{cases}$$

$$a) -e$$

$$b) 0$$

$$c) 1/e$$

$$d) -1/e$$

$$f'(n) = \begin{cases} e^n & , n < 0 \\ -e^{-n} & , n > 0 \end{cases} \quad f'(-1) = e^{-1} = \frac{1}{e}$$

Date

Q:) $y = \sin |u|$, $\left(\frac{dy}{du}\right)_{u = -\frac{\pi}{4}} = ?$

Sol:) $y = \sin |u| = \begin{cases} \sin(-u), & u < 0 \\ \sin(u), & u > 0 \end{cases} = \begin{cases} -\sin u, & u < 0 \\ \sin u, & u > 0 \end{cases}$

$$\frac{dy}{du} = \begin{cases} -\cos u, & u < 0 \\ \cos u, & u > 0 \end{cases}$$

$$= \left(\frac{dy}{du}\right)_{u = -\frac{\pi}{4}} = -\cos\left(-\frac{\pi}{4}\right) = -\cos\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

(ii) Also find $f'(0) = ?$

$$\text{L.H.D} = f'(0^-) = -\cos(0) = -1$$

$$\text{R.H.D} = f'(0^+) = \cos(0) = +1$$

$\therefore \text{L.H.D} \neq \text{R.H.D}$ so not Diff. at $x=0$.

$\therefore f'(0) = \text{DNE}$

5.) Chain Rule:

$$\frac{d}{du} [f(g(u))] = f'(g(u)) \cdot g'(u)$$

e.g. $\frac{d}{du} (\sin \sqrt{\tan u^3}) = ?$

$$= (\cos \sqrt{\tan u^3}) \cdot \frac{d}{du} (\tan u^3)^{\frac{1}{2}}$$

$$= \cos \sqrt{\tan u^3} \cdot \frac{1}{2} (\tan u^3)^{-\frac{1}{2}} \cdot \frac{d}{du} \tan u^3$$

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$$= \cos \sqrt{\tan n^3} \cdot \frac{1}{2\sqrt{\tan n^3}} \cdot \left[\sec^2 n^3 \right] \frac{d}{dn} n^3$$

$$= \cos \sqrt{\tan n^3} \cdot \frac{1}{2\sqrt{\tan n^3}} \cdot \sec^2 n^3 \cdot (3n^2)$$

Note:

$$1.) \frac{d}{dn}(ny) = n \frac{d}{dn}(y) + y \frac{d}{dn}(n) = n \left(\frac{dy}{dn} \right) + y(1)$$

$$2.) \frac{d}{dy}(ny) = n \frac{d}{dy}(y) + y \frac{d}{dy}(n) = n(1) + y \left(\frac{dn}{dy} \right)$$

$$3.) d(ny) = n dy + y dn$$

$$\text{e.g. } \frac{d}{dn}(y^3) = \frac{d}{dy}(y^3) \cdot \frac{dy}{dn} = 3y^2 \cdot \frac{dy}{dn}$$

Some Standard Result:

$$6.) \frac{d}{dn}(\sin n) = \cos n$$

$$7.) \frac{d}{dn}(\cos n) = -\sin n$$

$$8.) \frac{d}{dn}(\tanh) = \sec^2 n$$

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$$9.) \frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x$$

$$10.) \frac{d}{dx} (\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$11.) \frac{d}{dx} (\sec x) = \sec x \tan x$$

$$12.) \frac{d}{dx} (\sinh x) = \cosh x$$

$$13.) \frac{d}{dx} (\cosh x) = \sinh x$$

14.) Product formula:

$$a.) \frac{d}{dx} (fg) = fg' + gf'$$

$$b.) \frac{d}{dx} (f, g, h) = f'gh + fg'h + fgh'$$

15.) Quotient formula:

$$a.) \frac{d}{dx} \left(\frac{f}{g} \right) = \frac{gf' - fg'}{(g)^2}$$

or

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$$

Log diff

Date

Q:) if $n^y = e^{n-y}$ then $\frac{dy}{dn} = ?$

Sol:) $\log(n^y) = \log(e^{n-y})$

$$y \log n = (n-y) \log e$$

$$\log n = n-y$$

$$y(1 + \log n) = n$$

$$y = \frac{n}{1 + \log n}$$

$$\frac{dy}{dn} = \frac{(1 + \log n) \frac{d}{dn}(n) - n \frac{d}{dn}(1 + \log n)}{(1 + \log n)^2}$$

$$= \frac{(1 + \log n)(1) - n(\frac{1}{n})}{(1 + \log n)^2}$$

$$\boxed{\frac{dy}{dn} = \frac{\log n}{(1 + \log n)^2}} \text{ Ans.}$$

19.) Differentiation of Infinite series:

Q:) $y = \sqrt{\sin n} + \sqrt{\sin n} + \sqrt{\sin n} + \dots \infty$ then $\frac{dy}{dn} = ?$

Sol:) $y = \sqrt{\sin n} + y$

$$\frac{d}{dn}(y^2) = \frac{d}{dn}(y) = \frac{d}{dn}(\sin n)$$

$$y^2 = \sin n + y$$

$$y^2 - y = \sin n \text{ --- (1)}$$

$$= 2y \frac{dy}{dn} - \frac{dy}{dn} = \cos n$$

$$\boxed{\frac{dy}{dn} = \frac{\cos n}{2y-1}}$$

Spiral

Hint

Q:1) $y = x^x$ then $\frac{dy}{dx} = ?$

18.) Differentiation of parameter curve.

Curve	Parameter (vector)	Equation
Circle	$(a \cos \theta, a \sin \theta)$	$x^2 + y^2 = a^2$
Parabola	$(at^2, 2at)$	$y^2 = 4ax$
Ellipse	$(a \cos \theta, b \sin \theta)$	$x^2/a^2 + y^2/b^2 = 1$
Hyperbola	$(a \sec \theta, b \tan \theta)$	$x^2/a^2 - y^2/b^2 = 1$

* $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

$2\cos^2 \theta - 1 \longrightarrow \frac{1 + \cos 2\theta}{2} = \cos^2 \theta$

$1 - 2\sin^2 \theta \longrightarrow \frac{1 - \cos 2\theta}{2} = \sin^2 \theta$

* if $x = f(t)$, $y = g(t)$ then $\frac{dy}{dx} = ?$

$\frac{dy}{dt} \cdot \frac{dt}{dx} = \boxed{g'(t) \left(\frac{1}{f'(t)} \right)}$ $\frac{dx}{dt} = f'(t)$, $\frac{dy}{dt} = g'(t)$

Q:1) if $x = at^2$, $y = 2at$ then $\frac{dy}{dx} = ?$

Ans:1 $\frac{dx}{dt} = 2at$, $\frac{dy}{dt} = 2a$

So, $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = 2a \left(\frac{1}{2at} \right) = \frac{1}{t}$ Solved

(H.W.)

Date

Q:) if $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$, $\frac{dy}{dx} = ?$ a.) $\cot \theta/2$ ✓b.) $\tan \theta/2$ c.) $\theta/2$ d.) $\cot \theta$ 19.) Diff of funct w.r.t to another funct $\rightarrow$ if $u = f(x)$, $v = g(x)$ then $\frac{dv}{du} = ?$

$$= \frac{dv/dx}{du/dx} = \frac{f'(x)}{g'(x)}$$

(H.W.)

Q:) Diff x^n w.r.t to $x \log x$!Q:) a.) x^n ✓b.) $x^n(1 + \log x)$ c.) $1 + \log x$ d.) $x \log x$ **Note:**

$$1) \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

$$2) \frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$$

Date

$$3.) \frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$$

$$4.) \frac{d}{dx} (\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$5.) \sin^{-1} x + \cos^{-1} x = \pi/2$$

$$6.) \tan^{-1} x + \cot^{-1} x = \pi/2$$

$$Q.) \text{ Diff } \sin^{-1} \left(\frac{2x}{1+x^2} \right) \text{ w.r.to } \tan^{-1} x?$$

$$a.) \tan \theta$$

$$b.) \theta$$

$$c.) 2 \quad \checkmark$$

$$d.) 2\theta$$

Lecture no. 10

Partial diff: for $z = f(x, y)$

$$\frac{\partial z}{\partial x} = \lim_{h \rightarrow 0} \left[\frac{f(x+h, y) - f(x, y)}{h} \right]_{y = \text{const}}$$

$$\frac{\partial z}{\partial y} = \lim_{k \rightarrow 0} \left[\frac{f(x, y+k) - f(x, y)}{k} \right]_{x = \text{const}}$$

Note: In 2D.

Eq of x axis: $y = 0$

Eq of line // to x axis: $y = k$

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Eq of Y axis = 0

Eq of line $\parallel^r$ to Y axis: $x = h$ In 3D:Eq of XY plane: $z = 0$ " " YZ " " : $x = 0$ " " ZX " " : $y = 0$ Eq of plane $\parallel^r$ to xz plane $y = k$ " " " $\parallel^r$ to YZ plane $x = h$ $\Rightarrow$ Significance of $\frac{\partial z}{\partial x} \Rightarrow$ if we cut our surface $z = f(x, y)$ by the plane $\parallel$ to xz plane (i.e. from $y = \text{const}$)then we will get a curve of the type $z = f(x)$ Now $\frac{\partial z}{\partial x}$ = slope of tangent at any random point on this curveSimilarly we can define $\frac{\partial z}{\partial y}$ $z = f(x)$ (curve) = $\frac{\partial z}{\partial y}$ = slope of tangent on this curve

Note:

 $z = f(x, y)$ then $z_x = \frac{\partial z}{\partial x} = f_x = \frac{\partial f}{\partial x}$ all are same1) $z_{xx} = f_{xx} = \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 f}{\partial x^2}$

or

 $z_{xy} = f_{xy} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = 1$

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$$2.) \quad \frac{\partial^2 z}{\partial n \partial y} = \frac{\partial^2 z}{\partial y \partial n}$$

3.) All the result that are applicable in case of ordinary derivative are also valid in case of partial derivative keeping other variable constant.

Q:) if $f = y^n$ then $\left(\frac{\partial f}{\partial n \partial y} \right)_{y=1} = ? \quad n=2, y=1$

a.) 1 ✓

b.) -1 $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (y^n) = \therefore \frac{d}{dn} (n^n) = n n^{n-1}$

c.) 0

d.) 4

$$= n \cdot y^{n-1}$$

$$= \frac{\partial^2 f}{\partial n \partial y} = \frac{\partial}{\partial n} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial n} (n y^{n-1})$$

$$= n \frac{\partial}{\partial n} (y^{n-1}) + y^{n-1} \frac{\partial}{\partial n} (n)$$

$$= n \left[y^{n-1} \log_e y \right] + y^{n-1} (1)$$

$$\left(\frac{\partial^2 f}{\partial n \partial y} \right)_{y=1} = \left[n(0) + 1^{n-1}(1) \right] = 1$$

$$y=1$$

Q:) if $x^2 = n^2 + y^2 + z^2$ then evaluate $\frac{\partial x}{\partial n}, \frac{\partial x}{\partial y}, \frac{\partial x}{\partial z}$

Sol:) $\frac{\partial}{\partial n} (x^2) = \frac{\partial}{\partial n} (n^2 + y^2 + z^2)$

$$\frac{\partial}{\partial n} (x^2) \frac{\partial x}{\partial n} = 2n + 2y \frac{\partial y}{\partial n} + 2z \frac{\partial z}{\partial n} \quad \{ \text{blow} \}$$

$$= 2x \frac{\partial x}{\partial n} =$$

z = dependent variable.

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Independent variable = (x, y, z)

Date

$$= \frac{\partial z}{\partial x} = 2x + 0 + 0$$

Similarly

$\frac{\partial z}{\partial y} = \frac{y}{z}$	,	$\frac{\partial z}{\partial z} = \frac{z}{z}$
---	---	---

4) Don't assume dependent variable as constant if we are solving question based on partial derivatives.

Q:) if $U = (x^2 + y^2 + z^2)^{-1/2}$ then $U_{xx} + U_{yy} + U_{zz} = ?$

a.) $2(x^2 + y^2 + z^2)^{-3/2}$

b.) 4.5

c.) 0 ✓

d.) $6(x^2 + y^2 + z^2)^{-5/2}$

Sol:) M-1 $U_x = \frac{\partial U}{\partial x} = \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2x = -x(x^2 + y^2 + z^2)^{-3/2}$

$$U_x = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2x + 0 + 0)$$

$$U_x = -x(x^2 + y^2 + z^2)^{-3/2}$$

Similarly

you can use only this method w.o. l.

$$U_y = -y(x^2 + y^2 + z^2)^{-3/2}$$

$$U_z = -z(x^2 + y^2 + z^2)^{-3/2}$$

See in slide.

$$r^2 = x^2 + y^2 + z^2$$

2.8.1

$$r^2 = x^2 + y^2 + z^2$$

(2.8.1)

m-2 Let $U = (x^2 + y^2 + z^2)^{-1/2}$ then $U_{xx} + U_{yy} + U_{zz} = ?$

Let $r^2 = x^2 + y^2 + z^2$ then $\frac{\partial r}{\partial x} = \frac{x}{r}$ $\frac{\partial r}{\partial y} = \frac{y}{r}$ $\frac{\partial r}{\partial z} = \frac{z}{r}$

$$U = (r^2)^{-1/2} = r^{-1} = \frac{1}{r}$$

$$U_x = \frac{\partial U}{\partial x} = \frac{\partial}{\partial x} \left(\frac{1}{r} \right) = \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \frac{\partial r}{\partial x} = -\frac{1}{r^2} \left(\frac{x}{r} \right)$$

i.e. $U_x = -\frac{x}{r^3}$ $U_{xx} = -\frac{1}{r^3}$

Similarly

$U_{yy} = -\frac{1}{r^3}$, $U_{zz} = -\frac{1}{r^3}$ } P.A.P.

$U_{xx} + U_{yy} + U_{zz} = -\frac{3}{r^3}$ } $\nearrow$

$$U_{xx} = \frac{\partial^2 U}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial U}{\partial x} \right) = \frac{\partial}{\partial x} \left[-\frac{x}{r^3} \right]$$

$$= - \frac{r^3 \frac{\partial}{\partial x} (x) - x \frac{\partial}{\partial x} (r^3)}{(r^3)^2}$$

$$= - \frac{r^3 (1) - x \left(3r^2 \cdot \frac{\partial r}{\partial x} \right)}{r^6}$$

$$= - \frac{r^3 - x \left(3r^2 \cdot \frac{x}{r} \right)}{r^6}$$

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$$U_{xx} = - \left[\frac{x^2 - 2(3x^2)}{x^6} \right] = -\frac{1}{x^3} + \frac{1}{x^3} (3x^2)$$

Similarly

$$U_{yy} = -\frac{1}{y^3} + \frac{1}{y^3} (3y^2)$$

$$\therefore U_{zz} = -\frac{1}{z^3} + \frac{1}{z^3} (3z^2)$$

$$U_{xx} + U_{yy} + U_{zz} = -\frac{3}{x^3} + 3(x^2 + y^2 + z^2) \frac{1}{x^3}$$

$$= -\frac{3}{x^3} + 3(x^2) \frac{1}{x^3} = -\frac{3}{x^3} + \frac{3}{x^3} = 0 \text{ etc.}$$

Q: if $U = \log_e(x^2 + y^2)$ then $\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = ?$

a.) 2 $U_x = \frac{\partial U}{\partial x} = \frac{1}{\partial x} \left[\log(x^2 + y^2) \right]$

b.) 0 ✓

c.) -1

d.) $\frac{2(x+y)}{x^2+y^2}$ $= \frac{1}{x^2+y^2} \frac{\partial}{\partial x} (x^2+y^2) = \frac{2x}{(x^2+y^2)}$

$$U_{xx} = \frac{\partial^2 U}{\partial x^2} = \frac{\partial}{\partial x} \left[\frac{\partial U}{\partial x} \right] = \frac{\partial}{\partial x} \left[\frac{2x}{(x^2+y^2)} \right]$$

$$= 2 \left[\frac{(x^2+y^2) \cdot 1 - x(2x)}{(x^2+y^2)^2} \right] = \frac{2(y^2 - x^2)}{(x^2+y^2)^2}$$

$$\therefore \frac{2(y^2 - x^2)}{(x^2+y^2)^2} = U_{yy} = \frac{2(x^2 - y^2)}{(x^2+y^2)^2}$$

So final answer is $U_{xx} + U_{yy} = 0$

①

②

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Q:) if $z = x^2 + y - 3$ & $y^3 + z^3 + yz - xy = 1$
where x & y are independent variable then at (1,1)
evaluate $\frac{\partial z}{\partial x} = ?$

Sol:) a.) 4.0 $\frac{\partial z}{\partial x} = 2x + 0 - \frac{\partial z}{\partial x}$ — (1)
b.) 2.0

c.) 5.5 By eq (1)
d.) 4.5 ✓ $\frac{\partial}{\partial x} (y^3 + z^3 + yz - xy) = \frac{\partial}{\partial x} (1)$

$$= 0 + 3z^2 \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial x} - 1 \cdot y = 0$$

$$\Rightarrow \frac{\partial z}{\partial x} = \left(\frac{y}{3z^2 + y} \right)$$

$$\left(\frac{\partial z}{\partial x} \right)_{\substack{x=2 \\ y=-1 \\ z=1}} = 2x - \left[\frac{y}{3z^2 + y} \right] = 4 - \left[\frac{-1}{3 + (-1)} \right]$$

$$= 4.5 \quad \text{B}$$

Q:) let $f(x,y) = \frac{ax^2 + by^2}{xy}$, where a and b are constant

if $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ at $x=1$ & $y=2$ then the relation b/a

a) 4/3? M-L

a.) $a = b/4$
b.) $a = b/2$
c.) $a = 2b$
d.) $a = 4b$ ✓

$$f = \frac{ax^2 + by^2}{xy} = a \left(\frac{x}{y} \right) + b \left(\frac{y}{x} \right)$$

$$\frac{\partial f}{\partial x} = \frac{a}{y} + b \left(\frac{-y}{x^2} \right) \Rightarrow \left(\frac{\partial f}{\partial x} \right)_{x=1} = \frac{a}{2} - 2b$$

$$\frac{\partial f}{\partial y} = a \left(\frac{-x}{y^2} \right) + b \left(\frac{1}{x} \right) = -\frac{a}{4} + b \quad y=2$$

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A.T.O.:

$$= \frac{a}{2} - 2b = -\frac{a}{4} + b$$

$$= \boxed{a = 4b} \quad \text{or}$$

Homogeneous funct:

Consider a funct $U = U(x, y)$ then this funct is called homog. func of degree n , if each term is of same degree (n).

Trick:

if $\boxed{U(\lambda x, \lambda y) = \lambda^n U(x, y)}$ then U is called homog. func of degree n

when $n \in \mathbb{R}$

e.g.) $U = x^3 + 4xy^2 + 5x^2y$

$$U = (\lambda x)^3 + 4(\lambda x)(\lambda y)^2 + 5(\lambda x)^2(\lambda y)$$

$$= \lambda^3 [x^3 + 4xy^2 + 5x^2y]$$

$$= \lambda^3 \cdot U(x, y)$$

So, Homog. func of degree $n = 3$

e.g.) $U = \frac{x^2 + y^2}{\sqrt{x} + \sqrt{y}}$

sol.) $U = \frac{(\lambda x)^2 + (\lambda y)^2}{\sqrt{\lambda x} + \sqrt{\lambda y}} = \frac{\lambda^2}{\lambda^{1/2}} \left[\frac{x^2 + y^2}{\sqrt{x} + \sqrt{y}} \right] = \lambda^{3/2} U$

So, Homog. func of $n = 3/2$ *Ans*

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e.g.) $U = x^5 + 4x^2y^3 + \log(x^5)$

$$U(\lambda x, \lambda y) = (\lambda x)^5 + 4(\lambda x)^2(\lambda y)^3 + \log((\lambda x)^5)$$

$$= \lambda^5 \left[x^5 + 4x^2y^3 + \frac{1}{\lambda^5} \log(x^5) \right]$$

$$= \lambda^5 \neq \lambda^0 U(x, y)$$

So, it is non-homog. func. of ~~degree = 5~~ ^{degree = 0}

e.g.) $U = \log x - \log y$

$$U = \log\left(\frac{x}{y}\right)$$

$$U(\lambda x, \lambda y) = \log\left(\frac{\lambda x}{\lambda y}\right) = \log\left(\frac{x}{y}\right)$$

$$U(\lambda x, \lambda y) = U(x, y)$$

$$= \lambda^0 U(x, y)$$

So homog. func. of degree $n = 0$.

* Euler theorem for homog. func. :-
if u is homog. func. of degree n then

$$\boxed{x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu}$$

Note:

$$1) \quad x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

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Q:) if $z = \frac{x^4 + y^4}{x^3 + y^3}$ then $\begin{cases} x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \\ x^2 2x + 2xy 2y + y^2 2y \end{cases}$

Sol:) $z(\lambda x, \lambda y) = \frac{\lambda^4 (x^4 + y^4)}{\lambda^3 (x^3 + y^3)}$

$= \lambda^{1/3} z(x, y)$

z is homog func of $n = \frac{1}{12}$

1.) $n \cdot z = \frac{1}{12} z = \frac{1}{12} \left(\frac{1}{12} - 1 \right) z = \frac{+11}{144} z$ wrong

Q:) if $u = \tan^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$ then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = ?$

a.) $\sin 2u$ $u = u(\lambda x, \lambda y) \neq \lambda^n (u(x, y))$

b.) $\tan u$ So u is non-homog & can't apply Euler

c.) $\cot u$ theorem

d.) $\cos 2u$

Let $\tan u = v = \frac{x^3 + y^3}{x + y}$ where v is homog func of $n = 2$.

Hence apply Euler theorem for v

$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = n \cdot v$

$x \frac{\partial (\tan u)}{\partial x} + y \frac{\partial (\tan u)}{\partial y} = 2 \cdot \tan u$

$x \frac{\partial (\tan u)}{\partial x} \frac{du}{dx} + y \frac{\partial (\tan u)}{\partial y} \frac{du}{dy} = 2 \tan u$

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$$= x (\sec^2 u) \frac{\partial u}{\partial x} + y (\sec^2 u) \frac{\partial u}{\partial y} = 2 \tan u$$

$$= x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} - \frac{2 \tan u}{\sec^2 u} = \sin 2u \quad \text{Ans.}$$

Q:) if $u = \sin^{-1}\left(\frac{y}{x}\right) + \tan^{-1}\left(\frac{y}{x}\right)$ then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = ?$

Sol:) $\therefore$ u is homog funct of $n = 0$

So,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0 \cdot u = 0$$

(ii) Also evaluate $x^2 U_{xx} + 2xy U_{xy} + y^2 U_{yy} = ?$

$$= 0 \cdot (0-1) \cdot u = 0$$

Q:) if $z = x^2 y^4 \sin^{-1}\left(\frac{y}{x}\right) + \log x - \log y$ then $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = ?$

Sol:)

$$x^2 y^4 \sin^{-1}\left(\frac{y}{x}\right) + \log\left(\frac{x}{y}\right)$$

$$z = u + v \quad \text{--- (1)}$$

where z is non-homog funct But.

u is homog funct of $n_1 = 6$

v is " " " " $n_2 = 0$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 6u$$

$$\& x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 0 \cdot v = 0$$

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$$z = u + v \rightarrow \frac{\partial z}{\partial u} = \frac{\partial u}{\partial u} + \frac{\partial v}{\partial u}$$

$$\frac{\partial z}{\partial y} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

Now we need, $u \frac{\partial z}{\partial u} + y \frac{\partial z}{\partial y} = u \left[\frac{\partial u}{\partial u} + \frac{\partial v}{\partial u} \right] + y \left[\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \right]$

$$= \left(u \frac{\partial u}{\partial u} + y \frac{\partial u}{\partial y} \right) + \left(u \frac{\partial v}{\partial u} + y \frac{\partial v}{\partial y} \right)$$

$$= 6u + 0 = 6u \text{ Ans.}$$

(ii) Also evaluate $u^2 \frac{\partial^2 z}{\partial u^2} + 2uy \frac{\partial^2 z}{\partial u \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = ?$

Sol.) $(30u) \cdot \rho_2 = n_1(n_1-1)u + n_2(n_2-1)v$
 $= 6(6-1)u + 0(0-1)v$
 $= 30u + 0$

* S.p. note: -

if $z = uv$ where u & v are homog. func. of degree n_1 & n_2 . But z is non homog. func.

1.) $u \frac{\partial z}{\partial u} + y \frac{\partial z}{\partial y} = n_1 u + n_2 v$

(ii) $u^2 \frac{\partial^2 z}{\partial u^2} + 2uy \frac{\partial^2 z}{\partial u \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = n_1(n_1-1)u + n_2(n_2-1)v$

Sol: $n_1(n_1-1)u + n_2(n_2-1)v$

$$= 6(6-1)u + 0(0-1)v$$

$$= 30u \text{ Ans.}$$

Lecture no. 11

⇒ Derivative of an Implicit function:

Q:1 $x^3 + y^3 + 3xy = 1$ then $\frac{dy}{dx} = ?$

- (M-2)

Sol:1 $\frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) + 3 \frac{d}{dx}(xy) = \frac{d}{dx}(1)$

$$= 3x^2 + \frac{d}{dx}(y^3) \frac{dy}{dx} + 3 \left[x \frac{d}{dx}(y) + y \frac{d}{dx}(x) \right] = 0$$

$$= 3x^2 + 3y^2 \frac{dy}{dx} + 3 \left[x \frac{dy}{dx} + y(1) \right] = 0$$

$$\frac{dy}{dx} [3y^2 + 3x] = -(3x^2 + 3y)$$

$$\frac{dy}{dx} = \frac{-3(x^2 + y)}{3(y^2 + x)} = - \left[\frac{x^2 + y}{y^2 + x} \right]$$

⇒ Consider $f(x, y) = c$ then $\frac{dy}{dx} = \frac{-\partial f / \partial x}{\partial f / \partial y}$

Shortcut of finding derivative for implicit

M-2

$$f = x^3 + y^3 + 3xy = 1 = 0$$

$$\frac{\partial f}{\partial x} = 3x^2 + 0 + 3y(1) = 0 = 3(x^2 + y)$$

$$\frac{\partial f}{\partial y} = 0 + 3y^2 + 3x(1) = 0 = 3(y^2 + x)$$

$$\text{So } \frac{dy}{dx} = \frac{-\partial f / \partial x}{\partial f / \partial y} = \frac{-3(x^2 + y)}{3(y^2 + x)} = - \left[\frac{x^2 + y}{y^2 + x} \right]$$

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Total Derivative:-

let u is a function of $V = (x, y, z)$ where $x = x(t)$, $y = y(t)$, $z = z(t)$ that is

i.e. $V \longrightarrow (x, y, z) \longrightarrow t \text{ (alone) }$, Now total derivative exist.

Standard Result:

$$du = \left(\frac{\partial u}{\partial x} \right) dx + \left(\frac{\partial u}{\partial y} \right) dy + \left(\frac{\partial u}{\partial z} \right) dz$$

Hence:

$$\frac{du}{dt} = \left(\frac{\partial u}{\partial x} \right) \frac{dx}{dt} + \left(\frac{\partial u}{\partial y} \right) \frac{dy}{dt} + \left(\frac{\partial u}{\partial z} \right) \frac{dz}{dt}$$

Q: $d(x, y) = ?$ m-1 using common sense

a) $dx dy$

$$d(fg) = fg' + g f'$$

b) $dx + dy$

c) $x dx + y dy$

$$d(x, y) = x dy + y dx$$

d) $x dy + y dx$ ✓

m-2 let $V = xy$

then $\frac{\partial u}{\partial x} = y$ & $\frac{\partial u}{\partial y} = x$

Now standard result of T.O is

$$du = \left(\frac{\partial u}{\partial x} \right) dx + \left(\frac{\partial u}{\partial y} \right) dy = y dx + x dy \quad \text{A.}$$

Note: $\frac{d}{dx}(xy) = x \frac{d}{dx}(y) + y \frac{d}{dx}(x) = x \frac{dy}{dx} + y \cdot 1$

1) $\frac{d}{dy}(xy) = x \frac{d}{dy}(y) + y \frac{d}{dy}(x) = x + y \frac{dx}{dy}$

2) $d(xy) = x d(y) + y d(x) = x dy + y dx$ Simple

Date

Q:) if $U = x^2 - y^2 + 2 \sin y z$ then

$$\frac{\partial U}{\partial x} = 2x - 0 + 0 = 2x$$

$$\frac{\partial U}{\partial y} = 0 - 2y + 2 \{ \cos(yz) \cdot (z) \} = -2y + 2z \cos(yz)$$

$$\begin{aligned} \frac{\partial U}{\partial z} &= 0 - 0 + 2 \{ \cos(yz) \cdot (y) \} \\ &= 2y \cos(yz) \end{aligned}$$

Q:) if $y = e^n$ then $\frac{dy}{dn} = e^n$

Q:) if $z = \ln n$ then $\frac{dz}{dn} = \frac{1}{n}$

Q:) if $U = x^2 - y^2 + 2 \sin y z$ where $y = e^n$; $z = \ln n$ then $\frac{dU}{dn} = ?$

Sol:)

Here, $U \longrightarrow (x, y, z) \longrightarrow n \text{ (alone)}$

i.e. Question is based on T.D° concept.

$$\therefore \frac{dU}{dn} = \left(\frac{\partial U}{\partial x} \right) \frac{dx}{dn} + \left(\frac{\partial U}{\partial y} \right) \frac{dy}{dn} + \left(\frac{\partial U}{\partial z} \right) \frac{dz}{dn}$$

$$\begin{aligned} \Rightarrow \frac{dU}{dn} &= \left(\frac{\partial U}{\partial x} \right) + \left(\frac{\partial U}{\partial y} \right) \frac{dy}{dn} + \left(\frac{\partial U}{\partial z} \right) \frac{dz}{dn} \\ &= (2x) + (-2y + 2z \cos(yz)) e^n + (2y \cos(yz)) \frac{1}{n} \end{aligned}$$

Spiral 2

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$$\left(\frac{du}{dn}\right)_{2,0,1} = 4 + [0+2] \cdot 4 + 0 = 6 \cdot 4 = 4 + 2 \cdot 2^2 \text{ A}$$

Q:) if $U = n \log(n, y)$ then find.

$$1.) \frac{du}{dn} = \frac{\partial}{\partial n} [n \cdot \log(n, y)] = 1 + \log ny$$

$$2.) \frac{du}{dy} = n \frac{\partial}{\partial y} [\log(n, y)] = n \left[\frac{1}{ny} \frac{\partial}{\partial y} (ny) \right]$$

$$= n \left[\frac{1}{ny} (n) \right] = \frac{n}{y} \text{ A}$$

3.) Total derivative of $u =$

$$dU = \left(\frac{du}{dn}\right) dn + \left(\frac{du}{dy}\right) dy$$

$$= (1 + \log ny) dn + \left(\frac{n}{y}\right) dy \text{ A}$$

Q:) if $U = n \log(ny)$ where $x^3 + y^3 + 3ny = 1$ & n is an ind variable then $\frac{du}{dn} = ?$

sol:) $U \rightarrow (n, y) \rightarrow n \text{ (alone)}$

i.e. Question based on T.D. concept

$$dU = \left(\frac{\partial u}{\partial n}\right) dn + \left(\frac{\partial u}{\partial y}\right) dy$$

$$\frac{du}{dn} = \left(\frac{\partial u}{\partial n}\right) + \left(\frac{\partial u}{\partial y}\right) \frac{dy}{dn} = 1 + \log ny + \left(\frac{n}{y}\right) \left\{ -\frac{(n^3 + y^3)}{y^2 + n} \right\}$$

$$\frac{du}{dn} = -\frac{du}{dy} = -\left(\frac{n^3 + y^3}{y^2 + n}\right) \text{ A}$$

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Q:) Find the derivative of n^2y w.r to n where n and y are connected by the relation $n^2 + ny + y^2 = 1$

Sol:) let $u = n^2y$ — (1)

$$f = n^2 + ny + y^2 - 1 = 0 \quad \frac{dy}{dn} = -\frac{f_n}{f_y}$$

$$\frac{du}{dn} = \left(\frac{\partial u}{\partial n}\right) + \left(\frac{\partial u}{\partial y}\right) \frac{dy}{dn} = - \left(\frac{2n+y}{n+y} \right)$$

$$\frac{du}{dn} = \left(\frac{\partial u}{\partial n}\right) + \left(\frac{\partial u}{\partial y}\right) \frac{dy}{dn}$$

$$= (2ny + n^2) \frac{1}{n^2} - \left(\frac{2n+y}{n+y} \right) \frac{dy}{dn} = 0$$

Chain Rule of partial Derivative:-

let $u = u(x, y, z)$ where $x = x(t, s, t)$, $y = y(t, s, t)$
 $z = z(t, s, t)$

i.e

$$u \longrightarrow (x, y, z) \longrightarrow (t, s, t)$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \left(\frac{\partial x}{\partial t} \right) + \frac{\partial u}{\partial y} \left(\frac{\partial y}{\partial t} \right) + \frac{\partial u}{\partial z} \left(\frac{\partial z}{\partial t} \right)$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \left(\frac{\partial x}{\partial s} \right) + \frac{\partial u}{\partial y} \left(\frac{\partial y}{\partial s} \right) + \frac{\partial u}{\partial z} \left(\frac{\partial z}{\partial s} \right)$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \left(\frac{\partial x}{\partial t} \right) + \frac{\partial u}{\partial y} \left(\frac{\partial y}{\partial t} \right) + \frac{\partial u}{\partial z} \left(\frac{\partial z}{\partial t} \right)$$

⇒ Shortcut to learn above result :-

$$u \longrightarrow (x, y, z) \longrightarrow (t, s, t)$$

Date

$$1.) U_x = \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \left(\frac{\partial r}{\partial x} \right) + \frac{\partial u}{\partial y} \left(\frac{\partial y}{\partial x} \right) + \frac{\partial u}{\partial z} \left(\frac{\partial z}{\partial x} \right)$$

$$2.) U_y = \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \left(\frac{\partial r}{\partial y} \right) + \frac{\partial u}{\partial y} \left(\frac{\partial y}{\partial y} \right) + \frac{\partial u}{\partial z} \left(\frac{\partial z}{\partial y} \right)$$

$$3.) U_z = \frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \left(\frac{\partial r}{\partial z} \right) + \frac{\partial u}{\partial y} \left(\frac{\partial y}{\partial z} \right) + \frac{\partial u}{\partial z} \left(\frac{\partial z}{\partial z} \right)$$

Note:)

$$w \longrightarrow (x, y) \longrightarrow (r, \theta, z)$$

$$1.) \frac{\partial w}{\partial x} = \frac{\partial w}{\partial r} \left(\frac{\partial r}{\partial x} \right) + \frac{\partial w}{\partial \theta} \left(\frac{\partial \theta}{\partial x} \right) + \frac{\partial w}{\partial z} \left(\frac{\partial z}{\partial x} \right)$$

$$2.) \frac{\partial w}{\partial y} = \frac{\partial w}{\partial r} \left(\frac{\partial r}{\partial y} \right) + \frac{\partial w}{\partial \theta} \left(\frac{\partial \theta}{\partial y} \right) + \frac{\partial w}{\partial z} \left(\frac{\partial z}{\partial y} \right)$$

Q:) if $U = f(x, y)$ where $x = r + z$ & $y = r - z$ then
 $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z} = ?$

(a.) $2U_x$ $U \longrightarrow (x, y) \longrightarrow (r, z)$

b.) $2U_y$ i.e. Question Based on chain rule concept

c.) U_x

d.) $2(U_x + U_y)$ $U_x = \frac{\partial u}{\partial r} \left(\frac{\partial r}{\partial x} \right) + \frac{\partial u}{\partial y} \left(\frac{\partial y}{\partial x} \right) = \frac{\partial u}{\partial r}$

$$U_y = \frac{\partial u}{\partial r} \left(\frac{\partial r}{\partial y} \right) + \frac{\partial u}{\partial y} \left(\frac{\partial y}{\partial y} \right) = \frac{\partial u}{\partial z}$$

$$= U_x(1) + U_y(1)$$

$$= U_x(1) + U_y(-1) = \frac{\partial u}{\partial r} + \frac{\partial u}{\partial z} = 2 \frac{\partial u}{\partial r} + 0$$

$$= 2U_x$$

Spiral

Date

Q:) if $U = f\left(\frac{y \bar{z}}{xy}, \frac{z \bar{y}}{zn}\right)$ then $x^2 U_x + y^2 U_y + z^2 U_z =$

Sol:) a.) $U_x + U_y + U_z$

b.) 2

c.) 1

d.) 0

Here $U = f(r, s)$ where $r = \frac{y \bar{z}}{xy} = \frac{1}{x} \cdot \frac{1}{y}$ &

$$s = \frac{z \bar{y}}{zn} = \frac{1}{n} \cdot \frac{1}{z}$$

$\therefore U \longrightarrow (r, s) \longrightarrow (x, y, z)$

The question is based on chain rule concept.

$$U_x = \frac{\partial U}{\partial r} \left(\frac{\partial r}{\partial x} \right) + \frac{\partial U}{\partial s} \left(\frac{\partial s}{\partial x} \right) = U_x \left(-\frac{1}{x^2} \right) + U_s \left(-\frac{1}{x^2} \right)$$

$$U_y = \frac{\partial U}{\partial r} \left(\frac{\partial r}{\partial y} \right) + \frac{\partial U}{\partial s} \left(\frac{\partial s}{\partial y} \right) = U_x \left(+\frac{1}{y^2} \right) + U_s (0)$$

$$U_z = \frac{\partial U}{\partial r} \left(\frac{\partial r}{\partial z} \right) + \frac{\partial U}{\partial s} \left(\frac{\partial s}{\partial z} \right) = U_x (0) + U_s \left(+\frac{1}{zn} \right)$$

$$x^2 U_x = -U_x - U_s$$

$$y^2 U_y = U_x$$

$$z^2 U_z = U_s$$

$$x^2 U_x + y^2 U_y + z^2 U_z = 0 \quad \underline{A}$$

Spiral

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Q.) if $v = f(2x-3y, 3y-4z, 4z-2x)$ then
 $6v_x + 4v_y + 3v_z = ?$

- sol.) a.) 1 $v = f(r, s, t)$ where $r = 2x-3y$
 b.) -2 $s = 3y-4z$
 c.) 13 $t = 4z-2x$
 d.) 0 i.e $v \longrightarrow (r, s, t) \longrightarrow (x, y, z)$

Question is based on chain rule

Concept

$$v_x = \frac{\partial v}{\partial r} \left(\frac{\partial r}{\partial x} \right) + \frac{\partial v}{\partial s} \left(\frac{\partial s}{\partial x} \right) + \frac{\partial v}{\partial t} \left(\frac{\partial t}{\partial x} \right)$$

$$v_y = \frac{\partial v}{\partial r} \left(\frac{\partial r}{\partial y} \right) + \frac{\partial v}{\partial s} \left(\frac{\partial s}{\partial y} \right) + \frac{\partial v}{\partial t} \left(\frac{\partial t}{\partial y} \right)$$

$$v_z = \frac{\partial v}{\partial r} \left(\frac{\partial r}{\partial z} \right) + \frac{\partial v}{\partial s} \left(\frac{\partial s}{\partial z} \right) + \frac{\partial v}{\partial t} \left(\frac{\partial t}{\partial z} \right)$$

$$\left. \begin{aligned} &= v_x(2) + v_s(0) + v_t(-4) \\ &= v_x(-3) + v_s(3) + v_t(0) \\ &= v_s(0) + v_t(-4) + v_t(4) \end{aligned} \right\} =$$

$$= 6v_x + 4v_y + 3v_z = \dots = 0$$

* JACOBIAN:

if $u = u(x, y, z), v = v(x, y, z), w = w(x, y, z)$

i.e $(u, v, w) \longrightarrow (x, y, z)$

then derivative of u, v, w with respect to (x, y, z) is called jacobian.

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* it is provided by jacobian & is defined as

$$J = \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

Note: 1) if $(u,v) \longrightarrow (x,y)$ then $J = \frac{\partial(u,v)}{\partial(x,y)}$

$$= \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

2) Also Evaluate $\frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{J}$

i.e. if $(x,y) \longrightarrow (u,v) \longrightarrow J = \frac{\partial(u,v)}{\partial(x,y)}$

$$J' = \frac{\partial(x,y)}{\partial(u,v)}$$

$$\Rightarrow JJ' = 1 \quad \text{[here]}$$

Q: if $u = 3x + 5y$ & $v = 4x - 3y$ then $\frac{\partial(u,v)}{\partial(x,y)} = ?$

a.) 29

b.) -29

c.) 11

d.) -11

$(u,v) \longrightarrow (x,y)$

$$J = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 3 & 5 \\ 4 & -3 \end{vmatrix}$$

$$= -9 - 20 = -29$$

Q:) if $x = u$, $y = u \tan v$, $z = w$ then derivative of (u, v, w) w.r. to (x, y, z) will be?

a) $\cos^2 v / u$ ✓ $(x, y, z) \longrightarrow (u, v, w)$ $\frac{\partial(u, v, w)}{\partial(x, y, z)}$

b) $u \cos^2 v$

c) $u \sec^2 v$

d) 0

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ u \tan v & u \sec^2 v & 0 \\ 0 & 0 & 1 \end{vmatrix} = u \sec^2 v$$

$$J J' = 1 \Rightarrow J' = \frac{1}{J} = \frac{1}{u \sec^2 v} = \frac{\cos^2 v}{u}$$

Q:) if $u = \tan^{-1} x + \tan^{-1} y$ & $v = \frac{x+y}{1-xy}$ then $\frac{\partial(u, v)}{\partial(x, y)}$

Sol:) $u_x = \frac{1}{1+x^2}$, $u_y = \frac{1}{1+y^2}$

$$v_x = \frac{(1-xy)(1+0) - (x+y)(0+y)}{(1+xy)^2}$$

$$= \frac{1-y^2}{(1+xy)^2}$$

$$v_y = \frac{(1-xy)(0+1) - (x+y)(1+x)(0-x)}{(1+xy)^2}$$

$$= \frac{1+x^2}{(1+xy)^2}$$

Now $(u, v) \longrightarrow (x, y)$ so

$$J = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = \begin{vmatrix} \frac{1}{1+x^2} & \frac{1}{1+y^2} \\ \frac{1-y^2}{(1+xy)^2} & \frac{1+x^2}{(1+xy)^2} \end{vmatrix}$$

$$= \dots = 0$$

Note:)

Here in this question u, v are F.D

$$\therefore u = \tan^{-1} \left(\frac{x+y}{1-xy} \right) = \tan^{-1} v$$

$$v = \tan u$$

$\therefore u$ & v are connected by a relation hence dependent

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$$\text{So, } \frac{\partial h(x)}{\partial (x, y)} = \frac{\partial h(x)}{\partial (x, y)} = \frac{\partial (4xy^2)}{\partial (x, y)} = 4(x^2 + y^2) \cdot x$$

$$= 4(x^2)(x) = 4x^3 \text{ Ans.}$$

Maxima & Minima

Topic 3.

1.) Increasing Dec function & concavity of function.

2.) Max-min of curve.

3.) Max-min of surface $z = f(x, y)$

* **1st Derivative** Geometrical Significance $\rightarrow$ To know the slope of tangent.

Application: To know the Inc or Dec value

* **2nd Derivative**

Geometrical Significance $\rightarrow$ To know concavity of function

Application: To know Inflection point, point of max & point of min

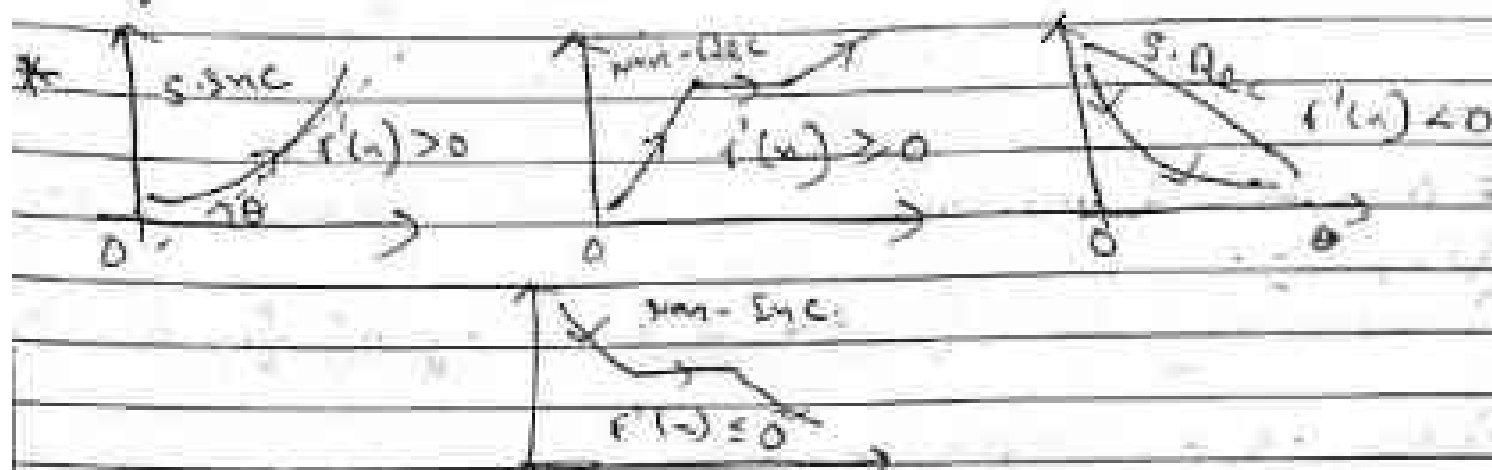
Increasing & Decreasing function:

Note that for $y = f(x)$, $\frac{dy}{dx} = f'(x) = \tan \theta$

II	I	1.) for Acute angle tangent, $f'(x) > 0$
sin +ve	all +ve	2.) " obtuse " " , $f'(x) < 0$
conc		3.) for horizontal tangent, $f'(x) = 0$
		4.) for vertical Tangent, $f'(x) = \infty$
tan +ve	inc	
at +ve	dec +ve	
III	IV	

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(in fact vertical tangent DNE otherwise $f(x)$ fails to follow V-L test)



1.) Monotonic function: Those function which are either S. Inc or S. Dec are called monotonic functions.

(eg:) if $f(x_1) < f(x_2)$ & $x_1 < x_2$; $x_1, x_2 \in D_f$ then $f(x)$ is S. Inc.

(eg:) if $f(x_1) > f(x_2)$ & $x_1 < x_2$; $x_1, x_2 \in D_f$ then $f(x)$ is Non-Inc or Dec.



2.) While solving question based on inc-dec function, 1st try to find write the domain of the function & then proceed.

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Q.9.) $f(n) = \log_a n$ is _____ where $a > 1$ $0 < n < \infty$

Ans. $f(n) = \log_a n = f'(n) = \frac{1}{n \log_e a}$
 $= \frac{\log_e n}{\log_e a}$

$\therefore f'(n) > 0 \forall n \in (0, \infty)$ So,
 $f(n)$ is S. increasing function.

Q.) $f(n) = \log_a n$ is _____ where $0 < a < 1$ $0 < n < \infty$

Ans. $f'(n) = \frac{1}{n \log_e a}$

But when a $0 < a < 1$, $\log_e a = -ve$ Don't
 $\& n > 0$ as per domain

Hence $f'(n) = \frac{1}{+ve(-ve)} = -ve < 0$

So $f(n)$ is S. Dec.

Q.) $f(n) = \frac{e^n}{1+e^n}$ is ?

- Monotonically Increase ✓
- Inc to max point & then Dec.
- Decrease.
- Can't say anything.

Dom = $(-\infty, \infty)$ i.e. all real numbers.

$$f'(n) = \frac{(1+e^n)(e^n) - e^n(e^n)}{(1+e^n)^2} = \frac{e^n}{(1+e^n)^2}$$

$\therefore f'(n) > 0 \because n \in \text{Domain}$ $f(n)$ is S. inc. Spiral

Q:) Find the interval in which $f(n) = n^n$ Increase & Dec

Sol:) $y = f(n) = n^n$, Domain: $(0, \infty)$

$$\log y = \log(n^n)$$

$$\log y = n \log n$$

$$\frac{d}{dn}(\log y) = \frac{d}{dn}(n \log n)$$

$$\frac{1}{y} \frac{dy}{dn} = n \cdot \frac{1}{n} + \log n(1)$$

$$\frac{dy}{dn} = y(1 + \log n) = n^n(1 + \log n)$$

$$f(n) = n^n \quad \boxed{f'(n) = n^n(1 + \log n)}$$

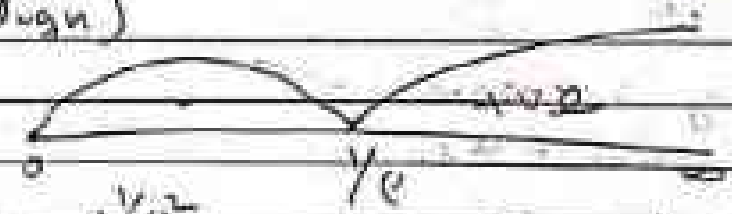
Turning points: put $f'(n) = 0$

$$n^n(1 + \log n) = 0 \quad \because n^n \neq 0 \text{ so } 1 + \log n = 0$$

$$\log n = -1$$

$$\boxed{n = \frac{1}{e}}$$

$$f'(n) = n^n(1 + \log n)$$



$$f\left(\frac{1}{e}\right) = \left(\frac{1}{e}\right)^{\frac{1}{e}} \left[1 + \log\left(\frac{1}{e}\right)\right] = \left(\frac{1}{e}\right)^{\frac{1}{e}} (1 - 1) = -ve$$

$$f'(1) = e^e(1 + \log e) = 2e^e = +ve$$

i.e.

$$f(n) < 0 \text{ for } n \in \left(0, \frac{1}{e}\right) \quad f'(n) > 0 \text{ for } n \in \left(\frac{1}{e}, \infty\right)$$

$$f(n) \text{ is } \downarrow \text{ Dec in } \left(0, \frac{1}{e}\right) \quad f(n) \text{ is } \uparrow \text{ Inc in } \left(\frac{1}{e}, \infty\right)$$

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Q.) Find the interval of inc & Dec for $y = \frac{\log n}{n}$.

a.) Inc $(0, e]$ ✓ $f(n) = \frac{\log n}{n}$, Dom $(0, \infty)$

b.) Dec $(0, e]$

c.) Inc $[e, \infty)$

d.) Dec $[e, \infty)$ ✓ $f'(n) = \frac{n \left(\frac{1}{n} \right) - \log n (1)}{(n)^2} = \frac{1 - \log n}{n^2}$

Turning point / critical points:

$$f'(n) = 0 = \frac{1 - \log n}{n^2} = 0 = 1 - \log_e n = 0$$

$$\log_e n = 1 \Rightarrow \boxed{n = e}$$

$$f'(n) = \frac{1 - \log n}{n^2} = f'\left(\frac{1}{e}\right) = \frac{1 - \log\left(\frac{1}{e}\right)}{\left(\frac{1}{e}\right)^2} = \frac{1 + 1}{\frac{1}{e^2}} = +ve$$

$$= f'(e^2) = \frac{1 - \log(e^2)}{(e^2)^2} = \frac{1 - 2}{e^4} = -ve$$



$$f'(n) = \frac{1 - \log n}{n^2} = f'\left(\frac{1}{e}\right) = +ve$$

$$= f'(e^2) = -ve$$

$$f(n) \text{ is } \swarrow \text{Inc in } (0, e) \because f'(n) > 0 \forall n \in (0, e)$$

$$f(n) \text{ is } \searrow \text{Dec in } (e, \infty) \because f'(n) < 0 \forall n \in (e, \infty)$$

Date

⇒ Information:

⇒ Concave upward curve:

if curve lies above the tangent always then curve is called concave upward curve.

$$f''(x) > 0$$

e.g. $f(x) = e^x$ is concave upward curve.

⇒ Concave downward curve:

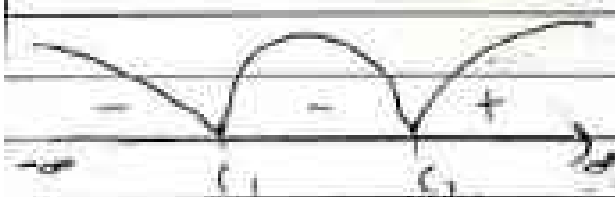
if curve lies below the tangent always then curve is called concave downward curve.

$$f''(x) < 0$$

e.g. $f(x) = \log x$ is concave downward curve.

1.) To find the interval in which $f(x)$ is inc. or Dec.

Put $f'(x) = 0$ & try to find T. Point (say $x = c_1, c_2$) then check sign of $f'(x)$ as follows:

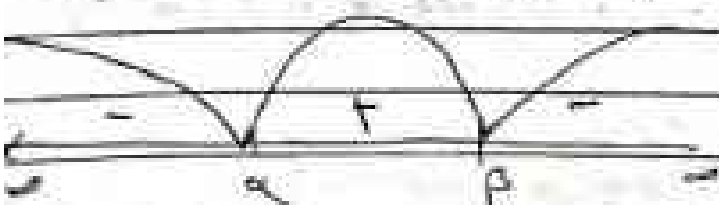


$f(x)$ is inc in $(-\infty, c_1) \cup (c_2, \infty)$

$f(x)$ is Dec in (c_1, c_2) .

2.) To find the interval where $f(x)$ is concave up or concave downward →

put $f''(x) = 0$ & try to find x (say $x = \alpha, \beta$) then check the sign of $f''(x)$ as follows.



$f(x)$ is Concave Down = $(-\infty, \alpha) \cup (\beta, \infty)$

$f(x)$ is Concave Up = (α, β)

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Q:) The function $f(x) = x^4$ is?

Domain = $(-\infty, \infty)$

a.) Strictly inc.

b.) " Dec.

c.) Concave Down.

d.) " Upward. ✓

$f(x) = x^4$

$f'(x) = 4x^3$

$f''(x) = 12x^2$

Ans $f'(x) = 0 \Rightarrow x = 0$



$f'(x) = 4x^3$

Ans $f''(x) = 0 \Rightarrow x = 0$



$f''(x) = 12x^2$

So $f(x)$ is always concave upward.Q:) Find the interval in which $f(x) = (x^2 - 4)^2$ is Concave upward & concave Downward?i) Also find the interval in which $f(x)$ inc or dec.

ii) Hence find point of inflexion (Saddle point).

Sol:) $f(x) = (x^2 - 4)^2$, Dom = $(-\infty, \infty)$

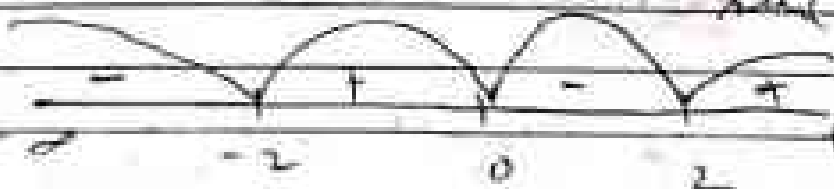
$$f'(x) = 2(x^2 - 4)(2x)$$

$$= 4x^3 - 16x$$

$f''(x) = 12x^2 - 16$

T. Points: $f'(x) = 0 \Rightarrow 4x^3 - 16x = 0$

$4x^2(x - 4) = 0 \Rightarrow x = -2, 0, 2$



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$f(x)$ is Dec in $(-\infty, -2) \cup (0, 2)$

$f(x)$ is Inc in $(-2, 0) \cup (2, \infty)$

$\Rightarrow$ For concavity: put $f''(x) = 0$

$$12x^2 - 16 = 0$$

$$x = \frac{-2}{\sqrt{3}}, + \frac{2}{\sqrt{3}}$$



$$f''(x) = 12x^2 - 16$$

$$f''(-2) = +ve$$

$$f''(0) = -ve$$

$$f''(2) = +ve$$

$f(x)$ is concave upward in

the intervals $(-\infty, -\frac{2}{\sqrt{3}}) \cup (\frac{2}{\sqrt{3}}, \infty)$

$f(x)$ is concave downward in

$(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}})$

(iii) point of inflexion is (Saddle point) = $-\frac{2}{\sqrt{3}}$ & $\frac{2}{\sqrt{3}}$

Saddle point / Point of Inflexion: $\rightarrow$

"Those points where curve change its concavity are called points of inflexion".

Note:

1) These point are neither maxima & neither minima are called points of inflexion $\Rightarrow$ False.

2) if $(x=a)$ is the point of inflexion $\Rightarrow$ we will get neither maxima nor minima at a .

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Shortcut:

Put $f''(n) = 0$ & try to get n (only if it is $n = 0$)if $f''(n) \neq 0$ then $n = 0$ is a saddle point.if $f'''(n) = 0$ then can't say anything.Q:) Find the inflexion points of $y = f(n) = (n^2 - 4)^3$!

$$\text{Sol:)} \quad f'(n) = 2(n^2 - 4)(2n) \\ = 4n^3 - 16n$$

$$f''(n) = 12n^2 - 16$$

$$f'''(n) = 24n$$

put

$$f''(n) = 0 \Rightarrow n = \pm \frac{2}{\sqrt{3}}$$

$$f'''(-\frac{2}{\sqrt{3}}) \neq 0$$

$$f'''(\frac{2}{\sqrt{3}}) \neq 0$$

i.e. $-\frac{2}{\sqrt{3}}$, $+\frac{2}{\sqrt{3}}$ are saddle points. Both.Q:) The no. of inflexion points of $f(n) = n + 12n^4$ is:

$$\text{Sol:)} \quad f'(n) = 1 + 48n^3$$

$$f''(n) = 144n^2$$

$$f'''(n) = 288n$$

$$f'(n) = 0 \Rightarrow 144n^2 = 0 \Rightarrow n = 0$$

$$f'''(0) = 288 \times 0 = 0$$

$$\text{a.) } 0 \quad \checkmark$$

$$\text{b.) } 1$$

$$\text{c.) } 2$$

$$\text{d.) } 3$$

So we have no saddle point for $f(n) = n + 12n^4$.

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Lecture no. 13

Dom $(-\infty, \infty)$

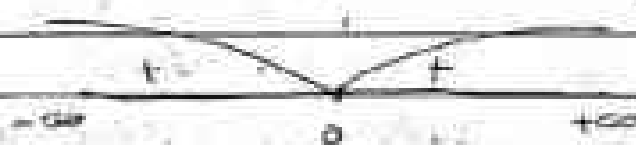
Q:) for the func $y = x^3$, $x=0$ is _____ Also find the turning point of $f(x) = x^3$

Sol:) $f(x) = x^3$, $f'(x) = 3x^2$, $f''(x) = 6x$

for ∇ points $f'(x) = 0 \Rightarrow 3x^2 = 0 \Rightarrow \boxed{x=0}$ or

for ∇ Saddle point

$f''(x) = 0 \Rightarrow 6x = 0 \Rightarrow \boxed{x=0}$ or



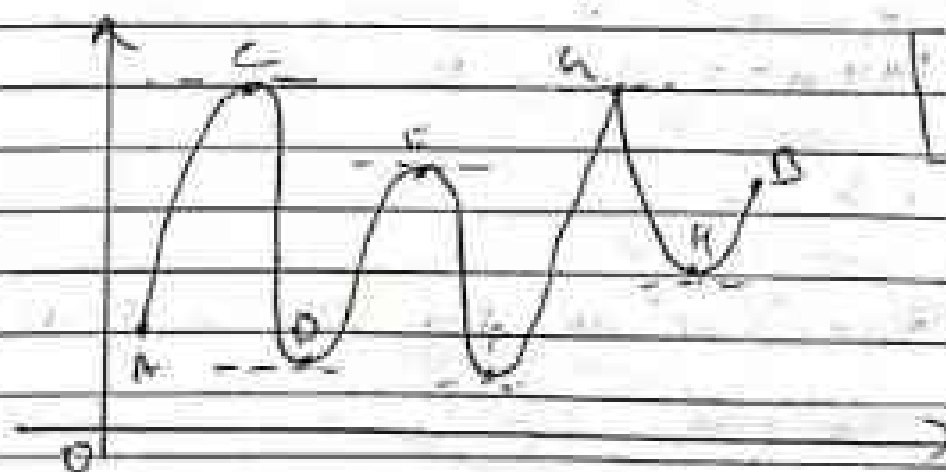
i.e. $f'(x) > 0$ always so $f(x)$ is S. inc

(i.e. ∇ max.)



Concavity change at $x=0$ point of inflection

Maxima - minima of function of single variable



i.e. of curve
 $y = f(x)$

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Point = x Value = y Local max points $\rightarrow C, E, G, B$ Absolute maxima $\rightarrow f(C)$ Local min points $\rightarrow A, D, F, H$ Local min value $\rightarrow f(A), f(D), f(F), f(H)$ Local max value $\rightarrow f(C), f(E), f(G), f(B)$ Absolute minimum/Global $\rightarrow f(F)$

Necessary Condition for maxima - minima -

$$1) \left[f'(x) = 0, \text{ or } f'(x) = \text{N.D.} \right]$$

i.e. either tangent is Horizontal or not defined.

2) Stationary point:

Those point where tangent is Horizontal are called stationary points

$$f'(x) = 0$$

e.g., C, D, E, F, H.

3) Turning points / critical points:

Those point where tangent is either Horizontal or not defined are called $\therefore$ points

e.g., C, D, E, F, H, G.

$$f'(x) = 0 \text{ or N.D.}$$

4) Extreme points / optimal points:

Those point where we get either maxima or minima are called Extreme points.

e.g., A, C, D, E, F, G, H, B.

Special

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→ Corner point are also Extrema point (T/F)?
 But take care; corner points are not turning points.

5.) Point of inflexion / Saddle point:-

Those point where curve change it's concavity are called Saddle point.

6.) Maxima / minima occurs alternately (T/F)?

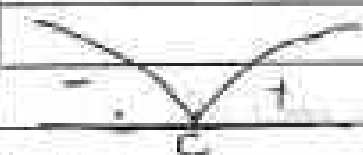
7.) Sufficient cond for maxima & minima:-
 (i) should satisfy either

1st derivative test or
 2nd derivative test.

8.) 1st derivative test:

let $x=c$ is the turning point.

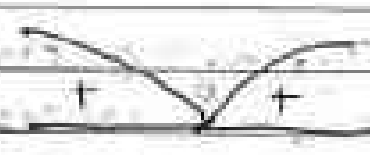
(it is obtained by solving eq. ①) if $f'(c) = 0$ or ∞ then we will check the sign of $f'(x)$ as follows.



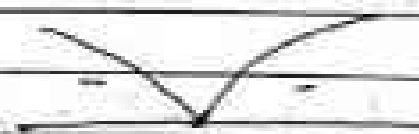
(minima)



(maxima)



(minima)

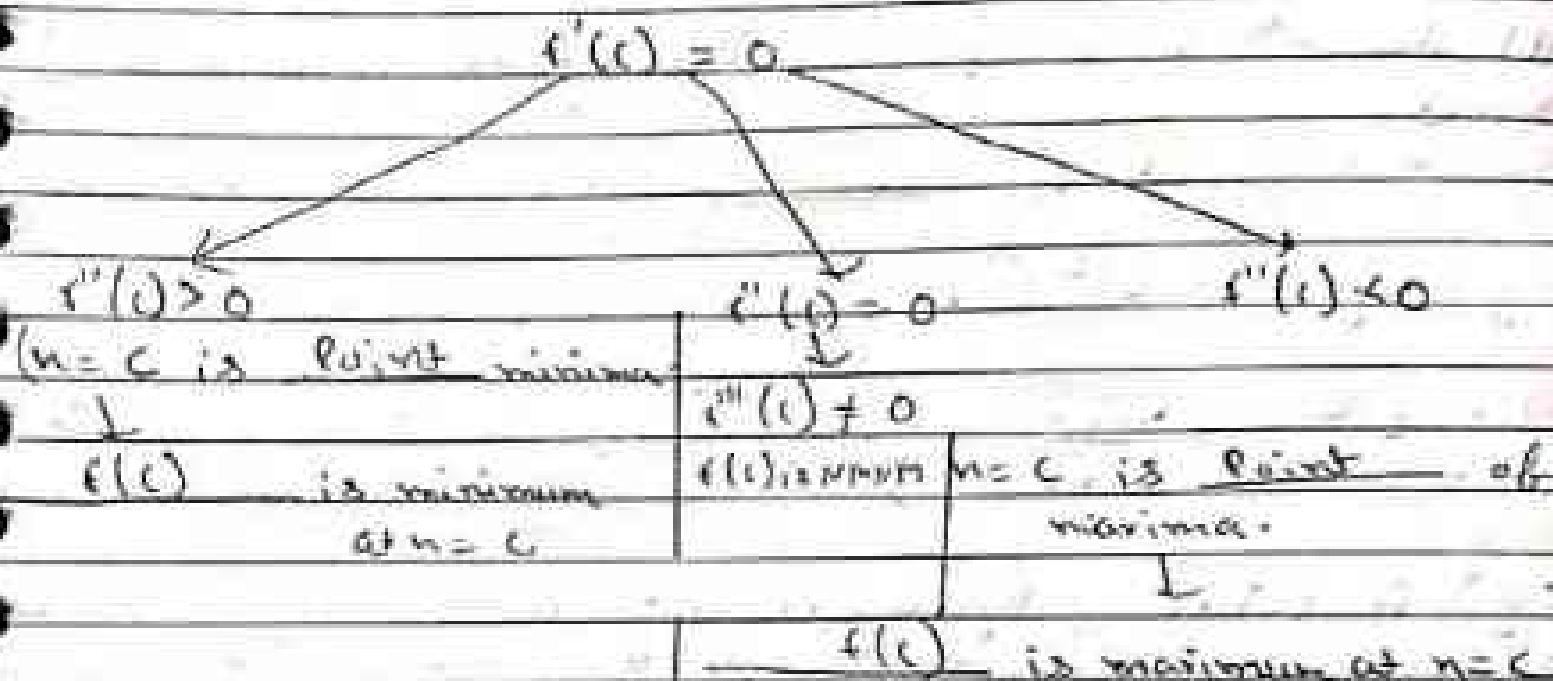


(minima)

wavy curve method / 1st derivative test.

9.) 2nd Derivative test?

let $x=c$ is critical point then



10.) if $f''(c) = 0$, & $f'''(c) \neq 0$

n is even then

$x=c$ is point of inflection

10.) if we want to check the nature of corner point then we have no shortcut method.

it's nature can be calculated by using following theorem only

"Maxima & minima occurs alternately"

for e.g:

(a) $f(x)$ is defined in $[a, b]$ & it's 4 points are $x = c_1, c_2, c_3$



∴ Here all three cases we have check the sign of $r'(u)$.

11) An n^{th} degree polynomial has at most $(n-1)$ times.

So, it has at most $(n-1)$ extremes (max & min).

e.g.) $y = (u^2 - u)^2$ (it is polynomial of degree 4).

∴ it has three extremes (one local maximum & two minima).

e.g.) A cubic polynomial with Real coefficients has at most two extremes & three zeros (crossing Axis).

Question Based on type 1: (Common sense Based Question)

Q:) Find maxima & minima of

1) $y = x^2 + 5$



min = $y = 5$
max = NONE

2) $y = x^2 + 5 \sin 3x + 4$

$$-1 \leq \sin 3x \leq 1$$

$$3 \leq \sin 3x + 4 \leq 5$$

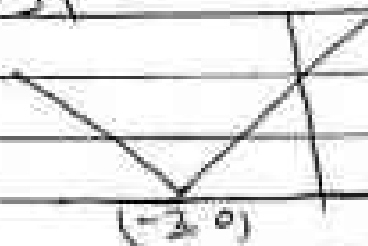
$$3 \leq y \leq 5$$

min value = 3

max value = 5

Date

3.) $y = |x+3|$



Point of minima $x = -3$

min value $= y = 0$

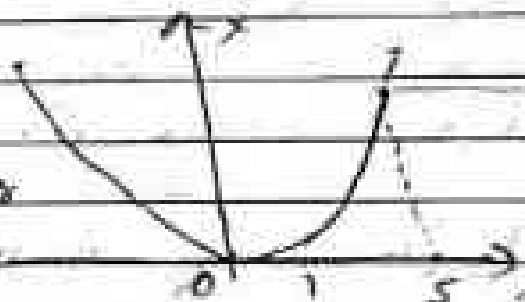
max value $= \text{O.P.E.}$

Q:) Max value of $f(x) = x^2$ is $[1, 5]$ will be?

Sol:) Q-1 $y = x^2$

min value $= (1)^2 = 1 = y$

max value $= (5)^2 = 25 = y$



Q-2

$f(x) = x^2$, $[1, 5]$

$f'(x) = 2x$

put $f'(x) = 0$

$2x = 0$, $x = 0$



But $0 \notin$ given Domain of $f(x)$

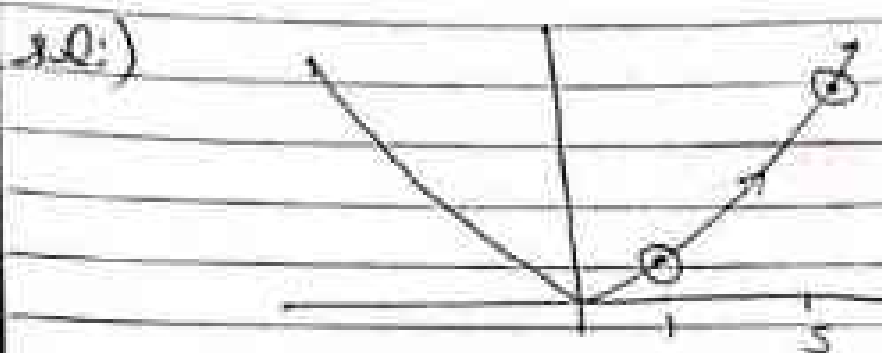
So $f(x)$ has no turning point

Maxima & min will occur only at corner point

$f(1) = 1$, min $f(5) = 25$ $\therefore$ maximum value

Date: _____

Q:) Find min & max value of $f(x) = x^2$ in $(1, 5)$?



∵ Domain is given in the form of open interval.
i.e. we have no corner point.
& in previous example, we have already shown that
 $f(x)$ does not have any turning point.
∴, min & max value on $\emptyset$

Type II (Basically Based on 1st Derivative test)

Q:) $y = (x^2 - 4)^2$, $(-\infty, \infty)$, i)

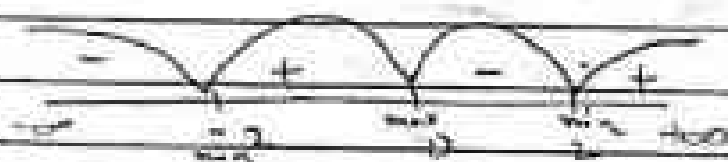
$$f(x) = (x^2 - 4)^2$$

$$f'(x) = 4x^2(x+2)(x-2)$$

∴ points are $x = -2, 0, 2$

No. of extrema are
three

1) Local max points are
 $x = 0$



Q:) $y = (x^2 - 4)^2$, $[-3, 4]$ 1) No. of extrema are
five

$$f'(x) = 4x^2(x+2)(x-2)$$

∴ points are $x = -2, 0, 2$

2) Local max points are
 $x = -3, 0, 4$



Spinal

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3.) Local max value is
 $f(0) = 16$

3.) Local max value is
 $f(-2) = 25, f(0) = 16, f(2) = 144$

4.) Absolute maxima =
 $= 0 \text{ NE}$

4.) Absolute maxima =
 $f(2) = 144$

5.) Local min points are
 $x = -1, 1$

5.) Local min points are
 $x = -2, 2$

6.) Local min value are
 $f(-1) = 0, f(1) = 0$

6.) Local min value
 $f(-2) = 0, f(2) = 0$

7.) Absolute min value
 $= 0$

7.) Absolute min value = 0

8.) Point of inflection.
 $x = -2/\sqrt{3}, 2/\sqrt{3}$

8.) Point of inflection are.
 $-2/\sqrt{3}, 2/\sqrt{3}$

Q.) $f(x) = 4x^2 + 1$ then find the interval in which
 $f(x)$ inc & dec.

Also find minimum & max value of $f(x)$ if exist!

Sol.) $f'(x) = \frac{x(f(x) - (4x^2 + 1))}{x^2} = \frac{x(4x^2 + 1 - 4x^2 - 1)}{x^2} = \frac{-1}{x^2}$

$$f'(x) = \frac{4x^2 - 1}{x^2} = \frac{(2x - 1)(2x + 1)}{x^2}$$

for T. points, $f'(x) = 0$, or undefined.

$$= x = -\frac{1}{2}, \frac{1}{2} \text{ or } x = 0 \quad \text{max value} = -4$$

$$\text{min value} = 4$$

Spiral



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Lecture no. 14

Note: $(+1)^{1/3} = 1$, $(-1)^{1/3} = -1 = 1, \omega, \omega^2$ & $\omega^2 = -1, -\omega$

Q: The minimum value of $y = (n-1)^{2/3}$ occur at $n=1$ & min value is 0

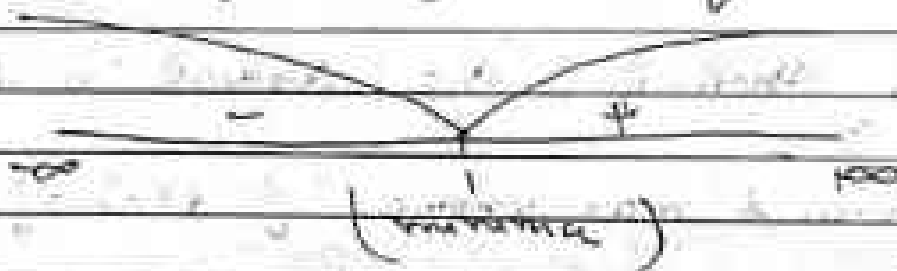
Sol: $f(n) = (n-1)^{2/3}$, Dom = $\mathbb{R} + \{ \}$

$$f'(n) = \frac{2}{3} (n-1)^{\frac{2}{3}-1} = \boxed{\frac{2}{3(n-1)^{1/3}}}$$

for turning points

$$f'(n) = 0 \text{ or } f'(n) = \text{undefined.}$$

not possible to find any value of n At $n=1$ it will be a turning point



$$f(1^-) = f(0) = \frac{2}{3}(0-1)^{2/3} = -\frac{2}{3} \quad (-ve)$$

$$f(1^+) = f(1) = \frac{2}{3}(1-1)^{2/3} = \frac{2}{3} \quad (+ve)$$

then

$$\text{min value} = f(1) = (1-1)^{2/3} = 0$$

Date

Q.) $f(x) = x^3 - 9x^2 + 24x + 5$, defined in $[1, 6]$ then minimum and maximum values are respectively

a.) 21, 41 ✓

b.) 4, 6

c.) 21, 25

d.) ONE, ONE

$$f(x) = x^3 - 9x^2 + 24x + 5$$

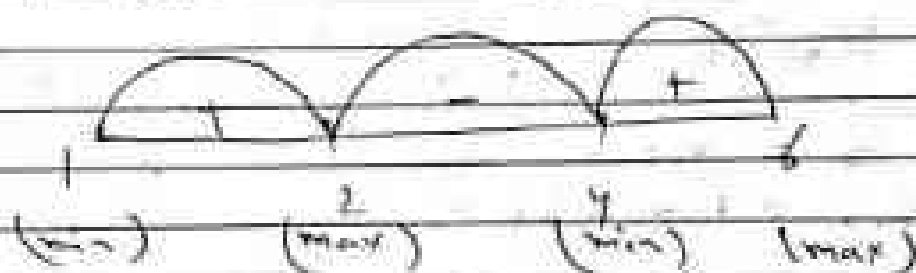
$$f'(x) = 3x^2 - 18x + 24$$

$$= 3(x^2 - 6x + 8)$$

$$f'(x) = 3(x-2)(x-4)$$

T.P are $= x=2, 4$

{T.P = Turning point}



$$f'(1.5) = +ve$$

$$f'(3) = -ve$$

$$f'(5) = +ve$$

Local minimum points are $= x=1$ ✓
 $= x=4$ ✓

Local maximum points are $= x=2$ ✓
 $= x=6$ ✓

$$\text{Local minimum values} = f(1) = \dots = 21 \text{ ✓}$$

$$f(4) = \dots = 21 \text{ ✓}$$

$$\text{Local maximum values} = f(2) = \dots = 25 \text{ ✓}$$

$$f(6) = \dots = 41 \text{ ✓}$$

Special

8.) $f(x) = x^3 - 9x^2 + 24x + 5$ then min & max value

is:

$$f'(x) = 3x^2 - 18x + 24$$

a.) 21, 41

b.) 4, 6

c.) 21, 25

d.) ONE, ONE, ✓

$$= 3(x^2 - 6x + 8)$$

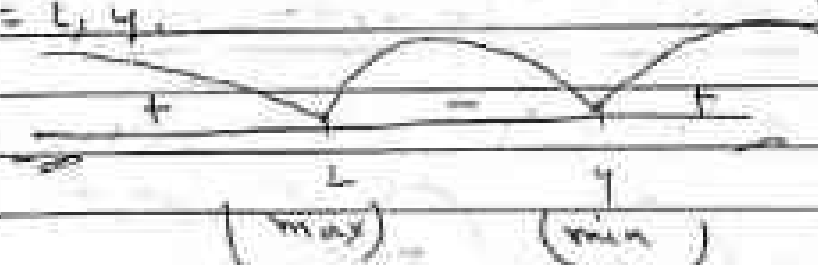
$$= 3(x-2)(x-4)$$

$$\text{Dom}(-\infty, \infty)$$

$$f'(x) = 0$$

$$f'(x) = x = 2, 4$$

$$I, p = x = 2, 4$$



Local minimum point = $x = 4$

" " value = $f(4) = 21$

Local max point = $x = 2$

" " value = $f(2) = 25$

max value = ONE.

min " = ONE.

9.) The maximum slope of $y = x^3 - 9x^2 + 24x + 5$ will be:

$$\text{let } f(x) = x^3 - 9x^2 + 24x + 5, \text{ Dom}(-\infty, \infty)$$

a.) 2

let

b.) 3

c.) 4

d.) ONE ✓

$$f'(x) = g(x) = 3x^2 - 18x + 24 \text{ (slope of)}$$

$$f'(x) = g(x) = 3(x^2 - 6x + 8)$$

Date

we will have to find max value of

$$f'(n)$$

DNE

i.e.

we will try to calculate max value of $g(n)$ & not of $f(n)$.

for turning points of $g(n)$, put $g'(n) = 0$

$$g'(n) = 3(2n - 6) = 0 \Rightarrow n = 3$$

$$g(n) = 3(n-2)(n-4)$$

L

$$g'(n) = 6(n-3)$$

$$g'(3^-) =$$

$$g'(3^+) =$$



i.e. $g(n)$ will be minimum at $n = 3$

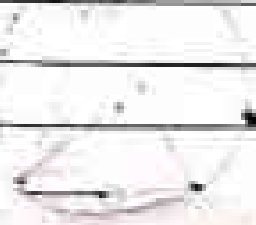
min slope - of $f(n) = -3$

min value of $g(n) = g(3) = -3$

or

max value of $g(n) = \text{DNE}$.

Q.) for the funct $e^y = n^y$, max value occurs at
 $n = e$ & max value will be
 $1/e$



Date

sol:) Taking log both side:-

$$\text{Dom} = (0, \infty)$$

$$\log(e^x) = \log(n^{\frac{1}{n}})$$

$$y(\log e^x) = \frac{1}{n} \log n$$

$$f(n) = \boxed{y = \frac{\log n}{n}}, \quad A-903$$

$$f'(n) = \frac{n \cdot \left(\frac{1}{n}\right) - (\log n)(1)}{n^2}$$

$$\boxed{f'(n) = \frac{1 - \log n}{n^2}}$$

$$\text{T.p} = f'(n) = 0 \Rightarrow \boxed{n = e}$$

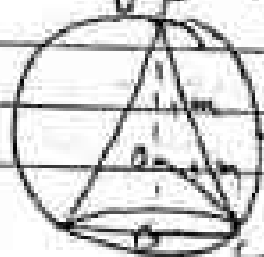
max value occurs at $n = e$

f

max value is $y = f(e) = \frac{\log e^e}{e} = \log e \cdot \frac{1}{e}$

9:) A Right circular cone of max vol is to be inscribed in a sphere of radius 1 mtr then find the height of such cone

sol:)



$$OA = OC = 1 \text{ mtr}$$

Date

let $AO = h$, $OC = x$

in $\triangle OAC$: $AO^2 + OC^2 = AC^2$

$$= (h-1)^2 + x^2 = 1^2 \quad \text{--- (1)}$$

vol = $\frac{1}{3} \pi x^2 h$ = { $v = f(x, h)$ } = surface.

$v = \frac{1}{3} \pi (1 - (h-1)^2) \cdot h$ { $\rightarrow v = f(h)$ curve }

$$v = \frac{\pi}{3} [1 - (h^2 + 1 - 2h)] h =$$

$$= \frac{\pi}{3} [-h^2 + 2h^2]$$

for max volume $\frac{dv}{dh} = 0$

~~$\frac{\pi}{3} h^2$~~ $\frac{\pi}{3} (-3h^2 + 4h) = 0$

$$h(-3h + 4) = 0$$

$$\boxed{h=0} \quad \boxed{h=\frac{4}{3}}$$

7.) In a right angle $\triangle$, if sum of the hypotenuse & one side is kept constant, in order to find max area then find the angle b/w them;

a.) 30°

b.) 60° ✓

c.) 45°

d.) 50°

Spiral

Sol:)



$$\text{At } \theta, n + \sqrt{n^2 + y^2} = K$$

$$n^2 + y^2 = (K - n)^2$$

$$n^2 + y^2 = K^2 + n^2 - 2Kn$$

$$\boxed{y^2 = K^2 - 2Kn} \quad \text{--- (1)}$$

$$A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times n \times y = \frac{1}{2} ny \quad \text{--- (2)}$$

$$\text{Let } u = A^2 = \frac{1}{4} n^2 y^2$$

$$= \frac{n^2}{4} (K^2 - 2Kn)$$

$$u = \frac{n^2}{4} (K^2 - 2Kn) = \frac{K^2}{4} n^2 - \frac{K}{2} n^3 \quad \text{--- (3)}$$

where

$$A = \sqrt{u} \Rightarrow A_{\max} = \sqrt{u_{\max}}$$

For

$$\max u, \frac{du}{dn} = 0 \Rightarrow \frac{K^2}{4} (2n) - \frac{3K}{2} n^2 = 0$$

or

$$K^2 n - 3K n^2 = 0$$

$$= \frac{K}{2} n [K - 3n] = 0 \quad n \neq 0 \quad \text{so } n = \frac{K}{3}$$

$$\text{2, (1)} \quad y^2 = K^2 - 2Kn = K^2 - 2K \left(\frac{K}{3} \right) = \frac{K^2}{3}$$

$$\boxed{y = \frac{K}{\sqrt{3}}} \Rightarrow \tan \theta = \frac{y}{n} = \frac{K/\sqrt{3}}{K/3} = \sqrt{3} = \frac{3}{1} \quad \text{Spiral 1st pt}$$

Date

Q.) A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in metres. The maximum possible height of the arch is

a.) 8 metres

b.) 10 "

c.) 12 "

d.) 14 "

$$y = 2x - 0.1x^2$$

$$\frac{dy}{dx} = 2 - 0.2x$$

$$\frac{d^2y}{dx^2} = -0.2$$

For critical point ; $\frac{dy}{dx} = 0$

$$2 - 0.2x = 0$$

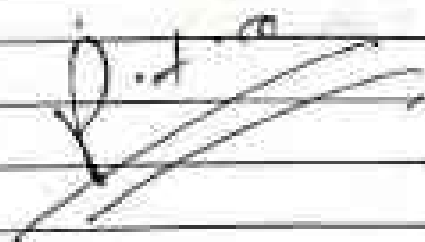
$$x = \frac{2}{0.2} = 10$$

$$\therefore \text{At } \left(\frac{d^2y}{dx^2} \right)_{x=10} = -0.2 = -ve$$

So At $x=10$ we will get max value of y

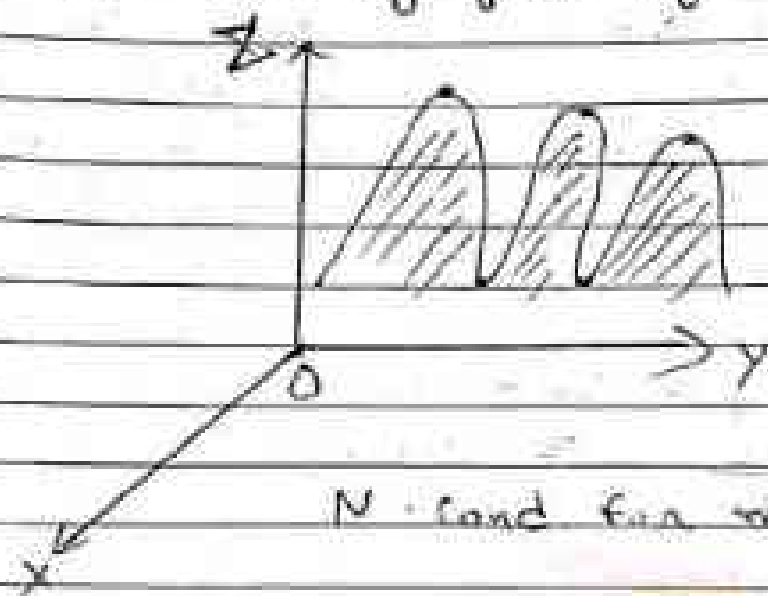
$$y_{\max} = (2x - 0.1x^2)_{x=10}$$

$$= 20 - 10 = 10 \text{ metres}$$



Date

Max-min of func of two variable (i.e. $z = f(x, y)$)



point = (x, y)

value = (z)

N. Cond for max-min: $\rightarrow$

$$\frac{\partial z}{\partial x} = 0, \quad \frac{\partial z}{\partial y} = 0$$

$$\text{①} \qquad \text{②}$$

Stationary point/critical point $\rightarrow$ points obtained by solving ① & ②
 for e.g. after solving $x = a$ & $y = b$ then $P(a, b)$ is critical point.

Sufficient Cond for max-min $\rightarrow$ function must satisfy Lagrange's condition.

Note:

let $P(a, b)$ is critical point then we can use following symbols.

$$\left(\frac{\partial^2 z}{\partial x^2} \right)_{P(a,b)} = x, \quad \left(\frac{\partial^2 z}{\partial y^2} \right)_{P(a,b)} = y, \quad \left(\frac{\partial^2 z}{\partial x \partial y} \right)_{P(a,b)} = z$$

Date

⇒ Lagrange's condition

1.) if $\Delta_1 - \Delta^2 > 0$ & $\Delta > 0$ then $P(a, b)$ is point of minima.

2.) if $\Delta_1 - \Delta^2 > 0$ & $\Delta < 0$ then $P(a, b)$ is point of maxima.

3.) if $\Delta_1 - \Delta^2 < 0$ then " " " " inflexion.

4.) if $\Delta_1 - \Delta^2 = 0$ then case fails i.e.

L-condition are unable to give an idea about maxima or minima & we need for their investigation.

Q:) The number of stationary point of $z = 2x^2 + 2xy$ is (are) two.

ii) Also find optimal if exist?

$$\text{Sol:)} \quad \frac{\partial z}{\partial x} = 4x + 2y$$

$$\frac{\partial^2 z}{\partial x^2} = 4$$

$$\frac{\partial z}{\partial y} = 2x - 3y^2$$

$$\frac{\partial^2 z}{\partial y^2} = -6y$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} (2x - 3y^2) = 2$$

Date

$$\text{put } \frac{\partial z}{\partial x} = 0 \quad \frac{\partial z}{\partial y} = 0$$

$$4x + y = 0, \quad \& \quad 2x - 3y^2 = 0$$

$$y = -2x \quad \& \quad 2x - 3(-2x)^2 = 0$$

$$\text{At } x = 0 \quad \bullet \quad 2x - 12x^2 = 0$$

$$\boxed{y = 0} \quad 2x(1 - 6x) = 0$$

$$\text{At } x = \frac{1}{6} \quad x = 0, \frac{1}{6}$$

$$\boxed{y = -\frac{1}{3}} \quad \frac{1}{6}$$

i.e.

$$O(0,0) \quad \& \quad P\left(\frac{1}{6}, -\frac{1}{3}\right)$$

are two stationary point

(ii)

$$f(x,y) = 2x^2 + 2xy - y^3 \quad \text{--- (1)}$$

To find optimal value at $O(0,0)$

$$\delta = (2xx)_0 = (4)_{(0,0)} = 4$$

$$\xi = (2xy)_0 = (2) = 2$$

$$\eta = (2yy)_0 = (-6y)_{(0,0)} = 0$$

$$\therefore H - 3\Delta^2 = (4)(0) - (2)^2 = -4$$

$$\delta + 3^2 < 0 \quad \text{so } O(0,0) \text{ is P.O.I.}$$

NO optimal value exist

pair of inflection

Date _____

To check optimal value at $P\left(\frac{1}{6}, -\frac{1}{3}\right)$

$$z_x(2mn)p = (4)p = 4$$

$$z_y = (2ny)p = (2)p = 2$$

$$z_{yy} = (2y)p = (-6)\left(\frac{1}{6}, -\frac{1}{3}\right) = 2$$

$$z_x^2 - z_y^2 = (4)(2) - (2)^2 = 4 > 0$$

$$\& z = 4 > 0$$

i.e. $P\left(\frac{1}{6}, -\frac{1}{3}\right)$ is point of maxima

$$\text{Hence } (z)_p = (f(x, y))_p = (2x^2 + 2xy - y^3)\left(\frac{1}{6}, -\frac{1}{3}\right)$$

$$= -\frac{1}{54} \quad \square$$

Q: $z = f(x, y)$ s.t. $f_x(a, b) = 0$, $f_y(a, b) = 0$

and also we have

$(f_{xy}(a, b) - f_{yx}(a, b)) f_{yy}(a, b) < 0$ & $f_{xx}(a, b) < 0$

then $P(a, b)$ is point of

a) minima

$$S^2: z_x^2 < 0 \text{ & } z < 0$$

b) maxima

c) inflexion

$$z_x^2 - z_y^2 > 0 \text{ & } z < 0$$

d) Data insufficient

$P(a, b)$ is point of
maxima.

Lecture no. 15

Date

Q: $f(x, y) = (x+y-1)^2 + (x+y)^2$ & $S = \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}\}$
 where S contain set of points which minimizes $f(x, y)$ then:

- a) S contain exactly one points.
- b) S is empty
- c) S contain infinitely many points. ✓
- d) S contain finite number of points

Sol: $f_x = 2(x+y-1) + 2(x+y)$
 $f_{xx} = 2 + 2 = 4 = \text{+}$
 $f_y = 2(x+y-1) + 2(x+y)$
 $f_{yy} = 2 + 2 = 4 = \text{+}$
 $f_{xy} = \frac{\partial}{\partial x}(f_y) = \frac{\partial}{\partial x}(2+2) = 2+2 = 4 = \text{+}$

for T: points $f_x = 0$ & $f_y = 0$

$$(x+y-1) + (x+y) = 0 \Rightarrow 4(x+y) = 0 \Rightarrow x+y = 0 \quad Y_1$$

$$(x+y-1) + (x+y) = 0 \Rightarrow 4(x+y) = 0 \Rightarrow x+y = 0 \quad Y_2$$

i.e. we have only one eq. in terms of $x+y$,
 which $\boxed{x+y = 0}$. It is kind of underdetermined system.

$$D^2 f = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix} \Rightarrow \det(D^2 f) = 0$$

Case 4: for Lagrange condition

i.e. not possible to use Lagrange's condition.

→ Now we have as sol

for e.g. $f(x, y) = (1, 1), (1, 0), (0, 1), \left(\frac{1}{2}, \frac{1}{2}\right)$
(S_{optimal})

Date

i.e. Set of critical points is given as
 $S = \{(x, y) : x, y \in \mathbb{R} \text{ s.t. } x+y = \frac{1}{2}\}$

Set consist ∞ points

$\Rightarrow$ Also find the minimum value of f .

$$f(x, y) = (x+y-1)^2 + (x+y)^2$$

$$f(x, y) \geq 0 \text{ s.t. } f(x, y) = \left(\frac{1}{2} - 1\right)^2 + \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \quad \text{--- (3)}$$

$$f(x, y) \geq \frac{1}{2} \quad \text{Ans}$$

Q.) Given a function $f(x, y) = 4x^2 + 6y^2 - 8x - 9y + 8$
 The optimum value of $f(x, y)$ is.

- a.) a minimum equal to $10/3$ ✓
- b.) a maximum equal to $10/3$
- c.) a minimum equal to $8/3$
- d.) a maximum equal to $8/3$

$$f_x = 8x - 8$$

$$f_{xx} = 8$$

$$f_y = 12y - 9$$

$$f_{yy} = 12$$

$$f(x, y) = \frac{\partial}{\partial x}(f_x) = 0$$

$f_x = 0$ put $f_x = 0$, $f_y = 0$

$$f_x = 8x - 8 = 0$$

$$f_x = 8$$

$$\boxed{x = 1}$$

$$f_y = 12y - 9 = 0$$

$$y = \frac{9}{12} = \frac{3}{4}$$

$$\boxed{y = \frac{3}{4}}$$

$\therefore (1, \frac{3}{4})$ is C.P.

Spiral

Date

$$x = f(x, y) \text{ at } (1, 8) = 8 \quad x > 0$$

$$y = f(x, y) \text{ at } (0) = 0 \quad y$$

$$z = f(x, y) \text{ at } (1, 8) = 12$$

$$x^2 - y^2 = (8)(12) - (0)^2 \\ = 96 > 0$$

$\therefore$

$x^2 - y^2 > 0$ & $x > 0$ so $P(1, 8)$ is point of minima

By (1) min value

$$= f(x, y) \text{ at } (1, 8)$$

$$= (4x^2 + 6y^2 - 8x - 4y + 8) \text{ at } (1, 8)$$

$$= \dots = 10/3 \quad (2)$$

Bolzano Theorem

Roots / Solutions / Zero crossing / ZEROS $\rightarrow$
 $\approx$ (Real roots) $\approx$ (Complex roots)

Those points where graph of $f(x)$ cuts x-axis called
 Roots of $f(x) = 0$



$$\because f(\alpha) = f(\beta) = f(\gamma) = 0$$

$\Rightarrow \alpha, \beta, \gamma$ are the roots of $f(x) = 0$

* complex roots occur in pair only when coefficients are real.
 Spiral

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e.g) $x^2 - (i+1)x + i = 0$

$$x^2 - ix - x + i = 0$$

$$x(x-i) - 1(x-i) = 0$$

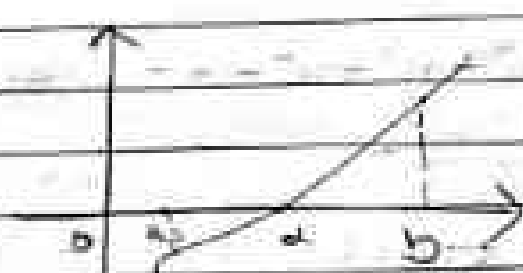
$$(x-i)(x-1) = 0$$

$$\boxed{x = i, 1}$$

e.g.) Consider a 7th degree polynomial with Real Coeff then various Ans.

7th degree poly $\begin{cases} \rightarrow \text{Total roots} = 7 \text{ (whether Real or complex)} \\ \rightarrow \text{min no. of Real Roots} = 1 \\ \rightarrow \text{max no. of comp. roots} = 6 \text{ (}\because \text{comp roots will occur in pair root).} \end{cases}$

Bolzano Theorem:



$$f(a) = -ve$$

$$f(b) = +ve$$

$$\therefore f(a) \cdot f(b) < 0$$

$$\Rightarrow \alpha \in (a, b)$$



$$f(a) = +ve$$

$$f(b) = -ve$$

$$\therefore f(a) \cdot f(b) < 0$$

$$\Rightarrow \alpha \in (a, b)$$

if $f(a) \cdot f(b) < 0$ then there exist atleast one Root α of $f(x)$ in (a, b) s.t. $f(\alpha) = 0$

OR

if product of $f(a)$ and $f(b)$ of opposite signs then there is one root α of $f(x)$ in (a, b) Spiral

if $f(a)$ & $f(b)$ are of opposite sign then $\exists$ at least one root α of $f(x)$ b/w a & b

Q:) one of the root of $xe^x - \cos x = 0$ lies in b/w

- a.) $(-1, 0)$ $f(x) = xe^x - \cos x$ $\rightarrow f(-1) = -e^{-1} - \cos(-1) = -1/e - \cos 1 = -ve$
 b.) $(0, 1)$ ✓ checking option A $\rightarrow f(0) = 0 - \cos 0 = -1 = -ve$
 c.) $(1, 2)$
 d.) $(2, 5)$ i.e. $\alpha \notin (-1, 0)$

checking option B $\rightarrow f(0) = -ve$ (subsets)
 $f(1) = e - \cos(1) = e - (735/10^4) = +ve$

$\therefore f(0)$ & $f(1)$ are of opposite sign. so by
 B theorem $\alpha \in (0, 1)$

Q:) The root of the equation $x^4 - x^3 - x^2 - 4 = 0$ that lies in the interval $[1, 9]$ will be

sol:) let $f(x) = x^4 - x^3 - x^2 - 4$ $\rightarrow f(1) = 1^4 - 1^3 - 1^2 - 4 = -ve$
 $f(9) = 9^4 - 9^3 - 9^2 - 4 = +ve$

By Bolzano the $\alpha \in (1, 9)$
 let

$$x_1 = \frac{1+9}{2} = 5, \text{ \& } f(5) = 5^4 - 5^3 - 5^2 - 4 = +ve$$

so By Bolzano the $(1, 5)$

$x_2 = \frac{1+5}{2} = 3$ & $f(3) = 3^4 - 3^3 - 3^2 - 4 = +ve$
 so By Bolzano the $\alpha \in (1, 3)$ now

$$x_3 = \frac{1+3}{2} = 2 \text{ \& } f(2) = 2^4 - 2^3 - 2^2 - 4 = 0 \text{ \& } \alpha = 2$$

(ii) How many times we have used B. In to find the sol?

Ans.) Three times.

Indefinite Integration:

Collection of all the antiderivatives is called ind.

Antiderivative.

Derivative

$$\frac{d}{dx}(x^3) = 3x^2$$

$$(x^3 + 11) = 3x^2$$

$$\frac{d}{dx}(x^3 + c) = 3x^2$$

$$x^3 - 7 = 3x^2$$

$$\Rightarrow \int 3x^2 dx = (x^3 + c)$$

= family of antiderivatives

Methods of solving Indefinite Integration.

M-1 By Standard Result. (Easiest).

M-2 By Substitution. (Most Important).

M-3 By Parts. i.e. $\int u v dx = u \int v dx - \int \left\{ \frac{du}{dx} \int v dx \right\} dx + c$

M-4 By partial fraction.



Standard Result:-

$$1.) \text{ Power formula: } \int x^a dx = \frac{x^{a+1}}{a+1}, a \neq -1$$

$$2.) \int \frac{1}{x} dx = \log x + C$$

$$3.) \int a^x dx = \frac{a^x}{\log_e a} + C$$

$$4.) \int e^x dx = e^x + C$$

$$5.) \int \sin x dx = -\cos x + C$$

$$6.) \int \cos x dx = \sin x + C$$

$$7.) \int \tanh x dx = \log \cosh x + C$$

$$8.) \int \coth x dx = \log \sinh x + C$$

$$9.) \int \operatorname{sech} x dx = \log (\cosh x + \sinh x) + C$$

$$10.) \int (\operatorname{cosech} x) dx = \log (\cosh x - \sinh x) + C$$

$$11.) \int \operatorname{sech}^2 x dx = \tanh x + C$$

$$12.) \int \operatorname{cosech}^2 x dx = -\coth x + C$$

$$13.) \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$14.) \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$$

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$$15.) \int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$16.) \int \frac{dx}{x^2-a^2} = \frac{1}{2a} \log \left(\frac{x-a}{x+a} \right) + C$$

$$17.) \int \frac{dx}{a^2-x^2} = \frac{1}{2a} \log \left(\frac{a+x}{a-x} \right) + C$$

$$18.) \int \frac{dx}{\sqrt{x^2+a^2}} = \log \left(x + \sqrt{x^2+a^2} \right) + C$$

$$19.) \int \frac{dx}{\sqrt{x^2-a^2}} = \log \left(x + \sqrt{x^2-a^2} \right) + C$$

$$20.) \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$$

$$21.) \int \sqrt{a^2+x^2} dx = \frac{x}{2} \sqrt{x^2+a^2} + \frac{a^2}{2} \log \left(x + \sqrt{x^2+a^2} \right) + C$$

$$22.) \int \sqrt{x^2-a^2} dx = \frac{x}{2} \sqrt{x^2-a^2} - \frac{a^2}{2} \log \left(x + \sqrt{x^2-a^2} \right) + C$$

$$23.) \int \sqrt{a^2-x^2} dx = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \left(\sin^{-1} \left(\frac{x}{a} \right) \right) + C$$

$$24.) \int \sec x \tan x = \sec x + C$$

$$25.) \int (\sec x \cot x) dx = -\cot x + C$$

$$26.) \int \frac{f'(x)}{f(x)} dx = \log f(x) + C$$

$$27.) \int e^{mx} \{f(x) + f'(x)\} dx = e^{mx} f(x) + C$$

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